Worked Solutions to Concentration Inequalities (Boucheron, Lugosi, Massart)

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Chapter 2 — Basic Inequalities

Exercise 2.1

Central idea. A median minimizes expected absolute deviation: $\mathbb{E}|Z - M_Z| \le \mathbb{E}|Z - m|$ for all m, in particular for $m = \mathbb{E}Z$. Combining this with Jensen's inequality ($|M_Z - \mathbb{E}Z| \le \mathbb{E}|Z - M_Z|$) and Cauchy–Schwarz ($\mathbb{E}|Z - \mathbb{E}Z| \le \sqrt{\mathrm{Var}(Z)}$) gives the bound.

Solution. Let $M = M_Z$ be a median of Z, so $\mathbb{P}\{Z \ge M\} \ge 1/2$ and $\mathbb{P}\{Z \le M\} \ge 1/2$. Since Z is square-integrable, $\mathbb{E}|Z| < \infty$.

First, $|M - \mathbb{E}Z| = |\mathbb{E}(M - Z)| \le \mathbb{E}|Z - M|$ by Jensen's inequality.

Next, we show M minimizes $f(m) = \mathbb{E}|Z - m|$ over $m \in \mathbb{R}$. Let $m > M$. Decomposing on $\{Z \le M\}$,

$\{Z \geq m\}$, $\{M < Z < m\}$:

$$|Z-m| - |Z-M| = \begin{cases} m-M & Z \leq M \\ -(m-M) & Z \geq m \\ m+M-2Z & M < Z < m \end{cases}$$

and $|m+M-2Z| \leq m-M$ on $\{M < Z < m\}$. Hence

$$\begin{aligned} f(m) - f(M) &\geq (m-M)\mathbb{P}\{Z \leq M\} - (m-M)\mathbb{P}\{Z \geq m\} - (m-M)\mathbb{P}\{M < Z < m\} \\ &= (m-M)\big[\mathbb{P}\{Z \leq M\} - \mathbb{P}\{Z > M\}\big] \\ &= (m-M)\big[2\mathbb{P}\{Z \leq M\} - 1\big] \geq 0, \end{aligned}$$

using $\{Z \geq m\} \cup \{M < Z < m\} = \{Z > M\}$ (disjoint) and $\mathbb{P}\{Z \leq M\} \geq 1/2$. Similarly, for $m < M$,

$$f(m) - f(M) \geq (M-m)\big[2\mathbb{P}\{Z \geq M\} - 1\big] \geq 0.$$

Thus $\mathbb{E}|Z-M| \leq \mathbb{E}|Z-\mathbb{E}Z|$.

Finally, by Cauchy–Schwarz (or Jensen), $\mathbb{E}|Z-\mathbb{E}Z| \leq \sqrt{\mathbb{E}[(Z-\mathbb{E}Z)^2]} = \sqrt{\mathrm{Var}(Z)}$.

Chaining the three inequalities:

$$|M_Z - \mathbb{E}Z| \leq \mathbb{E}|Z-M_Z| \leq \mathbb{E}|Z-\mathbb{E}Z| \leq \sqrt{\mathrm{Var}(Z)}.$$

Exercise 2.2

Central idea. The sub-Gaussian-type tail bound makes X integrable, so $|M_X - \mathbb{E}X| \leq \mathbb{E}|X-M_X|$. Write the mean absolute deviation as a tail integral $\int_0^\infty \mathbb{P}\{|X-M_X| > t\}\,dt$ and bound it two ways: directly (giving $a\sqrt{b\pi}/2$) and via $\mathbb{E}[Y] \leq \sqrt{\mathbb{E}[Y^2]}$ with $\mathbb{E}[Y^2] = \int_0^\infty 2t\,\mathbb{P}\{Y > t\}\,dt$ (giving $\sqrt{ab}$).

Solution. Let $M = M_X$ and $Y = |X - M_X| \geq 0$. By hypothesis, $\mathbb{P}\{Y > t\} \leq ae^{-t^2/b}$ for all $t > 0$.

First, $\mathbb{E}Y = \int_0^\infty \mathbb{P}\{Y > t\}\,dt \leq \int_0^\infty ae^{-t^2/b}\,dt < \infty$, so $\mathbb{E}|X| \leq |M| + \mathbb{E}Y < \infty$ and $\mathbb{E}X$ exists. By Jensen,

$$|M_X - \mathbb{E}X| = |\mathbb{E}(M_X - X)| \leq \mathbb{E}|X - M_X| = \mathbb{E}Y.$$

Bound 1. Using $\mathbb{P}\{Y > t\} \leq ae^{-t^2/b}$ directly:

$$\mathbb{E}Y \leq \int_0^\infty ae^{-t^2/b}\,dt = a\sqrt{b}\int_0^\infty e^{-u^2}\,du = \frac{a\sqrt{b\pi}}{2} \leq a\sqrt{\frac{b\pi}{2}},$$

via $u = t/\sqrt{b}$ and $\int_0^\infty e^{-u^2}du = \sqrt{\pi}/2$.

Bound 2. By Cauchy–Schwarz, $\mathbb{E}Y \leq \sqrt{\mathbb{E}[Y^2]}$, and

$$\mathbb{E}[Y^2] = \int_0^\infty 2t\,\mathbb{P}\{Y > t\}\,dt \leq \int_0^\infty 2t\,ae^{-t^2/b}\,dt = 2a \cdot \frac{b}{2} = ab,$$

since $\int_0^\infty 2te^{-t^2/b}dt = b$ (substitute $s = t^2/b$). Hence $\mathbb{E}Y \leq \sqrt{ab}$.

Therefore $|M_X - \mathbb{E}X| \leq \min\big(\sqrt{ab},\, a\sqrt{b\pi}/2\big) \leq \min\big(\sqrt{ab},\, a\sqrt{b\pi/2}\big)$.

Exercise 2.3

Central idea. (Chebyshev–Cantelli.) Shift Y by a free parameter $a > 0$ before applying Chebyshev: $\mathbb{P}\{Y - \mu \geq t\} = \mathbb{P}\{Y - \mu + a \geq t + a\} \leq (\sigma^2 + a^2)/(t + a)^2$. Optimizing over a (best choice $a = \sigma^2/t$) yields $\sigma^2/(\sigma^2 + t^2)$, strictly better than Chebyshev's σ^2/t^2.

Solution. If $\mathrm{Var}(Y) = \infty$, the right-hand side is 1 and there is nothing to prove. If $\mathrm{Var}(Y) = 0$, then $Y = \mathbb{E}Y$ a.s. and both sides are 0. Assume $0 < \sigma^2 := \mathrm{Var}(Y) < \infty$ (hence $\mathbb{E}|Y| < \infty$), and write $\mu = \mathbb{E}Y$.

For any $a > 0$,

$$\mathbb{P}\{Y - \mu \geq t\} = \mathbb{P}\{Y - \mu + a \geq t + a\} \leq \frac{\mathbb{E}[(Y - \mu + a)^2]}{(t + a)^2}$$

by Markov's inequality applied to $(Y - \mu + a)^2$. Since $\mathbb{E}[Y - \mu] = 0$,

$$\mathbb{E}[(Y - \mu + a)^2] = \mathbb{E}[(Y - \mu)^2] + 2a\,\mathbb{E}[Y - \mu] + a^2 = \sigma^2 + a^2.$$

Thus $\mathbb{P}\{Y - \mu \geq t\} \leq \frac{\sigma^2 + a^2}{(t+a)^2}$ for all $a > 0$.

Minimize $f(a) = \frac{\sigma^2 + a^2}{(t+a)^2}$ over $a > 0$:

$$f'(a) = \frac{2a(t + a)^2 - 2(t + a)(\sigma^2 + a^2)}{(t + a)^4} = \frac{2(at - \sigma^2)}{(t + a)^3},$$

so f decreases on $(0, \sigma^2/t)$ and increases on $(\sigma^2/t, \infty)$, with minimum at $a = \sigma^2/t$. Substituting,

$$f(\sigma^2/t) = \frac{\sigma^2 + \sigma^4/t^2}{(t + \sigma^2/t)^2} = \frac{\sigma^2(t^2 + \sigma^2)/t^2}{(t^2 + \sigma^2)^2/t^2} = \frac{\sigma^2}{\sigma^2 + t^2}.$$

Hence $\mathbb{P}\{Y - \mathbb{E}Y \geq t\} \leq \frac{\mathrm{Var}(Y)}{\mathrm{Var}(Y) + t^2}$.

Exercise 2.4

Central idea. (Paley–Zygmund.) Split $\mathbb{E}Y$ according to whether $Y \geq a\mathbb{E}Y$: the small part contributes at most $a\mathbb{E}Y$, and the large part is controlled by Cauchy–Schwarz as $\sqrt{\mathbb{E}[Y^2]\,\mathbb{P}\{Y \geq a\mathbb{E}Y\}}$. Rearranging isolates the tail probability.

Solution. If $\mathbb{E}Y = 0$, then $Y = 0$ a.s. (as $Y \geq 0$), so $\mathbb{P}\{Y \geq a\mathbb{E}Y\} = 1 \geq 0$ and the inequality holds. If $\mathbb{E}[Y^2] = \infty$, the right-hand side is 0 and there is nothing to prove. Assume $0 < \mathbb{E}Y < \infty$ and $\mathbb{E}[Y^2] < \infty$.

Write

$$\mathbb{E}Y = \mathbb{E}[Y\mathbf{1}_{\{Y \geq a\mathbb{E}Y\}}] + \mathbb{E}[Y\mathbf{1}_{\{Y < a\mathbb{E}Y\}}].$$

The second term is at most $a\mathbb{E}Y$. By Cauchy–Schwarz,

$$\mathbb{E}[Y\mathbf{1}_{\{Y \geq a\mathbb{E}Y\}}] \leq \sqrt{\mathbb{E}[Y^2]\,\mathbb{P}\{Y \geq a\mathbb{E}Y\}}.$$

Hence

$$\mathbb{E}Y \leq \sqrt{\mathbb{E}[Y^2]\,\mathbb{P}\{Y \geq a\mathbb{E}Y\}} + a\mathbb{E}Y,$$

i.e.,

$$(1 - a)\mathbb{E}Y \leq \sqrt{\mathbb{E}[Y^2]\,\mathbb{P}\{Y \geq a\mathbb{E}Y\}}.$$

Squaring both sides (all terms nonnegative, $a \in (0, 1)$) gives

$$\mathbb{P}\{Y \geq a\mathbb{E}Y\} \geq \frac{(1 - a)^2(\mathbb{E}Y)^2}{\mathbb{E}[Y^2]}.$$

Exercise 2.5

Central idea. The inequality is false as stated: for $Y \equiv 2t$ the moment side equals 2 while the Chernoff side collapses to 1. The flaw is that the minimum is taken over $q \geq 1$ only, excluding the trivial bound $\mathbb{P}\{Y \geq t\} \leq 1$ (the $q = 0$ case). Including $q = 0$ fixes it: with $m = \min(1, \inf_{q\geq 1} \mathbb{E}[Y^q]t^{-q})$, term-by-term integration of the exponential series gives $\mathbb{E}e^{\lambda(Y-t)} \geq m$ for every $\lambda > 0$.

Solution. The stated inequality

$$\min_{q\geq 1} \mathbb{E}[Y^q]t^{-q} \leq \inf_{\lambda>0} \mathbb{E}e^{\lambda(Y-t)}$$

is false in general. Counterexample: $Y \equiv 2t$ constant. Then $\mathbb{E}[Y^q]t^{-q} = 2^q$, so the left side is 2, while $\mathbb{E}e^{\lambda(Y-t)} = e^{\lambda t} \to 1$ as $\lambda \downarrow 0$, so the right side is 1.

The corrected statement includes $q = 0$. Set $m = \inf_{q\in\{0,1,2,\dots\}} \mathbb{E}[Y^q]t^{-q} = \min(1, \inf_{q\geq 1} \mathbb{E}[Y^q]t^{-q})$. Then $\mathbb{E}[Y^q] \geq mt^q$ for every integer $q \geq 0$ (for $q = 0$ since $m \leq 1$). As $Y \geq 0$, monotone convergence justifies term-by-term integration:

$$\mathbb{E}e^{\lambda(Y-t)} = e^{-\lambda t} \sum_{q=0}^{\infty} \frac{\lambda^q \, \mathbb{E}[Y^q]}{q!} \geq e^{-\lambda t} \, m \sum_{q=0}^{\infty} \frac{(\lambda t)^q}{q!} = m, \qquad \lambda > 0.$$

Taking $\inf_{\lambda>0}$ gives $m \leq \inf_{\lambda>0} \mathbb{E}e^{\lambda(Y-t)}$. Since the Chernoff bound tends to 1 as $\lambda \downarrow 0$ (dominated convergence, when the mgf is finite near 0), this proves the stated inequality whenever the Chernoff bound is nontrivial (< 1).

Sharper form. With $m_1 = \inf_{q\geq 1} \mathbb{E}[Y^q]t^{-q}$, splitting the $q = 0$ term gives $\mathbb{E}e^{\lambda(Y-t)} \geq e^{-\lambda t} + m_1(1 - e^{-\lambda t})$. So whenever the Chernoff bound is below 1, m_1 is below it too — the $q \geq 1$ moment bounds win exactly where it matters. ■

Exercise 2.6

Central idea. Finiteness of $\mathbb{E}e^{\lambda Z}$ propagates downward ($\lambda_0 \in S$, $0 < \lambda < \lambda_0 \Rightarrow \lambda \in S$) via $e^{\lambda Z} \leq e^{\lambda_0 Z} + 1$. Log-convexity of the mgf follows from Hölder; C^∞ regularity from differentiation under the integral, dominated using that exponentials beat polynomials ($|Z|^k \leq C(e^{\delta Z} + e^{-\delta Z})$). At $\lambda = 0$, dominated convergence gives right-differentiability with derivative $\mathbb{E}Z = 0$. The Cramér transform's supremum can be restricted to I since $\psi_Z = +\infty$ beyond b and the value at 0 is a limit from I.

Solution. Let $M(\lambda) = \mathbb{E}e^{\lambda Z}$ and $\psi_Z(\lambda) = \log M(\lambda)$.

The set S. If $\lambda_0 \in S$ and $0 < \lambda < \lambda_0$, then for $Z \geq 0$, $e^{\lambda Z} \leq e^{\lambda_0 Z}$, and for $Z < 0$, $e^{\lambda Z} \leq 1$. Hence $e^{\lambda Z} \leq e^{\lambda_0 Z} + 1$, so $\mathbb{E}e^{\lambda Z} \leq \mathbb{E}e^{\lambda_0 Z} + 1 < \infty$, i.e., $\lambda \in S$. Thus S is either empty or an interval of the form $(0, b)$ or $(0, b]$ with $b = \sup S \in (0, \infty]$.

Convexity on $I = (0, b)$. For $\lambda_1, \lambda_2 \in I$ and $\theta \in (0, 1)$, Hölder's inequality gives

$$M(\theta\lambda_1 + (1-\theta)\lambda_2) = \mathbb{E}\big[e^{\theta\lambda_1 Z} e^{(1-\theta)\lambda_2 Z}\big] \leq M(\lambda_1)^\theta M(\lambda_2)^{1-\theta}.$$

Taking logarithms, ψ_Z is convex on I (all terms finite and $M > 0$).

C^∞ on I. Fix $\lambda_0 \in I$ and choose $\delta > 0$ with $[\lambda_0 - 2\delta, \lambda_0 + 2\delta] \subset I$. Since $|z|^k \leq C_{k,\delta}(e^{\delta z} + e^{-\delta z})$ for all $z \in \mathbb{R}$ (exponentials dominate polynomials), for $|\lambda - \lambda_0| \leq \delta$,

$$|Z|^k e^{\lambda Z} \leq C_{k,\delta}(e^{(\lambda_0+2\delta)Z} + e^{(\lambda_0-2\delta)Z}),$$

whose expectation is finite as both exponents lie in S. By dominated convergence, differentiation under the integral is justified to all orders:

$$M^{(k)}(\lambda_0) = \mathbb{E}\big[Z^k e^{\lambda_0 Z}\big], \qquad k = 1, 2, \dots$$

Hence $M \in C^\infty(I)$, and since $M > 0$, $\psi_Z = \log M \in C^\infty(I)$.

Behavior at 0 when $\mathbb{E}Z = 0$. (This assumes $\mathbb{E}|Z| < \infty$.) Clearly $\psi_Z(0) = \log 1 = 0$. For continuity at 0: if $0 < \lambda \le \delta < b$, then $e^{\lambda Z} \le e^{\delta Z} + 1$ (splitting on $\{Z \ge 0\}$, $\{Z < 0\}$), which is integrable, so $M(\lambda) \to 1$ by dominated convergence, i.e., $\psi_Z(\lambda) \to 0$.

For right-differentiability at 0:

$$\frac{M(h) - 1}{h} = \mathbb{E}\left[\frac{e^{hZ} - 1}{h}\right], \qquad 0 < h \le \delta.$$

For $Z \ge 0$, $h \mapsto \frac{e^{hZ}-1}{h}$ is nondecreasing (derivative $\frac{hZe^{hZ} - e^{hZ} + 1}{h^2} \ge 0$), so $\frac{e^{hZ}-1}{h} \le \frac{e^{\delta Z}-1}{\delta}$; for $Z < 0$, $\left|\frac{e^{hZ}-1}{h}\right| = \frac{1-e^{hZ}}{h} \le |Z|$ since $1 - e^{-y} \le y$ for $y \ge 0$. Thus $\left|\frac{e^{hZ}-1}{h}\right| \le \frac{e^{\delta Z}}{\delta} + |Z|$, integrable. By dominated convergence, $\frac{M(h)-1}{h} \to \mathbb{E}Z = 0$. Hence M is right-differentiable at 0 with $M'_+(0) = 0$, so ψ_Z is right-differentiable at 0 with $\psi'_{Z,+}(0) = M'_+(0)/M(0) = 0$. A similar domination shows $\psi'_Z(\lambda) = \mathbb{E}[Ze^{\lambda Z}]/\mathbb{E}[e^{\lambda Z}] \to 0$ as $\lambda \downarrow 0$, so $\psi_Z \in C^1[0, b)$ with $\psi'_Z(0) = 0$.

Cramér transform. The Cramér transform is $\sup_{\lambda \ge 0}(\lambda t - \psi_Z(\lambda))$. If $b < \infty$ and $\lambda > b$, then $\lambda \notin S$ (as $b = \sup S$), so $\psi_Z(\lambda) = +\infty$ and $\lambda t - \psi_Z(\lambda) = -\infty$, contributing nothing. If $b \in S$ (i.e. $S = (0, b]$), then M is left-continuous at b: for $\lambda \uparrow b$, $e^{\lambda Z} \to e^{bZ}$ pointwise with dominating integrable function $1 + e^{bZ}$, so $M(\lambda) \to M(b)$ by dominated convergence, and $\lambda t - \psi_Z(\lambda)$ at b is a limit of values from $I = (0, b)$. At $\lambda = 0$ the value is $0 - 0 = 0$, which equals $\lim_{\lambda \downarrow 0}(\lambda t - \psi_Z(\lambda))$ by continuity at 0. Hence

$$\sup_{\lambda \ge 0}\left(\lambda t - \psi_Z(\lambda)\right) = \sup_{\lambda \in (0,b)}\left(\lambda t - \psi_Z(\lambda)\right) = \psi_Z^*(t).$$

(If $b = \infty$, then $I = (0, \infty)$ and there is nothing to exclude.)

Exercise 2.7

Central idea. By scaling, reduce to $\sigma = 1$. Write $\mathbb{P}\{Z \ge t\}e^{t^2/2} = \int_0^\infty \phi(u)e^{-ut}\,du$ (shift $x = u + t$), which is $\le \int_0^\infty \phi = 1/2$ since $e^{-ut} \le 1$, with equality in the limit $t \to 0^+$ by dominated convergence. Hence the supremum is $1/2$, showing the Chernoff exponent $t^2/(2\sigma^2)$ is sharp up to the constant $1/2$.

Solution. If $Z \sim N(0, \sigma^2)$, then $W = Z/\sigma \sim N(0, 1)$ and

$$\mathbb{P}\{Z \ge t\}\exp\!\left(\tfrac{t^2}{2\sigma^2}\right) = \mathbb{P}\{W \ge t/\sigma\}\exp\!\left(\tfrac{(t/\sigma)^2}{2}\right).$$

As t ranges over $(0, \infty)$, so does $s = t/\sigma$; hence it suffices to prove the claim for $\sigma = 1$. Let $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

For $t > 0$,

$$\begin{aligned}
\mathbb{P}\{Z \ge t\}e^{t^2/2} &= e^{t^2/2}\int_t^\infty \phi(x)\,dx = \int_t^\infty \frac{1}{\sqrt{2\pi}}e^{-(x^2-t^2)/2}\,dx \\
&= \int_0^\infty \frac{1}{\sqrt{2\pi}}e^{-((u+t)^2-t^2)/2}\,du \quad (u = x - t) \\
&= \int_0^\infty \phi(u)\,e^{-ut}\,du \le \int_0^\infty \phi(u)\,du = \frac{1}{2},
\end{aligned}$$

since $e^{-ut} \le 1$ for $u, t \ge 0$. Thus $\sup_{t>0}\left(\mathbb{P}\{Z \ge t\}e^{t^2/2}\right) \le 1/2$.

For the reverse, as $t \downarrow 0$, $\phi(u)e^{-ut} \to \phi(u)$ pointwise with dominating integrable function ϕ, so by dominated convergence

$$\lim_{t \downarrow 0}\mathbb{P}\{Z \ge t\}e^{t^2/2} = \int_0^\infty \phi(u)\,du = \frac{1}{2}.$$

Hence the supremum over $t > 0$ is at least $1/2$. Combined with the upper bound, the supremum equals $1/2$ (not attained, but approached as $t \to 0^+$).

Exercise 2.8

Central idea. Each inequality is proved by the "derivative comparison" method: set $f = \text{LHS} - \text{RHS}$, verify $f(0) = f'(0) = 0$, and show $f''(u) \geq 0$ on the domain, which forces $f' \geq 0$ and hence $f \geq 0$. The second-derivative checks reduce to elementary polynomial inequalities.

Solution. In each case define $f(u) = \text{LHS} - \text{RHS}$; we show $f(0) = f'(0) = 0$ and $f''(u) \geq 0$.

(a) $f(u) = \frac{u^2}{2(1-u)} + \log(1-u) + u$ on $(0,1)$. Then $f(0) = 0$ and

$$f'(u) = \frac{u(2-u)}{2(1-u)^2} - \frac{1}{1-u} + 1 = \frac{u(2-u) - 2(1-u) + 2(1-u)^2}{2(1-u)^2} = \frac{u^2}{2(1-u)^2} \geq 0.$$

Since $f(0) = 0$ and $f' \geq 0$, $f(u) \geq 0$ on $(0,1)$, i.e., $-\log(1-u) - u \leq \frac{u^2}{2(1-u)}$.

(b) $f(u) = (1+u)\log(1+u) - u - \frac{u^2}{2(1+u/3)}$ for $u > 0$. Then $f(0) = 0$,

$$f'(u) = \log(1+u) - \frac{3u(6+u)}{2(3+u)^2}, \qquad f'(0) = 0,$$

$$f''(u) = \frac{1}{1+u} - \frac{27}{(3+u)^3}.$$

Now $f''(u) \geq 0 \iff (3+u)^3 \geq 27(1+u)$. With $g(u) = (3+u)^3 - 27(1+u)$: $g(0) = 0$ and $g'(u) = 3(3+u)^2 - 27 = 3u(u+6) \geq 0$ for $u \geq 0$, so $g(u) \geq 0$. Hence $f''(u) \geq 0$, so f' is nondecreasing with $f'(0) = 0$, giving $f'(u) \geq 0$, and f is nondecreasing with $f(0) = 0$, giving $f(u) \geq 0$. That is, $h(u) \geq \frac{u^2}{2(1+u/3)}$.

(c) $f(u) = 1 + u - \sqrt{1+2u} - \frac{u^2}{2(1+u)}$ for $u > 0$. Then $f(0) = 0$,

$$f'(u) = 1 - (1+2u)^{-1/2} - \frac{u(2+u)}{2(1+u)^2}, \qquad f'(0) = 0,$$

$$f''(u) = (1+2u)^{-3/2} - (1+u)^{-3}.$$

Now $f''(u) \geq 0 \iff (1+u)^3 \geq (1+2u)^{3/2} \iff (1+u)^2 \geq 1+2u \iff u^2 \geq 0$, true. Hence $f''(u) \geq 0$, so $f'(u) \geq f'(0) = 0$, so $f(u) \geq f(0) = 0$. That is, $h_1(u) \geq \frac{u^2}{2(1+u)}$.

Exercise 2.9

Central idea. For $X \geq 0$, the one-sided mgf $\mathbb{E}e^{-\lambda(X-\mathbb{E}X)}$ is controlled by the quadratic Taylor bound $e^{-y} \leq 1 - y + y^2/2$ $(y \geq 0)$, giving a sub-Gaussian-type estimate with variance proxy $\mathbb{E}[X^2]$ rather than $\text{Var}(X)$. Independence factorizes the mgf of the sum, and optimizing the Chernoff bound over λ yields the Gaussian tail $\exp(-t^2/(2v))$.

Solution. Let $X \geq 0$ with $\mathbb{E}[X^2] < \infty$ (hence $\mathbb{E}X \leq \sqrt{\mathbb{E}[X^2]} < \infty$). For $y \geq 0$, $e^{-y} \leq 1 - y + y^2/2$: indeed $g(y) = 1 - y + y^2/2 - e^{-y}$ satisfies $g(0) = g'(0) = 0$ and $g''(y) = 1 - e^{-y} \geq 0$, so $g \geq 0$ on $[0,\infty)$.

For $\lambda > 0$, with $y = \lambda X \geq 0$,

$$\mathbb{E}e^{-\lambda X} \leq \mathbb{E}\left[1 - \lambda X + \tfrac{\lambda^2}{2}X^2\right] = 1 - \lambda\mathbb{E}X + \tfrac{\lambda^2}{2}\mathbb{E}[X^2].$$

Therefore, using $1 + z \leq e^z$,

$$\mathbb{E}e^{-\lambda(X-\mathbb{E}X)} = e^{\lambda\mathbb{E}X}\mathbb{E}e^{-\lambda X} \leq e^{\lambda\mathbb{E}X}(1 - \lambda\mathbb{E}X + \tfrac{\lambda^2}{2}\mathbb{E}[X^2]) \leq e^{\lambda^2\mathbb{E}[X^2]/2}.$$

Now let $X_1, \ldots, X_n \geq 0$ be independent, $S = \sum_{i=1}^n (X_i - \mathbb{E}X_i)$, $v = \sum_{i=1}^n \mathbb{E}[X_i^2]$. If $v = 0$, then $X_i = 0$ a.s. for all i, so $S = 0$ and $\mathbb{P}\{S \leq -t\} = 0$ for $t > 0$. Assume $v > 0$. By independence,

$$\mathbb{E}e^{-\lambda S} = \prod_{i=1}^n \mathbb{E}e^{-\lambda(X_i - \mathbb{E}X_i)} \leq \prod_{i=1}^n e^{\lambda^2\mathbb{E}[X_i^2]/2} = e^{\lambda^2 v/2}.$$

By Chernoff's bound, for $t > 0$ and $\lambda > 0$,

$$\mathbb{P}\{S \le -t\} = \mathbb{P}\{e^{-\lambda S} \ge e^{\lambda t}\} \le e^{-\lambda t}\,\mathbb{E}e^{-\lambda S} \le \exp\Big(\tfrac{\lambda^2 v}{2} - \lambda t\Big).$$

Minimizing $\frac{\lambda^2 v}{2} - \lambda t$ over $\lambda > 0$ gives $\lambda = t/v$ and minimum value $-\frac{t^2}{2v}$. Hence

$$\mathbb{P}\{S \le -t\} \le \exp\left(-\frac{t^2}{2v}\right).$$

Exercise 2.10

Central idea. (Angluin–Valiant.) Direct Chernoff bounds using $1 - p_i + p_i e^{\pm\lambda} \le \exp(p_i(e^{\pm\lambda} - 1))$. The upper tail optimizes to $\exp(-np \cdot h(\varepsilon/p))$ with $h(\delta) = (1+\delta)\log(1+\delta) - \delta$, and Exercise 2.8(b) gives $h(\delta) \ge \delta^2/(2(1+\delta/3)) \ge \varepsilon^2/3$ after using $p \le 1$, $\varepsilon \le 1 - p$. The lower tail similarly gives $\exp(-np \cdot k(\varepsilon/p))$ with $k(\delta) = \delta + (1-\delta)\log(1-\delta) \ge \delta^2/2$, and $p \le 1$ converts $n\varepsilon^2/(2p)$ to the claimed $np\varepsilon^2/2$.

Solution. Let $\varepsilon > 0$. If $p = 0$, then $S_n = 0$ a.s. and both probabilities are 0. Assume $p > 0$.

Upper tail. If $\varepsilon > 1 - p$, then $np + n\varepsilon > n \ge S_n$, so $\mathbb{P}\{S_n - np \ge n\varepsilon\} = 0$. Assume $0 < \varepsilon \le 1 - p$. For $\lambda > 0$, $\mathbb{E}e^{\lambda X_i} = 1 - p_i + p_i e^{\lambda} \le \exp(p_i(e^{\lambda} - 1))$ (using $1 + x \le e^x$). By independence and Chernoff,

$$\begin{aligned}\mathbb{P}\{S_n - np \ge n\varepsilon\} &\le e^{-\lambda n\varepsilon}\,\mathbb{E}e^{\lambda(S_n - np)} \\ &\le \exp(np(e^{\lambda} - 1) - \lambda np - \lambda n\varepsilon).\end{aligned}$$

Minimizing $\phi(\lambda) = np[e^{\lambda} - 1 - \lambda(1+\varepsilon/p)]$: $\phi'(\lambda) = 0$ gives $e^{\lambda} = 1 + \varepsilon/p$, i.e., $\lambda = \log(1+\delta)$ with $\delta = \varepsilon/p > 0$; since $\phi'' > 0$ this is the minimizer, yielding

$$\mathbb{P}\{S_n - np \ge n\varepsilon\} \le \exp(-np \cdot h(\delta)), \quad h(\delta) = (1+\delta)\log(1+\delta) - \delta.$$

By Exercise 2.8(b), $h(\delta) \ge \frac{\delta^2}{2(1+\delta/3)}$. Hence

$$h(\varepsilon/p) \ge \frac{(\varepsilon/p)^2}{2(1 + \varepsilon/(3p))} = \frac{\varepsilon^2}{2p^2 + 2p\varepsilon/3} \ge \frac{\varepsilon^2}{3},$$

because $2p^2 + 2p\varepsilon/3 \le 2 + 2/3 = 8/3 < 3$ (using $0 < p \le 1$, $0 < \varepsilon \le 1$). Therefore $\mathbb{P}\{S_n - np \ge n\varepsilon\} \le e^{-np\varepsilon^2/3}$.

Lower tail. For $\lambda > 0$, $\mathbb{E}e^{-\lambda X_i} = 1 - p_i + p_i e^{-\lambda} \le \exp(p_i(e^{-\lambda} - 1))$, so

$$\mathbb{P}\{S_n - np \le -n\varepsilon\} \le \exp(np(e^{-\lambda} - 1) + \lambda np - \lambda n\varepsilon).$$

Maximizing $g(\lambda) = -(e^{-\lambda} - 1 + \lambda) + \lambda\delta$ (with $\delta = \varepsilon/p$): $g'(\lambda) = e^{-\lambda} - (1-\delta) = 0$ gives $\lambda = -\log(1-\delta)$, valid for $\delta \in (0,1)$, i.e., $0 < \varepsilon < p$; $g'' < 0$ confirms a maximum, with value $g(-\log(1-\delta)) = \delta + (1-\delta)\log(1-\delta) =: k(\delta)$. Thus, for $0 < \varepsilon < p$,

$$\mathbb{P}\{S_n - np \le -n\varepsilon\} \le \exp(-np \cdot k(\varepsilon/p)).$$

Now $k(\delta) \ge \delta^2/2$ on $(0,1)$: with $\phi(\delta) = k(\delta) - \delta^2/2$, $\phi(0) = 0$ and $\phi'(\delta) = -\log(1-\delta) - \delta \ge 0$. So

$$\exp(-np\,k(\varepsilon/p)) \le \exp\left(-\frac{n\varepsilon^2}{2p}\right) \le \exp\left(-\frac{np\varepsilon^2}{2}\right),$$

using $1/(2p) \ge p/2 \iff p^2 \le 1$.

If $p \le \varepsilon \le 1$: then $\{S_n - np \le -n\varepsilon\} \subseteq \{S_n = 0\}$, and $\mathbb{P}\{S_n = 0\} = \prod_i (1 - p_i) \le e^{-\sum_i p_i} = e^{-np} \le e^{-np\varepsilon^2/2}$ since $\varepsilon^2 \le 1 < 2$. If $\varepsilon > 1$, the event is empty. This completes the lower tail.

Remark. The intermediate bounds $\exp(-np \cdot h(\varepsilon/p))$ and $\exp(-np \cdot k(\varepsilon/p))$ give the sharper multiplicative form (with $\delta = \varepsilon$ for $S_n \ge (1+\varepsilon)np$). The same proof works for $[0,1]$-valued X_i: the mgf bound $\mathbb{E}e^{\lambda X_i} \le 1 - p_i + p_i e^{\lambda}$ becomes an inequality by convexity of $x \mapsto e^{\lambda x}$.

Exercise 2.11

Central idea. The Chernoff bound $\mathbb{P}\{B \geq an\} \leq \inf_{\lambda>0}[(1-p+pe^{\lambda})e^{-\lambda a}]^n$, optimized explicitly: the minimizer $e^{\lambda} = a(1-p)/(p(1-a))$ yields the KL-divergence (method-of-types) form. The cruder e^{a-p} bound follows from $\log(1+z) \leq z$. The lower tail is the same argument with $\lambda < 0$.

Solution. Let $B \sim \text{Binomial}(n,p)$ and $p \leq a < 1$. If $p = 0$, then $B = 0$ a.s. and $\mathbb{P}\{B > an\} = 0$; if $a = p$, the first bound equals 1. Assume $0 < p \leq a < 1$ (the case $p < a$ below; $a = p$ is trivial).

For $\lambda > 0$, by Markov's inequality, $\mathbb{P}\{B \geq an\} \leq \mathbb{E}[e^{\lambda B}]/e^{\lambda an} = [(1-p+pe^{\lambda})e^{-\lambda a}]^n$ (the same proof works for $\geq$, hence for $>$). Minimize $f(\lambda) = \log(1-p+pe^{\lambda}) - \lambda a$:

$$f'(\lambda) = \frac{pe^{\lambda}}{1-p+pe^{\lambda}} - a = 0 \iff pe^{\lambda}(1-a) = a(1-p) \iff e^{\lambda} = \frac{a(1-p)}{p(1-a)} > 1,$$

valid since $a > p$. Since $f'' > 0$, this is the minimizer. Substituting,

$$1-p+pe^{\lambda^*} = 1-p+\frac{a(1-p)}{1-a} = \frac{1-p}{1-a}, \qquad e^{\lambda^* a} = \left(\frac{a(1-p)}{p(1-a)}\right)^a,$$

so with $D(a\|p) = a\log\frac{a}{p} + (1-a)\log\frac{1-a}{1-p}$,

$$\mathbb{P}\{B > an\} \leq \mathbb{P}\{B \geq an\} \leq \left[\left(\frac{p}{a}\right)^a \left(\frac{1-p}{1-a}\right)^{1-a}\right]^n = e^{-nD(a\|p)}.$$

For the second inequality, $(1-a)\log\frac{1-p}{1-a} \leq a-p$: indeed $\psi(a) = a-p-(1-a)\log\frac{1-p}{1-a}$ satisfies $\psi(p) = 0$ and $\psi'(a) = \log\frac{1-p}{1-a}$, so ψ decreases on $(0,p]$ and increases on $[p,1)$, giving $\psi \geq 0$. Exponentiating yields $\left(\frac{1-p}{1-a}\right)^{1-a} \leq e^{a-p}$, hence

$$\mathbb{P}\{B > an\} \leq \left[\left(\frac{p}{a}\right)^a e^{a-p}\right]^n.$$

For $0 < a < p$ (if $p = 1$, then $B = n$ a.s. and $\mathbb{P}\{B \leq an\} = 0$; assume $p < 1$), use $\lambda = -\mu < 0$: $\mathbb{P}\{B \leq an\} \leq [(1-p+pe^{-\mu})e^{\mu a}]^n$. Minimizing $g(\mu) = \log(1-p+pe^{-\mu})+\mu a$ gives $e^{\mu^*} = p(1-a)/(a(1-p)) > 1$ (since $p > a$), and the same algebra yields $\mathbb{P}\{B \leq an\} \leq [(p/a)^a((1-p)/(1-a))^{1-a}]^n$. The bound $((1-p)/(1-a))^{1-a} \leq e^{a-p}$ holds for $a < p$ as well since $\psi(a) > \psi(p) = 0$ there.

Exercise 2.12

Central idea. For $p \geq 1/2$, the Bernoulli log-MGF is sub-Gaussian on $\lambda \geq 0$: $\log \mathbb{E}e^{\lambda(X-p)} \leq \lambda^2 p(1-p)/2$. Writing the deficit as $\phi(\lambda)$, $\phi(0) = \phi'(0) = 0$ and $\phi''(\lambda) \leq 0$ reduces to $(1-p+pe^{\lambda})^2 \geq e^{\lambda}$, which holds for $p \geq 1/2$ since $g(s) = (1-p+ps)^2 - s$ is convex with $g(1) = 0$, $g'(1) = 2p-1 \geq 0$. The lower tail follows by applying this to $1 - X_i \sim \text{Bernoulli}(1-p)$.

Solution. Let $\varepsilon > 0$ and $X \sim \text{Bernoulli}(p)$ with $p \geq 1/2$. If $p = 1$, then $B = n$ a.s. and $\mathbb{P}\{B - np \geq n\varepsilon\} = 0$; assume $p < 1$. Define

$$\phi(\lambda) = \log(1-p+pe^{\lambda}) - \lambda p - \frac{\lambda^2}{2}p(1-p), \qquad \lambda \geq 0,$$

so that $\log \mathbb{E}e^{\lambda(X-p)} \leq \lambda^2 p(1-p)/2$ is $\phi(\lambda) \leq 0$. Now $\phi(0) = 0$,

$$\phi'(\lambda) = \frac{pe^{\lambda}}{1-p+pe^{\lambda}} - p - \lambda p(1-p), \quad \phi'(0) = 0,$$

$$\phi''(\lambda) = p(1-p)\left[\frac{e^{\lambda}}{(1-p+pe^{\lambda})^2} - 1\right].$$

Thus $\phi''(\lambda) \leq 0$ iff $e^{\lambda} \leq (1-p+pe^{\lambda})^2$. With $s = e^{\lambda} \geq 1$ and $g(s) = (1-p+ps)^2 - s$: $g(1) = 0$, $g'(s) = 2p(1-p+ps) - 1$ with $g'(1) = 2p-1 \geq 0$, and $g''(s) = 2p^2 > 0$. Since g' is strictly increasing and

$g'(1) \geq 0$, g is increasing on $[1, \infty)$, so $g(s) \geq 0$ for $s \geq 1$, with strict inequality for $s > 1$. Hence $\phi''(\lambda) < 0$ for $\lambda > 0$, so ϕ' is strictly decreasing, $\phi'(\lambda) < 0$ for $\lambda > 0$, and $\phi(\lambda) < 0$ for $\lambda > 0$:

$$\mathbb{E}e^{\lambda(X-p)} < e^{\lambda^2 p(1-p)/2}, \qquad \lambda > 0.$$

For $B = \sum_{i=1}^{n} X_i$ with X_i i.i.d. Bernoulli(p), $\mathbb{E}e^{\lambda(B-np)} < e^{n\lambda^2 p(1-p)/2}$, so by Chernoff,

$$\mathbb{P}\{B - np \geq n\varepsilon\} < \exp\Big(\tfrac{n\lambda^2 p(1-p)}{2} - \lambda n\varepsilon\Big).$$

With $\lambda = \varepsilon/(p(1-p)) > 0$, this is $< \exp(-n\varepsilon^2/(2p(1-p)))$.

If $p \leq 1/2$ and $Y_i = 1 - X_i \sim$ Bernoulli($1-p$) with $1-p \geq 1/2$, then $\{B - np \leq -n\varepsilon\} = \{\sum_i Y_i - n(1-p) \geq n\varepsilon\}$, and the upper-tail bound applied to $\sum_i Y_i$ gives $\mathbb{P}\{B - np \leq -n\varepsilon\} < \exp(-n\varepsilon^2/(2(1-p)p))$.

Exercise 2.13

Central idea. $\{\sqrt{B} - \sqrt{np} \geq \varepsilon\sqrt{n}\} = \{B \geq n(\sqrt{p} + \varepsilon)^2\}$ reduces each tail to a binomial deviation at $a = (\sqrt{p} \pm \varepsilon)^2$, controlled by the KL Chernoff bound of Exercise 2.11. The analytic core is $D((\sqrt{p}+\varepsilon)^2 \| p) > 2\varepsilon^2$ (upper) and $D((\sqrt{p} - \varepsilon)^2 \| p) > \varepsilon^2$ (lower), proved from $\ln(1+x) \geq \frac{2x}{2+x}$ $(x > 0)$ and $\ln(1-y) \geq \frac{-y}{1-y}$ $(0 \leq y < 1)$.

Solution. Let $\varepsilon > 0$, $u = \sqrt{p} \in [0, 1]$, $B \sim$ Binomial(n, p). Write $D(a\|p) = a \ln \frac{a}{p} + (1-a) \ln \frac{1-a}{1-p}$. Exercise 2.11 gives $\mathbb{P}\{B \geq na\} \leq e^{-nD(a\|p)}$ (its Chernoff proof works for $\geq$) and $\mathbb{P}\{B \leq na\} \leq e^{-nD(a\|p)}$.

Upper tail. $\{\sqrt{B} - \sqrt{np} \geq \varepsilon\sqrt{n}\} = \{B \geq n(u+\varepsilon)^2\}$. If $u = 0$ or $u + \varepsilon > 1$, this probability is 0. Assume $0 < u < 1$ and $u + \varepsilon \leq 1$, and set $a = (u+\varepsilon)^2 \in (p, 1]$. If $a = 1$ (i.e. $\varepsilon = 1 - u$), $D(1\|p) = -\ln p = -2\ln u > 2(1-u)^2 = 2\varepsilon^2$ since $-\ln(1-v) > v \geq v^2$. Assume $a < 1$. Then

$$D(a\|p) = 2(u+\varepsilon)^2 \ln(1 + \tfrac{\varepsilon}{u}) + (1-a) \ln\Big(\tfrac{1-a}{1-p}\Big).$$

Since $\ln(1+z) < z$ for $z > 0$, $(1-a) \ln \frac{1-a}{1-p} = -(1-a) \ln \frac{1-p}{1-a} > -(a-p) = -2u\varepsilon - \varepsilon^2$. Since $\ln(1+x) \geq \frac{2x}{2+x}$ for $x > 0$,

$$2(u+\varepsilon)^2 \ln(1 + \tfrac{\varepsilon}{u}) \geq \frac{4\varepsilon(u+\varepsilon)^2}{2u+\varepsilon}.$$

Hence $D(a\|p) > \frac{4\varepsilon(u+\varepsilon)^2}{2u+\varepsilon} - 2u\varepsilon - \varepsilon^2 \geq 2\varepsilon^2$, because the difference equals $\frac{\varepsilon^3}{2u+\varepsilon} > 0$. Therefore

$$\mathbb{P}\{\sqrt{B} - \sqrt{np} \geq \varepsilon\sqrt{n}\} \leq e^{-nD(a\|p)} < e^{-2n\varepsilon^2}.$$

Lower tail. $\{\sqrt{B} - \sqrt{np} \leq -\varepsilon\sqrt{n}\} = \{\sqrt{B} \leq \sqrt{n}(u - \varepsilon)\}$. If $\varepsilon > u$ or $p \in \{0, 1\}$, this probability is 0. Assume $0 < \varepsilon \leq u < 1$ and set $a' = (u-\varepsilon)^2 < p$. If $a' = 0$ (i.e. $\varepsilon = u$), $D(0\|p) = -\ln(1-p) = -\ln(1-\varepsilon^2) > \varepsilon^2$. Assume $a' > 0$. Then

$$D(a'\|p) = 2(u-\varepsilon)^2 \ln(1 - \tfrac{\varepsilon}{u}) + (1-a') \ln\Big(\tfrac{1-a'}{1-p}\Big).$$

Since $\ln(1-y) \geq \frac{-y}{1-y}$ for $0 \leq y < 1$, $2(u-\varepsilon)^2 \ln(1 - \varepsilon/u) \geq -2\varepsilon(u-\varepsilon)$. Writing $\frac{1-a'}{1-p} = 1 + \frac{p-a'}{1-p}$ and using $\ln(1+z) \geq \frac{2z}{2+z}$,

$$(1-a') \ln\Big(\tfrac{1-a'}{1-p}\Big) \geq \frac{2(1-a')(p-a')}{2-p-a'}.$$

Thus $D(a'\|p) > -2\varepsilon(u-\varepsilon) + \frac{2(1-a')(p-a')}{2-p-a'} \geq \varepsilon^2$, with gap $\frac{(p-a')^2}{2-p-a'} > 0$ (using $p - a' = \varepsilon(2u - \varepsilon)$). Therefore

$$\mathbb{P}\{\sqrt{B} - \sqrt{np} \leq -\varepsilon\sqrt{n}\} = \mathbb{P}\{B \leq na'\} \leq e^{-nD(a'\|p)} < e^{-n\varepsilon^2}.$$

Exercise 2.14

Central idea. The left-hand side is 2^n times the lower-tail probability $\mathbb{P}\{B \le D\}$ of a symmetric binomial $B \sim \mathrm{Bin}(n, 1/2)$. Apply the Cramér–Chernoff method with the exactly optimized exponential parameter $\lambda^* = \log \frac{n-D}{D}$, then simplify using $1 + x \le e^x$. The case $D \ge n/2$ is handled separately by monotonicity of $(en/D)^D$.

Solution. Let $B \sim \mathrm{Bin}(n, 1/2)$. Then

$$\sum_{j=0}^{D} \binom{n}{j} = 2^n \mathbb{P}\{B \le D\}.$$

Case $D \ge n/2$. Since $\sum_{j=0}^{D} \binom{n}{j} \le 2^n$, it suffices to show $(en/D)^D \ge 2^n$. The function $f(D) = \log(en/D)^D = D(1 + \log n - \log D)$ satisfies $f'(D) = \log(n/D) \ge 0$ for $1 \le D \le n$, so f is increasing and

$$\left(\frac{en}{D}\right)^D \ge \left(\frac{en}{n/2}\right)^{n/2} = (2e)^{n/2} \ge 2^n,$$

where $(2e)^{n/2} \ge 2^n$ follows from $2e \ge 4$.

Case $1 \le D < n/2$. By Chernoff's inequality (the Cramér–Chernoff method), for any $\lambda > 0$,

$$\mathbb{P}\{B \le D\} = \mathbb{P}\{e^{-\lambda B} \ge e^{-\lambda D}\} \le e^{\lambda D}\, \mathbb{E}[e^{-\lambda B}] = e^{\lambda D} \left(\frac{1 + e^{-\lambda}}{2}\right)^n.$$

Minimize $\phi(\lambda) = \lambda D + n \log(1 + e^{-\lambda}) - n \log 2$ over $\lambda > 0$. Since

$$\phi'(\lambda) = D - n \frac{e^{-\lambda}}{1 + e^{-\lambda}},$$

the unique minimizer satisfies $\frac{e^{-\lambda}}{1+e^{-\lambda}} = \frac{D}{n}$, i.e. $e^{-\lambda^*} = \frac{D}{n-D}$, $\lambda^* = \log \frac{n-D}{D} > 0$ (using $D < n/2$). Substituting,

$$\phi(\lambda^*) = D \log \frac{n-D}{D} + n \log \frac{n}{n-D} - n \log 2 = n \log n - n \log 2 - D \log D - (n-D) \log(n-D),$$

hence

$$\sum_{j=0}^{D} \binom{n}{j} \le 2^n e^{\phi(\lambda^*)} = \frac{n^n}{D^D\, (n-D)^{\,n-D}}.$$

It remains to show $\frac{n^n}{D^D (n-D)^{n-D}} \le \left(\frac{en}{D}\right)^D = \frac{e^D n^D}{D^D}$, i.e.

$$n^{\,n-D} \le e^D (n-D)^{\,n-D} \quad \Longleftrightarrow \quad \left(1 + \frac{D}{n-D}\right)^{n-D} \le e^D.$$

This follows from $1 + x \le e^x$ with $x = D/(n-D)$:

$$\left(1 + \frac{D}{n-D}\right)^{n-D} \le \exp\left(\frac{D}{n-D} \cdot (n-D)\right) = e^D.$$

This completes the proof.

Exercise 2.15

Central idea. Work with $m(\lambda) = \mathbb{E}e^{\lambda Y} = e^{\psi_Y(\lambda)}$ rather than ψ_Y itself: the tilted-distribution computation gives $m''(\lambda) \le v e^{\lambda}$, and comparing the ODE $m'' \le v e^{\lambda}$ against $u(\lambda) = 1 + v(e^{\lambda} - \lambda - 1)$ (which satisfies $u'' = v e^{\lambda}$ with matching initial data) yields $m \le u$, i.e. the refined Bennett bound. Note the estimate only holds for $\lambda \ge 0$; the claim is false for $\lambda < 0$ (see remark).

Solution. Let $m(\lambda) = \mathbb{E}[e^{\lambda Y}]$, so $\psi_Y(\lambda) = \log m(\lambda)$. Since Y is centered with $\mathrm{Var}(Y) \le v < \infty$, m is twice differentiable (differentiation under the expectation is justified by dominated convergence on compact λ-intervals, using $Y^2 e^{\lambda Y} \le e^{\lambda} Y^2$ for $\lambda \ge 0$, $Y \le 1$), with

$$m(0) = 1, \qquad m'(0) = \mathbb{E}[Y] = 0, \qquad m''(\lambda) = \mathbb{E}[Y^2 e^{\lambda Y}].$$

For $\lambda \ge 0$, since $Y \le 1$ almost surely, $e^{\lambda Y} \le e^{\lambda}$, so

$$m''(\lambda) \le e^{\lambda}\, \mathbb{E}[Y^2] \le v e^{\lambda}.$$

(This is the estimate behind the hint's check: with P_λ the tilted distribution of density $e^{-\psi_Y(\lambda)} e^{\lambda x}$ w.r.t. P and $Z \sim P_\lambda$, one has $\psi_Y''(\lambda) = \mathrm{Var}_{P_\lambda}(Z)$, and the same computation gives the bound.)

Now set $u(\lambda) = 1 + v(e^{\lambda} - \lambda - 1)$. Then $u(0) = 1 = m(0)$, $u'(0) = v(e^0 - 1) = 0 = m'(0)$, and $u''(\lambda) = v e^{\lambda}$. Define $g(\lambda) = u(\lambda) - m(\lambda)$ for $\lambda \ge 0$. Then

$$g(0) = 0, \qquad g'(0) = 0, \qquad g''(\lambda) = v e^{\lambda} - m''(\lambda) \ge 0.$$

Hence $g'(\lambda) = \int_0^{\lambda} g''(t)\, dt \ge 0$ and $g(\lambda) = \int_0^{\lambda} g'(t)\, dt \ge 0$ for $\lambda \ge 0$. That is,

$$\mathbb{E}[e^{\lambda Y}] = m(\lambda) \le 1 + v(e^{\lambda} - \lambda - 1), \qquad \lambda \ge 0,$$

and taking logarithms (both sides ≥ 1),

$$\psi_Y(\lambda) \le \log(1 + v(e^{\lambda} - \lambda - 1)), \qquad \lambda \ge 0.$$

Since $\log(1 + x) \le x$, this refines Lemma 2.2's bound $\psi_Y(\lambda) \le v(e^{\lambda} - \lambda - 1)$.

Remark. The inequality does not hold for all $\lambda \in \mathbb{R}$ as stated. For a Rademacher Y ($\mathbb{P}\{Y = \pm 1\} = 1/2$, so $v = 1$), $\psi_Y(-1) = \log\cosh(1) \approx 0.434$ while $\log(1 + (e^{-1} + 1 - 1)) = \log(1 + e^{-1}) \approx 0.313$. The step $m''(\lambda) = \mathbb{E}[Y^2 e^{\lambda Y}] \le e^{\lambda} \mathbb{E}[Y^2]$ uses $e^{\lambda Y} \le e^{\lambda}$, which needs both $Y \le 1$ and $\lambda \ge 0$; for $\lambda < 0$ the monotonicity reverses. The correct statement is for $\lambda \ge 0$, which is all Bennett's inequality requires.

Exercise 2.16

Central idea. The sub-Gaussian MGF bound controls the second-order Taylor expansion of $\mathbb{E}e^{\lambda(X - \mathbb{E}X)}$ at $\lambda = 0$: comparing coefficients of λ^2 as $\lambda \to 0$ forces $\mathrm{Var}(X) \le v$. (We assume X centered, i.e. $\mathbb{E}X = 0$; otherwise apply the argument to $X - \mathbb{E}X$, whose variance factor is the same.)

Solution. Let $\mu = \mathbb{E}X$ and assume $\mu = 0$ (replacing X by $X - \mu$ otherwise). Sub-Gaussianity gives $\mathbb{E}[e^{t|X|}] < \infty$ for all $t \in \mathbb{R}$, since $e^{t|X|} \le e^{tX} + e^{-tX}$ and both terms have finite expectation by the MGF bound; in particular $\mathrm{Var}(X) = \mathbb{E}[X^2] < \infty$.

For $|\lambda| \le 1$, Taylor's formula with integral remainder gives

$$e^{\lambda X} = 1 + \lambda X + \lambda^2 X^2 \int_0^1 (1 - s)\, e^{s\lambda X}\, ds.$$

Taking expectations and using $\mathbb{E}X = 0$,

$$\mathbb{E}[e^{\lambda X}] = 1 + \lambda^2\, \mathbb{E}\left[X^2 \int_0^1 (1 - s)\, e^{s\lambda X}\, ds \right].$$

As $\lambda \to 0$, the bracket converges almost surely to $X^2 \int_0^1 (1-s)\, ds = X^2/2$, and is dominated by $X^2 e^{|X|}$ (since $|s\lambda X| \le |X|$ for $|\lambda| \le 1$), which is integrable because $X^2 e^{|X|} \le e^{2|X|}$ (using $x^2 \le e^{|x|}$ for all x). By dominated convergence,

$$\mathbb{E}\left[X^2 \int_0^1 (1 - s)\, e^{s\lambda X}\, ds \right] \longrightarrow \frac{\mathbb{E}[X^2]}{2} = \frac{\mathrm{Var}(X)}{2}.$$

Hence

$$\mathbb{E}[e^{\lambda X}] = 1 + \frac{\lambda^2}{2}\operatorname{Var}(X) + o(\lambda^2) \qquad (\lambda \to 0).$$

The sub-Gaussian hypothesis gives $\mathbb{E}[e^{\lambda X}] \le e^{v\lambda^2/2} = 1 + \frac{v\lambda^2}{2} + o(\lambda^2)$. Therefore

$$\frac{\lambda^2}{2}\operatorname{Var}(X) + o(\lambda^2) \le \frac{v\lambda^2}{2} + o(\lambda^2),$$

and dividing by $\lambda^2/2$ and letting $\lambda \to 0$ yields $\operatorname{Var}(X) \le v$.

Exercise 2.17

Central idea. The upper bound $\mathbb{E}\max_i G_i \le \sqrt{2\log N}$ is the book's maximal inequality (Jensen + union bound, optimized over λ). Sharpness follows from a Gaussian lower-tail bound: with $t_N = \sqrt{2\log N - 2\log\log N}$, the union bound in reverse shows $\mathbb{P}\{\max_i G_i \ge t_N\} \to 1$, so $\mathbb{E}\max_i G_i \ge t_N(1-o(1))$.

Solution. *Upper bound.* For $\lambda > 0$, by Jensen's inequality and the union bound,

$$\exp\Big(\lambda\,\mathbb{E}\max_{1\le i\le N} G_i\Big) \le \mathbb{E}\big[e^{\lambda \max_i G_i}\big] = \mathbb{E}\Big[\max_i e^{\lambda G_i}\Big] \le \sum_{i=1}^{N}\mathbb{E}[e^{\lambda G_i}] = Ne^{\lambda^2/2},$$

using $\mathbb{E}[e^{\lambda G_i}] = e^{\lambda^2/2}$ for standard normal G_i. Taking logarithms,

$$\mathbb{E}\max_{1\le i\le N} G_i \le \frac{\log N}{\lambda} + \frac{\lambda}{2}.$$

Minimizing over $\lambda > 0$ (attained at $\lambda = \sqrt{2\log N}$) gives

$$\mathbb{E}\max_{1\le i\le N} G_i \le \sqrt{2\log N}, \qquad \text{hence} \qquad \frac{\mathbb{E}\max_i G_i}{\sqrt{2\log N}} \le 1.$$

Lower bound. For $t \ge 1$, bounding the Gaussian density from below on $[t, t+1/t]$,

$$\mathbb{P}\{G_1 \ge t\} \ge \int_t^{t+1/t} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx \ge \frac{1}{t\sqrt{2\pi}}\, e^{-(t+1/t)^2/2} \ge c_0\,\frac{e^{-t^2/2}}{t},$$

with $c_0 = e^{-3/2}/\sqrt{2\pi}$, since $(t+1/t)^2 = t^2 + 2 + 1/t^2 \le t^2 + 3$.

Take $t_N = \sqrt{2\log N - 2\log\log N}$ (for $N \ge 3$). Then $e^{-t_N^2/2} = \frac{\log N}{N}$, so with $p_N = \mathbb{P}\{G_1 \ge t_N\}$,

$$Np_N \ge c_0\,\frac{\log N}{t_N} \ge c_0\sqrt{\frac{\log N}{2}} \longrightarrow \infty.$$

By independence,

$$\mathbb{P}\Big\{\max_{1\le i\le N} G_i \ge t_N\Big\} = 1 - (1-p_N)^N \ge 1 - e^{-Np_N} \longrightarrow 1.$$

Since $\max_i G_i \ge t_N \mathbf{1}_{\{\max_i G_i \ge t_N\}} - (\max_i G_i)^-$ pointwise (the subtracted term corrects the case $\max_i G_i < 0$), and $(\max_i G_i)^- \le G_1^-$ gives $\mathbb{E}[(\max_i G_i)^-] \le 1/\sqrt{2\pi}$,

$$\mathbb{E}\max_{1\le i\le N} G_i \ge t_N\,(1 - e^{-Np_N}) - \frac{1}{\sqrt{2\pi}},$$

and therefore

$$\frac{\mathbb{E}\max_i G_i}{\sqrt{2\log N}} \ge \frac{\sqrt{2\log N - 2\log\log N}}{\sqrt{2\log N}}\,(1 - e^{-Np_N}) - \frac{1}{\sqrt{4\pi\log N}} \longrightarrow 1.$$

Combined with the upper bound, the limit equals 1.

Exercise 2.18

Central idea. The maximal inequality (Theorem 2.5, cited as Theorem 2.4 in the exercise: Jensen + union bound) gives $\mathbb{E}\max_i X_i \le \inf_{\lambda>0} \frac{\log n + e^\lambda - 1}{\lambda}$ using the Poisson MGF $\mathbb{E}e^{\lambda X} = e^{e^\lambda - 1}$. The exact optimizer solves $e^\lambda(\lambda - 1) = \log(n/e)$, i.e. $\lambda = 1 + W(\log(n/e)/e)$, yielding the Lambert W bound. The "manageable" bound is just this inequality evaluated at the convenient (non-optimal) choice $\lambda = \log\log(en)$.

Erratum. The statement's hypothesis "$n \ge e^3$" for the explicit (second) bound is wrong as printed, and the hint's "Theorem 2.4" should read Theorem 2.5 (the maximal inequality; the book has only a Lemma 2.4). Write $z = \log(n/e)/e$. At $n = e^3$, $z = 2/e < 1$, so $\log(\log(n/e)/e) = \log(2/e) < 0$ and the logarithm in the denominator is undefined. For $1 < z < e$ the right side is defined but the bound is *false*: at $n = 160$ ($z \approx 1.499$) the explicit right side equals 3.113, while the true value is $\mathbb{E}\max_{i\le 160} X_i = \sum_{k\ge 0}(1 - F(k)^{160}) = 4.503$ (F the Poisson(1) cdf; the Lambert W bound there is 5.616, consistent). The lower bound $W(z) \ge \log z - \log\log z$ requires $z \ge e$, so the corrected statement, proved below, is: the explicit bound holds for $n \ge e^{e^2+1} \approx 4399$. (The Lambert W bound and the "manageable" bound hold as printed.)

Solution. For $X \sim \text{Poisson}(1)$, $\mathbb{E}[e^{\lambda X}] = \sum_{k\ge 0} e^{\lambda k} e^{-1}/k! = e^{e^\lambda - 1}$ for $\lambda \in \mathbb{R}$. By the maximal inequality (Theorem 2.5, cited as Theorem 2.4 in the exercise),

$$\exp\Big(\lambda\, \mathbb{E}\max_{1\le i\le n} X_i\Big) \le \sum_{i=1}^n \mathbb{E}[e^{\lambda X_i}] = n\, e^{e^\lambda - 1}, \qquad \lambda > 0,$$

so with $f(\lambda) = \frac{\log n + e^\lambda - 1}{\lambda}$,

$$\mathbb{E}\max_{1\le i\le n} X_i \le \inf_{\lambda>0} f(\lambda). \tag{1}$$

Lambert W bound. $f'(\lambda) = \frac{\lambda e^\lambda - (\log n + e^\lambda - 1)}{\lambda^2}$, so $f'(\lambda) = 0$ iff

$$e^\lambda(\lambda - 1) = \log n - 1 = \log(n/e).$$

Writing $u = \lambda - 1$ gives $ue^u = \log(n/e)/e$, i.e. $u = W(\log(n/e)/e)$, where $W = W_0$ is the principal branch of Lambert W (defined since $\log(n/e)/e \ge -1/e$ for $n \ge 1$; for $1 < n < e$ the argument lies in $(-1/e, 0)$ where W has two real branches, and $W_0 > -1$ is the one giving $\lambda^* > 0$). Thus the unique minimizer ($f \to +\infty$ at both ends) is $\lambda^* = 1 + W(\log(n/e)/e) > 0$ for $n > 1$ (for $n = 1$ both sides equal 1). At the optimum, $\log n - 1 = e^{\lambda^*}(\lambda^* - 1)$, so

$$f(\lambda^*) = \frac{e^{\lambda^*}(\lambda^* - 1) + e^{\lambda^*}}{\lambda^*} = e^{\lambda^*} = e \cdot e^W = e \cdot \frac{\log(n/e)}{e\, W(\log(n/e)/e)} = \frac{\log(n/e)}{W(\log(n/e)/e)},$$

using $We^W = \log(n/e)/e$. With (1),

$$\mathbb{E}\max_{1\le i\le n} X_i \le \frac{\log(n/e)}{W(\log(n/e)/e)}.$$

Lower bound for W. Let $z \ge e$ and $w = W(z)$, so $we^w = z$. Set $w_0 = \log z - \log\log z \ge 1 > -1$ (the minimum of $\log z - \log\log z$ on $[e, \infty)$ is 1, attained at $z = e$). Then

$$w_0 e^{w_0} = (\log z - \log\log z)\,\frac{z}{\log z} = z\left(1 - \frac{\log\log z}{\log z}\right) \le z = we^w.$$

Since $x \mapsto xe^x$ is strictly increasing on $[-1, \infty)$, $w_0 \le w$, i.e.

$$W(z) \ge \log z - \log\log z, \qquad z \ge e.$$

Combined bound. Applying this with $z = \log(n/e)/e$ gives, whenever $z \ge e$ (i.e. $n \ge e^{e^2+1}$),

$$\mathbb{E}\max_{1\le i\le n} X_i \le \frac{\log(n/e)}{\log(\log(n/e)/e) - \log(\log(\log(n/e)/e))}.$$

(See the Erratum: for $n \geq e^3$ alone this fails — e.g. at $n = 160$ the right side is $3.113 < 4.503 = \mathbb{E}\max_i X_i$.)

Manageable bound. Evaluate (1) directly at $\lambda = \log\log(en) > 0$ (for $n \geq 2$): since $e^\lambda = \log(en) = 1 + \log n$,

$$\mathbb{E}\max_{1\leq i\leq n} X_i \leq \frac{\log n + (1+\log n) - 1}{\log\log(en)} = \frac{2\log n}{\log(\log(en))}.$$

Exercise 2.19

Central idea. A Binomial(m,p) moment-generating function is dominated by that of a Poisson(mp) variable, since $1 - p + pe^\lambda \leq e^{p(e^\lambda - 1)}$. Feeding this into the maximal-inequality Chernoff argument (Jensen's inequality plus $\max_i a_i \leq \sum_i a_i$ — the estimate at the heart of Theorem 2.5, cited as Theorem 2.4 in the exercise) gives $\mathbb{E}\max_i X_i \leq \inf_{\lambda>0} \frac{\log n + mp(e^\lambda - 1)}{\lambda}$; the minimizer satisfies $ue^u = \frac{\log n - mp}{emp}$ with $u = \lambda - 1$, which is exactly the Lambert W function.

Solution. Write $\mu = mp$. For $X \sim \text{Binomial}(m,p)$ and $\lambda > 0$,

$$\mathbb{E}e^{\lambda X} = (1 - p + pe^\lambda)^m \leq e^{mp(e^\lambda - 1)} = e^{\mu(e^\lambda - 1)},$$

using $1 + x \leq e^x$ with $x = p(e^\lambda - 1)$.

For any $\lambda > 0$, since $0 \leq \max_i X_i \leq m$ (so Jensen applies to the bounded variable $\log e^{\lambda \max_i X_i}$),

$$\mathbb{E}\max_{1\leq i\leq n} X_i = \frac{1}{\lambda}\mathbb{E}\log e^{\lambda\max_i X_i} \leq \frac{1}{\lambda}\log \mathbb{E}e^{\lambda\max_i X_i} = \frac{1}{\lambda}\log \mathbb{E}\max_i e^{\lambda X_i} \leq \frac{1}{\lambda}\log\sum_{i=1}^n \mathbb{E}e^{\lambda X_i} \leq \frac{\log n + \mu(e^\lambda - 1)}{\lambda},$$

using $\max_i a_i \leq \sum_i a_i$ for $a_i \geq 0$. Hence, with $f(\lambda) = \frac{\log n + \mu(e^\lambda - 1)}{\lambda}$,

$$\mathbb{E}\max_i X_i \leq \inf_{\lambda>0} f(\lambda).$$

For $n = 1$, $\mathbb{E}\max_i X_i = \mu$ while the claimed bound equals $\mu e^{1+W(-1/e)} = \mu$; assume $n \geq 2$. Then f is continuous on $(0,\infty)$ with $f(\lambda) \to +\infty$ as $\lambda \to 0^+$ (numerator $\to \log n > 0$) and as $\lambda \to \infty$, so the infimum is attained. Moreover

$$f'(\lambda) = \frac{\mu e^\lambda \lambda - \log n - \mu e^\lambda + \mu}{\lambda^2},$$

so $f'(\lambda) = 0$ iff $\mu e^\lambda(\lambda - 1) = \log n - \mu$. Since $h(\lambda) = \mu e^\lambda(\lambda - 1)$ satisfies $h'(\lambda) = \mu\lambda e^\lambda > 0$ on $(0,\infty)$, it is strictly increasing, hence there is a unique critical point, which is the global minimizer.

Put $u = \lambda - 1$: then $\mu e^{u+1}u = \log n - \mu$, i.e.

$$ue^u = \frac{\log n - \mu}{e\mu}.$$

Since $\log n \geq 0$, $\frac{\log n - \mu}{e\mu} \geq -\frac{1}{e}$ (strictly $>$ for $n \geq 2$), so with $W = W_0$ the principal branch of Lambert W (the relevant branch when the argument lies in $(-1/e, 0)$, i.e. when $\log n < \mu$), $u = W\left(\frac{\log n - \mu}{e\mu}\right) > -1$ and $\lambda^* = 1 + u > 0$. Now

$$f(1+u) = \frac{\log n + \mu e^{1+u} - \mu}{1+u}.$$

From $ue^u = \frac{\log n - \mu}{e\mu}$ we get $\log n - \mu = e\mu ue^u$, so the numerator equals $e\mu ue^u + \mu e \cdot e^u = \mu e^{1+u}(u+1)$, and

$$f(1+u) = \mu e^{1+u} = \mu e^{1+W((\log n - \mu)/(e\mu))}.$$

With $\mu = mp$ this is

$$\mathbb{E}\max_{i=1,\ldots,n} X_i \leq mpe^{1+W((\log(n) - mp)/(emp))},$$

as claimed.

Exercise 2.20

Central idea. The n bin counts B_i are multinomial (hence dependent), but each is marginally Binomial$(m, 1/n)$, and the maximal-inequality mgf step $\mathbb{E}e^{\lambda \max_i B_i} \le \sum_i \mathbb{E}e^{\lambda B_i}$ uses only marginal mgfs — no independence is needed. The Exercise 2.19 optimization therefore applies verbatim with $\mu = m/n$. The two special cases are direct substitutions; the last uses the lower bound $W(z) \ge \log z - \log\log z$ $(z \ge e)$ from Exercise 2.18.

Solution. Let B_i be the number of balls in bin i; marginally $B_i \sim$ Binomial$(m, 1/n)$, and $M = \max_{1\le i\le n} B_i$ (so $0 \le M \le m$). For $\lambda > 0$,

$$\mathbb{E}e^{\lambda M} = \mathbb{E}\max_i e^{\lambda B_i} \le \sum_{i=1}^{n} \mathbb{E}e^{\lambda B_i} = n\,\mathbb{E}e^{\lambda B_1} \le n\,e^{\frac{m}{n}(e^\lambda - 1)},$$

using the binomial mgf bound $(1 - \frac{1}{n} + \frac{1}{n}e^\lambda)^m \le e^{\frac{m}{n}(e^\lambda-1)}$ from Exercise 2.19. Hence, by Jensen's inequality as in 2.19 (the maximal inequality, Theorem 2.5, cited as Theorem 2.4),

$$\mathbb{E}M \le \inf_{\lambda>0} \frac{\log n + \frac{m}{n}(e^\lambda - 1)}{\lambda} = \frac{m}{n} e^{1+W((\log(n)-m/n)/(em/n))},$$

by the identical optimization with $\mu = m/n$.

If $m = cn\log n$, then $\mu = m/n = c\log n$ and

$$\frac{\log n - \mu}{e\mu} = \frac{\log n - c\log n}{ec\log n} = \frac{1-c}{ec},$$

so $\mathbb{E}M \le c\log(n)\, e^{1+W((1-c)/(ce))}$.

If $m = cn$, then $\mu = c$ and, using $\mu e^{1+w} = \frac{\log n - \mu}{w}$ for $w = W\big(\frac{\log n-\mu}{e\mu}\big)$,

$$\mathbb{E}M \le ce^{1+w} = \frac{\log n - c}{w}, \qquad w = W(z),\ z = \frac{\log n - c}{ec}.$$

By Exercise 2.18, $W(z) \ge \log z - \log\log z$ for $z \ge e$ (the hypothesis $z \ge e$ is exactly what the Exercise 2.18 estimate requires). This requires $\frac{\log n - c}{ec} \ge e$, i.e. $\log n \ge (1+e^2)c$ — see the remark below. Under this condition, $\log z - \log\log z$ is positive and increasing for $z \ge e$ (it equals 1 at $z = e$), and since $\log n > c$,

$$\mathbb{E}M \le \frac{\log n - c}{w} \le \frac{\log n - c}{\log z - \log\log z} = \frac{\log(n) - c}{\log((\log(n)-c)/(ec)) - \log(\log((\log(n)-c)/(ec)))}.$$

Remark. The hypothesis "$\log n > c$" as stated does not suffice for the last conclusion: the right-hand side is not even real unless $\frac{\log n - c}{ec} > 1$, and the appeal to Exercise 2.18 needs $\frac{\log n - c}{ec} \ge e$. The bound holds as written whenever $\log n \ge (1+e^2)c$.

Exercise 2.21

Central idea. Forward: the variance bound comes from Fatou applied to $(\mathbb{E}e^{\lambda X} - 1)/\lambda^2$ (no finite-variance assumption needed); moments follow from Bernstein's tail bound via $\mathbb{E}[Z^q] = q\int_0^\infty t^{q-1}P(Z > t)\,dt$ split into Gaussian and exponential regimes. Converse: write $\mathbb{E}e^{\lambda X} = 1 + \mathbb{E}[\phi(\lambda X)]$ with $\phi(x) = e^x - 1 - x$ so the linear term vanishes exactly ($\mathbb{E}X = 0$); use $\phi(x) \le x^2/2$ for $x \le 0$ and the power series for $x \ge 0$, summing with $q! \ge (q/e)^q$.

Solution. Recall $X \in \Gamma_+(v, c)$, centered, means $\log \mathbb{E}e^{\lambda X} \le \frac{\lambda^2 v}{2(1-c\lambda)}$ for $0 < \lambda < 1/c$.

Forward. For the variance bound, let $\phi(u) = e^u - 1 - u \ge 0$. Since $\mathbb{E}X = 0$,

$$\frac{\mathbb{E}e^{\lambda X} - 1}{\lambda^2} = \mathbb{E}\frac{\phi(\lambda X)}{\lambda^2} \le \frac{e^{\lambda^2 v/(2(1-c\lambda))} - 1}{\lambda^2} \xrightarrow[\lambda\downarrow 0]{} \frac{v}{2}.$$

As $\phi(\lambda X)/\lambda^2 \to X^2/2$ pointwise, Fatou gives $\mathbb{E}[X^2]/2 \le v/2$, i.e. $\mathrm{Var}(X) \le v$ (no a priori finite-variance assumption needed).

The mgf bound implies Bernstein's inequality $P(X > t) \le \exp(-\frac{t^2}{2(v+ct)})$ for $t > 0$ (standard Chernoff optimization). Since $\frac{t^2}{2(v+ct)} \ge \min(\frac{t^2}{4v}, \frac{t}{4c})$ (if $ct \le v$ then $v + ct \le 2v$; else $v + ct < 2ct$),

$$P(X_+ > t) \le e^{-t^2/(4v)} + e^{-t/(4c)}.$$

For $q \ge 2$, $\mathbb{E}[X_+^q] = q\int_0^\infty t^{q-1} P(X_+ > t)\, dt$, so with $(a+b)^{1/q} \le a^{1/q} + b^{1/q}$,

$$(\mathbb{E}[X_+^q])^{1/q} \le \Big[q\int_0^\infty t^{q-1} e^{-t^2/(4v)} dt\Big]^{1/q} + \Big[q\int_0^\infty t^{q-1} e^{-t/(4c)} dt\Big]^{1/q}.$$

First integral: $u = t^2/(4v)$ gives $q 2^{q-1} v^{q/2}\Gamma(q/2) \le q 2^{q-1}(qv)^{q/2}$ (using $\Gamma(q/2) \le q^{q/2}$); $1/q$ power $\le 4\sqrt{qv}$ for $q \ge 2$. Second: $q(4c)^q\Gamma(q) = (4c)^q q!$; $1/q$ power $\le 4cq$ (using $q! \le q^q$). Hence $(\mathbb{E}[X_+^q])^{1/q} \le 4(\sqrt{qv} + cq)$, i.e. $C = 4$.

Converse. Write $\mathbb{E}e^{\lambda X} = 1 + \mathbb{E}[\phi(\lambda X)]$ with $\phi(x) = e^x - 1 - x$; the linear term vanishes exactly since $\mathbb{E}X = 0$. For $x \le 0$, $\phi(x) \le x^2/2$; for $x \ge 0$, $\phi(x) = \sum_{q\ge 2} x^q/q!$. Hence, with $\mathbb{E}[X^2] = \mathrm{Var}(X) \le A$,

$$\mathbb{E}[\phi(\lambda X)] \le \frac{\lambda^2}{2}\mathbb{E}[X^2 \mathbf{1}_{X\le 0}] + \sum_{q\ge 2} \frac{\lambda^q\, \mathbb{E}[X_+^q]}{q!} \le \frac{\lambda^2 A}{2} + \sum_{q\ge 2} \frac{\lambda^q(\sqrt{qA} + Bq)^q}{q!}.$$

Now $(\sqrt{qA} + Bq)^q \le 2^{q-1}((qA)^{q/2} + (qB)^q)$, and $q! \ge (q/e)^q$. The B-terms give $\frac{1}{2}\sum_{q\ge 2}(2eB\lambda)^q \le \frac{2e^2B^2\lambda^2}{1-2eB\lambda}$ for $0 < \lambda < 1/(2eB)$. For the A-terms, $q! \ge (q/e)^q$ gives $\frac{\lambda^q 2^{q-1}(qA)^{q/2}}{q!} \le \frac{1}{2}\frac{w^q}{q^{q/2}}$ with $w = 2e\lambda\sqrt{A}$. Since $q! \le q^q$, $\frac{w^q}{q^{q/2}} \le \frac{w^q q^{q/2}}{q!}$, and splitting even and odd terms,

$$\sum_{q\ge 2} \frac{w^q q^{q/2}}{q!} \le \sum_{k\ge 1} \frac{(2w^2)^k}{k!} + 2w \sum_{k\ge 1} \frac{(2w^2)^k}{k!} = (1+2w)(e^{2w^2} - 1), \qquad w \ge 0,$$

using $(2k)! \ge k!\, k^k$ for the even terms ($q = 2k$) and $(2k+1)! \ge k!(k+1)^{k+1}$ for the odd terms ($q = 2k+1 \ge 3$). A calculus check ($h(0) = h'(0) = 0$, $h'' \ge 0$) gives $(1+2w)(e^{2w^2} - 1) \le e^{6w^2} - 1$ for $w \ge 0$. Hence the A-terms are at most $\frac{1}{2}(e^{6w^2} - 1) = \frac{1}{2}(e^{24e^2A\lambda^2} - 1)$, valid for every $\lambda \ge 0$. With $z = A\lambda^2$,

$$1 + \frac{\lambda^2 A}{2} + (A\text{-terms}) \le 1 + \frac{z}{2} + \frac{e^{24e^2 z} - 1}{2} = \frac{1 + e^{24e^2 z}}{2} + \frac{z}{2} \le e^{(24e^2+1/2)z},$$

using $\frac{1+e^u}{2} \le e^u$ and $1 + v \le e^v$. Thus $1 + \frac{\lambda^2 A}{2} + (A\text{-terms}) \le e^{K_1 A\lambda^2}$ for all $\lambda \ge 0$ with $K_1 = 24e^2 + \frac{1}{2}$. Consequently, for $0 < \lambda < \frac{1}{2eB}$,

$$\mathbb{E}e^{\lambda X} \le e^{K_1 A\lambda^2} + \frac{2e^2B^2\lambda^2}{1-2eB\lambda} \le \exp\Big(K_1 A\lambda^2 + \frac{2e^2B^2\lambda^2}{1-2eB\lambda}\Big) \le \exp\Big(\frac{\lambda^2 v}{2(1-c\lambda)}\Big),$$

with $c = 2eB$ and $v = (48e^2 + 1)A + 4e^2B^2$, using $1 + u \le e^u$ and $K_1 A\lambda^2 \le K_1 A\lambda^2/(1 - c\lambda)$. Taking $C = 48e^2 + 1$ gives $v \le C(A + B^2)$ and $c \le CB$, i.e. $X \in \Gamma_+(v, c)$. ■

Exercise 2.22

Central idea. For the exponential law the moments come from the gamma integral $\int_0^\infty u^q e^{-u}\, du = q!$ after the change of variables $u = ax$. For a sub-exponential X, expand the mgf as a power series and use monotone convergence (all terms nonnegative since $X \ge 0$) to read off each moment from the series coefficients; each single term is bounded by the whole sum, and evaluating at $\lambda = a/2$ gives the constant 2^{q+1}.

Solution. Let X have exponential distribution with parameter $a > 0$, i.e. density ae^{-ax} on $x \geq 0$. For a positive integer q,

$$\mathbb{E}[X^q] = \int_0^\infty x^q a e^{-ax}\, dx.$$

Substituting $u = ax$ ($dx = du/a$),

$$\mathbb{E}[X^q] = a \int_0^\infty \left(\frac{u}{a}\right)^q e^{-u}\, \frac{du}{a} = \frac{1}{a^q} \int_0^\infty u^q e^{-u}\, du.$$

Writing $I_q = \int_0^\infty u^q e^{-u}\, du$, integration by parts gives

$$I_q = \Big[-u^q e^{-u}\Big]_0^\infty + q \int_0^\infty u^{q-1} e^{-u}\, du = qI_{q-1},$$

since $u^q e^{-u} \to 0$ as $u \to \infty$ and vanishes at $u = 0$ for $q \geq 1$. With $I_0 = \int_0^\infty e^{-u}\, du = 1$, induction yields $I_q = q!$. Hence $\mathbb{E}[X^q] = q!/a^q$.

Now suppose $X \geq 0$ is sub-exponential: $\mathbb{E}e^{\lambda X} \leq 1/(1 - \lambda/a)$ for all $0 < \lambda < a$. Fix such a λ. Since $X \geq 0$, the partial sums of $e^{\lambda X} = \sum_{q=0}^\infty \lambda^q X^q/q!$ are nonnegative and increase monotonically to $e^{\lambda X}$, so by monotone convergence (in $[0, \infty]$),

$$\mathbb{E}e^{\lambda X} = \sum_{q=0}^\infty \frac{\lambda^q}{q!}\, \mathbb{E}[X^q].$$

Combining with the sub-exponential bound,

$$\sum_{q=0}^\infty \frac{\lambda^q}{q!}\, \mathbb{E}[X^q] \leq \frac{1}{1 - \lambda/a} < \infty, \qquad 0 < \lambda < a.$$

Every term of the series is nonnegative, so each term is bounded by the sum: for every positive integer q and every $0 < \lambda < a$,

$$\frac{\lambda^q}{q!}\, \mathbb{E}[X^q] \leq \frac{1}{1 - \lambda/a}, \qquad \text{i.e.} \qquad \mathbb{E}[X^q] \leq \frac{q!}{\lambda^q}\, \frac{1}{1 - \lambda/a}.$$

(The finiteness of the right-hand side also shows $\mathbb{E}[X^q] < \infty$.) Taking $\lambda = a/2$, for which $1 - \lambda/a = 1/2$,

$$\mathbb{E}[X^q] \leq \frac{q!}{(a/2)^q} \cdot 2 = 2^{q+1}\, \frac{q!}{a^q},$$

for every positive integer q, as required.

Exercise 2.23

Central idea. Expand $e^{\lambda X}$ as a power series and take expectations term by term. The hypothesis bounds each signed moment $\mathbb{E}[X^q]$, turning the series into a geometric one. The technical work is justifying the interchange: dominate the partial sums by $e^{\lambda |X|}$ and show it is integrable using the even-moment bounds plus Cauchy–Schwarz for the odd moments.

Solution. Fix $0 < \lambda < a$. For even $q \geq 2$, $|X|^q = X^q \geq 0$, so the hypothesis gives $\mathbb{E}|X|^q = \mathbb{E}[X^q] \leq q!/a^q < \infty$. For odd $q \geq 1$, Lyapunov's inequality gives $\mathbb{E}|X|^q \leq (\mathbb{E}|X|^{q+1})^{q/(q+1)} = (\mathbb{E}[X^{q+1}])^{q/(q+1)} \leq ((q+1)!/a^{q+1})^{q/(q+1)} < \infty$. Hence all absolute moments of X are finite.

We show $\mathbb{E}e^{\lambda |X|} < \infty$. By Tonelli's theorem,

$$\mathbb{E}e^{\lambda |X|} = \sum_{q=0}^\infty \frac{\lambda^q}{q!}\, \mathbb{E}|X|^q.$$

The even terms satisfy $\lambda^{2k}\mathbb{E}|X|^{2k}/(2k)! \le (\lambda/a)^{2k}$. For the odd terms, Cauchy–Schwarz gives $\mathbb{E}|X|^{2k+1} \le \sqrt{\mathbb{E}[X^{2k}]\,\mathbb{E}[X^{2k+2}]} \le \sqrt{(2k)!(2k+2)!}/a^{2k+1}$, so

$$\frac{\lambda^{2k+1}}{(2k+1)!}\,\mathbb{E}|X|^{2k+1} \le \left(\frac{\lambda}{a}\right)^{2k+1} \frac{\sqrt{(2k)!(2k+2)!}}{(2k+1)!} = \left(\frac{\lambda}{a}\right)^{2k+1} \sqrt{\frac{2k+2}{2k+1}} \le \sqrt{2}\left(\frac{\lambda}{a}\right)^{2k+1}.$$

Since $\lambda/a < 1$, both the even and odd series converge, so $\mathbb{E}e^{\lambda|X|} < \infty$.

Now $S_N := \sum_{q=0}^{N} \lambda^q X^q/q! \to e^{\lambda X}$ pointwise, and $|S_N| \le e^{\lambda|X|}$, which is integrable. By dominated convergence,

$$\mathbb{E}e^{\lambda X} = \sum_{q=0}^{\infty} \frac{\lambda^q}{q!}\,\mathbb{E}[X^q],$$

the series converging absolutely since $|\lambda^q\mathbb{E}[X^q]/q!| \le \lambda^q\mathbb{E}|X|^q/q!$ is summable. For $q \ge 1$, the hypothesis and $\lambda^q/q! > 0$ give $\lambda^q\mathbb{E}[X^q]/q! \le (\lambda/a)^q$; for $q = 0$ the term is $\mathbb{E}[1] = 1 = (\lambda/a)^0$. Therefore

$$\mathbb{E}e^{\lambda X} \le \sum_{q=0}^{\infty} \left(\frac{\lambda}{a}\right)^q = \frac{1}{1-\lambda/a},$$

the geometric series converging because $\lambda < a$. This is the sub-exponential moment-generating bound (cf. Exercise 2.22).

Exercise 2.24

Central idea. The hypothesis $\mathbb{E}[(X-a)_+] \le \mathbb{E}[(Y-a)_+]$ for all a is the increasing convex order, so any upper bound on $\mathbb{P}\{X \ge t\}$ obtained from $\mathbb{E}[(X-s)_+]$ via Markov's inequality transfers to Y, whose exponential tail integrates explicitly. The shift $s = t - 1/b$ is chosen to optimize the resulting bound, and the e^1 factor is the price of that optimization.

Solution. Let $t \ge 0$. If $t < 1/b$, then $1 - bt > 0$, so $\kappa e^{1-bt} > \kappa \ge 1 \ge \mathbb{P}\{X \ge t\}$ and there is nothing to prove. Assume henceforth $t \ge 1/b$ and set $s := t - 1/b \ge 0$.

Since $t - s = 1/b > 0$, we have $\{X \ge t\} = \{(X-s)_+ \ge t - s\}$: indeed, $X \ge t$ implies $X - s \ge t - s > 0$, so $(X-s)_+ = X - s \ge t - s$; conversely, $(X-s)_+ \ge t - s > 0$ forces $X - s > 0$ and hence $X \ge t$. By Markov's inequality and the hypothesis,

$$\mathbb{P}\{X \ge t\} = \mathbb{P}\{(X-s)_+ \ge t-s\} \le \frac{\mathbb{E}[(X-s)_+]}{t-s} \le \frac{\mathbb{E}[(Y-s)_+]}{t-s}.$$

(The expectations are finite: for $s \ge 0$, $\mathbb{E}[(Y-s)_+] \le \int_s^\infty \kappa e^{-bu}\,du < \infty$ by the layer-cake identity below, hence $\mathbb{E}[(X-s)_+] < \infty$ too.)

By the layer-cake representation, for any real s,

$$\mathbb{E}[(Y-s)_+] = \int_0^\infty \mathbb{P}\{(Y-s)_+ > v\}\,dv = \int_s^\infty \mathbb{P}\{Y \ge u\}\,du.$$

Since $s \ge 0$, the tail bound on Y applies throughout the range of integration:

$$\mathbb{E}[(Y-s)_+] \le \int_s^\infty \kappa e^{-bu}\,du = \frac{\kappa e^{-bs}}{b}.$$

Therefore, with $t - s = 1/b$,

$$\mathbb{P}\{X \ge t\} \le \frac{\kappa e^{-bs}/b}{1/b} = \kappa e^{-bs} = \kappa e^{-b(t-1/b)} = \kappa e^{1-bt}.$$

Remark. The choice $s = t - 1/b$ is optimal: minimizing $\phi(s) := e^{-bs}/(t-s)$ over $s \in [0, t)$ gives $\phi'(s) = e^{-bs}(1 - b(t-s))/(t-s)^2 = 0$ exactly at $s = t - 1/b$. This is Panchenko's argument (Panchenko, 2003).

Exercise 2.25

Central idea. Diagonalize A (symmetric, zero diagonal) to $Z \overset{d}{=} \sum_k \mu_k(Y_k^2-1)$ with Y_k iid $N(0,1)$. The *upper* bound comes from the mgf: $\log \mathbb{E}e^{\lambda Z} \le \lambda^2\|A\|_{\mathrm{HS}}^2/(1-2|\lambda|\,\|A\|)$, placing Z in $\Gamma_+(2\|A\|_{\mathrm{HS}}^2, 2\|A\|)$, then Exercise 2.21. The *lower* bound uses decoupling: with Y' an independent copy, $Z-Z' = 2\sum_k \mu_k U_k V_k$ for iid Gaussians (U_k, V_k); conditioning on V makes the sum Gaussian, and Jensen gives the $\sqrt{q}\,\|A\|_{\mathrm{HS}}$ term while the top eigendirection gives the $q\,\|A\|$ term.

Solution. Write $a = \|A\|_{\mathrm{HS}}$, $b = \|A\|$. Since A is symmetric with zero diagonal, $A = \sum_k \mu_k u_k u_k^\top$ in an orthonormal basis; with $Y_k = u_k^\top X$ iid $N(0,1)$ and $\sum_k \mu_k = \mathrm{tr}(A) = 0$,

$$Z = \sum_k \mu_k(Y_k^2-1), \qquad a^2 = \sum_k \mu_k^2, \qquad b = \max_k |\mu_k|.$$

Let $\gamma_q = (\mathbb{E}|G|^q)^{1/q}$ for $G \sim N(0,1)$; then $\gamma_q \ge \sqrt{q}/2$ for $q \ge 2$ (from $\mathbb{E}|G|^q = 2^{q/2}\Gamma(\frac{q+1}{2})/\sqrt{\pi}$ and Stirling's lower bound $\Gamma(x) \ge \sqrt{2\pi/x}\,(x/e)^x$).

Upper bound. For $|u|<1$, $-\log(1-u)-u \le \frac{u^2}{2(1-|u|)}$ (for $u>0$ this is Exercise 2.8(a); for $u \le 0$ put $\psi(u) = \frac{u^2}{2} - (-\log(1-u)-u)$, so $\psi'(u) = -\frac{u^2}{1-u} \le 0$ and $\psi(u) \ge \psi(0) = 0$). With $\mathbb{E}e^{\lambda\mu(Y^2-1)} = e^{-\lambda\mu}(1-2\lambda\mu)^{-1/2}$, for $|\lambda| < 1/(2b)$,

$$\log \mathbb{E}e^{\lambda Z} = \sum_k \tfrac{1}{2}\big(-\log(1-2\lambda\mu_k) - 2\lambda\mu_k\big) \le \frac{\lambda^2 a^2}{1-2|\lambda|\,b}.$$

So Z and $-Z$ lie in $\Gamma_+(2a^2, 2b)$. By Exercise 2.21 (direct part, $C=4$), for $q \ge 2$,

$$(\mathbb{E}|Z|^q)^{1/q} \le 2^{1/q}\cdot 4\big(\sqrt{2q}\,a + 2q\,b\big) \le 12\big(\sqrt{q}\,a + q\,b\big).$$

Lower bound (decoupling). Let Y' be an independent copy of Y and $Z' = \sum_k \mu_k({Y'_k}^2 - 1)$. By Minkowski, $(\mathbb{E}|Z-Z'|^q)^{1/q} \le 2(\mathbb{E}|Z|^q)^{1/q}$. Set $U_k = (Y_k - Y'_k)/\sqrt{2}$, $V_k = (Y_k + Y'_k)/\sqrt{2}$: these are iid $N(0,1)$ (orthogonal transformation), and $Y_k^2 - {Y'_k}^2 = 2U_kV_k$, so $Z - Z' = 2\sum_k \mu_k U_k V_k$. Conditionally on V, the sum $\sum_k \mu_k U_k V_k$ is $N(0, \sum_k \mu_k^2 V_k^2)$, hence

$$\mathbb{E}\Big|\sum_k \mu_k U_k V_k\Big|^q = \gamma_q^q\,\mathbb{E}\Big(\sum_k \mu_k^2 V_k^2\Big)^{q/2}.$$

By Jensen ($q/2 \ge 1$), $\mathbb{E}(\sum_k \mu_k^2 V_k^2)^{q/2} \ge (\sum_k \mu_k^2)^{q/2} = a^q$. Keeping only k^* with $|\mu_{k^*}| = b$ gives $\mathbb{E}(\sum_k \mu_k^2 V_k^2)^{q/2} \ge b^q \gamma_q^q$. Therefore, since $(\mathbb{E}|Z|^q)^{1/q} \ge (\mathbb{E}|\sum_k \mu_k U_k V_k|^q)^{1/q}$ (the factor of 2 cancels),

$$(\mathbb{E}|Z|^q)^{1/q} \ge \max\big(\gamma_q a,\ \gamma_q^2 b\big) \ge \tfrac{1}{2}\Big(\tfrac{\sqrt{q}}{2}a + \tfrac{q}{4}b\Big) \ge \tfrac{1}{8}\big(\sqrt{q}\,a + q\,b\big).$$

The two-sided estimate holds with $c = 1/8$, $C = 12$, uniformly in n and $q \ge 2$. ■

Exercise 2.26

Central idea. As stated the claim is **false**: with g allowed to change sign, no conclusion follows. A two-point counterexample kills it. The intended statement needs $g \ge 0$ (or $\mathbb{E}[h(X)g(X)] \ge 0$); then tilting by h and applying Chebyshev's association inequality to $\phi = 1 - f$ and g gives the result.

Solution. *The statement as written is false.* Take $X \in \{1,2\}$ with $\mathbb{P}\{X=1\} = \mathbb{P}\{X=2\} = 1/2$, $h \equiv 1$, $f(1) = 1$, $f(2) = 0$ (nonnegative, nonincreasing), and $g(1) = -10$, $g(2) = -9$ (nondecreasing, real-valued). Then $\mathbb{E}[h(X)f(X)] = 1/2 \le 1 = \mathbb{E}[h(X)]$, but

$$\mathbb{E}[f(X)g(X)h(X)] = -5 > -9.5 = \mathbb{E}[h(X)g(X)],$$

contradicting the claim. The sign of g is the obstruction.

Corrected statement. Assume additionally $\mathbb{E}[h(X)g(X)] \geq 0$ (this holds in particular if $g \geq 0$), and that the expectations below are absolutely convergent. Then $\mathbb{E}[f(X)g(X)h(X)] \leq \mathbb{E}[h(X)g(X)]$.

Proof. If $\mathbb{E}[h(X)] = 0$ then $h(X) = 0$ a.s. (as $h \geq 0$), and both sides equal 0. Otherwise, define the probability measure $d\mu = h\, d\mathbb{P}/\mathbb{E}[h]$. The claim is equivalent to $\mathbb{E}_\mu[fg] \leq \mathbb{E}_\mu[g]$. Now $\mathbb{E}_\mu[f] = \mathbb{E}[hf]/\mathbb{E}[h] \leq 1$, so $\phi := 1 - f$ is nondecreasing with $\mathbb{E}_\mu[\phi] \geq 0$. For independent copies $X, X' \sim \mu$,

$$\mathbb{E}_\mu[\phi g] - \mathbb{E}_\mu[\phi]\mathbb{E}_\mu[g] = \tfrac{1}{2}\,\mathbb{E}_{\mu\otimes\mu}\big[(\phi(X) - \phi(X'))(g(X) - g(X'))\big] \geq 0,$$

since ϕ and g are both nondecreasing, so the integrand is a.s. nonnegative (this is Chebyshev's association inequality, Theorem 2.14). Hence

$$\mathbb{E}_\mu[(1-f)g] \geq \mathbb{E}_\mu[1-f]\,\mathbb{E}_\mu[g] \geq 0,$$

using $\mathbb{E}_\mu[1-f] \geq 0$ and $\mathbb{E}_\mu[g] = \mathbb{E}[hg]/\mathbb{E}[h] \geq 0$. That is, $\mathbb{E}_\mu[g] \geq \mathbb{E}_\mu[fg]$, i.e. $\mathbb{E}[h(X)g(X)] \geq \mathbb{E}[f(X)g(X)h(X)]$. ■

Exercise 2.27

Central idea. As stated the claim is **false** without centering: a positive mean shifts the whole sum and no n-independent L can work (explicit Exp(1) counterexample below). Under $\mathbb{E}[X_i] = 0$, the tail $\mathbb{P}\{|X_i| \geq u\} \leq e^{-u^p}$ yields the mgf bound $\log \mathbb{E}e^{\lambda Y_i} \leq C(\lambda^2 + |\lambda|^q)$ for all λ ($q = p/(p-1)$) — by power series for $|\lambda| \leq 1$ and a Laplace-type integral estimate for $|\lambda| \geq 1$ — and Chernoff optimization over λ produces exactly the $\min(t^2/\|s\|_2^2,\, t^p/\|s\|_q^p)$ exponent. The case $p = 1$ is Bernstein's inequality.

Solution. *The statement needs* $\mathbb{E}[X_i] = 0$. Counterexample: let $p = 1$, $X_i \sim \mathrm{Exp}(1)$ iid (so $\mathbb{P}\{|X_i| \geq u\} = e^{-u}$), $s_i \equiv 1$. Then $\mathbb{E}[X_i] = 1$, $Z = \sum_{i=1}^n X_i$, and for $t = n/2$, $\mathbb{P}\{Z \geq n/2\} \to 1$ by the law of large numbers, whereas the claimed bound gives $L\exp(-\frac{1}{L}\min(n/4, n/2)) \to 0$. No L independent of n works. We henceforth assume the X_i are centered; write $Y_i = X_i$.

Moment bound. For $k \geq 1$,

$$\mathbb{E}|Y_i|^k = \int_0^\infty k u^{k-1}\mathbb{P}\{|Y_i| \geq u\}\, du \leq \int_0^\infty k u^{k-1} e^{-u^p}\, du = \frac{k}{p}\Gamma\Big(\frac{k}{p}\Big) =: m_k,$$

via $v = u^p$. In particular $v := \mathbb{E}[Y_i^2] \leq m_2 < \infty$. For $p > 1$, $\sum_{k\geq 1} m_k/k! < \infty$ by the ratio test, since $\frac{m_{k+1}/(k+1)!}{m_k/k!} = \frac{\Gamma((k+1)/p)}{\Gamma(k/p)}\frac{1}{k} \lesssim k^{1/p-1} \to 0$ (using $\Gamma(x+\alpha) \lesssim x^\alpha\Gamma(x)$ for $\alpha = 1/p \in (0,1)$). (For $p = 1$, $m_k = k!$ and the series diverges; that case is handled separately below.)

Case $p = 1$ (Bernstein). Here $m_k = k!$, so for $|\lambda| < 1$,

$$\mathbb{E}e^{\lambda Y_i} \leq 1 + \sum_{k=2}^\infty |\lambda|^k = 1 + \frac{\lambda^2}{1-|\lambda|} \leq \exp\Big(\frac{\lambda^2}{1-|\lambda|}\Big).$$

Chernoff with $\lambda = \min\big(\frac{t}{4\|s\|_2^2}, \frac{1}{4\|s\|_\infty}\big) < 1/\|s\|_\infty$ gives

$$\mathbb{P}\Big\{\sum s_i Y_i \geq t\Big\} \leq \exp\Big(-\lambda t + \frac{\lambda^2\|s\|_2^2}{1-\lambda\|s\|_\infty}\Big) \leq \exp\Big(-\tfrac{1}{6}\min\Big(\frac{t^2}{\|s\|_2^2}, \frac{t}{\|s\|_\infty}\Big)\Big),$$

the two cases $\lambda = t/(4\|s\|_2^2)$ and $\lambda = 1/(4\|s\|_\infty)$ each yielding one term. Since $q = \infty$ for $p = 1$, $\|s\|_q = \|s\|_\infty$, matching the claim.

Case $p > 1$: mgf bound for all λ. Let $q = p/(p-1)$. For $|\lambda| \leq 1$, since $\mathbb{E}[Y_i] = 0$,

$$\mathbb{E}e^{\lambda Y_i} \leq 1 + \frac{v\lambda^2}{2} + \sum_{k=3}^\infty \frac{|\lambda|^k m_k}{k!}.$$

Split the sum at $\max(3,q)$: for $3 \le k < q$, $|\lambda|^k \le |\lambda|^3 \le \lambda^2$; for $k \ge q$, $|\lambda|^k \le |\lambda|^q$. With $\sum m_k/k! < \infty$,

$$\mathbb{E}e^{\lambda Y_i} \le 1 + A\lambda^2 + B|\lambda|^q \le \exp(A\lambda^2 + B|\lambda|^q), \qquad |\lambda| \le 1,$$

for A, B depending only on p. For $|\lambda| \ge 1$,

$$\mathbb{E}e^{\lambda Y_i} \le \mathbb{E}e^{|\lambda||Y_i|} \le 1 + |\lambda| \int_0^\infty e^{|\lambda|u - u^p} du.$$

With $u = |\lambda|^{q-1}v$ (using $(q-1)p = q$), the integral equals $|\lambda|^q \int_0^\infty e^{c(v-v^p)}dv$ with $c = |\lambda|^q \ge 1$. Since $\int_0^2 e^{c(v-v^p)}dv \le 2e^{cm_0}$ ($m_0 = \max_{[0,2]}(v-v^p) > 0$) and, for $v \ge 2$, $v - v^p \le -(2^{p-1}-1)v$, giving $\int_2^\infty \le 1/(cc_p)$, we get $\mathbb{E}e^{\lambda Y_i} \le C_1 e^{C_2|\lambda|^q}$, i.e. $\log \mathbb{E}e^{\lambda Y_i} \le C|\lambda|^q$ for $|\lambda| \ge 1$. Combined,

$$\log \mathbb{E}\exp(\lambda Y_i) \le C(\lambda^2 + |\lambda|^q) \quad \forall\, \lambda \in \mathbb{R}, \tag{$*$}$$

with $C = C(p)$.

Chernoff optimization. For $t > 0$ (else trivial) and $s \ne 0$,

$$\mathbb{P}\Big\{ \sum_{i=1}^n s_i Y_i \ge t \Big\} \le \inf_{\lambda \ge 0} \exp(-\lambda t + C\lambda^2 \|s\|_2^2 + C\lambda^q \|s\|_q^q).$$

Take $\lambda = \min(\lambda_1, \lambda_2)$ with $\lambda_1 = \frac{t}{4C\|s\|_2^2}$ and $\lambda_2 = \big(\frac{t}{4C\|s\|_q^q}\big)^{1/(q-1)}$. Since $\lambda \le \lambda_1$ gives $C\lambda^2\|s\|_2^2 \le \lambda t/4$ and $\lambda \le \lambda_2$ gives $C\lambda^q\|s\|_q^q \le \lambda t/4$, the exponent is at most $-\lambda t/2$. Moreover

$$\frac{\lambda t}{2} = \frac{1}{2}\min\Big(\frac{t^2}{4C\|s\|_2^2}, t\Big(\frac{t}{4C\|s\|_q^q}\Big)^{1/(q-1)}\Big), \qquad t\Big(\frac{t}{\|s\|_q^q}\Big)^{1/(q-1)} = \frac{t^p}{\|s\|_q^p}$$

because $\frac{q}{q-1} = p$. Hence

$$\mathbb{P}\Big\{ \sum_{i=1}^n s_i Y_i \ge t \Big\} \le \exp\Big(-c\min\Big(\frac{t^2}{\|s\|_2^2}, \frac{t^p}{\|s\|_q^p}\Big)\Big)$$

for $c = c(p) > 0$, which is of the required form $L\exp(-\frac{1}{L}\min(\cdots))$ with $L = \max(1, 1/c)$.

Remark. The constants A, B, C, c depend on p and blow up as $p \downarrow 1$; this is permitted since L may depend on p. For a two-sided bound on $|\sum s_i X_i|$, apply the result to $-X_i$ (which satisfy the same hypotheses) and double L. ■

Exercise 2.28

Central idea. The mgf of the Gumbel law is out of reach directly, but Rényi's representation writes $\max(Y_1, \dots, Y_n)$ of iid Exp(1) variables as $\sum_{i=1}^n E_i/i$, a sum of *independent* variables. Means, variances, and mgf bounds all pass through this sum explicitly; convergence in distribution plus uniform integrability (from bounded 4th moments) then transfers them to the Gumbel limit. The mgf bound itself needs only Fatou's lemma, not uniform integrability.

Erratum. The statement's range "for all $\lambda \ge 0$" is wrong as printed (the hint's "$\max(X_1, \dots, X_n)$" is also a typo for $\max(Y_1, \dots, Y_n)$). Here $\mathrm{Var}(X) = \pi^2/6$ and $\mathbb{E}e^{\lambda X} = \Gamma(1-\lambda)$ only for $\lambda < 1$; for $\lambda \ge 1$, $\mathbb{E}e^{\lambda(X - \mathbb{E}X)} = +\infty$. For $\lambda > 1$ the printed right side $\mathrm{Var}(X)\lambda^2/(2(1-\lambda))$ is finite and negative — e.g. at $\lambda = 2$ it equals $(\pi^2/6)\cdot 4/(2\cdot(-1)) = -\pi^2/3 \approx -3.29$ — so the inequality is false there; at $\lambda = 1$ the right side is undefined. The corrected statement, proved below, is the bound for $0 \le \lambda < 1$ (sub-gamma with variance factor $\pi^2/6$ and scale 1).

Solution. Let $Y_1, \dots, Y_n$ be iid Exp(1) and $M_n = \max(Y_1, \dots, Y_n)$. By Rényi's representation (given in the hint),

$$M_n \overset{d}{=} \sum_{i=1}^n \frac{E_i}{i},$$

where the E_i are iid Exp(1), and $M_n - \log n \Rightarrow X$ where X is Gumbel. (The hint's "$\max(X_1, \dots, X_n)$" is a typo for $\max(Y_1, \dots, Y_n)$.) Hence, with $H_n = \sum_{i=1}^n 1/i$,

$$\mathbb{E}[M_n] = H_n, \qquad \mathrm{Var}(M_n) = \sum_{i=1}^n \frac{1}{i^2},$$

using $\mathbb{E}[E_i] = 1$, $\mathrm{Var}(E_i) = 1$, and independence.

Mean. $\mathbb{E}[(M_n - \log n)^2] = \sum_{i=1}^n 1/i^2 + (H_n - \log n)^2 \le \pi^2/6 + C < \infty$ uniformly in n, since $H_n - \log n \to \gamma$ is bounded. Bounded second moments imply uniform integrability, so with $M_n - \log n \Rightarrow X$,

$$\mathbb{E}[X] = \lim_{n\to\infty} \mathbb{E}[M_n - \log n] = \lim_{n\to\infty} (H_n - \log n) = \gamma,$$

the Euler–Mascheroni constant.

Variance. Put $Z_i = (E_i - 1)/i$ (centered, independent). Then $M_n - \log n = \sum_{i=1}^n Z_i + (H_n - \log n)$, and

$$\mathbb{E}\Big[\Big(\sum_{i=1}^n Z_i\Big)^4\Big] = \sum_{i=1}^n \mathbb{E}[Z_i^4] + 6\sum_{i<j} \mathbb{E}[Z_i^2]\mathbb{E}[Z_j^2],$$

odd cross-terms vanishing. Since $\mathbb{E}[(E_1 - 1)^4] = 9$ and $\mathbb{E}[Z_i^2] = 1/i^2$,

$$\mathbb{E}\Big[\Big(\sum_{i=1}^n Z_i\Big)^4\Big] = 9\sum_{i=1}^n \frac{1}{i^4} + 3\Big(\sum_{i=1}^n \frac{1}{i^2}\Big)^2 - 3\sum_{i=1}^n \frac{1}{i^4} \le C < \infty$$

uniformly in n. Hence $\sup_n \mathbb{E}[(M_n - \log n)^4] < \infty$, so $(M_n - \log n)^2$ is uniformly integrable and

$$\mathbb{E}[X^2] = \lim_{n\to\infty} \mathbb{E}[(M_n - \log n)^2] = \lim_{n\to\infty} \Big(\sum_{i=1}^n \frac{1}{i^2} + (H_n - \log n)^2\Big) = \frac{\pi^2}{6} + \gamma^2,$$

using $\sum_{i=1}^\infty 1/i^2 = \pi^2/6$. Therefore $\mathrm{Var}(X) = \pi^2/6$.

Sub-gamma mgf bound. For $E_1 \sim \mathrm{Exp}(1)$ and $t < 1$, $\mathbb{E}[e^{t(E_1 - 1)}] = e^{-t}/(1-t)$, and for $t \in [0, 1)$,

$$-t - \log(1-t) = \sum_{k=2}^\infty \frac{t^k}{k} \le \frac{1}{2}\sum_{k=2}^\infty t^k = \frac{t^2}{2(1-t)}.$$

For $\lambda \in [0, 1)$ and every $i \ge 1$ (so $\lambda/i < 1$),

$$\log \mathbb{E}\exp\Big(\lambda \frac{E_i - 1}{i}\Big) \le \frac{(\lambda/i)^2}{2(1 - \lambda/i)} \le \frac{\lambda^2/i^2}{2(1-\lambda)}.$$

By independence and Rényi's representation,

$$\log \mathbb{E}\exp(\lambda(M_n - \mathbb{E}M_n)) \le \frac{\lambda^2}{2(1-\lambda)} \sum_{i=1}^n \frac{1}{i^2} \le \frac{\lambda^2}{2(1-\lambda)} \cdot \frac{\pi^2}{6}, \qquad 0 \le \lambda < 1.$$

Now $M_n - \mathbb{E}M_n = (M_n - \log n) - (H_n - \log n) \Rightarrow X - \gamma = X - \mathbb{E}X$ by Slutsky's theorem. Since $x \mapsto e^{\lambda x}$ is continuous and nonnegative, Fatou's lemma gives

$$\mathbb{E}e^{\lambda(X - \mathbb{E}X)} \le \liminf_{n\to\infty} \mathbb{E}e^{\lambda(M_n - \mathbb{E}M_n)},$$

hence

$$\log \mathbb{E}\exp(\lambda(X - \mathbb{E}X)) \le \frac{\mathrm{Var}(X)\,\lambda^2}{2(1-\lambda)}, \qquad 0 \le \lambda < 1.$$

Remark on the range. As noted in the Erratum, the bound cannot extend to $\lambda \ge 1$: the left side is $+\infty$ there, while the right side is negative for $\lambda > 1$ ($-\pi^2/3 \approx -3.29$ at $\lambda = 2$) and undefined at $\lambda = 1$. The correct range is $0 \le \lambda < 1$, i.e., X is sub-gamma with variance factor $\mathrm{Var}(X) = \pi^2/6$ and scale 1.

Chapter 3 — Bounding the Variance

Exercise 3.1

Central idea. Write $Z = \|G\|_2$ with $G \sim N(0, I_D)$. The upper bound is Jensen ($\sqrt{\cdot}$ concave). For the lower bound, the Gaussian Poincaré inequality gives $\mathrm{Var}(Z) \le \mathbb{E}\|\nabla\|G\|_2\|^2 = 1$, since the Euclidean norm is 1-Lipschitz with unit gradient a.e.; then $(\mathbb{E}Z)^2 = D - \mathrm{Var}(Z) \ge D - 1$, and $\sqrt{D-1} \ge \sqrt{D} - 1$ for $D \ge 1$.

Solution. Since $Z^2 \sim \chi^2_D$, we have $Z \overset{d}{=} \|G\|_2$ where $G = (G_1, \dots, G_D)$ has i.i.d. $N(0,1)$ coordinates; in particular $\mathbb{E}[Z^2] = D$.

Upper bound. By Jensen's inequality ($\sqrt{\cdot}$ concave),

$$\mathbb{E}Z = \mathbb{E}\sqrt{Z^2} \le \sqrt{\mathbb{E}[Z^2]} = \sqrt{D}.$$

Lower bound. Let $f(x) = \|x\|_2$. Then $\nabla f(x) = x/\|x\|_2$ for $x \neq 0$, so $\|\nabla f(x)\|_2 = 1$ a.e. The Gaussian Poincaré inequality gives

$$\mathrm{Var}(Z) = \mathrm{Var}(f(G)) \le \mathbb{E}\|\nabla f(G)\|_2^2 = 1.$$

Hence $(\mathbb{E}Z)^2 = \mathbb{E}[Z^2] - \mathrm{Var}(Z) \ge D - 1$, and since $\mathbb{E}Z \ge 0$,

$$\mathbb{E}Z \ge \sqrt{D-1}.$$

For $D \ge 1$, $\sqrt{D-1} \ge \sqrt{D} - 1$: both sides are nonnegative (as $\sqrt{D} \ge 1$), and squaring gives $D - 1 \ge D - 2\sqrt{D} + 1$, i.e. $2\sqrt{D} \ge 2$. Therefore $\mathbb{E}Z \ge \sqrt{D} - 1$. ■

Exercise 3.2

Central idea. Apply the Efron–Stein inequality $\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Var}_i(Z)]$, where Var_i is the variance conditionally on all variables except X_i. Given X_{-i}, Z is a function ϕ of the binary X_i alone; for $\mathbb{P}\{X_i = 1\} = p_i$, $\mathrm{Var}(\phi(X_i)) = p_i(1-p_i)(\phi(1) - \phi(-1))^2$, and bounded differences gives $(\phi(1) - \phi(-1))^2 \le c_i^2$.

Solution. Let $\mathrm{Var}_i(Z) = \mathrm{Var}(Z \mid X_{-i})$ denote conditional variance given all variables except X_i. The Efron–Stein inequality gives

$$\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Var}_i(Z)].$$

Fix i and condition on $X_{-i} = x_{-i}$. Write $\phi(t) = f(x_{-i}, t)$ for $t \in \{-1, 1\}$, so that conditionally $Z = \phi(X_i)$ with X_i binary, $\mathbb{P}\{X_i = 1\} = p_i$. For $a = \phi(1)$, $b = \phi(-1)$,

$$\begin{aligned}\mathrm{Var}(\phi(X_i)) &= p_i a^2 + (1-p_i)b^2 - \big(p_i a + (1-p_i)b\big)^2 \\ &= p_i(1-p_i)(a-b)^2 = p_i(1-p_i)\big(\phi(1) - \phi(-1)\big)^2.\end{aligned}$$

The bounded differences property gives $|\phi(1) - \phi(-1)| = |f(x_{-i}, 1) - f(x_{-i}, -1)| \le c_i$, uniformly in x_{-i}. Hence $\mathrm{Var}_i(Z) \le p_i(1-p_i)c_i^2$ a.s., and taking expectations,

$$\mathbb{E}[\mathrm{Var}_i(Z)] \le p_i(1-p_i)c_i^2.$$

Summing over i yields $\mathrm{Var}(Z) \le \sum_{i=1}^n c_i^2 p_i(1-p_i)$. ■

Exercise 3.3

Central idea. Use the Efron–Stein inequality in the form $\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}[(Z - Z_i')_+^2]$, with $Z = \max_j X_j$ and Z_i' the max after resampling X_i. Since $(Z - Z_i')_+ \le (X_i - \max_{j \neq i} X_j)_+$ and at most one index can be the unique maximizer, the sum collapses to $(X_{(n)} - X_{(n-1)})^2$. Both sides are then computed explicitly: via Rényi's representation of exponential spacings, and via the joint density of the top two uniform order statistics.

Solution. Let $Z = X_{(n)} = \max_{1\le j\le n} X_j$ and let Z_i' be Z with X_i replaced by an independent copy X_i'. Write $M_{-i} = \max_{j\neq i} X_j$, so $Z = \max(X_i, M_{-i})$ and $Z_i' = \max(X_i', M_{-i})$. Then

$$(Z - Z_i')_+ = \big(\max(X_i, M_{-i}) - \max(X_i', M_{-i})\big)_+ \le (X_i - M_{-i})_+,$$

since if $X_i \le M_{-i}$ both sides are 0, and if $X_i > M_{-i}$ then $Z = X_i$ and $Z_i' \ge M_{-i}$. The Efron–Stein inequality gives

$$\operatorname{Var}(X_{(n)}) \le \sum_{i=1}^n \mathbb{E}[(Z - Z_i')_+^2] \le \sum_{i=1}^n \mathbb{E}[(X_i - M_{-i})_+^2].$$

Now $(X_i - M_{-i})_+ > 0$ implies $X_i > \max_{j\neq i} X_j$, i.e. X_i is the unique maximizer; two distinct indices cannot both be unique maximizers. Hence at most one term of the sum is positive, and when $(X_{i_0} - M_{-i_0})_+ > 0$ we have $X_{i_0} = X_{(n)}$, $M_{-i_0} = X_{(n-1)}$. Therefore

$$\sum_{i=1}^n (X_i - M_{-i})_+^2 \le (X_{(n)} - X_{(n-1)})^2 \quad \text{a.s.},$$

and $\operatorname{Var}(X_{(n)}) \le \mathbb{E}[(X_{(n)} - X_{(n-1)})^2]$, with no assumption on the laws of the X_i.

Exponential case. Let $S_k = X_{(k)} - X_{(k-1)}$ ($X_{(0)} := 0$); the S_k are independent exponentials with means $1/n, 1/(n-1), \dots, 1$. Then $X_{(n)} = \sum_{k=1}^n S_k$, so

$$\operatorname{Var}(X_{(n)}) = \sum_{k=1}^n \frac{1}{(n-k+1)^2} = \sum_{j=1}^n \frac{1}{j^2},$$

while $X_{(n)} - X_{(n-1)} = S_n \sim \mathrm{Exp}(1)$, so $\mathbb{E}[(X_{(n)} - X_{(n-1)})^2] = 2$. Indeed $\sum_{j=1}^n 1/j^2 \le \pi^2/6 < 2$.

Uniform case. $X_{(n)}$ has d.f. t^n on $[0,1]$, so $\mathbb{E}X_{(n)} = n/(n+1)$, $\mathbb{E}[X_{(n)}^2] = n/(n+2)$, and

$$\operatorname{Var}(X_{(n)}) = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n}{(n+1)^2(n+2)}.$$

The pair $(X_{(n-1)}, X_{(n)})$ has density $n(n-1)s^{n-2}$ on $0 \le s \le t \le 1$, so with $u = s/t$,

$$\begin{aligned}
\mathbb{E}[(X_{(n)} - X_{(n-1)})^2] &= \int_0^1 \int_0^t (t-s)^2\, n(n-1)s^{n-2}\, ds\, dt \\
&= n(n-1) \int_0^1 t^{n+1} \int_0^1 (1-u)^2 u^{n-2}\, du\, dt \\
&= n(n-1) \cdot \frac{2}{(n+1)n(n-1)} \cdot \frac{1}{n+2} = \frac{2}{(n+1)(n+2)},
\end{aligned}$$

using $\int_0^1 (1-u)^2 u^{n-2}\, du = B(n-1,3) = 2/[(n+1)n(n-1)]$. Again $\frac{n}{(n+1)^2(n+2)} \le \frac{2}{(n+1)(n+2)}$. ■

Exercise 3.4

Central idea. Assume the X_i i.i.d., so each Z_i has the law of f_{n-1} on $n-1$ samples. Write $b_m = \mathbb{E}f_m - \theta$ for the bias at sample size m. Then $\mathbb{E}\tilde{Z} - \theta = nb_n - (n-1)b_{n-1}$; the c/n terms cancel exactly ($n \cdot c/n - (n-1) \cdot c/(n-1) = 0$), which is the jackknife's bias correction. (Remark below on the remainder: the $O(n^{-2})$ conclusion as stated needs the bias expansion to second order.)

Solution. Assume $X_1, \dots, X_n$ i.i.d. By symmetry $\mathbb{E}Z_i$ is independent of i, each Z_i having the distribution of f_{n-1} applied to $n-1$ i.i.d. copies. Let $b_m = \mathbb{E}f_m(X_1, \dots, X_m) - \theta$, so $b_m = c/m + O(m^{-2})$. Then

$$\begin{aligned}
\mathbb{E}\tilde{Z} - \theta &= \mathbb{E}Z - (n-1)\Big(\frac{1}{n}\sum_{i=1}^n \mathbb{E}Z_i - \mathbb{E}Z\Big) - \theta \\
&= n(\mathbb{E}Z - \theta) - (n-1)(\mathbb{E}Z_1 - \theta) \\
&= nb_n - (n-1)b_{n-1}.
\end{aligned}$$

Inserting $b_m = c/m + \rho_m$ with $\rho_m = O(m^{-2})$,

$$nb_n - (n-1)b_{n-1} = \Big(c + n\rho_n\Big) - \Big(c + (n-1)\rho_{n-1}\Big) = n\rho_n - (n-1)\rho_{n-1},$$

so the order-$1/n$ bias cancels: this is Quenouille's correction.

Remark on the remainder. As stated, the hypothesis $b_n = c/n + O(n^{-2})$ only yields $n\rho_n - (n-1)\rho_{n-1} = O(n^{-1})$ in the worst case: e.g. $\rho_n = (-1)^n/n^2$ gives $n\rho_n - (n-1)\rho_{n-1} = (-1)^n(2n-1)/(n(n-1))$, which is $\asymp 1/n$, not $O(n^{-2})$. The $O(n^{-2})$ conclusion holds under the classical smooth two-term expansion $b_n = c/n + d/n^2 + O(n^{-3})$ (Quenouille's setting), for which

$$nb_n - (n-1)b_{n-1} = d\Big(\frac{1}{n} - \frac{1}{n-1}\Big) + O(n^{-2}) = -\frac{d}{n(n-1)} + O(n^{-2}) = O(n^{-2}).$$

In applications the bias is smooth in $1/n$, so the jackknife indeed reduces the bias from order $1/n$ to order $1/n^2$. ■

Exercise 3.5

Central idea. Apply Efron–Stein: $\mathrm{Var}(f(X)) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Var}_i(f(X))]$. Each coordinate slice $t \mapsto f(x_{-i}, t)$ is 1-Lipschitz (from ℓ^1-Lipschitzness of f), and a 1-Lipschitz map cannot increase variance: $\mathrm{Var}(\phi(X_i)) = \frac{1}{2}\mathbb{E}[(\phi(X_i) - \phi(X_i'))^2] \le \frac{1}{2}\mathbb{E}[(X_i - X_i')^2] = \mathrm{Var}(X_i)$ for X_i' an independent copy. Summing gives $\mathrm{Var}(f(X)) \le \sum_i \mathrm{Var}(X_i) = \mathrm{Var}(\sum_i X_i)$, attained at $f(x) = \sum_i x_i \in \mathcal{F}$.

Solution. Let $X_1, \dots, X_n$ be independent with $\mathrm{Var}(X_i) < \infty$ and $f \in \mathcal{F}$. Since $|f(x)| \le |f(0)| + \sum_i |x_i|$, $f(X)$ has finite second moment, so the Efron–Stein inequality applies:

$$\mathrm{Var}(f(X)) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Var}_i(f(X))],$$

where Var_i conditions on $X_{-i} = (X_j)_{j \ne i}$.

Fix x_{-i} and set $\phi_{x_{-i}}(t) = f(x_{-i}, t)$. From ℓ^1-Lipschitzness,

$$|\phi_{x_{-i}}(s) - \phi_{x_{-i}}(t)| = |f(x_{-i}, s) - f(x_{-i}, t)| \le |s - t|,$$

so each slice is 1-Lipschitz. Conditionally on $X_{-i} = x_{-i}$, $f(X) = \phi_{x_{-i}}(X_i)$ with X_i independent of x_{-i}. For any 1-Lipschitz ϕ and X_i' an independent copy of X_i,

$$\mathrm{Var}(\phi(X_i)) = \tfrac{1}{2}\,\mathbb{E}\big[(\phi(X_i) - \phi(X_i'))^2\big] \le \tfrac{1}{2}\,\mathbb{E}\big[(X_i - X_i')^2\big] = \mathrm{Var}(X_i),$$

using $\mathrm{Var}(W) = \frac{1}{2}\mathbb{E}[(W - W')^2]$. Hence $\mathrm{Var}_i(f(X)) \le \mathrm{Var}(X_i)$ a.s., and

$$\mathrm{Var}(f(X)) \le \sum_{i=1}^n \mathrm{Var}(X_i).$$

By independence, $\sum_i \mathrm{Var}(X_i) = \mathrm{Var}(\sum_i X_i)$. The function $f_0(x) = \sum_{i=1}^n x_i$ satisfies $|f_0(x) - f_0(y)| \le \sum_i |x_i - y_i|$, so $f_0 \in \mathcal{F}$, and $\mathrm{Var}(f_0(X)) = \sum_i \mathrm{Var}(X_i)$. Thus the maximum of $\mathrm{Var}(f(X))$ over $\mathcal{F}$ is attained at the sum. ■

Exercise 3.6

Central idea. Apply the Efron–Stein inequality in jackknife (leave-one-out) form. Deleting X_i shifts the median $X_{(m)}$ ($m = n/2$) to $X_{(m+1)}$ exactly when $\mathrm{rank}(X_i) \le m$, and leaves it unchanged otherwise; precisely $m = n/2$ deletions are effective, giving $\mathrm{Var}(X_{(m)}) \le \frac{n}{2}\mathbb{E}[(X_{(m+1)} - X_{(m)})^2]$, and exchangeability of uniform spacings converts the upper gap to the lower gap. Both sides are then explicit via Beta/Dirichlet laws:

$\mathrm{Var}(X_{(m)}) = n/[4(n+1)^2] \sim 1/(4n)$ while the jackknife bound is $n/[(n+1)(n+2)] \sim 1/n$. The jackknife is inconsistent: n times the estimate converges to $\frac{1}{2}E^2$ ($E \sim \mathrm{Exp}(1)$), a non-degenerate law with mean $1 \neq 1/4$.

Solution. Write $m = n/2$ and $Z = X_{(m)}$. For each i, let $Z^{(i)}$ be the median of the $n-1$ observations with X_i deleted, i.e. the m-th order statistic of the reduced sample (which has odd size $n-1$, so the median is uniquely defined). By the Efron–Stein inequality in jackknife form,

$$\mathrm{Var}(Z) \leq \sum_{i=1}^{n} \mathbb{E}[(Z - Z^{(i)})^2].$$

Let $k = \mathrm{rank}(X_i)$ in the full sample (a.s. unique). If $k > m$, deleting X_i does not affect the m smallest values, so $Z^{(i)} = X_{(m)} = Z$. If $k \leq m$, the m-th smallest of the remaining values is $X_{(m+1)}$, so $Z^{(i)} = X_{(m+1)} \geq Z$. Hence, with $\mathbf{1}_i = \mathbf{1}_{\{\mathrm{rank}(X_i) \leq m\}}$,

$$(Z - Z^{(i)})^2 = (X_{(m+1)} - X_{(m)})^2 \, \mathbf{1}_i,$$

and since exactly m observations have rank $\leq m$,

$$\sum_{i=1}^{n} (Z - Z^{(i)})^2 = (X_{(m+1)} - X_{(m)})^2 \sum_{i=1}^{n} \mathbf{1}_i = \frac{n}{2} \, (X_{(m+1)} - X_{(m)})^2 \quad \text{a.s.}$$

Taking expectations and using exchangeability of the uniform spacings (the $n+1$ gaps $(X_{(1)}, X_{(2)} - X_{(1)}, \dots, 1 - X_{(n)})$ are exchangeable, Dirichlet$(1, \dots, 1)$),

$$\mathrm{Var}(X_{(m)}) \leq \frac{n}{2} \, \mathbb{E}[(X_{(m+1)} - X_{(m)})^2] = \frac{n}{2} \, \mathbb{E}[(X_{(m)} - X_{(m-1)})^2].$$

Explicit values. $X_{(m)} \sim \mathrm{Beta}(m, n-m+1) = \mathrm{Beta}(n/2, n/2+1)$, so

$$\mathrm{Var}(X_{(m)}) = \frac{\frac{n}{2} \cdot \frac{n+2}{2}}{(n+1)^2(n+2)} = \frac{n}{4(n+1)^2} \sim \frac{1}{4n}.$$

The gap $D = X_{(m)} - X_{(m-1)} \sim \mathrm{Beta}(1, n)$, so $\mathbb{E}[D^2] = \frac{2}{(n+1)(n+2)}$ and

$$\frac{n}{2} \mathbb{E}[(X_{(m)} - X_{(m-1)})^2] = \frac{n}{(n+1)(n+2)} \sim \frac{1}{n}.$$

Indeed $\frac{n}{4(n+1)^2} \leq \frac{n}{(n+1)(n+2)}$, confirming the inequality.

Consistency. The jackknife estimate $\widehat{V}_n = \frac{n}{2}(X_{(m)} - X_{(m-1)})^2$ satisfies $n\widehat{V}_n = \frac{1}{2}(nD)^2$. Since $D \sim \mathrm{Beta}(1, n)$, $\mathbb{P}\{nD > t\} = (1 - t/n)^n \to e^{-t}$, i.e. $nD \Rightarrow E \sim \mathrm{Exp}(1)$, so $n\widehat{V}_n \Rightarrow \frac{1}{2}E^2$. This limit is non-degenerate, hence $n\widehat{V}_n \not\xrightarrow{\mathbb{P}} 1/4$ (the true asymptotic variance, as $\sqrt{n}\,(X_{(m)} - \frac{1}{2}) \Rightarrow N(0, \frac{1}{4})$). The jackknife is **inconsistent** for the median: it neither concentrates nor has the right mean ($\mathbb{E}[\frac{1}{2}E^2] = 1 \neq \frac{1}{4}$, a fourfold overestimate). ■

Exercise 3.7

Central idea. For $Z = \mathbb{E}_\sigma \sup_{f \in \mathcal{F}} \sum_i \sigma_i f(X_i)$ with $0 \leq f \leq 1$, take Z_i to be the Rademacher average with coordinate i dropped. Then $0 \leq Z - Z_i \leq 1$: the upper bound because $\sup_f \sigma_i f(X_i) \leq 1$, the lower bound by Jensen ($\mathbb{E}_{\sigma_i} \sup_f \geq \sup_f \mathbb{E}_{\sigma_i}$, killing the i-th term since $\mathbb{E}\sigma_i = 0$). For the sum bound, couple through the σ-wise (near-)maximizer $\hat{f}$: $Z - Z_i \leq \mathbb{E}_\sigma[\sigma_i \hat{f}(X_i)]$, and $\sum_i \sigma_i \hat{f}(X_i) = \sup_f \sum_i \sigma_i f(X_i)$, so $\sum_i (Z - Z_i) \leq Z$. Hence Z is $(1, 0)$-self-bounding.

Solution. Let $\mathcal{F}$ be a countable class of measurable $f : \mathcal{X} \to [0, 1]$, $X_1, \dots, X_n$ independent, and $\sigma_1, \dots, \sigma_n$ iid Rademacher independent of X. The conditional Rademacher average is

$$Z = Z(X) = \mathbb{E}_\sigma \Big[\sup_{f \in \mathcal{F}} \sum_{i=1}^{n} \sigma_i f(X_i) \Big],$$

a nonnegative function of $X = (X_1, \dots, X_n)$. For each i put

$$Z_i = \mathbb{E}_\sigma\Big[\sup_{f\in\mathcal{F}} \sum_{j\neq i} \sigma_j f(X_j)\Big],$$

which does not depend on X_i or σ_i. We show $0 \le Z - Z_i \le 1$ for all i and $\sum_i (Z - Z_i) \le Z$.

Upper bound. For fixed σ,

$$\sup_f \sum_{j=1}^n \sigma_j f(X_j) \le \sup_f [\sigma_i f(X_i)] + \sup_f \sum_{j\neq i} \sigma_j f(X_j) \le 1 + \sup_f \sum_{j\neq i} \sigma_j f(X_j),$$

since $\sigma_i f(X_i) \le 1$ ($0 \le f \le 1$). Taking $\mathbb{E}_\sigma$ gives $Z \le Z_i + 1$.

Lower bound. By Jensen, for fixed $\sigma^{(i)} = (\sigma_j)_{j\neq i}$,

$$\mathbb{E}_{\sigma_i}\Big[\sup_f \sum_{j=1}^n \sigma_j f(X_j)\Big] \ge \sup_f \mathbb{E}_{\sigma_i}\Big[\sum_{j=1}^n \sigma_j f(X_j)\Big] = \sup_f \sum_{j\neq i} \sigma_j f(X_j),$$

as $\mathbb{E}[\sigma_i] = 0$. Taking $\mathbb{E}_{\sigma^{(i)}}$ yields $Z \ge Z_i$. Thus $0 \le Z - Z_i \le 1$.

Sum bound. Fix σ and $\epsilon > 0$, and choose $f_{\epsilon,\sigma} \in \mathcal{F}$ with $\sum_j \sigma_j f_{\epsilon,\sigma}(X_j) \ge \sup_f \sum_j \sigma_j f(X_j) - \epsilon$. Then

$$\sup_f \sum_{j=1}^n \sigma_j f(X_j) - \sup_g \sum_{j\neq i} \sigma_j g(X_j) \le \sigma_i f_{\epsilon,\sigma}(X_i) + \epsilon,$$

using the near-maximizer for the first supremum and evaluating the second at $f_{\epsilon,\sigma}$. Taking $\mathbb{E}_\sigma$,

$$Z - Z_i \le \mathbb{E}_\sigma[\sigma_i f_{\epsilon,\sigma}(X_i)] + \epsilon.$$

Summing over i and noting $\sum_i \sigma_i f_{\epsilon,\sigma}(X_i) \le \sup_f \sum_i \sigma_i f(X_i)$ pointwise,

$$\sum_{i=1}^n (Z - Z_i) \le \mathbb{E}_\sigma\Big[\sup_f \sum_{i=1}^n \sigma_i f(X_i)\Big] + n\epsilon = Z + n\epsilon.$$

Since $\epsilon > 0$ is arbitrary, $\sum_i (Z - Z_i) \le Z$.

Therefore Z is $(1, 0)$-self-bounding: with the functions Z_i above, $0 \le Z - Z_i \le 1$ and $\sum_i (Z - Z_i) \le Z$. In particular the Efron–Stein inequality gives $\mathrm{Var}(Z) \le \mathbb{E}[Z]$. ■

Exercise 3.8

Central idea. Write $Z = \sum_{k\ge1} I_k$ with $I_k = \mathbf{1}_{\{k \text{ appears}\}}$. Then $\mathbb{E}[Z]/n = \frac{1}{n}\sum_k (1 - (1-p_k)^n) \to 0$ by dominated convergence, since $(1-(1-p_k)^n)/n \le p_k$ is summable. The variance is explicit from $\mathrm{Var}(I_k)$ and $\mathrm{Cov}(I_k, I_l) = (1 - p_k - p_l)^n - (1-p_k)^n(1-p_l)^n \le 0$ (distinct values compete, so indicators are negatively correlated). Z is $(1, 0)$-self-bounding (deleting one observation removes at most one distinct value; the number of effective deletions is at most Z), so Efron–Stein gives $\mathrm{Var}(Z) \le \mathbb{E}[Z]$ — a bound that ignores the negative correlations and is therefore strictly looser than the explicit formula.

Solution. Let $X_1, \dots, X_n$ be iid with $\mathbb{P}\{X_1 = k\} = p_k$, $k = 1, 2, \dots$, $\sum_k p_k = 1$. Let $Z = |\{X_1, \dots, X_n\}|$ and $I_k = \mathbf{1}_{\{\exists i: X_i = k\}}$, so $Z = \sum_{k\ge1} I_k$.

Mean. $\mathbb{E}[I_k] = 1 - (1-p_k)^n$, hence $\mathbb{E}[Z] = \sum_k (1 - (1-p_k)^n)$. Since $1 - (1-x)^n \le nx$ for $x \in [0, 1]$,

$$\frac{1 - (1-p_k)^n}{n} \le p_k, \qquad \sum_k p_k = 1 < \infty,$$

and $\frac{1-(1-p_k)^n}{n} \to 0$ for each k. By dominated convergence,

$$\frac{\mathbb{E}[Z]}{n} = \sum_{k\geq 1} \frac{1-(1-p_k)^n}{n} \longrightarrow 0.$$

Variance. $\mathrm{Var}(I_k) = (1-(1-p_k)^n)(1-p_k)^n$. For $k \neq l$,

$$\begin{aligned}\mathbb{E}[I_k I_l] &= \mathbb{P}\{k \text{ and } l \text{ both appear}\} \\ &= 1-(1-p_k)^n-(1-p_l)^n+(1-p_k-p_l)^n,\end{aligned}$$

so with $\mathbb{E}[I_k]\mathbb{E}[I_l] = (1-(1-p_k)^n)(1-(1-p_l)^n)$,

$$\mathrm{Cov}(I_k, I_l) = (1-p_k-p_l)^n - \big((1-p_k)(1-p_l)\big)^n \leq 0,$$

because $(1-p_k)(1-p_l) = 1-p_k-p_l+p_kp_l \geq 1-p_k-p_l$. Hence

$$\mathrm{Var}(Z) = \sum_{k\geq 1}(1-(1-p_k)^n)(1-p_k)^n + \sum_{k\neq l}\Big[(1-p_k-p_l)^n - \big((1-p_k)(1-p_l)\big)^n\Big],$$

the second sum being nonpositive.

Efron–Stein comparison. Let Z_i be the number of distinct values with X_i deleted. Then $0 \leq Z - Z_i \leq 1$ (deleting one observation destroys at most one distinct value), and $\sum_i (Z - Z_i)$ equals the number of values occurring exactly once, hence $\leq Z$. So Z is $(1,0)$-self-bounding and Efron–Stein gives $\mathrm{Var}(Z) \leq \mathbb{E}[Z] = \sum_k (1-(1-p_k)^n)$. The explicit formula refines this: $\mathrm{Var}(Z) \leq \sum_k \mathrm{Var}(I_k) \leq \mathbb{E}[Z]$, with strict inequality whenever some covariance is nonzero — the Efron–Stein bound does not exploit the negative dependence between distinct values. ■

Exercise 3.9

Central idea. Write $Z = \sum_t T_t$ over the $\binom{n}{3}$ triples, T_t the indicator that triple t is a triangle. Then $\mathrm{Var}(Z)$ has only two non-zero covariance types: $\mathrm{Var}(T_t) = p^3(1-p^3)$, and for triples sharing exactly an edge, $\mathrm{Cov}(T_t, T_s) = p^5 - p^6$ (five distinct edges involved); triples sharing ≤ 1 vertex are independent. Counting $\binom{n}{2}(n-2)(n-3)$ ordered edge-sharing pairs gives the exact variance. For Efron–Stein, resampling one edge e changes Z by $(X_e - X_e')\sum_{t\ni e} M_t$ with M_t the product of the other two edges; independence of X_e from the M_t's and $\mathbb{E}|X_e - X_e'| = 2p(1-p)$ yield a bound whose n^4-order (covariance) term matches the truth exactly, the looseness being confined to the n^3-order term.

Solution. Let $X_e \sim \mathrm{Bernoulli}(p)$ be iid over the $N = \binom{n}{2}$ potential edges. For a triple $t = \{i,j,k\}$ put $T_t = X_{ij}X_{jk}X_{ik}$; then $Z = \sum_t T_t$ over the $\binom{n}{3}$ triples, $\mathbb{E}[Z] = \binom{n}{3}p^3$.

Exact variance. $\mathrm{Var}(T_t) = p^3(1-p^3)$. For $t \neq s$, $\mathrm{Cov}(T_t, T_s)$ depends on the shared edges: if t, s share at most one vertex they involve disjoint edge sets, so the covariance is 0. If they share exactly two vertices, i.e. an edge e (say $t = \{1,2,3\}$, $s = \{1,2,4\}$), then with five distinct edges,

$$\mathbb{E}[T_tT_s] = \mathbb{E}[X_{12}^2 X_{23}X_{13}X_{24}X_{14}] = p^5,$$

so $\mathrm{Cov}(T_t, T_s) = p^5 - p^6 = p^5(1-p)$. The number of ordered pairs (t,s), $t \neq s$, sharing an edge: choose the edge ($\binom{n}{2}$), the third vertex of t ($n-2$), the third vertex of s ($n-3$, must differ). Hence

$$\mathrm{Var}(Z) = \binom{n}{3}p^3(1-p^3) + \binom{n}{2}(n-2)(n-3)\,p^5(1-p).$$

Efron–Stein. Resample edge e to X_e'. Only the $n-2$ triangles containing e can change: with M_t the product of the two edges of t other than e,

$$Z - Z_e' = (X_e - X_e')\sum_{t\ni e} M_t.$$

By (X_e, X'_e)-symmetry, $\mathbb{E}[(Z - Z'_e)_+^2] = \frac{1}{2}\mathbb{E}[(Z - Z'_e)^2]$, and $(X_e - X'_e)^2 = |X_e - X'_e|$ is independent of the M_t's with mean $2p(1-p)$. For $e = \{i, j\}$, $\sum_{t \ni e} M_t = \sum_{k \neq i,j} X_{ik} X_{jk}$, so

$$\mathbb{E}\Big[\Big(\sum_{t \ni e} M_t\Big)^2\Big] = (n-2)p^2 + (n-2)(n-3)p^4.$$

Efron–Stein ($\mathrm{Var}(Z) \le \sum_e \mathbb{E}[(Z - Z'_e)_+^2]$) gives

$$\mathrm{Var}(Z) \le \binom{n}{2} p(1-p)\Big[(n-2)p^2 + (n-2)(n-3)p^4\Big].$$

The n^4 term is $\frac{n(n-1)(n-2)(n-3)}{2} p^5(1-p)$, exactly matching the true covariance contribution; the n^3 term exceeds the truth by the factor $\frac{3}{1+p+p^2} \in [1, 3]$. Thus Efron–Stein is sharp on the dominant (dependent-pair) term and loose only on the lower-order diagonal term. ■

Exercise 3.10

Central idea. Sandwich the weight of a minimizing path γ: below by $a|\gamma|$ (every edge costs at least $a > 0$), above by the weight of the deterministic ℓ^1-shortest path from 0 to v (exactly $\|v\|_1$ edges, each costing at most b). Minimality of γ forces $a|\gamma| \le T(0, v) \le b\|v\|_1$, giving $|\gamma| \le (b/a)\|v\|_1$ deterministically, uniformly over the randomness.

Solution. Let $\{X_e\}$ be the edge weights, $a \le X_e \le b$ a.s. with $0 < a < b < \infty$. For $v \in \mathbb{Z}^d$ let

$$T(0, v) = \min_{\gamma: 0 \to v} \sum_{e \in \gamma} X_e$$

be the first passage time, the minimum over all lattice paths γ from 0 to v, and let γ^* be a minimizing path (which exists a.s. since $X_e \ge a > 0$ forces the infimum to be attained over paths with boundedly many edges). Write $|\gamma^*|$ for its number of edges.

Lower bound. Every edge of γ^* has weight at least a, so

$$T(0, v) = \sum_{e \in \gamma^*} X_e \ge a\,|\gamma^*|.$$

Upper bound. Consider the deterministic path from 0 to v that moves $|v_1|$ steps along the first coordinate axis, $|v_2|$ along the second, etc. It uses exactly $\|v\|_1 = |v_1| + \cdots + |v_d|$ edges, each of weight at most b. Since γ^* minimizes,

$$T(0, v) \le b\,\|v\|_1.$$

Combining, $a|\gamma^*| \le b\|v\|_1$, i.e.

$$|\gamma^*| \le \frac{b}{a}\,\|v\|_1.$$

The bound is deterministic (holds for every realization of the weights) and uses only $0 < a \le X_e \le b$; no integrability or independence is needed. In particular geodesics cannot be much longer than the ℓ^1 distance: their length is at most a constant factor (b/a) times $\|v\|_1$. ■

Exercise 3.11

Central idea. For a real symmetric matrix, $\lambda_{\max}(A) = \max_{\|v\|=1} v^T A v$ (Rayleigh quotient), so the largest eigenvalue dominates the quadratic form at *any* fixed unit vector. Test with the uniform vector $v = n^{-1/2}(1, \ldots, 1)$: $\mathbb{E}[v^T A v]$ is just the average entry of A, which is $(n-1)p$.

Solution. The adjacency matrix A of $\mathcal{G}(n, p)$ is real symmetric ($X_{i,j} = X_{j,i}$, $X_{i,i} = 0$), with $\mathbb{E}[X_{i,j}] = p$ for $i \neq j$. For a real symmetric matrix, the largest eigenvalue is the maximum of the Rayleigh quotient over unit vectors:

$$\lambda_{\max}(A) = \max_{\|v\|=1} v^T A v.$$

Hence for every fixed unit vector v, $\lambda_{\max}(A) \geq v^T A v$ pointwise, and taking expectations,

$$\mathbb{E}[\lambda_{\max}(A)] \geq \mathbb{E}[v^T A v].$$

Take $v = \frac{1}{\sqrt{n}}(1, \ldots, 1)$, so $\|v\| = 1$. Then

$$v^T A v = \frac{1}{n} \sum_{i,j=1}^{n} X_{i,j}.$$

Since the diagonal is identically zero and there are $n(n-1)$ off-diagonal entries, each with expectation p,

$$\mathbb{E}[v^T A v] = \frac{1}{n} \cdot n(n-1)p = (n-1)p.$$

Therefore $\mathbb{E}[\lambda_{\max}(A)] \geq (n-1)p$. ■

Exercise 3.12

Central idea. If $G(n,p)$ is disconnected, some $k \leq n/2$ vertices form a union of components, i.e. no edges cross the cut — a union bound over $\binom{n}{k}$ sets gives $\sum_{k \leq n/2} \binom{n}{k}(1-p)^{k(n-k)}$. Split at $k_0 = \frac{(c-1)n}{2c}$: for small k, $\binom{n}{k} \leq n^k/k!$ turns the sum into the exponential series $e^{n^{(1-c)/2}} - 1$; for large k, $\binom{n}{k} \leq 2^n$ and $k(n-k) \gtrsim n^2$ makes $(1-p)^{k(n-k)}$ superpolynomially small. (Note: the book writes the bound with m where it means n — a typo; we use n throughout.)

Solution. Let $p = c \log n/n$ with $c > 1$. If $G(n,p)$ is disconnected, its smallest component has k vertices for some $1 \leq k \leq n/2$. For a fixed set S of k vertices, $\mathbb{P}\{\text{no edges between } S \text{ and } S^c\} = (1-p)^{k(n-k)}$ by independence. Union bounding over the $\binom{n}{k}$ choices of S and over k,

$$\mathbb{P}\{G(n,p) \text{ disconnected}\} \leq \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{k}(1-p)^{k(n-k)}. \tag{1}$$

Put $k_0 = \frac{(c-1)n}{2c} \leq n/2$ and split the sum at k_0.

Small $k \leq k_0$. Using $\binom{n}{k} \leq n^k/k!$,

$$\binom{n}{k}(1-p)^{k(n-k)} \leq \frac{[n(1-p)^{\,n-k}]^k}{k!}.$$

We claim $n(1-p)^{n-k} \leq n^{(1-c)/2}$ for $k \leq k_0$. Indeed, with $(1-p) \leq e^{-p}$ it suffices that $ne^{-p(n-k)} \leq n^{(1-c)/2}$, i.e. $\log n - p(n-k) \leq \frac{1-c}{2}\log n$, i.e. $p(n-k) \geq \frac{1+c}{2}\log n$. Since $p = \frac{c \log n}{n}$, this is $\frac{c(n-k)}{n} \geq \frac{1+c}{2}$, i.e. $k \leq n - \frac{(1+c)n}{2c} = \frac{(c-1)n}{2c} = k_0$. Hence

$$\sum_{1 \leq k \leq k_0} \binom{n}{k}(1-p)^{k(n-k)} \leq \sum_{k \geq 1} \frac{(n^{(1-c)/2})^k}{k!} = e^{\,n^{(1-c)/2}} - 1.$$

Large $k_0 < k \leq n/2$. Here $\binom{n}{k} \leq 2^n$, so it suffices to bound $(1-p)^{k(n-k)}$. For $k \leq n/2$, $n-k \geq n/2$, so $k(n-k) \geq kn/2 > k_0 n/2 = \frac{(c-1)n^2}{4c}$. With $(1-p) \leq e^{-p}$,

$$(1-p)^{k(n-k)} \leq \exp(-pk(n-k)) < \exp\left(-\frac{c \log n}{n} \cdot \frac{(c-1)n^2}{4c}\right) = n^{-(c-1)n/4}.$$

Therefore

$$\sum_{k_0 < k \leq n/2} \binom{n}{k}(1-p)^{k(n-k)} \leq n^{-(c-1)n/4} \sum_{k} \binom{n}{k} \leq 2^n\, n^{-(c-1)n/4}.$$

Combining with (1) and padding by factors of 2 (both terms are nonnegative),

$$\mathbb{P}\{G(n,p) \text{ disconnected}\} \le 2(e^{\,n^{(1-c)/2}} - 1) + 2^{\,n+1} n^{-(c-1)n/4},$$

which is the claimed bound with $m = n$. Since $c > 1$, $n^{(1-c)/2} \to 0$ gives $e^{n^{(1-c)/2}} - 1 \to 0$, and $\frac{(c-1)n}{4} \log n \gg (n+1)\log 2$ gives $2^{n+1} n^{-(c-1)n/4} \to 0$; hence $G(n, c\log n/n)$ is connected with high probability. ■

Exercise 3.13

Central idea. Follow the two ingredients of the MST argument (Example 3.15), but do not truncate.

1. **Efron–Stein.** Efron–Stein over the m^2 independent entries, computed exactly one entry at a time, shows that only the entries used by the optimal permutation π^* contribute: $\mathrm{Var}(Z_m) \le \mathbb{E} \sum_i X^2_{i,\pi^*(i)}$.
2. **Random-graph step.** This plays the role that connectivity of $G(n, c\log n/n)$ plays for the MST. LP duality gives potentials $a_k - b_l \le X_{kl}$, with equality on π^*. An expansion property of the random bipartite graphs $\{X_{kl} \le s\}$, valid off an event of probability $O(m^{-2})$, then forces every entry of π^* to be at most $T_m = O(\log^2 m/m)$.

Hence $\sum_i X^2_{i,\pi^*(i)} \le T_m Z_m$ on the good event. Walkup's classical theorem that $\mathbb{E}Z_m$ is bounded uniformly in m (by 3), an external result, gives $\mathrm{Var}(Z_m) = O(\log^2 m/m)$. Without it, the same argument gives $O(\log^4 m/m)$ self-containedly.

The naive transfer of the MST proof, which caps entries at δ_m and bounds $\mathbb{P}\{Z_m \ne \bar{Z}_m\}$, does fail; a remark at the end shows this. That failure concerns the truncation device only. The claimed variance bound is true.

Solution. Let $X = (X_{kl})_{k,l \le m}$ be i.i.d. Uniform$[0,1]$. Since the entries have a continuous distribution, the optimal permutation π^* is a.s. unique. Write $\rho = (\pi^*)^{-1}$.

Step 1: $\mathrm{Var}(Z_m) \le \mathbb{E} \sum_{i=1}^m X^2_{i,\pi^*(i)}$.

Fix an entry (i,j). Define two auxiliary optimal costs, neither of which depends on X_{ij}: - Z^{in} is the optimal assignment cost of the $(m-1)\times(m-1)$ minor obtained by deleting row i and column j; - Z^{out} is the minimum of $\sum_k X_{k,\pi(k)}$ over permutations with $\pi(i) \ne j$ (with $Z^{\mathrm{out}} = +\infty$ if $m = 1$).

Then

$$Z_m = \min\left(X_{ij} + Z^{\mathrm{in}},\, Z^{\mathrm{out}}\right) = Z^{\mathrm{in}} + \min(X_{ij}, D_{ij}), \qquad D_{ij} := Z^{\mathrm{out}} - Z^{\mathrm{in}}.$$

Moreover $(i,j) \in \pi^*$ iff $X_{ij} < D_{ij}$, outside the null event $\{X_{ij} = D_{ij}\}$ (null because D_{ij} is independent of X_{ij}).

Let $\mathbb{E}^{(ij)}$ denote integration over X_{ij} only. Then $\mathbb{E}^{(ij)}(Z_m - \mathbb{E}^{(ij)}Z_m)^2 = v(D_{ij})$, where $v(d) := \mathrm{Var}(\min(U,d))$ with $U \sim$ Uniform$[0,1]$ and d treated as a constant. Direct computation gives: - for $d \le 0$: $v(d) = 0$; - for $0 < d < 1$: $\mathbb{E}\min(U,d) = d - d^2/2$ and $\mathbb{E}\min(U,d)^2 = d^2 - 2d^3/3$, so $v(d) = \frac{d^3}{3} - \frac{d^4}{4}$; - for $d \ge 1$: $v(d) = \frac{1}{12}$.

In every case $v(d) \le \mathbb{E}[U^2 \mathbf{1}_{\{U<d\}}]$, which equals 0, $\frac{d^3}{3}$ and $\frac{1}{3}$ respectively.

By the Efron–Stein inequality (Theorem 3.1) applied to the m^2 independent entries, and by independence of X_{ij} and D_{ij},

$$\mathrm{Var}(Z_m) \le \sum_{i,j} \mathbb{E}\, v(D_{ij}) \le \sum_{i,j} \mathbb{E}\big[X^2_{ij} \mathbf{1}_{\{X_{ij} < D_{ij}\}}\big] = \mathbb{E} \sum_{(i,j)\in\pi^*} X^2_{ij}. \tag{1}$$

Step 2: LP duality and a monotonicity lemma.

By the Birkhoff–von Neumann theorem, the assignment LP $\min\{\sum_{k,l} X_{kl} y_{kl} : y \text{ doubly stochastic}\}$ has optimal value Z_m, attained at the permutation matrix of π^*. LP duality and complementary slackness give reals $a_1, \dots, a_m$ (rows) and $b_1, \dots, b_m$ (columns) with

$$a_k - b_l \le X_{kl} \ \text{ for all } (k,l), \qquad a_k - b_l = X_{kl} \ \text{ if } l = \pi^*(k).$$

(Take $a_k = u_k$ and $b_l = -v_l$ for an optimal dual solution (u, v) of $\max\{\sum u_k + \sum v_l : u_k + v_l \le X_{kl}\}$.)

For $s > 0$ let G_s be the bipartite graph with an edge (k, l) iff $X_{kl} \le s$. Write $N_s(S)$ for the neighbourhood in G_s of a set S of rows (a set of columns), and likewise for a set of columns. Put

$$R(\theta) = \{k : a_k \ge \theta\}, \qquad C(\varphi) = \{l : b_l \le \varphi\}.$$

Lemma. $|R(\theta - s)| \ge |N_s(R(\theta))|$ and $|C(\varphi + s)| \ge |N_s(C(\varphi))|$.

Proof. Let $l \in N_s(R(\theta))$, say $X_{kl} \le s$ with $a_k \ge \theta$. Then $b_l \ge a_k - X_{kl} \ge \theta - s$. Also $a_{\rho(l)} = b_l + X_{\rho(l),l} \ge b_l \ge \theta - s$. So the injective map ρ sends $N_s(R(\theta))$ into $R(\theta - s)$.

Symmetrically, let $k \in N_s(C(\varphi))$, say $X_{kl} \le s$ with $b_l \le \varphi$. Then $a_k \le b_l + X_{kl} \le \varphi + s$, and $b_{\pi^*(k)} = a_k - X_{k,\pi^*(k)} \le a_k \le \varphi + s$. So π^* maps $N_s(C(\varphi))$ injectively into $C(\varphi + s)$. $\square$

Step 3: the good event.

For integers $\sigma \ge 1$ put $s_\sigma = 12\log(em/\sigma)/m$, which is decreasing in σ. Define two events: - $\mathcal{E}_1$: for every integer $1 \le \sigma \le m/4$, every set of σ rows has at least 2σ neighbours in G_{s_σ}, and every set of σ columns has at least 2σ neighbours in G_{s_σ}. - $\mathcal{E}_2$: for every set S of more than $m/4$ rows and every set T of more than $m/4$ columns, some $(k, l) \in S \times T$ has $X_{kl} \le 32/m$.

Bound on $\mathbb{P}(\mathcal{E}_1^c)$. Suppose a set S of $\sigma \le m/4$ rows has fewer than 2σ neighbours. Since $2\sigma \le m$, its neighbourhood lies inside some set T of exactly 2σ columns. Then all $\sigma(m - 2\sigma) \ge \sigma m/2$ entries of $S \times T^c$ exceed s_σ. If $s_\sigma \ge 1$ this is impossible; otherwise it has probability $(1 - s_\sigma)^{\sigma(m-2\sigma)} \le e^{-s_\sigma \sigma m/2} = (em/\sigma)^{-6\sigma}$. A union bound, using $\binom{m}{\sigma} \le (em/\sigma)^\sigma$ and $\binom{m}{2\sigma} \le (em/\sigma)^{2\sigma}$ and doubling for columns, gives

$$\mathbb{P}(\mathcal{E}_1^c) \le 2 \sum_{1 \le \sigma \le m/4} \Big(\frac{\sigma}{em}\Big)^{3\sigma} \le 2 \cdot \frac{m}{4} \cdot (em)^{-3} \le \frac{1}{2e^3 m^2}.$$

Here we used that $\sigma \mapsto (\sigma/em)^{3\sigma}$ is decreasing on $[1, m]$: the derivative of $3\sigma \log(\sigma/em)$ is $3\log(\sigma/m) \le 0$.

Bound on $\mathbb{P}(\mathcal{E}_2^c)$. If $m \le 32$, every entry is $\le 1 \le 32/m$, so $\mathcal{E}_2$ holds surely. For $m > 32$ it suffices to consider $|S| = |T| = q := \lfloor m/4 \rfloor + 1 \ge m/4$. There are at most 4^m such pairs, each failing with probability $(1 - 32/m)^{q^2} \le e^{-(32/m)(m^2/16)} = e^{-2m}$. Hence

$$\mathbb{P}(\mathcal{E}_2^c) \le 4^m e^{-2m} = e^{-(2 - \log 4)m} \le e^{-0.6m}.$$

Step 4: on $\mathcal{E}_1 \cap \mathcal{E}_2$*, every entry of* π^* *is at most* T_m. Put

$$\Delta_m = \frac{12(1 + \log_2 m)(1 + \log m)}{m}, \qquad T_m = 2\Delta_m + \frac{32}{m} = O\Big(\frac{\log^2 m}{m}\Big).$$

Fix (i, j) with $j = \pi^*(i)$. Build a row chain: - start with $\theta_0 = a_i$, so $i \in R(\theta_0)$; - with $\sigma_k = |R(\theta_k)|$, while $\sigma_k \le m/4$ set $\theta_{k+1} = \theta_k - s_{\sigma_k}$.

By the Lemma and $\mathcal{E}_1$, $\sigma_{k+1} \ge |N_{s_{\sigma_k}}(R(\theta_k))| \ge 2\sigma_k$, so $\sigma_k \ge 2^k$. Let K be the first index with $\sigma_K > m/4$. Then $2^{K-1} \le \sigma_{K-1} \le m/4$, so $K \le \log_2 m$. Since s is decreasing, each decrement is at most $s_1 = 12(1 + \log m)/m$, hence

$$a_i - \theta_K \le K s_1 \le \Delta_m.$$

Therefore $|R(a_i - \Delta_m)| \ge |R(\theta_K)| > m/4$.

The same chain on columns, starting from b_j (so $j \in C(b_j)$) and using the second half of the Lemma, gives $|C(b_j + \Delta_m)| > m/4$.

By $\mathcal{E}_2$ there are $k \in R(a_i - \Delta_m)$ and $l \in C(b_j + \Delta_m)$ with $X_{kl} \le 32/m$. Then

$$a_i - \Delta_m \le a_k \le b_l + X_{kl} \le b_j + \Delta_m + \frac{32}{m},$$

so $X_{ij} = a_i - b_j \le T_m$.

Step 5: conclusion. On $\mathcal{E}_1 \cap \mathcal{E}_2$, each used entry satisfies $X_{ij}^2 \le T_m X_{ij}$, so $\sum_{(i,j)\in\pi^*} X_{ij}^2 \le T_m Z_m$. On the complement the sum is at most m, since there are m entries, each at most 1. By (1) and Steps 3–4,

$$\mathrm{Var}(Z_m) \le T_m\, \mathbb{E}Z_m + m\Big(\frac{1}{2e^3 m^2} + e^{-0.6m}\Big).$$

External input (cited, not proved here). Walkup (1979, "On the expected value of a random assignment problem", *SIAM J. Comput.* 8, 440–442) proved that the expected optimal cost is bounded independently of m, with bound 3 (stated in the literature as $\mathbb{E}Z_m < 3$, or as $\limsup_m \mathbb{E}Z_m \le 3$). Karp (1987) later improved the bound to 2, and Aldous (2001) proved $\mathbb{E}Z_m \to \pi^2/6$. Only $C := \sup_m \mathbb{E}Z_m < \infty$ is used; either form of Walkup's bound gives this, since trivially $\mathbb{E}Z_m \le m$ for each fixed m. With it,

$$\mathrm{Var}(Z_m) \le C\, T_m + O(m^{-1}) = O\Big(\frac{\log^2 m}{m}\Big).$$

Without the external bound, Step 4 alone gives $\sum_{\pi^*} X_{ij}^2 \le m T_m^2$ on the good event, hence the fully self-contained $\mathrm{Var}(Z_m) = O(\log^4 m/m)$.

Comparison with the MST. For the MST, connectivity of $G(n, c\log n/n)$ (Exercise 3.12) shows directly that every tree edge is at most $c\log n/n$. The tree analogue of (1) then gives $(n-1)(c\log n/n)^2 = O(\log^2 n/n)$. Here, connectivity is replaced by the expansion events $\mathcal{E}_1, \mathcal{E}_2$ combined with LP duality.

Remark (why naive truncation does not transfer). One might cap the entries, $\bar{X}_{kl} = \min(X_{kl}, \delta)$, and hope that $Z_m = \bar{Z}_m$ whenever the optimal matching uses only entries at most δ, or whenever G_δ has a perfect matching. Both implications are false. Take $m = 2$, $\delta = 1/2$ and

$$X = \begin{pmatrix} 0.4 & 0.9 \\ 0.1 & 0.4 \end{pmatrix}.$$

Here $Z_2 = 0.8$, attained by the identity, which uses only entries at most δ. But $\bar{X} = \begin{pmatrix} 0.4 & 0.5 \\ 0.1 & 0.4 \end{pmatrix}$ gives $\bar{Z}_2 = \min(0.8, 0.6) = 0.6 < Z_2$. Capping can push a suboptimal matching below the optimum. The argument above avoids this entirely: it never truncates, and bounds the entries of π^* itself. This 2×2 example is consistent with Step 4. ■

Exercise 3.14

Central idea. Write $N = 2m$ and vertices as $(x, y) \in \{0,1\}^m \times \{0,1\}^m$. The "swap-halves" permutation $\sigma(x, y) = (y, x)$ forces every shortest path to pass through the diagonal vertex (y, y) (dimension-order routing fixes the first m coordinates first). A fixed edge leaving the diagonal in dimension $m + 1$ is then traversed by all 2^{m-1} paths with the right midpoint and first flip, giving congestion $2^{N/2-1} \ge 2^{N/2}/N$.

Solution. Identify the hypercube with $\{0,1\}^N$ and assume $N = 2m$ is even (the odd case is analogous). Write vertices as $u = (x, y)$ with $x, y \in \{0,1\}^m$. Let $\sigma(x, y) = (y, x)$, an involution, hence a permutation.

The shortest-path strategy flips, in increasing coordinate order, each coordinate where the endpoints differ. For $(x, y) \to (y, x) = \sigma(x, y)$, the differing coordinates are $\{i \le m : x_i \ne y_i\} \cup \{m + i : x_i \ne y_i\}$. Flipping the first batch (in order) reaches (y, y); flipping the second batch reaches (y, x). Hence every routed path passes through the diagonal vertex (y, y).

Fix $z \in \{0,1\}^m$ and let e be the edge from (z, z) to (z, z'), where z' is z with the first bit flipped (an edge in dimension $m + 1$). The path $(x, y) \to (y, x)$ traverses e iff its midpoint is (z, z) (i.e. $y = z$) and its first flip in coordinates $\{m + 1, \dots, N\}$ is coordinate $m + 1$ (i.e. $x_1 \ne y_1$, since flips proceed in increasing order). Thus e is used exactly by the pairs (x, y) with $y = z$ and $x_1 \ne z_1$: there are 2^{m-1} such pairs.

So some edge has congestion $2^{m-1} = 2^{N/2}/2 \ge 2^{N/2}/N$ (for $N \ge 2$). Hence the maximal congestion over all edges is at least $2^{N/2}/N$. ■

Exercise 3.15

Central idea. Fix a directed edge $\vec{e}$ in dimension i. A leg $u \to W(u)$ traverses $\vec{e}$ only if u matches the edge's tail on coordinates $i, \dots, N$ (else 0) and then $W(u)$ satisfies i independent coordinate constraints (probability 2^{-i}); summing over u gives mean $\le 1/2$, as a sum of 2^N independent indicators. The same holds for legs $W(u) \to \sigma(u)$ (using that σ is a permutation to count preimages). Chernoff plus a union bound over the $N2^{N-1}$ edges then gives $\mathbb{E}[\max_e C_e] = O(N)$.

Solution. Work on $\{0,1\}^N$. Each u chooses $W(u)$ iid uniform over $\{0,1\}^N$; legs $u \to W(u)$ and $W(u) \to \sigma(u)$ follow dimension-order shortest paths. For an edge e, let C_e be the number of the 2^{N+1} legs traversing e (either direction); $M = \max_e C_e \le 2^{N+1}$ always.

Fix a directed edge $\vec{e}$ in dimension i, from x to $x \oplus \delta_i$. Under dimension-order routing, when coordinate i is flipped the current vertex has the target's values on coordinates $1, \dots, i-1$ and the source's values on $i+1, \dots, N$. So a leg $a \to b$ traverses $\vec{e}$ iff $a_i = x_i$, $b_i = \bar{x}_i$, $b_j = x_j$ $(j < i)$, $a_j = x_j$ $(j > i)$.

For first legs ($a = u$, $b = W(u)$): if u disagrees with x on any coordinate $\ge i$ the indicator is 0; for the 2^{i-1} vertices u agreeing with x on coordinates $i, \dots, N$, the conditions on $W(u)$ are i independent coordinate constraints, so the probability is 2^{-i}. Hence, with independent indicators across u (the $W(u)$ are independent),

$$A_{\vec{e}} := \sum_u \mathbf{1}\{\text{leg } u \to W(u) \text{ uses } \vec{e}\}, \qquad \mathbb{E}[A_{\vec{e}}] = 2^{i-1} \cdot 2^{-i} = \tfrac{1}{2}.$$

For second legs ($a = W(u)$, $b = \sigma(u)$): the indicator is 0 unless $\sigma(u)$ agrees with x on coordinates $1, \dots, i-1$ with $\sigma(u)_i = \bar{x}_i$; since σ is a permutation there are 2^{N-i} such u, and for each the $N - i + 1$ constraints on $W(u)$ hold with probability $2^{-(N-i+1)}$. Thus

$$B_{\vec{e}} := \sum_u \mathbf{1}\{\text{leg } W(u) \to \sigma(u) \text{ uses } \vec{e}\}, \qquad \mathbb{E}[B_{\vec{e}}] = 2^{N-i} \cdot 2^{-(N-i+1)} = \tfrac{1}{2},$$

again a sum of 2^N independent indicators.

Now $C_e \le \sum (A_{\vec{e}} + B_{\vec{e}})$ over the 2 directed versions of e. For a sum S of independent indicators with $\mu = \mathbb{E}S \le 1/2$, Chernoff gives $\mathbb{P}\{S \ge t\} \le (e\mu/t)^t \le (e/(2t))^t$ for $t \ge 1$. Hence

$$\mathbb{P}\{C_e \ge t\} \le 4\left(\frac{2e}{t}\right)^{t/4},$$

by a union bound over the 4 directed sums (each must reach $t/4$). With $t = KN$ and a union bound over the $N2^{N-1}$ edges,

$$\mathbb{P}\{M \ge KN\} \le N2^{N-1} \cdot 4\left(\frac{2e}{KN}\right)^{KN/4} \le 2^{-N}$$

for K a sufficiently large absolute constant (the exponent $\frac{KN}{4}\log\frac{KN}{2e}$ dominates $2N\log 2 + \log N$). Finally,

$$\mathbb{E}[M] \le KN + 2^{N+1}\mathbb{P}\{M > KN\} \le KN + 2 = O(N). \qquad \blacksquare$$

Exercise 3.16

Central idea. Diagonalize $A = U\Lambda U^T$, $B = VMV^T$; unitary invariance of the Hilbert–Schmidt norm gives $\|A - B\|_{\mathrm{HS}}^2 = \sum_{i,j} q_{ij}^2(\lambda_i - \mu_j)^2$ where $Q = U^TV$ is orthogonal, so $S = (q_{ij}^2)$ is doubly stochastic. The problem's claim that $\sum_{i,j} p_{ij}^2(\lambda_i - \mu_j)^2$ is minimized at $P = I$ over doubly stochastic P is **false as stated** (counterexample below); the correct statement is the linear one, $\sum_{i,j} p_{ij}(\lambda_i - \mu_j)^2 \ge \sum_i (\lambda_i - \mu_i)^2$, proved via Birkhoff–von Neumann (reducing to permutation matrices) and the rearrangement inequality.

Solution. Write $\mu_j = \lambda_j(B)$. Since A, B are symmetric, $A = U\Lambda U^T$ and $B = VMV^T$ with U, V orthogonal, $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$, $M = \mathrm{diag}(\mu_1, \dots, \mu_n)$, $\lambda_1 \ge \cdots \ge \lambda_n$, $\mu_1 \ge \cdots \ge \mu_n$.

The hint's identity. By unitary invariance of $\|\cdot\|_{\mathrm{HS}}$,

$$\|A - B\|_{\mathrm{HS}}^2 = \|U\Lambda U^T - VMV^T\|_{\mathrm{HS}}^2 = \|\Lambda - QMQ^T\|_{\mathrm{HS}}^2, \qquad Q := U^TV,$$

multiplying by U^T on the left and U on the right. Right-multiplying by the orthogonal Q,

$$\|\Lambda - QMQ^T\|_{\mathrm{HS}}^2 = \|\Lambda Q - QM\|_{\mathrm{HS}}^2.$$

Since $(\Lambda Q - QM)_{ij} = \lambda_i q_{ij} - q_{ij}\mu_j = q_{ij}(\lambda_i - \mu_j)$,

$$\|A - B\|_{\mathrm{HS}}^2 = \sum_{i,j=1}^{n} q_{ij}^2(\lambda_i - \mu_j)^2. \tag{1}$$

The matrix $S = (q_{ij}^2)$ is doubly stochastic: rows and columns of the orthogonal Q are unit vectors.

The p_{ij}^2 claim is false. The problem asks to show $\sum_{i,j} p_{ij}^2 d_{ij}$, $d_{ij} := (\lambda_i - \mu_j)^2$, is minimized at $P = I$ over doubly stochastic P. This is false. Counterexample: $n = 2$, $\lambda = (2, 0)$, $\mu = (1, 1)$, so $d_{ij} \equiv 1$. For the uniform $P = \frac{1}{2}\mathbf{1}\mathbf{1}^T$, $\sum_{i,j} p_{ij}^2 d_{ij} = 4 \cdot \frac{1}{4} = 1$, whereas at $P = I$ it equals 2. Hence I is not even a minimizer. The intended claim is the linear version.

Correct lemma. For P doubly stochastic, $\sum_{i,j} p_{ij} d_{ij} \geq \sum_i d_{ii}$.

Proof. $F(P) := \sum_{i,j} p_{ij} d_{ij}$ is linear in P, hence attains its minimum over the convex compact set of doubly stochastic matrices at an extreme point. By the Birkhoff–von Neumann theorem, extreme points are permutation matrices, so $\min_P F(P) = \min_\pi \sum_i d_{i,\pi(i)}$. For a permutation π,

$$\sum_i d_{i,\pi(i)} - \sum_i d_{ii} = \sum_i (\lambda_i - \mu_{\pi(i)})^2 - \sum_i (\lambda_i - \mu_i)^2 = 2\left(\sum_i \lambda_i \mu_i - \sum_i \lambda_i \mu_{\pi(i)}\right) \geq 0$$

by the rearrangement inequality, since (λ_i) and (μ_i) are both nonincreasing. ■

Remark on exchanges. The hint's exchange works cleanly for the linear version: for $k < \ell$, the 2×2 swap moving $\epsilon > 0$ from $(k, \ell), (\ell, k)$ to $(k, k), (\ell, \ell)$ preserves double stochasticity and changes $\sum_{i,j} p_{ij} d_{ij}$ by $\epsilon(d_{kk} + d_{\ell\ell} - d_{k\ell} - d_{\ell k}) \leq 0$, since $d_{kk} + d_{\ell\ell} - d_{k\ell} - d_{\ell k} = -2(\lambda_k - \lambda_\ell)(\mu_k - \mu_\ell) \leq 0$ (Monge property). For the p_{ij}^2 version the linear terms do not cancel and no such exchange exists — consistent with the counterexample.

Conclusion. Applying the lemma to $S = (q_{ij}^2)$ in (1),

$$\|A - B\|_{\mathrm{HS}}^2 = \sum_{i,j} q_{ij}^2(\lambda_i - \mu_j)^2 \geq \sum_{i=1}^{n} (\lambda_i - \mu_i)^2,$$

which is Lidskii's (Hoffman–Wielandt) inequality. ■

Exercise 3.17

Central idea. The Section 3.6 proof bounds $b - a$ by comparing a variance lower bound for the truncated function $g_{a,b}$ (coming from $\mathbb{P}\{Z \geq b\}$) against an Efron–Stein upper bound. Here the hypothesis controls *upward* jumps, $\sum_i (Z_i' - Z)_+^2 \leq v$, so by the exchangeability of the pair $(X, \tilde{X}^{(i)})$ we flip the Efron–Stein sum to $\sum_i \mathbb{E}(g_{a,b}(\tilde{X}^{(i)}) - g_{a,b}(X))_+^2$. The uniform bound B implies each summand vanishes on $\{Z \leq a - B\}$; shifting the lower truncation level to $a = Q_{1-\gamma} + B$ then turns the leftover indicator into $\mathbb{P}\{Z > Q_{1-\gamma}\} \leq \gamma$, at the price of the extra B in the final estimate.

Solution. Write $Z = f(X)$ and $Z_i' = f(\tilde{X}^{(i)})$, where $\tilde{X}^{(i)}$ is X with the i-th coordinate replaced by an independent copy. By definition of B, $|Z - Z_i'| \leq B$ almost surely for every i, and by hypothesis

$$\sum_{i=1}^{n} (Z_i' - Z)_+^2 = \sum_{i=1}^{n} (Z - Z_i')_-^2 \leq v \quad \text{a.s.}$$

Fix $0 < \delta < \gamma \leq 1/2$ with $Q_{1-\gamma} \geq MZ + B$, and set

$$a = Q_{1-\gamma} + B, \qquad b = Q_{1-\delta}.$$

If $b \le a$ there is nothing to prove, since then $Q_{1-\delta} - Q_{1-\gamma} \le B$. Assume $b > a$, so that with $g_{a,b}$ defined as in Section 3.6 ($g_{a,b}(x) = b$ if $f(x) \ge b$, $f(x)$ if $a < f(x) < b$, a if $f(x) \le a$), the truncation is non-degenerate. Note that $a \ge MZ$ by hypothesis.

Lower bound. Since $\mathbb{P}\{Z > a\} \le \mathbb{P}\{Z > Q_{1-\gamma}\} \le \gamma \le 1/2$,

$$\mathbb{E}g_{a,b}(X) \le a\,\mathbb{P}\{Z \le a\} + b\,\mathbb{P}\{Z > a\} \le \frac{a+b}{2},$$

and therefore, as in Section 3.6,

$$\mathrm{Var}(g_{a,b}(X)) \ge \mathbb{P}\{g_{a,b}(X) = b\}\,(b - \mathbb{E}g_{a,b}(X))^2 \ge \frac{\mathbb{P}\{Z \ge b\}}{4}\,(b-a)^2.$$

Upper bound. By the Efron–Stein inequality and the exchangeability of $(X, \tilde{X}^{(i)})$,

$$\mathrm{Var}(g_{a,b}(X)) \le \sum_{i=1}^{n} \mathbb{E}(g_{a,b}(X) - g_{a,b}(\tilde{X}^{(i)}))^2 = 2\sum_{i=1}^{n} \mathbb{E}(g_{a,b}(\tilde{X}^{(i)}) - g_{a,b}(X))_+^2.$$

Since $g_{a,b}$ is nondecreasing and 1-Lipschitz, $(g_{a,b}(\tilde{X}^{(i)}) - g_{a,b}(X))_+ \le (Z_i' - Z)_+$ pointwise. Moreover, on the event $\{Z \le a - B\}$ we have $Z_i' \le Z + B \le a$ for every i, hence $g_{a,b}(\tilde{X}^{(i)}) = a \le g_{a,b}(X)$ and the positive part vanishes. Consequently,

$$\sum_{i=1}^{n} (g_{a,b}(\tilde{X}^{(i)}) - g_{a,b}(X))_+^2 \le \mathbf{1}_{\{Z > a-B\}} \sum_{i=1}^{n} (Z_i' - Z)_+^2 \le v\,\mathbf{1}_{\{Z > a-B\}} \quad \text{a.s.},$$

and taking expectations,

$$\mathrm{Var}(g_{a,b}(X)) \le 2v\,\mathbb{P}\{Z > a - B\} = 2v\,\mathbb{P}\{Z > Q_{1-\gamma}\} \le 2v\gamma.$$

Conclusion. Comparing the two bounds and using $\mathbb{P}\{Z \ge b\} = \mathbb{P}\{Z \ge Q_{1-\delta}\} \ge \delta$,

$$\frac{\delta}{4}(b-a)^2 \le 2v\gamma, \qquad \text{i.e.} \qquad b - a \le \sqrt{\frac{8v\gamma}{\delta}}.$$

Since $b - a = Q_{1-\delta} - Q_{1-\gamma} - B$, this is exactly

$$Q_{1-\delta} - Q_{1-\gamma} \le B + \sqrt{\frac{8v\gamma}{\delta}},$$

as claimed.

Exercise 3.18

Central idea. For self-bounding $Z \ge 0$, the square root $W = \sqrt{Z}$ itself satisfies condition (3.4) with the *universal* constant $v = 1$: indeed $(\sqrt{Z} - \sqrt{Z_i'})_+ \le (Z - Z_i')_+ / \sqrt{Z}$, while self-bounding gives $\sum_i (Z - Z_i')_+^2 \le \sum_i (Z - Z_i')_+ \le Z$. Applying the Section 3.6 quantile argument to W — whose quantiles are $\sqrt{a_k}$ — yields $\sqrt{a_{k+1}} - \sqrt{a_k} \le 4$. (The problem's "$ck$" is a typo for a universal constant c: with a growing bound the claimed $e^{-\sqrt{t}/C}$ tail would not follow.) Telescoping the quantile gaps then gives the sub-exponential upper tail.

Erratum. The statement is wrong as printed in two respects. (i) Rate: a gap bound $\sqrt{a_{k+1}} - \sqrt{a_k} \le ck$ telescopes only to $\sqrt{a_m} \lesssim m^2$, which gives an $e^{-t^{1/4}}$-type tail, so "which implies" fails; the provable gap bound is $\sqrt{a_{k+1}} - \sqrt{a_k} \le c$ with a universal c (here $c = 4$). (ii) Constant: the tail $\mathbb{P}\{Z > MZ + t\} \le Ce^{-\sqrt{t}/C}$ cannot hold with C universal; C must be allowed to depend on MZ. Counterexample: $Z = \sum_{i=1}^{n} X_i$ with X_i iid Bernoulli(1/2) is self-bounding ($f_i = \sum_{j \ne i} X_j$), $MZ = n/2$, and $\sqrt{n}$ is two standard deviations, so

$\mathbb{P}\{Z > MZ + \sqrt{n}\} \to \mathbb{P}\{N(0,1) > 2\} \approx 0.023$ (exact binomial values: 0.0231 at $n = 10^3$, 0.0227 at $n = 10^6$), whereas a universal C would force $Ce^{-n^{1/4}/C} \to 0$. The corrected statement, proved below, is: $\sqrt{a_{k+1}} - \sqrt{a_k} \le 4$ for $k \ge 1$, and $\mathbb{P}\{Z > MZ + t\} \le e^{-\frac{\ln 2}{8}\sqrt{t}}$ for $t \ge \max(64, 16MZ)$, hence $\mathbb{P}\{Z > MZ + t\} \le Ce^{-\sqrt{t}/C}$ for all $t > 0$ with C depending on MZ.

Solution. Recall that f self-bounding means there are functions f_i of all coordinates but the i-th with $0 \le f(x) - f_i(x^{(i)}) \le 1$ for all x, i and $\sum_{i=1}^n (f(x) - f_i(x^{(i)})) \le f(x)$; in particular $Z = f(X) \ge 0$. Write $Z_i' = f(\tilde{X}^{(i)})$ with $\tilde{X}^{(i)}$ the vector X with X_i replaced by an independent copy.

We first record two consequences. Since f_i does not depend on coordinate i,

$$f(\tilde{X}^{(i)}) \ge f_i((\tilde{X}^{(i)})^{(i)}) = f_i(X^{(i)}),$$

so that

$$Z - Z_i' = f(X) - f(\tilde{X}^{(i)}) \le f(X) - f_i(X^{(i)}) \le 1 \quad \text{a.s.}$$

Hence $(Z - Z_i')_+ \le 1$ and $(Z - Z_i')_+^2 \le (Z - Z_i')_+$. Moreover, $(Z - Z_i')_+ \le (f(X) - f_i(X^{(i)}))_+ = f(X) - f_i(X^{(i)})$, so

$$\sum_{i=1}^n (Z - Z_i')_+ \le \sum_{i=1}^n \big(f(X) - f_i(X^{(i)})\big) \le Z \quad \text{a.s.}$$

Now set $W = \sqrt{Z} \ge 0$ and $W_i' = \sqrt{Z_i'}$. For $Z > 0$,

$$0 \le (W - W_i')_+ = (\sqrt{Z} - \sqrt{Z_i'})_+ = \frac{(Z - Z_i')_+}{\sqrt{Z} + \sqrt{Z_i'}} \le \frac{(Z - Z_i')_+}{\sqrt{Z}},$$

and for $Z = 0$ the left side is 0. Therefore, with the convention $0/0 = 0$,

$$\sum_{i=1}^n (W - W_i')_+^2 \le \frac{1}{Z} \sum_{i=1}^n (Z - Z_i')_+^2 \le \frac{1}{Z} \sum_{i=1}^n (Z - Z_i')_+ \le 1 \quad \text{a.s.},$$

i.e. W satisfies condition (3.4) of Section 3.6 with the universal constant $v = 1$.

Apply the Section 3.6 argument to W. With $b_k = Q_{1-2^{-k}}(W) = \sqrt{a_k}$ (quantiles commute with the increasing map $z \mapsto \sqrt{z}$) and $k \ge 1$,

$$b_{k+1} - b_k \le \sqrt{\frac{8v \cdot 2^{-k}}{2^{-(k+1)}}} = 4\sqrt{v} = 4,$$

that is,

$$\sqrt{a_{k+1}} - \sqrt{a_k} \le 4 \qquad (k \ge 1).$$

Telescoping from $a_1 = Q_{1/2} = MZ$ gives $\sqrt{a_m} \le \sqrt{MZ} + 4(m-1)$ for $m \ge 1$.

For the tail bound, let $t > 0$ and take $m = \lfloor \sqrt{t}/8 \rfloor$. If $t \ge \max(64, 16MZ)$ then $m \ge 1$ and

$$a_{m+1} \le \big(\sqrt{MZ} + 4m\big)^2 = MZ + 8m\sqrt{MZ} + 16m^2 \le MZ + \sqrt{tMZ} + \frac{t}{4} \le MZ + t,$$

using $m \le \sqrt{t}/8$ and $\sqrt{tMZ} \le t/4$. Hence $\{Z > MZ + t\} \subseteq \{Z > a_{m+1}\}$ and

$$\mathbb{P}\{Z > MZ + t\} \le \mathbb{P}\{Z > a_{m+1}\} \le 2^{-(m+1)} \le e^{-\frac{\ln 2}{8}\sqrt{t}}.$$

For $0 < t < \max(64, 16MZ)$ the same bound holds up to a prefactor depending only on MZ. Thus for a constant C (depending at most on MZ; by the Erratum, no universal C exists),

$$\mathbb{P}\{Z > MZ + t\} \le Ce^{-\sqrt{t}/C} \qquad \text{for all } t > 0,$$

as claimed. In particular the upper tail of a self-bounding function is sub-exponential in $\sqrt{t}$.

Exercise 3.19

Central idea. Bound $\mathrm{Var}(\sqrt{X})$ by $\mathbb{E}[(\sqrt{X}-\sqrt{\mu})^2]$ with $\mu=\mathbb{E}X$, rewrite $(\sqrt{X}-\sqrt{\mu})^2=\frac{(X-\mu)^2}{(\sqrt{X}+\sqrt{\mu})^2}\le\frac{(X-\mu)^2}{X+\mu}$, and observe $\mathbb{E}\left[\frac{(X-\mu)^2}{X+\mu}\right]\le\frac{\mathrm{Var}(X)}{\mu}$ because $\frac{1}{\mu}-\frac{1}{X+\mu}=\frac{X}{\mu(X+\mu)}\ge 0$ for $X\ge 0$.

Solution. If $\mu:=\mathbb{E}X=0$ then $X=0$ a.s. (as $X\ge 0$) and there is nothing to prove. Assume $\mu>0$. Since $\mathrm{Var}(Y)=\min_c\mathbb{E}[(Y-c)^2]$,

$$\mathrm{Var}(\sqrt{X})\le\mathbb{E}\left[(\sqrt{X}-\sqrt{\mu})^2\right].$$

Now $(\sqrt{X}-\sqrt{\mu})^2(\sqrt{X}+\sqrt{\mu})^2=(X-\mu)^2$, so

$$(\sqrt{X}-\sqrt{\mu})^2=\frac{(X-\mu)^2}{(\sqrt{X}+\sqrt{\mu})^2}\le\frac{(X-\mu)^2}{X+\mu},$$

using $(\sqrt{X}+\sqrt{\mu})^2=X+\mu+2\sqrt{X\mu}\ge X+\mu$ (and $X+\mu\ge\mu>0$). Hence

$$\mathrm{Var}(\sqrt{X})\le\mathbb{E}\left[\frac{(X-\mu)^2}{X+\mu}\right].$$

Moreover,

$$\frac{\mathbb{E}[(X-\mu)^2]}{\mu}-\mathbb{E}\left[\frac{(X-\mu)^2}{X+\mu}\right]=\mathbb{E}\left[(X-\mu)^2\left(\frac{1}{\mu}-\frac{1}{X+\mu}\right)\right]=\frac{1}{\mu}\,\mathbb{E}\left[(X-\mu)^2\cdot\frac{X}{X+\mu}\right]\ge 0,$$

since $X\ge 0$. Therefore $\mathrm{Var}(\sqrt{X})\le\mathrm{Var}(X)/\mu\le a$, the last step by the hypothesis $\mathrm{Var}(X)\le a\mu$. ■

Exercise 3.20

Central idea. Apply the Efron–Stein inequality to $\sqrt{Z}$ with the $X^{(i)}$-measurable approximants $\sqrt{Z_i}$: $\mathrm{Var}(\sqrt{Z})\le\sum_i\mathbb{E}[(\sqrt{Z}-\sqrt{Z_i})^2]$, and on $\{Z>0\}$, $(\sqrt{Z}-\sqrt{Z_i})^2=\frac{(Z-Z_i)^2}{(\sqrt{Z}+\sqrt{Z_i})^2}\le\frac{(Z-Z_i)^2}{Z}$. Summing gives $\mathrm{Var}(\sqrt{Z})\le\mathbb{E}[W\mathbf{1}_{\{Z>0\}}]$. This is at most $\mathbb{E}W$ once $W\ge 0$. The statement leaves that hypothesis out, and it cannot be dropped.

Remark (erratum: missing hypothesis $W\ge 0$). As printed, the hypothesis $V\le WZ$ puts no constraint on W on the event $\{Z=0\}$, and the claim can then fail. Take $n=1$ and $Z=f(X_1)=X_1\sim\mathrm{Bernoulli}(1/2)$. Then $Z_1=\inf_x f(x)=0$, so $V=Z^2=Z$. Let $W=1$ on $\{Z=1\}$ and $W=-100$ on $\{Z=0\}$. Then $V\le WZ$ holds ($1\le 1$ on $\{Z=1\}$, $0\le 0$ on $\{Z=0\}$), yet

$$\mathbb{E}W=\tfrac{1}{2}\cdot 1+\tfrac{1}{2}\cdot(-100)=-49.5<\tfrac{1}{4}=\mathrm{Var}(\sqrt{Z}).$$

On $\{Z>0\}$ the condition $W\ge 0$ holds automatically, since $W\ge V/Z\ge 0$ there.

The corrected statement reads: *if $V\le WZ$ and $W\ge 0$ on $\{Z=0\}$ (equivalently, $W\ge 0$ a.s.), then* $\mathrm{Var}(\sqrt{Z})\le\mathbb{E}W$. With no sign condition at all, the bound $\mathrm{Var}(\sqrt{Z})\le\mathbb{E}[W\mathbf{1}_{\{Z>0\}}]$ still holds; both are proved below.

Solution. Since $f\ge 0$ and the infimum defining Z_i includes $x_i'=X_i$, we have $0\le Z_i\le Z$. Moreover $\sqrt{Z_i}$ is a function of $X^{(i)}=(X_1,\dots,X_{i-1},X_{i+1},\dots,X_n)$ only.

Step 1 (Efron–Stein with arbitrary $X^{(i)}$-measurable approximants). Write $\mathbb{E}^{(i)}$ for integration with respect to X_i only. For square-integrable $Y=g(X_1,\dots,X_n)$, the Efron–Stein inequality (Theorem 3.1) gives $\mathrm{Var}(Y)\le\sum_{i=1}^n\mathbb{E}[(Y-\mathbb{E}^{(i)}Y)^2]$. For any $X^{(i)}$-measurable Y_i, the conditional mean minimizes the conditional mean-square error: $\mathbb{E}^{(i)}[(Y-\mathbb{E}^{(i)}Y)^2]\le\mathbb{E}^{(i)}[(Y-Y_i)^2]$. Hence

$$\mathrm{Var}(Y)\le\sum_{i=1}^n\mathbb{E}[(Y-Y_i)^2].\tag{1}$$

Step 2 (the pointwise bound). Put

$$S = \sum_{i=1}^{n} (\sqrt{Z} - \sqrt{Z_i})^2.$$

On $\{Z > 0\}$, using $(\sqrt{Z} + \sqrt{Z_i})^2 \geq Z$,

$$S = \sum_{i=1}^{n} \frac{(Z - Z_i)^2}{(\sqrt{Z} + \sqrt{Z_i})^2} \leq \frac{1}{Z} \sum_{i=1}^{n} (Z - Z_i)^2 = \frac{V}{Z} \leq W.$$

On $\{Z = 0\}$ every $Z_i = 0$ (as $0 \leq Z_i \leq Z$), so $S = 0$. Thus $S \leq W \mathbf{1}_{\{Z>0\}}$ pointwise, and

$$B := \mathbb{E}S \leq \mathbb{E}[W \mathbf{1}_{\{Z>0\}}].$$

Step 3 (square integrability). Assume $B < \infty$; otherwise the right-hand side is $+\infty$ and there is nothing to prove. For $K > 0$ let $Y_K = \min(\sqrt{Z}, K)$ and $Y_{K,i} = \min(\sqrt{Z_i}, K)$. Then $Y_{K,i}$ is $X^{(i)}$-measurable and $|Y_K - Y_{K,i}| \leq |\sqrt{Z} - \sqrt{Z_i}|$, since $x \mapsto \min(x, K)$ is 1-Lipschitz. By (1), $\mathrm{Var}(Y_K) \leq B$ for every K.

Let Z' be an independent copy of Z. Then $\mathrm{Var}(Y_K) = \frac{1}{2}\mathbb{E}(Y_K - Y_K')^2$. By Fatou, as $K \to \infty$,

$$\tfrac{1}{2}\mathbb{E}(\sqrt{Z} - \sqrt{Z'})^2 \leq B < \infty.$$

By Fubini, $\mathbb{E}(\sqrt{Z} - c)^2 < \infty$ for some constant c, so $\sqrt{Z} \in L^2$ and $\mathrm{Var}(\sqrt{Z}) = \frac{1}{2}\mathbb{E}(\sqrt{Z} - \sqrt{Z'})^2 \leq B$. (When $\mathbb{E}Z < \infty$ is known in advance, (1) with $Y = \sqrt{Z}$, $Y_i = \sqrt{Z_i}$ gives this directly.)

Step 4 (conclusion). With no sign assumption on W,

$$\mathrm{Var}(\sqrt{Z}) \leq \mathbb{E}\left[\frac{V}{Z}\mathbf{1}_{\{Z>0\}}\right] \leq \mathbb{E}\left[W \mathbf{1}_{\{Z>0\}}\right].$$

Under the hypothesis $W \geq 0$ on $\{Z = 0\}$, which is necessary by the Remark,

$$\mathbb{E}\left[W \mathbf{1}_{\{Z>0\}}\right] = \mathbb{E}W - \mathbb{E}\left[W \mathbf{1}_{\{Z=0\}}\right] \leq \mathbb{E}W.$$

Hence $\mathrm{Var}(\sqrt{Z}) \leq \mathbb{E}W$. ■

Exercise 3.21

Central idea. Write X as a sum of n iid Poisson(μ/n) variables and apply the Efron–Stein inequality (Theorem 3.1): resampling one coordinate changes f by a telescoping sum of discrete derivatives, and Cauchy–Schwarz reduces the problem to an exact Poisson computation where the cross-term collapses to $2\lambda p_\ell$ via the identity $k p_k = \lambda p_{k-1}$.

Solution. If $\mathbb{E}[(Df(X))^2] = \infty$ there is nothing to prove. We first prove the inequality for bounded f; the general case follows by truncation (see below).

Let $n \geq 1$, $\lambda = \mu/n$, and write $X \overset{d}{=} \sum_{i=1}^{n} X_i$ with X_i iid $\sim$ Poisson(λ). By the Efron–Stein inequality (Theorem 3.1), in the resampling form $\mathrm{Var}(Z) \leq \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}[(Z - Z_i')^2]$ (since $\mathbb{E}[(Z - \mathbb{E}_i Z)^2] = \frac{1}{2}\mathbb{E}[(Z - Z_i')^2]$ for Z_i' resampling coordinate i),

$$\mathrm{Var}(f(X)) \leq \frac{1}{2} \sum_{i=1}^{n} \mathbb{E}[(f(X) - f(X^{(i)}))^2],$$

where $X^{(i)} = X - X_i + X_i'$ with X_i' an independent copy of X_i.

Fix i and write $S_i = \sum_{j \neq i} X_j$. Conditional on $S_i = s$, with Y, Y' iid $\sim$ Poisson(λ) and $p_k = e^{-\lambda}\lambda^k/k!$,

$$\mathbb{E}[(f(s + Y) - f(s + Y'))^2] = \sum_{j,k \geq 0} p_j p_k (f(s + j) - f(s + k))^2.$$

Telescoping, $f(s+j)-f(s+k)=\sum_{\ell=\min(j,k)}^{\max(j,k)-1}\sigma_{j,k}\,Df(s+\ell)$ with $\sigma_{j,k}=\operatorname{sgn}(j-k)$, so by Cauchy–Schwarz,

$$\big(f(s+j)-f(s+k)\big)^2\le|j-k|\sum_{\ell=\min(j,k)}^{\max(j,k)-1}(Df(s+\ell))^2.$$

Hence, by Tonelli's theorem,

$$\mathbb{E}\big[(f(s+Y)-f(s+Y'))^2\big]\le\sum_{\ell\ge0}(Df(s+\ell))^2\,c_\ell,\qquad c_\ell:=\sum_{\substack{j,k\ge0\\ \min(j,k)\le\ell<\max(j,k)}}p_jp_k|j-k|.$$

By symmetry, $c_\ell=2\sum_{j\le\ell<k}p_jp_k(k-j)$. Using $kp_k=\lambda p_{k-1}$,

$$\sum_{k>\ell}p_k(k-j)=\lambda\,\mathbb{P}\{Y\ge\ell\}-j\,\mathbb{P}\{Y>\ell\},\qquad\sum_{j\le\ell}jp_j=\lambda\,\mathbb{P}\{Y\le\ell-1\},$$

so $c_\ell=2\lambda\big[\mathbb{P}\{Y\ge\ell\}\mathbb{P}\{Y\le\ell\}-\mathbb{P}\{Y>\ell\}\mathbb{P}\{Y\le\ell-1\}\big]$. Writing $a=\mathbb{P}\{Y\le\ell-1\}$, $b=p_\ell$, $c=\mathbb{P}\{Y>\ell\}$ (so $a+b+c=1$), the bracket equals $(b+c)(a+b)-ca=b(a+b+c)=b=p_\ell$. Thus $c_\ell=2\lambda p_\ell$, and

$$\mathbb{E}\big[(f(s+Y)-f(s+Y'))^2\big]\le2\lambda\sum_{\ell\ge0}p_\ell(Df(s+\ell))^2=2\lambda\,\mathbb{E}\big[(Df(s+Y))^2\big].$$

Unconditioning, $\mathbb{E}[(f(X)-f(X^{(i)}))^2]\le2\lambda\,\mathbb{E}[(Df(X))^2]$. Summing over i,

$$\operatorname{Var}(f(X))\le\frac12\cdot n\cdot2\lambda\,\mathbb{E}\big[(Df(X))^2\big]=\mu\,\mathbb{E}\big[(Df(X))^2\big].$$

For general f with $\mathbb{E}[(Df(X))^2]<\infty$, truncate via $f_m(k)=f(\min(k,m))$. Then f_m is bounded, $(Df_m(k))^2\le(Df(k))^2$ for all k (equality for $k<m$, both sides 0 for $k\ge m$ up to the single point $k=m$ where $Df_m(m)=0$), so $\operatorname{Var}(f_m(X))\le\mu\,\mathbb{E}[(Df(X))^2]$. With X' an independent copy of X, $(f_m(X)-f_m(X'))^2\to(f(X)-f(X'))^2$ a.s., so Fatou's lemma gives $\operatorname{Var}(f(X))=\frac12\mathbb{E}[(f(X)-f(X'))^2]\le\liminf_m\operatorname{Var}(f_m(X))\le\mu\,\mathbb{E}[(Df(X))^2]$. ■

Exercise 3.22

Central idea. Apply the given integration-by-parts identity to $h(x)=(f(x)-f(0))^2$ rather than to f itself: then the boundary term $h(0)$ vanishes, and Cauchy–Schwarz turns $2\mathbb{E}[\operatorname{sgn}(X)(f(X)-f(0))f'(X)]$ into the sharp bound $\mathbb{E}[(f(X)-f(0))^2]\le4\mathbb{E}[(f'(X))^2]$. Since variance is the minimum mean squared error over all constants, the claim follows.

Solution. If $\mathbb{E}[(f'(X))^2]=\infty$ there is nothing to prove; assume it is finite. Since $\operatorname{Var}(f(X))<\infty$, also $\mathbb{E}[(f(X)-f(0))^2]=\operatorname{Var}(f(X))+(\mathbb{E}f(X)-f(0))^2<\infty$.

The identity $\mathbb{E}[h(X)]=h(0)+\mathbb{E}[\operatorname{sgn}(X)h'(X)]$ follows by integration by parts on $(0,\infty)$ and $(-\infty,0)$: for instance, $\int_0^A h(x)\frac12e^{-x}\,dx=\frac12h(0)-\frac12e^{-A}h(A)+\int_0^A h'(x)\frac12e^{-x}\,dx$, and the boundary term $e^{-A}h(A)\to0$ as $A\to\infty$ whenever $\int_0^\infty e^{-x}|h(x)|\,dx$ and $\int_0^\infty e^{-x}|h'(x)|\,dx$ are finite (then $e^{-x}h(x)$ is Cauchy at ∞, hence convergent, and the limit must be 0 lest the first integral diverge).

Apply the identity to $h(x)=(f(x)-f(0))^2$, for which $h(0)=0$ and $h'(x)=2(f(x)-f(0))f'(x)$. The integrability holds since, by Cauchy–Schwarz,

$$\mathbb{E}\big[|f(X)-f(0)|\cdot|f'(X)|\big]\le\sqrt{\mathbb{E}\big[(f(X)-f(0))^2\big]\,\mathbb{E}\big[(f'(X))^2\big]}<\infty.$$

We obtain

$$\mathbb{E}\big[(f(X)-f(0))^2\big]=2\,\mathbb{E}\big[\operatorname{sgn}(X)(f(X)-f(0))f'(X)\big].$$

Applying Cauchy–Schwarz to the right-hand side (using $\operatorname{sgn}(X)^2 = 1$ a.s.),

$$\mathbb{E}[(f(X) - f(0))^2] \le 2\sqrt{\mathbb{E}[(f(X) - f(0))^2]\,\mathbb{E}[(f'(X))^2]}.$$

If $\mathbb{E}[(f(X) - f(0))^2] = 0$ the claim is trivial; otherwise divide by its square root and square both sides to get

$$\mathbb{E}[(f(X) - f(0))^2] \le 4\,\mathbb{E}[(f'(X))^2].$$

Finally, $\operatorname{Var}(f(X)) = \min_{c\in\mathbb{R}} \mathbb{E}[(f(X)-c)^2] \le \mathbb{E}[(f(X)-f(0))^2]$, which yields $\operatorname{Var}(f(X)) \le 4\,\mathbb{E}[(f'(X))^2]$. ■

Exercise 3.23

Central idea. Apply the Poisson Poincaré inequality (Exercise 3.21) to $f(x) = \sqrt{x}$; the discrete derivative rationalizes to $(\sqrt{x+1} - \sqrt{x})^2 = 1/(\sqrt{x+1} + \sqrt{x})^2$, and $(\sqrt{x+1} + \sqrt{x})^2 = 2x + 1 + 2\sqrt{x^2+x} \ge 4x + 1$ gives the pointwise bound $(Df(x))^2 \le 1/(4x+1)$.

Solution. Let $X \sim \text{Poisson}(\mu)$ with $\mu = \mathbb{E}X$, and set $f(x) = \sqrt{x}$ for integers $x \ge 0$. By the Poisson Poincaré inequality (Exercise 3.21),

$$\operatorname{Var}(\sqrt{X}) = \operatorname{Var}(f(X)) \le \mu\,\mathbb{E}[(Df(X))^2], \qquad Df(x) = f(x+1) - f(x).$$

For $x \ge 0$, rationalizing the difference,

$$Df(x) = \sqrt{x+1} - \sqrt{x} = \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{\sqrt{x+1} + \sqrt{x}} = \frac{1}{\sqrt{x+1} + \sqrt{x}},$$

so that

$$(Df(x))^2 = \frac{1}{(\sqrt{x+1} + \sqrt{x})^2}.$$

Now $(\sqrt{x+1} + \sqrt{x})^2 = (x+1) + x + 2\sqrt{(x+1)x} = 2x + 1 + 2\sqrt{x^2+x}$. Since $\sqrt{x^2+x} \ge \sqrt{x^2} = x$ for $x \ge 0$,

$$(\sqrt{x+1} + \sqrt{x})^2 \ge 2x + 1 + 2x = 4x + 1,$$

hence

$$(Df(x))^2 \le \frac{1}{4x+1} \qquad \text{for every } x \ge 0.$$

Therefore

$$\operatorname{Var}(\sqrt{X}) \le \mu\,\mathbb{E}[(Df(X))^2] \le \mu\,\mathbb{E}\left[\frac{1}{4X+1}\right] = (\mathbb{E}X)\,\mathbb{E}\left[\frac{1}{4X+1}\right].$$

■

Exercise 3.24

Central idea. Write the centered Gaussian vector as a linear image of iid $N(0,1)$ variables, so $Z = \max_t X_t$ becomes the maximum of linear forms — a Lipschitz function whose gradient, at points where the maximum is uniquely attained, is exactly the coefficient vector of the maximizing coordinate. The Gaussian Poincaré inequality (Theorem 3.2) then bounds $\operatorname{Var}(Z)$ by the expected squared gradient norm, which is at most the largest coordinate variance.

Solution. Let $m = |\mathcal{T}| < \infty$ and identify $(X_t)_{t\in\mathcal{T}}$ with a centered Gaussian vector in $\mathbb{R}^m$ with covariance matrix $\Sigma \succeq 0$. Write $\Sigma = AA^\top$; with $g = (g_1, \dots, g_m)$ iid $N(0,1)$, $(X_t)_{t\in\mathcal{T}} \overset{d}{=} (Ag)_{t\in\mathcal{T}}$. Denoting by $a_t \in \mathbb{R}^m$ the t-th row of A, we have $X_t \overset{d}{=} \langle a_t, g\rangle$ and $\operatorname{Var}(X_t) = \|a_t\|^2$.

Thus $Z \overset{d}{=} f(g)$ where $f(x) = \max_{t\in\mathcal{T}} \langle a_t, x\rangle$. The function f is convex and Lipschitz with constant $\max_t \|a_t\|$ (since $|f(x) - f(y)| \le \max_t |\langle a_t, x - y\rangle| \le \max_t \|a_t\| \cdot \|x - y\|$), hence differentiable Lebesgue-a.e. by

Rademacher's theorem. If at x the maximum is uniquely attained at $t^*(x)$, then, $\mathcal{T}$ being finite, f coincides with the linear map $\langle a_{t^*}, \cdot \rangle$ in a neighborhood of x, so $\nabla f(x) = a_{t^*(x)}$ and

$$\|\nabla f(x)\|^2 = \|a_{t^*(x)}\|^2 = \mathrm{Var}(X_{t^*(x)}) \le \max_{t \in \mathcal{T}} \mathrm{Var}(X_t) \quad \text{for a.e. } x.$$

By the Gaussian Poincaré inequality (Theorem 3.2), extended from C^1 to Lipschitz f by mollification (which preserves the Lipschitz constant and gives a.e. convergence of gradients, so dominated convergence applies),

$$\mathrm{Var}(Z) = \mathrm{Var}(f(g)) \le \mathbb{E}[\|\nabla f(g)\|^2] \le \max_{t \in \mathcal{T}} \mathrm{Var}(X_t).$$

■

Chapter 4 — Basic Information Inequalities

Exercise 4.1

Central idea. Raise the Kraft sum to the n-th power and count: the n-fold product expands into $\sum_k N_k D^{-k}$ where N_k counts n-tuples of codewords with total length k. Unique decodability makes the concatenation map injective, so $N_k \le D^k$ (the number of length-k strings over $\mathcal{Y}$); the n-th root then kills the polynomial factor as $n \to \infty$. The entropy bound follows by applying Gibbs' inequality to p against the normalized code-length distribution $q(x) = D^{-|\phi(x)|}/S$, where the Kraft inequality forces the normalizing constant $S \le 1$ to only help.

Solution. Let $D = |\mathcal{Y}|$ and $\ell_x = |\phi(x)|$.

Kraft–McMillan. First let $\mathcal{X} = \{x_1, \dots, x_m\}$ be finite, with $\ell_i = \ell_{x_i}$ and $\ell_{\max} = \max_i \ell_i$. Then

$$\left(\sum_{i=1}^{m} D^{-\ell_i} \right)^n = \sum_{i_1, \dots, i_n} D^{-(\ell_{i_1} + \cdots + \ell_{i_n})} = \sum_{k=1}^{n\ell_{\max}} N_k D^{-k},$$

where N_k is the number of n-tuples $(x_{i_1}, \dots, x_{i_n})$ with $\ell_{i_1} + \cdots + \ell_{i_n} = k$. By unique decodability, distinct n-tuples give distinct concatenated codewords, and a concatenation of total length k is a string of length k over $\mathcal{Y}$, of which there are exactly D^k. Hence $N_k \le D^k$, so

$$\left(\sum_{i=1}^{m} D^{-\ell_i} \right)^n \le \sum_{k=1}^{n\ell_{\max}} 1 = n\ell_{\max}.$$

Taking n-th roots and letting $n \to \infty$ gives $\sum_{i=1}^{m} D^{-\ell_i} \le 1$. For countable $\mathcal{X}$, every finite subset satisfies the bound; the inequality extends by monotone convergence.

Entropy lower bound. If $\mathbb{E}\,|\phi(X)| = \infty$ there is nothing to prove; assume it finite. Let $S = \sum_x D^{-\ell_x} \le 1$ (so $S > 0$) and $q(x) = D^{-\ell_x}/S$, a probability distribution on $\mathcal{X}$. By Gibbs' inequality (Jensen on $t \mapsto t \log t$), with $0 \log 0 := 0$,

$$0 \le \sum_x p(x) \log_D \frac{p(x)}{q(x)} = \sum_x p(x) \log_D p(x) - \sum_x p(x) \log_D q(x).$$

Since $\log_D q(x) = -\ell_x - \log_D S$,

$$\sum_x p(x) \log_D q(x) = -\mathbb{E}\,|\phi(X)| - \log_D S.$$

Hence $-\frac{H(X)}{\log D} \ge -\mathbb{E}\,|\phi(X)| - \log_D S$, i.e.

$$\frac{H(X)}{\log D} \le \mathbb{E}\,|\phi(X)| + \log_D S \le \mathbb{E}\,|\phi(X)|,$$

because $S \le 1$ gives $\log_D S \le 0$. The left side is $\mathbb{E}[-\log_D p(X)]$. ■

Exercise 4.2

Central idea. Build a prefix code (hence uniquely decodable) greedily: process lengths in nondecreasing order, and at each step pick a codeword of the required length avoiding all extensions of previously chosen codewords. The Kraft condition guarantees an unblocked string exists, because prior codewords of length $\le \ell$ block at most $D^\ell(1 - D^{-\ell}) < D^\ell$ of the D^ℓ candidates. For the entropy bound, the Shannon lengths $\lceil -\log_D p(x) \rceil$ satisfy Kraft's inequality since $D^{-\lceil -\log_D p(x) \rceil} \le p(x)$, and rounding up costs at most 1 bit per symbol on average.

Solution. Let $D = |\mathcal{Y}|$ and order $\mathcal{X} = \{x_1, x_2, \dots\}$ so that $\ell_1 \le \ell_2 \le \cdots$, where $\ell_j = \ell(x_j)$. We choose codewords c_j of length ℓ_j inductively so that no c_j is an extension of (or equal to) any earlier c_i.

Suppose $c_1, \dots, c_{j-1}$ chosen, all of length $\le \ell_j$. A previous codeword c_i blocks exactly $D^{\ell_j - \ell_i}$ strings of length ℓ_j (its extensions, including itself). The total number of blocked length-ℓ_j strings is at most

$$\sum_{i<j} D^{\ell_j - \ell_i} = D^{\ell_j} \sum_{i<j} D^{-\ell_i} \le D^{\ell_j} \left(1 - D^{-\ell_j}\right) < D^{\ell_j},$$

using $\sum_{i \le j} D^{-\ell_i} \le 1$. Hence some length-$\ell_j$ string is unblocked; choose it as c_j. Since earlier codewords are no longer than c_j, none is a proper extension of c_j, and c_j is not an extension of any earlier one by construction — so $\{c_j\}$ is a prefix code, hence uniquely decodable, with $|\phi(x_j)| = \ell_j$.

For the second claim, let $\ell(x) = \lceil -\log_D p(x) \rceil$ for $p(x) > 0$ (symbols with $p(x) = 0$ need no codeword). Then $D^{-\ell(x)} \le D^{\log_D p(x)} = p(x)$, so $\sum_x D^{-\ell(x)} \le \sum_x p(x) = 1$. The construction above yields ϕ with $|\phi(x)| = \ell(x)$, and

$$\mathbb{E}\,|\phi(X)| = \sum_x p(x) \lceil -\log_D p(x) \rceil \le \sum_x p(x)(-\log_D p(x) + 1) = \frac{H(X)}{\log D} + 1,$$

since $\lceil u \rceil \le u + 1$. ■

Exercise 4.3

Central idea. Normalize both sequences to probability distributions ($\bar{a}_i = a_i/a$, $\bar{b}_i = b_i/b$ with $a = \sum a_i$, $b = \sum b_i$): the difference between the two sides is exactly a times the relative entropy $D(\bar{a} \| \bar{b})$, which is nonnegative by Jensen's inequality applied to the concave logarithm (Gibbs' inequality). The statement holds for positive reals, not just integers.

Solution. Let $a = \sum_i a_i > 0$, $b = \sum_i b_i > 0$, and set $\bar{a}_i = a_i/a$, $\bar{b}_i = b_i/b$, so $\sum_i \bar{a}_i = \sum_i \bar{b}_i = 1$. Then

$$\begin{aligned}\sum_i a_i \log \frac{a_i}{b_i} - a \log \frac{a}{b} &= \sum_i a_i \left[\log \frac{a_i}{b_i} - \log \frac{a}{b}\right] = \sum_i a_i \log \frac{a_i/a}{b_i/b} \\ &= a \sum_i \bar{a}_i \log \frac{\bar{a}_i}{\bar{b}_i}.\end{aligned}$$

By Jensen's inequality (concavity of log),

$$\sum_i \bar{a}_i \log \frac{\bar{b}_i}{\bar{a}_i} \le \log \sum_i \bar{a}_i \frac{\bar{b}_i}{\bar{a}_i} = \log \sum_i \bar{b}_i = \log 1 = 0,$$

so $\sum_i \bar{a}_i \log(\bar{a}_i/\bar{b}_i) \ge 0$. Multiplying by $a > 0$ gives

$$\sum_i a_i \log \frac{a_i}{b_i} \ge a \log \frac{a}{b} = \left(\sum_i a_i\right) \log \frac{\sum_i a_i}{\sum_i b_i}. \qquad \blacksquare$$

Exercise 4.4

Central idea. Factor the joint densities into conditionals, $p(x_{1:n}) = \prod_{i=1}^{n} p(x_i|x_{1:i-1})$ (and likewise for q): the log-likelihood ratio splits into a sum of n conditional log-likelihood ratios. The tower property of expectation then turns the i-th term's P-expectation into the $P_{1:i-1}$-expectation of the conditional relative entropy $D(P_{X_i|1:i-1}\|Q_{X_i|1:i-1})$.

Solution. Assume $P \ll Q$ (otherwise $D(P\|Q) = \infty$ and the identity holds trivially with the convention $\infty = \infty$ whenever some term is infinite). Write $p_{1:n}$, $q_{1:n}$ for the joint densities and $p_{i|1:i-1}$, $q_{i|1:i-1}$ for the conditional densities of X_i given $X_{1:i-1}$ (with the $i = 1$ case understood as the marginal). Then

$$\frac{p_{1:n}(x_{1:n})}{q_{1:n}(x_{1:n})} = \prod_{i=1}^{n} \frac{p_{i|1:i-1}(x_i|x_{1:i-1})}{q_{i|1:i-1}(x_i|x_{1:i-1})},$$

so with logarithms,

$$D(P\|Q) = \mathbb{E}_P\left[\log \frac{p_{1:n}(X_{1:n})}{q_{1:n}(X_{1:n})}\right] = \sum_{i=1}^{n} \mathbb{E}_P\left[\log \frac{p_{i|1:i-1}(X_i|X_{1:i-1})}{q_{i|1:i-1}(X_i|X_{1:i-1})}\right].$$

For fixed i, conditioning on $X_{1:i-1}$ and using the tower property,

$$\begin{aligned}\mathbb{E}_P\left[\log \frac{p_{i|1:i-1}(X_i|X_{1:i-1})}{q_{i|1:i-1}(X_i|X_{1:i-1})}\right] &= \mathbb{E}_{P_{1:i-1}}\left[\mathbb{E}_{P_{X_i|1:i-1}}\left[\log \frac{p_{i|1:i-1}(X_i|x_{1:i-1})}{q_{i|1:i-1}(X_i|x_{1:i-1})}\,\middle|\, X_{1:i-1} = x_{1:i-1}\right]\right] \\ &= \mathbb{E}_{P_{1:i-1}}\left[D\left(P_{X_i|1\ldots i-1}\,\|\,Q_{X_i|1\ldots i-1}\right)\right],\end{aligned}$$

since the inner expectation is exactly the relative entropy of the conditional distributions given $x_{1:i-1}$. Summing over i gives the chain rule. ■

Exercise 4.5

Central idea. The set $A^* = \{P \geq Q\}$ maximizes $P(A) - Q(A)$ because every point outside it contributes negatively; the ℓ^1 identity follows since $P-Q$ sums to zero, so the positive and negative parts have equal mass. For the coupling characterization, any coupling has $\mathbb{P}\{X \neq Y\} \geq V(P,Q)$ (restrict to $\{X \in A, Y \notin A\}$), and equality is attained by the maximal coupling: with probability $r = \sum_x \min(P(x), Q(x)) = 1 - V(P,Q)$ set $X = Y \sim R/r$ where $R = \min(P,Q)$, otherwise draw X, Y independently from the disjointly supported remainders $P - R$ and $Q - R$, forcing $X \neq Y$ there.

Solution. Recall $V(P,Q) = \sup_A (P(A) - Q(A))$.

Scheffé's identity. For any $A \subseteq \mathcal{X}$,

$$P(A) - Q(A) = \sum_{x \in A} (P(x) - Q(x)) \leq \sum_{x \in A^*} (P(x) - Q(x)) = P(A^*) - Q(A^*),$$

since terms with $x \notin A^*$ are strictly negative and can only decrease the sum. Hence $V(P,Q) = P(A^*) - Q(A^*)$. Moreover, as $\sum_x (P(x) - Q(x)) = 0$,

$$\sum_{x \in A^*} (P(x) - Q(x)) = -\sum_{x \notin A^*} (P(x) - Q(x)) = \sum_{x \notin A^*} (Q(x) - P(x)),$$

so

$$\sum_x |P(x) - Q(x)| = \sum_{x \in A^*} (P(x) - Q(x)) + \sum_{x \notin A^*} (Q(x) - P(x)) = 2(P(A^*) - Q(A^*)),$$

i.e. $V(P,Q) = \frac{1}{2}\sum_x |P(x) - Q(x)|$.

Coupling lower bound. Let (X, Y) have marginals P, Q. For any A,

$$P(A) - Q(A) = \mathbb{P}\{X \in A\} - \mathbb{P}\{Y \in A\} \leq \mathbb{P}\{X \in A,\ Y \notin A\} \leq \mathbb{P}\{X \neq Y\}.$$

Taking $\sup_A$ gives $V(P,Q) \le \mathbb{P}\{X \neq Y\}$ for every coupling.

Maximal coupling. Set $R(x) = \min(P(x), Q(x))$ and $r = \sum_x R(x)$. Since $\min(p,q) = \frac{1}{2}(p+q-|p-q|)$,

$$r = \tfrac{1}{2} \sum_x \left(P(x) + Q(x) - |P(x) - Q(x)|\right) = 1 - V(P,Q).$$

If $r = 1$ then $P = Q$ and $X = Y \sim P$ gives $\mathbb{P}\{X \neq Y\} = 0 = V$. Assume $r < 1$. Note $P - R$ is supported on $\{P > Q\}$ and $Q - R$ on $\{P < Q\}$ (where $P = Q$, both vanish), so these supports are disjoint. Define the coupling: with probability r, draw $Z \sim R/r$ and set $X = Y = Z$; with probability $1 - r$, draw $X \sim (P-R)/(1-r)$ and $Y \sim (Q-R)/(1-r)$ independently. Then $X \sim r \cdot \frac{R}{r} + (1-r) \cdot \frac{P-R}{1-r} = P$, and similarly $Y \sim Q$. On the second branch $X \neq Y$ a.s. by disjoint supports, so $\mathbb{P}\{X \neq Y\} = 1 - r = V(P,Q)$. Hence $\min \mathbb{P}\{X \neq Y\} = V(P,Q)$. ■

Exercise 4.6

Central idea. Apply Han's inequality (Theorem 4.1) to X uniform on A: then $H(X) = \log|A|$, while each $(d-1)$-dimensional marginal $X^{(i)}$ is supported on the projection A_i, so $H(X^{(i)}) \le \log|A_i|$. Han's inequality $\sum_{i=1}^d H(X^{(i)}) \ge (d-1)H(X)$ then yields $(d-1)\log|A| \le \sum_{i=1}^d \log|A_i|$.

Solution. If $A = \emptyset$ the inequality is trivial for $d \ge 2$; assume $A \neq \emptyset$ (the case $d = 1$ is trivial since A_1 is then a singleton). Let $X = (X_1, \dots, X_d)$ be uniformly distributed on the finite set $A \subset \mathbb{Z}^d$, so with logarithms to any fixed base > 1,

$$H(X) = \log|A|.$$

For $i \le d$, let $X^{(i)} = (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_d)$ be X with the i-th coordinate dropped. By definition $X^{(i)}$ takes values in A_i, the projection of A along the i-th coordinate, hence

$$H(X^{(i)}) \le \log|A_i|, \qquad i = 1, \dots, d,$$

since the uniform distribution maximizes entropy on a finite set. Han's inequality (Theorem 4.1) gives

$$H(X) \le \frac{1}{d-1} \sum_{i=1}^d H(X^{(i)}).$$

Combining,

$$\log|A| \le \frac{1}{d-1} \sum_{i=1}^d \log|A_i| = \frac{1}{d-1} \log \prod_{i=1}^d |A_i|,$$

and exponentiating yields $|A|^{d-1} \le \prod_{i=1}^d |A_i|$. ■

Exercise 4.7

Central idea. Split ∂A by direction: for fixed i, each nonempty fiber of A over $z \in A_i$ has a maximum, contributing $(x, e_i, +1) \in \partial A$, and a minimum, contributing $(x, e_i, -1) \in \partial A$ — at least $2|A_i|$ boundary points in direction i. Summing over i and applying AM–GM and Loomis–Whitney (Exercise 4.6) gives $|\partial A| \ge 2\sum_i |A_i| \ge 2d|A|^{(d-1)/d}$.

Solution. Write $B = \{e_1, \dots, e_d\}$ and partition $\partial A = \bigsqcup_{i=1}^d S_i$ with $S_i = \{(x, e_i, s) \in \partial A\}$. Fix i and $z \in A_i$, where A_i is the projection of A along coordinate i. The fiber

$$F_z = \{t \in \mathbb{Z} : (z_1, \dots, z_{i-1}, t, z_{i+1}, \dots, z_d) \in A\}$$

is a finite nonempty subset of $\mathbb{Z}$. Let $t_z^+ = \max F_z$, $t_z^- = \min F_z$, and let $x_z^+, x_z^- \in A$ be the corresponding points. Then $x_z^+ + e_i \notin A$ (otherwise $t_z^+ + 1 \in F_z$) and $x_z^- - e_i \notin A$ (otherwise $t_z^- - 1 \in F_z$), so

$$(x_z^+, e_i, +1) \in S_i, \qquad (x_z^-, e_i, -1) \in S_i.$$

For fixed z these two triples are distinct (they differ in s), and for distinct z the triples are distinct (they differ in the projection). Hence $|S_i| \geq 2|A_i|$, and

$$|\partial A| = \sum_{i=1}^{d} |S_i| \geq 2 \sum_{i=1}^{d} |A_i|.$$

By AM–GM and the discrete Loomis–Whitney inequality (Exercise 4.6),

$$\sum_{i=1}^{d} |A_i| \geq d\Big(\prod_{i=1}^{d} |A_i| \Big)^{1/d} \geq d(|A|^{d-1})^{1/d} = d|A|^{\frac{d-1}{d}}.$$

Therefore $|\partial A| \geq 2d|A|^{(d-1)/d}$. ■

Exercise 4.8

Central idea. Take h_i to be h itself on $\mathcal{X}^{n-1}$ (the log-trace on $n-1$ points). Since $\mathrm{tr}(x^{(i)})$ is the coordinate projection of $\mathrm{tr}(x)$, we get $0 \leq h(x) - h_i(x^{(i)}) \leq 1$ from projection-monotonicity and the hypothesis. The sum bound $\sum_i (h - h_i) \leq h$ is exactly Loomis–Whitney (Exercise 4.6) applied to $A = \mathrm{tr}(x) \subset \{0,1\}^n$: $|T|^{n-1} \leq \prod_i |T_i|$ becomes $(n-1)\log_b |T| \leq \sum_i \log_b |T_i|$.

Solution. Let $\mathrm{tr}(x) \subset \{0,1\}^n$ be the trace on $x = (x_1, \dots, x_n)$ and $T(x) = |\mathrm{tr}(x)| \geq 1$, so that $h(x) = \log_b T(x)$ is well defined and nonnegative; the requirement $0 \leq h(x) - h(x^{(i)})$ forces $b > 1$, which we assume. For $y \in \mathcal{X}^{n-1}$ define $h_i(y) = \log_b |\mathrm{tr}(y)|$, i.e. h_i is the same combinatorial entropy on $n-1$ points.

Fix x and i. The trace $\mathrm{tr}(x^{(i)})$ is precisely the projection of $\mathrm{tr}(x) \subset \{0,1\}^n$ along the i-th coordinate: dropping the i-th entry of each pattern $(1_{x_1 \in A}, \dots, 1_{x_n \in A})$ gives exactly the patterns of $\mathrm{tr}(x^{(i)})$. Hence $|\mathrm{tr}(x^{(i)})| \leq |\mathrm{tr}(x)|$, so with $b > 1$,

$$h(x) - h_i(x^{(i)}) = \log_b \frac{|\mathrm{tr}(x)|}{|\mathrm{tr}(x^{(i)})|} \geq 0,$$

and $h(x) - h_i(x^{(i)}) \leq 1$ is the hypothesis. Thus $0 \leq h(x) - h_i(x^{(i)}) \leq 1$ for all x, i.

For the sum, apply Exercise 4.6 (discrete Loomis–Whitney) to the finite set $A = \mathrm{tr}(x) \subset \mathbb{Z}^n$: with A_i the projection along coordinate i (so $|A_i| = |\mathrm{tr}(x^{(i)})|$),

$$|\mathrm{tr}(x)|^{n-1} \leq \prod_{i=1}^{n} |\mathrm{tr}(x^{(i)})|.$$

Taking $\log_b$ gives $(n-1)h(x) \leq \sum_{i=1}^{n} h_i(x^{(i)})$, i.e.

$$\sum_{i=1}^{n} (h(x) - h_i(x^{(i)})) = nh(x) - \sum_{i=1}^{n} h_i(x^{(i)}) \leq h(x).$$

Hence h has the self-bounding property. ■

Exercise 4.9

Central idea. The product $A \times B$ is compact (Tychonoff's theorem), and the Minkowski sum $A + B$ is its continuous image under the addition map $(a, b) \mapsto a + b$; continuous images of compact sets are compact. (This is needed for the Brunn–Minkowski inequality in Section 4.14.)

Solution. Let $A, B \subset \mathbb{R}^d$ be compact, hence with the subspace topology. The product $A \times B$ with the product topology is compact by Tychonoff's theorem (for a finite product this is elementary). The addition map

$$s : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d, \qquad s(a, b) = a + b,$$

is continuous. By definition of the Minkowski sum,

$$A + B = \{a + b : a \in A,\ b \in B\} = s(A \times B).$$

A continuous image of a compact set is compact: if $\{U_\alpha\}$ is an open cover of $s(A \times B)$, then $\{s^{-1}(U_\alpha)\}$ is an open cover of $A \times B$, which has a finite subcover $s^{-1}(U_{\alpha_1}), \dots, s^{-1}(U_{\alpha_m})$, and then $U_{\alpha_1}, \dots, U_{\alpha_m}$ cover $s(A \times B)$. Hence $A + B$ is compact. ■

Exercise 4.10

Central idea. The duality formula (Corollary 4.15) writes each atomic divergence $D(P\|Q|\mathcal{G})$ as a supremum over $\mathcal{G}$-measurable test functions, so shrinking the σ-algebra shrinks the supremum (data processing), and quantizing test functions shows $D(P\|Q)$ is the supremum over finite partitions. To lift Han's inequality (Theorem 4.9) to uncountable $\mathcal{X}$, discretize each coordinate by finite partitions and push forward to a finite product space where the countable 4.9 applies; a diagonal common-refinement argument makes one discretization serve all n marginals at once. Theorem 4.10 then follows exactly as in the countable case, using $dQ^{(i)}/dP^{(i)} = \mathbb{E}^{(i)}Z$.

Solution. Recall the duality formula (Corollary 4.15): for probability measures P, Q,

$$D(P\|Q) = \sup_Z \{\mathbb{E}_P Z - \log \mathbb{E}_Q e^Z\}$$

over all measurable Z with $\mathbb{E}_Q e^Z < \infty$. If $\mathcal{G}$ has countably many atoms, $D(P\|Q|\mathcal{G})$ is the divergence of the restrictions to $(\mathrm{atom}(\mathcal{G}), \mathcal{G})$, so

$$D(P\|Q|\mathcal{G}) = \sup \{\mathbb{E}_P Z - \log \mathbb{E}_Q e^Z : Z\ \mathcal{G}\text{-measurable},\ \mathbb{E}_Q e^Z < \infty\}. \tag{1}$$

Data processing. If $\mathcal{H} \subset \mathcal{G}$ (both with countably many atoms), every $\mathcal{H}$-measurable Z is $\mathcal{G}$-measurable, so the supremum in (1) for $\mathcal{H}$ is over a smaller set: $D(P\|Q|\mathcal{H}) \le D(P\|Q|\mathcal{G})$.

Supremum over finite partitions. Clearly $D(P\|Q|\mathcal{G}) \le D(P\|Q)$ for every finite $\mathcal{G}$, so $\sup_{\mathcal{G}} D(P\|Q|\mathcal{G}) \le D(P\|Q)$. For the reverse, let Z be bounded, $|Z| \le M$. Quantizing $[-M, M]$ into intervals of length $< \delta$ gives a finite $\mathcal{G} = \sigma(\{Z \in I_k\})$ and a $\mathcal{G}$-measurable Z' with $\|Z - Z'\|_\infty < \delta$. Then $|\mathbb{E}_P Z - \mathbb{E}_P Z'| < \delta$ and, $\log$ being e^M-Lipschitz on $[e^{-M}, e^M]$, $|\log \mathbb{E}_Q e^Z - \log \mathbb{E}_Q e^{Z'}| \le e^{2M}(e^\delta - 1)$. Hence $D(P\|Q|\mathcal{G}) \ge \mathbb{E}_P Z - \log \mathbb{E}_Q e^Z - \epsilon(\delta)$ with $\epsilon(\delta) \to 0$. Thus $\sup_{\mathcal{G}} D(P\|Q|\mathcal{G}) \ge \sup_{\text{bounded } Z}\{\mathbb{E}_P Z - \log \mathbb{E}_Q e^Z\}$, and the latter equals $D(P\|Q)$ by truncation ($Z_n = (-n) \vee (\log \frac{dP}{dQ} \wedge n)$; if $D(P\|Q) < \infty$ then $\mathbb{E}_P|\log \frac{dP}{dQ}| < \infty$ since $\int_{\{f<1\}} |\log f|\, dP = \int_{\{g>1\}} \frac{\log g}{g}\, dQ \le 1/e$, so dominated convergence gives $\mathbb{E}_P Z_n \to D(P\|Q)$, $\mathbb{E}_Q e^{Z_n} \to 1$; if $D(P\|Q) = \infty$ the same truncations, or $Z_n = n1_A$ with $P(A) > 0 = Q(A)$, drive the functional to ∞). Therefore

$$D(P\|Q) = \sup \{D(P\|Q|\mathcal{G}) : \mathcal{G} \text{ has finitely many atoms}\}. \tag{2}$$

Product partitions are cofinal. Let μ, ν be probabilities on $\mathcal{X}^m$ with the product σ-algebra. Then $D(\mu\|\nu) = \sup D(\mu\|\nu|\mathcal{G})$ over finite product partitions $\mathcal{G} = \bigotimes_{j=1}^m \mathcal{P}_j$. Indeed, "$\le$" is clear; for "$\ge$", given bounded Z ($|Z| \le M$) and $\delta > 0$, approximate Z in $L^1(\mu + \nu)$ by $W_0 = \sum_k c_k 1_{R_k}$ with R_k measurable rectangles (finite linear combinations of rectangles are L^1-dense), truncate to $W = (-M - 1) \vee (W_0 \wedge (M + 1))$ (still $\|Z - W\|_{L^1} \le \delta$), and let $\mathcal{G}$ be the product partition generated by the sides of the R_k, so W is $\mathcal{G}$-measurable. By (1), $D(\mu\|\nu|\mathcal{G}) \ge \mathbb{E}_\mu W - \log \mathbb{E}_\nu e^W \ge \mathbb{E}_\mu Z - \log \mathbb{E}_\nu e^Z - C\delta$. Letting $\delta \to 0$ and taking sup over bounded Z gives the claim.

Theorem 4.9 for general $\mathcal{X}$. Let $P = \bigotimes_{j=1}^n P_j$ on $\mathcal{X}^n$ and Q arbitrary. For each i and $\epsilon > 0$, the lemma gives finite partitions $\mathcal{F}_j^{(i)}$ ($j \ne i$) of $\mathcal{X}$ with $D(Q^{(i)}\|P^{(i)}|\bigotimes_{j\ne i} \mathcal{F}_j^{(i)}) \ge D(Q^{(i)}\|P^{(i)}) - \epsilon$ (with the obvious $M \to \infty$ modification if the divergence is infinite). Put $\mathcal{F}_j = \bigvee_{i\ne j} \mathcal{F}_j^{(i)}$; then $\bigotimes_{j\ne i} \mathcal{F}_j \supset \bigotimes_{j\ne i} \mathcal{F}_j^{(i)}$, so by data processing $D(Q^{(i)}\|P^{(i)}|\bigotimes_{j\ne i} \mathcal{F}_j) \ge D(Q^{(i)}\|P^{(i)}) - \epsilon$ for every i. Let $\pi_j : \mathcal{X} \to \mathcal{Y}_j := \mathrm{atom}(\mathcal{F}_j)$

and $\pi = \bigotimes_j \pi_j : \mathcal{X}^n \to \mathcal{Y} := \prod_j \mathcal{Y}_j$, with pushforwards $\bar{Q} = Q \circ \pi^{-1}$, $\bar{P} = P \circ \pi^{-1} = \bigotimes_j \bar{P}_j$ (a product since P is). The countable Theorem 4.9 on the finite space $\mathcal{Y}$ gives $D(\bar{Q}\|\bar{P}) \geq \frac{1}{n-1}\sum_i D(\bar{Q}^{(i)}\|\bar{P}^{(i)})$. Here $D(\bar{Q}\|\bar{P}) = D(Q\|P|\mathcal{G}) \leq D(Q\|P)$ for $\mathcal{G} = \bigotimes_j \sigma(\mathcal{F}_j)$, and $D(\bar{Q}^{(i)}\|\bar{P}^{(i)}) = D(Q^{(i)}\|P^{(i)}|\bigotimes_{j\neq i}\sigma(\mathcal{F}_j))$. Hence

$$D(Q\|P) \geq \frac{1}{n-1}\sum_{i=1}^{n}\left(D(Q^{(i)}\|P^{(i)}) - \epsilon\right),$$

and $\epsilon \to 0$ yields Theorem 4.9 for general $\mathcal{X}$.

Theorem 4.10 for general $\mathcal{X}$. Let $X_1, \dots, X_n$ be independent, $Z = f(X_1, \dots, X_n) \geq 0$ measurable. If $\mathbb{E}Z = 0$ there is nothing to prove; otherwise scale to $\mathbb{E}Z = 1$ (both sides are homogeneous of degree one). Define Q on $\mathcal{X}^n$ by $dQ/dP = Z$, so $D(Q\|P) = \mathbb{E}_P[Z\log Z] = \mathrm{Ent}(Z)$. By the general Theorem 4.9, in its equivalent form,

$$D(Q\|P) \leq \sum_{i=1}^{n}\left[D(Q\|P) - D(Q^{(i)}\|P^{(i)})\right].$$

For $B \subset \mathcal{X}^{n-1}$, $Q^{(i)}(B) = \int_{B\times\mathcal{X}} Z\,dP = \int_B \mathbb{E}^{(i)}Z\,dP^{(i)}$ (using that P is a product), so $dQ^{(i)}/dP^{(i)} = \mathbb{E}^{(i)}Z$. Thus $D(Q^{(i)}\|P^{(i)}) = \mathbb{E}_P[(\mathbb{E}^{(i)}Z)\log(\mathbb{E}^{(i)}Z)]$, while $\mathbb{E}[\mathrm{Ent}^{(i)}(Z)] = \mathbb{E}_P[\mathbb{E}^{(i)}[Z\log Z]] - \mathbb{E}_P[(\mathbb{E}^{(i)}Z)\log(\mathbb{E}^{(i)}Z)] = \mathbb{E}_P[Z\log Z] - D(Q^{(i)}\|P^{(i)})$. Hence $D(Q\|P) - D(Q^{(i)}\|P^{(i)}) = \mathbb{E}[\mathrm{Ent}^{(i)}(Z)]$, and

$$\mathrm{Ent}(Z) \leq \sum_{i=1}^{n}\mathbb{E}[\mathrm{Ent}^{(i)}(Z)],$$

which is Theorem 4.10 for general $\mathcal{X}$. ■

Exercise 4.11

Central idea. The Donsker–Varadhan duality representation (Lemma 4.14) writes $D(Q\|P)$ as a supremum, over bounded measurable f, of functionals that are affine in Q; a supremum of affine functions is convex, which is exactly the claim.

Solution. Recall the duality representation of the Kullback–Leibler divergence (Lemma 4.14): for probability measures Q, P on $\mathcal{X}$,

$$D(Q\|P) = \sup_f\left\{\int f\,dQ - \log\int e^f\,dP\right\},$$

where the supremum is taken over all bounded measurable functions $f : \mathcal{X} \to \mathbb{R}$ (the supremum equals $+\infty$ when $Q \not\ll P$, in agreement with $D(Q\|P) = \infty$). For each fixed f, the map

$$Q \longmapsto \int f\,dQ - \log\int e^f\,dP$$

is affine in Q, since $\int f\,dQ$ is linear in Q and the second term does not depend on Q.

Let Q_1, Q_2 be probability measures on $\mathcal{X}$ and $\lambda \in [0,1]$, and set $Q_\lambda = \lambda Q_1 + (1-\lambda)Q_2$. Then

$$\begin{aligned}
D(Q_\lambda\|P) &= \sup_f\left\{\int f\,dQ_\lambda - \log\int e^f\,dP\right\}\\
&= \sup_f\left\{\lambda\left(\int f\,dQ_1 - \log\int e^f\,dP\right) + (1-\lambda)\left(\int f\,dQ_2 - \log\int e^f\,dP\right)\right\}\\
&\leq \lambda\sup_f\left\{\int f\,dQ_1 - \log\int e^f\,dP\right\} + (1-\lambda)\sup_f\left\{\int f\,dQ_2 - \log\int e^f\,dP\right\}\\
&= \lambda D(Q_1\|P) + (1-\lambda)D(Q_2\|P),
\end{aligned}$$

using linearity of the integral in Q and the elementary inequality $\sup_f[\lambda a_f + (1-\lambda)b_f] \leq \lambda\sup_f a_f + (1-\lambda)\sup_f b_f$, valid in $[-\infty,\infty]$. Thus $Q \mapsto D(Q\|P)$ is convex on the set of probability distributions over $\mathcal{X}$. ■

Exercise 4.12

Central idea. Split the log-likelihood ratio $\log \frac{d\mathbf{P}}{d(Q_X \otimes Q_Y)}$ into the "dependence" part $\log \frac{d\mathbf{P}}{d(P_X \otimes P_Y)}$ plus the two marginal log-ratios $\log \frac{dP_X}{dQ_X}(x) + \log \frac{dP_Y}{dQ_Y}(y)$; integrating against $\mathbf{P}$, the marginal terms collapse to $D(P_X \| Q_X)$ and $D(P_Y \| Q_Y)$ because the marginals of $\mathbf{P}$ are P_X and P_Y.

Solution. The identity is understood in $[0, \infty]$. If $D(\mathbf{P} \| Q_X \otimes Q_Y) = \infty$, then at least one term on the right is infinite: finiteness of all three would give $\mathbf{P} \ll P_X \otimes P_Y$ and $P_X \otimes P_Y \ll Q_X \otimes Q_Y$ (the latter by the π-λ theorem from $P_X \ll Q_X$, $P_Y \ll Q_Y$), hence $\mathbf{P} \ll Q_X \otimes Q_Y$ by transitivity, a contradiction. Assume therefore $\mathbf{P} \ll Q_X \otimes Q_Y$.

Then $P_X \ll Q_X$ and $P_Y \ll Q_Y$: if $Q_X(A) = 0$ then $(Q_X \otimes Q_Y)(A \times \mathcal{Y}) = 0$, so $P_X(A) = \mathbf{P}(A \times \mathcal{Y}) = 0$, and likewise for P_Y. Write $f = \frac{d\mathbf{P}}{d(Q_X \otimes Q_Y)}$, $g_X = \frac{dP_X}{dQ_X}$, $g_Y = \frac{dP_Y}{dQ_Y}$. On rectangles, $(P_X \otimes P_Y)(A \times B) = \int_{A \times B} g_X(x) g_Y(y)\, d(Q_X \otimes Q_Y)$, so by the π-λ theorem

$$\frac{d(P_X \otimes P_Y)}{d(Q_X \otimes Q_Y)} = g_X \otimes g_Y.$$

Moreover, $\frac{d\mathbf{P}}{d(P_X \otimes P_Y)} = \frac{f}{g_X \otimes g_Y}$ (with $\frac{f}{0} := 0$): for any measurable E,

$$\int_E \frac{f}{g_X g_Y}\, d(P_X \otimes P_Y) = \int_E \frac{f}{g_X g_Y}(g_X g_Y)\, d(Q_X \otimes Q_Y) = \int_E f\, d(Q_X \otimes Q_Y) = \mathbf{P}(E),$$

since on $\{g_X g_Y = 0\}$ one has $f = 0$ a.e. $[Q_X \otimes Q_Y]$ (e.g. $\mathbf{P}(\{g_X = 0\} \times \mathcal{Y}) = P_X(\{g_X = 0\}) = 0$, and likewise for g_Y).

Hence, with $h = \frac{f}{g_X \otimes g_Y} = \frac{d\mathbf{P}}{d(P_X \otimes P_Y)}$ a.e.,

$$\begin{aligned} D(\mathbf{P} \| Q_X \otimes Q_Y) &= \int \log f\, d\mathbf{P} = \int \log h\, d\mathbf{P} + \int \log\left(g_X(x) g_Y(y)\right) d\mathbf{P}(x, y) \\ &= D(\mathbf{P} \| P_X \otimes P_Y) + \int \log g_X(x)\, dP_X(x) + \int \log g_Y(y)\, dP_Y(y) \\ &= D(\mathbf{P} \| P_X \otimes P_Y) + D(P_X \| Q_X) + D(P_Y \| Q_Y), \end{aligned}$$

where the marginal collapse uses that the x-marginal of $\mathbf{P}$ is P_X and the y-marginal is P_Y, and $\log f = \log h + \log(g_X g_Y)$ holds $\mathbf{P}$-a.e. since $f > 0$ $\mathbf{P}$-a.e. ■

Exercise 4.13

Central idea. This is Fenchel–Legendre duality for the log-moment-generating function.

Direction $\leq$. The Gibbs variational inequality $\mathbb{E}_Q f \leq D(Q \| P) + \log \mathbb{E}_P e^f$, applied with $f = \lambda(Z - \mathbb{E}Z)$, gives $\psi^*(t) \leq \inf\{D(Q \| P) : \mathbb{E}_Q Z - \mathbb{E}Z \geq t\}$.

Direction $\geq$. The exponentially tilted laws $dQ_\lambda / dP = e^{\lambda(Z - \mathbb{E}Z) - \psi(\lambda)}$ satisfy $D(Q_\lambda \| P) = \lambda m(\lambda) - \psi(\lambda)$, where $m(\lambda) = \psi'(\lambda)$ is the tilted mean. Whenever t is a tilted mean, Q_λ attains $\psi^*(t)$. The remaining values of t fall into two cases: - *Atom at the essential supremum.* Condition on it. - *No exponential moments beyond the abscissa b of convergence.* Use a heavy-tail lemma: a law R with no exponential moments has $\liminf_M \frac{1}{M} \log \frac{1}{R(W > M)} = 0$. Mixing R with $R(\cdot \mid W > M)$ along such M raises the mean to t at vanishing extra divergence.

(An earlier version claimed $\varepsilon_M \log(1/R(W \geq M)) \to 0$ along *all* M via a Chernoff bound. That uses Chernoff in the wrong direction; for a Gaussian the quantity grows like $M/2$. The subsequence argument below replaces it.)

Solution. The statement involves $\mathbb{E}Z$, so Z is assumed integrable. Let P be the underlying probability measure and set $W := Z - \mathbb{E}Z$, so $\mathbb{E}_P W = 0$. Let $\psi(\lambda) := \log \mathbb{E}_P e^{\lambda W} \in [0, \infty]$; it is convex, $\psi(0) = 0$, and

$\psi \geq 0$ by Jensen. The constraint set consists of the Q for which $\mathbb{E}_Q W$ exists (in $(-\infty, \infty]$) and is $\geq t$. Only $Q \ll P$ matter, since otherwise $D(Q\|P) = \infty$. Write $I(t)$ for the infimum of $D(Q\|P)$ over this set and fix $t > 0$.

For $\lambda < 0$, $\lambda t - \psi(\lambda) < 0 = 0 \cdot t - \psi(0)$, hence

$$\psi^*(t) = \sup_{\lambda \geq 0} (\lambda t - \psi(\lambda)) \geq 0.$$

(a) $\psi^*(t) \leq I(t)$**.**

Gibbs inequality. Let $Q \ll P$ with $D(Q\|P) < \infty$, $h = dQ/dP$, and let f satisfy $\mathbb{E}_P e^f < \infty$. Then

$$\mathbb{E}_Q f \leq D(Q\|P) + \log \mathbb{E}_P e^f.$$

Proof. First, $\mathbb{E}_Q f^+ < \infty$. Young's inequality $ab \leq a \log a - a + e^b$ (valid for $a \geq 0$) gives $h f^+ \leq h \log h - h + e^{f^+}$. Moreover $\mathbb{E}_P |h \log h| < \infty$, because $h \log h \geq -1/e$ and $\mathbb{E}_P h \log h = D(Q\|P) < \infty$. Hence

$$\mathbb{E}_Q f^+ \leq D(Q\|P) - 1 + \mathbb{E}_P e^{f^+} \leq D(Q\|P) + \mathbb{E}_P e^f < \infty.$$

If $\mathbb{E}_Q f = -\infty$, there is nothing to prove. Otherwise $f \in L^1(Q)$. Also $\log h \in L^1(Q)$: we have $\mathbb{E}_Q (\log h)^- = \mathbb{E}_P [h (\log h)^-] \leq 1/e$, and $\mathbb{E}_Q \log h = D(Q\|P)$. Jensen for the concave log under Q then gives

$$\mathbb{E}_Q f - D(Q\|P) = \mathbb{E}_Q \log \frac{e^f}{h} \leq \log \mathbb{E}_Q \frac{e^f}{h} = \log \mathbb{E}_P [e^f \mathbf{1}_{\{h>0\}}] \leq \log \mathbb{E}_P e^f. \qquad \square$$

Applying it. Let Q be feasible with $D(Q\|P) < \infty$; otherwise the bound $\psi^*(t) \leq D(Q\|P)$ is trivial. Let $\lambda \geq 0$; if $\psi(\lambda) = \infty$, then $\lambda t - \psi(\lambda) = -\infty$. Otherwise apply the Gibbs inequality to $f = \lambda W$. Since $\mathbb{E}_Q W$ exists and is $\geq t$,

$$\lambda t - \psi(\lambda) \leq \lambda \, \mathbb{E}_Q W - \psi(\lambda) \leq D(Q\|P).$$

(For $\lambda > 0$ the Gibbs step also shows $\mathbb{E}_Q W < \infty$.) Take the supremum over $\lambda \geq 0$, then the infimum over feasible Q.

(b) $I(t) \leq \psi^*(t)$**.** Put $L := \psi^*(t)$ and assume $L < \infty$; otherwise there is nothing to prove.

Tilted laws. Let $b := \sup\{\lambda \geq 0 : \psi(\lambda) < \infty\} \in [0, \infty]$. For $0 \leq \lambda < b$: - $\mathbb{E}_P[|W| e^{\lambda W}] < \infty$. On $\{W \leq 0\}$ the integrand is at most $|W|$. On $\{W > 0\}$ it is at most $C\, e^{\lambda' W}$ for any $\lambda < \lambda' < b$. - Hence, by dominated convergence, ψ is differentiable on $[0, b)$ (right derivative at 0), with

$$\psi'(\lambda) = m(\lambda) := \mathbb{E}_{Q_\lambda} W, \qquad \frac{dQ_\lambda}{dP} := e^{\lambda W - \psi(\lambda)}.$$

- $m(0) = \mathbb{E}_P W = 0$. - m is nondecreasing and continuous on $[0, b)$: it is the derivative of a convex function, and right-continuity at 0 follows by dominated convergence. - $D(Q_\lambda\|P) = \mathbb{E}_{Q_\lambda}[\lambda W - \psi(\lambda)] = \lambda m(\lambda) - \psi(\lambda)$.

Set $t_{\max} := \sup_{[0,b)} m = \lim_{\lambda \uparrow b} m(\lambda) \in [0, \infty]$, with $t_{\max} := 0$ if $b = 0$.

Case A: $0 < t < t_{\max}$. By continuity and $m(0) = 0$, there is $\lambda \in (0, b)$ with $m(\lambda) = t$. Then Q_λ is feasible ($\mathbb{E}_{Q_\lambda} W = t$), and

$$D(Q_\lambda\|P) = \lambda t - \psi(\lambda) \leq L.$$

So assume from now on that $t \geq t_{\max}$.

Heavy-tail lemma. Let R be a probability measure with $\mathbb{E}_R |W| < \infty$ and $\mathbb{E}_R e^{\varepsilon W} = \infty$ for every $\varepsilon > 0$. Then $R(W > M) > 0$ for all M, and

$$\liminf_{M \to \infty} \frac{1}{M} \log \frac{1}{R(W > M)} = 0.$$

Proof. If $R(W > M_0) = 0$, then $W \le M_0$ R-a.s. and every exponential moment is finite. The lim inf is ≥ 0 since $R(W > M) \le 1$. If it were strictly positive (possibly $+\infty$), pick a finite $c > 0$ with $2c$ below it; then $R(W > u) \le e^{-cu}$ for all $u \ge M_0$, for some M_0. With $\varepsilon = c/2$, the layer-cake formula gives

$$\mathbb{E}_R e^{\varepsilon W} \le e^{\varepsilon M_0} + \int_{M_0}^{\infty} \varepsilon e^{\varepsilon u} R(W > u)\, du < \infty,$$

a contradiction. □

Mixture construction. Let $R \ll P$ satisfy the lemma's hypotheses, with $r := \mathbb{E}_R W < t$. Choose $M_j \to \infty$ with $M_j > t$ and $\frac{1}{M_j} \log \frac{1}{R(A_j)} \to 0$, where $A_j = \{W > M_j\}$. Define: - $R_j := R(\cdot \mid A_j)$ and $\mu_j := \mathbb{E}_{R_j} W \in [M_j, \infty)$; finiteness holds because $\mathbb{E}_R |W| < \infty$; - $\varepsilon_j := \frac{t-r}{\mu_j - r} \in (0, 1)$; - $Q_j := (1 - \varepsilon_j)R + \varepsilon_j R_j$.

Then $\mathbb{E}_{Q_j} W = r + \varepsilon_j(\mu_j - r) = t$, so Q_j is feasible.

$D(\cdot\|P)$ is convex: integrate the pointwise inequality $\phi((1-\varepsilon)h_1 + \varepsilon h_2) \le (1-\varepsilon)\phi(h_1) + \varepsilon\phi(h_2)$ for $\phi(x) = x \log x$. Hence

$$D(Q_j\|P) \le (1 - \varepsilon_j) D(R\|P) + \varepsilon_j D(R_j\|P).$$

Also $\frac{dR_j}{dP} = \frac{dR}{dP} \cdot \frac{\mathbf{1}_{A_j}}{R(A_j)}$, so

$$D(R_j\|P) = \mathbb{E}_{R_j}\Big[\log \frac{dR}{dP}\Big] + \log \frac{1}{R(A_j)}.$$

Finally,

$$\varepsilon_j \log \frac{1}{R(A_j)} \le \frac{t - r}{M_j - r} \log \frac{1}{R(A_j)} \longrightarrow 0. \tag{$*$}$$

Case B: $b = 0$. Then $\psi(\lambda) = \infty$ for every $\lambda > 0$, so $L = 0$. Take $R = P$; then $r = 0 < t$, and $\mathbb{E}_P e^{\varepsilon W} = \infty$ for all $\varepsilon > 0$. Here $dR/dP = 1$, so $D(R\|P) = 0$, and by $(*)$

$$D(Q_j\|P) \le \varepsilon_j \log \frac{1}{P(A_j)} \to 0.$$

Hence $I(t) = 0 = L$.

Case C: $0 < b < \infty$ and $t \ge t_{\max}$.

Step 1: $\psi(b) < \infty$. On $[0, b)$, $\psi(\lambda) = \int_0^\lambda m \le \lambda t_{\max} \le bt$. As $\lambda \uparrow b$, $\mathbb{E}_P e^{\lambda W} \to \mathbb{E}_P e^{bW}$ by monotone convergence on $\{W > 0\}$ and dominated convergence on $\{W \le 0\}$. So $\psi(b) = \lim_{\lambda \uparrow b} \psi(\lambda) \le bt < \infty$.

Step 2: $L = bt - \psi(b)$. The map $\lambda \mapsto \lambda t - \psi(\lambda)$ has derivative $t - m(\lambda) \ge 0$ on $[0, b)$, is continuous up to b, and equals $-\infty$ for $\lambda > b$.

Step 3: the law $R := Q_b$. Define $dQ_b/dP = e^{bW - \psi(b)}$. As $\lambda \uparrow b$,

$$\mathbb{E}_P[W^+ e^{\lambda W}] \uparrow \mathbb{E}_P[W^+ e^{bW}] \text{ (monotone convergence)}, \qquad \mathbb{E}_P[W^- e^{\lambda W}] \to \mathbb{E}_P[W^- e^{bW}] \text{ (dominated by } W^-).$$

Hence $\mathbb{E}_{Q_b} W = \lim_{\lambda \uparrow b} m(\lambda) = t_{\max}$, which is finite, so $\mathbb{E}_{Q_b} |W| < \infty$. Moreover $D(Q_b\|P) = bt_{\max} - \psi(b)$.

If $t = t_{\max}$, then Q_b is feasible with $D(Q_b\|P) = bt - \psi(b) = L$.

If $t > t_{\max}$, apply the mixture construction with $R = Q_b$ and $r = t_{\max}$. The lemma applies, since $\mathbb{E}_{Q_b} e^{\varepsilon W} = e^{-\psi(b)} \mathbb{E}_P e^{(b+\varepsilon)W} = \infty$ for $\varepsilon > 0$. Here $\log \frac{dR}{dP} = bW - \psi(b)$, so

$$D(R_j\|P) = b\mu_j - \psi(b) + \log \frac{1}{R(A_j)},$$

and therefore

$$D(Q_j\|P) \le (1 - \varepsilon_j)(bt_{\max} - \psi(b)) + \varepsilon_j(b\mu_j - \psi(b)) + \varepsilon_j \log \frac{1}{R(A_j)} = bt - \psi(b) + \varepsilon_j \log \frac{1}{R(A_j)}.$$

By $(*)$ this tends to $bt - \psi(b) = L$. Hence $I(t) \le L$.

Case D: $b = \infty$ and $t \ge t_{\max}$. Let $w^* := \operatorname{ess\,sup} W \in [0, \infty]$. If $w^* = 0$, then $W = 0$ a.s., so $\psi \equiv 0$ and $L = \infty$, which is excluded. So $w^* > 0$.

Step 1: $t_{\max} = w^*$. Trivially $m(\lambda) \le w^*$. Conversely, fix $0 < c' < c'' < w^*$. Then $P(W > c'') > 0$, and $\mathbb{E}_P e^{\lambda W} \ge e^{\lambda c''} P(W > c'') \to \infty$. Hence

$$Q_\lambda(W \le c') \le \frac{e^{\lambda c'}}{e^{\lambda c''} P(W > c'')} \to 0, \qquad \mathbb{E}_{Q_\lambda} W^- \le \frac{\mathbb{E}_P W^-}{\mathbb{E}_P e^{\lambda W}} \to 0,$$

using $e^{\lambda W} \le 1$ on $\{W < 0\}$. Since $m(\lambda) \ge c' Q_\lambda(W > c') - \mathbb{E}_{Q_\lambda} W^-$, we get $\liminf_{\lambda \to \infty} m(\lambda) \ge c'$. As $c' < w^*$ was arbitrary, $t_{\max} = w^*$.

Step 2: the remaining values of t. - If $w^* = \infty$, no finite $t \ge t_{\max}$ exists. - If $t > w^*$, then $\psi(\lambda) \le \lambda w^*$, so $\lambda t - \psi(\lambda) \ge \lambda(t - w^*) \to \infty$. Thus $L = \infty$, which is excluded. - If $t = w^* < \infty$, write $\lambda t - \psi(\lambda) = -\log \mathbb{E}_P e^{-\lambda(w^* - W)}$. The integrand decreases to $\mathbf{1}_{\{W = w^*\}}$, so $\lambda t - \psi(\lambda) \to \log \frac{1}{p_0}$ with $p_0 := P(W = w^*)$. - If $p_0 = 0$, then $L = \infty$, which is excluded. - If $p_0 > 0$, then $Q^* := P(\cdot \mid W = w^*)$ is feasible ($\mathbb{E}_{Q^*} W = w^* = t$), and $D(Q^* \| P) = \log \frac{1}{p_0} \le L$.

Cases A–D exhaust all possibilities for $t > 0$ (Case A covers $t < t_{\max}$ whenever $b > 0$). Hence $I(t) \le \psi^*(t)$. Together with (a), for all $t > 0$,

$$\psi^*(t) = \inf\{D(Q\|P) : \ \mathbb{E}_Q Z - \mathbb{E} Z \ge t\}. \qquad \blacksquare$$

Exercise 4.14

Central idea. Couple each Bernoulli summand with an independent Poisson of the same mean so that the i-th pair disagrees with probability $V(\mathrm{Ber}(p_i), \mathrm{Po}(p_i))$ (Exercise 4.5); a union bound tensorizes the one-coordinate errors into a bound on $V(\mathbf{P}, \mathrm{Po}(\mu))$, because sums of independent Poissons are Poisson (infinite divisibility). The one-coordinate distance is computed explicitly: $V(\mathrm{Ber}(p), \mathrm{Po}(p)) = p(1 - e^{-p}) \le p^2$, using only $e^{-p} \ge 1 - p$. For $p_i = \mu/n$ this yields $V(\mathrm{Binomial}(n, \mu/n), \mathrm{Po}(\mu)) \le \mu^2/n$, the law of rare events with an explicit $O(1/n)$ rate.

Solution. For each i, let (X_i, Y_i) be a coupling with $X_i \sim \mathrm{Ber}(p_i)$, $Y_i \sim \mathrm{Po}(p_i)$, and $\mathbb{P}\{X_i \ne Y_i\} = V(\mathrm{Ber}(p_i), \mathrm{Po}(p_i))$, which exists by the coupling characterization of total variation (Exercise 4.5). Take the n pairs independent across i. Then $\sum_{i=1}^n X_i \sim \mathbf{P}$ and, by stability of the Poisson law, $\sum_{i=1}^n Y_i \sim \mathrm{Po}(\sum_i p_i) = \mathrm{Po}(\mu)$. Since $\{\sum_i X_i \ne \sum_i Y_i\} \subseteq \bigcup_{i=1}^n \{X_i \ne Y_i\}$,

$$V(\mathbf{P}, \mathrm{Po}(\mu)) \le \mathbb{P}\Big\{ \sum_{i=1}^n X_i \ne \sum_{i=1}^n Y_i \Big\} \le \sum_{i=1}^n \mathbb{P}\{X_i \ne Y_i\} = \sum_{i=1}^n V(\mathrm{Ber}(p_i), \mathrm{Po}(p_i)).$$

It remains to bound the one-coordinate distance. For discrete laws, $V(P, Q) = \frac{1}{2} \sum_k |P(k) - Q(k)|$. Let $X \sim \mathrm{Ber}(p)$, $Y \sim \mathrm{Po}(p)$. Since $e^{-p} \ge 1 - p$ for all p,

$$\begin{aligned} k = 0 : &\quad |(1 - p) - e^{-p}| = e^{-p} - (1 - p), \\ k = 1 : &\quad |p - p e^{-p}| = p(1 - e^{-p}), \\ k \ge 2 : &\quad |0 - \mathbb{P}\{Y \ge 2\}| = 1 - e^{-p}(1 + p) \ge 0, \end{aligned}$$

the last inequality because $e^p \ge 1 + p$. Summing and halving,

$$V(\mathrm{Ber}(p), \mathrm{Po}(p)) = \tfrac{1}{2}\big[(e^{-p} - 1 + p) + p(1 - e^{-p}) + (1 - e^{-p}(1 + p))\big] = p(1 - e^{-p}) \le p^2,$$

using $1 - e^{-p} \le p$. Combining with the tensorization gives $V(\mathbf{P}, \mathrm{Po}(\mu)) \le \sum_{i=1}^n p_i^2$.

Interpretation. With $p_i = \mu/n$, $\mathbf{P} = \mathrm{Binomial}(n, \mu/n)$ and $V(\mathrm{Binomial}(n, \mu/n), \mathrm{Po}(\mu)) \le \mu^2/n \to 0$ as $n \to \infty$: the sum of many rare independent Bernoulli trials is asymptotically Poisson in total variation, at the explicit rate $O(1/n)$ — the classical law of rare events.

Exercise 4.15

Central idea. Pass from the dependent vector $(X_1, \dots, X_n)$ to an independent Poisson vector via the chain rule: the KL divergence of the joint law Q from the product Poisson law $\mathbf{R} = \bigotimes_i \mathrm{Po}(p_i)$ splits into the sum of the one-coordinate divergences $D(\mathrm{Ber}(p_i)\|\mathrm{Po}(p_i))$ plus the entropy gap $\sum_i H(X_i) - H(X_1, \dots, X_n)$ measuring the dependence. Each one-coordinate term is at most p_i^2, and the data processing inequality (Exercise 4.10) transfers the bound from the vector to its sum S_n, whose image under $\mathbf{R}$ is $\mathrm{Po}(\mu)$ by infinite divisibility.

Solution. Let Q be the joint law of $(X_1, \dots, X_n)$ on $\{0,1\}^n$, with marginals $Q_i = \mathrm{Ber}(p_i)$, and let $\mathbf{R} = \bigotimes_{i=1}^n R_i$ where $R_i = \mathrm{Po}(p_i)$. Since $\mathbf{R}$ charges every point of $\mathbb{N}^n \supset \{0,1\}^n$, $Q \ll \mathbf{R}$ and

$$D(Q\|\mathbf{R}) = \sum_{x \in \{0,1\}^n} Q(x) \ln \frac{Q(x)}{\prod_{i=1}^n R_i(x_i)} = -H(X_1, \dots, X_n) - \sum_{i=1}^n \mathbb{E}_Q[\ln R_i(X_i)],$$

with entropy in nats. Similarly, for each marginal,

$$D(Q_i\|R_i) = \sum_{x_i \in \{0,1\}} Q_i(x_i) \ln \frac{Q_i(x_i)}{R_i(x_i)} = -H(X_i) - \mathbb{E}_{Q_i}[\ln R_i(X_i)].$$

Since $\mathbb{E}_Q[\ln R_i(X_i)] = \mathbb{E}_{Q_i}[\ln R_i(X_i)]$ (the integrand depends only on coordinate i), subtracting gives the chain rule

$$D(Q\|\mathbf{R}) = \sum_{i=1}^n D(Q_i\|R_i) + \Big[\sum_{i=1}^n H(X_i) - H(X_1, \dots, X_n)\Big].$$

Next, the one-coordinate divergence: with $R_i(0) = e^{-p_i}$ and $R_i(1) = p_i e^{-p_i}$,

$$\begin{aligned} D(\mathrm{Ber}(p)\|\mathrm{Po}(p)) &= (1-p)\ln\frac{1-p}{e^{-p}} + p\ln\frac{p}{pe^{-p}} \\ &= (1-p)\ln(1-p) + p(1-p) + p^2. \end{aligned}$$

Since $\ln(1-p) \le -p$ for $p \in [0,1]$, $(1-p)\ln(1-p) \le -p(1-p)$, whence $D(\mathrm{Ber}(p)\|\mathrm{Po}(p)) \le p^2$. (This is the previous exercise, 4-14, in KL form.)

Finally, let $T(x_1, \dots, x_n) = \sum_i x_i$. By the data processing inequality (Exercise 4.10),

$$D(\mathbf{P}\|\mathrm{Po}(\mu)) = D(Q \circ T^{-1}\|\mathbf{R} \circ T^{-1}) \le D(Q\|\mathbf{R}),$$

where $Q \circ T^{-1} = \mathbf{P}$ is the law of S_n, and $\mathbf{R} \circ T^{-1}$ is the law of a sum of independent $\mathrm{Po}(p_i)$ variables, hence $\mathrm{Po}(\mu)$ with $\mu = \sum_i p_i$ by infinite divisibility of the Poisson law. Combining with the chain rule and the one-coordinate bound,

$$D(\mathbf{P}\|\mathrm{Po}(\mu)) \le \sum_{i=1}^n p_i^2 + \sum_{i=1}^n H(X_i) - H(X_1, \dots, X_n),$$

as claimed. The entropy gap is nonnegative by subadditivity of entropy, vanishing when the X_i are independent.

Exercise 4.16

Central idea. (Barthe's transport proof.) Push the uniform distribution on $(0,1)$ forward to f/F and g/G via the quantile maps $x(t), y(t)$. For a.e. t these maps are differentiable with $f(x(t))\,x'(t) = F$ and $g(y(t))\,y'(t) = G$ (Lebesgue differentiation at the point $x(t)$, legitimate because the image of Lebesgue measure under x is the absolutely continuous measure $f\,du/F$). The interpolant $z(t) = (1-\lambda)x(t) + \lambda y(t)$ satisfies $z' \ge (x')^{1-\lambda}(y')^{\lambda}$ by weighted AM–GM, so $h(z(t))z'(t) \ge F^{1-\lambda}G^{\lambda}$ a.e.; the change-of-variables inequality $\int_0^1 h(z)z' \le \int_{\mathbb{R}} h$ for increasing z finishes the proof. No smoothness of f, g and no approximation step is needed. (Note: the hint's second formula should integrate g, not f.)

Solution. Let $F = \int_{\mathbb{R}} f$, $G = \int_{\mathbb{R}} g$. If $F = 0$ or $G = 0$ the right-hand side is 0 (convention $0 \cdot \infty = 0$) and there is nothing to prove.

Reduction to $0 < F, G < \infty$. For $R > 0$ put $f_R = \min(f, R)\,\mathbf{1}_{[-R,R]}$ and $g_R = \min(g, R)\,\mathbf{1}_{[-R,R]}$. Since $0 \le f_R \le f$ and $0 \le g_R \le g$, we have $f_R(x)^{1-\lambda} g_R(y)^{\lambda} \le f(x)^{1-\lambda} g(y)^{\lambda} \le h((1-\lambda)x + \lambda y)$, so (f_R, g_R, h) satisfies the hypothesis. Moreover $\int f_R \le 2R^2 < \infty$, and $\int f_R \uparrow F$, $\int g_R \uparrow G$ by monotone convergence, so $\int f_R, \int g_R > 0$ for R large. If the result holds whenever $0 < F, G < \infty$, then $\int h \ge (\int f_R)^{1-\lambda} (\int g_R)^{\lambda}$ for all large R, and letting $R \to \infty$ gives the claim. Hence assume $0 < F, G < \infty$.

Reduction to Borel h. h is Lebesgue measurable, so there is a Borel h_1 with $h_1 = h$ off a Lebesgue-null set E, and a Borel null set $N \supseteq E$. Put $\tilde{h} = h_1$ on N^c and $\tilde{h} = +\infty$ on N. Then $\tilde{h}$ is Borel, $\tilde{h} \ge h$ everywhere (so the hypothesis holds with $\tilde{h}$ in place of h), and $\int \tilde{h} = \int h$. It therefore suffices to prove $\int \tilde{h} \ge F^{1-\lambda} G^{\lambda}$; we write h for $\tilde{h}$ from now on.

Transport maps. Let $\Phi(u) = \int_{-\infty}^{u} f$, a continuous nondecreasing function with $\Phi(-\infty) = 0$ and $\Phi(+\infty) = F$. For $t \in (0,1)$ let

$$x(t) = \inf\{u : \Phi(u) \ge tF\}, \qquad y(t) = \inf\Big\{u : \int_{-\infty}^{u} g \ge tG\Big\},$$

so that $\int_{-\infty}^{x(t)} f = tF$ and $\int_{-\infty}^{y(t)} g = tG$ (the second defining integral is over g, correcting the hint's typo). Since $tF \in (0, F)$, $x(t)$ is finite; as $\{u : \Phi(u) \ge tF\}$ is a closed up-ray (continuity and monotonicity of Φ), it equals $[x(t), \infty)$. Hence $\Phi(x(t)) = tF$ and

$$x(t) \le u \iff \Phi(u) \ge tF \iff t \le \Phi(u)/F. \tag{1}$$

If $t < s$ then $\Phi(x(t)) = tF < sF = \Phi(x(s))$, so $x(t) < x(s)$: x is strictly increasing. The same holds for y.

Derivatives a.e. By Lebesgue's theorem the monotone functions x, y are differentiable (with finite derivative) at a.e. $t \in (0,1)$. By the Lebesgue differentiation theorem, the set $N_f = \{u : \Phi \text{ is not differentiable at } u \text{ with } \Phi'(u) = f(u)\}$ is Lebesgue-null; let $B \supseteq N_f$ be a Borel null set. By (1), the image of Lebesgue measure on $(0,1)$ under x has distribution function $u \mapsto \Phi(u)/F$, i.e. it is the measure $f(u)\,du/F$. Hence

$$\big|\{t \in (0,1) : x(t) \in N_f\}\big| \le \big|\{t \in (0,1) : x(t) \in B\}\big| = \frac{1}{F}\int_B f = 0.$$

So for a.e. t, x is differentiable at t and Φ is differentiable at $x(t)$ with $\Phi'(x(t)) = f(x(t))$. At such t the one-variable chain rule applied to $\Phi(x(t)) = tF$ gives

$$f(x(t))\,x'(t) = F.$$

In particular $x'(t) > 0$ and $f(x(t)) = F/x'(t) > 0$. Likewise $g(y(t))\,y'(t) = G$ with $y'(t) > 0$ for a.e. t. Let $T \subseteq (0,1)$ be the full-measure set on which both identities hold.

AM–GM. $z = (1-\lambda)x + \lambda y$ is strictly increasing, and for $t \in T$ it is differentiable with

$$z'(t) = (1-\lambda)x'(t) + \lambda y'(t) \;\ge\; x'(t)^{1-\lambda} y'(t)^{\lambda} = \frac{F^{1-\lambda} G^{\lambda}}{f(x(t))^{1-\lambda} g(y(t))^{\lambda}}$$

by weighted AM–GM. With the hypothesis $h(z(t)) \ge f(x(t))^{1-\lambda} g(y(t))^{\lambda}$ this yields, for every $t \in T$,

$$h(z(t))\,z'(t) \;\ge\; F^{1-\lambda} G^{\lambda}. \tag{2}$$

Change of variables for increasing z. We show $\int_0^1 h(z(t))\,z'(t)\,dt \le \int_{\mathbb{R}} h$ for Borel $h \ge 0$. Here z' (defined a.e.) is a nonnegative measurable function, being an a.e. limit of difference quotients. For nondecreasing z and $u < v$ in $(0,1)$, $\int_u^v z' \le z(v) - z(u)$ (Fatou's lemma applied to the difference quotients). Define the Borel measure $\mu(B) = \int_{z^{-1}(B)} z'(t)\,dt$ (countably additive since preimages of disjoint sets are disjoint). For

$a < b$, $I = z^{-1}((a,b])$ is an interval (as z is monotone) with endpoints $s \le s'$; for $s < u < v < s'$ we have $u, v \in I$, so $\int_u^v z' \le z(v) - z(u) \le b - a$, and letting $u \downarrow s$, $v \uparrow s'$ (monotone convergence) gives $\mu((a,b]) \le b - a$. Every open subset of $\mathbb{R}$ is a countable disjoint union of intervals of the form $(a,b]$ (each open interval (α,β) is the disjoint union of $(c_k, c_{k+1}]$ for a sequence c_k increasing from α to β, two-sided if $\alpha = -\infty$), so $\mu(O) \le |O|$ for open O, and by outer regularity of Lebesgue measure $\mu(B) \le |B|$ for every Borel B. Therefore $\int h\,d\mu \le \int h\,dx$ for simple Borel $h \ge 0$ and then, by monotone convergence, for all Borel $h \ge 0$; and $\int h\,d\mu = \int_0^1 h(z(t))z'(t)\,dt$ by the definition of μ (simple h first, then monotone convergence).

Conclusion. Combining with (2), and since T has full measure,

$$\int_{\mathbb{R}} h \;\ge\; \int_0^1 h(z(t))\,z'(t)\,dt \;\ge\; \int_0^1 F^{1-\lambda}G^{\lambda}\,dt \;=\; \left(\int_{\mathbb{R}} f\right)^{1-\lambda}\left(\int_{\mathbb{R}} g\right)^{\lambda}.$$

This is the Prékopa–Leindler inequality on $\mathbb{R}$, proved without Brunn–Minkowski. (When f, g are continuous and strictly positive, x, y are C^1 by the inverse function theorem, as the hint suggests; the argument above treats general measurable f, g directly, so no approximation of f, g is required.) ■

Exercise 4.17

Central idea. The key geometric inclusion $\frac{2}{t+1}(tA)^c + \frac{t-1}{t+1}A \subseteq A^c$ (which uses symmetry to write x/t as a convex combination of $z \in A$ and $-y \in A$) feeds into Brunn–Minkowski; weighted AM–GM then eliminates the dimension n, yielding the dimension-free exponential bound. Convexity is essential (counterexample below); symmetry is needed for the inclusion.

Solution. Write $p = \mathbf{P}\{A\} > 1/2$ and $q = \mathbf{P}\{(tA)^c\}$, where complements are in C. Note $\frac{2}{t+1} + \frac{t-1}{t+1} = 1$.

The inclusion. Let $t > 1$, $x \in (tA)^c = C \setminus tA$, $y \in A$. Since $x, y \in C$ and C is convex, $z := \frac{2}{t+1}x + \frac{t-1}{t+1}y \in C$. Suppose $z \in A$. Then

$$\frac{x}{t} \;=\; \frac{t+1}{2t}\,z \;+\; \frac{t-1}{2t}\,(-y).$$

By symmetry $-y \in A$, and $\frac{t+1}{2t} + \frac{t-1}{2t} = 1$ with both coefficients nonnegative, so convexity of A gives $x/t \in A$, i.e. $x \in tA$ — contradiction. Hence $z \notin A$, so $z \in A^c$, proving

$$\frac{2}{t+1}(tA)^c + \frac{t-1}{t+1}A \;\subseteq\; A^c.$$

Brunn–Minkowski. Both sides are Borel, so with $\mathbf{P}$ uniform on C,

$$(1-p)^{1/n} \;\ge\; \mathbf{P}\big(\tfrac{2}{t+1}(tA)^c + \tfrac{t-1}{t+1}A\big)^{1/n} \;\ge\; \frac{2}{t+1}\,q^{1/n} + \frac{t-1}{t+1}\,p^{1/n}.$$

By weighted AM–GM,

$$(1-p)^{1/n} \;\ge\; \frac{2}{t+1}q^{1/n} + \frac{t-1}{t+1}p^{1/n} \;\ge\; \big(q^{1/n}\big)^{\frac{2}{t+1}}\big(p^{1/n}\big)^{\frac{t-1}{t+1}}.$$

Raising to the power $(t+1)/2$ gives $(1-p)^{(t+1)/2} \ge q\,p^{(t-1)/2}$, i.e.

$$\mathbf{P}\{(tA)^c\} \;\le\; p\left(\frac{1-p}{p}\right)^{(t+1)/2},$$

the dimension n having cancelled. Since $\frac{1-p}{p} < 1$, the right side decays exponentially in t, uniformly in n and C.

On the hypotheses. Convexity cannot be dropped: with $C = [-1,1]$, $A = [-1,-0.4] \cup [0.4,1]$ is symmetric with $p = 0.6$, but for $t = 2$, $(tA)^c = (-0.8, 0.8)$ has $\mathbf{P}\{(tA)^c\} = 0.8 \not\le 0.6\,(2/3)^{3/2} \approx 0.327$. Symmetry is needed for the inclusion: with $C = [-1,1]$, $A = [-0.1,1]$ convex, $t = 2$, $x = -1/2 \in (tA)^c$, $y = 1 \in A$ gives $\frac{2}{3}x + \frac{1}{3}y = 0 \in A \not\subseteq A^c$. ■

Exercise 4.18

Central idea. For $Z = \sum_i y_i X_i$ with $\sigma^2 = \mathbb{E}Z^2 > 0$, take the symmetric convex slab $S = \{x : |y \cdot x| \le 2\sigma\}$ and $A = C \cap S$, so $\mathbf{P}(A) \ge 3/4$ by Chebyshev. A itself need not be symmetric (only $\mathbb{E}X = 0$ is assumed, not $C = -C$), so Borell's lemma (Exercise 4.17) is not quoted but its proof is rerun: the key inclusion $\frac{2}{t+1}(C \setminus tA) + \frac{t-1}{t+1}A \subseteq C \setminus A$ still holds using only the symmetry of S and the fact $0 = \mathbb{E}X \in C$ (so $C \subseteq tC$). Brunn–Minkowski then gives the exponential tail $\mathbf{P}\{|Z| > u\} \le C_0 e^{-cu/\sigma}$ for $u > 2\sigma$, and integrating pu^{p-1} against it yields $\|Z\|_p < 4p\,\sigma$. For special bodies (e.g. the cube) the linear form is sub-Gaussian and the bound tightens to order $\sqrt{p}\,\sigma$.

Solution. Fix $y \in \mathbb{R}^n$ and put $Z = \sum_{i=1}^n y_i X_i = y \cdot X$. Since $\mathbb{E}X = 0$, $\mathbb{E}Z = 0$; write $\sigma^2 = \mathbb{E}Z^2$ (finite, C being bounded) and $\|Z\|_p = (\mathbb{E}|Z|^p)^{1/p}$. If $\sigma = 0$ then $Z = 0$ a.s. and both sides vanish; we read the claim for $\sigma > 0$ (with $\sigma = 0$ the strict inequality degenerates to $0 < 0$, so the statement is meant for nondegenerate Z). Assume $\sigma > 0$.

Step 1: $0 \in C$ and $C \subseteq tC$ for $t \ge 1$. If $0 \notin C$, then since C is compact and convex there is w and $\delta > 0$ with $w \cdot x \ge \delta$ for all $x \in C$ (strict separation), whence $w \cdot \mathbb{E}X \ge \delta > 0$, contradicting $\mathbb{E}X = 0$. So $0 \in C$, and for $t \ge 1$ and $x \in C$, $x/t = \frac{1}{t}x + (1 - \frac{1}{t})\,0 \in C$, i.e. $x \in tC$.

Step 2: the set A. Let $S = \{x \in \mathbb{R}^n : |y \cdot x| \le 2\sigma\}$ (closed, convex, symmetric) and $A = C \cap S$ (compact, convex). By Chebyshev, $\mathbf{P}\{|Z| > 2\sigma\} \le \sigma^2/(4\sigma^2) = 1/4$, so $p_0 := \mathbf{P}(A) = \mathbf{P}\{|Z| \le 2\sigma\} \ge 3/4$. For $t \ge 1$, $tA = tC \cap tS$ and, by Step 1,

$$C \setminus tA = C \setminus tS = \{x \in C : |y \cdot x| > 2t\sigma\}, \qquad \text{so} \qquad \mathbf{P}(C \setminus tA) = \mathbf{P}\{|Z| > 2t\sigma\}.$$

Step 3: the inclusion. Let $t > 1$, $x \in C \setminus tA$ and $a \in A$. By convexity of C, $w := \frac{2}{t+1}x + \frac{t-1}{t+1}a \in C$. Suppose $w \in A$. Then

$$\frac{x}{t} \;=\; \frac{t+1}{2t}\,w \;+\; \frac{t-1}{2t}\,(-a),$$

with nonnegative coefficients summing to 1. Here $w \in A \subseteq S$ and $-a \in S$ (as $a \in S$ and S is symmetric), so convexity of S gives $x/t \in S$, i.e. $x \in tS$. Also $x \in C \subseteq tC$ by Step 1. Thus $x \in tC \cap tS = tA$, a contradiction. Hence $w \in C \setminus A$, proving

$$\tfrac{2}{t+1}\,(C \setminus tA) + \tfrac{t-1}{t+1}\,A \;\subseteq\; C \setminus A. \tag{1}$$

Step 4: Brunn–Minkowski. Fix $t > 1$ and let $q = \mathbf{P}(C \setminus tA)$; if $q = 0$ there is nothing to prove. Otherwise let $K \subseteq C \setminus tA$ be any nonempty compact set (the latter set is Borel and bounded). A is nonempty compact, so $\frac{2}{t+1}K + \frac{t-1}{t+1}A$ is compact and, by (1), contained in $C \setminus A$. Writing $|\cdot|$ for Lebesgue measure and dividing the Brunn–Minkowski inequality by $|C|^{1/n}$,

$$(1 - p_0)^{1/n} \;\ge\; \frac{2}{t+1}\Big(\frac{|K|}{|C|}\Big)^{1/n} + \frac{t-1}{t+1}\,p_0^{1/n} \;\ge\; \Big(\frac{|K|}{|C|}\Big)^{\frac{2}{n(t+1)}} p_0^{\frac{t-1}{n(t+1)}}$$

by weighted AM–GM. Taking the supremum over K (inner regularity) and raising to the power $n(t+1)/2$ gives $(1 - p_0)^{(t+1)/2} \ge q\,p_0^{(t-1)/2}$, i.e.

$$\mathbf{P}\{|Z| > 2t\sigma\} = q \;\le\; \varphi(p_0), \qquad \varphi(p) := p\Big(\frac{1-p}{p}\Big)^{(t+1)/2}.$$

Since $\frac{d}{dp}\log\varphi(p) = \frac{1-t}{2p} - \frac{t+1}{2(1-p)} < 0$ on $(0,1)$ for $t > 1$, φ is decreasing, and $p_0 \ge 3/4$ gives

$$\mathbf{P}\{|Z| > 2t\sigma\} \;\le\; \tfrac{3}{4}\big(\tfrac{1}{3}\big)^{(t+1)/2}, \qquad t > 1.$$

Writing $u = 2t\sigma$, this is the exponential tail

$$\mathbf{P}\{|Z| > u\} \;\le\; C_0\,e^{-cu/\sigma} \quad (u > 2\sigma), \qquad C_0 = \tfrac{3}{4\sqrt{3}} \approx 0.433, \quad c = \tfrac{1}{4}\ln 3 \approx 0.2747,$$

with C_0, c universal (independent of n, C, y).

Step 5: moments. We use the elementary bound $\Gamma(r+1) \le 2\,(2/e)^r r^r$ for $r > 0$: from $\ln s \le s-1$, $\tau^r = (2r)^r(\tau/2r)^r \le (2r)^r e^{\tau/2-r}$, so $\Gamma(r+1) = \int_0^\infty \tau^r e^{-\tau} d\tau \le (2r)^r e^{-r} \int_0^\infty e^{-\tau/2} d\tau = 2(2/e)^r r^r$. Now for $p \ge 2$,

$$\begin{aligned}\mathbb{E}|Z|^p &= \int_0^\infty p u^{p-1}\, \mathbf{P}\{|Z| > u\}\, du \;\le\; \int_0^{2\sigma} p u^{p-1}\, du + C_0 \int_0^\infty p u^{p-1} e^{-cu/\sigma}\, du \\ &= (2\sigma)^p + C_0\,\Gamma(p+1)\,(\sigma/c)^p \;\le\; (2\sigma)^p + 2C_0\,(2/e)^p\, p^p\,(\sigma/c)^p.\end{aligned}$$

With $(a+b)^{1/p} \le a^{1/p} + b^{1/p}$, $(2C_0)^{1/p} \le 1$ (as $2C_0 = \sqrt{3}/2 < 1$) and $2 \le p$,

$$\|Z\|_p \;\le\; 2\sigma + \frac{2}{e\,c}\,p\,\sigma \;\le\; \Big(1 + \frac{2}{e\,c}\Big) p\,\sigma \;\approx\; 3.679\,p\,\sigma \;<\; 4\,p\,\sigma.$$

Hence the inequality holds with the universal constant $\kappa = 4$:

$$\mathbb{E}\Big[\Big|\sum_{i=1}^n y_i X_i\Big|^p\Big]^{1/p} \;<\; 4\,p\,\mathbb{E}\Big[\Big|\sum_{i=1}^n y_i X_i\Big|^2\Big]^{1/2} \qquad (p \ge 2,\ y \in \mathbb{R}^n).$$

Tightening for special bodies. Yes. For the cube $C = [-1,1]^n$ (which has $\mathbb{E}X = 0$), the coordinates X_i are independent and uniform on $[-1,1]$, so $Z = \sum_i y_i X_i$ is a sum of independent centred terms with $y_i X_i \in [-|y_i|, |y_i|]$. With $v = \sum_i y_i^2 = 3\sigma^2$ (as $\mathbb{E}X_i^2 = 1/3$), Hoeffding's inequality gives $\mathbf{P}\{|Z| > u\} \le 2e^{-u^2/(2v)}$, so (substituting $w = u^2/(2v)$)

$$\mathbb{E}|Z|^p \le \int_0^\infty p u^{p-1}\, 2e^{-u^2/(2v)}\, du = 2\,(2v)^{p/2}\,\Gamma(p/2+1) \le 4\,(2v)^{p/2}(2/e)^{p/2}(p/2)^{p/2},$$

using the Gamma bound above with $r = p/2$. Hence, for $p \ge 2$, $\|Z\|_p \le 4^{1/p}\sqrt{6/e}\,\sqrt{p}\,\sigma \le 2\sqrt{6/e}\,\sqrt{p}\,\sigma \approx 2.97\sqrt{p}\,\sigma$: the growth in p improves from linear to $\sqrt{p}$, uniformly in n. ■

Exercise 4.19

Central idea. For boxes the Minkowski sum is coordinatewise: $A+B = \prod_i [a_i + c_i,\, b_i + d_i]$. Normalizing by $\mathrm{Vol}(A+B)^{1/n}$ reduces the claim to $1 \ge \prod_i (\frac{\alpha_i}{\alpha_i+\delta_i})^{1/n} + \prod_i (\frac{\delta_i}{\alpha_i+\delta_i})^{1/n}$, which follows from AM–GM applied to each geometric mean, since the two fractions in each coordinate sum to 1.

Solution. Write $A = \prod_{i=1}^n [a_i, b_i]$ and $B = \prod_{i=1}^n [c_i, d_i]$ with $a_i \le b_i$, $c_i \le d_i$. The Minkowski sum is

$$A + B = \prod_{i=1}^n [a_i + c_i,\; b_i + d_i],$$

since $[a_i, b_i] + [c_i, d_i] = [a_i + c_i,\, b_i + d_i]$ in each coordinate. Put $\alpha_i = b_i - a_i \ge 0$ and $\delta_i = d_i - c_i \ge 0$. Then

$$\mathrm{Vol}(A) = \prod_{i=1}^n \alpha_i, \qquad \mathrm{Vol}(B) = \prod_{i=1}^n \delta_i, \qquad \mathrm{Vol}(A+B) = \prod_{i=1}^n (\alpha_i + \delta_i).$$

If $\alpha_i + \delta_i = 0$ for some i, then $\alpha_i = \delta_i = 0$, so $\mathrm{Vol}(A) = \mathrm{Vol}(B) = \mathrm{Vol}(A+B) = 0$ and the inequality holds trivially. Otherwise all $\alpha_i + \delta_i > 0$; dividing through by $\mathrm{Vol}(A+B)^{1/n}$ the claim becomes

$$1 \;\ge\; \left(\prod_{i=1}^n \frac{\alpha_i}{\alpha_i + \delta_i}\right)^{1/n} + \left(\prod_{i=1}^n \frac{\delta_i}{\alpha_i + \delta_i}\right)^{1/n}.$$

By the arithmetic–geometric mean inequality,

$$\left(\prod_{i=1}^n \frac{\alpha_i}{\alpha_i + \delta_i}\right)^{1/n} \le \frac{1}{n}\sum_{i=1}^n \frac{\alpha_i}{\alpha_i + \delta_i}, \qquad \left(\prod_{i=1}^n \frac{\delta_i}{\alpha_i + \delta_i}\right)^{1/n} \le \frac{1}{n}\sum_{i=1}^n \frac{\delta_i}{\alpha_i + \delta_i}.$$

Adding, the right-hand side is $\frac{1}{n}\sum_{i=1}^n \frac{\alpha_i+\delta_i}{\alpha_i+\delta_i} = 1$, which proves the claim. ■

Chapter 5 — Logarithmic Sobolev Inequalities

Exercise 5.1

Central idea. Interpolate between variance ($p = 1$) and entropy ($p \uparrow 2$) via $\Psi_p(Z) = \|Z\|_2^2 - \|Z\|_p^2$. The normalized ratio $\Psi_p(Z)/((1/p) - (1/2))$ is nondecreasing in p by H"older's log-convexity of L^p norms, which reduces to the scalar inequality $(1-u)\gamma \le 1 - u^\gamma$ for $u \in [0,1]$, $\gamma \in (0,1)$. Its limit as $p \uparrow 2$ is $2\,\mathrm{Ent}(Z^2)$ (l'H$\hat{}$opital), while at $p = 1$ it equals $2\,\mathrm{Var}(Z)$. A Rademacher Z has $Z^2 \equiv 1$, so $\mathrm{Ent}(Z^2) = 0 < 1 = \mathrm{Var}(Z)$, showing nonnegativity is essential.

Solution. Let $Z \ge 0$ with $\mathbb{E}[Z^2] < \infty$ (otherwise there is nothing to prove) and put $\Psi_p(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z^p])^{2/p}$ for $p \in [1,2)$. Note $\Psi_1(Z) = \mathrm{Var}(Z)$.

Monotonicity. Fix $1 \le p < q < 2$ and write $\alpha = 1/p - 1/2 > \beta = 1/q - 1/2 > 0$, $X = \|Z\|_p^2$, $Y = \|Z\|_q^2$, $a = \|Z\|_2^2$. We show $\frac{a-X}{\alpha} \le \frac{a-Y}{\beta}$, i.e. $(a-X)\beta \le (a-Y)\alpha$. By log-convexity of L^p norms on a probability space, with $\frac{1}{q} = \frac{1-\theta}{p} + \frac{\theta}{2}$,

$$\|Z\|_q \le \|Z\|_p^{1-\theta}\|Z\|_2^{\theta}, \qquad \theta = \frac{\alpha-\beta}{\alpha} \in (0,1),$$

so $Y \le X^{\beta/\alpha} a^{(\alpha-\beta)/\alpha}$. It suffices that $(a-X)\beta \le (a - X^{\beta/\alpha}a^{(\alpha-\beta)/\alpha})\alpha$. Dividing by $a > 0$ (if $a = 0$ all terms vanish) and setting $u = X/a \in [0,1]$ (since $\|Z\|_p \le \|Z\|_2$), $\gamma = \beta/\alpha \in (0,1)$, this is $(1-u)\gamma \le 1 - u^\gamma$. Indeed $f(u) := 1 - u^\gamma - \gamma(1-u)$ satisfies $f(1) = 0$ and $f'(u) = \gamma(1 - u^{\gamma-1}) < 0$ on $(0,1)$ (as $\gamma < 1$ gives $u^{\gamma-1} > 1$), hence $f \ge 0$ on $[0,1]$. So $\Psi_p(Z)/((1/p) - (1/2))$ is nondecreasing in p.

Limit. (Note: the hint as stated omits the normalization — $\Psi_p(Z) \to \Psi_2(Z) = 0$ as $p \uparrow 2$. The correct statement, proved here, is for the normalized ratio $\Psi_p(Z)/((1/p)-(1/2))$.) Put $F(p) = \mathbb{E}[Z^p]$. By l'H$\hat{}$opital (justified e.g. when $\mathbb{E}[Z^2(1 + |\log Z|)] < \infty$; the general case follows by truncation),

$$\lim_{p\uparrow 2} \frac{\Psi_p(Z)}{1/p - 1/2} = \lim_{p\to 2} p^2 e^{G(p)} G'(p) = 4\,\mathbb{E}[Z^2]\,G'(2),$$

where $G(p) = \frac{2}{p}\log F(p)$ and $G'(2) = -\frac{1}{2}\log \mathbb{E}[Z^2] + \frac{\mathbb{E}[Z^2 \log Z]}{\mathbb{E}[Z^2]}$. Hence the limit equals $4\mathbb{E}[Z^2 \log Z] - 2\mathbb{E}[Z^2]\log \mathbb{E}[Z^2] = 2\,\mathrm{Ent}(Z^2)$, since $\mathrm{Ent}(Z^2) = 2\mathbb{E}[Z^2 \log Z] - \mathbb{E}[Z^2]\log \mathbb{E}[Z^2]$.

Conclusion. At $p = 1$, $\Psi_1(Z)/((1)-(1/2)) = 2\,\mathrm{Var}(Z)$. Monotonicity gives $2\,\mathrm{Var}(Z) \le \Psi_p(Z)/((1/p)-(1/2))$ for $p \in [1,2)$; letting $p \uparrow 2$ yields $2\,\mathrm{Var}(Z) \le 2\,\mathrm{Ent}(Z^2)$, i.e. $\mathrm{Var}(Z) \le \mathrm{Ent}(Z^2)$.

Nonnegativity is needed. Let Z be Rademacher (± 1 with probability $1/2$). Then $\mathrm{Var}(Z) = 1$ but $Z^2 \equiv 1$ gives $\mathrm{Ent}(Z^2) = 0$. ∎

Exercise 5.2

Central idea. Apply the logarithmic Sobolev inequality to the perturbation $1 + \varepsilon f$ and Taylor-expand in ε: $\mathrm{Ent}((1+\varepsilon f)^2) = 2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3)$ while the Dirichlet form is exactly quadratic, $\mathcal{E}(1+\varepsilon f) = \varepsilon^2 \mathcal{E}(f)$. Dividing by $2\varepsilon^2$ and letting $\varepsilon \to 0$ turns LSI into Poincar'e. The same computation with $\mathbb{E}\|\nabla f\|^2$ in place of $\mathcal{E}(f)$ derives Gaussian Poincar'e from Gaussian LSI.

Solution. Let $m_1 = \mathbb{E}[f]$, $m_2 = \mathbb{E}[f^2]$, so $\mathrm{Var}(f) = m_2 - m_1^2$. For $|\varepsilon|$ small put $Y_\varepsilon = (1+\varepsilon f)^2 = 1 + 2\varepsilon f + \varepsilon^2 f^2 \ge 0$. With $\phi(y) = y\log y = \delta + \delta^2/2 + O(\delta^3)$ near $y = 1 + \delta$,

$$\mathbb{E}[\phi(Y_\varepsilon)] = \mathbb{E}[2\varepsilon f + \varepsilon^2 f^2] + \tfrac{1}{2}\mathbb{E}[(2\varepsilon f)^2] + O(\varepsilon^3) = 2\varepsilon m_1 + 3\varepsilon^2 m_2 + O(\varepsilon^3),$$
$$\phi(\mathbb{E}[Y_\varepsilon]) = \phi(1 + 2\varepsilon m_1 + \varepsilon^2 m_2) = 2\varepsilon m_1 + \varepsilon^2 m_2 + \tfrac{1}{2}(2\varepsilon m_1)^2 + O(\varepsilon^3),$$

using $\mathbb{E}[Y_\varepsilon] = 1 + 2\varepsilon m_1 + \varepsilon^2 m_2$. Subtracting,

$$\mathrm{Ent}(Y_\varepsilon) = \mathbb{E}[\phi(Y_\varepsilon)] - \phi(\mathbb{E}[Y_\varepsilon]) = 2\varepsilon^2(m_2 - m_1^2) + O(\varepsilon^3) = 2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3),$$

valid for bounded f (hence on $\{-1,1\}^n$), and for f with $\mathbb{E}|f|^3 < \infty$ in general.

Hypercube. Theorem 5.1 gives $\mathrm{Ent}(g^2) \le 2\mathcal{E}(g)$ for all g. Take $g = 1 + \varepsilon f$. Since $\mathcal{E}$ is a quadratic form vanishing on constants, $\mathcal{E}(1+\varepsilon f) = \varepsilon^2 \mathcal{E}(f)$. Hence

$$2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3) = \mathrm{Ent}((1+\varepsilon f)^2) \le 2\varepsilon^2 \mathcal{E}(f).$$

Dividing by $2\varepsilon^2$ and letting $\varepsilon \to 0$ yields $\mathrm{Var}(f(X)) \le \mathcal{E}(f)$.

Gaussian. Theorem 5.4 gives $\mathrm{Ent}(g^2) \le 2\,\mathbb{E}\|\nabla g\|^2$. With $g = 1 + \varepsilon f$ (for smooth f with the requisite integrability; the general case follows by approximation), $\mathbb{E}\|\nabla(1+\varepsilon f)\|^2 = \varepsilon^2 \mathbb{E}\|\nabla f\|^2$, so

$$2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3) \le 2\varepsilon^2\, \mathbb{E}\|\nabla f\|^2,$$

and letting $\varepsilon \to 0$ gives $\mathrm{Var}(f) \le \mathbb{E}\|\nabla f\|^2$, the Gaussian Poincar'e inequality (Theorem 3.20). ■

Exercise 5.3

Central idea. A logarithmic Sobolev inequality with constant c linearizes (as in Exercise 5.2) to a Poincar'e inequality with constant $c/2$. But Poincar'e with constant 1 is best possible: the dictator $f(x) = x_1$ satisfies $\mathrm{Var}(f) = \mathcal{E}(f) = 1$. Hence $c/2 \ge 1$, so no $c < 2$ can work in Theorem 5.1.

Solution. Suppose, to the contrary, that $\mathrm{Ent}(g^2) \le c\,\mathcal{E}(g)$ held for all $g : \{-1,1\}^n \to \mathbb{R}$ with some $c < 2$. Applying this to $g = 1 + \varepsilon f$ and using $\mathrm{Ent}((1+\varepsilon f)^2) = 2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3)$ (Exercise 5.2) together with $\mathcal{E}(1+\varepsilon f) = \varepsilon^2 \mathcal{E}(f)$ gives

$$2\varepsilon^2 \mathrm{Var}(f) + O(\varepsilon^3) \le c\,\varepsilon^2 \mathcal{E}(f) \qquad \forall\, f,$$

whence, dividing by ε^2 and letting $\varepsilon \to 0$,

$$\mathrm{Var}(f) \le \frac{c}{2}\,\mathcal{E}(f) \qquad \forall\, f.$$

Take the dictator $f(x) = x_1$. Then $\mathbb{E}[f(X)] = 0$ and $\mathbb{E}[f(X)^2] = 1$, so $\mathrm{Var}(f) = 1$. Moreover, flipping coordinate 1 changes f by $2X_1$ while flipping any other coordinate leaves f unchanged, so with the usual Dirichlet form $\mathcal{E}(f) = \frac{1}{4}\sum_{i=1}^n \mathbb{E}[(f(X) - f(X^{(i)}))^2] = \frac{1}{4}\mathbb{E}[(2X_1)^2] = 1$. Thus $1 = \mathrm{Var}(f) \le \frac{c}{2}\mathcal{E}(f) = \frac{c}{2}$, forcing $c \ge 2$, a contradiction. Hence the constant 2 in Theorem 5.1 cannot be improved. ■

Exercise 5.4

Central idea. By subadditivity of entropy (Theorem 4.10) it suffices to prove the one-dimensional inequality. For strictly positive f (reduction via $|f|$ and $f + \varepsilon$), the duality formula (Theorem 4.13) writes the one-dimensional entropy as a supremum over the curve $pe^u + qe^v = 1$; the desired bound then becomes a uniform 2×2 quadratic-form inequality, i.e. a negative-semidefiniteness condition reducing to three elementary estimates verified by calculus. The constant $c(p)$ is best possible, as testing the third estimate at $s = 1 - p$ forces $c \ge \frac{\log((1-p)/p)}{1-2p}$.

Solution. Let $q = 1 - p$ and assume $0 < p \le \frac{1}{2}$ (the case $p > \frac{1}{2}$ follows by symmetry since $c(1-p) = c(p)$; $p = \frac{1}{2}$ gives $c = 2$). Write $\mathbb{E}$ for expectation under $\mathbb{P}_p^{\otimes n}$ and $\mathcal{E}(f) = pq\,\mathbb{E}[\sum_{i=1}^n (f(X) - f(\bar{X}^{(i)}))^2]$ where $\bar{X}^{(i)}$ flips the i-th coordinate.

By Theorem 4.10, $\mathrm{Ent}(f^2) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Ent}^{(i)}(f^2)]$ with $\mathrm{Ent}^{(i)}$ conditional on $X^{(i)}$. It suffices to show, for each i,

$$\mathrm{Ent}^{(i)}(f^2) \le c(p)pq\,\mathbb{E}^{(i)}[(f - \bar{f}^{(i)})^2], \tag{1}$$

for summing then gives $\mathrm{Ent}(f^2) \le c(p)\mathcal{E}(f)$. Since Ent depends only on f^2 and $(|a| - |b|)^2 \le (a-b)^2$, replacing f by $|f|$ can only decrease $\mathcal{E}$; approximating $f \ge 0$ by $f + \varepsilon > 0$ and letting $\varepsilon \downarrow 0$, it suffices to prove (1) for $f > 0$.

Fix i and $X^{(i)}$, and write $a = f|_{X_i=1} > 0$, $b = f|_{X_i=-1} > 0$. Then (1) is the one-dimensional inequality

$$pa^2 \log a^2 + qb^2 \log b^2 - (pa^2 + qb^2)\log(pa^2 + qb^2) \le c(p)pq(a-b)^2 \tag{$*$}$$

for all $a, b > 0$. By Theorem 4.13 applied to $Y = f^2 \geq 0$ on the two-point space,

$$\mathrm{Ent}_p(f^2) = \max\{pa^2u + qb^2v : pe^u + qe^v = 1\}.$$

Hence (*) follows if, with $c = c(p)$, for every (u, v) with $pe^u + qe^v = 1$ and every $a, b > 0$,

$$pa^2u + qb^2v \leq cpq(a-b)^2. \tag{**}$$

For fixed (u, v), (**) is negative-semidefiniteness of $\left(\begin{smallmatrix} p(u-cq) & cpq \\ cpq & q(v-cp) \end{smallmatrix}\right)$, i.e.

$$\text{(i) } u \leq cq, \qquad \text{(ii) } v \leq cp, \qquad \text{(iii) } (cq-u)(cp-v) \geq c^2pq.$$

(i),(ii): On $pe^u + qe^v = 1$, $\max u = \log(1/p)$ and $\max v = \log(1/q)$. Put $t = q/p \geq 1$, so $c(p) = \frac{(t+1)\log t}{t-1}$. Then (i) is $\log(1+t) \leq \frac{t}{t-1}\log t$, i.e. $N(t) := t\log t - (t-1)\log(t+1) \geq 0$. Since $N(1) = 0$ and $N'(t) = \frac{2}{t+1} - \log(1+\frac{1}{t}) \geq 0$ (as $\frac{2s}{1+s} \geq \log(1+s)$ for $s = \frac{1}{t} \in (0,1]$, since $\phi(s) = \frac{2s}{1+s} - \log(1+s)$ satisfies $\phi(0) = 0$, $\phi'(s) = \frac{1-s}{(1+s)^2} \geq 0$), (i) holds. Similarly (ii) is $\log(1+\frac{1}{t}) \leq \frac{\log t}{t-1}$, i.e. $\psi(t) := \log t - (t-1)\log(1+\frac{1}{t}) \geq 0$, with $\psi(1) = 0$, $\psi'(t) = \frac{2}{t+1} - \log(1+\frac{1}{t}) \geq 0$.

(iii): Parametrize $s = pe^u \in (0,1)$, so $u = \log(s/p)$, $v = \log((1-s)/q)$, and set $H(s) = (cq - \log(s/p))(cp - \log((1-s)/q))$; we need $H(s) \geq c^2pq$. Now $H(p) = c^2pq$ and $H'(s) = K(s)/(s(1-s))$ where

$$K(s) = s(cq - \log(s/p)) - (1-s)(cp - \log((1-s)/q)).$$

Differentiating,

$$K'(s) = c - 2 - \log\frac{s(1-s)}{pq}, \qquad K''(s) = \frac{2s-1}{s(1-s)},$$

so K' strictly decreases on $(0, \frac{1}{2}]$ and strictly increases on $[\frac{1}{2}, 1)$; K has at most one critical point in each, hence at most three distinct zeros in $(0,1)$ by Rolle's theorem. For $p < \frac{1}{2}$ there are three: $K(p) = pcq - qcp = 0$; $K(\frac{1}{2}) = \frac{1}{2}[c(q-p) - \log(q/p)] = 0$ since $c(q-p) = \log(q/p)$; and with $L = \log((1-p)/p)$,

$$K(1-p) = (1-p)(cq - L) - p(cp + L) = c(q-p) - L = 0.$$

These are therefore all the zeros of K on $(0,1)$. Since $\lim_{s\downarrow 0} K(s) = -(cp + \log q) < 0$ (as $cp + \log q \geq 2p + \log(1-p) > 0$ on $(0, \frac{1}{2})$) and K has no zero in $(0,p)$, $K < 0$ there. Moreover $K'(p) = K'(1-p) = c - 2 > 0$ for $p < \frac{1}{2}$ (as $c(p) = \frac{\log(q/p)}{q-p} > 2$, from $\log t > \frac{2(t-1)}{t+1}$ for $t = q/p > 1$), so $K > 0$ on $(p, \frac{1}{2})$ and on $(1-p, 1)$, and $K < 0$ on $(\frac{1}{2}, 1-p)$. Thus H strictly decreases on $(0,p)$, strictly increases on $(p, \frac{1}{2})$, strictly decreases on $(\frac{1}{2}, 1-p)$, and strictly increases on $(1-p, 1)$; its minima on $(0,1)$ are at p and $1-p$, where $H(p) = c^2pq$ and $H(1-p) = (cq-L)(cp+L) = c^2pq + L[c(q-p) - L] = c^2pq$. Hence $H(s) \geq c^2pq$ on $(0,1)$, which is (iii). For $p = \frac{1}{2}$ ($c = 2$), $K'(s) = -\log(4s(1-s)) \geq 0$ with equality only at $s = \frac{1}{2}$, so K is strictly increasing, $K < 0$ on $(0, \frac{1}{2})$ and $K > 0$ on $(\frac{1}{2}, 1)$; H decreases then increases with minimum $H(\frac{1}{2}) = 1 = c^2pq$.

Hence (**) holds, proving (*) and therefore Theorem 5.2.

Optimality. If (*) holds with constant $c > 0$, then by the duality equivalence above, (iii) holds with c. Take $s_0 = 1 - p$ (i.e. $u = \log((1-p)/p)$, $v = -\log((1-p)/p)$). Then (iii) gives

$$[c(1-p) - L][cp + L] \geq c^2p(1-p), \qquad L := \log\frac{1-p}{p} > 0,$$

i.e. $cL(1-2p) - L^2 \geq 0$, so $c \geq \frac{L}{1-2p} = c(p)$. Thus no smaller constant works. ■

Exercise 5.5

Central idea. This is the entropy-method (Herbst) proof of Theorem 5.3, but the full-square hypothesis lets us avoid the positive part. The key is the elementary inequality $(e^{z/2} - e^{y/2})^2 \leq \frac{(z-y)^2}{8}(e^z + e^y)$ (proved

via Cauchy–Schwarz on $\int_y^z \frac{1}{2}e^{s/2}\,ds$ plus $\tanh(u/2) \le u/2$), applied to $z = \lambda f(x)$, $y = \lambda f(x^{(i)})$ inside the discrete log-Sobolev inequality; the $e^{\lambda f(x)} + e^{\lambda f(x^{(i)})}$ symmetrizes to $2e^{\lambda f(x)}$ since $X^{(i)} \overset{d}{=} X$, closing the Herbst differential inequality $\frac{d}{d\lambda}(H(\lambda)/\lambda) \le v/8$.

Solution. Write $f_i(x) = f(x^{(i)})$ for the i-th coordinate flip, and $Z = f(X)$ with X uniform on $\{-1,1\}^n$.

Step 1: the elementary inequality. Both sides of

$$(e^{z/2} - e^{y/2})^2 \le \frac{(z-y)^2}{8}(e^z + e^y) \tag{1}$$

are symmetric in z, y, so assume $z \ge y$. By the fundamental theorem of calculus and Cauchy–Schwarz,

$$(e^{z/2} - e^{y/2})^2 = \left(\int_y^z \tfrac{1}{2}e^{s/2}\,ds\right)^2 \le (z-y)\int_y^z \tfrac{1}{4}e^s\,ds = \frac{z-y}{4}(e^z - e^y).$$

It remains to show $\frac{z-y}{4}(e^z - e^y) \le \frac{(z-y)^2}{8}(e^z + e^y)$. For $z = y$ both sides vanish; for $z > y$, dividing by $\frac{z-y}{4}$ and setting $u = z - y > 0$ gives the equivalent $\frac{e^u - 1}{e^u + 1} \le \frac{u}{2}$, i.e. $\tanh(u/2) \le u/2$. This holds since $\tanh 0 = 0$ and $\frac{d}{dv}\tanh v = \operatorname{sech}^2 v \le 1$ for $v \ge 0$. This proves (1).

Step 2: the entropy bound. Apply the discrete log-Sobolev inequality (Theorem 5.1) to $g = e^{\lambda f/2}$, $\lambda > 0$:

$$\operatorname{Ent}(e^{\lambda f(X)}) \le \tfrac{1}{2}\sum_{i=1}^n \mathbb{E}\left[\left(e^{\lambda f(X)/2} - e^{\lambda f_i(X)/2}\right)^2\right].$$

By (1) with $z = \lambda f(x)$, $y = \lambda f_i(x)$,

$$\left(e^{\lambda f(x)/2} - e^{\lambda f_i(x)/2}\right)^2 \le \frac{\lambda^2}{8}(f(x) - f_i(x))^2\left(e^{\lambda f(x)} + e^{\lambda f_i(x)}\right).$$

Now $X^{(i)} \overset{d}{=} X$ (a coordinate flip preserves the uniform law), and $(f(x) - f_i(x))^2$ is symmetric under the flip, so

$$\mathbb{E}\left[(f(X) - f_i(X))^2 e^{\lambda f_i(X)}\right] = \mathbb{E}\left[(f(X) - f_i(X))^2 e^{\lambda f(X)}\right].$$

Hence

$$\mathbb{E}\left[\left(e^{\lambda f(X)/2} - e^{\lambda f_i(X)/2}\right)^2\right] \le \frac{\lambda^2}{4}\,\mathbb{E}\left[(f(X) - f_i(X))^2 e^{\lambda f(X)}\right],$$

and summing over i,

$$\operatorname{Ent}(e^{\lambda f(X)}) \le \frac{\lambda^2}{8}\,\mathbb{E}\left[e^{\lambda f(X)}\sum_{i=1}^n (f(X) - f_i(X))^2\right] \le \frac{\lambda^2 v}{8}\,\mathbb{E}[e^{\lambda f(X)}], \tag{2}$$

using the hypothesis $\sum_i (f(x) - f_i(x))^2 \le v$ for all x. This is exactly the hint's chain.

Step 3: Herbst's argument. Let $M(\lambda) = \mathbb{E}e^{\lambda f(X)}$, $K(\lambda) = \log M(\lambda)$, and $H(\lambda) = K(\lambda) - \lambda\mathbb{E}Z = \log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)}$. Then

$$\operatorname{Ent}(e^{\lambda f(X)}) = \mathbb{E}[\lambda f e^{\lambda f}] - M(\lambda)K(\lambda) = M(\lambda)(\lambda K'(\lambda) - K(\lambda)).$$

With (2), $\lambda K'(\lambda) - K(\lambda) \le \lambda^2 v/8$. Since $\frac{d}{d\lambda}\frac{H(\lambda)}{\lambda} = \frac{\lambda K'(\lambda) - K(\lambda)}{\lambda^2}$ (the $\mathbb{E}Z$ terms cancel),

$$\frac{d}{d\lambda}\frac{H(\lambda)}{\lambda} \le \frac{v}{8}, \qquad \lambda > 0.$$

Integrating from ε to λ and letting $\varepsilon \downarrow 0$ (using $H(\varepsilon)/\varepsilon \to H'(0) = 0$) gives $\frac{H(\lambda)}{\lambda} \le \frac{v\lambda}{8}$, i.e.

$$\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \le \frac{v\lambda^2}{8}, \qquad \lambda > 0. \tag{3}$$

Step 4: Chernoff. For $t > 0$ and any $\lambda > 0$,

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{-\lambda t}\,e^{H(\lambda)} \le \exp\!\left(-\lambda t + \tfrac{v\lambda^2}{8}\right).$$

Optimizing at $\lambda = 4t/v$ yields $\exp(-2t^2/v)$, as claimed. ■

Exercise 5.6

Central idea. Apply the asymmetric log-Sobolev inequality (Theorem 5.2) to $e^{\lambda f/2}$ and use $(e^u - e^v)^2 \le (u-v)_+^2 e^{2u} + (u-v)_-^2 e^{2v}$. For nondecreasing f, conditioning on $X^{(i)}$ shows the two resulting terms combine to $(f(X^{(i)},1) - f(X^{(i)},-1))^2 e^{\lambda f(X^{(i)},1)}$; rewriting via $\mathbf{1}_{\{X_i=1\}}$ and using the hypothesis gives a factor $(1-p)$ after the $p(1-p)$ from the Dirichlet form cancels. Herbst's argument then yields the tail bound.

Solution. Theorem 5.2 states $\mathrm{Ent}_\mu(g^2) \le c(p)\mathcal{E}(g)$ with $\mathcal{E}(g) = p(1-p)\mathbb{E}[\sum_{i=1}^n (g(X) - g(\bar X^{(i)}))^2]$ and $c(p) = \frac{1}{1-2p}\log\frac{1-p}{p}$. Take $g = e^{\lambda f/2}$ ($\lambda > 0$), so $\mathrm{Ent}_\mu(e^{\lambda f}) \le c(p)p(1-p)\mathbb{E}[\sum_i (e^{\lambda f(X)/2} - e^{\lambda f(\bar X^{(i)})/2})^2]$.

For $u, v \in \mathbb{R}$, $(e^u - e^v)^2 \le (u-v)_+^2 e^{2u} + (u-v)_-^2 e^{2v}$ (if $u \ge v$, $(e^u - e^v)^2 = e^{2v}(e^{u-v}-1)^2 \le (u-v)^2 e^{2u}$ since $e^w - 1 \le we^w$ for $w \ge 0$; symmetrically if $u < v$). With $u = \lambda f(X)/2$, $v = \lambda f(\bar X^{(i)})/2$,

$$\mathbb{E}\Big[\sum_i (e^{\lambda f/2} - e^{\lambda f^{(i)}/2})^2\Big] \le \frac{\lambda^2}{4}\sum_i \mathbb{E}[(f - f^{(i)})_+^2 e^{\lambda f} + (f - f^{(i)})_-^2 e^{\lambda f^{(i)}}],$$

writing $f^{(i)} = f(\bar X^{(i)})$.

Let f be nondecreasing. Condition on $X^{(i)}$ and set $a = f(X^{(i)},1) \ge b = f(X^{(i)},-1)$. Then $\mathbb{E}[(f - f^{(i)})_+^2 e^{\lambda f} \mid X^{(i)}] = p(a-b)^2 e^{\lambda a}$ (only $X_i = 1$ contributes) and $\mathbb{E}[(f - f^{(i)})_-^2 e^{\lambda f^{(i)}} \mid X^{(i)}] = (1-p)(a-b)^2 e^{\lambda a}$ (only $X_i = -1$ contributes, where $(f - f^{(i)})_- = a - b$ and $f^{(i)} = a$). Hence

$$\sum_i \mathbb{E}[(e^{\lambda f/2} - e^{\lambda f^{(i)}/2})^2] \le \frac{\lambda^2}{4}\sum_i \mathbb{E}[(f(X^{(i)},1) - f(X^{(i)},-1))^2 e^{\lambda f(X^{(i)},1)}].$$

Since $\mathbb{E}_{X^{(i)}}[h] = \frac{1}{p}\mathbb{E}[\mathbf{1}_{\{X_i=1\}}h]$ and on $\{X_i = 1\}$, $(f(X^{(i)},1) - f(X^{(i)},-1))^2 = (f(X) - f(\bar X^{(i)}))^2 = (f(X) - f(\bar X^{(i)}))_+^2$ (as f is nondecreasing),

$$\sum_i \mathbb{E}[\cdots] = \frac{1}{p}\,\mathbb{E}\Big[e^{\lambda f(X)}\sum_i (f(X) - f(\bar X^{(i)}))_+^2\Big] \le \frac{v}{p}\,\mathbb{E}e^{\lambda f}$$

by the hypothesis. Therefore

$$\mathrm{Ent}(e^{\lambda f}) \le c(p)p(1-p)\cdot\frac{\lambda^2}{4}\cdot\frac{v}{p}\,\mathbb{E}e^{\lambda f} = \frac{(1-p)c(p)v\lambda^2}{4}\,\mathbb{E}e^{\lambda f}.$$

Herbst's argument: with $H(\lambda) = \log \mathbb{E}e^{\lambda(f-\mathbb{E}f)}$, $\lambda H' - H = \mathrm{Ent}(e^{\lambda(f-\mathbb{E}f)})/\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \le \frac{(1-p)c(p)v\lambda^2}{4}$, so $(H/\lambda)' \le \frac{(1-p)c(p)v}{4}$ and $H(\lambda) \le \frac{(1-p)c(p)v\lambda^2}{4}$. Chernoff's bound gives $\mathbb{P}\{f - \mathbb{E}f > t\} \le \exp(-\lambda t + \frac{(1-p)c(p)v\lambda^2}{4})$; optimizing $\lambda = \frac{2t}{(1-p)c(p)v}$ yields $\exp(\frac{-t^2}{(1-p)c(p)v})$.

If f is nonincreasing, then with $a = f(X^{(i)},1) \le b = f(X^{(i)},-1)$ the same computation gives $\mathbb{E}[(f - f^{(i)})_+^2 e^{\lambda f} + (f - f^{(i)})_-^2 e^{\lambda f^{(i)}} \mid X^{(i)}] = (b-a)^2 e^{\lambda b}$, and $\mathbb{E}_{X^{(i)}}[h] = \frac{1}{1-p}\mathbb{E}[\mathbf{1}_{\{X_i=-1\}}h]$ with $(f(X) - f(\bar X^{(i)}))^2 = (f(X) - f(\bar X^{(i)}))_+^2$ on $\{X_i = -1\}$. Hence $\mathrm{Ent}(e^{\lambda f}) \le \frac{pc(p)v\lambda^2}{4}\mathbb{E}e^{\lambda f}$, and Herbst gives $\exp(\frac{-t^2}{pc(p)v})$. ■

Exercise 5.7

Central idea. Adapt the proof of Lemma 5.19 (rather than its statement): for $s \le t$, write $Z_t = e^{-(t-s)}Z_s + (1 - e^{-(t-s)})\alpha_0$ and dominate $\|Z_t\|$ by the scalar $Y_t = e^{-(t-s)}\|Z_s\| + (1 - e^{-(t-s)})\mathbb{E}\|Z_s\| \ge 0$, which satisfies $dY_t/dt = \mathbb{E}Y_t - Y_t$. Since Y_t is a scalar function of the Rademacher X, the differentiation/entropy argument in Lemma 5.19's proof applies verbatim, giving monotonicity of $\frac{1}{q(t)}\log \mathbb{E}Y_t^{q(t)}$, hence of $\frac{1}{q(t)}\log\mathbb{E}\|Z_t\|^{q(t)}$.

Solution. Fix $0 \le s < t$ and write $Z_u = \alpha_0 + e^{-u}\alpha_1 X$ for $u \ge s$. Then $Z_t = e^{-(t-s)}Z_s + (1 - e^{-(t-s)})\alpha_0$, since $\mathbb{E}Z_s = \alpha_0$. By the triangle inequality and Jensen ($\|\mathbb{E}Z_s\| \le \mathbb{E}\|Z_s\|$),

$$\|Z_t\| \le e^{-(t-s)}\|Z_s\| + (1 - e^{-(t-s)})\|\alpha_0\| \le e^{-(t-s)}\|Z_s\| + (1 - e^{-(t-s)})\mathbb{E}\|Z_s\| =: Y_t.$$

Put $A = \|Z_s\| \geq 0$ (a scalar function of X) and $b = \mathbb{E}A$, so $Y_u = e^{-(u-s)}A + (1 - e^{-(u-s)})b$ for $u \geq s$. Then $Y_u \geq 0$, $Y_s = A = \|Z_s\|$, $\|Z_t\| \leq Y_t$, and differentiating,

$$\frac{dY_u}{du} = -e^{-(u-s)}A + e^{-(u-s)}b = e^{-(u-s)}(b - A) = \mathbb{E}Y_u - Y_u,$$

since $\mathbb{E}Y_u = b$. Thus Y_u is a scalar nonnegative function of the Rademacher variable X satisfying the same differential relation as in Lemma 5.19's proof. The differentiation there gives, with $q = q(u)$,

$$\frac{d}{du}\Big(\frac{1}{q}\log \mathbb{E}Y_u^q\Big) = \frac{q'}{q^2 \mathbb{E}Y_u^q}\Big(\mathrm{Ent}(Y_u^q) + \frac{q^2}{q'}\mathbb{E}[Y_u^{q-1}(\mathbb{E}Y_u - Y_u)]\Big),$$

and the log-Sobolev estimate in that proof (which uses only that $Y_u \geq 0$ is a function of X and $q(u) = \beta e^{2u}+1$) shows the right-hand side is ≤ 0. Hence $\frac{1}{q(u)}\log \mathbb{E}Y_u^{q(u)}$ is nonincreasing for $u \geq s$. Consequently,

$$\frac{1}{q(t)}\log \mathbb{E}\|Z_t\|^{q(t)} \leq \frac{1}{q(t)}\log \mathbb{E}Y_t^{q(t)} \leq \frac{1}{q(s)}\log \mathbb{E}Y_s^{q(s)} = \frac{1}{q(s)}\log \mathbb{E}\|Z_s\|^{q(s)},$$

i.e.

$$\big(\mathbb{E}\|Z_t\|^{q(t)}\big)^{1/q(t)} \leq \big(\mathbb{E}\|Z_s\|^{q(s)}\big)^{1/q(s)}.$$

■

Remark on the hint. The hint's identities are misstated: in fact $\alpha_0 + e^{-t}\alpha_1 = v(1 + e^{-t})/2 + w(1 - e^{-t})/2$ and $\alpha_0 - e^{-t}\alpha_1 = v(1 - e^{-t})/2 + w(1 + e^{-t})/2$ (the hint has e^t where e^{-t} belongs), $|Z_t|$ should be $\|Z_t\|$, and β_1 should be $(\|v\| - \|w\|)/2$. Rather than repair the hint, the proof above adapts the argument of Lemma 5.19 directly to the dominating scalar process Y_t.

Exercise 5.8

Central idea. Expand $S^4 = (\sum_i b_i X_i)^4$: only even-multiplicity index patterns survive, giving $\mathbb{E}S^4 = 3(\mathbb{E}S^2)^2 - 2\sum_i b_i^4 \leq 3(\mathbb{E}S^2)^2$. Hölder with exponents $3/2, 3$ on $Z^2 = Z^{2/3}Z^{4/3}$ yields $\mathbb{E}Z^2 \leq (\mathbb{E}Z)^{2/3}(\mathbb{E}Z^4)^{1/3}$; combining and cancelling $(\mathbb{E}Z^2)^{2/3}$ gives $\mathbb{E}Z^2 \leq 3(\mathbb{E}Z)^2$. The constant 3 in the fourth-moment bound is best possible (attained asymptotically by $b_i \equiv 1$), though Szarek's inequality improves the final constant to 2.

Solution. Write $S = \sum_{i=1}^n b_i X_i$, so $Z = |S|$, $Z^2 = S^2$, $Z^4 = S^4$. Since $\mathbb{E}[X_i X_j] = \delta_{ij}$, $\mathbb{E}S^2 = \sum_i b_i^2$.

Expanding $S^4 = \sum_{i,j,k,l} b_i b_j b_k b_l X_i X_j X_k X_l$, $\mathbb{E}[X_i X_j X_k X_l] \neq 0$ only when each index has even multiplicity: (i) $i = j = k = l$, contributing $\sum_i b_i^4$; (ii) two distinct pairs, e.g. $i = j \neq k = l$, of which there are 3 pairings, contributing $3\sum_{i \neq k} b_i^2 b_k^2$. Hence

$$\mathbb{E}S^4 = \sum_i b_i^4 + 3\sum_{i \neq k} b_i^2 b_k^2 = 3\Big(\sum_i b_i^2\Big)^2 - 2\sum_i b_i^4 \leq 3(\mathbb{E}S^2)^2,$$

i.e. $\mathbb{E}[Z^4] \leq 3(\mathbb{E}[Z^2])^2$.

By Hölder with $p = 3/2$, $q = 3$ applied to $Z^2 = Z^{2/3} \cdot Z^{4/3}$,

$$\mathbb{E}[Z^2] \leq \big(\mathbb{E}[(Z^{2/3})^{3/2}]\big)^{2/3}\big(\mathbb{E}[(Z^{4/3})^3]\big)^{1/3} = (\mathbb{E}Z)^{2/3}(\mathbb{E}[Z^4])^{1/3}.$$

Combining,

$$\mathbb{E}[Z^2] \leq (\mathbb{E}Z)^{2/3}\big(3(\mathbb{E}[Z^2])^2\big)^{1/3} = 3^{1/3}(\mathbb{E}Z)^{2/3}(\mathbb{E}[Z^2])^{2/3}.$$

If $\mathbb{E}[Z^2] = 0$ there is nothing to prove; otherwise dividing by $(\mathbb{E}[Z^2])^{2/3}$ and cubing gives $\mathbb{E}[Z^2] \leq 3(\mathbb{E}Z)^2$.

The constant 3 in $\mathbb{E}[Z^4] \leq 3(\mathbb{E}[Z^2])^2$ cannot be improved: with $b_1 = \cdots = b_n = 1$, $\mathbb{E}S^4/(\mathbb{E}S^2)^2 = 3 - 2/n \to 3$. However, the resulting $(\mathbb{E}Z^2)^{1/2} \leq \sqrt{3}\,\mathbb{E}Z$ is not sharp: Szarek's inequality (Theorem 5.20) gives the optimal $(\mathbb{E}Z^2)^{1/2} \leq \sqrt{2}\,\mathbb{E}Z$. ■

Exercise 5.9

Central idea. The first inequality is Theorem 5.20 (Szarek's sharp Rademacher lower bound $\mathbb{E}_X|\sum a_iX_i| \geq \|a\|_2/\sqrt{2}$) applied conditionally on Y. The second inequality is false as stated — $Y_1 \equiv 1$, $Y_2 \equiv -1$ gives $\sqrt{2} \leq 0$ — and needs $\mathbb{E}Y_i = 0$. Under centering, symmetrization does it directly: Jensen over an independent copy Y' gives $\mathbb{E}|\sum X_iY_i| \leq \mathbb{E}|\sum X_i(Y_i - Y_i')|$; since each $Y_i - Y_i'$ is symmetric, the signs X_i can be removed, and the triangle inequality gives $\mathbb{E}|\sum X_iY_i| \leq 2\,\mathbb{E}|\sum Y_i|$, which is the claim after multiplying by $\sqrt{2}$.

Solution. Assume the X_i are independent of the Y_i. Write $\mathbb{E}_X$ for expectation over X only.

First inequality. Condition on $Y = (Y_1, \dots, Y_n)$. Theorem 5.20 (Szarek: $\sqrt{\mathbb{E}Z^2} \leq \sqrt{2}\,\mathbb{E}Z$ for $Z = |\sum_i b_iX_i|$) applied to the Rademacher sum with frozen coefficients $b_i = Y_i$, together with $\mathbb{E}_X(\sum_i X_iY_i)^2 = \sum_i Y_i^2$, gives

$$\mathbb{E}_X\Big[\Big|\sum_{i=1}^n X_iY_i\Big|\Big] \geq \frac{1}{\sqrt{2}}\Big(\sum_{i=1}^n Y_i^2\Big)^{1/2} \quad \text{a.s.}$$

Taking $\mathbb{E}_Y$ yields $\mathbb{E}|\sum_i X_iY_i| \geq \frac{1}{\sqrt{2}}\mathbb{E}[(\sum_i Y_i^2)^{1/2}]$, i.e.

$$\mathbb{E}\Big[\Big(\sum_{i=1}^n Y_i^2\Big)^{1/2}\Big] \leq \sqrt{2}\,\mathbb{E}\Big[\Big|\sum_{i=1}^n X_iY_i\Big|\Big].$$

Remark on the second inequality. As stated it is false without centering: for $Y_1 \equiv 1$, $Y_2 \equiv -1$ (independent, finite variance), $\sqrt{2}\,\mathbb{E}|X_1 - X_2| = \sqrt{2}$ (as $|X_1 - X_2|$ equals 0 or 2 with probability 1/2 each) while $2\sqrt{2}\,\mathbb{E}|Y_1 + Y_2| = 0$. Assume henceforth $\mathbb{E}Y_i = 0$ for all i; we prove the second inequality in this corrected form.

Second inequality (centered case). All expectations below are finite since the Y_i have finite variance. Let $Y_1', \dots, Y_n'$ be independent copies of $Y_1, \dots, Y_n$, independent of everything else, and put $Z_i = Y_i - Y_i'$.

(a) *Jensen.* By centering, $\sum_i X_iY_i = \mathbb{E}_{Y'}[\sum_i X_i(Y_i - Y_i')]$ (conditionally on X, Y). Jensen's inequality in Y' gives $|\sum_i X_iY_i| \leq \mathbb{E}_{Y'}|\sum_i X_iZ_i|$, and taking $\mathbb{E}_{X,Y}$,

$$\mathbb{E}\Big|\sum_i X_iY_i\Big| \leq \mathbb{E}\Big|\sum_i X_iZ_i\Big|.$$

(b) *Removing the signs.* Each Z_i is symmetric ($Y_i - Y_i' \overset{d}{=} Y_i' - Y_i$) and independent of X_i, so $X_iZ_i \overset{d}{=} Z_i$: conditionally on $X_i = \pm 1$, X_iZ_i is $\pm Z_i$, which has the law of Z_i. The pairs (X_i, Z_i), $i = 1, \dots, n$, are independent, hence the vector $(X_1Z_1, \dots, X_nZ_n)$ has the same law as $(Z_1, \dots, Z_n)$, and

$$\mathbb{E}\Big|\sum_i X_iZ_i\Big| = \mathbb{E}\Big|\sum_i Z_i\Big| = \mathbb{E}\Big|\sum_i Y_i - \sum_i Y_i'\Big| \leq \mathbb{E}\Big|\sum_i Y_i\Big| + \mathbb{E}\Big|\sum_i Y_i'\Big| = 2\,\mathbb{E}\Big|\sum_i Y_i\Big|,$$

by the triangle inequality and $Y' \overset{d}{=} Y$.

Combining (a) and (b), $\mathbb{E}|\sum_i X_iY_i| \leq 2\,\mathbb{E}|\sum_i Y_i|$, and multiplying by $\sqrt{2}$,

$$\sqrt{2}\,\mathbb{E}\Big[\Big|\sum_{i=1}^n X_iY_i\Big|\Big] \leq 2\sqrt{2}\,\mathbb{E}\Big[\Big|\sum_{i=1}^n Y_i\Big|\Big] \qquad (\mathbb{E}Y_i = 0).$$

Together with the first inequality this proves Marcinkiewicz's inequalities for centred Y_i. ■

Exercise 5.10

Central idea. Induction on n: writing $S_n = S_{n-1} + \varepsilon_n\alpha_n$ and averaging over $\varepsilon_n = \pm 1$ kills odd powers in the binomial expansion, leaving a recurrence in the even moments. The induction closes via the combinatorial inequality $\binom{2p}{2j}(2j-1)!! \leq (2p-1)!!\binom{p}{j}$, i.e. $2^k k!/(2k)! \leq 1$. Optimality: with $\alpha_i = 1/\sqrt{n}$ the CLT gives $\mathbb{E}|S_n|^{2p} \to \mathbb{E}|G|^{2p} = (2p)!/(2^p p!)$.

Solution. Let $S_n = \sum_{i=1}^n \varepsilon_i \alpha_i$ and $M_p^{(n)} = \mathbb{E}|S_n|^{2p}$. Since $|S_n|^{2p} = S_n^{2p}$,

$$\mathbb{E}|S_n|^{2p} = \tfrac{1}{2}\mathbb{E}[(S_{n-1}+\alpha_n)^{2p}] + \tfrac{1}{2}\mathbb{E}[(S_{n-1}-\alpha_n)^{2p}].$$

Binomial expansion gives $(S_{n-1}\pm\alpha_n)^{2p} = \sum_{k=0}^{2p}\binom{2p}{k} S_{n-1}^k (\pm\alpha_n)^{2p-k}$; averaging over the signs, odd powers of $\pm\alpha_n$ cancel:

$$M_p^{(n)} = \sum_{j=0}^{p} \binom{2p}{2j} M_j^{(n-1)}\, \alpha_n^{2p-2j}, \qquad M_0^{(n)} = 1.$$

We prove $M_p^{(n)} \le (2p-1)!!\,(\sum_{i=1}^n \alpha_i^2)^p$ by induction on n, where $(2p-1)!! = (2p)!/(2^p p!)$. For $n=1$, $M_p^{(1)} = |\alpha_1|^{2p} \le (2p-1)!!\,|\alpha_1|^{2p}$ since $(2p-1)!! \ge 1$. Let $A = \sum_{i=1}^{n-1}\alpha_i^2$ and assume $M_j^{(n-1)} \le (2j-1)!!\,A^j$ for $j \le p$. Then

$$M_p^{(n)} \le \sum_{j=0}^{p}\binom{2p}{2j}(2j-1)!!\,A^j \alpha_n^{2p-2j}.$$

Since $(A+\alpha_n^2)^p = \sum_{j=0}^p \binom{p}{j} A^j \alpha_n^{2(p-j)}$, it suffices that $\binom{2p}{2j}(2j-1)!! \le (2p-1)!!\binom{p}{j}$ for all j. Indeed,

$$\frac{\binom{2p}{2j}(2j-1)!!}{(2p-1)!!\binom{p}{j}} = \frac{(2p)!/(2p-2j)!\,2^j j!}{(2p)!/2^p j!(p-j)!} = \frac{2^{p-j}(p-j)!}{(2(p-j))!} = \frac{1}{(2k-1)!!} \le 1,$$

with $k = p-j$ and $(2j)!/(2^j j!) = (2j-1)!!$. This completes the induction.

Optimality. Take $\alpha_i = 1/\sqrt{n}$, so $\sum_i \alpha_i^2 = 1$ and $S_n = \frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i$. By the CLT, $S_n \Rightarrow G \sim N(0,1)$; Hoeffding's inequality gives uniform exponential tails, hence $\mathbb{E}|S_n|^{2p} \to \mathbb{E}|G|^{2p}$. Since $\mathbb{E}[G^{2p}] = (2p-1)!! = (2p)!/(2^p p!)$,

$$\frac{\mathbb{E}|\sum_i \varepsilon_i\alpha_i|^{2p}}{(\sum_i \alpha_i^2)^p} \longrightarrow \frac{(2p)!}{2^p p!} \qquad (n\to\infty),$$

so no smaller dimension-free constant works. ■

Exercise 5.11

Central idea. The Herbst argument turns a logarithmic Sobolev inequality with constant c into the Gaussian mgf bound $\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \le e^{c\lambda^2/4}$ for 1-Lipschitz f. For the linear function $f(x) = x_1$ this mgf bound is an equality ($e^{\lambda^2/2}$), forcing $c/4 \ge 1/2$, i.e. $c \ge 2$; since Theorem 5.4 attains $c = 2$, the constant is best possible.

Solution. Suppose the Gaussian logarithmic Sobolev inequality held with some constant $c > 0$ in place of 2, i.e.

$$\mathrm{Ent}(f^2) \le c\,\mathbb{E}\|\nabla f(X)\|^2$$

for all suitable f, where X is standard normal in $\mathbb{R}^n$. Let f be 1-Lipschitz and, for $\lambda > 0$, apply this to $F = e^{\lambda f/2}$. Since $\nabla F = \frac{\lambda}{2} e^{\lambda f/2}\nabla f$,

$$\mathrm{Ent}(e^{\lambda f}) = \mathrm{Ent}(F^2) \le c\,\mathbb{E}\|\nabla F\|^2 = \frac{c\lambda^2}{4}\,\mathbb{E}[e^{\lambda f}\|\nabla f\|^2] \le \frac{c\lambda^2}{4}\,\mathbb{E}e^{\lambda f}.$$

Writing $\phi(\lambda) = \mathbb{E}e^{\lambda f}$, we have $\mathrm{Ent}(e^{\lambda f}) = \lambda\phi'(\lambda) - \phi(\lambda)\log\phi(\lambda)$, so

$$\lambda\phi'(\lambda) - \phi(\lambda)\log\phi(\lambda) \le \frac{c\lambda^2}{4}\phi(\lambda), \qquad \lambda > 0.$$

Set $\psi(\lambda) = \log \mathbb{E}e^{\lambda(f-\mathbb{E}f)} = \log\phi(\lambda) - \lambda\mathbb{E}f$. Then $\psi(0) = \psi'(0) = 0$ and a short computation gives

$$\lambda\psi'(\lambda) - \psi(\lambda) \le \frac{c\lambda^2}{4}, \qquad \text{i.e.} \quad \frac{d}{d\lambda}\frac{\psi(\lambda)}{\lambda} \le \frac{c}{4}.$$

Since $\psi(\lambda)/\lambda \to \psi'(0) = 0$ as $\lambda \downarrow 0$, integrating yields $\psi(\lambda) \le c\lambda^2/4$, i.e.

$$\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \le e^{c\lambda^2/4} \qquad \text{for all 1-Lipschitz } f,\ \lambda > 0.$$

Now take the linear function $f(x) = x_1$, which is 1-Lipschitz with $f(X) - \mathbb{E}f(X) = X_1 \sim N(0,1)$. Here the bound is an equality:

$$\mathbb{E}e^{\lambda(f-\mathbb{E}f)} = \mathbb{E}e^{\lambda X_1} = e^{\lambda^2/2} \qquad \forall\, \lambda.$$

Hence $e^{\lambda^2/2} \le e^{c\lambda^2/4}$ for all $\lambda > 0$, forcing $1/2 \le c/4$, i.e. $c \ge 2$. Since Theorem 5.4 establishes the inequality with $c = 2$, the constant 2 is the best possible. ■

Exercise 5.12

Central idea. Prove the inequality for C_c^∞ functions, then approximate a general C^1 function f (with $\mathbb{E}_\gamma f^2, \mathbb{E}_\gamma\|\nabla f\|^2 < \infty$) in two stages: truncation by smooth cutoffs η_k gives compactly supported C^1 approximants converging in $H^1(\gamma_n)$, and mollification makes them smooth; a diagonal sequence $f_k \in C_c^\infty$ then satisfies $f_k \to f$, $\nabla f_k \to \nabla f$ in $L^2(\gamma_n)$. The entropy passes to the limit by lower semicontinuity (via its variational formula), while the Dirichlet energy converges directly.

Solution. Let γ_n be the standard Gaussian law on $\mathbb{R}^n$ and assume the logarithmic Sobolev inequality

$$\mathrm{Ent}_{\gamma_n}(g^2) \le 2\,\mathbb{E}_{\gamma_n}\|\nabla g\|^2 \tag{$*$}$$

proved for $g \in C_c^\infty(\mathbb{R}^n)$. Let f be continuously differentiable with $\mathbb{E}_{\gamma_n}[f^2] < \infty$ and $\mathbb{E}_{\gamma_n}\|\nabla f\|^2 < \infty$ (if the latter is infinite there is nothing to prove).

Step 1: truncation. Fix $\eta \in C_c^\infty(\mathbb{R}^n)$ with $0 \le \eta \le 1$, $\eta \equiv 1$ on the unit ball, $\operatorname{supp}\eta \subset B(0,2)$, and set $\eta_k(x) = \eta(x/k)$, so $\|\nabla\eta_k\|_\infty \le C_0/k$. Put $g_k = f\eta_k \in C_c^1$. Then

$$\|g_k - f\|_{L^2(\gamma_n)}^2 = \int (1-\eta_k)^2 f^2\, d\gamma_n \le \int_{|x|>k} f^2\, d\gamma_n \longrightarrow 0,$$

and, with $\nabla g_k = \eta_k \nabla f + f\nabla\eta_k$,

$$\|\nabla g_k - \nabla f\|_{L^2(\gamma_n)} \le \|(\eta_k - 1)\nabla f\|_{L^2(\gamma_n)} + \frac{C_0}{k}\|f\|_{L^2(\gamma_n)} \longrightarrow 0,$$

the first term by dominated convergence ($|\eta_k\nabla f| \le |\nabla f| \in L^2(\gamma_n)$).

Step 2: mollification and diagonalization. Let ρ_ε be a standard mollifier and $g_{k,j} = g_k * \rho_{1/j} \in C_c^\infty$. For fixed k, g_k and ∇g_k are uniformly continuous with compact support, so $g_{k,j} \to g_k$ and $\nabla g_{k,j} = (\nabla g_k) * \rho_{1/j} \to \nabla g_k$ uniformly, hence in $L^2(\gamma_n)$. Choosing j_k with $\|g_{k,j_k} - g_k\|_{H^1(\gamma_n)} < 1/k$ and setting $f_k = g_{k,j_k} \in C_c^\infty$ gives

$$f_k \longrightarrow f, \qquad \nabla f_k \longrightarrow \nabla f \quad \text{in } L^2(\gamma_n),$$

in particular $\mathbb{E}_{\gamma_n}\|\nabla f_k\|^2 \to \mathbb{E}_{\gamma_n}\|\nabla f\|^2$ and $f_k^2 \to f^2$ in $L^1(\gamma_n)$.

Step 3: passage to the limit. By $(*)$, $\mathrm{Ent}_{\gamma_n}(f_k^2) \le 2\,\mathbb{E}_{\gamma_n}\|\nabla f_k\|^2$. The entropy is L^1-lower semicontinuous: from the variational formula $\mathrm{Ent}(g) = \sup\{\mathbb{E}[gh] : h \text{ bounded},\ \mathbb{E}e^h \le 1\}$ (Section 5.1), Ent is a supremum of L^1-continuous functionals, so with $f_k^2 \to f^2$ in $L^1(\gamma_n)$,

$$\mathrm{Ent}_{\gamma_n}(f^2) \le \liminf_{k\to\infty} \mathrm{Ent}_{\gamma_n}(f_k^2) \le \lim_{k\to\infty} 2\,\mathbb{E}_{\gamma_n}\|\nabla f_k\|^2 = 2\,\mathbb{E}_{\gamma_n}\|\nabla f\|^2.$$

This is the logarithmic Sobolev inequality for f. ■

Exercise 5.13

Central idea. Write $X = \Gamma^{1/2}Z$ with $Z \sim N(0, I_n)$ and transfer the standard Gaussian inequalities to $g = f \circ \Gamma^{1/2}$; the chain rule gives $\nabla g(z) = \Gamma^{1/2}\nabla f(\Gamma^{1/2}z)$, so $\|\nabla g(z)\|^2 = \langle \Gamma \nabla f(\Gamma^{1/2}z), \nabla f(\Gamma^{1/2}z)\rangle$, and since $f(X) \overset{d}{=} g(Z)$ the variance and entropy are preserved.

Solution. Let Γ be symmetric positive semidefinite and $\Gamma^{1/2}$ its symmetric positive semidefinite square root. For $Z \sim N(0, I_n)$, $X := \Gamma^{1/2}Z$ satisfies $\mathbb{E}e^{i\langle t, X\rangle} = e^{-\|\Gamma^{1/2}t\|^2/2} = e^{-\langle \Gamma t, t\rangle/2}$, so $X \sim N(0, \Gamma)$ (this holds also when Γ is singular).

Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and put $g = f \circ \Gamma^{1/2}$, i.e. $g(z) = f(\Gamma^{1/2}z)$, which is continuously differentiable. By the chain rule and symmetry of $\Gamma^{1/2}$,

$$\nabla g(z) = \Gamma^{1/2}\, \nabla f(\Gamma^{1/2}z), \qquad z \in \mathbb{R}^n,$$

hence

$$\|\nabla g(z)\|^2 = \langle \Gamma^{1/2}\nabla f(\Gamma^{1/2}z), \Gamma^{1/2}\nabla f(\Gamma^{1/2}z)\rangle = \langle \Gamma\, \nabla f(\Gamma^{1/2}z), \nabla f(\Gamma^{1/2}z)\rangle.$$

Since $f(X) \overset{d}{=} g(Z)$, we have $\mathrm{Var}(f(X)) = \mathrm{Var}(g(Z))$ and $\mathrm{Ent}(f(X)^2) = \mathrm{Ent}(g(Z)^2)$ (both depend only on the law), and

$$\mathbb{E}\|\nabla g(Z)\|^2 = \mathbb{E}\big[\langle \Gamma\, \nabla f(\Gamma^{1/2}Z), \nabla f(\Gamma^{1/2}Z)\rangle\big] = \mathbb{E}\big[\langle \Gamma \nabla f(X), \nabla f(X)\rangle\big].$$

Applying the standard Gaussian Poincaré inequality to g (the estimate being trivial if the right-hand side is infinite),

$$\mathrm{Var}(f(X)) = \mathrm{Var}(g(Z)) \le \mathbb{E}\|\nabla g(Z)\|^2 = \mathbb{E}\big[\langle \Gamma \nabla f(X), \nabla f(X)\rangle\big],$$

and applying the Gaussian logarithmic Sobolev inequality (Theorem 5.4),

$$\mathrm{Ent}(f(X)^2) = \mathrm{Ent}(g(Z)^2) \le 2\,\mathbb{E}\|\nabla g(Z)\|^2 = 2\,\mathbb{E}\big[\langle \Gamma \nabla f(X), \nabla f(X)\rangle\big].$$

This is the claim. ■

Exercise 5.14

Central idea. A.s. continuity on the totally bounded index set lets us replace $\mathcal{T}$ by a countable dense $\mathcal{D}$; for finite $F \subset \mathcal{T}$ the Gaussian Poincaré inequality (Exercise 5.13, via mollification of the max function) gives $\mathrm{Var}(\max_F X_t) \le \sigma^2 := \sup_t \mathrm{Var}(X_t)$. With $F_k \uparrow \mathcal{D}$, the maxima $Z_k \uparrow Z$; monotone medians $M_k \uparrow\le M$ and Exercise 2.1 ($|\mathbb{E}Z_k - M_k| \le \sigma$) bound $\mathbb{E}Z_k \le M + \sigma$, so monotone convergence yields $\mathbb{E}Z < \infty$. The finite-set concentration bound then passes to the limit by monotone convergence.

Solution. Let $\{X_t\}_{t\in\mathcal{T}}$ be centered Gaussian, $\mathcal{T}$ totally bounded for $d(s,t) = (\mathbb{E}(X_s - X_t)^2)^{1/2}$, with a.s. continuous (hence a.s. bounded) sample paths. Put $\sigma^2 = \sup_{t\in\mathcal{T}} \mathbb{E}X_t^2 < \infty$ (finiteness: $t \mapsto \|X_t\|_2$ is 1-Lipschitz for d, hence bounded on the totally bounded set $\mathcal{T}$). Let $\mathcal{D} = \{t_1, t_2, \dots\}$ be countable dense in $\mathcal{T}$.

Reduction to $\mathcal{D}$. For a.e. ω the path $t \mapsto X_t(\omega)$ is d-continuous, so $\sup_{t\in\mathcal{T}} X_t(\omega) = \sup_{t\in\mathcal{D}} X_t(\omega)$: if $t_n \in \mathcal{D}$, $d(t_n, t) \to 0$ then $X_{t_n}(\omega) \to X_t(\omega)$. Hence $Z := \sup_{\mathcal{T}} X_t = \sup_{\mathcal{D}} X_t$ a.s.; $Z < \infty$ a.s. and $Z \ge X_{t_1} > -\infty$ a.s., so Z is real-valued.

Finite subsets. For finite $F \subset \mathcal{T}$ put $Z_F = \max_{t\in F} X_t = g(X_F)$ with $g(x) = \max_{t\in F} x_t$, $X_F \sim N(0, \Sigma_F)$. The function g is 1-Lipschitz, differentiable a.e. with $\nabla g(x) = e_{t^*(x)}$ (the maximizing coordinate, a.e. unique). Mollifying, $g_\varepsilon = g * \rho_\varepsilon$ is smooth with $\nabla g_\varepsilon \to \nabla g$ a.e. and $\|\nabla g_\varepsilon\| \le 1$; applying Exercise 5.13 to g_ε and letting $\varepsilon \downarrow 0$ (dominated convergence, using $\langle \Sigma_F v, v\rangle \le \|\Sigma_F\|_{\mathrm{op}}$ for $\|v\| \le 1$) gives

$$\mathrm{Var}(Z_F) \le \mathbb{E}\big[\langle \Sigma_F \nabla g(X_F), \nabla g(X_F)\rangle\big] = \mathbb{E}\big[(\Sigma_F)_{t^*(X_F), t^*(X_F)}\big] \le \max_{t\in F} \mathrm{Var}(X_t) \le \sigma^2.$$

Finiteness of $\mathbb{E}Z$. Let $F_k = \{t_1, \dots, t_k\}$ and $Z_k = Z_{F_k} \uparrow Z$ a.s. Since $Z_k \ge X_{t_1} \in L^1$, monotone convergence applied to $Z_k - X_{t_1} \ge 0$ gives $\mathbb{E}Z_k \uparrow \mathbb{E}Z \in (-\infty, +\infty]$. Let $M_k = \inf\{t : \mathbb{P}\{Z_k \le t\} \ge 1/2\}$ (a median

of Z_k) and M the corresponding median of Z (finite since Z is real-valued). From $Z_k \le Z_{k+1}$ we get $\{Z_{k+1} \le t\} \subset \{Z_k \le t\}$, whence $M_k \le M_{k+1}$; from $Z_k \le Z$, $M_k \le M$. Each Z_k is square-integrable, so by Exercise 2.1, $|\mathbb{E}Z_k - M_k| \le \sqrt{\mathrm{Var}(Z_k)} \le \sigma$, i.e. $\mathbb{E}Z_k \le M_k + \sigma \le M + \sigma$. Passing to the limit,

$$\mathbb{E}Z = \lim_{k\to\infty} \mathbb{E}Z_k \le M + \sigma < \infty,$$

so Z has finite expectation.

Conclusion. For finite F, the Herbst argument from Exercise 5.13 (using $\langle \Sigma_F \nabla g, \nabla g\rangle \le \sigma^2$) gives the Gaussian concentration $\mathbb{P}\{|Z_F - \mathbb{E}Z_F| \ge t\} \le e^{-t^2/(2\sigma^2)}$. Since $Z_k \uparrow Z$ a.s. and $\mathbb{E}Z_k \uparrow \mathbb{E}Z < \infty$, monotone (indeed setwise) convergence gives for the upper tail

$$\mathbb{P}\{Z - \mathbb{E}Z \ge t\} = \lim_{k\to\infty} \mathbb{P}\{Z_k - \mathbb{E}Z \ge t\} = \lim_{k\to\infty} \mathbb{P}\{Z_k - \mathbb{E}Z_k \ge t + (\mathbb{E}Z - \mathbb{E}Z_k)\} \le e^{-t^2/(2\sigma^2)},$$

and the lower tail follows by applying the same to $\{-X_t\}$. ■

Exercise 5.15

Central idea. Direct computation: $\mathbb{E}Z^2 = D + \delta^2$, $\mathrm{Var}(Z^2) = 2D + 4\delta^2$, and the noncentral chi-square mgf $\mathbb{E}e^{\lambda Z^2} = (1-2\lambda)^{-D/2} e^{\lambda\delta^2/(1-2\lambda)}$ satisfies the sub-gamma bound with $(v, c) = (2\mathbb{E}Z^2 + 2\delta^2, 2)$ by a second-derivative comparison. Since $Z(x) = \|x + \delta e_1\|$ is 1-Lipschitz with $\|\nabla Z\| = 1$ a.e., Poincaré gives $\mathrm{Var}(Z) < 1$ (strict, as Z is not affine) and Herbst gives sub-Gaussianity with variance factor 1; running Herbst on Z^2 itself, using $\|\nabla(Z^2)\|^2 = 4Z^2$, yields the sub-gamma bound with variance factor $4\mathbb{E}Z^2$ and scale 2.

Solution. Write $Z^2 = (X_1 + \delta)^2 + \sum_{i=2}^{D} X_i^2$ with X_i iid $N(0,1)$.

Mean and variance. $\mathbb{E}[(X_1+\delta)^2] = 1 + \delta^2$ and $\mathbb{E}[X_i^2] = 1$, so

$$\mathbb{E}Z^2 = 1 + \delta^2 + (D-1) = D + \delta^2.$$

Moreover $\mathrm{Var}((X_1+\delta)^2) = \mathrm{Var}(X_1^2 + 2\delta X_1) = \mathrm{Var}(X_1^2) + 4\delta^2\mathrm{Var}(X_1) = 2 + 4\delta^2$ (using $\mathbb{E}X_1^3 = 0$), and by independence

$$\mathrm{Var}(Z^2) = (2 + 4\delta^2) + 2(D-1) = 2D + 4\delta^2.$$

Z^2 is sub-gamma. For $\lambda < 1/2$, completing the square gives $\mathbb{E}e^{\lambda(X_1+\delta)^2} = (1-2\lambda)^{-1/2}\exp(\frac{\lambda\delta^2}{1-2\lambda})$, whence by independence

$$\mathbb{E}e^{\lambda Z^2} = (1-2\lambda)^{-D/2}\exp\Big(\frac{\lambda\delta^2}{1-2\lambda}\Big), \qquad \lambda < \tfrac{1}{2}.$$

Put $\psi(\lambda) = \log \mathbb{E}e^{\lambda(Z^2 - \mathbb{E}Z^2)} = -\frac{D}{2}\log(1-2\lambda) + \frac{\lambda\delta^2}{1-2\lambda} - \lambda(D+\delta^2)$. Then $\psi(0) = \psi'(0) = 0$ and

$$\psi''(\lambda) = \frac{2D}{(1-2\lambda)^2} + \frac{4\delta^2}{(1-2\lambda)^3}.$$

With $v = 2D + 4\delta^2 = 2\mathbb{E}Z^2 + 2\delta^2$ and $u(\lambda) = \frac{\lambda^2 v}{2(1-2\lambda)}$, we have $u(0) = \psi(0)$, $u'(0) = \psi'(0) = 0$ and $u''(\lambda) = \frac{v}{(1-2\lambda)^3} \ge \psi''(\lambda)$ on $[0, \frac{1}{2})$, since $v - 2D(1-2\lambda) - 4\delta^2 = 4D\lambda \ge 0$. Integrating twice gives $\psi \le u$, i.e. Z^2 is sub-gamma with variance factor $v = 2\mathbb{E}Z^2 + 2\delta^2$ and scale factor 2.

The function Z. As a function of $x \in \mathbb{R}^D$, $Z(x) = \|x + \delta e_1\|$ is 1-Lipschitz and $\|\nabla Z(x)\| = 1$ for $x \ne -\delta e_1$ (hence a.e.). By the Gaussian Poincaré inequality, $\mathrm{Var}(Z) \le \mathbb{E}\|\nabla Z\|^2 = 1$, with strict inequality since equality in Poincaré holds only for affine functions and Z is strictly convex; thus $\mathrm{Var}(Z) < 1$. By the Gaussian concentration inequality (Herbst's argument), the 1-Lipschitz Z satisfies $\mathbb{E}e^{t(Z-\mathbb{E}Z)} \le e^{t^2/2}$ for all t, i.e. Z is sub-Gaussian with variance factor 1.

Sub-gamma bound with $4\mathbb{E}Z^2$. Apply Herbst to Z^2 itself. $F = e^{\lambda Z^2/2}$ is smooth (Z^2 is a polynomial in x), $\nabla F = \lambda Z e^{\lambda Z^2/2}\nabla Z$, so $\|\nabla F\|^2 = \lambda^2 Z^2 e^{\lambda Z^2}$ a.e. The logarithmic Sobolev inequality gives, for $0 < \lambda < 1/2$,

$$\mathrm{Ent}(e^{\lambda Z^2}) \le 2\,\mathbb{E}\|\nabla F\|^2 = 2\lambda^2\,\mathbb{E}\big[Z^2 e^{\lambda Z^2}\big].$$

With $\phi(\lambda) = \mathbb{E}e^{\lambda Z^2}$ this reads $\lambda\phi' - \phi\log\phi \le 2\lambda^2\phi'$. Setting $\psi(\lambda) = \log\phi(\lambda) - \lambda\mathbb{E}Z^2$ and $E = \mathbb{E}Z^2$,

$$\lambda\psi' - \psi \le 2\lambda^2\psi' + 2\lambda^2 E.$$

Put $u = \psi/\lambda$ (so $u(0) = \psi'(0) = 0$); then $u' - \frac{2}{1-2\lambda}u \le \frac{2E}{1-2\lambda}$. With integrating factor $1-2\lambda$,

$$\frac{d}{d\lambda}\big[(1-2\lambda)u(\lambda)\big] \le 2E,$$

whence $(1-2\lambda)u(\lambda) \le 2E\lambda$ and therefore

$$\log\mathbb{E}e^{\lambda(Z^2-\mathbb{E}Z^2)} = \psi(\lambda) \le \frac{2\lambda^2 E}{1-2\lambda} = \frac{\lambda^2(4\mathbb{E}Z^2)}{2(1-2\lambda)}, \qquad 0 < \lambda < \tfrac{1}{2}.$$

Thus Z^2 is sub-gamma with variance factor $4\mathbb{E}Z^2$ and scale factor 2. ■

Exercise 5.16

Central idea. Herbst's argument via the Gaussian logarithmic Sobolev inequality: applying the LSI to $e^{\lambda Z/2}$ gives the differential inequality $\lambda H' - H \le \frac{\lambda^2}{2}R(\lambda)$ for the log-mgf $H(\lambda) = \log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$, where $R(\lambda)$ is the tilted mean of $\|\nabla f\|^2$. The entropy variational formula $\mathbb{E}_Q[W] \le \log\mathbb{E}_P[e^W] + D(Q\|P)$ (Corollary 4.15) bounds $R(\lambda)$ through $K(\lambda) = \log\mathbb{E}e^{(\lambda/\theta)\|\nabla f\|^2}$, closing the ODE as $(1-\lambda\theta/2)(\lambda H' - H) \le \frac{\lambda\theta}{2}K(\lambda)$; integrating with monotonicity of $K(\lambda)/\lambda$ and $-\log(1-x) \le x/(1-x)$ yields the claim.

Solution. We argue formally, assuming the expectations below are finite (justified by applying the argument to the truncations $f_M = (-M)\vee(f\wedge M)$, for which $\|\nabla f_M\| \le \|\nabla f\|$ a.e., and passing to the limit). Put $H(\lambda) = \log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$, so $H(0) = H'(0) = 0$.

Apply the Gaussian logarithmic Sobolev inequality to $g = e^{\lambda Z/2}$:

$$\mathrm{Ent}(e^{\lambda Z}) = \mathrm{Ent}(g^2) \le 2\mathbb{E}\|\nabla g\|^2 = \frac{\lambda^2}{2}\mathbb{E}[\|\nabla f\|^2 e^{\lambda Z}].$$

Since $\frac{d}{d\lambda}\log\mathbb{E}e^{\lambda Z} = \mathbb{E}[Ze^{\lambda Z}]/\mathbb{E}e^{\lambda Z}$,

$$\mathrm{Ent}(e^{\lambda Z}) = \mathbb{E}e^{\lambda Z}\,(\lambda H'(\lambda) - H(\lambda)),$$

so with $R(\lambda) = \mathbb{E}[\|\nabla f\|^2 e^{\lambda Z}]/\mathbb{E}e^{\lambda Z}$,

$$\lambda H'(\lambda) - H(\lambda) \le \frac{\lambda^2}{2}R(\lambda). \tag{1}$$

Let $dQ_\lambda = e^{\lambda(Z-\mathbb{E}Z)}dP/e^{H(\lambda)}$, so $R(\lambda) = \mathbb{E}_{Q_\lambda}[\|\nabla f\|^2]$. By the entropy variational formula, for $W = (\lambda/\theta)\|\nabla f\|^2$,

$$\mathbb{E}_{Q_\lambda}[W] \le \log\mathbb{E}_P[e^W] + D(Q_\lambda\|P),$$

and $D(Q_\lambda\|P) = \mathbb{E}_{Q_\lambda}[\lambda(Z-\mathbb{E}Z) - H(\lambda)] = \lambda H'(\lambda) - H(\lambda)$. Hence, with $K(\lambda) = \log\mathbb{E}\exp\big(\frac{\lambda}{\theta}\|\nabla f\|^2\big)$ (finite for $\lambda/\theta < \lambda_0$),

$$R(\lambda) \le \frac{\theta}{\lambda}K(\lambda) + \theta H'(\lambda) - \frac{\theta}{\lambda}H(\lambda). \tag{2}$$

This is the content of Corollary 4.15. Combining (1)–(2),

$$\lambda H' - H \le \frac{\lambda\theta}{2}(K(\lambda) + \lambda H' - H),$$

i.e., for $\lambda\theta < 2$,

$$(1-\tfrac{\lambda\theta}{2})(\lambda H' - H) \le \tfrac{\lambda\theta}{2}K(\lambda). \tag{3}$$

Write $K(\lambda) = \lambda L(\lambda)$ with $L(\lambda) = K(\lambda)/\lambda$. Then (3) gives

$$\left(\frac{H(\lambda)}{\lambda}\right)' = \frac{\lambda H' - H}{\lambda^2} \le \frac{\theta}{2(1-\lambda\theta/2)} L(\lambda).$$

Now L is nondecreasing: with $M(u) = \log \mathbb{E} e^{uV}$, $V = \|\nabla f\|^2$, we have $L(\lambda) = \frac{1}{\theta}\frac{M(\lambda/\theta)}{\lambda/\theta}$, and $\frac{d}{du}\frac{M(u)}{u} = \frac{N(u)}{u^2}$ where $N(u) = uM'(u) - M(u)$ satisfies $N(0) = 0$, $N'(u) = uM''(u) \ge 0$ (as $M''(u)$ is a variance). Since $H(\lambda)/\lambda \to H'(0) = 0$,

$$\frac{H(\lambda)}{\lambda} \le \frac{\theta}{2} L(\lambda) \int_0^\lambda \frac{ds}{1 - s\theta/2} = -L(\lambda)\log(1 - \tfrac{\lambda\theta}{2}) \le L(\lambda)\frac{\lambda\theta/2}{1-\lambda\theta/2},$$

using $-\log(1-x) \le x/(1-x)$ for $x \in [0,1)$. Multiplying by λ and recalling $\lambda L(\lambda) = K(\lambda)$,

$$\log \mathbb{E} e^{\lambda(Z - \mathbb{E}Z)} \le \frac{\lambda\theta}{2(1-\lambda\theta/2)} \log \mathbb{E} \exp\!\big(\tfrac{\lambda}{\theta}\|\nabla f\|^2\big). \qquad \blacksquare$$

Application. Let P be the orthogonal projection onto a k-dimensional subspace and $f(x) = \|Px\|^2$. Then $\nabla f(x) = 2Px$, so $\|\nabla f\|^2 = 4\|Px\|^2 = 4f$, and $Z = \|PX\|^2$ is χ^2_k with $\mathbb{E}Z = k$. Moreover $\mathbb{E} e^{\lambda\|\nabla f\|^2} = \mathbb{E} e^{4\lambda Z} = (1-8\lambda)^{-k/2} < \infty$ for $\lambda < 1/8$, so $\lambda_0 = 1/8$. Taking e.g. $\theta = 8$ (so $\lambda\theta < 2$ and $\lambda/\theta < \lambda_0$ for $\lambda < 1/4$),

$$\log \mathbb{E} e^{\lambda(Z-k)} \le \frac{4\lambda}{1-4\lambda}\Big(-\frac{k}{2}\log(1-\lambda)\Big) \le 8k\lambda^2, \qquad 0 < \lambda \le 1/8,$$

using $-\log(1-\lambda) \le 2\lambda$. By Chernoff's bound, with $\lambda = t/(16k)$,

$$\mathbb{P}\{\|PX\|^2 - k \ge t\} \le \exp\!\left(-\frac{t^2}{32k}\right), \qquad 0 \le t \le 2k,$$

the usual sub-Gaussian tail for χ^2_k, obtained despite ∇f being unbounded.

Exercise 5.17

Central idea. Theorem 5.20 is the Rademacher version, $\mathbb{E}\|\sum c_j\varepsilon_j\|^2 \le 2(\mathbb{E}\|\sum c_j\varepsilon_j\|)^2$. Approximate each Gaussian X_i by a normalized Rademacher sum $X_i^{(m)} = m^{-1/2}\sum_{j=1}^m \varepsilon_{ij}$; then $Z^{(m)} = \|\sum_i b_i X_i^{(m)}\|$ is a Rademacher norm to which Theorem 5.20 applies. Uniform boundedness of fourth moments gives uniform integrability, so the inequality passes to the CLT limit $Z^{(m)} \xrightarrow{d} Z$.

Solution. Let ε_{ij}, $1 \le i \le n$, $1 \le j \le m$, be independent Rademacher variables and put $X_i^{(m)} = m^{-1/2}\sum_{j=1}^m \varepsilon_{ij}$. By the multivariate CLT, $(X_1^{(m)}, \dots, X_n^{(m)}) \xrightarrow{d} (X_1, \dots, X_n)$ with X_i iid $N(0,1)$. With $Z^{(m)} = \|\sum_{i=1}^n b_i X_i^{(m)}\|$ and $Z = \|\sum_{i=1}^n b_i X_i\|$, continuity of the norm gives $Z^{(m)} \xrightarrow{d} Z$.

Now $Z^{(m)} = \|\sum_{i,j} c_{ij}\varepsilon_{ij}\|$ with $c_{ij} = b_i/\sqrt{m}$. Theorem 5.20 applied to these mn vectors yields

$$\mathbb{E}\big[(Z^{(m)})^2\big] \le 2\big(\mathbb{E}Z^{(m)}\big)^2, \qquad m \ge 1. \tag{$*$}$$

To pass to the limit, note $\mathbb{E}|X_i^{(m)}|^4 = m^{-2}\mathbb{E}[(\sum_{j=1}^m \varepsilon_{ij})^4] = m^{-2}(m + 3m(m-1)) = 3 - 2/m \le 3$, since $\mathbb{E}[\varepsilon_j^4] = 1$, $\mathbb{E}[\varepsilon_j^2\varepsilon_k^2] = 1$ $(j \ne k)$, and all terms with an odd power vanish. Hence, with $(\sum_i a_i)^4 \le n^3 \sum_i a_i^4$,

$$\mathbb{E}\big[(Z^{(m)})^4\big] \le \mathbb{E}\Big[\Big(\sum_{i=1}^n \|b_i\|\,|X_i^{(m)}|\Big)^4\Big] \le 3n^3 \sum_{i=1}^n \|b_i\|^4 < \infty$$

uniformly in m, so $\{(Z^{(m)})^2\}$ is uniformly integrable. With $Z^{(m)} \xrightarrow{d} Z$ this gives $\mathbb{E}[(Z^{(m)})^2] \to \mathbb{E}[Z^2]$ and $\mathbb{E}Z^{(m)} \to \mathbb{E}Z$. Letting $m \to \infty$ in $(*)$,

$$\mathbb{E}[Z^2] \le 2(\mathbb{E}Z)^2, \qquad \text{i.e.} \qquad \sqrt{\mathbb{E}[Z^2]} \le \sqrt{2}\,\mathbb{E}Z. \qquad \blacksquare$$

Remark. The constant $\sqrt{2}$ is not optimal; a sharper argument gives $\sqrt{\pi/2}$ (see Latała–Oleszkiewicz (1999) for the best constants in moment comparison for Gaussian vectors).

Exercise 5.18

Central idea. Gross's proof of hypercontractivity via the Gaussian log-Sobolev inequality: for $f \ge 0$ smooth put $g_t = T_{e^{-t}} f$ and $q = q(t)$; differentiating $\Phi(t) = \frac{1}{q}\log \mathbb{E}[g_t^q]$ produces $\frac{q'}{q^2}\frac{\mathrm{Ent}(g_t^q)}{\mathbb{E}g_t^q} + \frac{\mathbb{E}[g_t^{q-1}Lg_t]}{\mathbb{E}g_t^q}$, and the LSI in the form $\mathrm{Ent}(g^r) \le -\frac{r^2}{2(r-1)}\mathbb{E}[g^{r-1}Lg]$ (from $\mathrm{Ent}(u^2) \le 2\mathbb{E}[(u')^2]$ with $u = g^{r/2}$ plus Gaussian integration by parts) makes this ≤ 0 precisely because $q'(t) = 2(q(t)-1)$. Hence $\|T_{e^{-t}}f\|_{q(t)}$ is nonincreasing, and $|T_{e^{-t}}f| \le T_{e^{-t}}|f|$ extends it to general f.

Solution. Let X, X' be independent $N(0,1)$. For $|\gamma| \le 1$, Jensen's inequality gives $(T_\gamma f(y))^2 \le \mathbb{E}[f(\gamma y + \sqrt{1-\gamma^2}X)^2]$, whence with X in place of y,

$$\mathbb{E}[(T_\gamma f(X))^2] \le \mathbb{E}[f(\gamma X + \sqrt{1-\gamma^2}X')^2] = \mathbb{E}[f(X)^2],$$

since $\gamma X + \sqrt{1-\gamma^2}X' \sim N(0,1)$: T_γ is an L^2-contraction.

It suffices to prove the claim for f smooth and strictly positive: for general f with $\mathbb{E}|f|^p < \infty$, Jensen gives $|T_{e^{-t}}f| \le T_{e^{-t}}|f|$, so $\|T_{e^{-t}}f\|_{q(t)} \le \|T_{e^{-t}}|f|\|_{q(t)}$, and $|f|$ is approximated in L^p by smooth strictly positive functions.

Let thus $g \ge 0$ be smooth (with enough decay) and put $g_t = T_{e^{-t}}g$. First,

$$\frac{d}{dt}T_{e^{-t}}g = L\,T_{e^{-t}}g, \qquad Lf = f'' - xf'.$$

Indeed, with $\gamma = e^{-t}$, $\sigma = \sqrt{1-\gamma^2}$, $u(t,y) = \mathbb{E}[g(\gamma y + \sigma X)]$,

$$\frac{\partial u}{\partial t} = \mathbb{E}[g'(\gamma y + \sigma X)(-\gamma y + \tfrac{\gamma^2}{\sigma}X)],$$

while $Lu = \gamma^2\mathbb{E}[g''(\gamma y + \sigma X)] - \gamma y\,\mathbb{E}[g'(\gamma y + \sigma X)]$; the two agree since Gaussian integration by parts ($\mathbb{E}[X\psi(X)] = \mathbb{E}[\psi'(X)]$) gives $\mathbb{E}[Xg'(\gamma y + \sigma X)] = \sigma\,\mathbb{E}[g''(\gamma y + \sigma X)]$.

Next, the LSI implies, for $r > 1$,

$$\mathrm{Ent}(g^r) \le -\frac{r^2}{2(r-1)}\mathbb{E}[g^{\,r-1}Lg]. \tag{1}$$

Indeed, with $h = \partial_x g^{r/2} = \frac{r}{2}g^{r/2-1}g'$, the Gaussian LSI $\mathrm{Ent}(u^2) \le 2\mathbb{E}[(u')^2]$ at $u = g^{r/2}$ gives $\mathrm{Ent}(g^r) \le \frac{r^2}{2}\mathbb{E}[g^{r-2}(g')^2]$. Gaussian integration by parts in the form $\mathbb{E}[\phi''\psi] = \mathbb{E}[X\phi'\psi] - \mathbb{E}[\phi'\psi']$ yields $\mathbb{E}[g^{r-1}g''] = \mathbb{E}[Xg^{r-1}g'] - (r-1)\mathbb{E}[g^{r-2}(g')^2]$, so $\mathbb{E}[g^{r-1}Lg] = -(r-1)\mathbb{E}[g^{r-2}(g')^2]$, and (1) follows.

Now let $q = q(t) = 1 + e^{2t}(p-1)$ and $\Phi(t) = \frac{1}{q}\log\mathbb{E}[g_t^q]$, so $e^{\Phi(t)} = \|g_t\|_{q(t)}$. Differentiating (using $\dot g_t = Lg_t$),

$$\begin{aligned}\Phi'(t) &= -\frac{q'}{q^2}\log\mathbb{E}[g_t^q] + \frac{1}{q}\frac{\mathbb{E}[qg_t^{q-1}Lg_t + q'g_t^q\log g_t]}{\mathbb{E}[g_t^q]}\\ &= \frac{q'}{q^2}\frac{\mathrm{Ent}(g_t^q)}{\mathbb{E}[g_t^q]} + \frac{\mathbb{E}[g_t^{q-1}Lg_t]}{\mathbb{E}[g_t^q]},\end{aligned}$$

since $\frac{q'}{q}\mathbb{E}[g^q \log g] = \frac{q'}{q^2}\mathbb{E}[g^q \log(g^q)]$. Applying (1) with $r = q$,

$$\Phi'(t) \le \frac{\mathbb{E}[g_t^{q-1} L g_t]}{\mathbb{E}[g_t^q]} \left[1 - \frac{q'(t)}{2(q(t)-1)}\right].$$

But $q'(t) = 2e^{2t}(p-1) = 2(q(t)-1)$, so the bracket vanishes and $\Phi'(t) \le 0$: $t \mapsto \|T_{e^{-t}} g\|_{q(t)}$ is nonincreasing. As $T_1 = \mathrm{id}$ and $q(0) = p$,

$$\left(\mathbb{E}|T_{e^{-t}} f(X)|^{q(t)}\right)^{1/q(t)} \le \left(\mathbb{E}|f(X)|^p\right)^{1/p}, \qquad t \ge 0,$$

first for smooth $f \ge 0$, then for general f via $|T_{e^{-t}} f| \le T_{e^{-t}}|f|$ and approximation. ■

Exercise 5.19

Central idea. Tensorization by induction on n, imitating Theorem 5.16: write $T_\gamma = T_\gamma^{(n)} \circ T_\gamma^{(<n)}$ with $T_\gamma^{(n)}$ acting on x_n and $T_\gamma^{(<n)}$ on the first $n-1$ coordinates; apply the induction hypothesis in the first $n-1$ variables, then Minkowski's integral inequality (using $q(t) \ge p$) to swap the integrals, then the one-dimensional Exercise 5.18 in x_n.

Solution. As in Exercise 5.18 it suffices to treat $f \ge 0$ smooth, since $|T_{e^{-t}} f| \le T_{e^{-t}}|f|$. Write $T_\gamma = T_\gamma^{(1)} \otimes \cdots \otimes T_\gamma^{(n)}$, where $T_\gamma^{(i)}$ acts on the i-th coordinate; these commute. Put $T_\gamma^{(<n)} = T_\gamma^{(1)} \cdots T_\gamma^{(n-1)}$, so $T_\gamma = T_\gamma^{(n)} \circ T_\gamma^{(<n)}$. Let $d\gamma_k$ denote the standard Gaussian law on $\mathbb{R}^k$ and $q = q(t) \ge p$.

We induct on n; $n = 1$ is Exercise 5.18. Assume the claim for $n-1$. For fixed x_n, the induction hypothesis applied to $g(\cdot) = (T_\gamma^{(n)} f)(\cdot, x_n)$ gives

$$\int_{\mathbb{R}^{n-1}} \left|T_\gamma^{(<n)} g\right|^q d\gamma_{n-1} \le \left(\int_{\mathbb{R}^{n-1}} |g|^p \, d\gamma_{n-1}\right)^{q/p}.$$

Hence, with $h = T_\gamma^{(n)} f$,

$$\|T_\gamma f\|_{L^q(\gamma_n)}^q \le \int_{\mathbb{R}} \left(\int_{\mathbb{R}^{n-1}} |h(x_{<n}, x_n)|^p \, d\gamma_{n-1}(x_{<n})\right)^{q/p} d\gamma_1(x_n). \tag{$*$}$$

Since $q/p \ge 1$, Minkowski's integral inequality yields

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}^{n-1}} |h|^p \, d\gamma_{n-1}\right)^{q/p} d\gamma_1 \le \left(\int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} |h|^q \, d\gamma_1\right)^{p/q} d\gamma_{n-1}\right)^{q/p}.$$

For each fixed $x_{<n}$, Exercise 5.18 applied in the x_n variable gives $\|h(x_{<n}, \cdot)\|_{L^q(\gamma_1)} = \|T_\gamma^{(n)} f(x_{<n}, \cdot)\|_{L^q(\gamma_1)} \le \|f(x_{<n}, \cdot)\|_{L^p(\gamma_1)}$. Therefore the right-hand side of $(*)$ is at most

$$\left(\int_{\mathbb{R}^{n-1}} \|f(x_{<n}, \cdot)\|_{L^p(\gamma_1)}^p \, d\gamma_{n-1}(x_{<n})\right)^{q/p} = \|f\|_{L^p(\gamma_n)}^q,$$

i.e.

$$\left(\mathbb{E}\left|T_{e^{-t}} f(X_1, \dots, X_n)\right|^{q(t)}\right)^{1/q(t)} \le \left(\mathbb{E}\left|f(X_1, \dots, X_n)\right|^p\right)^{1/p}, \qquad t \ge 0. \qquad \blacksquare$$

Exercise 5.20

Central idea. For $u(t) = T_{e^{-t}} h_n$, the heat equation $\dot{u} = L T_{e^{-t}} h_n$ from Exercise 5.18 combined with $L \circ T_{e^{-t}} = T_{e^{-t}} \circ L$ (Stein's lemma) and $L h_n = -n h_n$ gives the scalar ODE $\dot{u} = -nu$, whence $u(t) = e^{-nt} h_n$, i.e. $T_\gamma h_n = \gamma^n h_n$. Since T_γ is a tensor product over coordinates, it acts diagonally on product Hermite polynomials, and L^2-continuity (contraction) extends this to the Hermite expansion.

Solution. Let h_n be the n-th Hermite polynomial, $Lh_n = -nh_n$, and put $u(t) = T_{e^{-t}}h_n$ (the formal computations below are justified for polynomials by differentiation under the integral and Gaussian integration by parts). By Exercise 5.18, $\frac{d}{dt}T_{e^{-t}}g = L\,T_{e^{-t}}g$ for nice g, so $\dot{u}(t) = L\,T_{e^{-t}}h_n$. Moreover L commutes with T_γ: indeed,

$$T_\gamma Lg(y) - LT_\gamma g(y) = \mathbb{E}[\sigma^2 g''(\gamma y + \sigma X) - \sigma X g'(\gamma y + \sigma X)] = 0,$$

with $\sigma = \sqrt{1-\gamma^2}$, by Stein's lemma $\mathbb{E}[X\psi(X)] = \mathbb{E}[\psi'(X)]$ applied to $\psi(x) = \sigma g'(\gamma y + \sigma x)$. Hence

$$\dot{u}(t) = L\,T_{e^{-t}}h_n = T_{e^{-t}}Lh_n = T_{e^{-t}}(-nh_n) = -n\,u(t).$$

For each fixed x, $t \mapsto u(t,x)$ solves $\dot{u} = -nu$ with $u(0,x) = h_n(x)$, so $u(t,x) = e^{-nt}h_n(x)$. With $\gamma = e^{-t}$,

$$T_\gamma h_n = \gamma^n h_n, \qquad n = 0, 1, 2, ...$$

($n = 0$ since $h_0 \equiv 1$ and $T_\gamma 1 = 1$).

Now let $H_{\bar{k}}(x) = \prod_{i=1}^n h_{k_i}(x_i)$ for $\bar{k} = (k_1, ..., k_n) \in \mathbb{N}^n$, $|\bar{k}| = \sum_i k_i$. Since $T_\gamma = T_\gamma^{(1)} \otimes \cdots \otimes T_\gamma^{(n)}$ and the defining expectation factorizes over the independent X_i,

$$T_\gamma H_{\bar{k}} = \prod_{i=1}^n T_\gamma h_{k_i} = \prod_{i=1}^n \gamma^{k_i} h_{k_i} = \gamma^{|\bar{k}|} H_{\bar{k}}.$$

The $\{H_{\bar{k}}\}$ are orthonormal in $L^2(\gamma_n)$ (product of the one-dimensional orthonormality). For $f = \sum_{\bar{k}} \alpha_{\bar{k}} H_{\bar{k}}$ with $\sum_{\bar{k}} \alpha_{\bar{k}}^2 < \infty$, the partial sums converge to f in L^2, and T_γ, being an L^2-contraction (proved at the start of Exercise 5.18, tensored to n dimensions), is L^2-continuous. Thus

$$T_\gamma f = \lim_{N\to\infty} T_\gamma \sum_{|\bar{k}|\le N} \alpha_{\bar{k}} H_{\bar{k}} = \lim_{N\to\infty} \sum_{|\bar{k}|\le N} \gamma^{|\bar{k}|}\alpha_{\bar{k}} H_{\bar{k}} = \sum_{\bar{k}} \gamma^{|\bar{k}|}\alpha_{\bar{k}} \prod_{i=1}^n h_{k_i}$$

in L^2, since $\sum_{\bar{k}} |\gamma^{|\bar{k}|}\alpha_{\bar{k}}|^2 \le \sum_{\bar{k}} \alpha_{\bar{k}}^2 < \infty$. This is the exact Gaussian analog of Theorem 5.16: the Hermite polynomials diagonalize the hypercontractive semigroup. ■

Exercise 5.21

Central idea. The functions f_λ are eigenfunctions of the Ornstein–Uhlenbeck semigroup: $T_\gamma f_\lambda = f_{\lambda\gamma}$. Their $L^p(\gamma)$ norms are explicit Gaussians, so the operator norm ratio $\|T_{e^{-t}}f_\lambda\|_q / \|f_\lambda\|_2$ is $\exp(\lambda^2[(q-1)e^{-2t}-1]/2)$, which blows up as $\lambda \to \infty$ exactly when $q > 1 + e^{2t}$ — the threshold from Exercise 5.18 with $p = 2$.

Solution. Recall $T_\gamma f(y) = \mathbb{E}[f(\gamma y + \sqrt{1-\gamma^2}X)]$ with X standard Gaussian. For $f_\lambda(x) = e^{\lambda x - \lambda^2/2}$,

$$\begin{aligned} T_\gamma f_\lambda(y) &= \mathbb{E}\big[e^{\lambda(\gamma y + \sqrt{1-\gamma^2}X) - \lambda^2/2}\big] \\ &= e^{\lambda\gamma y - \lambda^2/2}\,\mathbb{E}\big[e^{\lambda\sqrt{1-\gamma^2}X}\big] \\ &= e^{\lambda\gamma y - \lambda^2/2}\,e^{\lambda^2(1-\gamma^2)/2} = e^{\lambda\gamma y - \lambda^2\gamma^2/2} = f_{\lambda\gamma}(y). \end{aligned}$$

Since $f_\lambda > 0$, for $p > 1$,

$$\mathbb{E}[|f_\lambda(X)|^p] = \mathbb{E}\big[e^{p\lambda X - p\lambda^2/2}\big] = e^{p^2\lambda^2/2 - p\lambda^2/2} = e^{p(p-1)\lambda^2/2},$$

using $\mathbb{E}[e^{aX}] = e^{a^2/2}$. Hence $\mathbb{E}[|f_\lambda(X)|^p]^{1/p} = e^{(p-1)\lambda^2/2} < \infty$, so each f_λ is admissible.

Take $f = f_\lambda$ with $\lambda \ge 0$ and $\gamma = e^{-t}$. Then $T_{e^{-t}}f_\lambda = f_{\lambda e^{-t}}$, and

$$\frac{\mathbb{E}[|T_{e^{-t}}f_\lambda(X)|^q]^{1/q}}{\mathbb{E}[|f_\lambda(X)|^2]^{1/2}} = \frac{e^{(q-1)(\lambda e^{-t})^2/2}}{e^{\lambda^2/2}} = \exp\!\left(\frac{\lambda^2}{2}\left[(q-1)e^{-2t} - 1\right]\right).$$

If $q > 1 + e^{2t}$, then $(q-1)e^{-2t} - 1 > 0$, and the right-hand side tends to $+\infty$ as $\lambda \to \infty$. Therefore

$$\sup_{f:\,\mathbb{E}[|f(X)|^2]<\infty} \frac{\mathbb{E}[|T_{e^{-t}} f(X)|^q]^{1/q}}{\mathbb{E}[|f(X)|^2]^{1/2}} = \infty \qquad \text{whenever } q > 1 + e^{2t}.$$

Exercise 5.18 with $p = 2$ gives $q(t) = 1 + e^{2t}$ with operator norm at most 1; the computation above shows no larger exponent works. Hence the hypercontractive bounds of Exercises 5.18 and 5.19 are tight. ■

Exercise 5.22

Central idea. With the Rodrigues definition $h_k(x) = \frac{(-1)^k}{\sqrt{k!}} e^{x^2/2} D^k e^{-x^2/2}$, the generating function is the Taylor series of a translated Gaussian (the operator $e^{-\lambda D}$ is translation). Orthonormality follows by differentiating $\mathbb{E}[F_\lambda(X) F_\mu(X)] = e^{\lambda\mu}$ under the expectation and matching Taylor coefficients. The recurrences come from the PDEs $\partial_\lambda F = (x - \lambda)F$ and $\partial_x F = \lambda F$ satisfied by $F(\lambda, x) = e^{\lambda x - \lambda^2/2}$; Christoffel–Darboux is a telescoping sum using the three-term recurrence.

Solution. We use the Rodrigues formula $h_k(x) = \frac{(-1)^k}{\sqrt{k!}} e^{x^2/2} \frac{d^k}{dx^k} e^{-x^2/2}$.

Generating function. For fixed x and $g(y) = e^{-y^2/2}$ (entire),

$$\begin{aligned}\sum_{k=0}^{\infty} \frac{\lambda^k}{\sqrt{k!}} h_k(x) &= e^{x^2/2} \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!}\, g^{(k)}(x) = e^{x^2/2} g(x - \lambda) \\ &= e^{x^2/2} e^{-(x-\lambda)^2/2} = e^{\lambda x - \lambda^2/2},\end{aligned}$$

where the Taylor series converges for all λ since g is entire. Thus, with $F(\lambda, x) = e^{\lambda x - \lambda^2/2}$,

$$F(\lambda, x) = \sum_{k=0}^{\infty} \frac{\lambda^k}{\sqrt{k!}} h_k(x) \qquad \forall\, \lambda, x \in \mathbb{R}. \tag{1}$$

The bivariate identity. Let (X_1, X_2) be Gaussian with standard Gaussian marginals and $\rho = \mathbb{E}[X_1 X_2]$. Then

$$\begin{aligned}\mathbb{E}\big[e^{\lambda X_1 - \lambda^2/2} e^{\mu X_2 - \mu^2/2}\big] &= e^{-(\lambda^2+\mu^2)/2}\, \mathbb{E}\big[e^{\lambda X_1 + \mu X_2}\big] \\ &= e^{-(\lambda^2+\mu^2)/2}\, e^{(\lambda^2 + 2\lambda\mu\rho + \mu^2)/2} = e^{\lambda\mu\rho},\end{aligned}$$

since $\lambda X_1 + \mu X_2 \sim N(0, \lambda^2 + 2\lambda\mu\rho + \mu^2)$.

Orthonormality. Take $X_1 = X_2 = X$ (so $\rho = 1$): $\mathbb{E}[F(\lambda, X) F(\mu, X)] = e^{\lambda\mu}$ for all λ, μ. Differentiating under the expectation is justified by dominated convergence: on compact (λ, μ)-sets, $|\partial_\lambda^i \partial_\mu^j e^{\lambda x - \lambda^2/2 + \mu x - \mu^2/2}| \le C(1 + |x|^{i+j}) e^{C|x|}$, integrable against the Gaussian law. From (1), $\partial_\lambda^i F(\lambda, x)|_{\lambda=0} = \sqrt{i!}\, h_i(x)$, so

$$\sqrt{i!\,j!}\, \mathbb{E}[h_i(X) h_j(X)] = \left.\frac{\partial^{i+j}}{\partial\lambda^i \partial\mu^j}\right|_0 e^{\lambda\mu}.$$

Since $e^{\lambda\mu} = \sum_{k \ge 0} \lambda^k \mu^k / k!$, the right-hand side is $i!$ if $i = j$ and 0 otherwise. Hence $\mathbb{E}[h_i(X) h_j(X)] = \delta_{ij}$.

Three-term recurrences. From $\partial_\lambda F = (x - \lambda)F$: writing $F = \sum_n c_n \lambda^n$ with $c_n = h_n / \sqrt{n!}$,

$$\sum_{n \ge 0} (n+1) c_{n+1} \lambda^n = \sum_{n \ge 0} \big[x c_n - c_{n-1}\big] \lambda^n \quad (c_{-1} := 0),$$

so $(n+1)c_{n+1} = x c_n - c_{n-1}$. Substituting $c_n = h_n/\sqrt{n!}$ and multiplying by $\sqrt{n!}$ gives $\sqrt{n+1}\, h_{n+1} = x h_n - \sqrt{n}\, h_{n-1}$, i.e.

$$x h_n(x) = \sqrt{n+1}\, h_{n+1}(x) + \sqrt{n}\, h_{n-1}(x) \tag{2}$$

(with $h_{-1} := 0$; for $n = 0$ this reads $x = h_1(x)$). From $\partial_x F = \lambda F$:

$$\sum_n \frac{h_n'(x)}{\sqrt{n!}}\lambda^n = \sum_{n\geq 1} \frac{h_{n-1}(x)}{\sqrt{(n-1)!}}\lambda^n,$$

so $h_n'(x) = \sqrt{n}\, h_{n-1}(x)$ for $n \geq 1$ (and $h_0' \equiv 0$). Inserting this into (2) yields

$$x h_n(x) = \sqrt{n+1}\, h_{n+1}(x) + h_n'(x). \tag{3}$$

Christoffel–Darboux. From (2),

$$\begin{aligned}(x-y)h_i(x)h_i(y) &= [xh_i(x)]h_i(y) - h_i(x)[yh_i(y)] \\ &= \sqrt{i+1}[h_{i+1}(x)h_i(y) - h_i(x)h_{i+1}(y)] \\ &\quad - \sqrt{i}[h_i(x)h_{i-1}(y) - h_{i-1}(x)h_i(y)].\end{aligned}$$

With $D_i = \sqrt{i+1}\,[h_{i+1}(x)h_i(y) - h_i(x)h_{i+1}(y)]$ (and $D_{-1} = 0$), this is $D_i - D_{i-1}$. Summing $i = 0$ to $n-1$ telescopes:

$$(x-y)\sum_{i=0}^{n-1} h_i(x)h_i(y) = D_{n-1} = \sqrt{n}\,[h_n(x)h_{n-1}(y) - h_{n-1}(x)h_n(y)],$$

and dividing by $x - y$ gives the formula.

Remark on completeness. Each h_k is a polynomial of exact degree k (leading coefficient $1/\sqrt{k!}$ from Rodrigues), so $\{h_0, \dots, h_n\}$ spans all polynomials of degree $\leq n$. Polynomials are dense in $L_2(\gamma)$: approximate $f \in L_2(\gamma)$ by bounded compactly supported continuous functions, then uniformly on large compacts by polynomials (Weierstrass), with Gaussian tails controlling the remainder. With orthonormality, $\{h_k\}$ is therefore an orthonormal basis of $L_2(\gamma)$. ■

Exercise 5.23

Central idea. As stated this is false: if W is unitary and H Hermitian, WH need not be Hermitian, so it cannot be GUE-distributed. The correct statement is invariance under unitary *conjugation* $H \mapsto WHW^*$. This follows because the GUE density is proportional to $\exp(-\mathrm{Tr}(H^2)/2$, and both the trace of H^2 and Lebesgue measure on Hermitian matrices are preserved by $H \mapsto WHW^*$.

Solution. *Remark on the statement.* The claim that WH is GUE-distributed cannot hold literally: the GUE is supported on Hermitian matrices, but WH is Hermitian only if $WH = HW^*$. For example, with $W = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (unitary) and $H = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ (Hermitian), $WH = \left(\begin{smallmatrix} 0 & 2 \\ 1 & 0 \end{smallmatrix}\right)$ is not Hermitian. The intended statement is invariance under unitary conjugation: WHW^* is GUE whenever H is.

Let H be GUE: $H = H^*$, with independent $H_{ii} \sim N(0,1)$ real and $H_{ij} = (X_{ij} + iY_{ij})/\sqrt{2}$, X_{ij}, Y_{ij} iid $N(0,1)$, for $i < j$. Since $\mathrm{Tr}(H^2) = \sum_i H_{ii}^2 + 2\sum_{i<j} |H_{ij}|^2$, the density of H with respect to Lebesgue measure dH on the real vector space of Hermitian matrices is

$$p(H) = Z^{-1} \exp\!\big(-\tfrac{1}{2}\mathrm{Tr}(H^2)\big)$$

for a normalizing constant Z.

Fix $W \in \mathcal{U}_n$ and consider $\Phi(H) = WHW^*$, a real-linear map on Hermitian matrices. It preserves the Hilbert–Schmidt inner product: $\mathrm{Tr}(\Phi(H_1)\Phi(H_2)) = \mathrm{Tr}(WH_1W^*WH_2W^*) = \mathrm{Tr}(H_1H_2)$. Hence Φ is an orthogonal transformation, so it preserves Lebesgue measure: $d(\Phi(H)) = dH$. Moreover $\mathrm{Tr}(\Phi(H)^2) = \mathrm{Tr}(H^2)$, so $p(\Phi(H)) = p(H)$.

For any bounded Borel test function φ on Hermitian matrices, with the change of variables $K = \Phi(H)$,

$$\mathbb{E}[\varphi(WHW^*)] = \int \varphi(\Phi(H))p(H)\,dH = \int \varphi(K)p(\Phi^{-1}(K))\,dK = \int \varphi(K)p(K)\,dK = \mathbb{E}[\varphi(H)],$$

since $\Phi^{-1} = \Phi_{W^*}$ also preserves $\mathrm{Tr}(K^2)$. Thus $WHW^* \overset{d}{=} H$: the GUE is invariant under unitary conjugation. ■

Exercise 5.24

Central idea. As stated this is false: for the nonzero constant polynomial $p \equiv 1$, $\{x : p(x) = 1\} = \mathbb{R}^n$ has infinite measure. The correct statement is that $\{q = 0\}$ has measure zero for every *nonzero* polynomial q (applied with $q = p - 1$, this covers all $p \not\equiv 1$). The proof is induction on n: write q as a polynomial in x_n with polynomial coefficients in $x' = (x_1, \dots, x_{n-1})$; where the leading coefficient vanishes use the induction hypothesis, and elsewhere each fiber has finitely many roots.

Solution. *Remark on the statement.* If $p \equiv 1$ then $\{x \in \mathbb{R}^n : p(x) = 1\} = \mathbb{R}^n$, which does not have Lebesgue measure 0. We prove the corrected claim: if q is a nonzero n-variate (real) polynomial, then $\{x \in \mathbb{R}^n : q(x) = 0\}$ has Lebesgue measure 0. For the original problem this gives the conclusion whenever $p - 1$ is nonzero, i.e. $p \not\equiv 1$.

We argue by induction on n, using Tonelli's theorem (zero sets of polynomials are closed, hence measurable). For $n = 1$, a nonzero univariate polynomial has finitely many roots, so $\{q = 0\}$ is finite and has measure 0.

Let $n \geq 2$ and write

$$q(x', x_n) = \sum_{k=0}^{d} q_k(x')\, x_n^k, \qquad x' = (x_1, \dots, x_{n-1}),$$

with q_k $(n-1)$-variate polynomials and $q_d \not\equiv 0$. Split $\{q = 0\} = A \cup B$ where

$$A = \{x' : q_d(x') = 0\} \times \mathbb{R}, \qquad B = \{(x', x_n) : q_d(x') \neq 0,\ q(x', x_n) = 0\}.$$

By the induction hypothesis, $N := \{x' : q_d(x') = 0\} \subset \mathbb{R}^{n-1}$ has $(n-1)$-dimensional measure 0. The x'-section of A is $\mathbb{R}$ on N and $\emptyset$ elsewhere, so by Tonelli $\lambda_n(A) = \int_N \infty\, dx' = 0$ (the integral over a null set is 0).

For x' with $q_d(x') \neq 0$, the univariate polynomial $x_n \mapsto q(x', x_n)$ is nonzero of degree at most d, hence has at most d real roots; the x'-section $B_{x'}$ is finite, so $\lambda_1(B_{x'}) = 0$. By Tonelli, $\lambda_n(B) = \int_{\mathbb{R}^{n-1}} \lambda_1(B_{x'})\, dx' = 0$.

Thus $\lambda_n(\{q = 0\}) = 0$, completing the induction. ■

Exercise 5.25

Central idea. If r is a common (possibly complex) root of P and Q, the vector $v = (r^{m+n-1}, \dots, r, 1)^T$ is annihilated by every row of the Sylvester matrix, since each row encodes the coefficients of $x^k P(x)$ or $x^\ell Q(x)$ evaluated at r. Hence $\det S_{P,Q} = 0$. A multiple root of P is a common root of P and P'. For a matrix A, a repeated eigenvalue is a multiple root of its characteristic polynomial χ_A, so $\mathcal{P}(A) = \det S_{\chi_A, \chi_A'}$ — a polynomial in the n^2 entries of A — vanishes exactly in that case.

Solution. Let $P(x) = \sum_{i=0}^{m} a_i x^i$ $(a_m \neq 0)$, $Q(x) = \sum_{j=0}^{n} b_j x^j$ $(b_n \neq 0)$. (The matrix as displayed has n rows built from the a_i's and m rows from the b_j's — i.e. $n-1$ and $m-1$ *further* shifts beyond the first row of each block — for a total of $m + n$ rows.)

Common root ⇒ vanishing discriminant. Let $r \in \mathbb{C}$ with $P(r) = Q(r) = 0$ and put $v = (r^{m+n-1}, r^{m+n-2}, \dots, r, 1)^T \neq 0$. Row k $(1 \leq k \leq n)$ of the a-block is $(0, \dots, 0, a_m, \dots, a_0, 0, \dots, 0)$ with a_m in column k; dotting with v,

$$\sum_{j=0}^{m} a_{m-j}\, r^{m+n-k-j} = r^{n-k} \sum_{i=0}^{m} a_i r^i = r^{n-k} P(r) = 0.$$

Similarly row ℓ $(1 \leq \ell \leq m)$ of the b-block, with b_n in column ℓ, dots with v to give $r^{m-\ell} Q(r) = 0$. Thus $S_{P,Q} v = 0$ with $v \neq 0$, so $D(P, Q) = \det S_{P,Q} = 0$.

Multiple roots. If r is a multiple root of P, i.e. $P(x) = (x - r)^k R(x)$ with $k \geq 2$, then $P(r) = P'(r) = 0$. By the first part applied to P, P', $D(P, P') = 0$.

Detecting repeated eigenvalues. Let $A = (a_{ij})$ be $n \times n$ and $\chi_A(t) = \det(tI - A) = t^n + c_{n-1}(A) t^{n-1} + \cdots + c_0(A)$ its characteristic polynomial; each c_k is a (fixed, integer-coefficient) polynomial in the n^2 entries a_{ij}. Set

$$\mathcal{P}(a_{11}, \dots, a_{nn}) = D(\chi_A, \chi_A') = \det S_{\chi_A, \chi_A'}.$$

Since the entries of $S_{\chi_A,\chi_A'}$ are among the $c_k(A)$ (and 0's and the integers $n, n-1, \ldots$ from differentiating), and the determinant is a polynomial in the matrix entries, $\mathcal{P}$ is an n^2-variate polynomial in the entries of A. If A has an eigenvalue of multiplicity > 1, then χ_A has a multiple (complex) root, so $\mathcal{P}(A) = 0$ by the second part. (For $n = 1$, $\mathcal{P} \equiv 1$, consistent with 1×1 matrices having no repeated eigenvalue; e.g. on diagonal matrices with distinct entries $\mathcal{P} \neq 0$, so $\mathcal{P}$ is not identically zero for $n \geq 2$.) ■

Exercise 5.26

Central idea. A repeated eigenvalue of A is a multiple root of its characteristic polynomial P_A, so by Exercise 5.25 the discriminant $Q(A) = D(P_A, P_A')$ — a polynomial in the entries — vanishes on $\mathcal{H}_n \setminus \mathcal{H}_n^d$. The coefficients of P_A are real polynomials in the real coordinates of $\mathcal{H}_n \cong \mathbb{R}^{n^2}$ (reality follows from $\overline{P_A(\bar{\lambda})} = P_A(\lambda)$ for Hermitian A), hence so is Q; and Q is not identically zero (it is nonzero on $\mathrm{diag}(1, \ldots, n)$). By Exercise 5.24 applied to the nonzero polynomial $Q + 1$, the zero set $\{Q = 0\} = \{Q + 1 = 1\}$ is Lebesgue-null, so $\mathcal{H}_n \setminus \mathcal{H}_n^d$ is null. Since the GUE law has a density with respect to Lebesgue measure on $\mathcal{H}_n$, a GUE matrix has distinct eigenvalues almost surely.

Solution. Identify $\mathcal{H}_n$ with $\mathbb{R}^{n^2}$ via the real diagonal entries and the real and imaginary parts of the strictly upper-triangular entries; Lebesgue measure on $\mathcal{H}_n$ means Lebesgue measure under this identification. The GUE distribution is absolutely continuous with respect to it (independent Gaussian entries), so it suffices to show $\mathcal{H}_n \setminus \mathcal{H}_n^d$ has Lebesgue measure zero.

For $A \in M_n(\mathbb{C})$, let $P_A(\lambda) = \det(A - \lambda I) = \sum_{k=0}^{n} c_k(A)\lambda^k$ with $c_n = 1$. Each c_k is a polynomial with integer coefficients in the entries of A. If $A^* = A$, then

$$\overline{P_A(\bar{\lambda})} = \overline{\det(A - \bar{\lambda} I)} = \det(\bar{A} - \lambda I) = \det(A^T - \lambda I) = P_A(\lambda),$$

using $\bar{A} = A^T$; hence every $c_k(A)$ is real. Writing c_k as a complex polynomial in the real coordinates of $\mathcal{H}_n$ and splitting into real and imaginary parts, the imaginary part is a real polynomial vanishing on all of $\mathbb{R}^{n^2}$, hence identically zero. So each c_k is a real polynomial in the coordinates.

Let $Q(A) = D(P_A, P_A')$ be the discriminant from Exercise 5.25 (determinant of the Sylvester matrix). It is a polynomial with integer coefficients in the c_k, hence a real polynomial in the coordinates of $\mathcal{H}_n$. If A has an eigenvalue of multiplicity exceeding 1, then P_A has a multiple root, so $Q(A) = 0$ by Exercise 5.25. Thus

$$\mathcal{H}_n \setminus \mathcal{H}_n^d \subseteq \{A \in \mathcal{H}_n : Q(A) = 0\}.$$

Q is not the zero polynomial: for $A_0 = \mathrm{diag}(1, 2, \ldots, n)$, P_{A_0} has the distinct roots $1, \ldots, n$, and the standard product formula $D(P, P') = \pm \prod_{i<j} (\lambda_i - \lambda_j)^2$ for monic P gives $Q(A_0) \neq 0$. Hence $Q + 1$ is a nonzero real polynomial, and

$$\{Q = 0\} = \{Q + 1 = 1\}$$

has Lebesgue measure zero by Exercise 5.24. Therefore $\mathcal{H}_n \setminus \mathcal{H}_n^d$ has Lebesgue measure zero, and a random matrix from the GUE — whose law is absolutely continuous with respect to Lebesgue measure on $\mathcal{H}_n$ — has pairwise distinct eigenvalues almost surely. ■

Exercise 5.27

Central idea. For an eigenvalue λ of H, $A = H - \lambda I$ is singular, so $A \operatorname{Adj}(A) = 0$ forces every column of $\operatorname{Adj}(A)$ into the one-dimensional eigenspace $\ker A$; but the diagonal entries $\operatorname{Adj}(A)_{jj} = \det(H^{(j)} - \lambda I)$ are nonzero precisely because λ is not an eigenvalue of any $H^{(j)}$, so the corresponding eigenvector — and hence every column of U — has all entries nonzero. The hypotheses fail only on zero sets of the discriminant and resultant polynomials from Exercise 5.25, hence on a Lebesgue-null set by Exercise 5.24.

Solution. Let $H \in \mathcal{H}_n^d$ share no eigenvalue with any $H^{(k)}$, $1 \leq k \leq n$, and write $H = UDU^*$ with $U \in \mathcal{U}_n$ unitary and $D \in \mathcal{D}_n$ (strictly decreasing diagonal, possible since the eigenvalues are distinct). Fix an eigenvalue λ of H and let u be the corresponding column of U; put $A = H - \lambda I_n$.

Since λ is simple, $\ker A = \operatorname{span}\{u\}$ is one-dimensional. As $\det A = 0$, the adjugate identity $A\operatorname{Adj}(A) = \det(A) I_n = 0$ shows each column of $\operatorname{Adj}(A)$ lies in $\ker A$: for every j there is $\alpha_j \in \mathbb{C}$ with $[\operatorname{Adj}(A)]_{\cdot j} = \alpha_j u$. By the hint, $\operatorname{Adj}(A)_{i,j} = (-1)^{i+j} \det(A^{(ji)})$, so

$$\operatorname{Adj}(A)_{j,j} = \det\left(A^{(jj)}\right) = \det\left(H^{(j)} - \lambda I_{n-1}\right) \neq 0,$$

the inequality because λ is not an eigenvalue of $H^{(j)}$ by hypothesis. Since $\operatorname{Adj}(A)_{j,j} = \alpha_j u_j$, we get $u_j \neq 0$ for every j. As λ was arbitrary, every column of U has all entries nonzero.

It remains that a.e. $H \in \mathcal{H}_n$ satisfies these hypotheses. Identify $\mathcal{H}_n \cong \mathbb{R}^{n^2}$ with Lebesgue measure, and let $p_H(z) = \det(H - zI_n)$.

If H has a repeated eigenvalue, p_H and p_H' share a root, so $D(p_H, p_H') = 0$ by Exercise 5.25. Now $H \mapsto D(p_H, p_H')$ is a polynomial in the coordinates of H (the coefficients of p_H are polynomial in the entries of H, and the discriminant is polynomial in the coefficients), not identically zero: at $H_0 = \operatorname{diag}(1, \dots, n)$, p_{H_0} has n distinct roots, so $D(p_{H_0}, p_{H_0}') \neq 0$ by the resultant product formula. By Exercise 5.24, $\{H : H \text{ has a repeated eigenvalue}\}$ is Lebesgue-null.

Fix k. Then H and $H^{(k)}$ share an eigenvalue iff p_H and $p_{H^{(k)}}$ share a root, iff $\operatorname{Res}(p_H, p_{H^{(k)}}) = D(p_H, p_{H^{(k)}}) = 0$ by Exercise 5.25. This is again a polynomial in the coordinates of H, not identically zero: with distinct $d_1, \dots, d_{n-1}$ and H the arrowhead matrix with $H^{(k)} = D := \operatorname{diag}(d_1, \dots, d_{n-1})$, $H_{kk} = 0$ and $H_{ik} = H_{ki} = 1$ for $i \neq k$ (the spike in position k). Then $H^{(k)}$ has eigenvalues $\{d_j\}$, while $\det(H - d_j I_n) = -\prod_{i \neq j}(d_i - d_j) \neq 0$ for each j (arrowhead determinant, by permutation-similarity to the spike-at-n case), so the resultant is nonzero at H. By Exercise 5.24, $\{H : \sigma(H) \cap \sigma(H^{(k)}) \neq \emptyset\}$ is null.

Thus a.e. $H \in \mathcal{H}_n$ has distinct eigenvalues and shares no eigenvalue with any $H^{(k)}$. For such H the first part gives $H = UDU^*$ with $D \in \mathcal{D}_n^d$ and U entrywise nonzero; adjusting each column's phase (which preserves UDU^*) makes the diagonal entries real positive, i.e. $U \in \mathcal{U}_n^g$ with nonzero entries. With the book's definition of a minor ($U^{(ij)}$: delete row i and column j), all minors of U are then invertible: since $U^{-1} = U^*$ and $U^{-1} = \operatorname{Adj}(U)/\det U$, we have $(-1)^{i+j} \det U^{(ij)} = \operatorname{Adj}(U)_{ji} = \det(U)\,\overline{U_{ij}}$, and $|\det U| = 1$, so

$$|\det U^{(ij)}| = |U_{ij}| \neq 0 \qquad (1 \leq i, j \leq n).$$

Hence $U \in \mathcal{U}_n^{vg}$ and $H \in \mathcal{H}_n^{d,g}$. So $\mathcal{H}_n \setminus \mathcal{H}_n^{d,g}$ lies in a finite union of null sets, and it has Lebesgue measure 0.

Remark (all square minors). Inverting T in Exercise 5.28 uses the leading principal blocks of U. So there one wants the stronger reading of $\mathcal{U}_n^{vg}$ used by Anderson–Guionnet–Zeitouni: every square submatrix U_{IJ} ($I, J \subseteq \{1, \dots, n\}$, $|I| = |J| = m$) is invertible. This is still almost sure, and the polynomial method proves it as follows.

(a) Adjugate as a projector. Let $H \in \mathcal{H}_n^d$ with eigenvalues $\lambda_1 > \cdots > \lambda_n$ and unit eigenvectors $u_1, \dots, u_n$ (the columns of U). Put $c_j = \prod_{l \neq j}(\lambda_l - \lambda_j) \neq 0$. Then $\operatorname{Adj}(H - \lambda_j I) = c_j\, u_j u_j^*$. Indeed, $A = H - \lambda_j I$ is Hermitian, so $\operatorname{Adj}(A)$ is Hermitian: $(A^{(ji)})^* = A^{(ij)}$ gives $\det A^{(ji)} = \overline{\det A^{(ij)}}$. By the first part its columns lie in $\ker A = \operatorname{span}\{u_j\}$, so $\operatorname{Adj}(A) = \kappa\, u_j u_j^*$ for some real κ. Taking traces, $\kappa = \operatorname{tr}\operatorname{Adj}(A) = \sum_k \det A^{(k)}$. This is the elementary symmetric function e_{n-1} of the eigenvalues $\lambda_l - \lambda_j$ of A. Only the product omitting the zero eigenvalue survives, so $\kappa = c_j$.

(b) A polynomial detecting the bad set. Fix I with $|I| = m$. For an m-subset $J' \subseteq \{1, \dots, n\}$, put $P_{J'} = \sum_{j \in J'} u_j u_j^*$. Then $(P_{J'})_{II} = U_{IJ'} U_{IJ'}^*$, so $\det(P_{J'})_{II} = |\det U_{IJ'}|^2$. (Changing column phases does not affect this, so the condition $\det U_{IJ'} \neq 0$ depends only on H.) Treat $\lambda_1, \dots, \lambda_n$ as indeterminates and define

$$Q^{J'} = \sum_{j \in J'} \Big(\prod_{k \in J' \setminus \{j\}} c_k \Big) \operatorname{Adj}(H - \lambda_j I), \qquad F = \prod_{J'} \det\left(Q^{J'}_{II}\right),$$

with the product over all m-subsets J'. Each entry of $Q^{J'}$ is a polynomial in the real coordinates of H and in $\lambda_1, \dots, \lambda_n$ (complex coefficients). A permutation σ of the indeterminates sends $c_k \mapsto c_{\sigma(k)}$, hence

$Q^{J'} \mapsto Q^{\sigma(J')}$, so it permutes the factors of F. Thus F is symmetric in λ. By the fundamental theorem on symmetric polynomials (valid over any commutative ring), $F = G(e_1(\lambda), \dots, e_n(\lambda))$ for a polynomial G whose coefficients are polynomials in the coordinates of H. Substitute the true eigenvalues: $e_k(\lambda(H))$ is, up to sign, a coefficient of p_H, hence a polynomial in the coordinates of H. So $\mathcal{F}(H) := F|_{\lambda=\lambda(H)}$ is a polynomial in the n^2 real coordinates of H. On $\mathcal{H}_n^d$, (a) gives $Q^{J'} = (\prod_{k\in J'} c_k) P_{J'}$, and therefore

$$\mathcal{F}(H) = \prod_{J'} \Big(\prod_{k\in J'} c_k \Big)^m |\det U_{IJ'}|^2.$$

So for $H \in \mathcal{H}_n^d$: $\mathcal{F}(H) = 0$ iff $\det U_{IJ'} = 0$ for some J'.

(c) $\mathcal{F} \not\equiv 0$. Let A be the $m \times n$ real matrix $A_{rb} = b^{\,r-1}$. Its $m \times m$ minors are Vandermonde determinants with distinct nodes, hence nonzero. Gram–Schmidt on the (independent) rows gives $A = LQ$ with L invertible and the rows of Q orthonormal. Then every $m \times m$ minor satisfies $\det Q_{\cdot J'} = \det(A_{\cdot J'})/\det L \neq 0$. Complete the rows of Q to an orthonormal basis of $\mathbb{R}^n$. This gives an orthogonal W whose rows indexed by I are the rows of Q, so every $W_{IJ'}$ is invertible. Take $H_0 = W \operatorname{diag}(n, \dots, 1) W^* \in \mathcal{H}_n^d$. Its eigenvector matrix equals W up to column phases, so $\mathcal{F}(H_0) \neq 0$. (Checked numerically for $2 \le n \le 7$, all $m < n$.)

Since $\mathcal{F} \not\equiv 0$, its real or imaginary part is a nonzero real polynomial vanishing on $\{\mathcal{F} = 0\}$, and Exercise 5.24 makes that set null. There are finitely many I and J, and $\mathcal{H}_n \setminus \mathcal{H}_n^d$ is null. So for a.e. H, every square submatrix of U is invertible. The conclusion $|\mathcal{H}_n \setminus \mathcal{H}_n^{d,g}| = 0$ therefore holds under either reading of "all minors".

Exercise 5.28

Central idea. A matrix $H \in \mathcal{H}_n^{d,g}$ has a unique decomposition $H = UDU^*$ with $D \in \mathcal{D}_n^d$ (eigenvalues sorted decreasingly) and $U \in \mathcal{U}_n^{vg}$ (eigenvector phases pinned by the real-positive diagonal, possible since the diagonal entries are nonzero). The ratios $t_{ij} = U_{ij}/U_{ii}$ for $i < j$ then determine U completely: reconstructing the rows one by one, orthogonality to the earlier rows yields at step i a linear system whose coefficient matrix is (conjugate to) a leading principal minor of U — hence invertible by the very-generic hypothesis — while $U_{ii} > 0$ and unit norm fix the remaining scale. This reconstruction is a smooth inverse to T, and since every nonvanishing condition it uses is open, $T(\mathcal{U}_n^{vg})$ is open in $\mathbb{R}^{n(n-1)}$, so $H \mapsto (D, T(U))$ is a diffeomorphism onto its open image.

Solution. We use the definitions on p. 161 of the book: $\mathcal{U}_n^g$ is the set of $n \times n$ unitaries with real positive diagonal entries; $\mathcal{U}_n^{vg} \subset \mathcal{U}_n^g$ is the subset whose minors are all invertible; $\mathcal{D}_n^d$ is the set of real diagonal matrices with strictly decreasing diagonal entries; $\mathcal{H}_n^{d,g}$ is the set of Hermitian matrices admitting a decomposition UDU^* with $U \in \mathcal{U}_n^{vg}$ and $D \in \mathcal{D}_n^d$. Write $t_{ij} = U_{ij}/U_{ii} \in \mathbb{C}$ for $i < j$ (a pair of real numbers), so that $T : \mathcal{U}_n^{vg} \to \mathbb{R}^{n(n-1)}$; T is smooth since $U_{ii} > 0$.

Uniqueness of the decomposition. Let $H = UDU^* = VEV^*$ with $U, V \in \mathcal{U}_n^{vg}$ and $D, E \in \mathcal{D}_n^d$. The diagonal entries of D (resp. E) are the eigenvalues of H in strictly decreasing order, so $D = E = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1 > \cdots > \lambda_n$. Then $(V^*U)D = D(V^*U)$, and since D has distinct diagonal entries, V^*U is diagonal; being unitary it is a diagonal unitary D'. From $U = VD'$ we get $D'_{ii} = U_{ii}/V_{ii} > 0$ (a ratio of positive reals), and with $|D'_{ii}| = 1$ this forces $D'_{ii} = 1$ for all i, so $U = V$. Hence the decomposition is unique and

$$\Phi : \mathcal{H}_n^{d,g} \to \mathcal{D}_n^d \times \mathbb{R}^{n(n-1)}, \qquad \Phi(H) = (D, T(U)),$$

is well defined.

T is one-to-one on $\mathcal{U}_n^{vg}$. Let $U, V \in \mathcal{U}_n^{vg}$ with $T(U) = T(V) =: t$. We show the rows of U are forced by t. Write r_i for the i-th row; $(r_i)_j = t_{ij}(r_i)_i$ for $j > i$ with $(r_i)_i = U_{ii} > 0$. For $i = 1$, $r_1 = U_{11}(1, t_{12}, \dots, t_{1n})$ and $\|r_1\| = 1$ give $U_{11} = (1 + \sum_{j=2}^n |t_{1j}|^2)^{-1/2}$, so r_1 is determined by t. Suppose $r_1, \dots, r_{i-1}$ are determined and write $r_i = \phi_i(x)$ with $\phi_i(x_1, \dots, x_i) = (x_1, \dots, x_i, t_{i,i+1}x_i, \dots, t_{in}x_i)$ and $x_i = U_{ii} > 0$. Orthogonality $\langle r_i, r_k \rangle = 0$ for $k < i$ reads

$$\sum_{j=1}^{i-1} \overline{(r_k)_j}\, x_j + \left(\overline{(r_k)_i} + \sum_{j=i+1}^{n} t_{ij} \overline{(r_k)_j} \right) x_i = 0,$$

i.e. $Ax' + x_i c = 0$ with $A_{kj} = \overline{(r_k)_j}$ for $1 \le k, j \le i-1$. But A is the conjugate of the leading $(i-1)\times(i-1)$ principal submatrix of U, which is invertible because $U \in \mathcal{U}_n^{vg}$. Hence $x' = -x_i A^{-1}c$, i.e. $x = x_i v$ with $v = (-A^{-1}c, 1)$. Since $\phi_i(v) \neq 0$ (its i-th coordinate is 1) and $\phi_i(x) = x_i\phi_i(v)$, unit norm and $x_i > 0$ force $x_i = \|\phi_i(v)\|^{-1}$ and $x = \|\phi_i(v)\|^{-1}v$. Thus r_i is determined by t and the previous rows. By induction every row is determined by t, so $U = V$: T is injective on $\mathcal{U}_n^{vg}$.

$T(\mathcal{U}_n^{vg})$ is open, with smooth inverse. Fix $t^0 = T(U^0)$, $U^0 \in \mathcal{U}_n^{vg}$. The reconstruction above expresses each row as a rational function of t, with denominators the (conjugates of the) leading principal minors of the partially reconstructed matrix and the positive norms $\|\phi_i(v)\|$. At t^0 these minors are those of U^0, hence nonzero; by continuity they stay nonzero on a neighborhood Ω_0 of t^0, on which the reconstruction defines a smooth map $\tilde{S}$ with $\tilde{S}(t^0) = U^0$, $\tilde{S}(t)$ unitary with real positive diagonal entries, and $T(\tilde{S}(t)) = t$. Since U^0 has all minors invertible and there are finitely many minors, shrinking to $\Omega_1 \subset \Omega_0$ we may assume all minors of $\tilde{S}(t)$ are invertible for $t \in \Omega_1$, i.e. $\tilde{S}(t) \in \mathcal{U}_n^{vg}$ and $t = T(\tilde{S}(t)) \in T(\mathcal{U}_n^{vg})$. Hence every $t^0 \in T(\mathcal{U}_n^{vg})$ has a neighborhood contained in $T(\mathcal{U}_n^{vg})$: the image is open in $\mathbb{R}^{n(n-1)}$. On $\Omega_1 \subset T(\mathcal{U}_n^{vg})$, $\tilde{S}$ coincides with the set-theoretic inverse S of T (the preimage in $\mathcal{U}_n^{vg}$ is unique by injectivity), so $S : T(\mathcal{U}_n^{vg}) \to \mathcal{U}_n^{vg}$ is smooth near t^0; t^0 being arbitrary, S is smooth.

The diffeomorphism. Put $J(D,t) = S(t)DS(t)^*$ for $(D,t) \in \mathcal{D}_n^d \times T(\mathcal{U}_n^{vg})$. Then J takes values in $\mathcal{H}_n^{d,g}$, is smooth (as S is), and $\Phi(J(D,t)) = (D,t)$ by uniqueness of the decomposition, while $J(\Phi(H)) = H$ since $S(T(U)) = U$. Thus $\Phi : \mathcal{H}_n^{d,g} \to \mathcal{D}_n^d \times T(\mathcal{U}_n^{vg})$ is bijective with inverse J. It remains to see Φ is smooth. Let $H_0 = U_0 D_0 U_0^* \in \mathcal{H}_n^{d,g}$. For H Hermitian near H_0 the eigenvalues stay distinct; let $\lambda_1(H) > \cdots > \lambda_n(H)$ be the decreasing eigenvalues and $P_i(H)$ the spectral projections, both smooth in H by standard perturbation theory for simple eigenvalues (Riesz formula for P_i). With e_i the i-th standard basis vector, $u_i(H) = P_i(H)e_i/\|P_i(H)e_i\|$ is smooth near H_0 since $P_i(H_0)e_i = (U_0)_{ii}u_i^0 \neq 0$, where u_i^0 is the i-th column of U_0 and $(U_0)_{ii} > 0$. Since $(u_i(H))_i = e_i^* P_i(H)e_i/\|P_i(H)e_i\| = \|P_i(H)e_i\| > 0$ is real, the matrix $U(H)$ with columns $u_i(H)$ is exactly the normalized diagonalizing unitary, smooth in H, and $\Phi(H) = (\mathrm{diag}(\lambda_1(H), \dots, \lambda_n(H)), T(U(H)))$ is smooth near H_0. As H_0 was arbitrary, Φ is smooth. (The same local construction shows $\mathcal{H}_n^{d,g}$ is open in $\mathcal{H}_n$: for H near H_0, $U(H)$ is close to U_0, so its minors stay invertible and $H \in \mathcal{H}_n^{d,g}$.)

Therefore Φ is a diffeomorphism from $\mathcal{H}_n^{d,g}$ onto $\mathcal{D}_n^d \times T(\mathcal{U}_n^{vg})$, an open subset of $\mathcal{D}_n^d \times \mathbb{R}^{n(n-1)}$ of real dimension $n + n(n-1) = n^2$. This is the argument of Anderson–Guionnet–Zeitouni (2010, Lemma 2.5.5).

Remark. The image is $\mathcal{D}_n^d \times T(\mathcal{U}_n^{vg})$, and $T(\mathcal{U}_n^{vg}) \subsetneq \mathbb{R}^{n(n-1)}$ in general: e.g. for $n = 2$ the reconstruction gives $U = (1+|t|^2)^{-1/2}\begin{pmatrix} 1 & t \\ -\bar{t} & 1 \end{pmatrix}$, whose off-diagonal entries vanish iff $t = 0$, so $T(\mathcal{U}_2^{vg}) = \mathbb{C} \setminus \{0\}$. The "diffeomorphism with $\mathcal{D}_n^d \times \mathbb{R}^{n(n-1)}$" in the statement is thus to be understood as a diffeomorphism onto the open image in that ambient space.

Exercise 5.29

Central idea. All identities come from differentiating $H = U\Lambda U^*$ (with $U^*U = I$ and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ independent of the p_ℓ): unitarity forces S_ℓ skew-Hermitian, the product rule gives B_ℓ as a commutator, and $\partial H/\partial\lambda_i = u_i u_i^*$ because only the i-th eigenvalue moves.

Solution. Write $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ and $H = U\Lambda U^*$, where U depends on $p = (p_1, \dots, p_{n(n-1)})$ but not on Λ.

S_ℓ is skew-Hermitian. Differentiating $U^*U = I$ with respect to p_ℓ,

$$0 = \frac{\partial U^*}{\partial p_\ell}U + U^*\frac{\partial U}{\partial p_\ell} = S_\ell^* + S_\ell,$$

so $S_\ell^* = -S_\ell$.

The commutator formula. Since Λ is independent of p_ℓ,

$$\frac{\partial H}{\partial p_\ell} = \frac{\partial U}{\partial p_\ell}\Lambda U^* + U\Lambda\frac{\partial U^*}{\partial p_\ell}.$$

Conjugating by U,

$$U^*\frac{\partial H}{\partial p_\ell}U = U^*\frac{\partial U}{\partial p_\ell}\Lambda + \Lambda\frac{\partial U^*}{\partial p_\ell}U = S_\ell\Lambda + \Lambda S_\ell^* = S_\ell\Lambda - \Lambda S_\ell,$$

using $S_\ell^* = -S_\ell$. Entry-wise, $(S_\ell\Lambda)[i,j] = S_\ell[i,j]\lambda_j$ and $(\Lambda S_\ell)[i,j] = \lambda_i S_\ell[i,j]$, so with $B_\ell = U^*\frac{\partial H}{\partial p_\ell}U$,

$$B_\ell[i,j] = S_\ell[i,j](\lambda_j - \lambda_i), \tag{5.7}$$

for all i, j.

Derivative in λ_i. Write $H = \sum_{k=1}^n \lambda_k u_k u_k^*$ where u_k is the k-th column of U. Since U is independent of Λ,

$$\frac{\partial H}{\partial \lambda_i} = u_i u_i^*.$$

This is the orthogonal projection onto $\mathrm{span}(u_i)$: $(u_iu_i^*)^* = u_iu_i^*$ and $(u_iu_i^*)^2 = u_i(u_i^*u_i)u_i^* = u_iu_i^*$ as $u_i^*u_i = 1$.

Conjugating by U: since u_i is the i-th column of U, $U^*u_i = e_i$, so

$$U^*\frac{\partial H}{\partial \lambda_i}U = (U^*u_i)(u_i^*U) = e_ie_i^* = \mathrm{diag}(\underbrace{0,\dots,0}_{i-1},1,\underbrace{0,\dots,0}_{n-i}).$$

Entry-wise, $(U^*AU)[j,k] = \sum_{j',k'} \bar U[j',j]\,A[j',k']\,U[k',k]$ for any A, so with $A = \partial H/\partial\lambda_i$,

$$\left(U^*\frac{\partial H}{\partial \lambda_i}U\right)[j,k] = \sum_{j'=1}^n\sum_{k'=1}^n \frac{\partial H}{\partial\lambda_i}[j',k']\,\bar U[j',j]\,U[k',k] = \mathbb{1}_{\{i=j=k\}}, \tag{5.8}$$

for $1 \le j,k \le n$. ■

Exercise 5.30

Central idea. Right-multiplying $\mathrm{Jac}(J)$ by the real matrix M of the isometry $A \mapsto U^*AU$ block-diagonalizes $C = \mathrm{Jac}(J)M$: the λ-rows give I_n by (5.8), and each column of the p-block carries a factor $(\lambda_j - \lambda_i)$ by (5.7), so $\det C$ is $\Delta(\lambda)^2$ times a function of U alone. (The M printed in the exercise is wrong; see the Erratum. The proof below uses the exact real matrix of $A \mapsto U^*AU$.)

Erratum. The statement is wrong as printed: the displayed M is not the real matrix of $A \mapsto U^*AU$, so with it the claims of part 1 fail. For $j' < k'$, the true coefficients are, in the $\mathrm{Re}\,A[j',k']$ rows, Re/Im of $\bar U[j',j]U[k',k] + \bar U[k',j]U[j',k]$ (output entry $[j,k]$, $j \le k$), and in the $\mathrm{Im}\,A[j',k']$ rows, Re/Im of $i(\bar U[j',j]U[k',k] - \bar U[k',j]U[j',k])$. The printed M has two independent defects in the (j',k') rows: (a) in the first column block ($k = j$) it omits the factor 2 (printed $\mathrm{Re}\,\bar U[j',j]U[k',j]$ and $-\mathrm{Im}\,\bar U[j',j]U[k',j]$; true 2Re and -2Im); (b) in the off-diagonal column blocks ($j < k$) it doubles the first term, $2\bar U[j',j]U[k',k]$, in place of $\bar U[j',j]U[k',k] \pm \bar U[k',j]U[j',k]$. (The diagonal rows are correct.) Counterexample: $n = 2$, $U = 2^{-1/2}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$. Then the printed M gives, for $i = 1$, $C^{1,1}[1,\cdot] = (3/4,\ 1/4)$ and $C^{1,2}[1,1] = -1/2$ instead of $(1,0)$ and 0. Defect (a) alone accounts for $C^{1,1}$; defect (b) alone produces $C^{1,2}[1,1] = -1/2$: the $(\mathrm{Re}\,A[1,2], \mathrm{Re}\,[1,2])$ entry is printed $2\mathrm{Re}\,\bar U[1,1]U[2,2] = -1$ against the true 0. Doubling only the first column block restores $C^{1,1} = \mathrm{Id}_2$ but leaves $C^{1,2}[1,1] = -1/2$, and for a generic complex unitary U with $n = 3$ it leaves $C^{1,2}, C^{1,3} \neq 0$ (checked numerically). For the Hadamard U above, $\det M_{\text{printed}} = 1/2$, while the true matrix has $\det M = 1$. Corrected statement: with M the exact real matrix of $A \mapsto U^*AU$ (described above and in the Solution), parts 1–4 hold as stated; this is what is proved below.

Solution. Let M be the $n^2 \times n^2$ real matrix of the linear isometry $\Phi(A) = U^*AU$ on $\mathcal{H}_n$ in real coordinates (diagonal; Re/Im of upper triangle). Explicitly, with A Hermitian,

$$(U^*AU)[j,j] = \sum_a |U[a,j]|^2 A[a,a] + 2\sum_{a<b} \mathrm{Re}\big(A[a,b]\,\bar{U}[a,j]U[b,j]\big)\,,$$

so the first column block of M is $|U[a,j]|^2$, $2\mathrm{Re}\,\bar{U}[a,j]U[b,j]$, $-2\mathrm{Im}\,\bar{U}[a,j]U[b,j]$; similarly $(U^*AU)[j,k]$, $j<k$, gives the off-diagonal column blocks $\bar{U}[a,j]U[a,k]$ (diagonal rows) and $\bar{U}[a,j]U[b,k] + \bar{U}[b,j]U[a,k]$, $i(\bar{U}[a,j]U[b,k] - \bar{U}[b,j]U[a,k])$ (real and imaginary rows; take Re/Im). These are the corrections to the printed M (Erratum). Since Φ is a real-linear bijection, M is invertible with $\det M$ depending only on U.

1. For $A = \partial H/\partial\lambda_i$, $(U^*AU)[j,k] = \mathbb{1}_{\{i=j=k\}}$ by (5.8). Hence with the exact M,

   $$C^{1,1}[i,j] = (U^*\tfrac{\partial H}{\partial\lambda_i}U)[j,j] = \delta_{ij},$$

   i.e. $C^{1,1} = \mathrm{Id}_n$, and for $m = r(j,k)$, $j<k$,

   $$C^{1,2}[i,m] = \mathrm{Re}\,(U^*\tfrac{\partial H}{\partial\lambda_i}U)[j,k] = 0, \qquad C^{1,3}[i,m] = \mathrm{Im}\,(U^*\tfrac{\partial H}{\partial\lambda_i}U)[j,k] = 0.$$

2. For $A = \partial H/\partial p_\ell$ and $m = r(i,j)$, $i<j$, with $B_\ell = U^*AU$,

   $$C^{2,2}[\ell,m] = \mathrm{Re}\,B_\ell[i,j], \qquad C^{2,3}[\ell,m] = \mathrm{Im}\,B_\ell[i,j],$$

   and by (5.7), $B_\ell[i,j] = S_\ell[i,j](\lambda_j - \lambda_i)$. (The diagonal block $C^{2,1}[\ell,j] = B_\ell[j,j] = 0$ since $\lambda_j - \lambda_j = 0$.)

3. Thus $C = \left(\begin{smallmatrix} \mathrm{Id}_n & 0 \\ 0 & [C^{2,2}\ C^{2,3}] \end{smallmatrix}\right)$ is block diagonal, and $\det C = \det[C^{2,2}\ C^{2,3}]$. Each of the $n(n-1)$ columns of $[C^{2,2}\ C^{2,3}]$ (indexed by $m = r(i,j)$, $i<j$, in each half) carries the factor $(\lambda_j - \lambda_i)$, so

   $$\det[C^{2,2}\ C^{2,3}] = \prod_{1\le i<j\le n} (\lambda_j - \lambda_i)^2 \cdot g(U) = \Delta(\lambda_1,\dots,\lambda_n)^2\, g(U),$$

   where g depends on the $S_\ell[i,j]$ (hence on U) but not on λ. Since $\det C = \det(\mathrm{Jac}(J))\det(M)$,

   $$\det(\mathrm{Jac}(J)) = \Delta(\lambda_1,\dots,\lambda_n)^2 \cdot f(U), \qquad f(U) = g(U)/\det(M),$$

   a product of the squared Vandermonde determinant and a function of U alone.

4. Lemma 5.24 follows: by Exercise 5.28, $(\Lambda, p) \mapsto H = U\Lambda U^*$ is a diffeomorphism onto $\mathcal{H}_n^{d,g}$; the change-of-variables formula with the Jacobian just computed gives the joint density of $(\lambda_1,\dots,\lambda_n)$ as $\Delta(\lambda)^2$ times the pushforward density, and the U-dependent factor integrates to a constant independent of λ (by unitary invariance of the GUE law), yielding the eigenvalue density proportional to $\prod_{i<j}(\lambda_i - \lambda_j)^2$ times the radial part. ■

Exercise 5.31

Central idea. Write the Vandermonde determinant as $\det(x_i^{j-1})$ and use column operations to replace the monomials by monic Hermite polynomials (the book's orthonormal h_j rescaled), which are orthogonal under the standard Gaussian. Expanding the squared determinant, orthogonality kills every cross term; only the $n!$ diagonal terms survive, each contributing $\prod_{j=0}^{n-1} j!$. (Note: as written the stated identity is false — see the remark at the end — the correct value carries an extra $n!$; the claimed right-hand side is the integral over the ordered simplex $\{x_1 > \cdots > x_n\}$.)

Solution. Let γ be the standard Gaussian law on $\mathbb{R}$ and

$$I_n = \int_{\mathbb{R}^n} \Delta(x_1,\dots,x_n)^2\, \gamma^{\otimes n}(dx), \qquad \Delta(x) = \prod_{1\le i<j\le n}(x_i - x_j) = \det(x_i^{j-1})_{i,j=1}^n$$

up to a sign that disappears after squaring. Let h_j be the normalized (orthonormal) Hermite polynomials of Theorem 5.23, so $\int h_i h_j \, d\gamma = \delta_{ij}$, and let p_j be the monic Hermite polynomial of degree j. Since h_j has leading coefficient $1/\sqrt{j!}$ (it is $He_j/\sqrt{j!}$ with He_j monic), $p_j = \sqrt{j!}\, h_j$ and

$$\int p_i p_j \, d\gamma = j!\, \delta_{ij}.$$

Replacing, in the determinant, each column $(x_1^{j-1}, \dots, x_n^{j-1})^\top$ by $(p_{j-1}(x_1), \dots, p_{j-1}(x_n))^\top$ is a sequence of column operations (subtracting multiples of earlier columns), so

$$\Delta(x) = \det\left(p_{j-1}(x_i)\right)_{i,j=1}^{n}.$$

Expanding the determinant,

$$\Delta(x) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} p_{\sigma(i)-1}(x_i),$$

hence

$$\Delta(x)^2 = \sum_{\sigma,\tau \in S_n} \operatorname{sgn}(\sigma\tau) \prod_{i=1}^{n} p_{\sigma(i)-1}(x_i) p_{\tau(i)-1}(x_i).$$

Integrating against $\gamma^{\otimes n}$ and using independence across coordinates,

$$I_n = \sum_{\sigma,\tau} \operatorname{sgn}(\sigma\tau) \prod_{i=1}^{n} \langle p_{\sigma(i)-1}, p_{\tau(i)-1} \rangle_{L^2(\gamma)}.$$

By orthogonality, the i-th factor vanishes unless $\sigma(i) = \tau(i)$; the product over i is nonzero only when $\sigma = \tau$. For $\sigma = \tau$, $\operatorname{sgn}(\sigma\tau) = 1$ and

$$\prod_{i=1}^{n} \langle p_{\sigma(i)-1}, p_{\sigma(i)-1} \rangle = \prod_{i=1}^{n} (\sigma(i)-1)! = \prod_{j=0}^{n-1} j!,$$

independent of σ. With $n!$ permutations,

$$I_n = n! \prod_{j=0}^{n-1} j!.$$

Remark (correction to the stated identity). The identity as written is false: for $n = 2$, $\Delta(x_1, x_2) = x_2 - x_1$ and $I_2 = \mathbb{E}[(X_1 - X_2)^2] = 2$ for X_1, X_2 i.i.d. standard Gaussian, whereas $\prod_{j=0}^{1} j! = 0! \cdot 1! = 1$. The claimed value $\prod_{j=0}^{n-1} j!$ is instead the integral over the ordered region $\{x_1 > \cdots > x_n\}$ (the region of Lemma 5.24), since Δ^2 is permutation-symmetric and the $n!$ orderings partition $\mathbb{R}^n$ up to a null set.

Exercise 5.32

Central idea. Substitute $x = \sin\theta$ to turn the moment integral into a difference of sine-power integrals; the sine reduction formula then yields both hinted identities, and the Wallis value $I_{2k} = \pi(2k)!/(2^{2k}(k!)^2)$ produces the Catalan number.

Solution. With the semi-circular density,

$$m_{2k} = \frac{2}{\pi} \int_{-1}^{1} x^{2k} \sqrt{1 - x^2} \, dx.$$

Put $x = \sin\theta$, $dx = \cos\theta \, d\theta$; on $[-\pi/2, \pi/2]$, $\sqrt{1 - x^2} = \cos\theta \geq 0$, so

$$m_{2k} = \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin^{2k}\theta \cos^2\theta \, d\theta.$$

Writing $\cos^2\theta = 1-\sin^2\theta$ and $I_m = \int_{-\pi/2}^{\pi/2}\sin^m\theta\,d\theta$,

$$m_{2k} = \frac{2}{\pi}(I_{2k} - I_{2k+2}).$$

For $m \geq 2$, integration by parts gives the reduction formula $I_m = \frac{m-1}{m} I_{m-2}$. Hence $I_{2k+2} = \frac{2k+1}{2k+2} I_{2k}$, and

$$m_{2k} = \frac{2}{\pi} I_{2k}\left(1 - \frac{2k+1}{2k+2}\right) = \frac{2}{\pi(2k+2)} I_{2k},$$

the second hinted identity. Since $I_{2k} = \frac{2k+2}{2k+1} I_{2k+2}$, this also gives

$$m_{2k} = \frac{2}{\pi(2k+1)} I_{2k+2},$$

the first hinted identity. Iterating the reduction formula,

$$I_{2k} = \frac{2k-1}{2k}\cdot\frac{2k-3}{2k-2}\cdots\frac{1}{2}\cdot I_0, \qquad I_0 = \pi.$$

Now $\frac{1\cdot 3\cdots(2k-1)}{2\cdot 4\cdots(2k)} = \frac{(2k)!}{2^{2k}(k!)^2}$, since $(2k)! = (2^k k!)(1\cdot 3\cdots(2k-1))$. Thus $I_{2k} = \pi\frac{(2k)!}{2^{2k}(k!)^2}$, and

$$m_{2k} = \frac{2}{\pi(2k+2)}\cdot\pi\frac{(2k)!}{2^{2k}(k!)^2} = \frac{2(2k)!}{(2k+2)2^{2k}(k!)^2} = \frac{(2k)!}{(k+1)2^{2k}(k!)^2} = \frac{C_k}{2^{2k}},$$

where $C_k = \frac{1}{k+1}\binom{2k}{k}$ is the k-th Catalan number.

Exercise 5.33

Central idea. Parametrize the GUE by a standard Gaussian vector $g \in \mathbb{R}^{n^2}$ via the linear isometry $g \mapsto H(g)$ (possible because the book's GUE has $\mathbb{E}|H_{ij}|^2 = \sigma^2 = 1/(4n)$ for all i, j). For fixed 1-Lipschitz f, the map $g \mapsto L_n(f) = \frac{1}{n}\sum_i f(\lambda_i(H(g)))$ is $(1/2n)$-Lipschitz: f contributes the factor 1, Cauchy–Schwarz and Lidskii's inequality (Exercise 3.16) turn $\frac{1}{n}\sum_i|\lambda_i - \lambda_i'|$ into $\frac{1}{\sqrt{n}}\|H - H'\|_{\mathrm{HS}}$, and the parametrization contributes $\sigma = 1/\sqrt{4n}$. Theorem 5.6 (applied to $\pm F$) then gives a tail bound $2e^{-2n^2t^2}$, which implies the claim uniformly in $f \in \mathcal{B}$.

Solution. Recall from Section 5.10 that H is from the GUE if it is Hermitian with independent diagonal entries $H_{ii} \sim N(0,\sigma^2)$ and independent off-diagonal entries H_{ij}, $i<j$, whose real and imaginary parts are independent $N(0,\sigma^2/2)$, where $\sigma^2 = 1/(4n)$. Identify the space of $n\times n$ Hermitian matrices with $\mathbb{R}^{n^2}$ via

$$H_{ii} = \sigma g_i, \qquad H_{ij} = \frac{\sigma}{\sqrt{2}}(g_{ij}^R + i g_{ij}^I)\ \ (i<j), \qquad H_{ji} = \overline{H_{ij}},$$

so that $g = (g_i, g_{ij}^R, g_{ij}^I)$ is a standard Gaussian vector in $\mathbb{R}^{n^2}$. Since

$$\|H\|_{\mathrm{HS}}^2 = \sum_i H_{ii}^2 + 2\sum_{i<j}|H_{ij}|^2 = \sigma^2\|g\|_2^2,$$

the map $g \mapsto H(g)$ satisfies $\|H(g) - H(g')\|_{\mathrm{HS}} = \sigma\|g - g'\|_2$.

Fix $f \in \mathcal{B}$ and set $F(g) = L_n(f) = \frac{1}{n}\sum_{i=1}^n f(\lambda_i)$, where $\lambda_1 \geq \cdots \geq \lambda_n$ are the (continuously varying, hence measurable) eigenvalues of $H(g)$. For g, g' with corresponding eigenvalues λ_i, λ_i',

$$\begin{aligned}
|F(g) - F(g')| &\leq \frac{1}{n}\sum_{i=1}^n |f(\lambda_i) - f(\lambda_i')| \leq \frac{1}{n}\sum_{i=1}^n |\lambda_i - \lambda_i'| \\
&\leq \frac{1}{\sqrt{n}}\Big(\sum_{i=1}^n(\lambda_i - \lambda_i')^2\Big)^{1/2} \leq \frac{1}{\sqrt{n}}\|H(g) - H(g')\|_{\mathrm{HS}} \\
&= \frac{\sigma}{\sqrt{n}}\|g - g'\|_2 = \frac{1}{2n}\|g - g'\|_2,
\end{aligned}$$

using that f is 1-Lipschitz, Cauchy–Schwarz, Lidskii's inequality in the Hoffman–Wielandt form of Exercise 3.16 (valid for Hermitian matrices), and $\sigma/\sqrt{n} = 1/(2n)$. Thus F (and $-F$) is L-Lipschitz on $\mathbb{R}^{n^2}$ with $L = 1/(2n)$, independently of f.

Theorem 5.6 applied to F and to $-F$, with a union bound for the two one-sided tails, gives, for $t \geq 0$,

$$\mathbb{P}\big\{|L_n(f) - \mathbb{E}[L_n(f)]| \geq t\big\} \leq 2\exp\Big(-\frac{t^2}{2L^2}\Big) = 2e^{-2n^2t^2} \leq 2e^{-n^2t^2/2}.$$

The right-hand side does not depend on f, so taking the supremum over $f \in \mathcal{B}$ yields

$$\sup_{f\in\mathcal{B}} \mathbb{P}\big\{|L_n(f) - \mathbb{E}[L_n(f)]| \geq t\big\} \leq 2e^{-\frac{n^2t^2}{2}}, \qquad t \geq 0,$$

as claimed. (Only the 1-Lipschitz property of f was used; the bound $|f| \leq 1$ is not needed for the concentration itself.)

Exercise 5.34

Central idea. For fixed $f \in \mathcal{B}$, the map from the independent Gaussian entries of the GUE matrix to $L_n(f)$ is $(2n)^{-1}$-Lipschitz. Three facts give this: f is 1-Lipschitz, ordered eigenvalues are 1-Lipschitz in Frobenius norm (Hoffman–Wielandt), and the GUE entries are $(4n)^{-1/2}$ times standard normals. A supremum of $(2n)^{-1}$-Lipschitz functions is $(2n)^{-1}$-Lipschitz, so Theorem 5.6 gives the tail bound. For $\mathbb{E}Z$ we use a self-contained net argument instead of the semicircle-law rates. First, $\|H\| \leq 5$ except with probability e^{-2n}. Second, $\mathcal{B}$ restricted to $[-5,5]$ has an ε-net of log-cardinality $O(1/\varepsilon)$. Third, each $L_n(h) - \mathbb{E}L_n(h)$ is sub-Gaussian with variance factor $1/(4n^2)$. One-step chaining with $\varepsilon = n^{-2/3}$ gives $\mathbb{E}Z = O(n^{-2/3}) \leq \kappa/\sqrt{n}$.

Solution. Write the GUE matrix with the book's normalization ($\mathbb{E}|H_{ij}|^2 = \sigma^2 = 1/(4n)$). As in Exercise 5.33, $H = H(g)$ for a standard Gaussian vector $g \in \mathbb{R}^{n^2}$, through a linear map with $\|H(g) - H(g')\|_F = \sigma\|g - g'\|_2$. For $f \in \mathcal{B}$ put $X_f = L_n(f) - \mathbb{E}L_n(f)$, so $Z = \sup_{f\in\mathcal{B}} |X_f|$.

Lipschitz property. Fix $f \in \mathcal{B}$ and order eigenvalues decreasingly. Then

$$|L_n(f)(H) - L_n(f)(H')| \leq \frac{1}{n}\sum_{i=1}^{n} |f(\lambda_i) - f(\lambda_i')| \leq \frac{1}{n}\sum_{i=1}^{n} |\lambda_i - \lambda_i'| \leq \frac{1}{\sqrt{n}}\|H - H'\|_F.$$

This uses that f is 1-Lipschitz, Cauchy–Schwarz, and the Hoffman–Wielandt inequality $\sum_i(\lambda_i - \lambda_i')^2 \leq \|H - H'\|_F^2$. Hence $g \mapsto X_f$ is $(2n)^{-1}$-Lipschitz, since $\sigma/\sqrt{n} = 1/(2n)$. So is each $|X_f|$, and so is Z, because $|\sup_f a_f - \sup_f b_f| \leq \sup_f |a_f - b_f|$.

Measurability. All eigenvalues lie in $[-\|H\|, \|H\|]$. By Arzelà–Ascoli, $\mathcal{B}$ is separable for uniform convergence on compacts; let $\{f_k\}$ be dense. For $f \in \mathcal{B}$ and $f_{k_j} \to f$ uniformly on compacts, $L_n(f_{k_j}) \to L_n(f)$ pointwise. Also $\mathbb{E}L_n(f_{k_j}) \to \mathbb{E}L_n(f)$ by dominated convergence ($|f| \leq 1$). Hence $Z = \sup_k |X_{f_k}|$ is measurable, and $0 \leq Z \leq 2$.

Tail bound. Theorem 5.6 applied to the $(2n)^{-1}$-Lipschitz function Z gives, for $t \geq 0$,

$$\mathbb{P}\{Z \geq \mathbb{E}Z + t\} \leq \exp\Big(-\frac{t^2}{2(2n)^{-2}}\Big) = e^{-2n^2t^2} \leq 2e^{-nt^2/2},$$

since $2n^2 \geq n$. This is the claimed tail bound, and in fact a stronger one.

Expectation bound. Step 1: the operator norm. Let $\mathcal{N}$ be a $\frac{1}{4}$-net of the unit sphere of $\mathbb{C}^n \cong \mathbb{R}^{2n}$. The volumetric bound gives $|\mathcal{N}| \leq (1+8)^{2n} = 81^n$. Let v be a unit vector with $\|H\| = |v^*Hv|$, and let $w \in \mathcal{N}$ satisfy $|v - w| \leq \frac{1}{4}$. Then

$$|v^*Hv - w^*Hw| \leq |(v-w)^*Hv| + |w^*H(v-w)| \leq \tfrac{1}{2}\|H\|,$$

so $\|H\| \leq 2\max_{w\in\mathcal{N}} |w^*Hw|$. Each $w^*Hw = \langle H, ww^*\rangle_{\mathrm{HS}}$ is a linear functional of g with norm $\sigma\|ww^*\|_F = \sigma$, so it is $N(0,\sigma^2)$. The maximal inequality we use is $\mathbb{E}\max_{i\leq N} Y_i \leq \lambda^{-1}\log\sum_i \mathbb{E}e^{\lambda Y_i} \leq \lambda^{-1}\log N + \lambda v/2$

whenever $\log \mathbb{E} e^{\lambda Y_i} \le \lambda^2 v/2$; optimizing at $\lambda = \sqrt{2\log N/v}$ gives $\mathbb{E}\max_i Y_i \le \sqrt{2v\log N}$. Applied to the $2|\mathcal{N}|$ variables $\pm w^* H w$, it gives

$$\mathbb{E}\|H\| \le 2\sigma\sqrt{2\log(2\cdot 81^n)} = \frac{1}{\sqrt{n}}\sqrt{2\log 2 + 2n\log 81} \le \sqrt{2\log 2 + 2\log 81} < 3.2 \le 4.$$

Moreover $g \mapsto \|H(g)\|$ is σ-Lipschitz, because $\|\cdot\| \le \|\cdot\|_F$. Theorem 5.6 then gives

$$\mathbb{P}\{\|H\| > 5\} \le \mathbb{P}\{\|H\| - \mathbb{E}\|H\| > 1\} \le e^{-1/(2\sigma^2)} = e^{-2n}.$$

Step 2: a net of $\mathcal{B}$ on $I = [-5,5]$. Let $\varepsilon \in (0,1]$ and $\eta = \varepsilon/8$. Put $x_k = -5 + k\eta$ for $k = 0, \dots, K$, with $K = \lceil 10/\eta \rceil$. For $f \in \mathcal{B}$, let $r_k \in \eta\mathbb{Z}$ be a nearest point to $f(x_k)$, so $|r_k - f(x_k)| \le \eta/2$. Then $|r_{k+1} - r_k| \le \eta + \eta = 2\eta$, and $r_{k+1} - r_k \in \{0, \pm\eta, \pm 2\eta\}$. Let $\tilde f$ be the piecewise linear interpolant of (x_k, r_k). For $x \in [x_k, x_{k+1}]$,

$$|f(x) - \tilde f(x)| \le |f(x) - f(x_k)| + |f(x_k) - r_k| + |r_k - \tilde f(x)| \le \eta + \tfrac{\eta}{2} + 2\eta < 4\eta.$$

The interpolant is determined by r_0 and the K increments. Since $r_0 \in \eta\mathbb{Z} \cap [-1-\eta/2, 1+\eta/2]$ takes at most $2/\eta + 2$ values, there are at most $(2/\eta + 2)\,5^K$ interpolants. For each interpolant whose 4η-ball (sup norm on I) meets $\mathcal{B}$, choose one $h \in \mathcal{B}$ in that ball. The resulting set $\mathcal{N}_\varepsilon \subseteq \mathcal{B}$ has the property that every $f \in \mathcal{B}$ lies within $8\eta = \varepsilon$ of some $h \in \mathcal{N}_\varepsilon$ on I. Its size $N(\varepsilon)$ satisfies

$$\log N(\varepsilon) \le \log\Big(\frac{16}{\varepsilon} + 2\Big) + \Big(\frac{80}{\varepsilon} + 1\Big)\log 5 \le \frac{18}{\varepsilon} + \frac{81\cdot 1.61}{\varepsilon} \le \frac{150}{\varepsilon}.$$

Step 3: one-step chaining. For $f \in \mathcal{B}$ pick $\pi f \in \mathcal{N}_\varepsilon$ with $\sup_I |f - \pi f| \le \varepsilon$. Note $|f - \pi f| \le 2$ everywhere. Let $N_{\text{out}} = \#\{i : |\lambda_i| > 5\}$. Then

$$|L_n(f) - L_n(\pi f)| \le \varepsilon + \frac{2N_{\text{out}}}{n}, \qquad |X_f - X_{\pi f}| \le 2\varepsilon + \frac{2N_{\text{out}}}{n} + \frac{2\,\mathbb{E}N_{\text{out}}}{n},$$

so

$$Z \le \max_{h\in\mathcal{N}_\varepsilon} |X_h| + 2\varepsilon + \frac{2N_{\text{out}}}{n} + \frac{2\,\mathbb{E}N_{\text{out}}}{n}.$$

Each $\pm X_h$ ($h \in \mathcal{B}$) is a centered $(2n)^{-1}$-Lipschitz function of g. By Theorem 5.5, $\log \mathbb{E} e^{\lambda(\pm X_h)} \le \lambda^2/(8n^2)$ for all $\lambda \in \mathbb{R}$. The maximal inequality, with $v = 1/(4n^2)$ over $2N(\varepsilon)$ variables, then gives

$$\mathbb{E}\max_{h\in\mathcal{N}_\varepsilon} |X_h| \le \frac{1}{2n}\sqrt{2\log(2N(\varepsilon))} \le \frac{1}{2n}\sqrt{2 + \frac{300}{\varepsilon}}.$$

Since $N_{\text{out}} \le n\,\mathbb{1}\{\|H\| > 5\}$, Step 1 gives $\mathbb{E}N_{\text{out}}/n \le e^{-2n}$. Hence

$$\mathbb{E}Z \le \frac{1}{2n}\sqrt{2 + \frac{300}{\varepsilon}} + 2\varepsilon + 4e^{-2n}.$$

Take $\varepsilon = n^{-2/3} \in (0,1]$. Then $\sqrt{2 + 300n^{2/3}} \le \sqrt{302}\,n^{1/3}$, and therefore

$$\mathbb{E}Z \le \Big(\frac{\sqrt{302}}{2} + 2\Big)n^{-2/3} + 4e^{-2n} \le \frac{10.7}{\sqrt{n}} + \frac{4\sqrt{n}\,e^{-2n}}{\sqrt{n}} \le \frac{12}{\sqrt{n}},$$

using $n^{-2/3} \le n^{-1/2}$ and $\sqrt{n}\,e^{-2n} \le e^{-2} < 0.136$ for $n \ge 1$. Thus $\mathbb{E}Z \le \kappa/\sqrt{n}$ with the universal constant $\kappa = 12$ (indeed $\mathbb{E}Z = O(n^{-2/3})$). ■

Erratum (earlier version). An earlier version of this solution derived $\mathbb{E}Z \le \kappa/\sqrt{n}$ from the Götze–Tikhomirov Kolmogorov-distance rates. It relied on the lemma "$\pi \le$ Lévy $\le d_K$, hence $d_{BL} \le 3d_K$". That lemma is false in general: the Lévy metric is bounded by the Prokhorov metric, not conversely. Counterexample: μ uniform on $\{0, 2, \dots, 18\}$ and $\nu = \mu(\cdot - 1)$ (uniform on $\{1, 3, \dots, 19\}$). Here $d_K(\mu,\nu) = 0.1$. The

triangle wave $f(x) = \frac{1}{2} - \operatorname{dist}(x, 2\mathbb{Z})$ lies in $\mathcal{B}$ ($|f| \le \frac{1}{2}$, 1-Lipschitz), and $\mu(f) - \nu(f) = \frac{1}{2} - (-\frac{1}{2}) = 1$, so $d_{BL}(\mu,\nu) \ge 1 = 10\, d_K(\mu,\nu)$ (checked numerically). The proof above avoids this route entirely.

The citation route can be repaired when one of the two measures is the semicircle law, because that law is atomless with compact support. *Claim:* if ν is atomless with $\nu([-a,a]) = 1$, then $d_{BL}(\mu,\nu) \le (4 + 2a)\, d_K(\mu,\nu)$ for every probability μ. *Proof.* Let $G = F_\mu - F_\nu$, $\delta = \sup|G|$, $m = \mu - \nu$ and $f \in \mathcal{B}$. Since f is absolutely continuous with $|f'| \le 1$ a.e., Fubini gives

$$\int_{(-a,a]} f\, dm = f(-a)(G(a) - G(-a)) + \int_{-a}^{a} f'(t)(G(a) - G(t))\, dt = f(a)G(a) - f(-a)G(-a) - \int_{-a}^{a} Gf'\, dt,$$

which has absolute value at most $(2+2a)\delta$. Outside $(-a,a]$, ν has no mass (it has no atom at $-a$). Moreover $\mu((-\infty,-a]) = F_\mu(-a) = G(-a) \le \delta$ and $\mu((a,\infty)) = F_\nu(a) - F_\mu(a) = -G(a) \le \delta$. So $|\int_{\mathbb{R}\setminus(-a,a]} f\, dm| \le 2\delta$. In the book's normalization the semicircle law is supported on $[-2\sigma\sqrt{n}, 2\sigma\sqrt{n}] = [-1,1]$, so $d_{BL}(\cdot,\sigma_{sc}) \le 6\, d_K(\cdot,\sigma_{sc})$. Together with the cited rates and the triangle inequality, this gives $\mathbb{E}Z \le 6(C_1 + C_2)/\sqrt{n}$. The self-contained proof above does not need this.

Exercise 5.35

Central idea. Lemma 5.25 writes $\mathbb{E}L_n f$ as an average of $\mathbb{E}[f(\sigma X)h_i(X)^2]$; Christoffel–Darboux sums the Hermite functions in closed form into the kernel K_n (with the printed normalization $1/\sqrt{2\pi n}$, which is the correct one). The mgf $F_n(s)$ is then computed exactly via Mehler's formula, and dominated convergence reads off the termwise limit, which is the semicircular mgf with the Catalan moments $m_{2k} = C_k/4^k$ from Exercise 5.32.

Erratum. The hint is wrong as printed (the statement, including $K_n(x,x) = \frac{e^{-x^2/2}}{\sqrt{2\pi n}}(h_n' h_{n-1} - h_n h_{n-1}')(x)$, is correct: K_n equals the one-point density ρ_n below and $\int_{\mathbb{R}} K_n(x,x)\, dx = 1$). The hint's display $\frac{d}{dx}K_n(x,x) = -e^{-x^2/2}h_n(x)h_{n-1}(x) = -e^{-x^2/2}\frac{h_n(x)h_n'(x)}{\sqrt{n}}$ drops the factor $1/\sqrt{2\pi n}$; the correct identity is

$$\frac{d}{dx}K_n(x,x) = -\frac{e^{-x^2/2}}{\sqrt{2\pi n}}\, h_n(x)h_{n-1}(x) = -\frac{e^{-x^2/2}}{\sqrt{2\pi}\, n}\, h_n(x)h_n'(x).$$

For example, at $n = 2$ the true derivative is $1/(2\sqrt{\pi}) \approx 0.2821$ times the printed one (in general the ratio is $1/\sqrt{2\pi n}$; checked symbolically for $n = 1,2,3,5$). The second equality of the hint is correct, since $h_n' = \sqrt{n}\, h_{n-1}$.

Solution. The Christoffel–Darboux identity (Exercise 5.22, restated in 5.32) is, for $n \ge 1$,

$$\frac{1}{n}\sum_{i=0}^{n-1} h_i(x)^2 = \frac{1}{\sqrt{n}}(h_n'(x)h_{n-1}(x) - h_n(x)h_{n-1}'(x)), \qquad x \in \mathbb{R}.$$

Lemma 5.25 gives, with X standard Gaussian and $\sigma = 1/\sqrt{4n}$,

$$\mathbb{E}L_n f = \frac{1}{n}\sum_{i=0}^{n-1} \mathbb{E}[f(\sigma X)h_i(X)^2] = \int_{\mathbb{R}} f(\sigma x)\, \rho_n(x)\, dx, \quad \rho_n(x) = \frac{1}{n}\sum_{i=0}^{n-1} h_i(x)^2 \frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

By Christoffel–Darboux, $\rho_n(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}\sqrt{n}}(h_n'(x)h_{n-1}(x) - h_n(x)h_{n-1}'(x))$. Hence, with the kernel as printed,

$$K_n(x,x) = \frac{e^{-x^2/2}}{\sqrt{2\pi n}}(h_n'(x)h_{n-1}(x) - h_n(x)h_{n-1}'(x)) = \rho_n(x),$$

$$\mathbb{E}L_n f = \int_{\mathbb{R}} f(x/\sqrt{4n})\, K_n(x,x)\, dx \qquad \text{for bounded continuous } f.$$

(Consistency check: $\int K_n = \frac{1}{n}\sum_{i<n} \mathbb{E}[h_i(X)^2] = 1$ by orthonormality of the h_i.)

For the hint, differentiating K_n and using the Hermite ODE $h_k''-xh_k'+kh_k=0$ (the x-terms and the $h_n'h_{n-1}'$ terms cancel),

$$\frac{d}{dx}K_n(x,x)=-\frac{e^{-x^2/2}}{\sqrt{2\pi}\,\sqrt{n}}\,h_n(x)h_{n-1}(x)=-\frac{e^{-x^2/2}}{\sqrt{2\pi}\,n}\,h_n(x)h_n'(x),$$

since $h_n'=\sqrt{n}\,h_{n-1}$. (This is the corrected hint of the Erratum; the identity is not needed in what follows.) Now $F_n(s)=\mathbb{E}L_n[e^{s\cdot}]=\int e^{s\sigma x}K_n(x,x)\,dx=\frac{1}{n}\sum_{i=0}^{n-1}\mathbb{E}[e^{tX}h_i(X)^2]$ with $t=s\sigma=s/(2\sqrt{n})$. Mehler's formula for the orthonormal Hermite functions, $|r|<1$,

$$\sum_{i=0}^{\infty}h_i(x)h_i(y)r^i=\frac{1}{\sqrt{1-r^2}}\exp\!\left(\frac{2xyr-(x^2+y^2)r^2}{2(1-r^2)}\right),$$

with $y=x$ gives $\sum_i h_i(x)^2r^i=(1-r^2)^{-1/2}\exp(x^2r/(1+r))$. Multiplying by e^{tx} and taking $\mathbb{E}$ (Tonelli, terms nonnegative),

$$\sum_{i=0}^{\infty}\mathbb{E}[e^{tX}h_i(X)^2]r^i=\frac{1}{\sqrt{1-r^2}}\,\mathbb{E}\left[e^{tX+cX^2}\right],\quad c=\frac{r}{1+r}<\frac{1}{2},$$

and $\mathbb{E}[e^{tX+cX^2}]=e^{t^2/(2(1-2c))}/\sqrt{1-2c}$ with $1-2c=(1-r)/(1+r)$, so

$$\sum_{i=0}^{\infty}a_i(t)r^i=\frac{e^{t^2(1+r)/(2(1-r))}}{1-r},\qquad a_i(t):=\mathbb{E}[e^{tX}h_i(X)^2].$$

Expanding $e^{t^2(1+r)/(2(1-r))}=\sum_{k\ge0}\frac{(t^2/2)^k}{k!}\frac{(1+r)^k}{(1-r)^k}$ (uniform on $|r|\le\rho<1$) and extracting $[r^i]$,

$$a_i(t)=\sum_{k=0}^{\infty}\frac{(t^2/2)^k}{k!}\,c_{i,k},\qquad c_{i,k}=[r^i]\frac{(1+r)^k}{(1-r)^{k+1}}=\sum_{j=0}^{\min(i,k)}\binom{k}{j}\binom{i-j+k}{k}\ge0,$$

with the convention $\binom{a}{k}=0$ for $a<k$. Summing over $i<n$ (finite sum) and swapping with the k-sum,

$$F_n(s)=\sum_{k=0}^{\infty}\frac{(t^2/2)^k}{k!}\,b_{n,k},\qquad b_{n,k}=\frac{1}{n}\sum_{i=0}^{n-1}\sum_{j=0}^{k}\binom{k}{j}\binom{i-j+k}{k},$$

where $t^2/2=s^2/(8n)$. By hockey-stick, $\sum_{i=0}^{n-1}\binom{i-j+k}{k}=\binom{n-j+k}{k+1}$, so

$$\frac{(t^2/2)^k}{k!}b_{n,k}=\frac{(s^2/(8n))^k}{k!}\sum_{j=0}^{k}\binom{k}{j}\frac{1}{n}\binom{n-j+k}{k+1}.$$

This is dominated uniformly in n by $\frac{s^{2k}(1+k)^{k+1}}{4^k k!(k+1)!}$ (using $\binom{n+k}{k+1}\le(n+k)^{k+1}/(k+1)!$ and $n\ge1$), which is summable in k. Dominated convergence gives, for fixed k,

$$\frac{(s^2/(8n))^k}{k!}\sum_{j=0}^{k}\binom{k}{j}\frac{1}{n}\binom{n-j+k}{k+1}\longrightarrow\frac{(s^2/8)^k}{k!}\cdot\frac{2^k}{(k+1)!}=\frac{s^{2k}}{4^k k!(k+1)!},$$

since $\frac{1}{n}\binom{n-j+k}{k+1}\to\frac{n^k}{(k+1)!}$ and $\sum_{j=0}^{k}\binom{k}{j}=2^k$. Hence

$$\lim_{n\to\infty}F_n(s)=\sum_{k=0}^{\infty}\frac{s^{2k}}{4^k k!(k+1)!}=\sum_{k\in\mathbb{N}}\frac{m_{2k}s^{2k}}{(2k)!}\quad\text{for all } s\in\mathbb{R},$$

with $m_{2k}=C_k/4^k=\frac{(2k)!}{4^k k!(k+1)!}$ the even moments of the semicircular law of Exercise 5.32 (odd moments vanish by the symmetry $H\overset{d}{=}-H$). This is the moment-generating function of the semicircular distribution.

Exercise 5.36

Central idea. The GUE argument of Exercises 5.24–5.30 adapts to real symmetric matrices with one decisive change: eigenvectors are real orthogonal, so there are $n(n-1)/2$ real off-diagonal parameters (not $n(n-1)$), and the Jacobian contributes the Vandermonde $\Delta(\lambda) = \prod_{i<j}(\lambda_i - \lambda_j)$ to the first power, not squared. However, the formula as stated is false: with all entries $N(0, 1/n)$ the density of A is not orthogonally invariant (at $n = 1$ the true eigenvalue law is $N(0,1)$ while the formula predicts an $N(0, 1/4)$ shape), so no clean determinantal formula holds; we prove the corrected version for the orthogonally-invariant GOE normalization. (The prefactor D_n/σ^{2n} is not an error, since $\sigma = 1/\sqrt{4n}$ is fixed by n; see the closing remark.)

Erratum. The statement is false as printed. If all entries $(A_{ij})_{i\le j}$ have variance $1/n$, the law of A is not orthogonally invariant, and the eigenvalue density is not of the printed form. For $n = 1$, $\lambda_1 = A_{11} \sim N(0,1)$, but the formula with $\sigma = 1/2$ gives a density $\propto e^{-2\lambda_1^2}$, an $N(0, 1/4)$ shape that no choice of D_1 repairs. Corrected statement (proved below): the formula holds for the invariant normalization, where A has density $\propto \exp(-\mathrm{tr}(A^2)/(2\sigma^2))$, i.e. $A_{ii} \sim N(0,\sigma^2)$ and $A_{ij} \sim N(0,\sigma^2/2)$ for $i < j$; the density at $\lambda_1 > \cdots > \lambda_n$ is then $C_{n,\sigma}\prod_{i<j}(\lambda_i - \lambda_j)\exp(-\sum_i \lambda_i^2/(2\sigma^2))$.

Solution. *The stated formula is false.* For $n = 1$, $A = [A_{11}]$ with $A_{11} \sim N(0,1)$, so the eigenvalue $\lambda_1 = A_{11}$ has density $\frac{1}{\sqrt{2\pi}}e^{-\lambda_1^2/2}$. The claimed formula gives, with $\sigma = 1/2$, a density proportional to $e^{-\lambda_1^2/(2\sigma^2)} = e^{-2\lambda_1^2}$, i.e. an $N(0, 1/4)$ shape — a different distribution, and no choice of D_1 repairs the shape. More generally, the density $\prod_{i\le j}\sqrt{n/(2\pi)}\exp(-nA_{ij}^2/2) = \mathrm{const}\cdot\exp(-\frac{n}{2}\sum_{i\le j}A_{ij}^2)$ is not a function of $\mathrm{tr}(A^2) = \sum_i \lambda_i^2$ (since $\sum_{i\le j}A_{ij}^2 = \frac{1}{2}\mathrm{tr}(A^2) + \frac{1}{2}\sum_i A_{ii}^2$), hence not orthogonally invariant, so the O-integral below does not separate. We prove the corrected statement.

Corrected Exercise 5.36. Let A be real symmetric with density proportional to $\exp(-\mathrm{tr}(A^2)/(2\sigma^2))$ with respect to $dA = \prod_{i\le j} dA_{ij}$ (i.e. independent $A_{ii} \sim N(0,\sigma^2)$, $A_{ij} \sim N(0,\sigma^2/2)$ for $i < j$). Then at $\lambda_1 > \cdots > \lambda_n$ the eigenvalue density is $C_{n,\sigma}\,\Delta(\lambda)\exp(-\sum_{i=1}^n \lambda_i^2/(2\sigma^2))$.

Step 1: distinct eigenvalues a.s. Let $\mathcal{S}_n$ be the real symmetric $n \times n$ matrices and $\mathcal{S}_n^d$ those with pairwise distinct eigenvalues. As in Exercises 5.24–5.26, the coefficients of the characteristic polynomial are polynomials in the entries, and by Exercise 5.25 the discriminant $D(P, P')$ is a nonzero polynomial in the entries vanishing exactly when an eigenvalue repeats; by Exercise 5.24 its zero set has Lebesgue measure 0. Hence A has distinct eigenvalues a.s.

Step 2: the diffeomorphism (real case of 5.27–5.28). Let $\mathcal{O}_n^{vg}$ be the orthogonal matrices with positive diagonal entries and invertible minors, $\mathcal{D}_n^d$ the diagonal matrices with decreasing diagonal, and $\mathcal{S}_n^{d,g}$ the symmetric matrices $A = O\Lambda O^T$ with $O \in \mathcal{O}_n^{vg}$, $\Lambda \in \mathcal{D}_n^d$; as in 5.27, $\mathcal{S}_n \setminus \mathcal{S}_n^{d,g}$ has measure 0. The map

$$A = O\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)O^T \longmapsto (\mathrm{diag}(\lambda_1, \dots, \lambda_n),\, T(O)), \qquad T(O) = \left(\frac{O_{12}}{O_{11}}, \dots, \frac{O_{n-1,n}}{O_{n-1,n-1}}\right) \in \mathbb{R}^{n(n-1)/2},$$

is a diffeomorphism $\mathcal{S}_n^{d,g} \to \mathcal{D}_n^d \times T(\mathcal{O}_n^{vg})$ (real case of 5.28; no real/imaginary splitting is needed).

Step 3: the Jacobian (real case of 5.29–5.30). Write $A = O\Lambda O^T$; then $O^T dA\, O = [\delta O, \Lambda] + d\Lambda$ with $\delta O = O^T dO$ real skew-symmetric. Hence, with $B_\ell = O^T \frac{\partial A}{\partial p_\ell} O$ as in (5.7) (now real),

$$(O^T dA\, O)_{ij} = (\delta O)_{ij}(\lambda_j - \lambda_i) + \delta_{ij}\, d\lambda_i.$$

The $n(n+1)/2$ independent differentials $(d\lambda_i)_{i\le n}$, $((\delta O)_{ij})_{i<j}$ map block-diagonally to the entries of the symmetric matrix $O^T dA\, O$, with determinant $\prod_{i<j}|\lambda_j - \lambda_i|$. Since $dA \mapsto O^T dA\, O$ is a rotation (Jacobian 1),

$$dA = \Delta(\lambda_1, \dots, \lambda_n)\, d\Lambda\, d\mu(O) \qquad \text{on } \{\lambda_1 > \cdots > \lambda_n\},$$

where $d\mu(O)$ depends only on O: the Vandermonde appears to the *first* power (not squared as in the GUE case), because a real symmetric matrix has $n(n-1)/2$ independent real off-diagonal entries.

Step 4: change of variables. The joint density of (Λ, O) is therefore proportional to $\Delta(\lambda)\exp(-\mathrm{tr}(A^2)/(2\sigma^2))\,d\mu(O)\,d\Lambda$. By orthogonal invariance $\mathrm{tr}(A^2) = \mathrm{tr}(\Lambda^2) = \sum_i \lambda_i^2$, and integrating the O-factor gives a finite constant (it is, up to the Jacobian factor $h(p)$, the Haar volume of the compact orthogonal group, whose complement $\mathcal{O}_n \setminus \mathcal{O}_n^{vg}$ has measure 0). Absorbing constants, the eigenvalue density on $\{\lambda_1 > \cdots > \lambda_n\}$ is $C_{n,\sigma}\,\Delta(\lambda_1, \dots, \lambda_n)\exp(-\sum_{i=1}^n \lambda_i^2/(2\sigma^2))$ with $\Delta(\lambda) = \prod_{1\le i<j\le n}(\lambda_i - \lambda_j) > 0$ on this region (this equals $(-1)^{n(n-1)/2}\det[\lambda_i^{j-1}]_{i,j}$).

Remark (normalization). Substituting $\lambda_i = \sigma\mu_i$ gives $C_{n,\sigma}^{-1} = \sigma^{n(n+1)/2} I_n$ where $I_n = \int_{\mu_1>\cdots>\mu_n} \Delta(\mu) e^{-\sum \mu_i^2/2}\,d\mu$. So if σ is treated as a free scale parameter, the natural form of the prefactor is $D_n/\sigma^{n(n+1)/2}$ with $D_n = 1/I_n$ independent of σ. This is not an erratum in the printed statement: there $\sigma = 1/\sqrt{4n}$ is fixed by n, so D_n/σ^{2n} is merely one constant depending on n and D_n absorbs the difference ($D_n = \sigma^{2n-n(n+1)/2}/I_n$; e.g. for $n = 1$ the density $(2\pi\sigma^2)^{-1/2}e^{-\lambda_1^2/(2\sigma^2)}$ corresponds to $D_1 = \sigma/\sqrt{2\pi}$). ■

Exercise 5.37

Central idea. Match CDFs to write $Z = \max_{1\le i\le n}|X_i|$ as $f(Y)$, where $f(y) = U(2e^y)$, $Y = \max_{1\le i\le n} E_i$ and the $E_i \sim \mathrm{Exp}(1)$ are iid. The map f is concave, because its derivative is the Mills ratio $R = \bar\Phi/\phi$ evaluated at f, and R is decreasing. Concavity gives the tangent bound $f(Y) - \mathbb{E}f(Y) \le \delta_n + f'(H_n)(Y - H_n)$, where $\delta_n > 0$ is the Jensen gap. The bound $R(u) < 1/u$ gives $f'(H_n) \le 1/U(2n)$. Rényi's representation makes $Y - H_n$ a weighted sum of centered exponentials, so it is sub-gamma $\Gamma_+(2, 1)$. The scale factor is $c = 1$, not $1/3$. With $c = 1/3$ the statement as printed is **false**, and we prove it with denominator $2 + tU(2n)$. The rate $(2\log 2n)^{3/2}\delta_n \to \pi^2/12$ comes from the second-order Taylor remainder: $|f''| \sim f^{-3}$ and $\mathrm{Var}(Y) \to \pi^2/6$.

Solution. Erratum: the printed bound is false. The printed inequality, with denominator $2+tU(2n)/3$, fails for large n. For $x \ge 0$ we have $\mathbb{P}\{Z \le x\} = (1 - 2\bar\Phi(x))^n$, so tail probabilities and $\mathbb{E}Z = \int_0^\infty (1 - (1 - 2\bar\Phi(x))^n)\,dx$ can be computed to high precision. We used mpmath at 60 digits, and cross-checked $\mathbb{E}Z = \mathbb{E}f(Y)$ by integrating against the density of Y. Take $s := tU(2n) = 30$, so the printed right-hand side is $e^{-900/(2\cdot 12)} = e^{-37.5} \approx 5.18 \times 10^{-17}$ for every n. With $\delta_n = f(H_n) - \mathbb{E}Z$ as below:

n	$U(2n)$	$\mathbb{E}Z$	δ_n	$\mathbb{P}\{Z - \mathbb{E}Z \ge t + \delta_n\}$	printed bound	corrected bound $e^{-s^2/(2(2+s))}$
10^8	5.7307	5.82424	3.609×10^{-3}	1.90×10^{-20}	5.18×10^{-17}	7.81×10^{-7}
10^{20}	9.3360	9.39606	9.348×10^{-4}	1.85×10^{-16}	5.18×10^{-17} (**fails**)	7.81×10^{-7}
10^{75}	18.4148	18.44588	1.290×10^{-4}	1.22×10^{-14}	5.18×10^{-17} (**fails**)	7.81×10^{-7}
10^{200}	30.2285	30.24755	2.954×10^{-5}	3.05×10^{-14}	5.18×10^{-17} (**fails**)	7.81×10^{-7}

For example, at $n = 10^{20}$ we have $t = 30/9.3360 \approx 3.213 > 0$, and the true probability is 3.6 times the printed bound. A heuristic explains why: $U(2n)(Z - \mathbb{E}Z)$ is asymptotically a centered Gumbel variable, whose tail at s is about $e^{-s-\gamma}$, while the printed bound decays like $e^{9-3s/2}$. The error sits in the scale constant. $Y - H_n$ has an exponential right tail with scale 1. Even for $n = 1$, $\mathbb{P}\{Y - H_1 \ge 30\} = e^{-31} \approx 3.4 \times 10^{-14} > e^{-37.5}$, so the tail inequality $\mathbb{P}\{Y - H_n \ge s\} \le e^{-s^2/(2(2+s/3))}$ is false too. (An earlier version of this solution obtained it by plugging $c = 1$ into Theorem 2.10 and then writing $s/3$. In Bernstein's inequality the $1/3$ belongs to variables bounded by b, where $c = b/3$; it does not apply to exponentials.)

Corrected statement. For $t > 0$,

$$\mathbb{P}\{Z - \mathbb{E}Z \ge t + \delta_n\} \le \exp\left(-\frac{t^2 U(2n)^2}{2(2 + tU(2n))}\right),$$

where $\delta_n > 0$ and $\lim_n (2\log(2n))^{3/2}\delta_n = \pi^2/12$. We prove this below.

Notation and Gaussian facts. Let $\bar\Phi = 1-\Phi$ and $R = \bar\Phi/\phi$. Then $U(t) = \bar\Phi^{-1}(1/t)$ for $t > 1$, and U is increasing. We have $R'(u) = uR(u) - 1$, because $\phi' = -u\phi$. Integration by parts gives, for $u > 0$,

$$\bar\Phi(u) = \frac{\phi(u)}{u} - \int_u^\infty \frac{\phi}{t^2}\,dt, \qquad \int_u^\infty \frac{\phi}{t^2}\,dt = \frac{\phi(u)}{u^3} - 3\int_u^\infty \frac{\phi}{t^4}\,dt, \qquad \int_u^\infty \frac{\phi}{t^4}\,dt = \frac{\phi(u)}{u^5} - 5\int_u^\infty \frac{\phi}{t^6}\,dt.$$

Hence, for $u > 0$,

$$\frac{1}{u} - \frac{1}{u^3} < R(u) < \frac{1}{u}, \qquad R(u) < \frac{1}{u} - \frac{1}{u^3} + \frac{3}{u^5}. \tag{M}$$

In particular $R'(u) < 0$ for all $u \in \mathbb{R}$: for $u \le 0$ because $uR \le 0$, and for $u > 0$ because $uR < 1$.

Step 1: representation. Let $f(y) = U(2e^y) = \bar\Phi^{-1}(e^{-y}/2)$ for $y \ge 0$, and let $Y = \max_i E_i$. For $x \ge 0$,

$$f(Y) \le x \iff \tfrac{1}{2}e^{-Y} \ge \bar\Phi(x) \iff Y \le -\log(2\bar\Phi(x)),$$

so $\mathbb{P}\{f(Y) \le x\} = (1-2\bar\Phi(x))^n = \mathbb{P}\{Z \le x\}$. Both variables are nonnegative, hence $Z \stackrel{d}{=} f(Y)$. By Rényi's representation, $Y \stackrel{d}{=} \sum_{k=1}^n E'_k/k$ with E'_k iid Exp(1). So $\mathbb{E}Y = H_n = \sum_{k\le n} 1/k$, and $W := Y - H_n$ has variance $v_n = \sum_{k\le n} k^{-2} < \pi^2/6 < 2$.

Step 2: derivatives of f. Differentiating $\bar\Phi(f(y)) = e^{-y}/2$ gives $\phi(f)f' = e^{-y}/2 = \bar\Phi(f)$. Hence

$$f'(y) = R(f(y)) > 0, \qquad f''(y) = R'(f(y))\,R(f(y)) < 0.$$

So f is C^2, increasing and strictly concave on $[0,\infty)$, with $f(0) = 0$. Write $u = f(y) \ge 0$. Then $|f''(y)| = (1 - uR(u))R(u)$. Since $0 < 1 - uR \le 1$ and $R(u) \le R(0) = \sqrt{\pi/2}$, we get $|f''| \le \sqrt{\pi/2}$ everywhere. By (M), for $u > \sqrt{3}$,

$$u^{-3}(1-3u^{-2})(1-u^{-2}) \le |f''(y)| \le u^{-3}, \tag{D}$$

because $1 - uR \in (u^{-2} - 3u^{-4},\, u^{-2})$ and $R \in (u^{-1} - u^{-3},\, u^{-1})$. By (M), also $f'(y) < 1/f(y)$ for $y > 0$.

Growth. Let $g(x) = \bar\Phi^{-1}(e^{-x})$, so $f(y) = g(y + \log 2)$. If $u = g(x) \ge \sqrt{2}$, then $x = -\log\bar\Phi(u) = \frac{u^2}{2} + \log\sqrt{2\pi} - \log R(u)$. Also $\frac{1}{2u} \le \frac{1}{u} - \frac{1}{u^3} < R(u) < \frac{1}{u}$ by (M), which gives $\frac{u^2}{2} < x < \frac{u^2}{2} + \log(2\sqrt{2\pi}) + \log u < \frac{(u+1)^2}{2} + \frac{3}{2}$. Hence $\sqrt{2x-3} - 1 < g(x) < \sqrt{2x}$, so $g(x)/\sqrt{2x} \to 1$ as $x \to \infty$.

Step 3: sub-gamma tails of W. For $0 \le \lambda < 1$, $-\lambda - \log(1-\lambda) = \sum_{k\ge 2} \lambda^k/k \le \lambda^2/(2(1-\lambda))$. Therefore

$$\log \mathbb{E}e^{\lambda W} = \sum_{k=1}^n \Big(-\frac{\lambda}{k} - \log\big(1 - \tfrac{\lambda}{k}\big)\Big) \le \sum_{k=1}^n \frac{\lambda^2/k^2}{2(1-\lambda/k)} \le \frac{v_n\lambda^2}{2(1-\lambda)} \le \frac{2\lambda^2}{2(1-\lambda)}.$$

For $s > 0$, Chernoff's bound with $\lambda = s/(2+s)$ (so that $1 - \lambda = 2/(2+s)$) gives

$$\mathbb{P}\{W \ge s\} \le \exp\Big(-\lambda s + \frac{\lambda^2}{1-\lambda}\Big) = \exp\Big(-\frac{s^2}{2(2+s)}\Big). \tag{T}$$

For the lower tail, $u - \log(1+u) \le u^2/2$ for $u \ge 0$, so $\log \mathbb{E}e^{-\lambda W} \le \lambda^2 v_n/2 \le \lambda^2$ for $\lambda \ge 0$, and $\mathbb{P}\{W \le -s\} \le e^{-s^2/4}$.

Step 4: the tail bound. Let $\delta_n := f(H_n) - \mathbb{E}f(Y)$. It is strictly positive by Jensen's inequality, since f is strictly concave and Y is non-degenerate. By concavity, $f(Y) \le f(H_n) + f'(H_n)W$, so

$$f(Y) - \mathbb{E}f(Y) \le \delta_n + f'(H_n)W.$$

Since $H_n \ge 1$, $f(H_n) > 0$, and Step 2 gives $1/f'(H_n) > f(H_n) = U(2e^{H_n}) \ge U(2n)$; here $H_n > \log n$ and U is increasing. So if $f(Y) - \mathbb{E}f(Y) \ge t + \delta_n$ with $t > 0$, then $W \ge t/f'(H_n) \ge tU(2n)$. The map $s \mapsto s^2/(2(2+s))$ is increasing on $[0,\infty)$, so (T) gives

$$\mathbb{P}\{Z - \mathbb{E}Z \ge t + \delta_n\} \le \mathbb{P}\{W \ge tU(2n)\} \le \exp\!\left(-\frac{t^2U(2n)^2}{2(2+tU(2n))}\right).$$

(For $n=1$, $U(2)=0$ and the bound is trivial.)

Step 5: the rate of δ_n. Taylor's formula with integral remainder, between $H_n>0$ and $Y>0$, gives $f(Y)=f(H_n)+f'(H_n)W-W^2K_n$ with

$$K_n=\int_0^1(1-\theta)\,|f''(H_n+\theta W)|\,d\theta\in[0,\tfrac12\sqrt{\pi/2}].$$

All terms are integrable, since W has exponential moments. Because $\mathbb{E}W=0$, we get $\delta_n=\mathbb{E}[W^2K_n]$.

Let $\ell_n=f(H_n)$, $a_n=\sqrt{\log n}$ and $A_n=\{|W|\le a_n\}$. Put $m_n=f(H_n-a_n)$ and $M_n=f(H_n+a_n)$, which are defined once $H_n>a_n$. Since $H_n-\log n\in(0,1]$, we have $(H_n\pm a_n+\log 2)/(H_n+\log 2)\to 1$. The growth estimate then gives $m_n/\ell_n\to1$ and $M_n/\ell_n\to1$, and $m_n\to\infty$. On A_n, every point $y=H_n+\theta W$ satisfies $m_n\le f(y)\le M_n$, because f is increasing. For n large enough that $m_n>\sqrt{3}$, (D) gives

$$\alpha_n:=\Big(\frac{\ell_n}{M_n}\Big)^3(1-3m_n^{-2})(1-m_n^{-2})\le\ell_n^3|f''(y)|\le\Big(\frac{\ell_n}{m_n}\Big)^3=:\beta_n,$$

with $\alpha_n,\beta_n\to1$. Hence $\frac{\alpha_n}{2}\mathbb{E}[W^2;A_n]\le\ell_n^3\,\mathbb{E}[W^2K_n;A_n]\le\frac{\beta_n}{2}\mathbb{E}[W^2;A_n]$.

For $s\ge2$ we have $s^2/(2(2+s))\ge s/4$, so Step 3 gives $\mathbb{P}\{|W|>s\}\le2e^{-s/4}$. For $a\ge2$, therefore,

$$\mathbb{E}[W^2;|W|>a]=a^2\mathbb{P}\{|W|>a\}+\int_a^\infty 2s\,\mathbb{P}\{|W|>s\}\,ds\le(2a^2+16a+64)e^{-a/4}.$$

The growth estimate gives $\ell_n^3=O((\log n)^{3/2})$, and $(2\log n+16\sqrt{\log n}+64)\,e^{-\sqrt{\log n}/4}(\log n)^{3/2}\to0$. Hence $\ell_n^3\,\mathbb{E}[W^2K_n;A_n^c]\le\frac12\sqrt{\pi/2}\,\ell_n^3\,\mathbb{E}[W^2;A_n^c]\to0$. Also $\mathbb{E}[W^2;A_n]=v_n-\mathbb{E}[W^2;A_n^c]\to\pi^2/6$. Therefore

$$\ell_n^3\,\delta_n\to\frac12\cdot\frac{\pi^2}{6}=\frac{\pi^2}{12}.$$

Finally, $\ell_n=g(H_n+\log2)$ and $H_n+\log2=\log(2n)+O(1)$, so the growth estimate gives $\ell_n/\sqrt{2\log(2n)}\to1$. Hence $\lim_n(2\log(2n))^{3/2}\delta_n=\pi^2/12$. ■

Numerical check of the rate. The value of $(2\log2n)^{3/2}\delta_n$ is 0.852, 0.853, 0.845, 0.833, 0.828 for $n=10^4,10^8,10^{20},10^{75},10^{200}$. It converges slowly to $\pi^2/12\approx0.8225$ (computed with the same 60-digit quadrature).

Chapter 6 — The Entropy Method

Exercise 6.1

Central idea. Tensorize entropy and prove a sharp one-variable inequality with an independent copy: for i.i.d. bounded Z,Z', $\mathrm{Ent}(e^{\lambda Z})\le\frac{\lambda^2}{2}\mathbb{E}[e^{\lambda Z}(Z-Z')_+^2]$. Its proof is by exponential tilting: with $\psi(s)=\log\mathbb{E}e^{sZ}$ and tilted laws P_s, the left side divided by $\mathbb{E}e^{\lambda Z}$ is $\int_0^\lambda s\,\mathrm{Var}_{P_s}(Z)\,ds=\int_0^\lambda s\,\mathbb{E}_{P_s\otimes P_s}(Z-Z')_+^2\,ds$, the right side is $\int_0^\lambda s\,ds\cdot\mathbb{E}_{P_\lambda\otimes P_0}(Z-Z')_+^2$, and the stochastic ordering $P_0\le_{\mathrm{st}}P_s\le_{\mathrm{st}}P_\lambda$ compares the integrands. Summing over coordinates gives $\mathrm{Ent}(e^{\lambda Z})\le\frac{\lambda^2v}{2}\mathbb{E}e^{\lambda Z}$, and Herbst's argument yields the stated $e^{-t^2/(2v)}$.

Solution. As in Theorem 6.7, $X_1,\dots,X_n$ are independent. Write $Z=f(X_1,\dots,X_n)$, $Z_i'=f(X_1,\dots,X_{i-1},X_i',X_{i+1},\dots,X_n)$ with X_i' an independent copy of X_i, and $V_+=\mathbb{E}\big[\sum_{i=1}^n(Z-Z_i')_+^2\,\big|\,X\big]$. Assume $V_+\le v$ a.s. (Z is integrable, since $\mathbb{E}Z$ appears in the statement).

Reduction to bounded Z. For $K>0$ let $\tau_K(z)=(z\wedge K)\vee(-K)$ and $Z_K=\tau_K(Z)$. Since τ_K is nondecreasing and 1-Lipschitz, $(\tau_K(a)-\tau_K(b))_+\le(a-b)_+$, so $Z_K=(\tau_K\circ f)(X)$ satisfies the hypothesis with the same v. Suppose we have shown, for bounded Z, that

$$\log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}\le\frac{\lambda^2v}{2},\qquad\lambda\ge0.\tag{1}$$

Apply (1) to Z_K and let $K \to \infty$: $Z_K \to Z$ pointwise and $\mathbb{E}Z_K \to \mathbb{E}Z$ by dominated convergence ($|Z_K| \le |Z|$), so by Fatou's lemma $\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \liminf_K \mathbb{E}e^{\lambda(Z_K-\mathbb{E}Z_K)} \le e^{\lambda^2 v/2}$. Hence it suffices to prove (1) for bounded Z, which we assume from now on.

Lemma. If Z, Z' are i.i.d. bounded random variables, then for every $\lambda \ge 0$,

$$\mathrm{Ent}(e^{\lambda Z}) \le \frac{\lambda^2}{2}\,\mathbb{E}[e^{\lambda Z}(Z-Z')_+^2].$$

Proof of the lemma. Let P be the law of Z, $\psi(s) = \log \mathbb{E}e^{sZ}$ (finite and smooth since Z is bounded), $M(s) = e^{\psi(s)}$, and let P_s be the tilted law $dP_s(z) = e^{sz-\psi(s)}\,dP(z)$, so $P_0 = P$. Then $\psi'(s) = \mathbb{E}_{P_s} Z$ and $\psi''(s) = \mathrm{Var}_{P_s}(Z)$. If Z, Z' are i.i.d. with law P_s, then $\mathrm{Var}_{P_s}(Z) = \frac{1}{2}\mathbb{E}_{P_s\otimes P_s}(Z-Z')^2 = \mathbb{E}_{P_s\otimes P_s}(Z-Z')_+^2$, because $(Z-Z')^2 = (Z-Z')_+^2 + (Z'-Z)_+^2$ and the two terms have equal expectations by exchangeability.

Left side: $\mathrm{Ent}(e^{\lambda Z}) = \mathbb{E}[\lambda Z e^{\lambda Z}] - M(\lambda)\log M(\lambda) = M(\lambda)(\lambda\psi'(\lambda) - \psi(\lambda))$. Since $\frac{d}{ds}(s\psi'(s) - \psi(s)) = s\psi''(s)$ and the bracket vanishes at $s = 0$,

$$\frac{\mathrm{Ent}(e^{\lambda Z})}{M(\lambda)} = \int_0^\lambda s\,\psi''(s)\,ds = \int_0^\lambda s\,\mathbb{E}_{P_s\otimes P_s}(Z-Z')_+^2\,ds.$$

Right side: for Z, Z' independent with law P, $\mathbb{E}[e^{\lambda Z}(Z-Z')_+^2]/M(\lambda) = \mathbb{E}_{P_\lambda\otimes P_0}(Z-Z')_+^2$, so

$$\frac{\lambda^2}{2}\cdot\frac{\mathbb{E}[e^{\lambda Z}(Z-Z')_+^2]}{M(\lambda)} = \int_0^\lambda s\,ds\cdot\mathbb{E}_{P_\lambda\otimes P_0}(Z-Z')_+^2.$$

Comparison: for $0 \le s \le \lambda$ the likelihood ratio $\ell = dP_\lambda/dP_s \propto e^{(\lambda-s)z}$ is nondecreasing in z with $\mathbb{E}_{P_s}\ell = 1$. For any c, $\mathbb{E}_{P_s}[(\ell(Z)-1)\mathbf{1}_{\{Z>c\}}] \ge 0$, since $\ell - 1$ is nondecreasing with P_s-mean 0 (if $\ell(c) \ge 1$ the integrand is ≥ 0; otherwise the integral equals $-\mathbb{E}_{P_s}[(\ell-1)\mathbf{1}_{\{Z\le c\}}] \ge 0$). Thus $P_\lambda\{Z > c\} \ge P_s\{Z > c\}$ for all c, i.e. $P_\lambda \ge_{\mathrm{st}} P_s$; likewise $P_s \ge_{\mathrm{st}} P_0$. The function $h(z,z') = (z-z')_+^2$ is nondecreasing in z and nonincreasing in z'. For a product measure, replacing the first marginal by a stochastically larger one does not decrease $\mathbb{E}h$ (apply the stochastic order to the nondecreasing function $z \mapsto \mathbb{E}h(z,Z')$), and replacing the second marginal by a stochastically smaller one does not decrease it either. Therefore

$$\mathbb{E}_{P_s\otimes P_s}h \le \mathbb{E}_{P_\lambda\otimes P_s}h \le \mathbb{E}_{P_\lambda\otimes P_0}h, \qquad 0 \le s \le \lambda.$$

Integrating against $s\,ds$ over $[0,\lambda]$ and multiplying by $M(\lambda)$ proves the lemma. □

(The constant $\frac{1}{2}$ is sharp: as $\lambda \to 0$ both sides equal $\frac{\lambda^2}{2}\mathrm{Var}(Z) + O(\lambda^3)$.)

Tensorization. By the sub-additivity of entropy (Theorem 4.10), $\mathrm{Ent}(e^{\lambda Z}) \le \sum_{i=1}^n \mathbb{E}[\mathrm{Ent}^{(i)}(e^{\lambda Z})]$, where $\mathrm{Ent}^{(i)}$ is the entropy with respect to X_i with $X^{(i)} = (X_1,\dots,X_{i-1},X_{i+1},\dots,X_n)$ fixed. Conditionally on $X^{(i)}$, $Z = g(X_i)$ and $Z_i' = g(X_i')$ with $g = f(X_1,\dots,X_{i-1},\cdot,X_{i+1},\dots,X_n)$, so Z and Z_i' are conditionally i.i.d. and bounded, and the lemma gives $\mathrm{Ent}^{(i)}(e^{\lambda Z}) \le \frac{\lambda^2}{2}\mathbb{E}[e^{\lambda Z}(Z-Z_i')_+^2 \,|\, X^{(i)}]$. Summing, and conditioning on X (with respect to which $e^{\lambda Z}$ is measurable),

$$\mathrm{Ent}(e^{\lambda Z}) \le \frac{\lambda^2}{2}\sum_{i=1}^n \mathbb{E}\big[e^{\lambda Z}(Z-Z_i')_+^2\big] = \frac{\lambda^2}{2}\mathbb{E}\big[e^{\lambda Z}V_+\big] \le \frac{\lambda^2 v}{2}\,\mathbb{E}e^{\lambda Z}, \qquad \lambda \ge 0.$$

Herbst's argument. Let $F(\lambda) = \log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$. For $\lambda > 0$,

$$\frac{d}{d\lambda}\Big(\frac{F(\lambda)}{\lambda}\Big) = \frac{\lambda F'(\lambda) - F(\lambda)}{\lambda^2} = \frac{\mathrm{Ent}(e^{\lambda Z})}{\lambda^2\,\mathbb{E}e^{\lambda Z}} \le \frac{v}{2},$$

and $F(\lambda)/\lambda \to F'(0) = 0$ as $\lambda \downarrow 0$. Integrating gives $F(\lambda) \le \lambda^2 v/2$, which is (1). By Markov's inequality, for $t > 0$ and $\lambda > 0$, $\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{\lambda^2 v/2 - \lambda t}$; for $v > 0$ take $\lambda = t/v$, and for $v = 0$ let $\lambda \to \infty$. Hence

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{-t^2/(2v)}, \qquad t > 0$$

(read as 0 when $v = 0$).

Lower tail. Apply the upper-tail result to $-Z = (-f)(X)$. Since $\big((-Z)-(-Z_i')\big)_+ = (Z_i'-Z)_+ = (Z-Z_i')_-$, the hypothesis $\mathbb{E}[\sum_i (Z-Z_i')_-^2 \,\big|\, X] \le v$ is exactly the upper-tail hypothesis for $-Z$, whence

$$\mathbb{P}\{Z < \mathbb{E}Z - t\} = \mathbb{P}\{-Z > \mathbb{E}(-Z) + t\} \le e^{-t^2/(2v)}, \qquad t > 0. \ \blacksquare$$

Exercise 6.2

Central idea. (Courant–Fischer.) Diagonalize A in an orthonormal eigenbasis; the Rayleigh quotient on a subspace spanned by eigenvectors is bounded by the extreme eigenvalues. Dimension counting forces any subspace of complementary dimension to meet these spans, giving both min-max formulas. For $B = PAP$, $R_B = R_A$ on $\mathrm{Ran}(P)$, so the min-max over the smaller family of subspaces in $\mathrm{Ran}(P)$ is squeezed between the global min-max values.

Solution. Let $Av_k = \alpha_k v_k$ with $\{v_k\}$ orthonormal, $\alpha_1 \le \cdots \le \alpha_n$. For $x = \sum c_k v_k \ne 0$, $R_A(x) = \sum \alpha_k |c_k|^2 / \sum |c_k|^2$.

Min-max formulas. Let $\dim U = n-k+1$. With $W = \mathrm{span}\{v_1, \dots, v_k\}$ ($\dim W = k$), $\dim(U \cap W) \ge (n-k+1) + k - n = 1$, so some $0 \ne x \in U \cap W$ has $R_A(x) \le \alpha_k$ (as $x \in W$). Hence $\min_{x \in U} R_A(x) \le \alpha_k$ for all such U, so $\max_{\dim U = n-k+1} \min_{x \in U} R_A(x) \le \alpha_k$. With $U_0 = \mathrm{span}\{v_k, \dots, v_n\}$ ($\dim = n-k+1$), $R_A \ge \alpha_k$ on U_0, giving $\ge \alpha_k$. Thus

$$\alpha_k = \max\{\min\{R_A(x) : x \in U, x \ne 0\} : \dim U = n-k+1\}.$$

Similarly, for $\dim U = k$, U meets $\mathrm{span}\{v_k, \dots, v_n\}$ ($\dim = n-k+1$) nontrivially, giving $\max_{x \in U} R_A(x) \ge \alpha_k$; with $U = \mathrm{span}\{v_1, \dots, v_k\}$ we get $\le \alpha_k$. Hence

$$\alpha_k = \min\{\max\{R_A(x) : x \in U, x \ne 0\} : \dim U = k\}.$$

Interlacing. $P^* = P = P^2$, $\mathrm{rank}(P) = m$; $B^* = PAP = B$, and B preserves $\mathrm{Ran}(P)$, with eigenvalues $\beta_1 \le \cdots \le \beta_m$ (on $\mathrm{Ran}(P)$). For $0 \ne x \in \mathrm{Ran}(P)$, $Px = x$, so $R_B(x) = x^*PAPx/x^*x = x^*Ax/x^*x = R_A(x)$.

For the lower bound, Courant–Fischer on $\mathrm{Ran}(P)$ gives $\beta_j = \min\{\max_{x \in V} R_B(x) : V \subset \mathrm{Ran}(P), \dim V = j\}$. Since $R_B = R_A$ there,

$$\beta_j = \min_{\substack{V \subset \mathrm{Ran}(P) \\ \dim V = j}} \max_{x \in V} R_A(x) \ge \min_{\dim V = j} \max_{x \in V} R_A(x) = \alpha_j,$$

as the minimum over the smaller family $\{V \subset \mathrm{Ran}(P)\}$ exceeds that over all V.

For the upper bound, $\beta_j = \max\{\min_{x \in V} R_B(x) : V \subset \mathrm{Ran}(P), \dim V = m-j+1\}$, so

$$\beta_j = \max_{\substack{V \subset \mathrm{Ran}(P) \\ \dim V = m-j+1}} \min_{x \in V} R_A(x) \le \max_{\dim V = m-j+1} \min_{x \in V} R_A(x).$$

By the first min-max formula with $n-k+1 = m-j+1$, i.e. $k = n-m+j$, the right side is α_{n-m+j}. Hence $\alpha_j \le \beta_j \le \alpha_{n-m+j}$ for $j \le m$. $\blacksquare$

Exercise 6.3

Central idea. Since $\mathrm{rank}(A-B) = k$, a unitary congruence kills the last $n-k$ rows and columns of $A-B$, so A, B share their $(n-k) \times (n-k)$ bottom-right block A_{22}. Cauchy interlacing (Exercise 6.2) sandwiches the eigenvalue counting functions of A and B around that of A_{22}, each within k; hence F_A and F_B differ by at most k/n uniformly.

Solution. $L_A = \frac{1}{n}\sum_{i=1}^n \delta_{\alpha_i}$, $L_B = \frac{1}{n}\sum_{i=1}^n \delta_{\beta_i}$ with $\alpha_1 \le \cdots \le \alpha_n$, $\beta_1 \le \cdots \le \beta_n$ the eigenvalues; $F_A(t) = \frac{1}{n}\#\{i : \alpha_i \le t\}$, similarly F_B.

Reduction. Let $k = \operatorname{rank}(A-B)$. Choose unitary U with $U^*(A-B)U = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$, D $k \times k$. Eigenvalues are unitarily invariant, so replace (A, B) by (U^*AU, U^*BU); then with matching block sizes

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} B_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

A_{22} Hermitian $(n-k) \times (n-k)$, since $A - B$ vanishes outside the top-left $k \times k$ block.

Interlacing. Let $\gamma_1 \le \cdots \le \gamma_{n-k}$ be the eigenvalues of $A_{22} = PAP$ (P the coordinate projection of rank $n-k$). By Exercise 6.2 with $m = n-k$,

$$\alpha_j \le \gamma_j \le \alpha_{j+k}, \qquad \beta_j \le \gamma_j \le \beta_{j+k}, \qquad j = 1, \dots, n-k.$$

Counting functions. Put $N_A(t) = \#\{i : \alpha_i \le t\} = nF_A(t)$, N_B, N_γ analogously. From $\alpha_j \le \gamma_j$: $\gamma_j \le t \Rightarrow \alpha_j \le t$, so $N_\gamma(t) \le N_A(t)$. From $\gamma_j \le \alpha_{j+k}$: $\alpha_{j+k} \le t \Rightarrow \gamma_j \le t$, so

$$N_\gamma(t) \ge \#\{j \le n-k : \alpha_{j+k} \le t\} = \#\{i \ge k+1 : \alpha_i \le t\} \ge N_A(t) - k.$$

Thus $N_A(t) - k \le N_\gamma(t) \le N_A(t)$, and likewise $N_B(t) - k \le N_\gamma(t) \le N_B(t)$. Hence

$$N_A(t) \le N_\gamma(t) + k \le N_B(t) + k, \qquad N_B(t) \le N_A(t) + k,$$

so $|N_A(t) - N_B(t)| \le k$ for all t. Dividing by n,

$$\|F_A - F_B\|_\infty \le \frac{k}{n}. \blacksquare$$

Exercise 6.4

Central idea. f is the distance to A, hence 1-Lipschitz. For $X \in \{0,1\}^n$, $f(X) = \sqrt{(\sum X_i - n/2)_+}$ (flipping $(\sum X_i - n/2)_+$ ones to zeros attains it), so $Mf(X) = 0$ (as $P\{\sum X_i \le n/2\} \ge 1/2$) while $P\{f(X) > cn^{1/4}\} = P\{\sum X_i - n/2 > c^2\sqrt{n}\} \to P\{N(0,1) > 2c^2\} \ge 1/4$ for small c by the CLT — fluctuations of order $n^{1/4}$, not $O(1)$ as Theorem 6.10 would give.

Erratum. The set in the statement is wrong as printed. With $A = \{x \in [0,1]^n : \sum_i x_i \le n/2\}$, A is convex, so $f = d(\cdot, A)$ is convex (not "not convex"), Theorem 6.10 applies, and the claim is false. Indeed, for $x \in \{0,1\}^n$ with $m = \sum_i x_i > n/2$ ones, the nearest point of this convex A is y with $y_i = 1 - \delta$ on the m one-coordinates and $y_i = 0$ elsewhere, $\delta = (m - n/2)/m \in (0, 1/2]$ (KKT: on the ones the gradient $2(y_i - x_i) = -2\delta$ is balanced by the multiplier $\mu = 2\delta \ge 0$ of $\sum_i y_i \le n/2$; on the zeros $y_i = 0$ is at its lower bound with multiplier $\mu \ge 0$), so

$$f(x) = \frac{(\sum_i x_i - n/2)_+}{\sqrt{\sum_i x_i}}$$

(confirmed numerically against a constrained minimizer to 3×10^{-15} on 200 random cases). Since $(S_n - n/2)/\sqrt{S_n} \approx (S_n - n/2)/\sqrt{n/2}$ is $O_P(1)$, $f(X)$ has $O(1)$ fluctuations: $Mf(X) = 0$ and $\mathbb{P}\{f(X) > \frac{1}{2}n^{1/4}\} = 0.0060$ at $n = 100$ and 1.1×10^{-13} at $n = 10^4$ (exact binomial sums), not $\ge 1/4$. The example needs the discrete set $A = \{0,1\}^n \cap \{\sum_i x_i \le n/2\}$ (as in Ledoux–Talagrand 1991, p. 17), for which $f(X) = \sqrt{(\sum_i X_i - n/2)_+}$, $Mf(X) = 0$, and $c = 1/2$ works with limiting probability $\mathbb{P}\{N(0,1) > 1/2\} \approx 0.3085 > 1/4$. This corrected statement is what is proved below.

Solution. $f(x) = \inf_{y \in A} \|x - y\|$ is 1-Lipschitz: $|f(x) - f(x')| \le \|x - x'\|$ by the triangle inequality.

Let $S_n = \sum_{i=1}^n X_i \sim \text{Binomial}(n, 1/2)$. For $x \in \{0,1\}^n$ with $m = \sum x_i > n/2$, the closest $y \in A$ (with A the discrete set, see the Erratum) is obtained by changing $m - n/2$ ones to zeros, giving $f(x) = \sqrt{m - n/2}$; if $m \le n/2$ then $x \in A$ and $f(x) = 0$. Hence

$$f(X) = \sqrt{(S_n - n/2)_+} \quad \text{a.s.}$$

Since n is even, $P\{S_n \le n/2\} \ge 1/2$ (the median of Binomial$(n, 1/2)$ is $\ge n/2$), so $P\{f(X) = 0\} \ge 1/2$ and $Mf(X) = 0$.

Now, for $c > 0$,

$$\mathbb{P}\{f(X) > cn^{1/4}\} = \mathbb{P}\{(S_n - n/2)_+ > c^2\sqrt{n}\} = \mathbb{P}\{S_n - n/2 > c^2\sqrt{n}\}.$$

By the CLT, $\frac{S_n - n/2}{\sqrt{n}/2} \xrightarrow{d} N(0,1)$, so

$$\mathbb{P}\{S_n - n/2 > c^2\sqrt{n}\} \longrightarrow \mathbb{P}\{N(0,1) > 2c^2\}.$$

With $c = 1/2$, $2c^2 = 1/2$ and $\mathbb{P}\{N(0,1) > 1/2\} \approx 0.309 > 1/4$. Hence for all sufficiently large (even) n,

$$\mathbb{P}\{f(X) > Mf(X) + cn^{1/4}\} \ge 1/4,$$

with $c = 1/2$. The fluctuations are of order $n^{1/4} \to \infty$, whereas Theorem 6.10 would give $P\{Z > \mathbb{E}Z + t\} \le e^{-t^2/2}$ (sub-Gaussian, $O(1)$ fluctuations). Thus convexity cannot be dropped.

Remark. With the discrete A, f is genuinely non-convex: $u = (1, \dots, 1, 0, \dots, 0)$ ($n/2$ ones) and $\mathbf{1} - u$ both lie in A, so $f(u) = f(\mathbf{1} - u) = 0$, while their midpoint $(\frac{1}{2}, \dots, \frac{1}{2})$ is at distance $\sqrt{n}/2 > 0$ from every point of $\{0,1\}^n$. ■

Exercise 6.5

Central idea. This is Theorem 6.10 with $[0,1]^n$ replaced by $\mathcal{X}^n$. Apply Theorem 6.7 with $v = B^2$: the key is the vector analogue of the gradient estimate from Theorem 3.17. For $Z_i = \inf_{x_i'} f(X^{(i)}, x_i')$ attained at $x_i^* \in \mathcal{X}$, separate convexity gives $Z - Z_i \le \langle \nabla_{x_i} f(X), X_i - x_i^* \rangle$, whence $(Z - Z_i)^2 \le B^2 \|\nabla_{x_i} f(X)\|^2$ by Cauchy–Schwarz and $\|X_i - x_i^*\| \le B$. Summing over i and using $\|\nabla f(X)\| \le 1$ (1-Lipschitz) yields $\sum_i (Z - Z_i)^2 \le B^2$.

Solution. We may assume f is continuously differentiable on $\mathcal{X}^n$; otherwise approximate f uniformly by smooth separately convex 1-Lipschitz functions via mollification (which preserves separate convexity and does not increase the Lipschitz constant) and pass to the limit. Write $X = (X_1, \dots, X_n)$ with $X_i \in \mathcal{X}$, and for each i put $Z_i = \inf_{x_i' \in \mathcal{X}} f(X_1, \dots, x_i', \dots, X_n)$. Since $\mathcal{X}$ is compact and f continuous, the infimum is attained at some $x_i^* \in \mathcal{X}$.

Fix X and i, and consider $g_i(x_i) = f(X_1, \dots, x_i, \dots, X_n)$ as a function of $x_i \in \mathcal{X}$. By separate convexity g_i is convex, and it is 1-Lipschitz since f is 1-Lipschitz on $\mathcal{X}^n \subset \mathbb{R}^{dn}$ and $x_i \mapsto (X_1, \dots, x_i, \dots, X_n)$ is an isometry. We have $Z - Z_i = g_i(X_i) - g_i(x_i^*) \ge 0$. By the first-order condition for convex differentiable functions,

$$g_i(X_i) - g_i(x_i^*) \le \langle \nabla g_i(X_i),\, X_i - x_i^* \rangle.$$

Note $\nabla g_i(X_i)$ is the i-th d-dimensional block of $\nabla f(X)$. By Cauchy–Schwarz and $\|X_i - x_i^*\| \le B$ (diameter of $\mathcal{X}$),

$$(Z - Z_i)^2 \le \|\nabla g_i(X_i)\|^2 \, \|X_i - x_i^*\|^2 \le B^2 \, \|\nabla g_i(X_i)\|^2.$$

Summing over i,

$$\sum_{i=1}^n (Z - Z_i)^2 \le B^2 \sum_{i=1}^n \|\nabla g_i(X_i)\|^2 = B^2 \, \|\nabla f(X)\|^2 \le B^2 \quad \text{a.s.},$$

where $\|\nabla f(X)\| \le 1$ follows from f being 1-Lipschitz (at every point of differentiability, hence a.e. by Rademacher's theorem, and the bound extends by continuity). Theorem 6.7 applied with $v = B^2$ therefore gives, for all $t > 0$,

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{-t^2/(2B^2)}.$$

■

Exercise 6.6

Central idea. The largest singular value $f(X_1, \dots, X_n) = \|A\|_{\mathrm{op}} = \sup_{\|u\|=\|v\|=1} u^T A v$ is a supremum of functions affine in each column X_i separately, hence separately convex; it is 1-Lipschitz for the product Euclidean norm since $|\|A\|_{\mathrm{op}} - \|A'\|_{\mathrm{op}}| \le \|A - A'\|_{\mathrm{op}} \le \|A - A'\|_F$. Exercise 6.5 then applies directly.

Solution. Let $A = [X_1 \mid \cdots \mid X_n]$ and $f(X_1, \dots, X_n) = \|A\|_{\mathrm{op}} = \sup_{\|u\|_2=\|v\|_2=1} u^T A v$, the largest singular value.

Separate convexity. Fix X_j for $j \neq i$. Then $u^T A v = \sum_{j=1}^n v_j (u^T X_j) = v_i (u^T X_i) + C(u, v)$ where $C(u, v)$ does not depend on X_i. For each fixed (u, v) this is affine in X_i, so $X_i \mapsto f(X_1, \dots, X_n) = \sup_{\|u\|=\|v\|=1} [v_i (u^T X_i) + C(u, v)]$ is a supremum of affine functions, hence convex in X_i. Thus f is separately convex on $\mathcal{X}^n$.

1-*Lipschitz.* By the reverse triangle inequality, $|\|A\|_{\mathrm{op}} - \|A'\|_{\mathrm{op}}| \le \|A - A'\|_{\mathrm{op}}$. Since the operator norm is bounded by the Frobenius norm,

$$\|A - A'\|_{\mathrm{op}} \le \|A - A'\|_F = \Big(\sum_{i=1}^n \|X_i - X_i'\|_2^2 \Big)^{1/2},$$

which is exactly the Euclidean norm of $(X_1, \dots, X_n) - (X_1', \dots, X_n')$ in $\mathbb{R}^{dn}$. Hence $|f(x) - f(y)| \le \|x - y\|$ for all $x, y \in \mathcal{X}^n$.

Exercise 6.5 therefore applies and yields, for $Z = f(X_1, \dots, X_n)$ and all $t > 0$,

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{-t^2/(2B^2)}.$$

Comparison with Example 6.11. Example 6.11 assumes the *mn entries* $X_{i,j} \in [0, 1]$ are independent and applies Theorem 6.10 on $[0, 1]^{mn}$ to obtain $\mathbb{P}\{Z > \mathbb{E}Z + t\} \le e^{-t^2/2}$. Here only the n *columns* are assumed independent — the entries within each column may be arbitrarily dependent — and the conclusion is $e^{-t^2/(2B^2)}$. For instance, if $\mathcal{X} = [0, 1]^d$ then $B = \sqrt{d}$ and the bound is $e^{-t^2/(2d)}$, weaker than Example 6.11's $e^{-t^2/2}$ by the dimension factor d. The price of weakening entrywise independence to columnwise independence is the diameter B of each independent block appearing in the exponent: each column contributes fluctuations on the scale of $\mathrm{diam}(\mathcal{X}) = B$ rather than 1. ■

Exercise 6.7

Central idea. This is the $g(X)$-version of Theorem 6.27. For $\lambda < 0$, the third inequality of Theorem 6.15 (symmetrized modified log-Sobolev) bounds the Herbst deficit $\lambda\psi'(\lambda) - \psi(\lambda)$ by $\lambda^2 \mathbb{E}[e^{\lambda Z} V^-]/\mathbb{E}[e^{\lambda Z}]$ where $V^- = \sum_i ((Z_i' - Z)_+)^2 \le g(X)$, using $\tau(x) \le x^2$ for $x \le 0$. Since $\lambda < 0$ and f is nondecreasing, $e^{\lambda Z}$ is nonincreasing while g is nondecreasing, so Harris's inequality (Theorem 2.15) decouples $\mathbb{E}[e^{\lambda Z} g(X)] \le \mathbb{E}[e^{\lambda Z}]\mathbb{E}[g(X)]$. The Herbst argument then gives $\psi(\lambda) \le \lambda^2 \mathbb{E}[g(X)]$ and Chernoff yields the lower tail.

Solution. Let X_i' be an independent copy of X_i and $Z_i' = f(X_1, \dots, X_i', \dots, X_n)$. Put $V^- = \sum_{i=1}^n ((Z_i' - Z)_+)^2 = \sum_{i=1}^n ((Z - Z_i')_-)^2 \le g(X)$ a.s. Assume $0 < \mathbb{E}[g(X)] < \infty$ (if $\mathbb{E}[g(X)] = \infty$ the bound is trivially 1; the degenerate case $\mathbb{E}[g(X)] = 0$ is excluded). Write $F(\lambda) = \mathbb{E}[e^{\lambda Z}]$ and $\psi(\lambda) = \log \mathbb{E}[e^{\lambda(Z - \mathbb{E}Z)}] = \log F(\lambda) - \lambda \mathbb{E}Z$, so that $\lambda\psi'(\lambda) - \psi(\lambda) = \lambda F'(\lambda)/F(\lambda) - \log F(\lambda)$.

For $\lambda < 0$, the third inequality of Theorem 6.15 gives, with $\tau(x) = x(e^x - 1)$,

$$\lambda \frac{F'(\lambda)}{F(\lambda)} - \log F(\lambda) \le \frac{1}{F(\lambda)} \sum_{i=1}^n \mathbb{E}\big[e^{\lambda Z}\, \tau(\lambda (Z_i' - Z)_+)\big].$$

Since $\lambda < 0$ and $(Z_i' - Z)_+ \ge 0$, we have $\lambda(Z_i' - Z)_+ \le 0$. For $x \le 0$, writing $x = -y$ with $y \ge 0$, $\tau(x) = \tau(-y) = y(1 - e^{-y}) \le y^2 = x^2$ (using $1 - e^{-y} \le y$). Hence $\tau(\lambda(Z_i' - Z)_+) \le \lambda^2 ((Z_i' - Z)_+)^2$, and

$$\lambda\psi'(\lambda) - \psi(\lambda) \le \lambda^2 \frac{\mathbb{E}[e^{\lambda Z} V^-]}{F(\lambda)} \le \lambda^2 \frac{\mathbb{E}[e^{\lambda Z} g(X)]}{F(\lambda)}, \qquad \lambda < 0.$$

Because f is nondecreasing in each variable, $Z = f(X)$ is nondecreasing in X, so for $\lambda < 0$ the function $X \mapsto e^{\lambda Z}$ is nonincreasing, while $X \mapsto g(X)$ is nondecreasing by hypothesis. Harris's inequality (Theorem 2.15, opposite-monotonicity case) gives $\mathbb{E}[e^{\lambda Z} g(X)] \le \mathbb{E}[e^{\lambda Z}]\,\mathbb{E}[g(X)]$. Therefore

$$\lambda \psi'(\lambda) - \psi(\lambda) \le \lambda^2\, \mathbb{E}[g(X)], \qquad \lambda < 0.$$

This is the Herbst differential inequality: $\frac{d}{d\lambda}[\psi(\lambda)/\lambda] = (\lambda\psi'(\lambda) - \psi(\lambda))/\lambda^2 \le \mathbb{E}[g(X)]$. Integrating from $\lambda < 0$ to 0 and using $\psi(\lambda)/\lambda \to \psi'(0) = 0$ gives $-\psi(\lambda)/\lambda \le -\lambda\, \mathbb{E}[g(X)]$, i.e.

$$\psi(\lambda) \le \lambda^2\, \mathbb{E}[g(X)], \qquad \lambda < 0.$$

Finally, Chernoff's bound for the lower tail: for $\lambda < 0$ and $t > 0$,

$$\mathbb{P}\{Z - \mathbb{E}Z \le -t\} = \mathbb{P}\{e^{\lambda(Z - \mathbb{E}Z)} \ge e^{-\lambda t}\} \le e^{\lambda t + \psi(\lambda)} \le \exp\left(\lambda t + \lambda^2 \mathbb{E}[g(X)]\right).$$

Minimizing over $\lambda < 0$ at $\lambda = -t/(2\mathbb{E}[g(X)])$ yields

$$\mathbb{P}\{Z < \mathbb{E}Z - t\} \le \exp\left(\frac{-t^2}{4\,\mathbb{E}[g(X)]}\right), \qquad t > 0.$$

■

Exercise 6.8

Central idea. The natural proof applies the modified log-Sobolev inequality (Theorem 6.6) with $Z_i = \inf_{x_i'} f$, giving $\lambda\psi' - \psi \le \frac{\lambda^2}{2}\mathbb{E}[e^{\lambda Z} V(X)]/\mathbb{E}[e^{\lambda Z}]$ where $V(X) = \sum_i (Z - Z_i)^2 \le v + C\mathbf{1}_{A^c}(X)$. To close the Herbst argument one needs $\mathbb{E}[e^{\lambda Z}\mathbf{1}_{A^c}] \le \mathbb{E}[e^{\lambda Z}]\mathbb{P}(A^c)$, which via Harris (Theorem 2.15) requires $e^{\lambda Z}$ nondecreasing, i.e. f nondecreasing — a hypothesis missing from the statement (compare Exercise 6.7, which assumes it explicitly). We document the gap and prove the corrected version.

Erratum. The statement is missing a hypothesis, and without it the stated bound is false: f must be nondecreasing in each variable (the Harris step in the hint needs it; compare Exercise 6.7, which assumes it). Counterexample: $n = 1$, $\mathbb{P}\{X = 0\} = 0.1 = 1 - \mathbb{P}\{X = 1\}$, $A = [1, \infty)$ (monotone), $f(x) = \mathbf{1}_{\{x<1\}}$. For $x \in A$, $f(x) - \inf_{x'} f(x') = 0 \le v$; for $x \notin A$, it equals $1 \le C$; so the hypotheses hold with $v = 10^{-9}$, $C = 1$, and $\mathbb{P}\{X \notin A\} = 0.1$. Here $Z = \mathbf{1}_{\{X=0\}}$, $\mathbb{E}Z = 0.1$, and at $t = 0.9 - 10^{-6}$, $\mathbb{P}\{Z > \mathbb{E}Z + t\} = \mathbb{P}\{Z = 1\} = 0.1$, whereas the claimed bound is $\exp(-t^2/(2(10^{-9} + 0.1))) \approx 0.0174$. With "$f$ nondecreasing in each variable" added, the stated bound holds; that corrected statement is proved below.

Solution. *Remark on the statement.* As written, the Harris step of the proof is not justified: with $V(X) \le v\mathbf{1}_A(X) + C\mathbf{1}_{A^c}(X)$, one needs $\mathbb{E}[e^{\lambda Z}\mathbf{1}_{A^c}(X)] \le \mathbb{E}[e^{\lambda Z}]\,\mathbb{P}(X \notin A)$ for $\lambda > 0$, i.e. $\mathrm{Cov}(e^{\lambda Z}, \mathbf{1}_A) \ge 0$. By Harris this requires $e^{\lambda Z}$ nondecreasing, i.e. f nondecreasing in each variable, which is not assumed. Counterexample to the correlation step: $n = 1$, $X_1 \in \{0, 1\}$ with $\mathbb{P}\{X_1 = 0\} = \mathbb{P}\{X_1 = 1\} = 1/2$, $A = [1, \infty)$ (monotone), $f(x) = 100 \cdot \mathbf{1}_{\{x<1\}}$, so $f(0) = 100$, $f(1) = 0$. Then $V(0) = 10000$, $V(1) = 0$, so the hypotheses hold with any $v > 0$ and $C = 10000$, but $\mathbb{E}[e^{\lambda Z}\mathbf{1}_{A^c}]/\mathbb{E}[e^{\lambda Z}] = e^{100\lambda}/(e^{100\lambda} + 1) \to 1 > 1/2 = \mathbb{P}(A^c)$ as $\lambda \to \infty$. (This example breaks only the correlation step, not the final bound: here $\mathbb{E}Z = 50$, $\mathbb{P}\{Z > \mathbb{E}Z + t\} = 1/2$ for $t < 50$ and 0 for $t \ge 50$, while the bound $e^{-t^2/10^4}$ exceeds $e^{-1/4} \approx 0.78$ for $t < 50$; the Erratum gives an example that violates the bound itself.) We prove the result under the additional (evidently intended) hypothesis that f is nondecreasing in each variable.

Assume henceforth f is nondecreasing in each variable, and w.l.o.g. $C \ge v$ (otherwise $V \le v$ everywhere and Theorem 6.7 gives the claim since $v \le v + C\mathbb{P}(A^c)$). For $\lambda > 0$, Theorem 6.6 with $Z_i = \inf_{x_i'} f(X_1, \dots, x_i', \dots, X_n)$ gives

$$\lambda \frac{F'(\lambda)}{F(\lambda)} - \log F(\lambda) \le \frac{1}{F(\lambda)} \sum_{i=1}^{n} \mathbb{E}[e^{\lambda Z}\, \phi(-\lambda(Z - Z_i))],$$

where $F(\lambda) = \mathbb{E}[e^{\lambda Z}]$ and $\phi(x) = e^x - x - 1$. Since $Z \geq Z_i$ and $\lambda > 0$, $-\lambda(Z - Z_i) \leq 0$, and $\phi(x) \leq x^2/2$ for $x \leq 0$; hence $\phi(-\lambda(Z - Z_i)) \leq \frac{\lambda^2}{2}(Z - Z_i)^2$. With $V(X) = \sum_i (Z - Z_i)^2$ and $\psi(\lambda) = \log \mathbb{E}[e^{\lambda(Z - \mathbb{E}Z)}]$,

$$\lambda\psi'(\lambda) - \psi(\lambda) \leq \frac{\lambda^2}{2} \frac{\mathbb{E}[e^{\lambda Z} V(X)]}{F(\lambda)}, \qquad \lambda > 0.$$

By hypothesis $V(X) \leq v\mathbf{1}_A(X) + C\mathbf{1}_{A^c}(X) \leq v + C\mathbf{1}_{A^c}(X)$, so

$$\mathbb{E}[e^{\lambda Z} V(X)] \leq vF(\lambda) + C\,\mathbb{E}[e^{\lambda Z}\mathbf{1}_{A^c}(X)].$$

Since f is nondecreasing and $\lambda > 0$, $X \mapsto e^{\lambda f(X)}$ is nondecreasing, while $X \mapsto \mathbf{1}_{A^c}(X)$ is nonincreasing because A is upward closed. Harris's inequality (Theorem 2.15, opposite-monotonicity case) yields $\mathbb{E}[e^{\lambda Z}\mathbf{1}_{A^c}(X)] \leq F(\lambda)\,\mathbb{P}(X \notin A)$. Consequently, with $p = \mathbb{P}(X \notin A)$,

$$\lambda\psi'(\lambda) - \psi(\lambda) \leq \frac{\lambda^2}{2}(v + Cp), \qquad \lambda > 0.$$

The Herbst argument ($\frac{d}{d\lambda}[\psi(\lambda)/\lambda] \leq (v + Cp)/2$, integrate 0 to λ) gives $\psi(\lambda) \leq \frac{\lambda^2}{2}(v + Cp)$, and Chernoff's bound $\mathbb{P}\{Z - \mathbb{E}Z \geq t\} \leq \inf_{\lambda > 0} e^{-\lambda t + \psi(\lambda)}$, optimized at $\lambda = t/(v + Cp)$, yields for all $t > 0$

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \leq \exp\left(\frac{-t^2}{2(v + C\,\mathbb{P}\{X \notin A\})}\right).$$

■

Exercise 6.9

Central idea. For $Z = \max_{M \in \mathcal{T}} \sum_{i,j} M_{i,j}\varepsilon_i\varepsilon_j$, flipping $\varepsilon_k \to -\varepsilon_k$ changes the maximizing quadratic form by $4\varepsilon_k \sum_j M^*_{k,j}\varepsilon_j$ (symmetry, zero diagonal), so the Efron–Stein quantity V_Z^+ is at most $8Y^2$, giving $\mathrm{Var}(Z) \leq 8\mathbb{E}Y^2$. Writing $Y = \max_{M, \|\alpha\|_2 \leq 1} \sum_{i,j} \alpha_i M_{i,j}\varepsilon_j$ as the supremum of a Rademacher process, the same flip applied to $W = Y^2$ gives $\sum_k (W - W^{(k)})_+^2 \leq 16BW$ via $\|M^*\alpha^*\| \leq \sqrt{B}$ — i.e. W is $(16B, 0)$-weakly self-bounding. Theorem 6.16, through the differential inequality $\lambda G' - G \leq \frac{\lambda\theta}{1 - \lambda\theta}\rho(\lambda/\theta)$ established in its proof, converts $V_W^+ \leq 8BW$ into the Bernstein mgf bound for W. For Z, the sharp entropy inequality of Exercise 6.1 plus entropy duality gives $\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \frac{\lambda\theta}{2 - \lambda\theta} \log \mathbb{E}e^{\lambda V_Z^+/\theta}$; with $V_Z^+ \leq 8W$ and the choice $\theta = 8\sqrt{B}$ this yields the corrected (scale-invariant) fourth inequality — the printed one is false (see Erratum).

Erratum. The first three displays of the statement are correct, but the fourth, $\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \frac{16\lambda^2}{2(1 - 64B\lambda)}\mathbb{E}[Y^2]$, is false as printed (it is not scale invariant: $B\lambda$ scales like M, so replacing $\mathcal{T}$ by $c\mathcal{T}$, $c \to 0$, would make Z sub-Gaussian at every λ). Counterexample: $\mathcal{T} = \{c(J - I)\}$ with $n = 1000$, $c = 10^{-9}$, so $Z = c((\sum_i \varepsilon_i)^2 - n)$, $\mathbb{E}Z = 0$, $B = c^2(n-1)^2$, $\mathbb{E}[Y^2] = c^2 n(n-1)$; at $\lambda = 0.6/(cn)$ one has $64B\lambda \approx 3.8 \times 10^{-5} < 1$, the exact left side (binomial sum) is 24.77, and the printed right side is 2.88. The printed form arises from dropping the term $G(\lambda/8B)$ below, which Jensen's inequality does not allow. Corrected statement (proved below): for $0 \leq \lambda < 1/(4\sqrt{B})$,

$$\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \frac{4\lambda^2\,\mathbb{E}[Y^2]}{(1 - 4\sqrt{B}\lambda)^2}, \qquad \text{hence} \qquad \log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \frac{8\lambda^2}{2(1 - 8\sqrt{B}\lambda)}\,\mathbb{E}[Y^2] \quad (0 \leq \lambda < 1/(8\sqrt{B})),$$

i.e. the scale parameter is $8\sqrt{B} = 8\max_M \|M\|$, not $64B$. (The Theorem 6.16 route of the hint gives instead $\frac{8\lambda^2}{(1 - 64B\lambda)(1 - \lambda)}\mathbb{E}[Y^2]$ for $64B\lambda < 1$ and $\lambda < 1$; this is also valid but its range $\lambda < 1$ is not scale invariant — see the final remark.)

Solution. Write $\varepsilon^{(k)}$ for ε with ε_k flipped, and $Z^{(k)}, W^{(k)}$ for the corresponding values. For Rademacher variables an independent copy ε'_k equals ε_k or $-\varepsilon_k$ with probability $1/2$ each, so with $V^+ = \sum_k \mathbb{E}'[(Z - Z'_k)_+^2]$,

$$V^+ = \tfrac{1}{2}\sum_k (Z - Z^{(k)})_+^2,$$

and similarly for W.

Let M^* attain the maximum in Z. Since $Z^{(k)} \geq \sum_{i,j} M^*_{i,j}\varepsilon_i^{(k)}\varepsilon_j^{(k)}$,

$$Z - Z^{(k)} \leq \sum_{i,j} M^*_{i,j}\varepsilon_i\varepsilon_j - \sum_{i,j} M^*_{i,j}\varepsilon_i^{(k)}\varepsilon_j^{(k)}.$$

Only terms with $i = k$ or $j = k$ survive (the diagonal vanishes); using $M^*_{i,k} = M^*_{k,i}$,

$$Z - Z^{(k)} \leq 2\varepsilon_k\Big(\sum_{j\neq k} M^*_{k,j}\varepsilon_j + \sum_{i\neq k} M^*_{i,k}\varepsilon_i\Big) = 4\varepsilon_k \sum_j M^*_{k,j}\varepsilon_j.$$

Hence $(Z - Z^{(k)})_+ \leq 4\big|\sum_j M^*_{k,j}\varepsilon_j\big|$ and

$$\sum_k (Z - Z^{(k)})_+^2 \leq 16\sum_k \Big(\sum_j M^*_{k,j}\varepsilon_j\Big)^2 \leq 16 \max_M \sum_k \Big(\sum_j M_{k,j}\varepsilon_j\Big)^2 = 16Y^2.$$

Therefore $V_Z^+ \leq 8Y^2$ — i.e. $8Y^2$ upper-bounds the Efron–Stein estimate — and by Efron–Stein,

$$\mathrm{Var}(Z) \leq \mathbb{E}V_Z^+ \leq 8\mathbb{E}[Y^2].$$

For fixed M, $(\sum_i(\sum_j \varepsilon_j M_{i,j})^2)^{1/2} = \|M\varepsilon\|_2 = \max_{\|\alpha\|_2\leq 1}\sum_{i,j}\alpha_i M_{i,j}\varepsilon_j$, so with $\mathcal{T}$ finite,

$$Y = \max_{M\in\mathcal{T},\,\|\alpha\|_2\leq 1} \sum_{i,j} \alpha_i M_{i,j}\varepsilon_j,$$

a supremum of a Rademacher process. Let $W = Y^2$ and let (M^*, α^*) attain the maximum; put $S = \sum_{i,j}\alpha_i^* M^*_{i,j}\varepsilon_j$ so $W = S^2$. Since (M^*, α^*) is feasible for the flipped vector, $W^{(k)} \geq (S^{(k)})^2$ where $S^{(k)} = S - 2t_k\varepsilon_k$ with $t_k = \sum_i \alpha_i^* M^*_{i,k}$. Thus

$$W - W^{(k)} \leq S^2 - (S^{(k)})^2 = 4t_k\varepsilon_k S - 4t_k^2 \leq 4|t_k|\,|S| = 4|t_k|Y,$$

so $(W - W^{(k)})_+^2 \leq 16Wt_k^2$. Summing and using $\sum_k t_k^2 = \|M^*\alpha^*\|_2^2 \leq \|M^*\|^2\|\alpha^*\|_2^2 \leq B$,

$$\sum_k (W - W^{(k)})_+^2 \leq 16BW,$$

i.e. $W = Y^2$ is $(16B, 0)$-weakly self-bounding. Consequently $V_W^+ \leq 8BW$ and

$$\mathrm{Var}(Y^2) \leq \mathbb{E}V_W^+ \leq 8B\,\mathbb{E}[Y^2].$$

For the mgf bounds we use the differential inequality from the proof of Theorem 6.16: with $G(\lambda) = \log \mathbb{E}e^{\lambda(W-\mathbb{E}W)}$ and $\rho(u) = \log \mathbb{E}e^{uV^+}$,

$$\lambda G'(\lambda) - G(\lambda) \leq \frac{\lambda\theta}{1-\lambda\theta}\,\rho(\lambda/\theta), \qquad \lambda\theta < 1.$$

Since $V_W^+ \leq 8BW$, $\rho(u) \leq \log \mathbb{E}e^{8BuW} = G(8Bu) + 8Bu\,\mathbb{E}W$. Taking $\theta = 8B$ gives $\rho(\lambda/(8B)) \leq G(\lambda) + \lambda M$ with $M = \mathbb{E}W$, whence

$$\lambda G' - G \leq \frac{8B\lambda}{1-8B\lambda}(G + \lambda M).$$

With integrating factor $(1 - 8B\lambda)/\lambda$ (since $\int \frac{d\lambda}{\lambda(1-8B\lambda)} = \log\frac{\lambda}{1-8B\lambda}$) and $G(u) = O(u^2)$ as $u \to 0$,

$$\frac{d}{d\lambda}\Big(G(\lambda)\frac{1-8B\lambda}{\lambda}\Big) \leq 8BM, \qquad \text{so} \qquad G(\lambda) \leq \frac{8B\lambda^2}{1-8B\lambda}M,$$

i.e.

$$\log \mathbb{E}e^{\lambda(Y^2-\mathbb{E}Y^2)} \le \frac{\lambda^2}{1-8B\lambda}\, 8B\, \mathbb{E}[Y^2], \qquad 8B\lambda < 1.$$

The identity follows from $\sum_i (\sum_j \varepsilon_j M_{i,j})^2 = \sum_{j,k} \varepsilon_j \varepsilon_k \sum_i M_{i,j} M_{i,k}$ and $\sum_i M_{i,j} M_{i,k} = (M^T M)_{j,k} = (M^2)_{j,k}$ by symmetry, so

$$\mathbb{E}[Y^2] = \mathbb{E}\Big[\sup_{M\in\mathcal{T}} \sum_{i,j} \varepsilon_i \varepsilon_j (M^2)_{i,j}\Big].$$

Finally, the fourth inequality, in the corrected form of the Erratum. Let $H(\lambda) = \log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$, $F_Z(\lambda) = \mathbb{E}e^{\lambda Z}$, and $\rho_Z(u) = \log \mathbb{E}e^{uV_Z^+}$ (all finite: Z is bounded). By Exercise 6.1 (its tensorized entropy inequality, valid for bounded Z), for $\lambda \ge 0$,

$$\mathrm{Ent}(e^{\lambda Z}) \le \frac{\lambda^2}{2}\, \mathbb{E}[e^{\lambda Z} V_Z^+].$$

By the duality formula for entropy (Corollary 4.15), $\mathbb{E}[UY] \le \mathrm{Ent}(Y) + \mathbb{E}Y \log \mathbb{E}e^U$ for $Y > 0$; with $Y = e^{\lambda Z}$, $U = \lambda V_Z^+/\theta$ ($\theta > 0$),

$$\mathbb{E}\big[V_Z^+ e^{\lambda Z}\big] \le \frac{\theta}{\lambda}\Big[\mathrm{Ent}(e^{\lambda Z}) + F_Z(\lambda)\, \rho_Z(\lambda/\theta)\Big].$$

Since $\mathrm{Ent}(e^{\lambda Z}) = F_Z(\lambda)(\lambda H'(\lambda) - H(\lambda))$, combining the two displays gives $(1-\lambda\theta/2)(\lambda H' - H) \le \frac{\lambda\theta}{2}\rho_Z(\lambda/\theta)$, i.e.

$$\frac{d}{d\lambda}\Big(\frac{H(\lambda)}{\lambda}\Big) = \frac{\lambda H' - H}{\lambda^2} \le \frac{\theta}{2-\lambda\theta} \cdot \frac{\rho_Z(\lambda/\theta)}{\lambda}, \qquad 0 < \lambda < 2/\theta.$$

Fix θ and $\lambda_1 < 2/\theta$. Since ρ_Z is convex with $\rho_Z(0) = 0$, $u \mapsto \rho_Z(u)/u$ is nondecreasing, and $s \mapsto \theta/(2-s\theta)$ is increasing; so on $(0, \lambda_1]$ the right side is at most its value at λ_1. Integrating from 0 (where $H(s)/s \to 0$) to λ_1,

$$H(\lambda_1) \le \frac{\lambda_1\theta}{2-\lambda_1\theta}\, \rho_Z(\lambda_1/\theta).$$

Now $V_Z^+ \le 8W$ and, by the bound $\log \mathbb{E}e^{\mu W} \le \mu M + G(\mu)$ with the third inequality in its sharper intermediate form (obtained the same way from the entropy inequality of Exercise 6.1 and $V_W^+ \le 8BW$: $\mathrm{Ent}(e^{\mu W}) \le 4B\mu^2 \mathbb{E}[We^{\mu W}]$, i.e. $\mu(1-4B\mu)L' \le L$ for $L = \log \mathbb{E}e^{\mu W}$, which integrates to $L(\mu) \le \mu M/(1-4B\mu)$ for $4B\mu < 1$, using $L(\mu)/\mu \to M$ as $\mu \to 0$),

$$\rho_Z(\lambda/\theta) \le \log \mathbb{E}e^{8\lambda W/\theta} \le \frac{8\lambda M/\theta}{1 - 32B\lambda/\theta}.$$

Hence, for $\lambda\theta < 2$ and $32B\lambda < \theta$,

$$H(\lambda) \le \frac{8\lambda^2 M}{(2-\lambda\theta)(1-32B\lambda/\theta)}.$$

Take $\theta = 8\sqrt{B}$: both conditions become $\lambda < 1/(4\sqrt{B})$, and $(2 - 8\sqrt{B}\lambda)(1 - 4\sqrt{B}\lambda) = 2(1-4\sqrt{B}\lambda)^2$, so

$$\log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \frac{4\lambda^2\, \mathbb{E}[Y^2]}{(1-4\sqrt{B}\lambda)^2} \le \frac{8\lambda^2}{2(1-8\sqrt{B}\lambda)}\, \mathbb{E}[Y^2],$$

the first for $0 \le \lambda < 1/(4\sqrt{B})$, the second for $0 \le \lambda < 1/(8\sqrt{B})$ (using $(1-x)^2 \ge 1-2x$). Numerical check on $\mathcal{T} = \{J - I\}$ ($n = 10, 100, 1000$, exact binomial sums, $\lambda = \theta'/(4\sqrt{B})$ for $\theta' \in \{0.1, \dots, 0.99\}$): the ratio left/right of the first bound is at most 0.21.

Remark (the Theorem 6.16 route and the printed fourth inequality). Applying Theorem 6.16 directly with $\theta = 64B$ and $V_Z^+ \le 8W$ gives

$$\log \mathbb{E}e^{\lambda V_Z^+/(64B)} \le \log \mathbb{E}e^{\lambda W/(8B)} = \frac{\lambda}{8B} M + G\Big(\frac{\lambda}{8B}\Big) \le \frac{\lambda}{8B} M + \frac{\lambda^2/(8B)}{1-\lambda} M = \frac{\lambda}{8B(1-\lambda)} M,$$

by the third inequality, whence

$$H(\lambda) \le \frac{64B\lambda}{1-64B\lambda}\cdot\frac{\lambda}{8B(1-\lambda)}M = \frac{8\lambda^2}{(1-64B\lambda)(1-\lambda)}\,\mathbb{E}[Y^2], \qquad 64B\lambda<1,\ \lambda<1.$$

This is valid, but only for $\lambda<1$ in addition to $64B\lambda<1$, a restriction that is not scale invariant (for $c\mathcal{T}$ with $c\to 0$ the natural range of λ, of order $1/(c\max_M\|M\|)$, grows like $1/c$, while $\lambda<1$ stays fixed). The printed $\frac{16\lambda^2}{2(1-64B\lambda)}\mathbb{E}[Y^2]$ would require $\log\mathbb{E}e^{\lambda V_Z^+/(64B)}\le\frac{\lambda}{8B}\mathbb{E}[Y^2]$, i.e. dropping $G(\lambda/(8B))\ge 0$ (Jensen gives $\log\mathbb{E}e^X\ge\mathbb{E}X$), and the counterexample in the Erratum shows the printed inequality is in fact false. ■

Exercise 6.10

Central idea. A Poisson(μ) law is the limit of Binomial($n,\mu/n$) (the law of rare events). Apply the tensorized Bernoulli modified log-Sobolev inequality (Lemma 6.18) to $f(S_n)$ where S_n is a sum of n Bernoulli(μ/n) variables: flipping one coordinate changes the sum by exactly ±1, so the Bernoulli gradient ∇_i becomes the discrete derivative D, and passing to the limit $n\to\infty$ yields Theorem 6.17.

Solution. Let X be Poisson(μ), $\mu=\mathbb{E}X>0$, and let $f:\mathbb{N}\to(0,\infty)$. We first prove the inequalities for f bounded with $f\ge c>0$ and extend by truncation at the end.

For $n>\mu$ let $X_{n,1},\dots,X_{n,n}$ be i.i.d. Bernoulli(μ/n), $S_n=\sum_{i=1}^n X_{n,i}$ and $T_i=S_n-X_{n,i}=\sum_{j\ne i}X_{n,j}$. Put $g_n(x_1,\dots,x_n)=f(\sum_{i=1}^n x_i)$ on $\{0,1\}^n$, so $g_n(X_{n,1},\dots,X_{n,n})=f(S_n)$. By tensorization of entropy,

$$\operatorname{Ent}(f(S_n))=\operatorname{Ent}(g_n)\le\sum_{i=1}^n\mathbb{E}[\operatorname{Ent}_i(g_n)],$$

where Ent_i is the entropy with respect to $X_{n,i}$ conditionally on $(X_{n,j})_{j\ne i}$.

Fix i and condition on $(X_{n,j})_{j\ne i}$, so T_i is fixed. As a function of $x_i\in\{0,1\}$, g_n takes the values $f(T_i)$ at 0 and $f(T_i+1)$ at 1. With ∇_i denoting the coordinate flip as in Lemma 6.18, if $x_i=0$ then $\nabla_i g_n=f(T_i+1)-f(T_i)=Df(T_i)$, and if $x_i=1$ then $\nabla_i g_n=f(T_i)-f(T_i+1)=-Df(T_i)$; the same holds for $\log g_n$ with $D\log f$. Hence in both cases

$$\nabla_i g_n\cdot\nabla_i\log g_n = Df(T_i)\,D\log f(T_i),$$

the two factors always having the same sign. Applying the first inequality of Lemma 6.18 with $p=\mu/n$ gives

$$\operatorname{Ent}_i(g_n)\le\frac{\mu}{n}\Big(1-\frac{\mu}{n}\Big)\,\mathbb{E}_i[Df(T_i)\,D\log f(T_i)].$$

Summing over i and using that the T_i are identically distributed,

$$\operatorname{Ent}(f(S_n))\le\mu\Big(1-\frac{\mu}{n}\Big)\,\mathbb{E}[Df(T_1)\,D\log f(T_1)]. \tag{1}$$

Now let $n\to\infty$. By the law of rare events, $\mathcal{L}(S_n)\to$ Poisson(μ) in total variation, and likewise $\mathcal{L}(T_1)\to$ Poisson(μ) since $T_1\sim$ Binomial($n-1,\mu/n$). As f is bounded with $f\ge c>0$, the functions f, $f\log f$ and $Df\cdot D\log f$ are all bounded, so

$$\mathbb{E}[f(S_n)]\to\mathbb{E}[f(X)],\qquad \mathbb{E}[(f\log f)(S_n)]\to\mathbb{E}[(f\log f)(X)],$$

hence $\operatorname{Ent}(f(S_n))\to\operatorname{Ent}(f(X))$, and $\mathbb{E}[Df(T_1)D\log f(T_1)]\to\mathbb{E}[Df(X)D\log f(X)]$. Passing to the limit in (1) yields the first inequality,

$$\operatorname{Ent}(f(X))\le(\mathbb{E}X)\,\mathbb{E}[Df(X)\,D\log f(X)].$$

For the second inequality, apply the second bound of Lemma 6.18 instead:

$$\operatorname{Ent}_i(g_n)\le\frac{\mu}{n}\Big(1-\frac{\mu}{n}\Big)\,\mathbb{E}_i\Big[\frac{|\nabla_i g_n|^2}{g_n}\Big],\qquad \frac{|\nabla_i g_n|^2}{g_n}=\frac{|Df(T_i)|^2}{f(T_i+X_{n,i})}.$$

Write the summand as $\frac{|Df(T_i)|^2}{f(T_i)}\mathbf{1}_{\{X_{n,i}=0\}} + \frac{|Df(T_i)|^2}{f(T_i+1)}\mathbf{1}_{\{X_{n,i}=1\}}$. With $M = \|f\|_\infty$, each term involving $\mathbf{1}_{\{X_{n,i}=1\}}$ is at most $\frac{4M^2}{c}\mathbf{1}_{\{X_{n,i}=1\}}$, whose expectation is $O(1/n)$; summed over i with the prefactor μ/n this contributes $O(1/n) \to 0$. The main terms give $\mu(1-\mu/n)\,\mathbb{E}[|Df(T_1)|^2/f(T_1)] \to \mu\,\mathbb{E}[|Df(X)|^2/f(X)]$ by total-variation convergence as before. Hence

$$\mathrm{Ent}(f(X)) \le (\mathbb{E}X)\,\mathbb{E}\left[\frac{|Df(X)|^2}{f(X)}\right].$$

Finally, for general $f : \mathbb{N} \to (0,\infty)$ apply the result to the truncations $f_k = \min(\max(f, 1/k), k)$, which are bounded and bounded away from 0. Since the truncation map $y \mapsto \min(\max(y, 1/k), k)$ is nondecreasing and 1-Lipschitz, one checks directly that $0 \le Df_k\, D\log f_k \le Df\, D\log f$ pointwise (and similarly $|Df_k|^2/f_k$ is dominated), so dominated convergence passes the right-hand sides to the limit while monotone/dominated convergence passes $\mathrm{Ent}(f_k(X)) \to \mathrm{Ent}(f(X))$. This extends both inequalities to all positive f, completing the proof of Theorem 6.17.

Exercise 6.11

Central idea. An Exp(1) variable can be realized as $(X_1^2 + X_2^2)/2$ for i.i.d. standard Gaussians X_1, X_2. Apply the Gaussian logarithmic Sobolev inequality to $g(x_1, x_2) = f((x_1^2+x_2^2)/2)$: the chain rule turns $\|\nabla g\|^2$ into $2X\,(f'(X))^2$, and the constant 2 from the Gaussian LSI doubles to the claimed 4.

Solution. Let X_1, X_2 be independent standard Gaussian variables and put $R = (X_1^2 + X_2^2)/2$. Since $X_1^2 + X_2^2 \sim \chi_2^2$ has density $\frac{1}{2}e^{-x/2}$ on $(0,\infty)$, i.e. it is exponential with mean 2, the rescaled variable R has density e^{-x} on $(0,\infty)$: R is exponentially distributed, $R \overset{d}{=} X$.

Let $f : [0,\infty) \to \mathbb{R}$ be differentiable (we prove the claim for smooth f; the general case follows by approximation) and define $g(x_1, x_2) = f((x_1^2 + x_2^2)/2)$. Then $g(X_1, X_2) = f(R) \overset{d}{=} f(X)$, so $\mathrm{Ent}(g(X_1, X_2)^2) = \mathrm{Ent}(f(X)^2)$, the entropy depending only on the law. The Gaussian logarithmic Sobolev inequality applied to g gives

$$\mathrm{Ent}(g(X_1, X_2)^2) \le 2\,\mathbb{E}[\|\nabla g(X_1, X_2)\|^2].$$

By the chain rule, with $r = (x_1^2 + x_2^2)/2$,

$$\frac{\partial g}{\partial x_1} = f'(r)\,x_1, \qquad \frac{\partial g}{\partial x_2} = f'(r)\,x_2,$$

so that

$$\|\nabla g(x_1, x_2)\|^2 = (f'(r))^2\,(x_1^2 + x_2^2) = 2r\,(f'(r))^2.$$

Evaluating at (X_1, X_2) and writing $R = (X_1^2 + X_2^2)/2$,

$$\|\nabla g(X_1, X_2)\|^2 = 2R\,(f'(R))^2 \overset{d}{=} 2X\,(f'(X))^2.$$

Substituting into the Gaussian log-Sobolev inequality,

$$\mathrm{Ent}(f(X)^2) = \mathrm{Ent}(g(X_1, X_2)^2) \le 2\,\mathbb{E}\big[2R\,(f'(R))^2\big] = 4\,\mathbb{E}\big[X\,(f'(X))^2\big],$$

which is the claimed logarithmic Sobolev inequality for the exponential distribution. The right-hand side is interpreted as $+\infty$ when the expectation diverges, in which case the inequality is trivial.

Exercise 6.12

Central idea. For (a), approximate the Poisson by Bernoulli sums Z_n and apply Theorem 6.29 with the constant weight $W \equiv 1$, since for a Bernoulli sum $V^+ = (1-\mu/n)Z_n \le Z_n$; the uniform mgf bound passes to the Poisson limit. For (b), run the Herbst argument from the Poisson log-Sobolev inequality (Theorem 6.17) applied to $e^{\lambda\sqrt{X}}$: the discrete derivative $D\sqrt{X} = 1/(\sqrt{X+1} + \sqrt{X})$ satisfies $(D\sqrt{X})^2 \le 1/(4X+1)$, and Chebyshev association separates the decreasing factor $1/(4X+1)$ from the increasing $e^{\lambda\sqrt{X}}$, giving $\frac{d}{d\lambda}\frac{\psi(\lambda)}{\lambda} \le v\frac{e^\lambda - 1}{\lambda}$. Part (c) is Chernoff with $\lambda = \log(1 + t/(2v))$.

Solution. Let $X \sim \mathrm{Poisson}(\mu)$, $\mu = \mathbb{E}X$.

(a) Let $Z_n = \sum_{i=1}^n X_i$ with $X_i \sim \text{Bernoulli}(\mu/n)$ i.i.d. For $Z_n^{(i)}$ the leave-one-out copy, $Z_n - Z_n^{(i)} = X_i - X_i'$, and since the variables are $\{0,1\}$-valued, $\mathbb{E}'[(Z_n - Z_n^{(i)})_+^2] = \mathbb{P}'\{X_i = 1, X_i' = 0\} = X_i(1-\mu/n)$. Hence $V^+ = (1-\mu/n)Z_n \le Z_n$, i.e. $V^+ \le WZ_n$ with $W \equiv 1$. Theorem 6.29 with $\theta = 2$ gives, for $0 < \lambda < 1/2$,

$$\log \mathbb{E}e^{\lambda(\sqrt{Z_n} - \mathbb{E}\sqrt{Z_n})} \le \frac{2\lambda}{1-2\lambda} \log \mathbb{E}e^{\lambda/2} = \frac{\lambda^2}{1-2\lambda}.$$

By the law of rare events $Z_n \Rightarrow X$; $\{\sqrt{Z_n}\}$ is uniformly integrable (since $\mathbb{E}Z_n = \mu$), so $\mathbb{E}\sqrt{Z_n} \to \mathbb{E}\sqrt{X}$, and the uniform bound at any $\lambda' \in (\lambda, 1/2)$ gives uniform integrability of $e^{\lambda\sqrt{Z_n}}$, so $\mathbb{E}e^{\lambda\sqrt{Z_n}} \to \mathbb{E}e^{\lambda\sqrt{X}}$. The bound passes to the limit, proving the claim (both sides vanish at $\lambda = 0$).

(b) Let $f = \sqrt{X}$, $\psi(\lambda) = \log \mathbb{E}e^{\lambda(f - \mathbb{E}f)}$ for $\lambda \ge 0$, $F(\lambda) = \mathbb{E}e^{\lambda f}$. Apply Theorem 6.17 to $e^{\lambda f} > 0$. With $Df(x) = \sqrt{x+1} - \sqrt{x} = \frac{1}{\sqrt{x+1}+\sqrt{x}} \in [0,1]$,

$$\text{Ent}(e^{\lambda f}) \le \mu\, \mathbb{E}[e^{\lambda f}\, \lambda Df\, (e^{\lambda Df} - 1)].$$

Since $u \mapsto (e^u - 1)/u$ is increasing and $\lambda Df \le \lambda$, $e^{\lambda Df} - 1 \le Df\,(e^\lambda - 1)$. Moreover $(Df(x))^2 = \frac{1}{(\sqrt{x+1}+\sqrt{x})^2} \le \frac{1}{4x+1}$, because $(\sqrt{x+1} + \sqrt{x})^2 = 2x + 1 + 2\sqrt{x^2+x} \ge 4x + 1$. Hence

$$\text{Ent}(e^{\lambda f}) \le \mu\lambda(e^\lambda - 1)\, \mathbb{E}\left[\frac{e^{\lambda f}}{4X+1}\right].$$

The map $x \mapsto 1/(4x+1)$ is decreasing while $x \mapsto e^{\lambda\sqrt{x}}$ is increasing, so by Chebyshev's association inequality $\mathbb{E}[\frac{e^{\lambda f}}{4X+1}] \le \mathbb{E}[\frac{1}{4X+1}]\, \mathbb{E}e^{\lambda f}$. With $v = \mu\mathbb{E}[\frac{1}{4X+1}]$,

$$\text{Ent}(e^{\lambda f}) \le v\lambda(e^\lambda - 1)F(\lambda).$$

Since $\text{Ent}(e^{\lambda f}) = F(\lambda\psi'(\lambda) - \psi(\lambda))$, this gives $\frac{d}{d\lambda}\frac{\psi(\lambda)}{\lambda} \le v\frac{e^\lambda - 1}{\lambda}$. Integrating from 0 (where $\psi(\lambda)/\lambda \to \psi'(0) = 0$) and using $\frac{e^s - 1}{s} \le \frac{e^\lambda - 1}{\lambda}$ for $s \le \lambda$,

$$\frac{\psi(\lambda)}{\lambda} \le v \int_0^\lambda \frac{e^s - 1}{s}\, ds \le v(e^\lambda - 1),$$

i.e. $\log \mathbb{E}e^{\lambda(\sqrt{X} - \mathbb{E}\sqrt{X})} \le v\lambda(e^\lambda - 1)$.

(c) By Markov's inequality, for $\lambda \ge 0$,

$$\mathbb{P}\{\sqrt{X} - \mathbb{E}\sqrt{X} \ge t\} \le e^{-\lambda t}\mathbb{E}e^{\lambda(\sqrt{X} - \mathbb{E}\sqrt{X})} \le \exp\big(-\lambda t + v\lambda(e^\lambda - 1)\big).$$

Taking $\lambda = \log(1 + t/(2v))$, so $e^\lambda - 1 = t/(2v)$, the exponent becomes $-\lambda t + \lambda t/2 = -\frac{t}{2}\log(1 + \frac{t}{2v})$, which is the claimed bound.

Exercise 6.13

Central idea. Apply the Poisson log-Sobolev inequality (Theorem 6.17, second form) to the density $f = p/\pi$ of X with respect to Poisson(μ): the entropy $\text{Ent}_\pi(f)$ is exactly the KL divergence, and the energy term $\mu\, \mathbb{E}_\pi[(Df)^2/f]$ becomes, via $\pi(k+1)/\pi(k) = \mu/(k+1)$, the scaled Fisher information $K(X)$. Subadditivity follows from a projection identity: the centered score $r_S(S) - 1$ of the sum is the conditional expectation given S of $A = \sum_i (p_i/\mu)(r_i(X_i) - 1)$, a sum of independent centered terms. Conditional Jensen gives $\mathbb{E}(r_S - 1)^2 \le \mathbb{E}A^2 = \sum_i (p_i/\mu)^2\, \mathbb{E}(r_i - 1)^2$, and the squared weights produce exactly $\sum_i (p_i/\mu)K(X_i)$. For Bernoulli summands, $K(X_i) = p_i^2/(1-p_i)$. As printed, the statement is missing two hypotheses: supports without gaps (otherwise the first two claims fail), and Bernoulli summands for the final bound.

Solution. Erratum: two hypotheses are missing from the printed statement.

(a) *Support without gaps.* As printed, X is any random variable with nonnegative integer values. The definition of K only evaluates the ratio at points X actually takes, so mass at $k+1$ is ignored when $p(k)=0$, and then the claims fail. For example, $\mathbb{P}\{X=0\}=0.9$, $\mathbb{P}\{X=10\}=0.1$ gives $\mu=1$ and $K(X)=1\cdot(0.9+0.1)=1$ (the ratio is 0 at both support points), but $D(X\,\|\,\mathrm{Poisson}(1))=2.185\ldots>1$. Sub-additivity fails too: if X_1,X_2 are i.i.d. uniform on $\{0,2\}$, then $p_i=1$, $K(X_i)=1$, S takes values $0,2,4$ with probabilities $\frac{1}{4},\frac{1}{2},\frac{1}{4}$, and $K(S)=2>\sum_i \frac{p_i}{\mu}K(X_i)=1$. *Corrected hypothesis:* every variable Y whose scaled Fisher information appears has support of the form $\{0,1,\ldots,N\}$ with $N\le\infty$, i.e. $\mathbb{P}\{Y=k\}>0$ implies $\mathbb{P}\{Y=k-1\}>0$ for $k\ge 1$ (the setting of Kontoyiannis–Harremoës–Johnson). It holds for Bernoulli variables and is preserved under independent sums. We assume it throughout.

(b) *Bernoulli summands in the last claim.* The final bound $(1/\mu)\sum_i p_i^3/(1-p_i)$ uses $K(X_i)=p_i^2/(1-p_i)$, which holds for $X_i\sim\mathrm{Bernoulli}(p_i)$ but not for general X_i with $\mathbb{E}X_i=p_i$, even gap-free ones. For example, $n=1$ and $\mathbb{P}\{X_1=0\}=0.9$, $\mathbb{P}\{X_1=1\}=0.01$, $\mathbb{P}\{X_1=2\}=0.09$ give $\mu=p_1=0.19$ and $D(X_1\,\|\,\mathrm{Poisson}(0.19))=0.2103\ldots>p_1^2/(1-p_1)=0.0446\ldots$. *Corrected statement:* the last claim is for $X_i\sim\mathrm{Bernoulli}(p_i)$, $0<p_i<1$ (the law of rare events). The first two claims are proved below for general gap-free variables.

Let π be the Poisson(μ) mass function, $\pi(k)=e^{-\mu}\mu^k/k!$, and let p be the mass function of X, $\mu=\mathbb{E}X\in(0,\infty)$. We may assume $K(X)<\infty$.

KL divergence bounded by $K(X)$. Put $f(k)=p(k)/\pi(k)$. Then $\mathbb{E}_\pi f=1$ and

$$D(X\,\|\,\mathrm{Poisson}(\mu))=\sum_k p(k)\log\frac{p(k)}{\pi(k)}=\mathbb{E}_\pi[f\log f]=\mathrm{Ent}_\pi(f).$$

First assume $p(k)>0$ for all k, so $f:\mathbb{N}\to(0,\infty)$. The second inequality of Theorem 6.17 gives, with $Df(k)=f(k+1)-f(k)$,

$$\mathrm{Ent}_\pi(f)\le\mu\,\mathbb{E}_\pi\left[\frac{(Df)^2}{f}\right]=\mu\sum_k\pi(k)\,\frac{(f(k+1)-f(k))^2}{f(k)}.$$

Since $\pi(k+1)=\pi(k)\,\mu/(k+1)$,

$$f(k+1)=\frac{p(k+1)}{\pi(k+1)}=\frac{(k+1)p(k+1)}{\mu\,\pi(k)},\qquad f(k)=\frac{p(k)}{\pi(k)},$$

so that

$$\frac{(f(k+1)-f(k))^2}{f(k)}=\frac{\pi(k)}{p(k)}\cdot\frac{1}{\pi(k)^2}\left(\frac{(k+1)p(k+1)}{\mu}-p(k)\right)^2=\frac{1}{\pi(k)p(k)}\left(\frac{(k+1)p(k+1)}{\mu}-p(k)\right)^2.$$

Multiplying by $\pi(k)$ and summing,

$$\mu\,\mathbb{E}_\pi\left[\frac{(Df)^2}{f}\right]=\mu\sum_k p(k)\left(\frac{(k+1)p(k+1)}{\mu\,p(k)}-1\right)^2=K(X),$$

which is $D(X\,\|\,\mathrm{Poisson}(\mu))\le K(X)$.

If instead the support is $\{0,\ldots,N\}$ with $N<\infty$, let $p_\varepsilon=(1-\varepsilon)p+\varepsilon\pi$, $0<\varepsilon<1$. It has full support and the same mean μ, so $D(p_\varepsilon\,\|\,\pi)\le K(p_\varepsilon)$ by the case just proved. As $\varepsilon\downarrow 0$: the terms of $K(p_\varepsilon)$ with $k<N$ converge to those of $K(X)$; the $k=N$ term is $p_\varepsilon(N)\big(\frac{(N+1)\varepsilon\pi(N+1)}{\mu p_\varepsilon(N)}-1\big)^2\to p(N)$, which is the $k=N$ term of $K(X)$ (there $p(N+1)=0$); for $k>N$, $p_\varepsilon(k)=\varepsilon\pi(k)$ and the ratio equals $\frac{(k+1)\pi(k+1)}{\mu\pi(k)}=1$, so these terms vanish. Hence $K(p_\varepsilon)\to K(X)$. Also $D(p_\varepsilon\,\|\,\pi)=\sum_{k\le N}p_\varepsilon(k)\log\frac{p_\varepsilon(k)}{\pi(k)}+\sum_{k>N}\varepsilon\pi(k)\log\varepsilon\to D(X\,\|\,\pi)$. Therefore $D(X\,\|\,\mathrm{Poisson}(\mu))\le K(X)$ in this case too.

Subadditivity. Let $S = \sum_{i=1}^{n} X_i$ with the X_i independent, $p_i = \mathbb{E}X_i$, $\mu = \sum_i p_i$. Let m_i be the mass function of X_i, q that of S, and $q^{(i)}$ that of $S^{(i)} = S - X_i$. Summands with $p_i = 0$ vanish a.s. and may be dropped, so assume $p_i > 0$ for all i; we may also assume every $K(X_i) < \infty$. Each m_i has support $\{0, \dots, N_i\}$, so S has support $\{0, \dots, \sum_i N_i\}$ and $q(s) > 0$ there. Define the scores

$$r_S(s) = \frac{(s+1)q(s+1)}{\mu\, q(s)}, \qquad r_i(k) = \frac{(k+1)m_i(k+1)}{p_i\, m_i(k)} \quad (m_i(k) > 0),$$

so that $K(S) = \mu\, \mathbb{E}(r_S(S) - 1)^2$ and $K(X_i) = p_i\, \mathbb{E}(r_i(X_i) - 1)^2$.

Centering. By hypothesis (a), $m_i(j) > 0$ implies $m_i(j-1) > 0$ for $j \geq 1$, so

$$\mathbb{E}\, r_i(X_i) = \sum_{k:\, m_i(k)>0} \frac{(k+1)m_i(k+1)}{p_i} = \sum_{j \geq 1} \frac{j\, m_i(j)}{p_i} = 1.$$

Projection identity. On $\{S = s+1\}$ we have $\sum_i X_i = s+1$, hence

$$(s+1)q(s+1) = \sum_{i=1}^{n} \mathbb{E}[X_i \mathbf{1}_{\{S=s+1\}}] = \sum_{i=1}^{n} \sum_{j} j\, m_i(j)\, q^{(i)}(s+1-j).$$

For $q(s) > 0$, using $r_i(k)m_i(k) = (k+1)m_i(k+1)/p_i$, hypothesis (a) (the terms with $m_i(k) = 0$ have $m_i(k+1) = 0$), and the substitution $j = k+1$,

$$\begin{aligned}\mathbb{E}[r_i(X_i) \mid S = s] &= \frac{1}{q(s)} \sum_{k:\, m_i(k)>0} r_i(k)\, m_i(k)\, q^{(i)}(s-k) = \frac{1}{q(s)} \sum_{k \geq 0} \frac{(k+1)m_i(k+1)}{p_i}\, q^{(i)}(s-k) \\ &= \frac{1}{p_i\, q(s)} \sum_{j \geq 1} j\, m_i(j)\, q^{(i)}(s+1-j) = \frac{\mathbb{E}[X_i \mathbf{1}_{\{S=s+1\}}]}{p_i\, q(s)}.\end{aligned}$$

Consequently

$$r_S(s) = \frac{(s+1)q(s+1)}{\mu\, q(s)} = \sum_{i=1}^{n} \frac{p_i}{\mu}\, \mathbb{E}[r_i(X_i) \mid S = s],$$

and since $\sum_i p_i/\mu = 1$,

$$r_S(S) - 1 = \mathbb{E}[A \mid S], \qquad A = \sum_{i=1}^{n} \frac{p_i}{\mu}\, (r_i(X_i) - 1).$$

Variance bound. By the conditional Jensen inequality, $\mathbb{E}\big[(\mathbb{E}[A \mid S])^2\big] \leq \mathbb{E}A^2$. The summands of A are independent (functions of different X_i), centered (by the centering step), and square-integrable (as $K(X_i) < \infty$), so all cross terms vanish:

$$\mathbb{E}(r_S(S) - 1)^2 \leq \mathbb{E}A^2 = \sum_{i=1}^{n} \Big(\frac{p_i}{\mu}\Big)^2 \mathbb{E}(r_i(X_i) - 1)^2 = \sum_{i=1}^{n} \frac{p_i^2}{\mu^2} \cdot \frac{K(X_i)}{p_i} = \frac{1}{\mu} \sum_{i=1}^{n} \frac{p_i}{\mu} K(X_i).$$

Multiplying by μ gives

$$K(S) \leq \sum_{i=1}^{n} \frac{p_i}{\mu}\, K(X_i).$$

(The squared weights are essential: Jensen applied termwise to the convex combination would give only $\mathbb{E}(r_S - 1)^2 \leq \sum_i \frac{p_i}{\mu} \mathbb{E}(r_i - 1)^2$, i.e. $K(S) \leq \sum_i K(X_i)$, which is weaker.)

Bernoulli summands. Let $X_i \sim \mathrm{Bernoulli}(p_i)$ with $0 < p_i < 1$ (the law of rare events); its support $\{0, 1\}$ satisfies hypothesis (a). Then $m_i(0) = 1 - p_i$, $m_i(1) = p_i$, $r_i(0) = \frac{p_i}{p_i(1-p_i)} = \frac{1}{1-p_i}$, $r_i(1) = 0$, and

$$\mathbb{E}(r_i(X_i) - 1)^2 = (1 - p_i) \left(\frac{1}{1-p_i} - 1\right)^2 + p_i(0-1)^2 = \frac{p_i^2}{1-p_i} + p_i = \frac{p_i}{1-p_i}.$$

Hence $K(X_i) = p_i \cdot \frac{p_i}{1-p_i} = \frac{p_i^2}{1-p_i}$. Combining with the first two parts,

$$D(S \,\|\, \text{Poisson}(\mu)) \le K(S) \le \sum_{i=1}^{n} \frac{p_i}{\mu} K(X_i) = \frac{1}{\mu} \sum_{i=1}^{n} \frac{p_i^3}{1-p_i}.$$

(Check, $p = (0.1, 0.2)$: $K(S) = 0.03348\ldots \le \frac{1}{0.3}(\frac{0.001}{0.9} + \frac{0.008}{0.8}) = 0.03704\ldots$, and $D(S \,\|\, \text{Poisson}(0.3)) = 0.01005\ldots$.) ■

Exercise 6.14

Central idea. Write $D = \max_{1 \le i \le n} D_i$ with $D_i \sim \text{Bin}(n-1, p)$ (dependent), $\sigma^2 = p(1-p)$, $x = \sqrt{2\sigma^2 n \log n}$, $p \in (0,1)$ fixed. The upper tail holds at full strength by union bound plus Bernstein. The lower tail as stated is FALSE: a sharp binomial tail upper bound (Stirling plus a geometric ratio argument) gives $\mathbb{P}\{D \ge np + x - \delta\} \le nq \to 0$ whenever $\delta = a_n \sqrt{\sigma^2 n / \log n}$ with $a_n = o(\log\log n)$ — the Gumbel $\log\log n$ correction is real and unavoidable. The correct lower bound, via the second-moment method, requires $a_n \gg \log\log n$. Lemma 2.4 with Theorem 6.12 only yields a much weaker $\omega(\sqrt{n})$ window around $\mathbb{E}D$.

Solution. Let D_i be the degree of vertex i, so $D = \max_i D_i$ and each $D_i \sim \text{Bin}(n-1, p)$.

Upper tail. For $y > 0$, Bernstein's inequality gives $\mathbb{P}\{D_1 - (n-1)p \ge y\} \le \exp(-y^2/(2(\sigma^2 n + y/3)))$. With $y = x + \delta + p$ where $\delta = a_n \sqrt{\sigma^2 n / \log n}$ and $a_n \to \infty$ (since the event $\{D \ge np + x + \delta\}$ shrinks as a_n grows, we may replace a_n by $\min(a_n, \log\log n)$, so that $\delta^2 = o(n)$ and $a_n \sqrt{\log n / n} \to 0$; this makes the $o(1)$ terms below legitimate),

$$\frac{y^2}{2(\sigma^2 n + y/3)} = \frac{2\sigma^2 n \log n + 2\sqrt{2}\, a_n \sigma^2 n + o(n)}{2\sigma^2 n (1 + o(1))} = \log n + \sqrt{2}\, a_n + o(1),$$

since $2x\delta = 2\sqrt{2}\, a_n \sigma^2 n$. By the union bound,

$$\mathbb{P}\{D \ge np + x + \delta\} \le n \exp(-\log n - \sqrt{2}\, a_n + o(1)) = e^{-\sqrt{2}\, a_n + o(1)} \to 0.$$

Thus the upper bound holds as claimed, for every $a_n \to \infty$.

The lower bound as stated is false. We prove a sharp binomial upper-tail lemma. Let $S_N \sim \text{Bin}(N, p)$, $r_k = \mathbb{P}\{S_N = k\}$. By Stirling's formula, $r_k = \binom{N}{k} p^k (1-p)^{N-k} \le \frac{C_0}{\sqrt{N}} \exp(-N D(k/N \| p))$ uniformly for k/N in a compact subinterval of $(0,1)$. Since $q \mapsto D(q\|p)$ satisfies $D(p\|p) = D'(p\|p) = 0$ and $D''(p\|p) = 1/\sigma^2$, Taylor's theorem gives $N D(k/N\|p) = \frac{y^2}{2\sigma^2 N}(1 + o(1)) = \frac{y^2}{2\sigma^2 N} + o(1)$ uniformly for $k = Np + y$ with $y = o(N)$ (the relative error is $O(y/N)$ while $y^2/N = O(\log n)$ in the application below, so the absolute error is $o(1)$). For $k \ge Np + y$ ($1 \le y = o(N)$),

$$\frac{r_{k+1}}{r_k} = \frac{N-k}{k+1} \cdot \frac{p}{1-p} \le \frac{\sigma^2 N - py}{\sigma^2 N + y(1-p)} = 1 - \frac{y}{\sigma^2 N + y(1-p)} \le 1 - \frac{y}{2\sigma^2 N} =: \rho < 1$$

for large N. Hence, with $r_{\lceil Np+y \rceil} \le \frac{C_0}{\sqrt{N}} \exp(-\frac{y^2}{2\sigma^2 N} + o(1))$,

$$\mathbb{P}\{S_N \ge Np + y\} \le r_{\lceil Np+y \rceil} \sum_{j \ge 0} \rho^j \le \frac{C_0}{\sqrt{N}} e^{-y^2/(2\sigma^2 N) + o(1)} \cdot \frac{2\sigma^2 N}{y} = \frac{2C_0 \sigma^2 \sqrt{N}}{y}\, e^{-y^2/(2\sigma^2 N) + o(1)}. \qquad (*)$$

Now take $t_n = np + x - \delta_n$ with $\delta_n = a_n \sqrt{\sigma^2 n / \log n}$ and $a_n = o(\log\log n)$, $a_n \to \infty$ (e.g. $a_n = \log\log\log n$). By the union bound, $\mathbb{P}\{D \ge t_n\} \le n q_n$ with $q_n = \mathbb{P}\{\text{Bin}(n-1, p) \ge t_n\}$. Here $y_n = t_n - (n-1)p = x - \delta_n + O(1)$ satisfies $y_n^2/(2\sigma^2 n) = \log n - \sqrt{2}\, a_n + o(1)$, so by $(*)$,

$$n q_n \le n \cdot \frac{C\sqrt{n}}{y_n} \exp(-\log n + \sqrt{2}\, a_n + o(1)) \le \frac{C' e^{\sqrt{2}\, a_n + o(1)}}{\sqrt{\log n}} \to 0,$$

since $a_n = o(\log\log n)$. Thus $\mathbb{P}\{D \ge np + x - \delta_n\} \to 0$, whereas the problem claims this probability tends to 1. The stated lower bound is false; the obstruction is the Gumbel $\log\log n$ correction to the location of the maximum, which exceeds $a_n\sqrt{\sigma^2 n/\log n}$ when $a_n = o(\log\log n)$. (The threshold is sharp, see below.)

Correct lower bound. Let now $\delta = b_n\sqrt{\sigma^2 n/\log n}$ with $b_n/\log\log n \to \infty$ (the event $\{D \ge np + x - \delta\}$ grows with b_n, so we may assume $b_n \le (\log\log n)^2$, whence $\delta^2 = o(n)$ and $b_n^2/\log n \to 0$), and put $N = \sum_{i=1}^{n} \mathbf{1}_{\{D_i \ge t\}}$, $t = np + x - \delta$. By Paley–Zygmund, $\mathbb{P}\{D \ge t\} = \mathbb{P}\{N \ge 1\} \ge (nq)^2/(nq + n(n-1)q_2)$ where $q = \mathbb{P}\{D_1 \ge t\}$ and $q_2 = \mathbb{P}\{D_1 \ge t,\, D_2 \ge t\}$. Summing the Stirling lower bound $\mathbb{P}\{\mathrm{Bin}(N,p) = \lceil Np + y\rceil + j\} \ge c\,\exp(-\frac{y^2}{2\sigma^2 N}(1+o(1)))/\sqrt{N}$ over $0 \le j \le c'\sigma^2 N/y$ (the same Taylor expansion as above, relative error $O(y/N)$) gives $q \ge \frac{c''\sqrt{N}}{y} e^{-y^2/(2\sigma^2 N)+o(1)}$, whence with $y^2/(2\sigma^2 n) = \log n - \sqrt{2}\,b_n + o(1)$,

$$nq \ge \frac{c'' e^{\sqrt{2}\,b_n + o(1)}}{\sqrt{\log n}} \to \infty \qquad (b_n \gg \log\log n).$$

Writing $D_1 = I_{12} + B_1$, $D_2 = I_{12} + B_2$ with B_1, B_2 i.i.d. $\mathrm{Bin}(n-2,p)$ independent of I_{12} gives $q_2 - q^2 = p(1-p)\,\mathbb{P}\{B_1 = t-1\}^2 =: p(1-p)r^2$. Since $r \le \frac{C}{\sqrt{n}} e^{-y^2/(2\sigma^2 n)}$ while $q \ge \frac{c}{\sqrt{\log n}} e^{-y^2/(2\sigma^2 n)}$, $(r/q)^2 = O(\log n/n) \to 0$, so $q_2/q^2 \to 1$. Hence

$$\frac{(nq)^2}{nq + n(n-1)q_2} = \frac{1}{\frac{1}{nq} + \frac{(n-1)q_2}{nq^2}} \to 1,$$

i.e. $\mathbb{P}\{D \ge np + x - b_n\sqrt{\sigma^2 n/\log n}\} \to 1$ whenever $b_n/\log\log n \to \infty$. The $\log\log n$ threshold is sharp.

What Lemma 2.4 and Theorem 6.12 give. The centered degree $D_i - (n-1)p$ is a sum of $n-1$ independent centered variables $I - p \le 1 - p \le 1$ with variance σ^2 each, so by the moment-generating-function bound behind Bernstein's inequality (Theorem 2.10, with $c = 1/3$) $\psi_{D_i-(n-1)p}(\lambda) \le \frac{\lambda^2(n-1)\sigma^2}{2(1-\lambda/3)}$ for $0 < \lambda < 3$, i.e. $D_i - (n-1)p \in \Gamma_+((n-1)\sigma^2, 1/3)$. (It is not sub-Gaussian with variance factor $(n-1)\sigma^2$ when $p \ne 1/2$: for a single centered Bernoulli(0.1) and $\lambda = 1$, $\psi(1) = \log(0.9 + 0.1e) - 0.1 = 0.0586\ldots > 0.045 = \lambda^2\sigma^2/2$. Hoeffding's lemma gives the valid sub-Gaussian factor $(n-1)/4$, which is weaker unless $p = 1/2$.) Lemma 2.4 with Theorem 2.5, in the sub-gamma form of Corollary 2.6, yields

$$\mathbb{E}D = \mathbb{E}\max_i D_i \le (n-1)p + \sqrt{2(n-1)\sigma^2\log n} + \tfrac{1}{3}\log n \le np + x + \tfrac{1}{3}\log n,$$

while trivially $\mathbb{E}D \ge \mathbb{E}D_1 = (n-1)p$. Theorem 6.12 ($D$ is self-bounding by Example 6.13) gives $\mathbb{P}\{|D - \mathbb{E}D| \ge t\} \le 2\exp(-t^2/(2\mathbb{E}D + 2t/3))$, so, as $\mathbb{E}D \le n$, $|D - \mathbb{E}D| \le b_n\sqrt{n}$ with probability tending to 1 for any $b_n \to \infty$. Combined: with probability tending to 1, $(n-1)p - b_n\sqrt{n} \le D \le np + x + \frac{1}{3}\log n + b_n\sqrt{n}$. The upper bound has the correct leading terms $np + x$ but an error $\omega(\sqrt{n})$, far weaker than the sharp $a_n\sqrt{\sigma^2 n/\log n}$ upper tail above, and there is no lower bound capturing the $\sqrt{2\sigma^2 n\log n}$ term. The entropy method captures the correct order of $\mathbb{E}D$ but not the precise fluctuation scale. ■

Exercise 6.15

Central idea. Plant a clique: fix a set S of $m = \lceil Knp\rceil$ vertices with K a large constant depending on a. With probability $p^{\binom{m}{2}} = \exp(-\binom{m}{2}\log(1/p)) \ge \exp(-cp^2n^2\log(1/p))$ the set S induces a clique, contributing $\binom{m}{3} \ge (Knp-2)^3/6$ triangles; choosing K so that $(K-2)^3 > 1 + 6a$ makes this exceed $\mathbb{E}Z + an^3p^3$. The hypothesis $p \ge 1/n$ must be supplemented with p bounded away from 1: at $p = 1$ the left side is 0 while the right side is 1, so the inequality is false without an upper bound on p.

Solution. Let Z be the number of triangles, so $\mathbb{E}Z = \binom{n}{3}p^3 \le n^3p^3/6$. Fix $a > 0$ and choose an integer $K \ge 4$ with $(K-2)^3 \ge 1 + 6a$ (e.g. $K = 3 + \lceil(1+6a)^{1/3}\rceil$); K depends only on a. Assume $1/n \le p \le 1/(2K)$, so that $m := \lceil Knp\rceil \le n$ and $m \ge Knp \ge K \ge 4$.

Fix any set S of m vertices. On the event $C_S = \{S \text{ induces a clique}\}$ every triple in S is a triangle, so

$$Z \ge \binom{m}{3} \ge \frac{(m-2)^3}{6} \ge \frac{(Knp-2)^3}{6}.$$

Since $np \geq 1$, $Knp - 2 \geq (K-2)np$, and therefore

$$Z \geq \frac{(K-2)^3}{6}\, n^3p^3 \geq \frac{1+6a}{6}\, n^3p^3 = \frac{n^3p^3}{6} + an^3p^3 \geq \mathbb{E}Z + an^3p^3 \quad \text{on } C_S,$$

using $\mathbb{E}Z = \binom{n}{3}p^3 < n^3p^3/6$ for the strict inequality $Z > \mathbb{E}Z + an^3p^3$. Hence

$$\mathbb{P}\{Z > \mathbb{E}Z + an^3p^3\} \geq \mathbb{P}(C_S) = p^{\binom{m}{2}} = \exp\left(-\binom{m}{2}\log\frac{1}{p}\right).$$

Now $\binom{m}{2} \leq m^2/2 \leq (Knp+1)^2/2 \leq K^2n^2p^2 + 1 \leq (K^2+1)n^2p^2$, using $(x+1)^2 \leq 2x^2+2$ and $n^2p^2 \geq 1$. Thus

$$\mathbb{P}\{Z > \mathbb{E}Z + an^3p^3\} \geq \exp(-c\,p^2n^2\log(1/p)), \qquad c = K^2+1,$$

with c depending only on a (through K).

Remark on the range of p. The hypothesis $p \geq 1/n$ alone does not suffice: p must also be bounded away from 1. Indeed, at $p = 1$ we have $Z = \binom{n}{3} = \mathbb{E}Z$ a.s., so the left-hand side is $\mathbb{P}\{Z > \binom{n}{3} + an^3\} = 0$, while the right-hand side is $\exp(-c \cdot n^2 \cdot \log 1) = 1$ — the inequality $0 \geq 1$ is false. More generally, if $\binom{n}{3} \leq \binom{n}{3}p^3 + an^3p^3$ (which holds for p sufficiently close to 1), the event $\{Z > \mathbb{E}Z + an^3p^3\}$ is empty while the right-hand side is strictly positive. The planted-clique heuristic is a small-p argument; the proof above covers $1/n \leq p \leq 1/(2K)$. ■

Exercise 6.16

Central idea. Example 6.30 gives a sub-gamma tail for $Y = \sqrt{Z}$: $\mathbb{P}\{Y \geq \mathbb{E}Y + t\} \leq \exp(-t^2/(24(np^2 + \log n) + 20t))$. Write $\{Z > \mathbb{E}Z + t\} = \{\sqrt{Z} > \sqrt{\mathbb{E}Z + s}\}$ with $s = t/(\sqrt{\mathbb{E}Z + t} + \sqrt{\mathbb{E}Z})$; Jensen's inequality ($\mathbb{E}\sqrt{Z} \leq \sqrt{\mathbb{E}Z}$) lets us apply the Y-bound at deviation s. The hypothesis $t \leq (K^2-1)\mathbb{E}Z$ caps the denominator $D = \sqrt{\mathbb{E}Z + t} + \sqrt{\mathbb{E}Z} \leq (K+1)\sqrt{\mathbb{E}Z}$, and rewriting $s^2/(A + 20s) = t^2/(AD^2 + 20tD)$ yields the claimed Bernstein-type bound for Z.

Solution. Example 6.30 shows that $Y = \sqrt{Z}$ satisfies, with $A := 24(np^2 + \log n)$,

$$\mathbb{P}\{Y \geq \mathbb{E}Y + s\} \leq \exp\left(-\frac{s^2}{A + 20s}\right) \qquad \forall\, s > 0,$$

since $4\mathbb{E}W \leq 24(np^2 + \log n)$ where $W = 3\max_j B_j$. Let $K > 1$ and $0 < t \leq (K^2-1)\mathbb{E}Z$ (for $t \leq 0$ there is nothing to prove). Then

$$\{Z > \mathbb{E}Z + t\} = \{\sqrt{Z} > \sqrt{\mathbb{E}Z + t}\} = \{\sqrt{Z} > \sqrt{\mathbb{E}Z} + s\},$$

where $s := \sqrt{\mathbb{E}Z + t} - \sqrt{\mathbb{E}Z} = \frac{t}{\sqrt{\mathbb{E}Z+t} + \sqrt{\mathbb{E}Z}} > 0$. By Jensen's inequality $\mathbb{E}\sqrt{Z} \leq \sqrt{\mathbb{E}Z}$, so $\{\sqrt{Z} > \sqrt{\mathbb{E}Z} + s\} \subseteq \{Y \geq \mathbb{E}Y + s\}$, and

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \leq \exp\left(-\frac{s^2}{A + 20s}\right).$$

Put $D := \sqrt{\mathbb{E}Z + t} + \sqrt{\mathbb{E}Z}$, so $s = t/D$ and

$$\frac{s^2}{A + 20s} = \frac{t^2/D^2}{A + 20t/D} = \frac{t^2}{AD^2 + 20tD}.$$

From $t \leq (K^2-1)\mathbb{E}Z$ we get $\sqrt{\mathbb{E}Z + t} \leq K\sqrt{\mathbb{E}Z}$, hence $D \leq (K+1)\sqrt{\mathbb{E}Z}$. Therefore

$$\begin{aligned} AD^2 + 20tD &\leq A(K+1)^2\mathbb{E}Z + 20t(K+1)\sqrt{\mathbb{E}Z} \\ &= (K+1)^2\mathbb{E}Z\left(A + \frac{20t}{(K+1)\sqrt{\mathbb{E}Z}}\right), \end{aligned}$$

and consequently

$$\mathbb{P}\{Z > \mathbb{E}Z + t\} \leq \exp\left(-\frac{t^2}{(K+1)^2\mathbb{E}Z\left(24np^2 + 24\log n + \frac{20t}{(K+1)\sqrt{\mathbb{E}Z}}\right)}\right). \blacksquare$$

Chapter 7 — Concentration and Isoperimetry

Exercise 7.1

Central idea. Write $f(x) = d(x, B)$, which is 1-Lipschitz. First show $\mathbb{P}\{f \geq t_0\} \leq 1/2$: otherwise $A = \{f \geq t_0\}$ has $\mathbb{P}(A) \geq 1/2$, so $\mathbb{P}\{d(X, A) \geq t_0\} \leq \alpha(t_0)$, but $B \subset \{d(X, A) \geq t_0\}$ by the triangle inequality, contradicting $\mathbb{P}(B) \geq \varepsilon > \alpha(t_0)$. Hence $A' = \{f \leq t_0\}$ has $\mathbb{P}(A') \geq 1/2$, and $\{f \geq t_0 + t\} \subset \{d(X, A') \geq t\}$ (again by the triangle inequality), so the definition of $\alpha(t)$ gives the bound.

Solution. Let $f(x) = d(x, B)$, which satisfies $|f(x) - f(y)| \leq d(x, y)$, i.e., f is 1-Lipschitz.

We first claim $\mathbb{P}\{f \geq t_0\} \leq 1/2$. Suppose not, and set $A = \{x : d(x, B) \geq t_0\}$, so $\mathbb{P}(A) > 1/2 \geq 1/2$. By the definition of the concentration function, $\mathbb{P}\{d(X, A) \geq t_0\} \leq \alpha(t_0)$. But $B \subset \{x : d(x, A) \geq t_0\}$: indeed, for $b \in B$ and $a \in A$, the triangle inequality gives $d(b, a) \geq d(a, B) - d(b, B) = d(a, B) \geq t_0$, so $d(b, A) = \inf_{a \in A} d(b, a) \geq t_0$. Hence $\mathbb{P}(B) \leq \mathbb{P}\{d(X, A) \geq t_0\} \leq \alpha(t_0) < \varepsilon \leq \mathbb{P}(B)$, a contradiction. Therefore $\mathbb{P}\{f \geq t_0\} \leq 1/2$.

Consequently $A' = \{x : f(x) \leq t_0\}$ satisfies $\mathbb{P}(A') = 1 - \mathbb{P}\{f > t_0\} \geq 1 - \mathbb{P}\{f \geq t_0\} \geq 1/2$. We claim $\{f \geq t_0 + t\} \subset \{d(X, A') \geq t\}$. Indeed, if $f(x) \geq t_0 + t$ and $y \in A'$ (so $f(y) \leq t_0$), then $d(x, y) \geq f(x) - f(y) \geq t$ by the 1-Lipschitz property of f; hence $d(x, A') = \inf_{y \in A'} d(x, y) \geq t$. Therefore

$$\mathbb{P}\{d(X, B) \geq t_0 + t\} = \mathbb{P}\{f \geq t_0 + t\} \leq \mathbb{P}\{d(X, A') \geq t\} \leq \alpha(t),$$

the last inequality by the definition of $\alpha(t)$ since $\mathbb{P}(A') \geq 1/2$. ■

Exercise 7.2

Central idea. The stated inequality $\beta(t) \geq \alpha(t/2)$ is false in general: on the two-point space the minimal valid β is $(1/2)\mathbf{1}\{t \leq 1/2\}$ while $\alpha(1/2) = 1/2$. The obstruction is that a one-sided mean-centered bound cannot see the median; the mean of $d(\cdot, A)$ may be large. What is true is $\alpha(2t) \leq \beta(t)$: for $B = A_r^c$, the bounded 1-Lipschitz function $g = \min(d(\cdot, B), r)$ equals r on A, hence has mean at least $r/2$, and applying the hypothesis to $-g$ at level $s \leq \mathbb{E}g(X)$ bounds $\mathbb{P}(B)$ by $\beta(s)$. (Truncating at r guarantees that $\mathbb{E}g(X)$ is finite, which $d(\cdot, B)$ alone need not satisfy.)

Solution. The claim as stated is false. Consider $\mathcal{X} = \{0, 1\}$ with $d(0, 1) = 1$ and $\mathbb{P}$ uniform. For $A = \{0\}$ (so $\mathbb{P}(A) = 1/2$), $\mathbb{P}\{d(X, A) \geq t\} = (1/2)\mathbf{1}\{t \leq 1\}$, and the same for $A = \{1\}$; for $A = \{0, 1\}$ it is 0. Hence $\alpha(t) = (1/2)\mathbf{1}\{t \leq 1\}$, in particular $\alpha(1/2) = 1/2$. A function $f : \mathcal{X} \to \mathbb{R}$ is 1-Lipschitz iff $|f(0) - f(1)| \leq 1$. For such f, $\max f - \mathbb{E}f = |f(0) - f(1)|/2 \leq 1/2$, so $\{f \geq \mathbb{E}f + t\} \neq \emptyset$ requires $t \leq 1/2$, and $\mathbb{P}\{f \geq \mathbb{E}f + t\} \in \{0, 1/2\}$ (it cannot be 1 for $t > 0$). Thus $\beta(t) = (1/2)\mathbf{1}\{t \leq 1/2\}$ satisfies the hypothesis, but $\beta(1) = 0 \not\geq 1/2 = \alpha(1/2)$. So $\beta(t) \geq \alpha(t/2)$ fails in general.

The correct statement is $\alpha(2t) \leq \beta(t)$ for all $t > 0$ (equivalently, $\alpha(r) \leq \inf_{0 < s \leq r/2} \beta(s)$ for all $r > 0$). To see this, fix $r > 0$ and $A \subset \mathcal{X}$ with $\mathbb{P}(A) \geq 1/2$, and set $B = \{x : d(x, A) \geq r\} = A_r^c$. If $B = \emptyset$ then $\mathbb{P}(B) = 0$ and there is nothing to prove, so assume $B \neq \emptyset$; then $d(\cdot, B)$ is finite and 1-Lipschitz. Let

$$g(x) = \min\{d(x, B),\, r\}.$$

As a minimum of a 1-Lipschitz function and a constant, g is 1-Lipschitz; it is continuous (hence measurable) and takes values in $[0, r]$, so $\mathbb{E}g(X)$ is finite and the hypothesis may be applied to $-g$. (Without the truncation, $\mathbb{E}\, d(X, B)$ could be $+\infty$ on an unbounded space, and then $-d(\cdot, B)$ would have no finite mean.) For $a \in A$ and $b \in B$, $d(a, b) \geq d(b, A) - d(a, A) \geq r - 0 = r$, so $d(a, B) \geq r$ and hence $g = r$ on A. Since $g \geq 0$, $\mathbb{E}g(X) \geq r\,\mathbb{P}(A) \geq r/2$. Applying the hypothesis to the 1-Lipschitz function $-g$ gives, for every $s > 0$,

$$\mathbb{P}\{g(X) \leq \mathbb{E}g(X) - s\} \leq \beta(s).$$

Since $g = 0$ on B, we have $\mathbb{P}(B) \leq \mathbb{P}\{g(X) = 0\}$. Moreover $\{g(X) = 0\} \subset \{g(X) \leq \mathbb{E}g(X) - s\}$ whenever $0 < s \leq \mathbb{E}g(X)$. Hence $\mathbb{P}(B) \leq \beta(s)$ for all $0 < s \leq \mathbb{E}g(X)$, i.e.,

$$\mathbb{P}(B) \leq \inf_{0 < s \leq \mathbb{E}g(X)} \beta(s) \leq \inf_{0 < s \leq r/2} \beta(s),$$

using $\mathbb{E}g(X) \geq r/2$. Taking the supremum over A with $\mathbb{P}(A) \geq 1/2$ yields $\alpha(r) \leq \inf_{0<s\leq r/2} \beta(s)$. In particular, with $r = 2t$,

$$\alpha(2t) \leq \inf_{0<s\leq t} \beta(s) \leq \beta(t), \qquad t > 0.$$

■

Exercise 7.3

Central idea. Chernoff's bound turns the Laplace functional into a one-sided deviation inequality: for 1-Lipschitz f with $\mathbb{E}f(X) = 0$, $\mathbb{P}\{f(X) \geq s\} \leq e^{-\lambda s}\mathbb{E}e^{\lambda f(X)} \leq e^{-\lambda s}L(\lambda)$ for every $\lambda > 0$. Hence $\beta(s) = \inf_{\lambda>0} e^{-\lambda s}L(\lambda)$ satisfies the hypothesis of Exercise 7.2 (applied to $g - \mathbb{E}g(X)$ for general 1-Lipschitz g), and the corrected Exercise 7.2 gives $\alpha(t) \leq \beta(t/2) = \inf_{\lambda>0} e^{-\lambda t/2}L(\lambda)$.

Solution. Let $f : \mathcal{X} \to \mathbb{R}$ be 1-Lipschitz with $\mathbb{E}f(X) = 0$ (so $\mathbb{E}|f(X)| < \infty$). For $s > 0$ and $\lambda > 0$, by Markov's inequality applied to the nonnegative variable $e^{\lambda f(X)}$,

$$\mathbb{P}\{f(X) \geq s\} = \mathbb{P}\{e^{\lambda f(X)} \geq e^{\lambda s}\} \leq e^{-\lambda s}\,\mathbb{E}e^{\lambda f(X)} \leq e^{-\lambda s}L(\lambda),$$

where $\mathbb{E}e^{\lambda f(X)} \in [0, \infty]$ is well defined and bounded by $L(\lambda)$ by definition of the supremum. Taking the infimum over $\lambda > 0$,

$$\mathbb{P}\{f(X) \geq s\} \leq \inf_{\lambda>0} e^{-\lambda s}L(\lambda) =: \beta(s), \qquad s > 0.$$

Now let $g : \mathcal{X} \to \mathbb{R}$ be 1-Lipschitz with $\mathbb{E}g(X)$ existing and finite. Then $f = g - \mathbb{E}g(X)$ is 1-Lipschitz with $\mathbb{E}f(X) = 0$, and $\mathbb{P}\{g(X) \geq \mathbb{E}g(X) + s\} = \mathbb{P}\{f(X) \geq s\} \leq \beta(s)$. Thus β satisfies the hypothesis of Exercise 7.2. By the corrected form of Exercise 7.2 proved above, $\alpha(2t) \leq \beta(t)$ for all $t > 0$, i.e., $\alpha(u) \leq \beta(u/2)$ for all $u > 0$. Hence

$$\alpha(t) \leq \beta(t/2) = \inf_{\lambda>0} e^{-\lambda(t/2)}L(\lambda) = \inf_{\lambda>0} e^{-\lambda t/2}L(\lambda), \qquad t > 0.$$

■

Exercise 7.4

Central idea. A 1-Lipschitz f on a space of diameter D takes values in an interval $[a, b]$ with $b-a \leq D$ (since $|f(x) - f(y)| \leq d(x, y) \leq D$). Hoeffding's lemma (Lemma 2.2) then gives $\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \leq \exp(\lambda^2(b-a)^2/8) \leq \exp(\lambda^2 D^2/8)$; with $\mathbb{E}f(X) = 0$ and taking the supremum over f, this is $L(\lambda) \leq e^{D^2\lambda^2/8}$.

Solution. Let $f : \mathcal{X} \to \mathbb{R}$ be 1-Lipschitz with $\mathbb{E}f(X) = 0$. Since f is continuous, it is Borel measurable. Fix $x_0 \in \mathcal{X}$; then $|f(x)| \leq |f(x_0)| + d(x, x_0) \leq |f(x_0)| + D$ for all x, so f is bounded and all expectations below exist. Let $a = \inf_{\mathcal{X}} f$ and $b = \sup_{\mathcal{X}} f$, both finite. For any $x, y \in \mathcal{X}$, $f(x)-f(y) \leq |f(x)-f(y)| \leq d(x, y) \leq D$, so taking suprema gives $b - a \leq D$. Moreover $f(X) \in [a, b]$ almost surely.

Hoeffding's lemma (Lemma 2.2) states that a random variable taking values in $[a, b]$ satisfies $\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \leq \exp(\lambda^2(b-a)^2/8)$ for all $\lambda \in \mathbb{R}$. Applying this to $Z = f(X)$ and using $\mathbb{E}f(X) = 0$,

$$\mathbb{E}e^{\lambda f(X)} \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right) \leq \exp\left(\frac{D^2\lambda^2}{8}\right), \qquad \lambda \in \mathbb{R}.$$

Taking the supremum over all 1-Lipschitz f with $\mathbb{E}f(X) = 0$ yields $L(\lambda) \leq e^{D^2\lambda^2/8}$ for all λ. ■

Exercise 7.5

Central idea. Induction on n via conditional centering (tensorization). For 1-Lipschitz f on the ℓ^1-product, freeze the first $n-1$ coordinates: $x_n \mapsto f(x', x_n)$ is 1-Lipschitz on $\mathcal{X}_n$, so centering it by its $\mathbb{P}_n$-mean $m(x')$ and applying L_n gives $\mathbb{E}_{\mathbb{P}_n} e^{\lambda f} \leq e^{\lambda m(x')}L_n(\lambda)$; the conditional mean m is 1-Lipschitz on the $(n-1)$-fold product with mean zero, so induction applies. A truncation plus Fatou's lemma reduces to bounded f, where all conditional expectations are everywhere defined.

Solution. We prove $L(\lambda) \le \prod_{i=1}^n L_i(\lambda)$ by induction on n, the case $n=1$ being trivial. Let f be 1-Lipschitz on $(\mathcal{X}, d)$, $d(x,y) = \sum_{i=1}^n d_i(x_i, y_i)$, with $\mathbb{E} f(X) = 0$ under $\mathbb{P} = \bigotimes_{i=1}^n \mathbb{P}_i$ (so $\mathbb{E}|f(X)| < \infty$).

We first reduce to bounded f. Put $f_k = (f \wedge k) \vee (-k)$, which is 1-Lipschitz (composition of 1-Lipschitz maps) and bounded, and $g_k = f_k - \mathbb{E} f_k(X)$, which is 1-Lipschitz, bounded, with $\mathbb{E} g_k(X) = 0$. If the claim holds for bounded functions, $\mathbb{E} e^{\lambda g_k(X)} \le \prod_{i=1}^n L_i(\lambda)$ for all k. Since $f_k \to f$ pointwise and $|f_k| \le |f| \in L^1$, dominated convergence gives $\mathbb{E} f_k(X) \to 0$, so $g_k \to f$ pointwise; Fatou's lemma yields $\mathbb{E} e^{\lambda f(X)} \le \liminf_k \mathbb{E} e^{\lambda g_k(X)} \le \prod_{i=1}^n L_i(\lambda)$. Hence it suffices to prove $\mathbb{E} e^{\lambda f(X)} \le \prod_{i=1}^n L_i(\lambda)$ for bounded f.

Assume now f is bounded. Write $\mathcal{X} = \mathcal{X}' \times \mathcal{X}_n$ with $\mathcal{X}' = \prod_{i=1}^{n-1} \mathcal{X}_i$, $\mathbb{P} = \mathbb{P}' \otimes \mathbb{P}_n$ ($\mathbb{P}' = \bigotimes_{i=1}^{n-1} \mathbb{P}_i$), and $d = d' + d_n$. For fixed $x' \in \mathcal{X}'$, the map $f_{x'}(x_n) = f(x', x_n)$ is 1-Lipschitz on $(\mathcal{X}_n, d_n)$, since $d((x', x_n), (x', y_n)) = d_n(x_n, y_n)$, and bounded. Let $m(x') = \mathbb{E}_{\mathbb{P}_n}[f(x', X_n)]$, defined for every x'. Then $f_{x'} - m(x')$ is 1-Lipschitz on $\mathcal{X}_n$ with $\mathbb{P}_n$-mean zero, so

$$\mathbb{E}_{\mathbb{P}_n}\Big[e^{\lambda(f(x', X_n) - m(x'))}\Big] \le L_n(\lambda), \qquad \text{hence} \qquad \mathbb{E}_{\mathbb{P}_n}\Big[e^{\lambda f(x', X_n)}\Big] \le e^{\lambda m(x')} L_n(\lambda).$$

Moreover m is 1-Lipschitz on $(\mathcal{X}', d')$: indeed,

$$|m(x') - m(y')| \le \mathbb{E}_{\mathbb{P}_n} |f(x', X_n) - f(y', X_n)| \le \mathbb{E}_{\mathbb{P}_n}[d((x', X_n), (y', X_n))] = d'(x', y'),$$

and m is bounded with $\mathbb{E}_{\mathbb{P}'}[m(X')] = \mathbb{E}[f(X)] = 0$ by Fubini. By the induction hypothesis applied to m on the $(n-1)$-fold product,

$$\mathbb{E}_{\mathbb{P}'}[e^{\lambda m(X')}] \le \prod_{i=1}^{n-1} L_i(\lambda).$$

Therefore, by Fubini,

$$\mathbb{E}[e^{\lambda f(X)}] = \mathbb{E}_{\mathbb{P}'}\Big[\mathbb{E}_{\mathbb{P}_n}[e^{\lambda f(X', X_n)}]\Big] \le L_n(\lambda)\, \mathbb{E}_{\mathbb{P}'}[e^{\lambda m(X')}] \le \prod_{i=1}^{n} L_i(\lambda).$$

Taking the supremum over all 1-Lipschitz f with $\mathbb{E} f(X) = 0$ gives $L(\lambda) \le \prod_{i=1}^n L_i(\lambda)$. ∎

Exercise 7.6

Central idea. The hypothesis says f is 1-Lipschitz for the ℓ^1-product metric $d(x,y) = \sum_i d_i(x_i, y_i)$. By Exercise 7.5 the Laplace functional tensorizes, $L(\lambda) \le \prod_i L_i(\lambda)$, and by Exercise 7.4 each factor satisfies $L_i(\lambda) \le e^{D_i^2 \lambda^2/8}$, so $L(\lambda) \le e^{\lambda^2 V/8}$ with $V = \sum_i D_i^2$. A Chernoff bound (the mechanism behind Exercise 7.3) applied to $\pm(f - \mathbb{E} f)$ then gives sub-Gaussian tails; optimizing over λ yields $2e^{-2t^2/V}$, which implies the claimed bound.

Solution. The condition $|f(x) - f(y)| \le \sum_{i=1}^n d_i(x_i, y_i)$ says exactly that f is 1-Lipschitz for the product metric $d(x,y) = \sum_i d_i(x_i, y_i)$ of Exercise 7.5. Replacing f by $f - \mathbb{E} f$ (which preserves the Lipschitz condition), assume $\mathbb{E} f(X) = 0$.

Let $L(\lambda)$ be the Laplace functional of $(\mathcal{X}, d, \mathbb{P})$ and $L_i(\lambda)$ that of $(\mathcal{X}_i, d_i, \mathbb{P}_i)$. By Exercise 7.5, $L(\lambda) \le \prod_{i=1}^n L_i(\lambda)$, and by Exercise 7.4, $L_i(\lambda) \le \exp(D_i^2 \lambda^2/8)$ since $\operatorname{diam}(\mathcal{X}_i) = D_i$. Hence, with $V = \sum_{i=1}^n D_i^2$,

$$L(\lambda) \le \exp\left(\frac{\lambda^2 V}{8}\right), \qquad \lambda > 0.$$

For $\lambda > 0$, Chernoff's bound gives (this is the estimate underlying Exercise 7.3)

$$\mathbb{P}\{f(X) > t\} = \mathbb{P}\{e^{\lambda f(X)} > e^{\lambda t}\} \le e^{-\lambda t}\, \mathbb{E} e^{\lambda f(X)} \le e^{-\lambda t} L(\lambda) \le \exp\left(-\lambda t + \frac{\lambda^2 V}{8}\right),$$

using that f is 1-Lipschitz with $\mathbb{E} f(X) = 0$. Minimizing $-\lambda t + \lambda^2 V/8$ over $\lambda > 0$ (minimizer $\lambda = 4t/V$) yields

$$\mathbb{P}\{f(X) - \mathbb{E} f(X) > t\} \le e^{-2t^2/V}.$$

Applying the same to $-f$ (also 1-Lipschitz with mean zero) and using $\mathbb{P}\{|f-\mathbb{E}f|>t\} \le \mathbb{P}\{f-\mathbb{E}f > t\} + \mathbb{P}\{-(f-\mathbb{E}f)>t\}$,

$$\mathbb{P}\{|f(X_1,\dots,X_n)-\mathbb{E}f(X_1,\dots,X_n)|>t\} \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n D_i^2}\right) \le 2\exp\left(-\frac{t^2}{\sum_{i=1}^n D_i^2}\right),$$

since $e^{-2t^2/V} \le e^{-t^2/V}$. ■

Exercise 7.7

Central idea. Apply the divergence theorem to the vector field $F(x)=x$: $\operatorname{div} F = n$ on B, while on ∂B the outward unit normal is $\nu(x)=x$ itself, so $\langle F,\nu\rangle = 1$ and the boundary flux is exactly the surface area.

Solution. Let $B=\{x\in\mathbb{R}^n : \|x\|_2 \le 1\}$ and $\partial B = S^{n-1}$. By $\mathrm{Vol}(\partial B)$ we mean the $(n-1)$-dimensional surface (Hausdorff) measure $\sigma(\partial B)$, and by $\mathrm{Vol}(B)$ the n-dimensional Lebesgue measure.

Consider the vector field $F(x)=x$ on $\mathbb{R}^n$. Then $\operatorname{div} F = \sum_{i=1}^n \partial_{x_i} x_i = n$. By the divergence theorem,

$$\int_B \operatorname{div} F\,dx = \int_{\partial B} \langle F,\nu\rangle\,d\sigma,$$

where ν is the outward unit normal. On $\partial B = S^{n-1}$ we have $\nu(x)=x$ (the outward normal to the unit sphere at x is x itself), so $\langle F(x),\nu(x)\rangle = \langle x,x\rangle = \|x\|_2^2 = 1$ for $x\in\partial B$. Hence

$$n\,\mathrm{Vol}(B) = \int_B n\,dx = \int_{\partial B} 1\,d\sigma = \sigma(\partial B) = \mathrm{Vol}(\partial B).$$

That is, $\mathrm{Vol}(\partial B) = n\,\mathrm{Vol}(B)$. ■

Exercise 7.8

Central idea. A single integration by parts, $1-\Phi(t) = \int_t^\infty \phi = \phi(t)/t - \int_t^\infty \phi(x)/x^2\,dx$ (taking $u=1/x$, $dv = x\phi(x)\,dx$): dropping the subtracted integral gives the upper bound $\phi(t)/t$, while $1/x^2 \le 1/t^2$ on $[t,\infty)$ bounds the subtracted integral by $(1-\Phi(t))/t^2$, and rearranging yields the lower bound.

Solution. For $t>0$,

$$1-\Phi(t) = \int_t^\infty \phi(x)\,dx = \int_t^\infty \frac{1}{x}\cdot x\phi(x)\,dx.$$

Integrating by parts with $u=1/x$, $dv = x\phi(x)\,dx$ (so $du = -dx/x^2$, $v=-\phi(x)$, since $\phi'(x) = -x\phi(x)$):

$$1-\Phi(t) = \left[-\frac{\phi(x)}{x}\right]_t^\infty - \int_t^\infty \frac{\phi(x)}{x^2}\,dx = \frac{\phi(t)}{t} - \int_t^\infty \frac{\phi(x)}{x^2}\,dx. \tag{$*$}$$

Since the subtracted integral is nonnegative, $1-\Phi(t) \le \phi(t)/t$, i.e.

$$\frac{1-\Phi(t)}{\phi(t)} \le \frac{1}{t}.$$

For the lower bound, on $[t,\infty)$ we have $1/x^2 \le 1/t^2$, so

$$\int_t^\infty \frac{\phi(x)}{x^2}\,dx \le \frac{1}{t^2}\int_t^\infty \phi(x)\,dx = \frac{1-\Phi(t)}{t^2}.$$

Substituting in $(*)$,

$$1-\Phi(t) \ge \frac{\phi(t)}{t} - \frac{1-\Phi(t)}{t^2}.$$

Rearranging, $(1-\Phi(t))\left(1+\frac{1}{t^2}\right) \ge \frac{\phi(t)}{t}$, i.e.

$$\frac{1-\Phi(t)}{\phi(t)} \ge \frac{1/t}{1+1/t^2} = \frac{t}{t^2+1}.$$

This holds for all $t>0$. ■

Exercise 7.9

Central idea. By rotation invariance take $u = e_1$; then $\mu(C(u,s)) = c_n \int_s^1 (1-t^2)^{(n-3)/2}\,dt$ with $c_n = \sigma(S^{n-2})/\sigma(S^{n-1})$. Writing the integrand via $\frac{d}{dt}(1-t^2)^{(n-1)/2} = -(n-1)t(1-t^2)^{(n-3)/2}$ turns the integral into $(1-s^2)^{(n-1)/2}/(n-1)$ times $\mathbb{E}[1/t]$ under an explicit probability law on $[s,1]$. The upper bound uses $1/t \le 1/s$; the lower bound uses Jensen ($\mathbb{E}[s/t] \ge s/\mathbb{E}[t]$) together with an exponential tail estimate giving $\mathbb{E}[t] \le 2s$ (this is where $s^2 \ge 2/n$ enters). Log-convexity of Γ controls c_n from both sides.

Solution. By rotation invariance let $u = e_1$, so $C(u,s) = \{x \in S^{n-1} : x_1 \ge s\}$. Parametrize $x_1 = t \in (-1,1)$; by the coarea formula the surface element is $d\sigma_{n-1} = (1-t^2)^{(n-3)/2}\,dt\,d\sigma_{n-2}$, so with μ normalized,

$$\mu(C(u,s)) = c_n \int_s^1 (1-t^2)^{\frac{n-3}{2}}\,dt, \qquad c_n = \frac{\sigma(S^{n-2})}{\sigma(S^{n-1})} = \frac{\Gamma(n/2)}{\sqrt{\pi}\,\Gamma((n-1)/2)}.$$

For $n = 2$, $\sqrt{2/n} \le s \le 1$ forces $s = 1$, and both bounds are $0 = 0$; assume $n \ge 3$.

Since $\frac{d}{dt}(1-t^2)^{(n-1)/2} = -(n-1)t(1-t^2)^{(n-3)/2}$,

$$(1-t^2)^{\frac{n-3}{2}} = \frac{-\frac{d}{dt}(1-t^2)^{\frac{n-1}{2}}}{(n-1)t},$$

and with $I := \int_s^1 (1-t^2)^{(n-3)/2}\,dt$,

$$I = \frac{(1-s^2)^{\frac{n-1}{2}}}{n-1}\,\mathbb{E}\left[\frac{1}{t}\right], \tag{1}$$

where $\mathbb{E}$ is expectation under the probability density $p(t) = -\frac{d}{dt}(1-t^2)^{(n-1)/2}/(1-s^2)^{(n-1)/2}$ on $[s,1]$.

Gamma bounds. Log-convexity of Γ gives $\Gamma(x+\frac{1}{2})^2 \le \Gamma(x)\Gamma(x+1) = x\Gamma(x)^2$, so with $x = (n-1)/2$,

$$c_n = \frac{\Gamma(x+\frac{1}{2})}{\sqrt{\pi}\,\Gamma(x)} \le \frac{\sqrt{(n-1)/2}}{\sqrt{\pi}}.$$

Also $\Gamma(x+1)^2 \le \Gamma(x+\frac{1}{2})\Gamma(x+\frac{3}{2}) = (x+\frac{1}{2})\Gamma(x+\frac{1}{2})^2$ gives $\Gamma(x+\frac{1}{2})/\Gamma(x) \ge x/\sqrt{x+\frac{1}{2}}$, so

$$c_n \ge \frac{(n-1)/2}{\sqrt{\pi}\sqrt{n/2}} = \frac{n-1}{\sqrt{2\pi n}}. \tag{2}$$

Upper bound. On $[s,1]$, $1/t \le 1/s$, so from (1), $I \le (1-s^2)^{(n-1)/2}/((n-1)s)$. Hence

$$\mu(C(u,s)) \le \frac{(1-s^2)^{\frac{n-1}{2}}}{s} \cdot \frac{\sqrt{(n-1)/2}}{\sqrt{\pi}\,(n-1)} = \frac{(1-s^2)^{\frac{n-1}{2}}}{s\sqrt{n}} \cdot \sqrt{\frac{n}{2\pi(n-1)}}.$$

Since $\sqrt{n/(2\pi(n-1))} \le 1/2 \iff 2n \le \pi(n-1) \iff n \ge \pi/(\pi-2) \approx 2.75$, for $n \ge 3$,

$$\mu(C(u,s)) \le \frac{1}{2s\sqrt{n}}(1-s^2)^{\frac{n-1}{2}}.$$

Lower bound. By Jensen ($1/t$ convex), with $J := \mathbb{E}[s/t]$,

$$J = s\,\mathbb{E}\left[\frac{1}{t}\right] \ge \frac{s}{\mathbb{E}[t]}.$$

Moreover $\mathbb{E}[t-s] = \int_0^{1-s} \mathbb{P}\{t-s > u\}\,du$ with, for $u \in [0, 1-s]$,

$$\mathbb{P}\{t > s+u\} = \left(\frac{1-(s+u)^2}{1-s^2}\right)^{\frac{n-1}{2}} \le \exp\left(-\frac{(n-1)su}{1-s^2}\right),$$

using $1-\frac{(s+u)^2-s^2}{1-s^2} \le \exp(-\frac{2su}{1-s^2})$. Thus $\mathbb{E}[t-s] \le \frac{1-s^2}{(n-1)s}$, and

$$\mathbb{E}[t] \le s + \frac{1-s^2}{(n-1)s} \le 2s,$$

since $\frac{1-s^2}{(n-1)s} \le s \iff 1-s^2 \le (n-1)s^2 \iff 1 \le ns^2$, which holds as $s^2 \ge 2/n$. Hence $J \ge 1/2$. From (1),

$$\mu(C(u,s)) = c_n \frac{(1-s^2)^{\frac{n-1}{2}}}{n-1}\,\frac{J}{s} = \frac{(1-s^2)^{\frac{n-1}{2}}}{s\sqrt{n}}\left(c_n \frac{\sqrt{n}}{n-1} J\right) \ge \frac{(1-s^2)^{\frac{n-1}{2}}}{s\sqrt{n}} \cdot \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{2},$$

using (2). Since $2\sqrt{2\pi} \approx 5.01 < 6$,

$$\mu(C(u,s)) \ge \frac{1}{6s\sqrt{n}}(1-s^2)^{\frac{n-1}{2}}.$$

This completes the proof. ■

Exercise 7.10

Central idea. We read A_t^c as $(A_t)^c$, the complement of the t-blowup (for $(A^c)_t$ the stated upper bound would be false at large t, since $(A^c)_t \uparrow S$). The Section 7.2 argument generalizes: for $C, D \subset S$ with $\operatorname{dist}(C,D) \ge t$, Brunn–Minkowski applied to the cones $\tilde{C}, \tilde{D}$ gives $\operatorname{Vol}((\tilde{C}+\tilde{D})/2) \ge \sqrt{\mu(C)\mu(D)}\operatorname{Vol}(B)$, while the modulus of convexity — refined to general convex combinations — forces $(\tilde{C}+\tilde{D})/2 \subset \max(\frac{1}{2}, 1-\delta(t))\,B$, whence $\mu(C)\mu(D) \le 2e^{-2n\delta(t/2)}$. Apply this to $C = A$, $D = S \setminus A_t$.

Solution. Write $B = \{x : \|x\| \le 1\}$, $S = \{x : \|x\| = 1\}$, and for $E \subset S$ let $\tilde{E} = \bigcup_{a\in[0,1]} aE$ be the cone, so $\mu(E) = \operatorname{Vol}(\tilde{E})/\operatorname{Vol}(B)$. The t-blowup is $A_t = \{x \in S : \operatorname{dist}(x, A) \le t\}$ (distance in $\|\cdot\|$), and we prove the bound for $(A_t)^c = S \setminus A_t$.

The modulus δ is nondecreasing (a larger ε shrinks the constraint set), and $\delta(\varepsilon) \le \varepsilon/2$ for $\varepsilon \in [0,1]$: with $\|x\| = 1$, $y = (1-\varepsilon)x$ gives $\|x-y\| = \varepsilon$ and $1 - \|(x+y)/2\| = \varepsilon/2$. Since $\operatorname{dist} \le 2$ on S, we use $\delta(t/2) \le 1/2$ below.

Convex-combination refinement. Let $\|c\|, \|d\| \le 1$ and $\alpha \in [0,1]$. Assuming $\alpha \le 1/2$ (else swap),

$$\alpha c + (1-\alpha)d = (1-2\alpha)d + 2\alpha\frac{c+d}{2},$$

so $\|\alpha c + (1-\alpha)d\| \le (1-2\alpha) + 2\alpha(1-\delta(\|c-d\|)) = 1 - 2\alpha\,\delta(\|c-d\|)$. By symmetry,

$$\|\alpha c + (1-\alpha)d\| \le 1 - 2\min(\alpha, 1-\alpha)\,\delta(\|c-d\|). \tag{1}$$

Two-set lemma. Let $\emptyset \ne C, D \subset S$ be measurable with $\operatorname{dist}(C,D) \ge t$. For $\lambda c \in \tilde{C}$, $\mu d \in \tilde{D}$ ($\lambda, \mu \in [0,1]$, $c \in C$, $d \in D$), write $\frac{\lambda c + \mu d}{2} = s(\alpha c + (1-\alpha)d)$ with $s = \frac{\lambda+\mu}{2}$, $\alpha = \frac{\lambda}{\lambda+\mu}$ (if $\lambda + \mu = 0$ there is nothing to show). By (1) and $\|c-d\| \ge t$,

$$\left\|\frac{\lambda c + \mu d}{2}\right\| \le s(1 - 2\min(\alpha, 1-\alpha)\delta(t)) = \frac{\lambda+\mu}{2} - \min(\lambda,\mu)\,\delta(t).$$

With $m = \min(\lambda,\mu)$, $M = \max(\lambda,\mu)$, the right side is $h(m) := \frac{M-m}{2} + m(1-\delta(t))$. This is linear in $m \in [0, M]$, $h(m) = M/2 + m(\frac{1}{2} - \delta(t))$, hence maximized at an endpoint: $h(m) \le \max(h(0), h(M)) = \max(M/2,\, M(1-\delta(t))) \le \max(\frac{1}{2}, 1-\delta(t))$ (using $M \le 1$). Hence

$$\frac{\tilde{C}+\tilde{D}}{2} \subset \max\big(\tfrac{1}{2},\, 1-\delta(t)\big)\,B. \tag{2}$$

Brunn–Minkowski gives $\mathrm{Vol}((\tilde{C}+\tilde{D})/2) \geq \sqrt{\mathrm{Vol}(\tilde{C})\mathrm{Vol}(\tilde{D})} = \sqrt{\mu(C)\mu(D)}\,\mathrm{Vol}(B)$. With (2): - If $\delta(t) \geq 1/2$: $\sqrt{\mu(C)\mu(D)} \leq 2^{-n}$, so $\mu(C)\mu(D) \leq e^{-n\ln 4} \leq 2e^{-n} \leq 2e^{-2n\delta(t/2)}$ (as $\delta(t/2) \leq 1/2$). - If $\delta(t) \leq 1/2$: $\sqrt{\mu(C)\mu(D)} \leq (1-\delta(t))^n \leq e^{-n\delta(t)} \leq e^{-n\delta(t/2)}$, so $\mu(C)\mu(D) \leq e^{-2n\delta(t/2)} \leq 2e^{-2n\delta(t/2)}$. Thus, for $\mathrm{dist}(C,D) \geq t$,

$$\mu(C)\mu(D) \leq 2e^{-2n\delta(t/2)}. \tag{3}$$

Conclusion. Let $D_0 = S \setminus A_t = (A_t)^c$. For $a \in A$, $x \in D_0$: $\|a-x\| \geq \mathrm{dist}(x,A) > t$, so $\mathrm{dist}(A,D_0) \geq t$ (if $D_0 = \emptyset$ there is nothing to prove). Applying (3) with $C = A$, $D = D_0$ (if $\mu(A) = 0$ the claim is vacuous),

$$\mu((A_t)^c) = \mu(D_0) \leq \frac{2}{\mu(A)}\, e^{-2n\delta(t/2)}.$$

■

Exercise 7.11

Central idea. Write $A_0, A_1 \subset Q = \{-1,1\}^{n-1}$ for the two i-sections and S_a, S_b for the simplicial initial segments replacing them ($a = |A_0|$, $b = |A_1|$). The vertex boundary splits into within-face and across-face parts, giving the exact formula $|\partial_V(A)| = |\partial_V(A_0)| + |\partial_V(A_1)| + |A_1 \setminus N[A_0]| + |A_0 \setminus N[A_1]|$ where $N[X] = X \cup \partial_V(X)$. Arguing by induction on n (so Harper's inequality is available in dimension $n-1$), the within-face terms do not increase under compression. The leftover "far" term $|S_b \setminus N[S_a]|$ is controlled by the nested-neighborhoods property of the simplicial ordering ($N[S_k]$ is always an initial segment) via a short counting argument.

Solution. We prove by induction on n that $|\partial_V(C_i(A))| \leq |\partial_V(A)|$ for all $A \subset \{-1,1\}^n$ and all i. For $n = 1$ the sections are singletons or empty, hence already initial segments, so $C_i(A) = A$. Assume Harper's vertex isoperimetric inequality in dimension $n-1$: $|\partial_V(X)| \geq |\partial_V(S_{|X|})|$ for $X \subset \{-1,1\}^{n-1}$.

By symmetry take $i = n$. Write $Q = \{-1,1\}^{n-1}$ and

$$A_0 = \{y \in Q : (y,-1) \in A\}, \qquad A_1 = \{y \in Q : (y,+1) \in A\},$$

so $A = (A_0 \times \{-1\}) \cup (A_1 \times \{+1\})$. (The problem statement has two typos: in $A^{(i+)}$ it should be $x^{(i,1)} \in A$, and "i-th component is 0 (and 1)" should be -1 (and $+1$).) Put $a = |A_0|$, $b = |A_1|$, and let $S_a, S_b \subset Q$ be the simplicial initial segments. Swapping the two faces if necessary (an automorphism preserving $|\partial_V|$), assume $a \leq b$, so $S_a \subseteq S_b$. Then $C_n(A) = (S_a \times \{-1\}) \cup (S_b \times \{+1\})$.

For $X \subset Q$ let $N[X] = X \cup \partial_V(X)$ be the closed vertex-neighborhood in Q.

Boundary decomposition. A point $(y,-1) \notin A$ lies in $\partial_V(A)$ iff $y \in \partial_V(A_0)$ or $y \in A_1$. Hence, with L_A the lower-face part of $\partial_V(A)$,

$$L_A = \{y \notin A_0 : y \in \partial_V(A_0) \text{ or } y \in A_1\} = \partial_V(A_0) \,\dot\cup\, (A_1 \setminus N[A_0]),$$

disjoint because $\partial_V(A_0) \subseteq N[A_0]$. Indeed, if $y \in L_A$ and $y \notin \partial_V(A_0)$ then $y \in A_1 \setminus A_0$ and $y \notin \partial_V(A_0)$, so $y \notin N[A_0]$. Thus $|L_A| = |\partial_V(A_0)| + |A_1 \setminus N[A_0]|$, and similarly for the upper face. Therefore

$$|\partial_V(A)| = |\partial_V(A_0)| + |\partial_V(A_1)| + |A_1 \setminus N[A_0]| + |A_0 \setminus N[A_1]|. \tag{1}$$

For $C = C_n(A)$, the same argument gives $|L_C| = |\partial_V(S_a)| + |S_b \setminus N[S_a]|$ and $|U_C| = |\partial_V(S_b)| + |S_a \setminus N[S_b]|$. Since $S_a \subseteq S_b \subseteq N[S_b]$, the last term vanishes:

$$|\partial_V(C)| = |\partial_V(S_a)| + |\partial_V(S_b)| + |S_b \setminus N[S_a]|. \tag{2}$$

Induction. By the inductive hypothesis, $|\partial_V(A_0)| \geq |\partial_V(S_a)|$ and $|\partial_V(A_1)| \geq |\partial_V(S_b)|$. From (1)–(2) it suffices to prove

$$|S_b \setminus N[S_a]| \leq \delta_0 + \delta_1 + |A_1 \setminus N[A_0]| + |A_0 \setminus N[A_1]|, \tag{3}$$

where $\delta_0 = |\partial_V(A_0)| - |\partial_V(S_a)| \geq 0$ and $\delta_1 = |\partial_V(A_1)| - |\partial_V(S_b)| \geq 0$.

Nested neighborhoods. The simplicial ordering has the property that $N[S_k]$ is an initial segment for every k (the ordering sorts by Hamming weight, then reverse-lexicographically; a direct check shows the closed neighborhood of an initial segment is again initial). Write $N[S_a] = S_{a'}$. Then $|S_b \setminus N[S_a]| = \max(0, b - a')$, and $|\partial_V(S_a)| = a' - a$.

Proof of (3). If $b \leq a'$ the left side is 0. Assume $b > a'$, so the left side equals $b - a'$. Since $|A_1 \setminus N[A_0]| = b - |A_1 \cap N[A_0]|$, (3) becomes, after cancelling b,

$$|A_1 \cap N[A_0]| \leq a' + \delta_0 + \delta_1 + |A_0 \setminus N[A_1]|. \tag{4}$$

But $|A_1 \cap N[A_0]| \leq |N[A_0]| = a + |\partial_V(A_0)| = a + |\partial_V(S_a)| + \delta_0 = a' + \delta_0$, using $|\partial_V(S_a)| = a' - a$. Since $\delta_1 + |A_0 \setminus N[A_1]| \geq 0$, (4) follows. This proves (3).

Finally, from (1), (2), the inductive bounds, and (3):

$$\begin{aligned} |\partial_V(A)| - |\partial_V(C)| &= \delta_0 + \delta_1 + |A_1 \setminus N[A_0]| + |A_0 \setminus N[A_1]| - |S_b \setminus N[S_a]| \\ &\geq 0, \end{aligned}$$

so $|\partial_V(C_i(A))| \leq |\partial_V(A)|$. This completes the induction. ■

Exercise 7.12

Central idea. Use an integer-valued potential that every nontrivial compression strictly decreases: $\Phi(A) = \sum_{x \in A} r(x)$, where $r(x)$ is the rank of x in the simplicial ordering of $\{-1, 1\}^n$. On each fiber $\{x_i = \sigma\}$, the initial segment $S_N^{(i,\sigma)}$ is the set of N elements of smallest rank, and since ranks are distinct it is the *unique* minimizer of the rank sum among N-subsets of the fiber. Hence C_i strictly decreases Φ whenever $C_i(A) \neq A$; with $|A|$ preserved, Φ is bounded below, so the process cannot continue forever.

Solution. Let $r(x) \in \{1, \ldots, 2^n\}$ be the position of x in the simplicial ordering of $\{-1, 1\}^n$ (so r is injective), and define

$$\Phi(A) = \sum_{x \in A} r(x), \qquad A \subset \{-1, 1\}^n.$$

Fix i and $\sigma \in \{-1, 1\}$, and let $F_i^\sigma = \{x : x_i = \sigma\}$. The "simplicial ordering of all vectors whose i-th component is σ" is the restriction of the global simplicial ordering to F_i^σ: indeed, for $x, y \in F_i^\sigma$ the weight comparison $\|x\| < \|y\|$ is unaffected by the fixed coordinate, and the tie-break (at the smallest index where x and y differ, which is never i) is likewise unchanged. Thus

$$S_N^{(i,\sigma)} = \{\text{the } N \text{ elements of } F_i^\sigma \text{ of smallest } r\text{-value}\}.$$

Let $a_1 < a_2 < \cdots < a_{2^{n-1}}$ be the r-values on F_i^σ. For any N-subset with sorted r-values $b_1 < \cdots < b_N$, we have $b_j \geq a_j$ for each j (at least j fiber elements have r-value $\leq b_j$), so $\sum b_j \geq \sum_{j=1}^N a_j$ with equality iff the subset is exactly $S_N^{(i,\sigma)}$. That is, the initial segment is the unique minimizer of the rank sum over N-subsets of the fiber.

Now $C_i(A)$ agrees with A outside $F_i^- \cup F_i^+ = \{-1, 1\}^n$, and on each fiber replaces $A \cap F_i^\sigma$ (of size $N_\sigma = |A^{(i\sigma)}|$) by $S_{N_\sigma}^{(i,\sigma)}$. Hence

$$\Phi(C_i(A)) \leq \Phi(A),$$

with strict inequality iff $A \cap F_i^\sigma \neq S_{N_\sigma}^{(i,\sigma)}$ for some σ, i.e. iff $C_i(A) \neq A$.

By Exercise 7.11, $|C_i(A)| = |A|$, so every A_k has the same cardinality $m = |A|$. Since the r-values are distinct positive integers, $\Phi(A_k) \geq 1 + 2 + \cdots + m$ for all k. Each step of the process applies C_i with $C_i(A_k) \neq A_k$, strictly decreasing the positive integer Φ. A strictly decreasing sequence of integers bounded below is finite, so the process terminates: for some k, $C_i(A_k) = A_k$ for every i. ■

Exercise 7.13

Central idea. If every i-section of B is an initial segment, then any "out-of-order" pair $u \notin B$, $v \in B$ with u preceding v cannot share a coordinate (otherwise both lie in a common i-section, which would then fail to be initial); so $v = u^c$ is the complement of u. Applied to $u =$ the first non-element and $v =$ the last element of B, this forces $B = S_{|B|-1} \cup \{v\}$, where v is the immediate successor of u and $v = u^c$. Consecutive complementary pairs are rare: for odd n it is the cross-layer pair (last k-set, first $(k+1)$-set), for even n the pair (last k-set containing 1, first k-set not containing 1). These give exactly the two sets of the statement. A direct neighbourhood count shows they have vertex boundary at least that of $S_{|B|}$ — strictly larger for $n \geq 3$, but equal for $n \leq 2$ (so the book's "larger" needs $n \geq 3$) — which completes Harper's theorem.

Solution. *Conventions.* Identify $x \in \{-1,1\}^n$ with the set $\{i : x_i = 1\} \subseteq [n] = \{1, \dots, n\}$; then $\|x\| = |x|$, and the complement $x^c = [n] \setminus x$ corresponds to $-x$. Write $L_k = \{x : |x| = k\}$ and $x \prec y$ for the simplicial order: $|x| < |y|$, or $|x| = |y|$ and $\min(x \triangle y) \in x$ (this is the book's rule "$x_i = 1$, $y_i = -1$ at the first i with $x_i \neq y_i$"). It is a total order, and S_N is the set of its first N elements. Let $N[A] = A \cup \partial_V(A)$ be the closed neighbourhood, so $|\partial_V(A)| = |N[A]| - |A|$. For $\sigma \in \{-1,1\}$ let $F_i^\sigma = \{x : x_i = \sigma\}$.

Induced order. For $x, y \in F_i^\sigma$, the weight comparison is unaffected by the common coordinate i, and $\min(x \triangle y) \neq i$. Hence the simplicial order of the vectors with i-th coordinate σ (used to define $S_N^{(i,\pm)}$ in Exercise 7.11) is the restriction of $\prec$ to F_i^σ. Thus $C_i(B) = B$ means: for each σ, $B \cap F_i^\sigma$ is an initial segment of $(F_i^\sigma, \prec)$.

Assume $C_i(B) = B$ for all i, put $N = |B|$, and suppose $B \neq S_N$ (so $0 < N < 2^n$).

Step 1 (out-of-order pairs are complementary). Let $u \notin B$, $v \in B$, $u \prec v$. If $u_i = v_i = \sigma$ for some i, then $u, v \in F_i^\sigma$; since $B \cap F_i^\sigma$ is an initial segment of $(F_i^\sigma, \prec)$ containing v, it contains u, a contradiction. Hence u and v differ in every coordinate: $v = u^c$.

Step 2 (B is S_{N-1} plus the successor of its N-th element). Let $u = \min_\prec(\{-1,1\}^n \setminus B)$ and $v = \max_\prec B$. Since B is not an initial segment, some non-element precedes some element, so $u \prec v$, and by Step 1, $v = u^c$. - If $w \notin B$ and $w \prec v$, then by Step 1, $w = v^c = u$. Hence every element preceding v other than u lies in B, and (by maximality of v) nothing after v lies in B: $B = \{w : w \preceq v\} \setminus \{u\}$. - If $w \in B$ and $u \prec w$, then by Step 1, $w = u^c = v$. Any w with $u \prec w \prec v$ lies in B by the previous point, hence equals v, which is absurd. So v is the immediate successor of u.

Therefore u is the N-th element of the order, $v = u^c$ is the $(N+1)$-st, and

$$B = S_{N-1} \cup \{v\}, \qquad S_{N-1} = \{w : w \prec u\}.$$

Also $u \prec u^c$ forces $|u| \leq |u^c| = n - |u|$.

Step 3 (which consecutive pairs are complementary). Two facts about a level L_k ($0 \leq k \leq n$, all levels nonempty): (a) its last element is $\{n-k+1, \dots, n\}$ and its first is $\{1, \dots, k\}$. Indeed, for $w \in L_k$ with $w \neq T := \{n-k+1, \dots, n\}$, the elements of $w \setminus T$ are $\leq n-k$ and those of $T \setminus w$ are $\geq n-k+1$, and $w \setminus T \neq \emptyset$, so $\min(w \triangle T) \in w$ and $w \prec T$; the first element is handled symmetrically. (b) In L_k, every set containing 1 precedes every set not containing 1, since the minimum of their symmetric difference is 1.

Case $|u| < |v|$. As v is the immediate successor of u and every level is nonempty, u is the last element of $L_{|u|}$, v is the first element of $L_{|u|+1}$, and $|v| = |u| + 1$. With $|v| = n - |u|$ this gives $n = 2k+1$, $|u| = k$. By (a), $u = \{k+2, \dots, 2k+1\}$, and indeed $u^c = \{1, \dots, k+1\}$ is the first element of L_{k+1}. Since u is last in L_k, $S_{N-1} = L_{\leq k} \setminus \{u\}$, and as $k < n/2 < k+1$,

$$B = \{x : \|x\| < n/2\} \setminus \{u\} \cup \{u^c\},$$

where $u_i = 1$ iff $i \in \{(n+3)/2, \dots, n\}$ and $u_i^c = 1$ iff $i \in \{1, \dots, (n+1)/2\}$. These are the book's y and z: this is the stated odd exception. (For even n this case cannot occur.)

Case $|u| = |v|$. Then $|u| = n - |u|$, so $n = 2k$ with $k \geq 1$, and $u, u^c \in L_k$. Since $u \prec u^c$ and $\min(u \triangle u^c) = 1$, we get $1 \in u$, $1 \notin u^c$. By (b) and the fact that u^c is the immediate successor of u, u is the last element of

$L_k^{(1)} = \{w \in L_k : 1 \in w\}$ and u^c is the first element of $L_k \setminus L_k^{(1)}$. Comparing $w \in L_k^{(1)}$ through $w \setminus \{1\}$, a $(k-1)$-subset of $\{2, \dots, 2k\}$, fact (a) shows the last element of $L_k^{(1)}$ is $\{1, k+2, \dots, 2k\}$. Hence $u = \{1, k+2, \dots, 2k\}$ and $u^c = \{2, \dots, k+1\}$ (indeed the first k-set without 1). Here $S_{N-1} = L_{<k} \cup (L_k^{(1)} \setminus \{u\})$, so

$$B = \{x : \|x\| < n/2\} \cup \{x : \|x\| = n/2,\ x_1 = 1\} \setminus \{u\} \cup \{u^c\},$$

with $u_i = 1$ iff $i \in \{1, (n/2)+2, \dots, n\}$ and $u_i^c = 1$ iff $i \in \{2, \dots, (n/2)+1\}$ — the book's y and z: this is the stated even exception. (For odd n this case cannot occur.)

This proves the classification: $B = S_{|B|}$, or B is the exceptional set for the parity of n.

Converse (the exceptional set is indeed compressed). Let $u \prec v = u^c$ be the consecutive complementary pair found above and $B = \{w : w \preceq u\} \cup \{v\}$. Fix i and let $\sigma = v_i$, so $u_i = -\sigma$. In F_i^σ: the elements preceding v are $\{w : w \preceq u\} \cup \{u\}$, and $u \notin F_i^\sigma$, so $B \cap F_i^\sigma = \{w \in F_i^\sigma : w \preceq v\}$, an initial segment. In $F_i^{-\sigma}$: $v \notin F_i^{-\sigma}$, so $B \cap F_i^{-\sigma} = \{w \in F_i^{-\sigma} : w \prec u\}$, an initial segment. Hence $C_i(B) = B$ for all i, while $B \neq S_{|B|}$ since it omits u. So for every $n \geq 1$ there is exactly one compressed set that is not an initial segment. (A brute-force enumeration of all compressed sets for $n = 1, \dots, 9$, script `external_audit_2026-10-30/compare/repair_checks/check_07_13.py`, confirms exactly one non-initial compressed set per n, equal to the book's formula, and confirms Step 1 on every compressed set for $n \leq 6$.)

Step 4 (the exceptional set has boundary at least that of $S_{|B|}$). Let B be the exceptional set, $N = |B|$, $B_0 = S_N$. Since $|B| = |B_0|$, it suffices to compare $|N[B]|$ and $|N[B_0]|$.

Odd $n = 2k+1$ with $k \geq 1$ (i.e. $n \geq 3$). Here $|B| = |L_{\leq k}|$, so $B_0 = L_{\leq k}$ and $N[B_0] = L_{\leq k+1}$. We show $N[B] \supseteq L_{\leq k+1}$: $L_{<k} \subseteq B$; every element of L_k other than u is in B; u has a down-neighbour in $L_{k-1} \subseteq B$ (as $k \geq 1$); and each $w \in L_{k+1}$ has $k+1 \geq 2$ down-neighbours in L_k, at most one of which is u, so it has a neighbour in B. In addition, $N[B]$ contains the sets $u^c \cup \{j\}$, $j \notin u^c$, which lie in L_{k+2} (they exist because $|u^c| = k+1 < n$), hence outside $L_{\leq k+1}$. So $|N[B]| > |N[B_0]|$.

Even $n = 2k$ with $k \geq 2$ (i.e. $n \geq 4$). Here $B_0 = L_{<k} \cup L_k^{(1)}$, and $N[B_0] = L_{\leq k} \cup L_{k+1}^{(1)}$: every element of L_k has a down-neighbour in L_{k-1}; an element of L_{k+1} is adjacent to $L_k^{(1)}$ iff it contains 1 (removing an element other than 1 keeps 1, while if $1 \notin w$ no down-neighbour of w contains 1); and no element above L_{k+1} is reached. We show $N[B] \supseteq N[B_0]$: $L_{<k} \subseteq B$; every element of L_k is in B or has a down-neighbour in $L_{k-1} \subseteq B$; each $w \in L_{k+1}^{(1)}$ has $k \geq 2$ down-neighbours in $L_k^{(1)}$ (remove any of its k elements other than 1), at most one of which is u, and the others lie in $L_k^{(1)} \setminus \{u\} \subseteq B$. In addition, $N[B]$ contains $u^c \cup \{j\}$ for $j \in \{k+2, \dots, 2k\}$ (nonempty as $k \geq 2$), a $(k+1)$-set not containing 1, hence not in $N[B_0]$. So $|N[B]| > |N[B_0]|$.

Small cases — correction to the statement. For $n \leq 2$ the exceptional set does **not** have a larger boundary; the boundaries are equal. For $n = 1$: $B = \{(1)\}$ and $S_1 = \{(-1)\}$, both with $|\partial_V| = 1$. For $n = 2$: $B = \{(-1,-1), (-1,1)\}$ and $S_2 = \{(-1,-1), (1,-1)\}$, both with $|\partial_V| = 2$. So the book's "larger boundary" holds exactly for $n \geq 3$; for every n, however, $|\partial_V(B)| \geq |\partial_V(S_{|B|})|$, which is all that Harper's theorem needs. (Brute force, same script: $|\partial_V(B)|$ vs. $|\partial_V(S_{|B|})|$ for $n = 1, \dots, 9$ is 1/1, 2/2, 4/3, 7/6, 12/10, 22/20, 38/35, 73/70, 130/126.)

Conclusion: Harper's theorem. We prove $|\partial_V(A)| \geq |\partial_V(S_{|A|})|$ for all $A \subseteq \{-1,1\}^n$ by induction on n, as set up in Exercise 7.11. For $n = 1$, sets of size 0 or 2 are initial segments, and every set of size 1 has a boundary of size $1 = |\partial_V(S_1)|$. Assume the theorem in dimension $n-1$. Then Exercise 7.11 gives $|C_i(A')| = |A'|$ and $|\partial_V(C_i(A'))| \leq |\partial_V(A')|$ for all $A' \subseteq \{-1,1\}^n$ and all i, and Exercise 7.12 shows that iterating compressions from A terminates at a set B with $C_i(B) = B$ for all i, $|B| = |A|$ and $|\partial_V(B)| \leq |\partial_V(A)|$. By Steps 1–4, either $B = S_{|A|}$ or $|\partial_V(B)| \geq |\partial_V(S_{|A|})|$. Therefore

$$|\partial_V(A)| \geq |\partial_V(B)| \geq |\partial_V(S_{|A|})|,$$

which is Theorem 7.6 in dimension n. ■

Exercise 7.14

Central idea. Sion's theorem cannot be applied directly on $\mathcal{M}(A)$, since A is only assumed measurable and $\mathcal{M}(A)$ need not be weakly compact. Following the proof of Lemma 7.10, push measures forward through

$\chi(y) = (\mathbf{1}_{\{x_j \neq y_j\}})_{j \le n}$ to the finite-dimensional simplex $\mathcal{M}(\chi(A))$ of probability measures on $\{0,1\}^n$, which is convex and compact; there the objective $(\mu, \alpha) \mapsto \sum_j \alpha_j \mathbb{E}_\mu[\pi_j]$ is bilinear, hence convex-continuous in μ and concave-continuous in α, so Sion's theorem applies, and surjectivity of the pushforward transfers the identity back to $\mathcal{M}(A)$.

Solution. For fixed $x \in \mathcal{X}^n$ write

$$F(\nu, \alpha) = \sum_{j=1}^{n} \alpha_j \, \mathbb{E}_\nu \mathbf{1}_{\{x_j \neq Y_j\}}, \qquad \nu \in \mathcal{M}(A), \quad \alpha \in B := \{\alpha : \|\alpha\|_2 \le 1\}.$$

Since $\inf_{\nu \in \mathcal{M}(A)} \mathbb{E}_\nu[g(Y)] = \inf_{y \in A} g(y)$ for bounded measurable g (Dirac masses belong to $\mathcal{M}(A)$), the first line of (7.1) is $d_T(x, A) = \sup_{\alpha \in B} \inf_{\nu \in \mathcal{M}(A)} F(\nu, \alpha)$. We check Sion's conditions for F with the infimum taken over measures.

Let $\chi : A \to \{0,1\}^n$, $\chi(y) = (\mathbf{1}_{\{x_j \neq y_j\}})_{1 \le j \le n}$, and let π_j be the j-th coordinate projection on $\{0,1\}^n$. Then $\mathbb{E}_\nu \mathbf{1}_{\{x_j \neq Y_j\}} = \mathbb{E}_{\chi_* \nu}[\pi_j]$. Put $\mathcal{K} = \mathcal{M}(\chi(A))$, the probability measures on $\{0,1\}^n$ supported on $\chi(A)$, and

$$G(\mu, \alpha) = \sum_{j=1}^{n} \alpha_j \, \mathbb{E}_\mu[\pi_j], \qquad \mu \in \mathcal{K}, \ \alpha \in B.$$

(i) $\mathcal{K}$ is convex and compact: identifying $\mathcal{M}(\{0,1\}^n)$ with the probability simplex in $\mathbb{R}^{2^n}$ (Euclidean = weak topology), $\mathcal{K} = \{p : p_z = 0 \ \forall\, z \notin \chi(A)\}$ is a closed, hence compact, convex subset; mixtures of measures supported on $\chi(A)$ remain supported on $\chi(A)$.

(ii) For fixed α, $\mu \mapsto G(\mu, \alpha)$ is linear, hence convex, and continuous (each $\mathbb{E}_\mu[\pi_j]$ is a linear functional of the probability vector), hence lower semicontinuous.

(iii) For fixed μ, $\alpha \mapsto G(\mu, \alpha)$ is linear, hence concave, and continuous, hence upper semicontinuous; B is convex.

Thus Sion's minmax theorem applies with the infimum over $\mathcal{K}$:

$$\inf_{\mu \in \mathcal{K}} \sup_{\alpha \in B} G(\mu, \alpha) = \sup_{\alpha \in B} \inf_{\mu \in \mathcal{K}} G(\mu, \alpha).$$

Finally, for each fixed α,

$$\inf_{\nu \in \mathcal{M}(A)} F(\nu, \alpha) = \inf_{\mu \in \mathcal{K}} G(\mu, \alpha) :$$

indeed $F(\nu, \alpha) = G(\chi_* \nu, \alpha)$ gives $\ge$, and $\chi_* : \mathcal{M}(A) \to \mathcal{K}$ is onto — given μ supported on $\chi(A)$, pick $y(z) \in A$ with $\chi(y(z)) = z$ for each $z \in \chi(A)$ and set $\nu = \sum_z \mu(\{z\}) \delta_{y(z)} \in \mathcal{M}(A)$, so $\chi_* \nu = \mu$. Hence

$$\sup_{\alpha \in B} \inf_{\nu \in \mathcal{M}(A)} F(\nu, \alpha) = \inf_{\nu \in \mathcal{M}(A)} \sup_{\alpha \in B} F(\nu, \alpha),$$

which is (7.1). The infimum over the compact $\mathcal{K}$ of the continuous $G(\cdot, \alpha)$ is attained, and the supremum over the compact ball B is attained (the proof exhibits the maximizer $\hat{\alpha}$ via Cauchy–Schwarz), so the saddle point is achieved.

Exercise 7.15

Central idea. If $f(x) = k$ then k coordinates certify it: by heredity of Π, any z agreeing with x on more than a of those coordinates has $f(z) > a$, so every $z \in A_a$ differs from x on at least $k - a$ certificate coordinates. Taking α uniform on the certificate gives $d_T(x, A_a) \ge (f(x) - a)/\sqrt{f(x)}$; since $t \mapsto (t-a)/\sqrt{t}$ is increasing, large values of f force large convex distance, and Talagrand's convex distance inequality (Theorem 7.9) turns this into the two-sided tail bounds about a median.

Solution. Recall $d_T(x, A) = \sup_{\|\alpha\|_2 = 1, \, \alpha \ge 0} \inf_{y \in A} \sum_j \alpha_j \mathbf{1}_{\{x_j \neq y_j\}}$ and Theorem 7.9: $\mathbb{P}\{A\}\, \mathbb{P}\{d_T(X, A) \ge t\} \le e^{-t^2/4}$ for product $\mathbb{P}$. A median MZ satisfies $\mathbb{P}\{Z \le MZ\} \ge 1/2$ and $\mathbb{P}\{Z \ge MZ\} \ge 1/2$.

If $f(x) = k$, there are $i_1, \dots, i_k$ with $(x_{i_1}, \dots, x_{i_k})$ satisfying Π_k (Section 3.3); by heredity, any z agreeing with x on r of these coordinates has $f(z) \geq r$.

First inequality. Let $k = f(x)$. If $k \leq a$ there is nothing to prove. If $k > a$, take $\alpha_j = 1/\sqrt{k}$ on the certificate coordinates and 0 elsewhere, so $\|\alpha\|_2 = 1$. For $z \in A_a = \{f \leq a\}$: z agrees with x on at most a certificate coordinates (otherwise $f(z) > a$ by heredity), hence differs on at least $k - a$ of them, and

$$d_\alpha(x, z) = \sum_j \alpha_j \mathbf{1}_{\{x_j \neq z_j\}} \geq \frac{k-a}{\sqrt{k}}.$$

Thus $d_T(x, A_a) \geq \inf_{z \in A_a} d_\alpha(x, z) \geq (k-a)/\sqrt{k}$, i.e. $f(x) \leq a + \sqrt{f(x)}\, d_T(x, A_a)$.

Upper tail. Let $m = MZ$ and $A = A_m$, so $\mathbb{P}\{A\} \geq 1/2$. On $\{f(X) \geq m + t\}$, the first part gives $d_T(X, A) \geq \frac{f(X)-m}{\sqrt{f(X)}} = g(f(X))$ with $g(s) = (s-m)/\sqrt{s}$, which is increasing for $s \geq m$ since $g'(s) = (s+m)/(2s^{3/2}) > 0$ (the hint with $a = m$). Hence $d_T(X, A) \geq g(m+t) = t/\sqrt{m+t}$, and by Theorem 7.9,

$$\mathbb{P}\{Z \geq m + t\} \leq \mathbb{P}\{d_T(X, A) \geq t/\sqrt{m+t}\} \leq \frac{e^{-t^2/(4(m+t))}}{\mathbb{P}\{A\}} \leq 2e^{-\frac{t^2}{4(m+t)}}.$$

Lower tail. Let $A = \{f \leq m - t\}$ (if empty, $\mathbb{P}\{A\} = 0$ and there is nothing to prove). For y with $f(y) \geq m$, let J be a certificate with $|J| = \ell \geq m$ and take $\alpha_j = 1/\sqrt{\ell}$ on J, 0 elsewhere. For $x \in A$: x agrees with y on at most $f(x) \leq m - t$ coordinates of J by heredity, so differs on at least $\ell - m + t$ of them, giving $d_\alpha(y, x) \geq (\ell - m + t)/\sqrt{\ell} = h(\ell)$ with $h(s) = (s - m + t)/\sqrt{s}$. For $s \geq m$, $h'(s) = (s + m - t)/(2s^{3/2}) > 0$ (in the nontrivial case $t \leq m$, else $\{Z \leq m - t\} = \emptyset$ as $Z \geq 0$), so $h(\ell) \geq h(m) = t/\sqrt{m}$. Thus $\{f(X) \geq m\} \subseteq \{d_T(X, A) \geq t/\sqrt{m}\}$, and with $\mathbb{P}\{Z \geq m\} \geq 1/2$,

$$\mathbb{P}\{Z \leq m - t\} = \mathbb{P}\{A\} \leq \frac{e^{-(t^2/m)/4}}{\mathbb{P}\{f(X) \geq m\}} \leq 2e^{-\frac{t^2}{4m}}.$$

If $m = 0$, $\{Z \leq -t\} = \emptyset$ for $t > 0$ since $Z \geq 0$, so the lower bound holds trivially. ■

Exercise 7.16

Central idea. Follow the proof of Theorem 7.12, but replace the sublevel sets $A_s = \{f \leq s\}$ by $A_s \cap S$, which remain convex as intersections of convex sets. Since f is 1-Lipschitz on the convex set S, for $x \in S$ one still gets $f(x) \leq s + D(x, A_s \cap S) \leq s + d_T(x, A_s \cap S)$ via Lemma 7.11, so Talagrand's convex distance inequality (Theorem 7.9) applies to $A_s \cap S$. The mass $\mathbb{P}\{X \notin S\}$ outside S — where no Lipschitz control is available — is accounted for separately, and the median properties $\mathbb{P}\{f(X) \leq M\}, \mathbb{P}\{f(X) \geq M\} \geq 1/2$ give $\mathbb{P}\{X \in A_s \cap S\} \geq 1/2 - \mathbb{P}\{X \notin S\}$. Both tails come out as $p + \frac{1}{1/2-p} e^{-t^2/4}$ with $p = \mathbb{P}\{X \notin S\}$; the stated lower-tail form $2p + 2e^{-t^2/4}$ follows by a case split: when $e^{-t^2/4} \leq \frac{1}{2}(\frac{1}{2} - p)$ the first bound is smaller, and otherwise $2p + 2e^{-t^2/4} > \frac{1}{2} \geq \mathbb{P}\{f(X) < Mf(X)\}$.

Solution. Let $p = \mathbb{P}\{X \notin S\} < 1/2$ and let $M = Mf(X)$ be a median of $f(X)$, so $\mathbb{P}\{f(X) \leq M\} \geq 1/2$ and $\mathbb{P}\{f(X) \geq M\} \geq 1/2$. For $s \in \mathbb{R}$ put $A_s = \{x \in [0,1]^n : f(x) \leq s\}$, convex by quasi-convexity of f, and $B_s = A_s \cap S$, convex as an intersection of convex sets.

Key inequality. Let $x \in S$. For any $y \in B_s \subset S$, the 1-Lipschitz property on S gives $f(x) \leq f(y) + \|x - y\| \leq s + \|x - y\|$. Taking $\inf_{y \in B_s}$ yields $f(x) \leq s + D(x, B_s)$ for $x \in S$, where $D(x, B_s) = \inf_{y \in B_s} \|x - y\|$. By Lemma 7.11 ($B_s$ convex), $D(x, B_s) \leq d_T(x, B_s)$. Hence, for all $x \in S$ and all $s \in \mathbb{R}$,

$$f(x) \leq s + d_T(x, B_s). \tag{*}$$

In particular $\{f(X) \geq s + t\} \cap \{X \in S\} \subset \{d_T(X, B_s) \geq t\}$, so by Theorem 7.9 applied to the convex set B_s,

$$\mathbb{P}\{f(X) \geq s + t,\, X \in S\} \cdot \mathbb{P}\{X \in B_s\} \leq e^{-t^2/4}. \tag{**}$$

Consequently $\mathbb{P}\{f(X) \geq s+t\} \leq p + e^{-t^2/4}/\mathbb{P}\{X \in B_s\}$.

Upper tail. Take $s = M$. Since $\mathbb{P}\{X \in B_M\} \geq \mathbb{P}\{f(X) \leq M\} - p \geq 1/2 - p > 0$, (**) gives

$$\mathbb{P}\{f(X) > M+t\} \leq \mathbb{P}\{f(X) \geq M+t\} \leq p + \frac{1}{1/2-p}\, e^{-t^2/4}.$$

Lower tail. Take $s = M - t$, so $B_{M-t} = A_{M-t} \cap S$. From (**) with $s + t = M$,

$$\mathbb{P}\{f(X) \geq M,\, X \in S\} \cdot \mathbb{P}\{X \in B_{M-t}\} \leq e^{-t^2/4}.$$

Since $\mathbb{P}\{f(X) \geq M,\, X \in S\} \geq \mathbb{P}\{f(X) \geq M\} - p \geq 1/2 - p$, we obtain $\mathbb{P}\{X \in B_{M-t}\} \leq e^{-t^2/4}/(1/2-p)$. Hence

$$\mathbb{P}\{f(X) < M - t\} \leq \mathbb{P}\{f(X) \leq M - t\} \leq p + \mathbb{P}\{X \in B_{M-t}\} \leq p + \frac{1}{1/2-p}\, e^{-t^2/4}.$$

This is the bound $p + \frac{1}{1/2-p} e^{-t^2/4}$ for the lower tail; it remains to deduce the stated bound $2p + 2e^{-t^2/4}$ from it. Write $e = e^{-t^2/4}$ and split into two cases. - If $e \leq \frac{1}{2}\left(\frac{1}{2} - p\right)$, then

$$\frac{e}{1/2-p} = \frac{2e\left(\frac{1}{2}-p\right) + 2pe}{1/2-p} = 2e + \frac{2pe}{1/2-p} \leq 2e + p,$$

so $p + \frac{e}{1/2-p} \leq 2p + 2e$ and the display above gives $\mathbb{P}\{f(X) < M - t\} \leq 2p + 2e$. - If $e > \frac{1}{2}\left(\frac{1}{2} - p\right)$, then $2p + 2e > 2p + \frac{1}{2} - p = \frac{1}{2} + p \geq \frac{1}{2}$. On the other hand, since $\mathbb{P}\{f(X) \geq M\} \geq \frac{1}{2}$, we have $\mathbb{P}\{f(X) < M - t\} \leq \mathbb{P}\{f(X) < M\} \leq \frac{1}{2} < 2p + 2e$.

In both cases

$$\mathbb{P}\{f(X) < Mf(X) - t\} \leq 2\,\mathbb{P}\{X \notin S\} + 2e^{-t^2/4}.$$

Remark. The $A_s \cap S$ argument actually gives the lower-tail bound $\min\left(p + \frac{1}{1/2-p} e^{-t^2/4},\, \frac{1}{2}\right)$, which never exceeds the stated $2p + 2e^{-t^2/4}$ (the case split above; a numerical grid over $p \in [0, 0.4999]$, $t \in (0, 10]$ confirms it). When $p = 0$ both lower-tail bounds reduce to $2e^{-t^2/4}$, recovering Theorem 7.12. ■

Exercise 7.17

Central idea. Put $S = \sum_{i=1}^n X_i^2$ and $\mu = \mathbb{E}[S]$; the event is $\{S \geq 2\mu + t\} = \{S - \mu \geq \mu + t\}$. Since $X_i \in [0,1]$, the variables $Y_i := X_i^2 \in [0,1]$ are independent with $\mathrm{Var}(Y_i) \leq \mathbb{E}[Y_i^2] \leq \mathbb{E}[Y_i]$, so the total variance $v := \sum_i \mathrm{Var}(Y_i)$ satisfies $v \leq \mu$. Bernstein's inequality applied to $S - \mu$ at deviation $u = \mu + t$ gives $\exp(-u^2/(2(v + u/3)))$, and the algebra $\frac{(\mu+t)^2}{2(\mu+(\mu+t)/3)} \geq \frac{3}{8}(\mu + t)$ reduces exactly to $t \geq 0$. That hypothesis is missing from the printed statement and cannot be dropped: a binomial counterexample shows the bound fails for some $t < 0$.

Solution. Let $S = \sum_{i=1}^n X_i^2$ and $\mu = \mathbb{E}[S] \geq 0$. **Erratum.** As printed (with no restriction on t) the claim is false: it needs the hypothesis $t \geq 0$; see the counterexample at the end. We prove it for $t \geq 0$. If $\mu + t = 0$, i.e. $\mu = t = 0$, then $S = 0$ a.s. and both sides equal 1; assume $\mu + t > 0$. Since both sides are nonnegative,

$$\left\{\sqrt{S} \geq \sqrt{2\mu + t}\right\} = \{S \geq 2\mu + t\} = \{S - \mu \geq \mu + t\}.$$

Put $Y_i = X_i^2 \in [0,1]$, independent, and $u = \mu + t > 0$. As $X_i \in [0,1]$,

$$\mathrm{Var}(Y_i) \leq \mathbb{E}[Y_i^2] = \mathbb{E}[X_i^4] \leq \mathbb{E}[X_i^2] = \mathbb{E}[Y_i],$$

so with $v := \sum_{i=1}^n \mathrm{Var}(Y_i)$ we have $v \leq \mu$. If $\mu = 0$ then $S = 0$ a.s. and, as $u = t > 0$, the probability is 0; so assume $\mu > 0$. Put $Z_i = Y_i - \mathbb{E}Y_i$, so $|Z_i| \leq 1$ and $\sum_i \mathbb{E}Z_i^2 = v \leq \mu$. For $q \geq 3$, $\mathbb{E}(Z_i)_+^q \leq \mathbb{E}Z_i^2 \leq \frac{q!}{2} 3^{-(q-2)} \mathbb{E}Z_i^2$

because $q!/(2\cdot 3^{q-2}) \geq 1$. Hence the hypotheses of Theorem 2.10 hold with the positive constant μ in place of v and $c = 1/3$, and Corollary 2.11 gives

$$\mathbb{P}\{S - \mu \geq u\} \leq \exp\left(-\frac{u^2}{2(\mu + u/3)}\right) = \exp\left(-\frac{(\mu+t)^2}{2(\mu + (\mu+t)/3)}\right).$$

It remains to check $\frac{(\mu+t)^2}{2(\mu+(\mu+t)/3)} \geq \frac{3}{8}(\mu+t)$. Dividing by $\mu+t > 0$ and clearing denominators, this is equivalent to $8(\mu+t) \geq 6\mu + 2(\mu+t)$, i.e. $6t \geq 0$, which holds since $t \geq 0$. Hence

$$\mathbb{P}\left\{\sqrt{\sum_{i=1}^n X_i^2} \geq \sqrt{2\mathbb{E}\sum_{i=1}^n X_i^2 + t}\right\} \leq e^{-(3/8)(\mu+t)},$$

as claimed.

Counterexample for $t < 0$ (the printed claim is false without $t \geq 0$). The only place $t \geq 0$ was used is the final step $6t \geq 0$, and the claim genuinely fails for some negative t. Take $X_i \sim \text{Bernoulli}(0.02)$ i.i.d., $n = 10^4$. Then $X_i^2 = X_i$, $S = \sum_i X_i^2 \sim \text{Bin}(10^4, 0.02)$ and $\mu = \mathbb{E}S = 200$. For $t = -100$ the event is $\{\sqrt{S} \geq \sqrt{300}\} = \{S \geq 300\}$, and an exact binomial tail computation gives

$$\mathbb{P}\{S \geq 300\} \approx 1.62 \times 10^{-11} > 5.18 \times 10^{-17} \approx e^{-37.5} = e^{-(3/8)(200-100)}.$$

Likewise for $t = -25$: $\mathbb{P}\{S \geq 375\} \approx 3.85 \times 10^{-29} > 3.16 \times 10^{-29} \approx e^{-(3/8)\cdot 175}$. (For $t = -10$ and $t = -1$ the inequality happens to hold in this example, so the failure is not for every $t < 0$; and for $t \leq -\mu$ the right-hand side is ≥ 1, so the bound is trivially true there.) Hence the correct statement is: for all $t \geq 0$, $\mathbb{P}\{\sqrt{\sum_i X_i^2} \geq \sqrt{2\mathbb{E}\sum_i X_i^2 + t}\} \leq e^{-(3/8)(\mathbb{E}\sum_i X_i^2 + t)}$, which is what was proved above. (Numbers checked with exact binomial tails in `external_audit_2026-10-30/compare/repair_checks/check_07_17_16.py`.) ■

Chapter 8 — The Transportation Method

Exercise 8.1

Central idea. Apply Marton's transportation inequality (Theorem 8.2) to the conditional measures $\mathbb{Q}_A = \mathbb{P}(\cdot \mid A)$ and $\mathbb{Q}_B = \mathbb{P}(\cdot \mid B)$, for which $D(\mathbb{Q}_A\|\mathbb{P}) = \log(1/\mathbb{P}(A))$. Glue the two optimal couplings through a common $\mathbb{P}$-marginal X; then $d_H(A,B) \leq \mathbb{E}\, d_H(X,Y) + \mathbb{E}\, d_H(X,Z)$ by the triangle inequality, and Cauchy–Schwarz turns each expected Hamming distance into $\sqrt{n}$ times the root of Marton's cost. Taking $B = A_t^c$ gives $d_H(A,B) > t$, which rearranges into Marton's blowup lemma.

Solution. Assume $\mathbb{P}(A), \mathbb{P}(B) > 0$, else the right-hand side is $+\infty$. Set $\mathbb{Q}_A(\cdot) = \mathbb{P}(\cdot \mid A)$ and $\mathbb{Q}_B(\cdot) = \mathbb{P}(\cdot \mid B)$. Then

$$D(\mathbb{Q}_A\|\mathbb{P}) = \mathbb{E}_{\mathbb{Q}_A}\left[\log \frac{\mathbb{1}_A}{\mathbb{P}(A)}\right] = \log\frac{1}{\mathbb{P}(A)},$$

and similarly $D(\mathbb{Q}_B\|\mathbb{P}) = \log(1/\mathbb{P}(B))$.

Let $P_A \in \mathcal{P}(\mathbb{P}, \mathbb{Q}_A)$ and $P_B \in \mathcal{P}(\mathbb{P}, \mathbb{Q}_B)$ attain the minima in Theorem 8.2. Glue them: let (X, Y, Z) with $X \sim \mathbb{P}$ and, conditionally on X, $Y \sim P_A(\cdot \mid X)$ and $Z \sim P_B(\cdot \mid X)$ independent. Then $(X,Y) \sim P_A$, $(X,Z) \sim P_B$, so $Y \sim \mathbb{Q}_A$ (hence $Y \in A$ a.s.) and $Z \sim \mathbb{Q}_B$ (hence $Z \in B$ a.s.).

By the triangle inequality for Hamming distance, a.s.

$$d_H(A,B) \leq d_H(Y,Z) \leq d_H(X,Y) + d_H(X,Z).$$

Taking expectations and writing $d_H(X,Y) = \sum_i \mathbb{1}_{\{X_i \neq Y_i\}}$,

$$d_H(A,B) \leq \sum_{i=1}^n P_A\{X_i \neq Y_i\} + \sum_{i=1}^n P_B\{X_i \neq Z_i\}.$$

By Cauchy–Schwarz, $\sum_i P_A\{X_i \neq Y_i\} \le \sqrt{n \sum_i P_A\{X_i \neq Y_i\}^2}$, and Theorem 8.2 gives $\sum_i P_A\{X_i \neq Y_i\}^2 \le \frac{1}{2} D(\mathbb{Q}_A \| \mathbb{P}) = \frac{1}{2}\log(1/\mathbb{P}(A))$. Hence

$$\sum_{i=1}^n P_A\{X_i \neq Y_i\} \le \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(A)}},$$

and likewise for B. Adding yields

$$d_H(A,B) \le \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(A)}} + \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(B)}}.$$

Now take $B = A_t^c$, where $A_t = \{x : d_H(x,A) \le t\}$ is the t-blowup of A. For $x \in A$ and $y \in B$, $d_H(x,y) > t$ (else $y \in A_t$), so $d_H(A,B) > t$. If $\mathbb{P}(B) = 0$ then $\mathbb{P}(A_t) = 1$ and there is nothing to prove; else, with $a = \sqrt{\frac{n}{2}\log(1/\mathbb{P}(A))}$,

$$t < a + \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(B)}}.$$

For $t > a$ this gives $\frac{n}{2}\log(1/\mathbb{P}(B)) > (t-a)^2$, i.e. $\mathbb{P}(B) < \exp(-\frac{2}{n}(t-a)^2)$. Since $\mathbb{P}(B) = 1 - \mathbb{P}(A_t)$,

$$\mathbb{P}(A_t) > 1 - \exp\left(-\frac{2}{n}\left(t - \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(A)}}\right)^2\right), \qquad t > \sqrt{\frac{n}{2}\log\frac{1}{\mathbb{P}(A)}},$$

which is Marton's blowup lemma. In particular, if $\mathbb{P}(A) \ge 1/2$ then $\mathbb{P}(A_t) \ge 1 - \exp(-\frac{2}{n}(t - \sqrt{\frac{n}{2}\log 2})^2)$ for $t > \sqrt{\frac{n}{2}\log 2}$. ■

Exercise 8.2

Central idea. With $g = -f$, the key property reads $g(y) \le g(x) + \sum_i 2x_i \mathbb{1}_{\{x_i \neq y_i\}} + 1$. Repeating the proof of Theorem 8.6 but keeping the "+1" and using Marton's conditional transportation inequality (Theorem 8.5, dropping the Y-conditional term) gives $\mathbb{E}_{\mathbb{Q}} g - \mathbb{E}_{\mathbb{P}} g \le \sqrt{2vD(\mathbb{Q}\|\mathbb{P})} + 1$ with $v = 4\sum_i \mathbb{E}X_i^2$. Applying this to $\mathbb{Q} = \mathbb{P}(\cdot \mid \{g > \mathbb{E}g + t\})$ — the standard Lemma 4.18 deduction — turns the transportation bound into the tail bound, and $g = -f$ converts the upper tail of g into the left tail of Z.

Solution. Let $\mathbb{P}$ be the distribution of $X = (X_1, \dots, X_n)$ and $\mathbb{Q} \ll \mathbb{P}$. Put $g = -f$, so the key property (Section 7.6) becomes

$$g(y) \le g(x) + \sum_{i=1}^n c_i(x)\, \mathbb{1}_{\{x_i \neq y_i\}} + 1, \qquad c_i(x) = 2x_i.$$

Let $P \in \mathcal{P}(\mathbb{P}, \mathbb{Q})$ with $(X,Y) \sim P$, $X \sim \mathbb{P}$, $Y \sim \mathbb{Q}$. Then

$$\mathbb{E}_{\mathbb{Q}} g - \mathbb{E}_{\mathbb{P}} g = \mathbb{E}_P[g(Y) - g(X)] \le \mathbb{E}_P\left[\sum_{i=1}^n c_i(X) \mathbb{1}_{\{X_i \neq Y_i\}}\right] + 1.$$

Conditioning on X and applying Cauchy–Schwarz twice,

$$\begin{aligned}
\mathbb{E}_P\left[\sum_i c_i(X) \mathbb{1}_{\{X_i \neq Y_i\}}\right] &= \mathbb{E}_{\mathbb{P}}\left[\sum_i c_i(X)\, P\{X_i \neq Y_i \mid X\}\right] \\
&\le \mathbb{E}_{\mathbb{P}}\left[\sqrt{\sum_i c_i(X)^2}\,\sqrt{\sum_i P\{X_i \neq Y_i \mid X\}^2}\right] \\
&\le \sqrt{\mathbb{E}_{\mathbb{P}} \sum_i c_i(X)^2}\,\sqrt{\mathbb{E}_P \sum_i P\{X_i \neq Y_i \mid X\}^2}.
\end{aligned}$$

With $v = \mathbb{E}_{\mathbb{P}} \sum_i c_i(X)^2 = 4\sum_{i=1}^n \mathbb{E}X_i^2$ and Theorem 8.5 (dropping the nonnegative Y-conditional term, so $\min_P \mathbb{E}_P \sum_i P\{X_i \neq Y_i \mid X\}^2 \leq 2D(\mathbb{Q}\|\mathbb{P})$), minimizing over P gives

$$\mathbb{E}_{\mathbb{Q}} g - \mathbb{E}_{\mathbb{P}} g \leq \sqrt{2v\, D(\mathbb{Q}\|\mathbb{P})} + 1 \qquad \text{for all } \mathbb{Q} \ll \mathbb{P}.$$

Now fix $t > 0$ and put $p = \mathbb{P}\{g > \mathbb{E}g + t\}$; if $p = 0$ there is nothing to prove. Take $\mathbb{Q}$ with $d\mathbb{Q}/d\mathbb{P} = \mathbb{1}_{\{g > \mathbb{E}g + t\}}/p$, so $\mathbb{Q} \ll \mathbb{P}$, $\mathbb{E}_{\mathbb{Q}} g - \mathbb{E}g \geq t$, and $D(\mathbb{Q}\|\mathbb{P}) = \log(1/p)$. The transportation bound gives

$$t \leq \sqrt{2v \log \frac{1}{p}} + 1.$$

For $t > 1$ this yields $\log(1/p) \geq (t-1)^2/(2v)$, i.e.

$$\mathbb{P}\{g > \mathbb{E}g + t\} \leq \exp\left(-\frac{(t-1)^2}{2v}\right), \qquad t > 1.$$

Since $g = -f$ and $Z = f(X)$, $\{g > \mathbb{E}g + t\} = \{Z < \mathbb{E}Z - t\}$, and with $v = 4\sum_i \mathbb{E}X_i^2$,

$$\mathbb{P}\{Z < \mathbb{E}Z - t\} \leq \exp\left(-\frac{(t-1)^2}{8\sum_{i=1}^n \mathbb{E}X_i^2}\right), \qquad t > 1.$$

■

Erratum. An earlier version of this solution stated the bound for all $t > 0$, with the parenthetical claim that it was "trivial for $t \leq 1$ as the right-hand side exceeds 1". Both are false. The right-hand side is $\exp(-(t-1)^2/(8\sum_i \mathbb{E}X_i^2)) \leq 1$ for every t (the exponent is nonpositive, vanishing only at $t = 1$), so it never exceeds 1; and the inequality itself can fail for $t < 1$ (at $t = 1$ the right-hand side equals 1, so the bound holds trivially there). Counterexample: $n = 10$, X_i i.i.d. Bernoulli(0.01), $Z = \sum_{i=1}^n X_i$, which satisfies the key property (for $x_i, y_i \in \{0,1\}$, $x_i - y_i \leq 2x_i \mathbf{1}_{\{x_i \neq y_i\}}$ holds termwise: $-1 \leq 0$, $0 \leq 0$, $1 \leq 2$). Here $\mathbb{E}Z = 0.1$ and $\sum_i \mathbb{E}X_i^2 = 0.1$, and at $t = 0.05$,

$$\mathbb{P}\{Z < \mathbb{E}Z - t\} = \mathbb{P}\{Z = 0\} = 0.99^{10} \approx 0.9044 > 0.3236 \approx \exp\left(-\frac{(0.05-1)^2}{8 \cdot 0.1}\right).$$

The derivation above establishes the bound only for $t > 1$, which is the corrected statement (presumably the book's Example 8.9 as intended).

Exercise 8.3

Central idea. Put $s = \sqrt{y}$, $r = \sqrt{x}$ and complete the square: $s^2 - as \leq r^2$ becomes $(s - a/2)^2 \leq r^2 + a^2/4$. If $s \leq a/2$ the claim is immediate; else $s \leq a/2 + \sqrt{r^2 + a^2/4}$, and squaring plus the elementary estimate $\sqrt{r^2 + a^2/4} \leq r + a/2$ gives $s^2 \leq r^2 + ar + a^2$.

Solution. Let $s = \sqrt{y} > 0$ and $r = \sqrt{x} > 0$. The hypothesis is $s^2 - as \leq r^2$. Completing the square,

$$\left(s - \frac{a}{2}\right)^2 = s^2 - as + \frac{a^2}{4} \leq r^2 + \frac{a^2}{4}.$$

If $s \leq a/2$, then $s^2 \leq a^2/4 \leq r^2 + ar + a^2$ since $r^2 + ar \geq 0$. Otherwise $s > a/2$ and taking square roots gives $s - a/2 \leq \sqrt{r^2 + a^2/4}$, i.e.

$$s \leq \frac{a}{2} + \sqrt{r^2 + \frac{a^2}{4}}.$$

Squaring,

$$s^2 \leq \frac{a^2}{4} + a\sqrt{r^2 + \frac{a^2}{4}} + r^2 + \frac{a^2}{4} = r^2 + \frac{a^2}{2} + a\sqrt{r^2 + \frac{a^2}{4}}.$$

Since $\sqrt{r^2 + a^2/4} \le r + a/2$ (both sides nonnegative; squaring gives $r^2 + a^2/4 \le r^2 + ar + a^2/4$),

$$s^2 \le r^2 + \frac{a^2}{2} + a\left(r + \frac{a}{2}\right) = r^2 + ar + a^2.$$

That is, $y \le x + a\sqrt{x} + a^2$. ■

Exercise 8.4

Central idea. The layer-cake identity $|x-y| = \int_{-\infty}^{\infty} |\mathbf{1}_{\{x>t\}} - \mathbf{1}_{\{y>t\}}|\, dt$ turns $\mathbb{E}|X-Y|$ into an integral over t of $\mathbb{P}\{X>t\} + \mathbb{P}\{Y>t\} - 2\mathbb{P}\{X>t, Y>t\}$; the marginals fix the first two terms, so minimizing amounts to maximizing $\mathbb{P}\{X>t, Y>t\}$ pointwise, which the Fréchet–Hoechding upper bound $\min(\mathbb{P}\{X>t\}, \mathbb{P}\{Y>t\})$ caps — and the comonotonic coupling $X = F^{-1}(U)$, $Y = G^{-1}(U)$ attains it for every t simultaneously. (Note: the hint's formula has a sign error; the correct identity is $|X-Y| = 2\max(X,Y) - X - Y$, or equivalently $|X-Y| = X + Y - 2\min(X,Y)$.)

Solution. For real numbers x, y,

$$|x-y| = \int_{-\infty}^{\infty} \left|\mathbf{1}_{\{x>t\}} - \mathbf{1}_{\{y>t\}}\right| dt,$$

since the integrand equals 1 exactly when t lies strictly between x and y (an interval of length $|x-y|$), and 0 otherwise. By Tonelli's theorem (nonnegative integrand, no integrability needed),

$$\mathbb{E}|X-Y| = \int_{-\infty}^{\infty} \mathbb{E}\left|\mathbf{1}_{\{X>t\}} - \mathbf{1}_{\{Y>t\}}\right| dt.$$

For each fixed t,

$$\mathbb{E}\left|\mathbf{1}_{\{X>t\}} - \mathbf{1}_{\{Y>t\}}\right| = \mathbb{P}\{X>t\} + \mathbb{P}\{Y>t\} - 2\,\mathbb{P}\{X>t,\, Y>t\}.$$

The marginal constraints $X \sim F$, $Y \sim G$ fix $\mathbb{P}\{X>t\} = 1 - F(t)$ and $\mathbb{P}\{Y>t\} = 1 - G(t)$. Hence, for each t, minimizing the integrand over all couplings is equivalent to maximizing $\mathbb{P}\{X>t, Y>t\}$. By the Fréchet–Hoeffding bound,

$$\mathbb{P}\{X>t,\, Y>t\} \le \min\left(\mathbb{P}\{X>t\}, \mathbb{P}\{Y>t\}\right) = \min\left(1-F(t),\, 1-G(t)\right).$$

The comonotonic coupling attains this bound for every t simultaneously: with U uniform on $[0,1]$, $X = F^{-1}(U)$, $Y = G^{-1}(U)$, and using $F^{-1}(s) > x \iff s > F(x)$,

$$\mathbb{P}\{F^{-1}(U) > t,\, G^{-1}(U) > t\} = \mathbb{P}\{U > \max(F(t), G(t))\} = 1 - \max(F(t), G(t)),$$

which equals $\min(1-F(t), 1-G(t))$. Therefore the comonotonic coupling minimizes $\mathbb{E}|X-Y|$ over all couplings with marginals F, G (the argument is valid even when $\mathbb{E}|X| = \infty$, with both sides possibly $+\infty$).

The minimal value is

$$\mathbb{E}\left|F^{-1}(U) - G^{-1}(U)\right| = \int_0^1 \left|F^{-1}(t) - G^{-1}(t)\right| dt,$$

by the law of the unconscious statistician for $U \sim \text{Uniform}[0,1]$. ■

Exercise 8.5

Central idea. Apply the layer-cake identity $|a-b| = \int_{-\infty}^{\infty} |\mathbf{1}_{\{a>x\}} - \mathbf{1}_{\{b>x\}}|\, dx$ to $a = F^{-1}(t)$, $b = G^{-1}(t)$, swap the integrals by Tonelli, and use the generalized-inverse duality $F^{-1}(t) > x \iff t > F(x)$ to turn the inner t-integral into the measure of t strictly between $F(x)$ and $G(x)$, namely $|F(x) - G(x)|$.

Solution. For real a, b,

$$|a-b| = \int_{-\infty}^{\infty} \left|\mathbf{1}_{\{a>x\}} - \mathbf{1}_{\{b>x\}}\right| dx,$$

the integrand being 1 exactly on the interval between a and b. Hence, with F^{-1}, G^{-1} the generalized inverses,

$$\int_0^1 \left|F^{-1}(t) - G^{-1}(t)\right| dt = \int_0^1 \int_{-\infty}^{\infty} \left|\mathbf{1}_{\{F^{-1}(t)>x\}} - \mathbf{1}_{\{G^{-1}(t)>x\}}\right| dx\, dt.$$

By Tonelli's theorem (nonnegative integrand; both sides lie in $[0,\infty]$),

$$= \int_{-\infty}^{\infty} \int_0^1 \left|\mathbf{1}_{\{F^{-1}(t)>x\}} - \mathbf{1}_{\{G^{-1}(t)>x\}}\right| dt\, dx.$$

For the generalized inverse $F^{-1}(t) = \inf\{y : F(y) \geq t\}$ and fixed x,

$$F^{-1}(t) > x \iff t > F(x) :$$

indeed, if $\inf\{y : F(y) \geq t\} > x$ then $x \notin \{y : F(y) \geq t\}$, so $F(x) < t$; conversely, if $F(x) < t$, right-continuity of F gives $\delta > 0$ with $F < t$ on $[x, x+\delta)$, so the infimum exceeds x. Thus $\mathbf{1}_{\{F^{-1}(t)>x\}} = \mathbf{1}_{\{t>F(x)\}}$, and similarly for G. The inner integral becomes, for fixed x with $F(x), G(x) \in [0,1]$,

$$\int_0^1 \left|\mathbf{1}_{\{t>F(x)\}} - \mathbf{1}_{\{t>G(x)\}}\right| dt,$$

whose integrand is 1 exactly when t lies strictly between $F(x)$ and $G(x)$ — a subset of $[0,1]$ of Lebesgue measure $|F(x) - G(x)|$. Therefore

$$\int_0^1 \left|F^{-1}(t) - G^{-1}(t)\right| dt = \int_{-\infty}^{\infty} |F(x) - G(x)|\, dx. \blacksquare$$

Exercise 8.6

Central idea. Hoeffding's covariance identity writes $\mathrm{Cov}(X,Y)$ as a double integral of covariances of indicators $\mathbf{1}_{\{X>u\}}, \mathbf{1}_{\{Y>v\}}$; each is bounded by the strong mixing coefficient α and by the marginal tails (Fréchet–Hoeffding). The layer-cake identity $\min(c,a,b) = \int_0^c \mathbf{1}_{\{t<a\}}\mathbf{1}_{\{t<b\}}\, dt$ plus Tonelli converts the $\min(\alpha,\cdot,\cdot)$ double integral into the quantile integral $\int_0^\alpha F^{-1}(1-t)G^{-1}(1-t)\, dt$. For signed variables, split $X = X^+ - X^-$, $Y = Y^+ - Y^-$ and pair diagonal-with-diagonal and off-diagonal-with-off-diagonal; the elementary bound $\min(\alpha,a,b)+\min(\alpha,c,d) \leq \min(2\alpha, a+c, b+d)$ upgrades α to 2α while $\mathbb{P}\{X>u\}+\mathbb{P}\{X<-u\} \leq \mathbb{P}\{|X|>u\}$ folds both tails into the quantiles of $|X|, |Y|$.

Solution. Let $\bar{F}_X(u) = \mathbb{P}\{X > u\}$. For Borel A, B,

$$\left|\mathrm{Cov}(\mathbf{1}_A, \mathbf{1}_B)\right| \leq \min\left(\mathbb{P}(A), \mathbb{P}(B)\right), \tag{*}$$

since $\mathrm{Cov}(\mathbf{1}_A, \mathbf{1}_B) = \mathbb{P}(A\cap B) - \mathbb{P}(A)\mathbb{P}(B)$ lies between $-\mathbb{P}(A)\mathbb{P}(B)$ and $\mathbb{P}(A\cap B)$, both bounded in absolute value by $\min(\mathbb{P}(A), \mathbb{P}(B))$.

Fréchet's inequality. Let $X, Y \geq 0$ be square-integrable. By Tonelli,

$$\mathbb{E}[XY] = \int_0^\infty \int_0^\infty \mathbb{P}\{X > u,\, Y > v\}\, du\, dv \leq \int_0^\infty \int_0^\infty \min\left(\bar{F}_X(u), \bar{G}_Y(v)\right) du\, dv$$

by Fréchet–Hoeffding. For U uniform on $[0,1]$, using $F^{-1}(t) > x \iff t > F(x)$,

$$\mathbb{P}\{F^{-1}(U) > u,\, G^{-1}(U) > v\} = 1 - \max(F(u), G(v)) = \min\left(\bar{F}_X(u), \bar{G}_Y(v)\right),$$

so the right-hand side equals $\mathbb{E}[F^{-1}(U)G^{-1}(U)] = \int_0^1 F^{-1}(t)G^{-1}(t)\, dt$ (finite, $\leq \|X\|_2\|Y\|_2$). Since $\mathbb{E}X = \int_0^1 F^{-1}$ and $\mathbb{E}Y = \int_0^1 G^{-1}$ are fixed by the marginals,

$$\mathrm{Cov}(X,Y) \leq \int_0^1 F^{-1}(t)G^{-1}(t)\, dt - \int_0^1 F^{-1}(t)\, dt \int_0^1 G^{-1}(t)\, dt,$$

the covariance under the comonotonic coupling.

Rio's inequality, $X, Y \geq 0$. Hoeffding's identity (Tonelli, using $\mathbb{E}[XY] = \iint \mathbb{P}\{X > u, Y > v\}$ and $\mathbb{E}X\,\mathbb{E}Y = \iint \mathbb{P}\{X > u\}\mathbb{P}\{Y > v\}$):

$$\operatorname{Cov}(X, Y) = \int_0^\infty \int_0^\infty \operatorname{Cov}(\mathbf{1}_{\{X>u\}}, \mathbf{1}_{\{Y>v\}})\, du\, dv.$$

By $(*)$ and the definition of α (with $A = (u, \infty)$, $B = (v, \infty)$),

$$\big|\operatorname{Cov}(\mathbf{1}_{\{X>u\}}, \mathbf{1}_{\{Y>v\}})\big| \leq \min\big(\alpha,\, \mathbb{P}\{X > u\},\, \mathbb{P}\{Y > v\}\big),$$

hence

$$|\operatorname{Cov}(X, Y)| \leq \int_0^\infty \int_0^\infty \min\big(\alpha,\, \mathbb{P}\{X > u\},\, \mathbb{P}\{Y > v\}\big)\, du\, dv. \tag{1}$$

Quantile form. For $c, a, b \geq 0$, $\min(c, a, b) = \int_0^c \mathbf{1}_{\{t<a\}} \mathbf{1}_{\{t<b\}}\, dt$. With $\bar{F}(u) = \mathbb{P}\{X > u\}$, Tonelli gives

$$\iint_{u,v>0} \min(\alpha, \bar{F}(u), \bar{G}(v))\, du\, dv = \int_0^\alpha \Big[\int_0^\infty \mathbf{1}_{\{F(u)<1-t\}}\, du\Big]\Big[\int_0^\infty \mathbf{1}_{\{G(v)<1-t\}}\, dv\Big]\, dt.$$

Since $X \geq 0$, $\{u \geq 0 : F(u) < 1-t\} = [0, F^{-1}(1-t))$ up to a null set, so the bracket equals $F^{-1}(1-t)$. Thus

$$|\operatorname{Cov}(X, Y)| \leq \int_0^\alpha F^{-1}(1-t)\, G^{-1}(1-t)\, dt. \tag{2}$$

Signed X, Y. Now let F, G be the CDFs of $|X|, |Y|$, and write $X = X^+ - X^-$, $Y = Y^+ - Y^-$. Then

$$\operatorname{Cov}(X, Y) = \operatorname{Cov}(X^+, Y^+) + \operatorname{Cov}(X^-, Y^-) - \operatorname{Cov}(X^+, Y^-) - \operatorname{Cov}(X^-, Y^+).$$

For $u > 0$, $\{X^+ > u\} = \{X > u\}$ and $\{X^- > u\} = \{X < -u\}$, so Hoeffding's identity gives

$$\operatorname{Cov}(X^+, Y^+) = \iint_{u,v>0} \operatorname{Cov}(\mathbf{1}_{\{X>u\}}, \mathbf{1}_{\{Y>v\}})\, du\, dv,$$
$$\operatorname{Cov}(X^-, Y^-) = \iint_{u,v>0} \operatorname{Cov}(\mathbf{1}_{\{X<-u\}}, \mathbf{1}_{\{Y<-v\}})\, du\, dv,$$

and similarly for the off-diagonal pairs with $\{Y^- > v\} = \{Y < -v\}$. Pairing diagonal with diagonal,

$$\begin{aligned}&|\operatorname{Cov}(X^+, Y^+)| + |\operatorname{Cov}(X^-, Y^-)| \\ &\quad\leq \iint_{u,v>0} \Big[\min(\alpha, \mathbb{P}\{X > u\}, \mathbb{P}\{Y > v\}) + \min(\alpha, \mathbb{P}\{X < -u\}, \mathbb{P}\{Y < -v\})\Big]\, du\, dv.\end{aligned}$$

The elementary bound $\min(\alpha, a, b) + \min(\alpha, c, d) \leq \min(2\alpha, a + c, b + d)$, together with $\mathbb{P}\{X > u\} + \mathbb{P}\{X < -u\} \leq \mathbb{P}\{|X| > u\} =: \bar{F}(u)$ (disjoint union) and likewise for Y, yields

$$|\operatorname{Cov}(X^+, Y^+)| + |\operatorname{Cov}(X^-, Y^-)| \leq \iint_{u,v>0} \min\big(2\alpha, \bar{F}(u), \bar{G}(v)\big)\, du\, dv.$$

The same argument for the off-diagonal pair gives

$$|\operatorname{Cov}(X^+, Y^-)| + |\operatorname{Cov}(X^-, Y^+)| \leq \iint_{u,v>0} \min\big(2\alpha, \bar{F}(u), \bar{G}(v)\big)\, du\, dv.$$

By the Tonelli computation above with 2α in place of α (valid since $\alpha \leq \frac{1}{4}$, so $1 - t \in (0, 1]$ on $[0, 2\alpha]$),

$$\iint_{u,v>0} \min\big(2\alpha, \bar{F}(u), \bar{G}(v)\big)\, du\, dv = \int_0^{2\alpha} F^{-1}(1-t)\, G^{-1}(1-t)\, dt.$$

Consequently

$$|\operatorname{Cov}(X, Y)| \leq 2 \int_0^{2\alpha} F^{-1}(1-t)\, G^{-1}(1-t)\, dt. \ \blacksquare$$

Chapter 9 — Influences and Threshold Phenomena

Exercise 9.1

Central idea. On the symmetric difference $A \triangle B$, the binary-valued f and the thresholded indicator $\mathbb{1}_{\{g \geq 1/2\}}$ disagree, which forces $|f - g| \geq 1/2$: if $f(x) = 1$ then $g(x) < 1/2$, and if $f(x) = 0$ then $g(x) \geq 1/2$. Hence $\mathbb{1}_{A \triangle B} \leq 4(f-g)^2$ pointwise, and taking expectations gives the claim.

Solution. Write $A \triangle B = (A \setminus B) \cup (B \setminus A)$, where $A = \{x : f(x) = 1\}$ and $B = \{x : g(x) \geq 1/2\}$, so $A \triangle B = \{x : f(x) \neq \mathbb{1}_{\{g(x) \geq 1/2\}}\}$.

If $x \in A \setminus B$, then $f(x) = 1$ and $g(x) < 1/2$, so

$$(f(x) - g(x))^2 = (1 - g(x))^2 > \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}.$$

If $x \in B \setminus A$, then $f(x) = 0$ and $g(x) \geq 1/2$, so

$$(f(x) - g(x))^2 = g(x)^2 \geq \tfrac{1}{4}.$$

Thus $\mathbb{1}_{A \triangle B}(x) \leq 4(f(x) - g(x))^2$ for every $x \in \{-1, 1\}^n$. Taking expectations over the uniform measure,

$$\mathbb{P}(A \triangle B) \leq 4\, \mathbb{E}[(f-g)^2]. \qquad \blacksquare$$

Exercise 9.2

Central idea. For binary-valued f, Theorem 9.9 simplifies: $\mathbb{E}[(f(X) - f(X^{(i)}))^2] = I_i(f)$ and $\|g_i\|_2 / \|g_i\|_1 = 1/\sqrt{I_i(f)}$, giving $\mathrm{Var}(f) \leq C \sum_i I_i / (1 + \frac{1}{2}\log(1/I_i))$. The KKL bound $\sum_i I_i^2 \gtrsim \mathrm{Var}(f)^2 \log^2 n / n$ follows by Cauchy–Schwarz and contradiction: if $\sum_i I_i^2$ were too small, every I_i would be tiny, making the logarithmic denominators large and contradicting the variance lower bound.

Solution. Let $f : \{-1, 1\}^n \to \{0, 1\}$ be binary-valued. If $\mathrm{Var}(f) = 0$ there is nothing to prove; assume $\mathrm{Var}(f) > 0$. For $n = 1$ the claim is trivial since $\log 1 = 0$; assume $n \geq 2$.

For binary f, $(f(X) - f(X^{(i)}))^2 = \mathbb{1}_{\{f(X) \neq f(X^{(i)})\}}$, so $\mathbb{E}[(f(X) - f(X^{(i)}))^2] = I_i(f)$. Moreover, with $g_i = (f - f^{(i)})/2$, $\|g_i\|_2^2 = I_i(f)/4$ and $\|g_i\|_1 = I_i(f)/2$ (since $|f - f^{(i)}| = \mathbb{1}_{\{f \neq f^{(i)}\}}$), hence $\|g_i\|_2 / \|g_i\|_1 = 1/\sqrt{I_i(f)}$ whenever $I_i(f) > 0$ (the summand is 0 when $I_i(f) = 0$). Theorem 9.9 therefore gives, with a universal constant $C > 0$,

$$\mathrm{Var}(f) \leq C \sum_{i : I_i > 0} \frac{I_i}{1 + \frac{1}{2}\log(1/I_i)}. \tag{$*$}$$

We show $\sum_i I_i^2 \geq \frac{1}{K} \frac{\mathrm{Var}(f)^2 \log^2 n}{n}$ with $K = \max(64C^2, 1)$. Suppose, for contradiction,

$$\sum_i I_i^2 < \frac{\mathrm{Var}(f)^2 \log^2 n}{Kn}.$$

Then $I_i < \epsilon := \frac{\mathrm{Var}(f) \log n}{\sqrt{Kn}}$ for every i. By Cauchy–Schwarz,

$$\left(\sum_{i : I_i > 0} \frac{I_i}{1 + \frac{1}{2}\log(1/I_i)} \right)^2 \leq \left(\sum_{i : I_i > 0} I_i^2 \right) \left(\sum_{i : I_i > 0} \frac{1}{\left(1 + \frac{1}{2}\log(1/I_i)\right)^2} \right).$$

Since $x \mapsto \left(1 + \frac{1}{2}\log(1/x)\right)^{-2}$ is increasing on $(0, \infty)$ and $I_i \leq \epsilon$,

$$\sum_{i : I_i > 0} \frac{1}{\left(1 + \frac{1}{2}\log(1/I_i)\right)^2} \leq \frac{n}{\left(1 + \frac{1}{2}\log(1/\epsilon)\right)^2}.$$

Combining with $(*)$,

$$\frac{\mathrm{Var}(f)^2}{C^2} < \frac{\mathrm{Var}(f)^2 \log^2 n}{Kn} \cdot \frac{n}{\left(1 + \frac{1}{2}\log(1/\epsilon)\right)^2},$$

and cancelling $\mathrm{Var}(f)^2 > 0$ gives $1 + \frac{1}{2}\log(1/\epsilon) < \frac{C \log n}{\sqrt{K}}$, i.e.

$$\log(1/\epsilon) < \frac{2C \log n}{\sqrt{K}} - 2. \tag{1}$$

On the other hand, $\epsilon = \frac{\mathrm{Var}(f) \log n}{\sqrt{Kn}}$ and $\mathrm{Var}(f) \le 1/4$, so

$$\log(1/\epsilon) = \log \frac{\sqrt{Kn}}{\mathrm{Var}(f) \log n} \ge \log \frac{4\sqrt{Kn}}{\log n} = \tfrac{1}{2}\log(Kn) - \log\log n + \log 4. \tag{2}$$

From (1)–(2), $\frac{1}{2}\log K + (\frac{1}{2} - \frac{2C}{\sqrt{K}})\log n - \log\log n + \log 4 + 2 < 0$. Since $K \ge 64C^2$, $\frac{2C}{\sqrt{K}} \le \frac{1}{4}$, so $\frac{1}{2}\log K + \frac{1}{4}\log n - \log\log n + \log 4 + 2 < 0$. But $\frac{1}{4}\log n - \log\log n \ge -0.39$ for $n \ge 2$ (minimized at $n = e^4$), $\frac{1}{2}\log K \ge 0$, and $\log 4 + 2 > 3.38$, so the left side exceeds $3 > 0$, a contradiction.

Hence $\sum_{i=1}^n I_i(f)^2 \ge \frac{1}{K}\frac{\mathrm{Var}(f)^2 \log^2 n}{n}$, which is Theorem 9.5 up to the constant factor $1/K$. In particular, $\max_i I_i(f) \ge \frac{1}{\sqrt{K}}\frac{\mathrm{Var}(f)\log n}{n}$. ■

Exercise 9.3

Central idea. This is the Benjamini–Kalai–Schramm proof of Talagrand's L_1–L_2 inequality. Write $\mathrm{Var}(f)$ via the Fourier coefficients of the discrete derivatives $g_i = (f - f^{(i)})/2$, use $\frac{1}{|S|} \le \frac{3}{2|S|+1}$ to introduce the integral $\int_0^1 \gamma^{2|S|} d\gamma$, recognize the noise operator and apply Bonami–Beckner (Corollary 5.17) to get $\|g_i\|_{1+\gamma^2}^2$, then interpolate with Hölder and evaluate the resulting calculus integral. (Note: the hint's $\log \frac{\|g_i\|_1}{\|g_i\|_2}$ is a typo — see the remark at the end.)

Solution. Let $g_i(x) = \frac{f(x) - f(x^{(i)})}{2}$, so $\hat{g}_i(S) = \hat{f}(S)\mathbb{1}_{\{i \in S\}}$ and

$$\mathrm{Var}(f) = \sum_{S \ne \emptyset} \hat{f}(S)^2 = \sum_{i=1}^n \sum_{S \ne \emptyset} \frac{\hat{g}_i(S)^2}{|S|}.$$

For $|S| \ge 1$, $\frac{1}{|S|} \le \frac{3}{2|S|+1}$, and $\int_0^1 \gamma^{2|S|} d\gamma = \frac{1}{2|S|+1}$, so for each i,

$$\sum_{S \ne \emptyset} \frac{\hat{g}_i(S)^2}{|S|} \le 3 \sum_S \frac{\hat{g}_i(S)^2}{2|S|+1} = 3 \int_0^1 \sum_S \hat{g}_i(S)^2 \gamma^{2|S|} d\gamma.$$

Since $\sum_S \hat{g}_i(S)^2 \gamma^{2|S|} = \|T_\gamma g_i\|_2^2$ where T_γ is the noise operator, the Bonami–Beckner inequality (Corollary 5.17) gives $\|T_\gamma g_i\|_2 \le \|g_i\|_{1+\gamma^2}$, hence

$$\mathrm{Var}(f) \le 3 \sum_{i=1}^n \int_0^1 \|g_i\|_{1+\gamma^2}^2 \, d\gamma. \tag{1}$$

Fix i with $\|g_i\|_2 > 0$ (otherwise the summand is 0; note $\|g_i\|_2 > 0$ implies $\|g_i\|_1 > 0$). By Hölder (log-convexity of L^p norms), for $p = 1 + \gamma^2$,

$$\|g_i\|_{1+\gamma^2} \le \|g_i\|_1^{\frac{1-\gamma^2}{1+\gamma^2}} \|g_i\|_2^{\frac{2\gamma^2}{1+\gamma^2}},$$

so with $r_i = \|g_i\|_1 / \|g_i\|_2 \in (0, 1]$,

$$\int_0^1 \|g_i\|_{1+\gamma^2}^2 d\gamma \le \|g_i\|_2^2 \int_0^1 r_i^{\frac{2(1-\gamma^2)}{1+\gamma^2}} d\gamma. \tag{2}$$

Calculus lemma. For $t \geq 0$, $\int_0^1 e^{-2t\frac{1-\gamma^2}{1+\gamma^2}} d\gamma \leq \frac{6}{5(1+t)}$.

Proof. Since $\frac{1-\gamma^2}{1+\gamma^2} - (1-\gamma) = \frac{\gamma(1-\gamma)^2}{1+\gamma^2} \geq 0$ on $[0,1]$, the integral is at most $\int_0^1 e^{-2t(1-\gamma)} d\gamma = \frac{1-e^{-2t}}{2t}$ (and equals 1 at $t = 0$). For $t \leq 1/5$, $1 \leq \frac{6}{5(1+t)}$. For $t \geq 1/5$, $\frac{1-e^{-2t}}{2t} \leq \frac{6}{5(1+t)}$ is equivalent to $k(t) := 1 + t - e^{2t}(1 - 1.4t) \geq 0$; $k'(t) = 1 - e^{2t}(0.6 - 2.8t) > 0.4 > 0$ since $e^{2t}(0.6 - 2.8t)$ is strictly decreasing from 0.6, and $k(1/5) = 1.2 - 0.72e^{0.4} > 1.2 - 0.72 \cdot 1.5 > 0$. Hence $k \geq 0$ on $[1/5, \infty)$. ■

With $t_i = \log(1/r_i) = \log(\|g_i\|_2 / \|g_i\|_1) \geq 0$, the lemma gives

$$\int_0^1 r_i^{\frac{2(1-\gamma^2)}{1+\gamma^2}} d\gamma \leq \frac{6}{5(1 + \log \frac{\|g_i\|_2}{\|g_i\|_1})}.$$

Combined with (1)–(2),

$$\mathrm{Var}(f) \leq \frac{18}{5} \sum_{i=1}^n \frac{\|g_i\|_2^2}{1 + \log \frac{\|g_i\|_2}{\|g_i\|_1}}.$$

Since $\|g_i\|_2^2 = \frac{1}{4}\mathbb{E}[(f(X) - f(X^{(i)}))^2]$, this is Theorem 9.9 with $C = \frac{18}{5} \cdot \frac{1}{4} = \frac{9}{10}$. ■

Remark (typo in the hint). The hint writes the final denominator as $1 + \log \frac{\|g_i\|_1}{\|g_i\|_2}$. Since $\|g_i\|_1 \leq \|g_i\|_2$, this logarithm is ≤ 0, and for $\|g_i\|_1 / \|g_i\|_2 < 1/e$ the right-hand side would be negative while the left is positive — the inequality would be false. The correct form, consistent with the calculus above and with Theorem 9.9's proof, is $1 + \log \frac{\|g_i\|_2}{\|g_i\|_1}$.

Exercise 9.4

Central idea. Adapt the proof of Theorem 9.8 (truncation $g_{a,b}$ + Theorem 9.4) to the one-sided $(\cdot)_-$ sensitivity. The shift $a = Q_{1-\gamma} + B$ is the key: since single-coordinate changes are at most B, the negative part $(g_{a,b}(X) - g_{a,b}(X^{(i)}))_-$ can only be nonzero when $f(X) > Q_{1-\gamma}$, localizing both $\mathcal{E}(g_{a,b})$ and $\sum_j (\mathbb{E}|\Delta_j|)^2$ to a set of probability $\leq \gamma$. Theorem 9.4 then yields $A \log \frac{A}{4\gamma} \leq 1$ for $A = \frac{\delta(b-a)^2}{4v\gamma}$, whence $A \lesssim 1/\log(e^2/(2\gamma))$.

Solution. Let $0 < \delta < \gamma \leq 1/2$, $a = Q_{1-\gamma} + B$, $b = Q_{1-\delta}$, where $Q_\alpha = \inf\{z : \mathbb{P}\{f(X) \leq z\} \geq \alpha\}$ and $B = \max_{x,i} |f(x) - f(x^{(i)})|$. If $b \leq a$ then $Q_{1-\delta} - Q_{1-\gamma} \leq B$ trivially; assume $b > a$. Since $Q_{1-\delta} - Q_{1-\gamma} = (b-a) + B$, it suffices to bound $b - a$.

Define the truncation $g_{a,b}(x) = \max(a, \min(b, f(x)))$, nondecreasing and 1-Lipschitz in $f(x)$. Then $\mathrm{Var}(g_{a,b}) \geq \mathbb{P}\{f(X) \geq b\}(b-a)^2/4 \geq \delta(b-a)^2/4$, since $\mathbb{P}\{f \geq Q_{1-\delta}\} \geq \delta$ and $\mathbb{P}\{g_{a,b} = a\} \geq 1/2$.

Localization via the B-shift. For $\phi(s) = \max(a, \min(b, s))$, $(\phi(s) - \phi(t))_- \leq (s-t)_-$. If $f(x) \leq a - B = Q_{1-\gamma}$, then $f(x^{(i)}) \leq f(x) + B \leq a$ for all i, so $g_{a,b}(x) = g_{a,b}(x^{(i)}) = a$. Hence $(g_{a,b}(x) - g_{a,b}(x^{(i)}))_-$ is nonzero only if $f(x) > Q_{1-\gamma}$, and

$$(g_{a,b}(X) - g_{a,b}(X^{(i)}))_-^2 \leq \mathbb{1}_{\{f(X) > Q_{1-\gamma}\}} (f(X) - f(X^{(i)}))_-^2.$$

By the symmetry $(X, X^{(i)}) \overset{d}{=} (X^{(i)}, X)$, $\mathcal{E}(g_{a,b}) = \frac{1}{2}\sum_i \mathbb{E}[(g_{a,b}(X) - g_{a,b}(X^{(i)}))_-^2]$, so

$$\mathcal{E}(g_{a,b}) \leq \tfrac{1}{2}\,\mathbb{E}\Big[\mathbb{1}_{\{f(X) > Q_{1-\gamma}\}} \sum_i (f(X) - f(X^{(i)}))_-^2\Big] \leq \tfrac{v\gamma}{2}, \tag{1}$$

using $\sum_i (f(x) - f(x^{(i)}))_-^2 \leq v$ and $\mathbb{P}\{f(X) > Q_{1-\gamma}\} \leq \gamma$.

Martingale differences. With $\Delta_j = \mathbb{E}[g_{a,b}(X) \mid X_1, \dots, X_j] - \mathbb{E}[g_{a,b}(X) \mid X_1, \dots, X_{j-1}]$, $\mathbb{E}|\Delta_j| \leq \mathbb{E}|g_{a,b}(X) - g_{a,b}(X^{(j)})| = 2\mathbb{E}[(g_{a,b}(X) - g_{a,b}(X^{(j)}))_-]$ (by symmetry). Localizing as above and using Cauchy–Schwarz,

$$\mathbb{E}|\Delta_j| \leq 2\sqrt{\mathbb{E}[(g_{a,b}(X) - g_{a,b}(X^{(j)}))_-^2]}\,\sqrt{\gamma},$$

so

$$\sum_j (\mathbb{E}|\Delta_j|)^2 \le 4\gamma \sum_j \mathbb{E}[(\cdot)_-^2] = 8\gamma\, \mathcal{E}(g_{a,b}) \le 4v\gamma^2. \tag{2}$$

Applying Theorem 9.4. $\mathrm{Var}(g_{a,b}) \log \frac{\mathrm{Var}(g_{a,b})}{\sum_j (\mathbb{E}|\Delta_j|)^2} \le 2\mathcal{E}(g_{a,b}) \le v\gamma$. If $\mathrm{Var}(g_{a,b}) \le \sum_j (\mathbb{E}|\Delta_j|)^2$, then $\delta(b-a)^2/4 \le 4v\gamma^2$, so $b-a \le 4\gamma\sqrt{v/\delta} \le 4\sqrt{v\gamma/(\delta L)}$ where $L = \log \frac{e^2}{2\gamma} \ge 2$ (using $\gamma L \le 1$ for $\gamma \in (0, 1/2]$). Otherwise, with $A = \frac{\delta(b-a)^2}{4v\gamma}$: if $A < 4\gamma$ the same bound holds; if $A \ge 4\gamma$, monotonicity gives $A \log \frac{A}{4\gamma} \le 1$.

From $A \log \frac{A}{4\gamma} \le 1$ to the bound. We claim $A \le \frac{8}{L}$. Indeed, $x \mapsto x \log \frac{x}{4\gamma}$ is increasing for $x > 4\gamma/e$, and $\frac{8}{L} > \frac{4\gamma}{e}$ since $\gamma L < 2e$ on $(0, 1/2]$; if $A > 8/L$ then $1 \ge A \log \frac{A}{4\gamma} > \frac{8}{L} \log \frac{2}{\gamma L}$, but $\frac{8}{L} \log \frac{2}{\gamma L} > 1$ for $\gamma \in (0, 1/2]$ (with $t = \log(1/\gamma) \ge \log 2$, this is $9\log 2 - 2 + 7t - 8\log(2+t) > 0$, increasing in t and positive at $t = \log 2$). Hence $(b-a)^2 = \frac{4v\gamma A}{\delta} \le \frac{32 v\gamma}{\delta L}$.

In all cases $b - a \le 4\sqrt{2}\sqrt{\frac{v\gamma}{\delta \log(e^2/(2\gamma))}}$, so with $K = 4\sqrt{2}$,

$$Q_{1-\delta} - Q_{1-\gamma} = (b-a) + B \le B + K\sqrt{\frac{v\gamma}{\delta \log \frac{e^2}{2\gamma}}}. \qquad \blacksquare$$

Exercise 9.5

Central idea. Adapt the truncation + Theorem 9.4 proof of Theorem 9.8 (as in Exercise 9.4) to the one-sided $(\cdot)_+$ hypothesis $\sum_i (f - f_i)_+^2 \le \phi(f)$. Truncate f to $g = g_{a,b}$ with $a = Q_{1-\gamma}$, $b = Q_{1-\delta}$; since single-coordinate jumps are at most B, $(g - g_i)_+$ vanishes unless $Q_{1-\gamma} < f \le b + B$, so $\mathcal{E}(g)$ and the martingale differences localize to a set of probability $\le \gamma$ with ϕ evaluated at $b + B = Q_{1-\delta} + B$. Theorem 9.4 then yields $A \log \frac{A}{4\gamma} \le 1$ for $A \propto \frac{\delta(b-a)^2}{\gamma\phi(b+B)}$, whence the quantile gap bound. No B-shift of a is needed (unlike the $(\cdot)_-$ case), so $b - a = Q_{1-\delta} - Q_{1-\gamma}$ directly.

Solution. Let $0 < \delta < \gamma \le 1/2$, $a = Q_{1-\gamma}$, $b = Q_{1-\delta}$, where $Q_\alpha = \inf\{z : \mathbb{P}\{f(X) \le z\} \ge \alpha\}$. Since $\delta < \gamma$, $a \le b$; if $a = b$ there is nothing to prove. Assume $a < b$. Put $B = \max_{x,i} |f(x) - f(x^{(i)})|$ and $L = \log \frac{e^2}{2\gamma} \ge 2$.

Define the truncation $g(x) = g_{a,b}(x) = \max(a, \min(b, f(x)))$, nondecreasing and 1-Lipschitz in $f(x)$, so $(g(x) - g(x^{(i)}))_+ \le (f(x) - f(x^{(i)}))_+$.

Variance lower bound. $\mathbb{P}\{g = b\} = \mathbb{P}\{f \ge b\} \ge \delta$ and $\mathbb{P}\{g = a\} = \mathbb{P}\{f \le a\} \ge 1 - \gamma \ge 1/2$. For $g \in [a, b]$,

$$\mathrm{Var}(g) \ge \mathbb{P}\{g = a\}\mathbb{P}\{g = b\}(b-a)^2 \ge \tfrac{\delta}{2}(b-a)^2, \tag{1}$$

since with $m = \mathbb{E}g = a + t(b-a)$, $\mathrm{Var}(g) \ge (m-a)^2\mathbb{P}\{g = a\} + (b-m)^2\mathbb{P}\{g = b\} = (b-a)^2[t^2 q + (1-t)^2 p] \ge pq(b-a)^2$ (the bracket exceeds pq because $p + q \le 1$).

Localization of the $(\cdot)_+$ sensitivity. If $(g(x) - g(x^{(i)}))_+ > 0$ then $g(x) > a$, hence $f(x) > a = Q_{1-\gamma}$. If $f(x) > b + B$ then $f(x^{(i)}) \ge f(x) - B > b$ for every i, so $g \equiv b$ on $\{x\} \cup \{x^{(i)}\}$ and $(g(x) - g(x^{(i)}))_+ = 0$. Thus, with ϕ nondecreasing,

$$\sum_{i=1}^n (g(x) - g(x^{(i)}))_+^2 \le \mathbb{1}_{\{f(x) > Q_{1-\gamma}\}} \sum_i (f(x) - f(x^{(i)}))_+^2 \le \mathbb{1}_{\{f(x) > Q_{1-\gamma}\}}\, \phi(b + B). \tag{2}$$

By the flip symmetry $(X, X^{(i)}) \overset{d}{=} (X^{(i)}, X)$, $\mathbb{E}[(g - g_i)_+^2] = \mathbb{E}[(g - g_i)_-^2]$, so with $\mathcal{E}(g) = \frac{1}{2}\sum_i \mathbb{E}[(g - g_i)^2]$,

$$\mathcal{E}(g) = \sum_i \mathbb{E}[(g(X) - g(X^{(i)}))_+^2] \le \gamma\, \phi(b + B), \tag{3}$$

using $\mathbb{P}\{f > Q_{1-\gamma}\} \le \gamma$.

Martingale differences. With $\Delta_j = \mathbb{E}[g \mid X_1,\dots,X_j] - \mathbb{E}[g \mid X_1,\dots,X_{j-1}]$, $\mathbb{E}|\Delta_j| \le \mathbb{E}|g(X) - g(X^{(j)})| = 2\mathbb{E}[(g(X) - g(X^{(j)}))_+]$ by symmetry. By Cauchy–Schwarz and (2),

$$\mathbb{E}[(g-g_j)_+] \le \sqrt{\mathbb{E}[(g-g_j)_+^2]}\,\sqrt{\mathbb{P}\{f > Q_{1-\gamma}\}} \le \sqrt{\gamma}\,\sqrt{\mathbb{E}[(g-g_j)_+^2]},$$

since $(g(x) - g(x^{(j)}))_+ > 0$ implies $f(x) > Q_{1-\gamma}$. Hence

$$\sum_j (\mathbb{E}|\Delta_j|)^2 \le 4\gamma \sum_j \mathbb{E}[(g-g_j)_+^2] \le 4\gamma^2 \phi(b+B). \tag{4}$$

Applying Theorem 9.4. Theorem 9.4 gives $\mathrm{Var}(g)\log\frac{\mathrm{Var}(g)}{\sum_j(\mathbb{E}|\Delta_j|)^2} \le 2\mathcal{E}(g)$. Write $V = \mathrm{Var}(g)$, $S = \sum_j(\mathbb{E}|\Delta_j|)^2$. If $V \le S$, then by (1) and (4), $\frac{\delta}{2}(b-a)^2 \le 4\gamma^2\phi(b+B)$, so $(b-a)^2 \le \frac{8\gamma^2\phi(b+B)}{\delta} \le \frac{8\phi(b+B)\gamma}{\delta L}$, using $\gamma L \le 1$ on $(0,1/2]$ (since $\gamma \mapsto \gamma\log\frac{e^2}{2\gamma}$ is increasing there, with value 1 at $\gamma = 1/2$).

Otherwise $V > S$. Put $A = \frac{\delta(b-a)^2}{4\gamma\phi(b+B)}$. From $V\log(V/S) \le 2\mathcal{E}(g)$ with (1), (3), (4):

$$\frac{\delta}{2}(b-a)^2 \log\!\left(\frac{\delta(b-a)^2}{8\gamma^2\phi(b+B)}\right) \le 2\gamma\phi(b+B),$$

i.e. $A\log\frac{A}{2\gamma} \le 1$. If $A < 4\gamma$ then $(b-a)^2 = \frac{4\gamma\phi(b+B)A}{\delta} < \frac{16\gamma^2\phi(b+B)}{\delta} \le \frac{16\phi(b+B)\gamma}{\delta L}$. If $A \ge 4\gamma$, monotonicity gives $A\log\frac{A}{4\gamma} \le 1$.

From $A\log\frac{A}{4\gamma} \le 1$ to the bound. We claim $A \le \frac{8}{L}$. If $A \le 4\gamma$ this follows from $\gamma L \le 1 < 2$. Else $x \mapsto x\log\frac{x}{4\gamma}$ is increasing for $x > 4\gamma/e$, and $\frac{8}{L} > \frac{4\gamma}{e}$ (as $\gamma L \le 1 < 2e$); if $A > 8/L$ then $1 \ge A\log\frac{A}{4\gamma} > \frac{8}{L}\log\frac{2}{\gamma L}$. But $\frac{8}{L}\log\frac{2}{\gamma L} > 1$ on $(0,1/2]$: with $u = \frac{1}{2\gamma} \ge 1$, $w = \log u \ge 0$, $L = 2+w$, this is $\frac{8}{2+w}(\log 4 + w - \log(2+w)) = 8(1 - \frac{2-\log 4+\log s}{s}) \ge 8(1-0.654) > 2.7$ for $s = 2+w \ge 2$ (the fraction $\frac{0.614+\log s}{s}$ decreases on $[2,\infty)$). Contradiction. Hence $A \le 8/L$ and

$$(b-a)^2 = \frac{4\gamma\phi(b+B)A}{\delta} \le \frac{32\,\phi(b+B)\,\gamma}{\delta\,L}.$$

In all cases $b - a \le 4\sqrt{2}\sqrt{\frac{\phi(b+B)\gamma}{\delta\log(e^2/(2\gamma))}}$. Since $b - a = Q_{1-\delta} - Q_{1-\gamma}$ and $b + B = Q_{1-\delta} + B$, with $K = 4\sqrt{2}$,

$$Q_{1-\delta} - Q_{1-\gamma} \le K\sqrt{\frac{\phi(Q_{1-\delta}+B)\,\gamma}{\delta\,\log\frac{e^2}{2\gamma}}}, \qquad 0 < \delta < \gamma \le \tfrac{1}{2}.$$

In particular. With $a_k = Q_{1-2^{-k}}$, take $\delta = 2^{-(k+1)}$, $\gamma = 2^{-k}$: then $\gamma/\delta = 2$ and $L = \log(e^2 2^{k-1}) = 2 + (k-1)\log 2 \ge k\log 2$, so

$$a_{k+1} - a_k \le K\sqrt{\frac{2}{\log 2}}\sqrt{\frac{\phi(a_{k+1}+B)}{k}},$$

and absorbing $\sqrt{2/\log 2}$ into K gives $a_{k+1} - a_k \le K\sqrt{\phi(a_{k+1}+B)/k}$. ∎

Exercise 9.6

Central idea. Write $f = \mathbf{1}_A$ and $\phi(p) = \mathbb{P}_p(A)$. Couple two independent μ_p samples x, y by resampling coordinates one at a time from x to y: if $f(x) \ne f(y)$ then f must change value at some resampling step, and the change probability at step i is exactly $2p(1-p)I_i^p$. The union bound plus Russo's lemma (Theorem 9.15) yields the log-odds differential inequality $\frac{d}{dp}\log\frac{\phi}{1-\phi} \ge \frac{1}{p(1-p)}$; integrating from $p_{1/2}$ to $p_{1-\varepsilon}$ bounds $p_{1-\varepsilon}$ by a constant multiple of $p_{1/2}$. The case $p_{1/2} > 1/2$ reduces to this via the symmetry $g(x) = 1 - f(-x)$.

Solution. Let $f = \mathbf{1}_A : \{-1,1\}^n \to \{0,1\}$, monotone and non-constant (otherwise p_α is undefined), and $\phi(p) = \mathbb{P}_p(A) = \mathbb{E}_p[f]$, a polynomial in p. By Theorem 9.15 (Russo's lemma), $\phi'(p) = \sum_{i=1}^n I_i^p$ where $I_i^p = \mathbb{P}_p(\{x : f(x) \neq f(\bar{x}^{(i)})\})$. Monotonicity and non-constancy give $f(-1,\dots,-1) = 0$, $f(1,\dots,1) = 1$, so $\phi(0+) = 0$, $\phi(1-) = 1$; since some influence is strictly positive on $(0,1)$, $\phi' > 0$ there and ϕ is strictly increasing. Hence for each $\alpha \in (0,1)$ there is a unique $p_\alpha \in (0,1)$ with $\phi(p_\alpha) = \alpha$.

Reduction to $p_{1/2} \le 1/2$. Put $g(x) = 1 - f(-x)$, monotone and non-constant. Since $-x \sim \mu_{1-p}$ under $x \sim \mu_p$, $\mathbb{E}_p[g] = 1 - \phi(1-p)$, whence $p_\alpha(g) = 1 - p_{1-\alpha}(f)$. Thus $p_{1/2}(g) = 1 - p_{1/2}(f)$, $\min(p_{1/2}(g), 1 - p_{1/2}(g)) = \min(p_{1/2}(f), 1 - p_{1/2}(f))$, and $p_{1-\varepsilon}(g) - p_\varepsilon(g) = p_{1-\varepsilon}(f) - p_\varepsilon(f)$. It suffices to prove the claim under $p_{1/2} \le 1/2$.

Differential inequality. Let x, y be independent with law μ_p. Set $z^{(0)} = x$ and let $z^{(i)}$ be $z^{(i-1)}$ with its i-th coordinate replaced by y_i, so $z^{(n)} = y$. If $f(x) \neq f(y)$ then $f(z^{(i-1)}) \neq f(z^{(i)})$ for some i, hence

$$\mathbb{P}_p(f(x) \neq f(y)) \le \sum_{i=1}^n \mathbb{P}_p(f(z^{(i-1)}) \neq f(z^{(i)})). \tag{1}$$

By independence the left side is $2\phi(p)(1-\phi(p))$. For the right side, $z^{(i-1)}$ and $z^{(i)}$ share all coordinates but the i-th (x_i versus y_i); writing w for the shared part, $w \sim \mu_p^{n-1}$ is independent of the iid pair (x_i, y_i). Then f differs at step i iff $x_i \neq y_i$ (probability $2p(1-p)$) and i is pivotal for w, so

$$\mathbb{P}_p(f(z^{(i-1)}) \neq f(z^{(i)})) = 2p(1-p)\, I_i^p.$$

Summing over i and applying Russo's lemma to (1) gives, for all $p \in (0,1)$,

$$2\phi(p)(1-\phi(p)) \le 2p(1-p)\phi'(p), \qquad \text{i.e.} \qquad \frac{d}{dp}\log\frac{\phi(p)}{1-\phi(p)} = \frac{\phi'(p)}{\phi(p)(1-\phi(p))} \ge \frac{1}{p(1-p)}. \tag{2}$$

(Monotonicity is not used in (2), only for p_α to be well defined.)

Integration. Assume $p_{1/2} \le 1/2$. On $[p_{1/2}, p_{1-\varepsilon}]$ (nonempty since $1-\varepsilon > 1/2$) we have $\phi \in [1/2, 1-\varepsilon] \subset (0,1)$; integrating (2),

$$\log\frac{1-\varepsilon}{\varepsilon} = \left[\log\frac{\phi}{1-\phi}\right]_{p_{1/2}}^{p_{1-\varepsilon}} \ge \int_{p_{1/2}}^{p_{1-\varepsilon}} \frac{dq}{q(1-q)} = \log\frac{p_{1-\varepsilon}}{1-p_{1-\varepsilon}} - \log\frac{p_{1/2}}{1-p_{1/2}}.$$

Thus, with $K = (1-\varepsilon)/\varepsilon$,

$$\frac{p_{1-\varepsilon}}{1-p_{1-\varepsilon}} \le K\,\frac{p_{1/2}}{1-p_{1/2}} \le 2K\, p_{1/2},$$

using $p_{1/2} \le 1/2$. Since $\frac{q}{1-q} \ge q$ for $q \in [0,1)$,

$$p_{1-\varepsilon} - p_\varepsilon \le p_{1-\varepsilon} \le \frac{2(1-\varepsilon)}{\varepsilon}\, p_{1/2} = \frac{2(1-\varepsilon)}{\varepsilon}\min(p_{1/2}, 1-p_{1/2}),$$

the claim with $c = 2(1-\varepsilon)/\varepsilon$ depending only on ε. ■

Exercise 9.7

Central idea. This is the $\mathbb{P}_p$-generalization of the book's $p = 1/2$ computation that the total influence $I(A) = 2^{-n}\sum_{x\in A}(2\|x\| - n)$ is maximized by the Hamming ball. Differentiating $\mathbb{P}_p(A) = \sum_{x\in A} p^{\|x\|}(1-p)^{n-\|x\|}$ gives $\frac{d}{dp}\mathbb{P}_p(A) = \frac{1}{p(1-p)}\sum_{x\in A}(\|x\| - np)\mathbb{P}_p(x)$, and at any $p \in (0,1)$ with $\mathbb{P}_p(A) = \mathbb{P}_p(B)$, the Hamming ball $B = \{x : \|x\| \ge k\}$ strictly maximizes this derivative among all sets of the same $\mathbb{P}_p$-measure (unless $A = B$). Since $\mathbb{P}_q(A) = 1/2 \le \mathbb{P}_q(B)$ at $q = p_{1/2}(A) \ge p_{1/2}(B)$, a last-crossing argument shows A's curve can never overtake B's.

Solution. Write $\|x\|$ for the number of $+1$ coordinates of $x \in \{-1,1\}^n$, and $f(p) = \mathbb{P}_p(A)$, $g(p) = \mathbb{P}_p(B)$, where $\mathbb{P}_p$ is the product measure with $\mathbb{P}_p\{x_i = +1\} = p$. Both f and g are polynomials in p, hence

C^∞ on $[0,1]$, and nondecreasing since A and B are monotone. Let $q = p_{1/2}(A)$ and $r = p_{1/2}(B)$, so $f(q) = 1/2 = g(r)$ and $q \geq r$ by hypothesis. Since $k \geq 1$, $\mathbb{P}_0(B) = 0 < 1/2$, so $r > 0$ and hence $q > 0$. Moreover $f(q) = 1/2 \leq g(q)$, as g is nondecreasing and $q \geq r$.

For any $S \subset \{-1,1\}^n$ and $p \in (0,1)$, differentiating termwise,

$$\frac{d}{dp}\mathbb{P}_p(S) = \sum_{x \in S} \frac{d}{dp}\left[p^{\|x\|}(1-p)^{n-\|x\|}\right] = \frac{1}{p(1-p)} \sum_{x \in S} (\|x\| - np)\, \mathbb{P}_p(x), \tag{$*$}$$

since $\frac{d}{dp}[p^j(1-p)^{n-j}] = p^j(1-p)^{n-j}\left(\frac{j}{p} - \frac{n-j}{1-p}\right)$.

Key estimate. Let $p \in (0,1)$ with $\mathbb{P}_p(A) = \mathbb{P}_p(B) =: \alpha$. Then, with $\delta := \mathbb{P}_p(A \setminus B) = \mathbb{P}_p(B \setminus A)$,

$$\begin{aligned} &\frac{d}{dp}\mathbb{P}_p(A) - \frac{d}{dp}\mathbb{P}_p(B) \\ &\quad = \frac{1}{p(1-p)} \left[\sum_{x \in A \setminus B} (\|x\| - np)\mathbb{P}_p(x) - \sum_{x \in B \setminus A} (\|x\| - np)\mathbb{P}_p(x) \right]. \end{aligned}$$

For $x \in A \setminus B$, $\|x\| \leq k-1$ (as $x \notin B$), so the first sum is at most $(k - 1 - np)\,\delta$; for $x \in B \setminus A$, $\|x\| \geq k$, so the subtracted sum is at least $(k - np)\,\delta$. Hence

$$\frac{d}{dp}\mathbb{P}_p(A) - \frac{d}{dp}\mathbb{P}_p(B) \leq \frac{(k-1-np)\delta - (k-np)\delta}{p(1-p)} = \frac{-\delta}{p(1-p)} \leq 0,$$

with strict inequality iff $\delta > 0$. Since $\mathbb{P}_p$ charges every $x \in \{-1,1\}^n$ for $p \in (0,1)$, $\delta = 0$ iff $A = B$ as sets. Thus $f'(p) < g'(p)$ whenever $f(p) = g(p)$ and $A \neq B$; if $A = B$ then $f \equiv g$.

Now suppose, for contradiction, that $f(p_0) > g(p_0)$ for some $p_0 > q$. Let

$$p_1 = \sup\{p \in [q, p_0] : f(p) \leq g(p)\},$$

nonempty since $f(q) \leq g(q)$. By continuity $f(p_1) = g(p_1)$, and $p_1 < p_0 \leq 1$ since $f > g$ on $(p_1, p_0]$. If $p_1 = 1$ then $p_0 = 1$ and $f(1) > g(1) = 1$ (as $B \neq \emptyset$), impossible; so $p_1 \in (0,1)$, using $p_1 \geq q > 0$. Since $f(p_1) = g(p_1)$ and $f > g$ immediately to the right of p_1, $f'(p_1) \geq g'(p_1)$. But the key estimate gives $f'(p_1) < g'(p_1)$ unless $A = B$, in which case $f \equiv g$, contradicting $f(p_0) > g(p_0)$. This contradiction shows no such p_0 exists.

Therefore $f(p) \leq g(p)$ for all $p \geq q$, i.e. $\mathbb{P}_p(A) \leq \mathbb{P}_p(B)$ for all $p \geq p_{1/2}$. ■

Exercise 9.8

Central idea. In the proof of Theorem 9.4, replace the uniform log-Sobolev inequality by the p-biased one (Theorem 5.2: $\mathrm{Ent}_p(g^2) \leq c(p)\,\mathcal{E}(g)$ with $\mathcal{E}(g) = p(1-p)\sum_j \mathbb{E}_p(g - g^{(j)})^2$). Apply it to each martingale difference Δ_i of $f = \mathbf{1}_A$, and use $\mathbb{E}_p|\Delta_i| \leq 2p(1-p)I_i^p$ and $\sum_i \mathcal{E}(\Delta_i) = \mathcal{E}(f) = p(1-p)I^p$. This gives the p-biased (9.1): $\sum_i (I_i^p)^2 \geq \frac{V}{4p^2q^2} e^{-\beta I^p/V}$, where $\beta = c(p)pq \leq \frac{1}{2}$. A fixed-point argument, used instead of the two-case argument, then gives $\max_i I_i^p \geq \frac{y_n}{\beta}\frac{V}{n}$ and $\sum_i (I_i^p)^2 \geq (\frac{y_n}{\beta})^2 \frac{V^2}{n}$, where $y_n^2 e^{y_n} = c(p)^2 n/(4V)$.

For $p \neq \frac{1}{2}$ the inequality $\beta < \frac{1}{2}$ is strict. That leaves enough room to absorb the printed ε once $n \geq n_0(p)$. At $p = \frac{1}{2}$ we can only prove the bound with a larger $\varepsilon_n = O(\log\log n/\log n)$. Corollary 9.18 follows from the max-influence bound and Russo's lemma.

Solution.

Status: incomplete - Proved for $p \neq \frac{1}{2}$: both displayed inequalities with the *printed* ε, for every fixed $p \neq \frac{1}{2}$ and every $n \geq n_0(p)$. The threshold $n_0(p)$ is given by an explicit condition (Step 3). - **Proved for all p, including $p = \frac{1}{2}$:** a weaker form, with a larger $\varepsilon_n \to 0$ (Step 2). - **Open at $p = \frac{1}{2}$:** the printed bound. There $\varepsilon = \log(V\log^2 n/4)/(2\log n)$, which is exactly one quarter of the $\varepsilon = 2\log(V\log^2 n/4)/\log n$ used in the book's own proof of Theorem 9.5. We do **not** know whether the printed, stronger bound holds at $p = \frac{1}{2}$.

- Neither the two-case argument of Theorem 9.5 nor the fixed-point argument below proves it (Step 4). - We found no counterexample. Balanced tribes (width $w = 3, 5, 8, 12, 16, 20$ with $m \approx 2^w \log 2$ tribes, n from 18 to about $1.5 \cdot 10^7$) satisfy the sum-of-squares bound with a factor of at least 2.3 to spare. Over a fine grid of all tribe widths $w \le 29$ and tribe counts with $18 \le n \le 1.5 \cdot 10^7$, the smallest factor found is about 2.19 for the sum of squares (about 1.48 for the max influence); these are numerical checks, not a proof. - **Corollary 9.18:** proved completely, for all p.

Note on the previous version of this file. - It used the log-Sobolev inequality in the form $\mathrm{Ent}_p(f^2) \le c(p)p(1-p)\sum_i \mathbb{E}_p[((f - f^{(i)})/2)^2]$, which is 4× too strong. Take $n = 1$, $p = \frac{1}{2}$, $f = \mathbf{1}\{x = 1\}$: then $\mathrm{Ent}(f^2) = \frac{1}{2}\log 2 \approx 0.347$, while that right-hand side is 0.125. The book's right-hand side is 0.5. - It also contained an arithmetic slip: it wrote $2 - K(2-\varepsilon)$ where $1 - K(2-\varepsilon)$ was meant.

Both are corrected below.

Notation. - $q = 1 - p$, $f = \mathbf{1}_A$, and $V = P_p(A)(1 - P_p(A))$, assumed > 0 (otherwise ε is undefined). - For $h : \{-1, 1\}^n \to \mathbb{R}$, let $D_j h(x) = h(x^{(j,1)}) - h(x^{(j,-1)})$, where $x^{(j,\pm 1)}$ is x with coordinate j set to ± 1. Then $(h(x) - h(\bar{x}^{(j)}))^2 = (D_j h(x))^2$. - So the book's Dirichlet form (pp. 120 and 283) is $\mathcal{E}(h) = pq\sum_j \mathbb{E}_p(D_j h)^2$, and Theorem 5.2 reads $\mathrm{Ent}_p(h^2) \le c(p)\mathcal{E}(h)$. - For binary f, $(D_i f)^2 = \mathbf{1}\{i \text{ pivotal}\}$. Hence $I_i := I_i^p(A) = \mathbb{E}_p(D_i f)^2 = \mathbb{E}_p|D_i f|$, and $\mathcal{E}(f) = pqI$ with $I = \sum_i I_i$. - Put $\beta = c(p)pq$.

$\beta \le \frac{1}{2}$, with equality only at $p = \frac{1}{2}$. For $p < \frac{1}{2}$ put $t = q/p > 1$. Then $\beta = pq\log t/(q-p)$, and

$$\beta < \tfrac{1}{2} \iff \log t < \frac{q-p}{2pq} = \tfrac{1}{2}(t - t^{-1}).$$

The function $g(t) = \frac{1}{2}(t - t^{-1}) - \log t$ has $g(1) = 0$ and $g'(t) = \frac{1}{2}(1 - t^{-1})^2 > 0$, so $g > 0$ on $(1, \infty)$. The case $p > \frac{1}{2}$ follows from $c(p) = c(1-p)$. At $p = \frac{1}{2}$, $c(\frac{1}{2}) = 2$ and $\beta = \frac{1}{2}$. Numerically, $\max_{p \ne 1/2} \beta < \frac{1}{2}$ on a fine grid.

Step 1: the p-biased (9.1). Let $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$ and $\Delta_i = \mathbb{E}_p[f \mid \mathcal{F}_i] - \mathbb{E}_p[f \mid \mathcal{F}_{i-1}]$. Then $f - \mathbb{E}_p f = \sum_i \Delta_i$ and $\sum_i \mathbb{E}_p \Delta_i^2 = V$.

(a) We have $f(x) = f(x^{(i,-1)}) + \mathbf{1}\{x_i = 1\} D_i f(x)$, where $D_i f$ does not depend on x_i, and X_i is independent of the other coordinates. Hence

$$\Delta_i = (\mathbf{1}\{X_i = 1\} - p)\, \mathbb{E}_p[D_i f \mid \mathcal{F}_{i-1}], \qquad \text{so} \qquad \mathbb{E}_p|\Delta_i| = 2pq\, \mathbb{E}_p\big|\mathbb{E}_p[D_i f \mid \mathcal{F}_{i-1}]\big| \le 2pq\, I_i.$$

(b) We show $\sum_i \mathcal{E}(\Delta_i) = \mathcal{E}(f) = pqI$. Fix j.

- For $i < j$, Δ_i does not depend on x_j, so $D_j \Delta_i = 0$.
- For $i \ge j$, D_j acts on a coordinate that the conditioning keeps, so it commutes with $\mathbb{E}_p[\cdot \mid \mathcal{F}_i]$.
- For $i = j$: $\mathbb{E}_p[f \mid \mathcal{F}_{j-1}]$ does not depend on x_j, and $D_j f$ does not depend on x_j. Hence $D_j \Delta_j = \mathbb{E}_p[D_j f \mid \mathcal{F}_j] = \mathbb{E}_p[D_j f \mid \mathcal{F}_{j-1}]$.
- For $i > j$: $D_j \Delta_i = \mathbb{E}_p[D_j f \mid \mathcal{F}_i] - \mathbb{E}_p[D_j f \mid \mathcal{F}_{i-1}]$.

So the terms $D_j \Delta_i$, $i \ge j$, are an $\mathcal{F}_{j-1}$-measurable term followed by the later martingale differences of $D_j f$. They are mutually orthogonal and sum to $D_j f$. Hence $\sum_i \mathbb{E}_p(D_j \Delta_i)^2 = \mathbb{E}_p(D_j f)^2 = I_j$. Summing over j and multiplying by pq gives the claim. The same computation gives $\sum_i \mathcal{E}(\Delta_i(h)) = \mathcal{E}(h)$ for every real-valued h; this was checked numerically on random functions.

(c) For any g, $\mathrm{Ent}_p(g^2) \ge \mathbb{E}_p g^2 \log \frac{\mathbb{E}_p g^2}{(\mathbb{E}_p|g|)^2}$. To see this, let $dQ = g^2\, dP_p / \mathbb{E}_p g^2$. Jensen's inequality gives

$$\mathbb{E}_Q \log|g| \ge -\log \mathbb{E}_Q |g|^{-1} = \log \frac{\mathbb{E}_p g^2}{\mathbb{E}_p|g|},$$

so $\mathbb{E}_p[g^2 \log g^2] \ge 2\mathbb{E}_p g^2 \log \frac{\mathbb{E}_p g^2}{\mathbb{E}_p|g|}$. Subtracting $\mathbb{E}_p g^2 \log \mathbb{E}_p g^2$ gives the claim.

Now apply Theorem 5.2 to each nonzero Δ_i and use (c). Sum over i, then use (b), the log-sum inequality, and (a):

$$c(p)\,pq\,I = \sum_i c(p)\mathcal{E}(\Delta_i) \;\ge\; \sum_i \mathbb{E}_p\Delta_i^2 \log \frac{\mathbb{E}_p\Delta_i^2}{(\mathbb{E}_p|\Delta_i|)^2} \;\ge\; V \log \frac{V}{\sum_i (\mathbb{E}_p|\Delta_i|)^2} \;\ge\; V \log \frac{V}{4p^2q^2\sum_i I_i^2}.$$

Hence

$$\sum_{i=1}^n I_i^2 \;\ge\; \frac{V}{4p^2q^2}\exp\Big(-\frac{\beta I}{V}\Big). \tag{9.1_p}$$

At $p=\frac12$ this is exactly the book's (9.1). We verified (9.1_p) numerically, together with (a), (b) and Theorem 5.2 for each Δ_i, on 400 random sets with $2\le n\le 6$ and random p.

Step 2: fixed-point bounds. Let $y=y_n>0$ be the unique solution of $y^2e^y=\frac{c(p)^2n}{4V}$. Equivalently, $y+2\log y=s$ with $s:=\log\frac{c(p)^2n}{4V}$; the solution is unique because $y\mapsto y+2\log y$ increases from $-\infty$ to ∞. Note that

$$\frac{V}{4p^2q^2}e^{-y} = \frac{V}{4p^2q^2}\cdot\frac{4Vy^2}{c(p)^2n} = \frac{V^2y^2}{\beta^2 n}.$$

Max influence. Suppose $M:=\max_i I_i < \frac{yV}{\beta n}$. Then $I\le nM<\frac{yV}{\beta}$, so (9.1_p) gives $\sum_i I_i^2 > \frac{V^2y^2}{\beta^2 n}$. But also $\sum_i I_i^2 \le MI < \frac{yV}{\beta n}\cdot\frac{yV}{\beta}=\frac{V^2y^2}{\beta^2 n}$, a contradiction.

Sum of squares. Suppose $\sum_i I_i^2 < \frac{V^2y^2}{\beta^2 n}$. By Cauchy–Schwarz, $I\le\sqrt{n\sum_i I_i^2}<\frac{Vy}{\beta}$, and then (9.1_p) gives $\sum_i I_i^2>\frac{V^2y^2}{\beta^2 n}$, a contradiction.

Therefore, for every n, every p, and every A with $V>0$,

$$\max_i I_i^p(A) \;\ge\; \frac{y_n}{\beta}\cdot\frac{V}{n}, \qquad \sum_i I_i^p(A)^2 \;\ge\; \Big(\frac{y_n}{\beta}\Big)^2\frac{V^2}{n}. \tag{$*$}$$

Both bounds were checked numerically on the same random sets.

Since $c(p)\ge 2$ and $V\le\frac14$, we have $s\ge\log(4n)>1$. If $y<1$ then $y+2\log y<1<s$; hence $y\ge 1$, and therefore $y\le s$. So $(*)$ has the required form, with $2-\varepsilon$ replaced by $y_n/(\beta\log n)$.

To make this explicit, use $1/\beta\ge 2$ and $y_n=s-2\log y_n$. This gives $y_n/\beta\ge 2y_n=(2-\varepsilon_n')\log n$ with

$$\varepsilon_n' = \frac{2\log(4Vy_n^2/c(p)^2)}{\log n} \le \frac{2\log(s^2/4)}{\log n}.$$

This is $O(\frac{\log\log n}{\log n})$ for fixed p and V bounded below.

Step 3: the printed ε for $p\ne\frac12$, $n\ge n_0(p)$. Write $L=\log n$, $v=\log(1/V)\ge\log 4$, and $C=2\log(c(p)/2)\ge 0$. Then $s=L+v+C$ and

$$\varepsilon = \frac{\log(Vpq\,L^2)}{c(p)L} = \frac{-v+\log(pq)+2\log L}{c(p)L}.$$

The numerator is at most $2\log L-2\log 4$ and $c(p)L\ge 2L$. So $\varepsilon\le\max\big(0,\frac{\log L-\log 4}{L}\big)<1$, hence $2-\varepsilon>1>0$. It follows that both printed inequalities follow from $(*)$ once $y/\beta\ge(2-\varepsilon)L$.

Multiplying by $\beta=c(p)pq$, the condition $y/\beta\ge(2-\varepsilon)L$ becomes

$$y \ge 2\beta L + pq\,v - pq\log(pq) - 2pq\log L.$$

Substituting $y=L+v+C-2\log y$, this is equivalent to

$$D := (1-2\beta)L + (1-pq)v + C + pq\log(pq) + 2pq\log L - 2\log y \;\ge\; 0.$$

Now bound the terms of D: - $pq\log(pq) \geq \frac{1}{4}\log\frac{1}{4}$, because $x\log x$ is decreasing on $(0,\frac{1}{4}] \subset (0,e^{-1})$; - $2pq\log L \geq 0$ for $n \geq 3$; - $1-pq \geq \frac{3}{4}$; - $\log y \leq \log s$.

Let $a = 1-2\beta > 0$ and $m = \min(a, \frac{3}{4}) \leq 1$. Since $C \geq 0$, $aL + \frac{3}{4}v + C \geq m(L+v+C) = ms$. Hence

$$D \;\geq\; ms - 2\log s - \tfrac{\log 4}{4}.$$

The map $s \mapsto ms - 2\log s$ is increasing for $s \geq 2/m$, and $s \geq L$. Choose $L_0(p) \geq \max(2/m, 1)$ such that $mL_0 - 2\log L_0 \geq \frac{\log 4}{4}$. If $L \geq L_0(p)$, then $D \geq 0$ for **every** A, uniformly in V.

Hence, for $p \neq \frac{1}{2}$ and $n \geq n_0(p) = e^{L_0(p)}$,

$$\sum_{i=1}^{n} I_i^p(A)^2 \;\geq\; \frac{(2-\varepsilon)^2\big(P_p(A)(1-P_p(A))\big)^2\log^2 n}{n}, \qquad \max_i I_i^p(A) \;\geq\; \frac{(2-\varepsilon)P_p(A)(1-P_p(A))\log n}{n}.$$

The threshold $n_0(p)$ is huge near $p = \frac{1}{2}$: - For $p = 0.3$, $a \approx 0.110$ and $L_0 \approx 83.3$. - A numerical minimisation of the exact D over v is negative at $L = 40$ and positive at $L = 80$ for $p = 0.3$. - For $p = 0.45$ and $p = 0.49$ it is still negative at $L = 160$, consistent with $a \to 0$.

Step 4: what fails at $p = \frac{1}{2}$. Here $a = 0$, $C = 0$ and $pq = \frac{1}{4}$, so

$$D = \tfrac{3}{4}v - \tfrac{1}{4}\log 4 + \tfrac{1}{2}\log L - 2\log y,$$

with $y = L + v - 2\log y \sim L$. For fixed V and $L \to \infty$, $D \approx -\frac{3}{2}\log L \to -\infty$. So $(*)$ does not imply the printed bound at $p = \frac{1}{2}$; it gives only ε_n' from Step 2.

The book's two-case argument does not help either. In the case $I < (2-\varepsilon)V\log n$, (9.1) gives $\sum_i I_i^2 \geq 4Vn^{-1+\varepsilon/2}$. With the printed ε, $n^{\varepsilon/2} = z^{1/4}$, where $z = V\log^2 n/4$. The target is $(2-\varepsilon)^2V^2\log^2 n/n = 4(2-\varepsilon)^2Vz/n$. So this case yields the target only if $z^{1/4} \geq (2-\varepsilon)^2 z$, which fails as soon as $(2-\varepsilon)^2 z^{3/4} > 1$, for example for fixed V and large n.

With the book's $4\times$ larger ε, $n^{\varepsilon/2} = z$. The case then closes only for the weaker conclusion $\mathrm{Var}(f)^2\log^2 n/n$ of Theorem 9.5, using $\varepsilon < 1$ in the other case. Whether the printed inequality holds at $p = \frac{1}{2}$ is left open here.

Step 5: Corollary 9.18. We rename the corollary's ε to $\eta \in (0, \frac{1}{2})$, to avoid a clash with the ε above. We prove it for all p from $(*)$, that is, from the second inequality in the form proved above.

Let A be monotone and nontrivial, with $I_1^p(A) = \cdots = I_n^p(A)$ for all $p \in (0,1)$, and let $n > 16$. Put $u(p) = P_p(A) \in (0,1)$. By Russo's lemma (Theorem 9.15), $u'(p) = I^p(A) = n\max_i I_i^p(A)$. So by $(*)$ and $1/\beta \geq 2$,

$$u'(p) \;\geq\; 2y_n(p)\,u(1-u).$$

Since $V \leq \frac{1}{4}$ and $c(p) \geq 2$, we have $s \geq t := \log(4n)$. Let y^* solve $y^* + 2\log y^* = t$. Because $y \mapsto y + 2\log y$ is increasing, $y_n(p) \geq y^*$. As in Step 2, $1 \leq y^* \leq t$, hence $y^* \geq t - 2\log t$.

Also $y^* \geq w := \frac{1}{4}\log\frac{n}{16} > 0$: - if $w \leq 1$, then $w + 2\log w \leq w < t$; - if $w > 1$, then $w + 2\log w < 3w < t$.

Therefore $\frac{d}{dp}\log\frac{u}{1-u} = \frac{u'}{u(1-u)} \geq 2y^*$. Integrating over $[p_\eta, p_{1-\eta}]$,

$$2\log\frac{1-\eta}{\eta} \;\geq\; 2y^*(p_{1-\eta} - p_\eta), \qquad p_{1-\eta} - p_\eta \;\leq\; \frac{\log\frac{1-\eta}{\eta}}{y^*} \;\leq\; \frac{\log\frac{1-\eta}{\eta}}{\log(4n) - 2\log\log(4n)}.$$

Finally, $\frac{1-\eta}{\eta} \leq \frac{1}{4\eta^2}$, which is equivalent to $4\eta(1-\eta) \leq 1$. This gives $\log\frac{1-\eta}{\eta} \leq 2\log\frac{1}{2\eta}$. Combined with $y^* \geq \frac{1}{4}\log\frac{n}{16}$,

$$p_{1-\eta} - p_\eta \;\leq\; \frac{8\log\frac{1}{2\eta}}{\log\frac{n}{16}},$$

which is Corollary 9.18. The middle bound sharpens it to $(1+o(1))\log\frac{1-\eta}{\eta}/\log n$. ■

Exercise 9.9

Central idea. Run Friedgut's proof of Theorem 9.7 with the p-biased log-Sobolev inequality (Theorem 5.2). It reads $\mathrm{Ent}_p(h^2) \le c(p)\mathcal{E}(h)$, where $\mathcal{E}(h) = p(1-p)\sum_j \mathbb{E}_p(h - h^{(j)})^2$ is the book's Dirichlet form, *not* the variance. Order the variables by decreasing influence, and let g be the martingale truncated after the $m = \lfloor I/\varepsilon \rfloor$ most influential ones. Apply the log-Sobolev inequality to each discarded martingale difference Δ_i, and use three facts: - $\sum_i \mathcal{E}(\Delta_i) = \mathcal{E}(f) = p(1-p)I$; - $\mathbb{E}_p|\Delta_i| \le 2p(1-p)I_i$; - $I_i < \varepsilon$ for every discarded i.

These give $W \log(W/S) \le c(p)p(1-p)I$ for $W = \mathbb{E}_p(f-g)^2$ and $S \le 4p^2(1-p)^2\varepsilon I$. Inverting $x \log x \le y$ yields the bound. For a coarse threshold, Russo's lemma gives a point p^* in the window where $I^{p^*} \le 1/(2K_1)$. K_2 keeps p^* away from 0 and 1, so the bound at p^* tends to 0 as $\varepsilon \to 0$. Rounding g then gives a junta approximation of A.

Solution. Throughout, $f = \mathbf{1}_A : \{-1,1\}^n \to \{0,1\}$ and $q = 1-p$, and $\mathbb{E} = \mathbb{E}_p$ unless indicated. $I_i = I_i^p(f) = P_p\{f(X) \ne f(\bar{X}^{(i)})\}$ and $I = I(f) = \sum_i I_i$. For real h, $D_j h(x) = h(x^{(j,1)}) - h(x^{(j,-1)})$, so $(h - h^{(j)})^2 = (D_j h)^2$ and $\mathcal{E}(h) = pq\sum_j \mathbb{E}(D_j h)^2$ (book, pp. 120 and 283). For binary f, $\mathbb{E}(D_j f)^2 = I_j$, hence $\mathcal{E}(f) = pqI$. We also use $c(p) \ge 2$ and $\beta := c(p)pq \le \frac{1}{2}$; both are proved in solution 09-8.

Correction to the previous version. It set "$\mathcal{E}(f) = \sum_i \mathbb{E}[\Delta_i^2]$", which is $\mathrm{Var}_p(f)$, and used $\mathrm{Ent}_p(h^2) \le c(p)\mathcal{E}(h)$ with that $\mathcal{E}$. This is false. Take $n = 3$, $p = \frac{1}{2}$ and $f = \mathbf{1}\{x_1 = x_2 = x_3 = 1\}$: then $\mathrm{Ent}(f^2) = \frac{1}{8}\log 8 \approx 0.260$, but $c \cdot \mathrm{Var}(f) = 2 \cdot \frac{7}{64} \approx 0.219$. (The book's $c\,\mathcal{E}(f) = 0.375$.) Inequality (6) below remains true, by the argument given here.

Martingale facts. Fix an ordering $1, \dots, n$ of the coordinates. Let $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$ and $\Delta_i = \mathbb{E}[f \mid \mathcal{F}_i] - \mathbb{E}[f \mid \mathcal{F}_{i-1}]$. Steps 1(a) and 1(b) of solution 09-8 hold for any ordering:

$$\mathbb{E}|\Delta_i| \le 2pq\, I_i, \tag{1}$$

$$\sum_{i=1}^n \mathcal{E}(\Delta_i) = \mathcal{E}(f) = pq\, I, \qquad \mathcal{E}(\Delta_i) \ge 0. \tag{2}$$

Also, for every g (09-8, Step 1(c)),

$$\mathrm{Ent}(g^2) \;\ge\; \mathbb{E}g^2 \log \frac{\mathbb{E}g^2}{(\mathbb{E}|g|)^2}. \tag{3}$$

Junta construction. Order the coordinates so that $I_1 \ge I_2 \ge \dots \ge I_n$, and put $m = \lfloor I/\varepsilon \rfloor$. If $m \ge n$, take $g = f$. Otherwise $I_i < \varepsilon$ for every $i > m$: if $I_{m+1} \ge \varepsilon$, then $I \ge (m+1)\varepsilon > I$, since $m + 1 > I/\varepsilon$. Hence

$$\sum_{i>m} I_i^2 \;\le\; \varepsilon \sum_{i>m} I_i \;\le\; \varepsilon I. \tag{4}$$

Let $g = \mathbb{E}[f \mid \mathcal{F}_m]$. It depends only on the m most influential variables. Then $f - g = \sum_{i>m}\Delta_i$ has mean 0, and

$$W := \mathbb{E}(f-g)^2 = \sum_{i>m} \mathbb{E}\Delta_i^2, \qquad S := \sum_{i>m} (\mathbb{E}|\Delta_i|)^2 \;\le\; 4p^2q^2\varepsilon I \tag{5}$$

by orthogonality, (1) and (4). If $S = 0$, every Δ_i with $i > m$ vanishes, so $W = 0$ and there is nothing to prove. Otherwise: - apply Theorem 5.2 to each nonzero Δ_i with $i > m$, and use (3); - sum over $i > m$, and use (2) with $\mathcal{E}(\Delta_i) \ge 0$; - finally, apply the log-sum inequality.

This gives

$$W \log \frac{W}{S} \;\le\; \sum_{i>m} \mathbb{E}\Delta_i^2 \log \frac{\mathbb{E}\Delta_i^2}{(\mathbb{E}|\Delta_i|)^2} \;\le\; c(p) \sum_{i>m} \mathcal{E}(\Delta_i) \;\le\; c(p)pq\, I = \beta I. \tag{6}$$

Inversion. Put $x = W/S$ and $y = \beta I/S$, so that $x \log x \le y$. By (5), $y \ge \frac{\beta}{4p^2q^2\varepsilon} = \frac{c(p)}{4pq\,\varepsilon} \ge \frac{2}{\varepsilon} > 2$.

We claim: for $y > 1$, $x \log x \le y$ implies $x \le 2y/\log y$. Let $x_0 = 2y/\log y$. Since $y/\log y \ge e$ for $y > 1$, we have $x_0 \ge 2e > e^{-1}$, so $x \log x$ is increasing on $[x_0, \infty)$. Moreover, $\log x_0 \ge \frac{1}{2} \log y$ because $x_0^2 \ge y \iff 4y \ge \log^2 y$, which holds since $2\sqrt{y} - \log y \ge 2$ for $y > 0$. So $x_0 \log x_0 \ge y$, and any $x > x_0$ would give $x \log x > y$.

Therefore $W \le 2\beta I/\log y$, and with $\log y \ge \log \frac{1}{4\varepsilon} + \log \frac{c(p)}{pq}$ (a positive quantity, as shown above) we get

$$\mathbb{E}(f-g)^2 \;\le\; \frac{2c(p)\,pq\,I}{\log \frac{1}{4\varepsilon} + \log \frac{c(p)}{pq}}. \tag{7}$$

Since $\beta \le \frac{1}{2}$ and $y \ge 2/\varepsilon$, (7) also gives the p-free bound $\mathbb{E}(f-g)^2 \le I/\log(2/\varepsilon)$.

The printed bound. Let $T = \dfrac{2I\,c(p)}{pq\log(1/4\varepsilon) + \log(c(p)/(pq))}$. We treat two cases.

- **Case $\varepsilon \le \frac{1}{4}$.** Here $\log \frac{1}{4\varepsilon} \ge 0$. Since $c(p) \ge 2$ and $pq \le \frac{1}{4}$, also $\log \frac{c(p)}{pq} \ge \log 8 > 0$. Because $\frac{1}{pq} \ge 4 > 1 \ge pq$, the denominator of (7) divided by pq satisfies

$$\frac{\log \frac{1}{4\varepsilon} + \log \frac{c(p)}{pq}}{pq} \;\ge\; pq \log \tfrac{1}{4\varepsilon} + \log \tfrac{c(p)}{pq} > 0.$$

  Hence the right side of (7), which equals $2c(p)I$ divided by the left side above, is at most T.

- **Case $\varepsilon \in (\frac{1}{4}, 1)$.** The denominator of T satisfies $0 < -pq\log 4 + \log \frac{c(p)}{pq} \le \log \frac{c(p)}{pq} \le \frac{c(p)}{pq}$. (Positivity: $\log \frac{c(p)}{pq} \ge \log 8 > \frac{1}{4}\log 4 \ge pq \log 4$; numerically the minimum over p is about 1.73, at $p = \frac{1}{2}$. The last step uses $\log u \le u$.) So $T \ge 2pqI \ge pqI$. On the other hand, $W \le \mathrm{Var}_p(f) \le pqI$ by the Efron–Stein inequality, since $\mathrm{Var}_p(f) \le \sum_i \mathbb{E}\,\mathrm{Var}^{(i)}(f) = \sum_i pq\,\mathbb{E}(D_i f)^2$. Hence $W \le T$.

In both cases

$$\mathbb{E}[(f-g)^2] \;\le\; \frac{2I(f)c(p)}{p(1-p)\log(1/4\varepsilon) + \log(c(p)/(p(1-p)))},$$

with g depending only on $m = \lfloor I(f)/\varepsilon \rfloor$ variables. (On 300 random sets, with $2 \le n \le 6$, random p and $\varepsilon \in \{0.01, \ldots, 0.9\}$, W was checked numerically against (7), T, $I/\log(2/\varepsilon)$ and pqI.)

Threshold consequence. Let A be monotone and nontrivial, with $p_{3/4} - p_{1/4} \ge K_1$ and $\min(p_{1/2}, 1-p_{1/2}) \ge K_2$. Here p_t is the unique value with $P_{p_t}(A) = t$. It is unique because $p \mapsto P_p(A)$ is a polynomial with derivative $I^p(A) > 0$ on $(0,1)$ (Russo's lemma).

(i) *The window lies inside* $[K_2/3,\, 1 - K_2/3]$. Let $X^{(1)}, X^{(2)}, X^{(3)}$ be independent with law P_p, and let U be their coordinatewise maximum. Then $U \sim P_{p'}$ with $p' = 1 - (1-p)^3 \le 3p$, and $U \in A$ whenever some $X^{(k)} \in A$, because A is an up-set. So $P_{p'}(A) \ge 1 - (1 - P_p(A))^3$. At $p = p_{1/4}$ this gives $P_{p'}(A) \ge 1 - \frac{27}{64} > \frac{1}{2}$, hence $p_{1/2} < p' \le 3p_{1/4}$, i.e. $p_{1/4} > K_2/3$.

For the other end, apply the same argument to the dual up-set $A^* = \{x : -x \notin A\}$. It satisfies $P_p(A^*) = 1 - P_{1-p}(A)$, so its critical values are $p_t^* = 1 - p_{1-t}$. This gives $1 - p_{3/4} \ge (1 - p_{1/2})/3 \ge K_2/3$.

(ii) *A point with small influence.* By Russo's lemma and the mean value theorem, there is $p^* \in (p_{1/4}, p_{3/4})$ with

$$I^{p^*}(A) = \frac{P_{p_{3/4}}(A) - P_{p_{1/4}}(A)}{p_{3/4} - p_{1/4}} = \frac{1/2}{p_{3/4} - p_{1/4}} \le \frac{1}{2K_1}.$$

  By (i), $p^* \in [\delta, 1-\delta]$ with $\delta = K_2/3$. Hence $p^* q^* \ge \delta(1-\delta) \ge \frac{2\delta}{3}$, and $c(p^*) \le C_\delta := \max_{[\delta,1-\delta]} c < \infty$, since c is continuous there (its singularity at $\frac{1}{2}$ is removable).

(iii) *Junta.* Apply the first part at p^* with $\varepsilon \in (0, \frac{1}{4})$. It gives g depending on at most $m \le \frac{1}{2K_1\varepsilon}$ variables, with

$$\mathbb{E}_{p^*}(\mathbf{1}_A - g)^2 \;\le\; \frac{2C_\delta/(2K_1)}{(2\delta/3)\log \frac{1}{4\varepsilon}} =: \tau(\varepsilon) \;\xrightarrow[\varepsilon \to 0]{}\; 0.$$

To get the bound $\tau(\varepsilon)$, drop the positive term $\log(c/pq)$ from the denominator. Let $B = \{g \geq \frac{1}{2}\}$. It depends on the same variables, and $\mathbf{1}_{A \triangle B} \leq 4(\mathbf{1}_A - g)^2$ (Exercise 9.1): if $\mathbf{1}_A \neq \mathbf{1}_B$, then $|\mathbf{1}_A - g| \geq \frac{1}{2}$. Thus $P_{p^*}(A \triangle B) \leq 4\tau(\varepsilon)$.

Given $\eta > 0$, choose $\varepsilon = \varepsilon(\eta, K_1, K_2)$ with $4\tau(\varepsilon) \leq \eta$. Then A is η-close under P_{p^*}, where $P_{p^*}(A) \in [\frac{1}{4}, \frac{3}{4}]$, to a set determined by at most $\frac{1}{2K_1\varepsilon}$ variables. This number does not depend on n, so A is approximated by a junta.

(Using the p-free bound $\mathbb{E}(f-g)^2 \leq I/\log(2/\varepsilon)$ instead, the same conclusion holds with $\tau(\varepsilon) = \frac{1}{2K_1 \log(2/\varepsilon)}$, even without K_2. K_2 is needed only to make the printed bound small, because the $c(p)$ in its numerator is not multiplied by $p(1-p)$.) ■

Exercise 9.10

Central idea. Here p_t is the critical parameter: $P_{p_t}(A) = t$. By Russo's lemma, $\frac{1}{2} = P_{p_{3/4}}(A) - P_{p_{1/4}}(A) = \int_{p_{1/4}}^{p_{3/4}} I^p(A)\, dp$. So it suffices to bound the total influence from above on the window. For monotone A, $I_i^p = \mathrm{Cov}_p(\mathbf{1}_A, X_i)/(2p(1-p))$, and Cauchy–Schwarz then gives $I^p(A) \leq \frac{1}{2}\sqrt{n/(p(1-p))}$. A union-of-copies argument shows that the window $[p_{1/4}, p_{3/4}]$ lies inside $[\frac{1}{6}, \frac{5}{6}]$ when $p_{1/2} = \frac{1}{2}$, so $p(1-p) \geq \frac{5}{36}$ there. Together these give $p_{3/4} - p_{1/4} \geq \frac{\sqrt{5}}{6\sqrt{n}}$. Majority shows that the order $1/\sqrt{n}$ cannot be improved.

Solution. We use the notation of Section 9.5. P_p is the product measure on $\{-1,1\}^n$ with $P_p\{X_i = 1\} = p$, and $q = 1-p$. A is monotone, i.e. an up-set, and $f = \mathbf{1}_A$. $I_i^p = P_p\{i \text{ is pivotal}\}$ and $I^p = \sum_i I_i^p$. Since $p \mapsto P_p(A)$ is a polynomial with derivative $I^p(A)$ (Russo's lemma), and $P_{1/2}(A) = \frac{1}{2}$ means A is nontrivial, $I^p(A) > 0$ on $(0,1)$. So $P_p(A)$ increases strictly from 0 to 1, and p_t is well defined for every $t \in (0,1)$.

Correction to the previous version. That version read p_q as $P_q(A)$ and "proved" the n-free bound $p_{3/4} - p_{1/4} \geq 1 - e^{-1/2} \approx 0.3935$. With the correct reading this is false. Majority on $n = 101$ voters, $A = \{\sum_i x_i > 0\}$, has $p_{1/2} = \frac{1}{2}$ by symmetry, yet $p_{3/4} - p_{1/4} = 0.0669$. The statement of the exercise itself is correct.

Step 1: covariance identity. Fix i and write $f(x) = f(x^{(i,-1)}) + \mathbf{1}\{x_i = 1\} D_i f(x)$, where $D_i f = f(x^{(i,1)}) - f(x^{(i,-1)})$. By monotonicity $D_i f \in \{0,1\}$, and $D_i f = \mathbf{1}\{i \text{ pivotal}\}$. Neither $f(x^{(i,-1)})$ nor $D_i f$ depends on x_i, and X_i is independent of X_{-i}. Hence

$$\mathrm{Cov}_p(f, X_i) = \mathbb{E}_p[D_i f]\, \mathrm{Cov}_p(\mathbf{1}\{X_i = 1\}, X_i) = I_i^p \cdot 2pq,$$

because $X_i = 2 \cdot \mathbf{1}\{X_i = 1\} - 1$ gives $\mathrm{Cov}(\mathbf{1}\{X_i = 1\}, X_i) = 2\,\mathrm{Var}(\mathbf{1}\{X_i = 1\}) = 2pq$.

Step 2: upper bound on total influence. Sum Step 1 over i and apply Cauchy–Schwarz. Since $\mathrm{Var}_p(f) \leq \frac{1}{4}$ and $\mathrm{Var}_p(\sum_i X_i) = 4npq$,

$$I^p(A) = \frac{\mathrm{Cov}_p(f, \sum_i X_i)}{2pq} \leq \frac{\sqrt{\mathrm{Var}_p(f) \cdot 4npq}}{2pq} \leq \frac{1}{2}\sqrt{\frac{n}{pq}}.$$

Step 3: the window lies in $[\frac{1}{6}, \frac{5}{6}]$. Let $X^{(1)}, X^{(2)}, X^{(3)}$ be independent with law P_p, and let U be their coordinatewise maximum. Then $U \sim P_{p'}$ with $p' = 1 - q^3 \leq 3p$. Since A is an up-set, $U \in A$ whenever some $X^{(k)} \in A$, so

$$P_{p'}(A) \geq 1 - (1 - P_p(A))^3.$$

Take $p = p_{1/4}$. Then $P_{p'}(A) \geq 1 - (\frac{3}{4})^3 = \frac{37}{64} > \frac{1}{2}$, so $\frac{1}{2} = p_{1/2} < p' \leq 3p_{1/4}$, i.e. $p_{1/4} > \frac{1}{6}$.

For the other end, the dual set $A^* = \{x : -x \notin A\}$ is also an up-set. It satisfies $P_p(A^*) = 1 - P_{1-p}(A)$, so its critical values are $p_t^* = 1 - p_{1-t}$, and in particular $p_{1/2}^* = \frac{1}{2}$. Applying the same argument to A^* gives $1 - p_{3/4} = p_{1/4}^* > \frac{1}{6}$.

Hence $[p_{1/4}, p_{3/4}] \subset (\frac{1}{6}, \frac{5}{6})$. On this interval $pq \geq \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36}$, because pq is concave and symmetric about $\frac{1}{2}$.

Step 4: conclusion. By Russo's lemma, Step 2 and Step 3,

$$\frac{1}{2} = \int_{p_{1/4}}^{p_{3/4}} I^p(A)\,dp \le (p_{3/4} - p_{1/4}) \cdot \frac{1}{2}\sqrt{\frac{36\,n}{5}},$$

so

$$p_{3/4} - p_{1/4} \;\ge\; \frac{\sqrt{5}}{6} \cdot \frac{1}{\sqrt{n}} \approx \frac{0.373}{\sqrt{n}}.$$

This proves the claim with the universal constant $c = \sqrt{5}/6$. Using only the trivial window $[0,1]$ would not work, because $\sqrt{n/(pq)}$ blows up at the ends. That is why Step 3 is needed.

Checks and sharpness. Numerically, Step 1, Step 2 and the three-copy inequality of Step 3 hold on 300 random monotone sets with $n \le 6$ and random p. For majority, $\sqrt{n}\,(p_{3/4} - p_{1/4})$ is 0.50, 0.60, 0.65, 0.67, 0.674 for $n = 1, 3, 11, 101, 10001$. These values are always above $\sqrt{5}/6 \approx 0.373$. They also stay bounded, so the order $1/\sqrt{n}$ is best possible. ■

Exercise 9.11

Central idea. $\mathbb{E}f(X)$ is a polynomial in p (a finite sum of products of linear factors), hence differentiable; differentiating the product $\prod_{j=1}^{n} \mu_p(x_j)$ with $\mu_p(1) = p$, $\mu_p(-1) = 1 - p$ term-by-term, the derivative of the i-th factor is $x_i = \pm 1$, and summing over $x_i \in \{-1, 1\}$ against the frozen marginal of X_{-i} produces exactly the difference $f(X_{-i}, 1) - f(X_{-i}, -1)$.

Solution. For $p \in (0,1)$ write $\mu_p(1) = p$, $\mu_p(-1) = 1 - p$, so that

$$\mathbb{E}_p f(X) := \mathbb{E}f(X) = \sum_{x \in \{-1,1\}^n} f(x) \prod_{j=1}^{n} \mu_p(x_j).$$

This is a polynomial in p of degree at most n (each coordinate contributes a factor linear in p), hence C^∞ in p; we may differentiate term-by-term. Since $\frac{d}{dp}\mu_p(1) = 1$ and $\frac{d}{dp}\mu_p(-1) = -1$, i.e. $\frac{d}{dp}\mu_p(x_i) = x_i$ for $x_i \in \{-1, 1\}$, the product rule gives

$$\frac{d}{dp}\mathbb{E}_p f(X) = \sum_{x \in \{-1,1\}^n} f(x) \sum_{i=1}^{n} x_i \prod_{j \ne i} \mu_p(x_j).$$

Interchanging the sums and writing $x = (x_{-i}, x_i)$ with $x_{-i} \in \{-1, 1\}^{n-1}$,

$$\frac{d}{dp}\mathbb{E}_p f(X) = \sum_{i=1}^{n} \sum_{x_{-i}} \sum_{x_i \in \{-1,1\}} f(x_{-i}, x_i)\, x_i \prod_{j \ne i} \mu_p(x_j).$$

For fixed i and x_{-i}, the inner sum over x_i is

$$f(x_{-i}, 1) \cdot 1 + f(x_{-i}, -1) \cdot (-1) = f(x_{-i}, 1) - f(x_{-i}, -1),$$

while $\prod_{j \ne i} \mu_p(x_j) = \mathbb{P}_p\{X_{-i} = x_{-i}\}$ is the marginal of the remaining coordinates. Hence the i-th summand equals

$$\sum_{x_{-i}} \big(f(x_{-i}, 1) - f(x_{-i}, -1)\big)\, \mathbb{P}_p\{X_{-i} = x_{-i}\} = \mathbb{E}_p\big[f(X_1, \dots, X_{i-1}, 1, X_{i+1}, \dots, X_n) - f(X_1, \dots, X_{i-1}, -1, X_{i+1}, \dots, X_n)\big],$$

completing the proof. For $f = \mathbf{1}_A$ with A increasing, the bracketed difference is $\mathbf{1}\{i \text{ is pivotal for } A\}$, recovering the classical Russo's lemma. ■

Exercise 9.12

Central idea. Induct on the dimension by splitting on the last coordinate. If $A^{\pm} \subseteq \{-1,1\}^{n-1}$ are the two sections of A, the total influence decomposes exactly as $I(A) = \frac{1}{2}(I(A^+) + I(A^-)) + \mu(A^+ \triangle A^-)$, and the distance obeys $d((x',\pm 1), A) = \min\big(d(x', A^{\pm}),\, 1 + d(x', A^{\mp})\big)$. Keeping the minimum (via $\mathbb{E}\min \le \min \mathbb{E}$) rather than averaging the two sections is what closes the induction: the leftover term reduces to $\frac{(b-a)\delta}{2} \le |a-b|$ with $\delta \le 1$. (No greedy paths or injections are used; the earlier injectivity route was false and is abandoned.)

Solution. Throughout, $I_i(A) = \mathbb{P}\{\mathbb{1}_A(X) \neq \mathbb{1}_A(X^{(i)})\}$ and $I(A) = \sum_{i=1}^n I_i(A)$, as defined in Section 9.1 (recalled from Chapter 4), where $X^{(i)}$ is X with its i-th coordinate flipped. The statement only makes sense for $A \neq \emptyset$, which we assume. We prove, by induction on $n \ge 0$, that for every nonempty $A \subseteq \{-1,1\}^n$,

$$2\,\mathbb{P}(A)\,\mathbb{E}d(X,A) \le I(A). \tag{1}$$

Dividing by $2\mathbb{P}(A) > 0$ gives the claim.

Base case $n = 0$. The cube is a single point, A is the whole space, $d \equiv 0$ and both sides of (1) vanish.

Induction step. Let $n \ge 1$ and assume (1) in dimension $n-1$. Write $x = (x', x_n)$ with $x' \in \{-1,1\}^{n-1}$, let μ be the uniform probability on $\{-1,1\}^{n-1}$, and let X' be uniform on $\{-1,1\}^{n-1}$. Define the sections

$$A^+ = \{x' : (x',1) \in A\}, \qquad A^- = \{x' : (x',-1) \in A\}, \qquad a = \mu(A^+),\ b = \mu(A^-),$$

so $\mathbb{P}(A) = \frac{a+b}{2}$.

Influences. For $i < n$, conditioning on X_n,

$$I_i(A) = \tfrac{1}{2}\,\mu\{\mathbb{1}_{A^+}(x') \neq \mathbb{1}_{A^+}(x'^{(i)})\} + \tfrac{1}{2}\,\mu\{\mathbb{1}_{A^-}(x') \neq \mathbb{1}_{A^-}(x'^{(i)})\} = \tfrac{1}{2}(I_i(A^+) + I_i(A^-)),$$

and $I_n(A) = \mu\{x' : \mathbb{1}_{A^+}(x') \neq \mathbb{1}_{A^-}(x')\} = \mu(A^+ \triangle A^-)$. Hence

$$I(A) = \tfrac{1}{2}(I(A^+) + I(A^-)) + \mu(A^+ \triangle A^-), \tag{2}$$

with the convention $I(\emptyset) = 0$. Note $\mu(A^+ \triangle A^-) \ge |\mu(A^+) - \mu(A^-)| = |a-b|$.

Distances. A point $y \in A$ is either $(y',1)$ with $y' \in A^+$, at distance $d(x',y') + \mathbb{1}\{x_n \neq 1\}$ from x, or $(y',-1)$ with $y' \in A^-$, at distance $d(x',y') + \mathbb{1}\{x_n \neq -1\}$. Therefore, with $d(\cdot,\emptyset) = +\infty$,

$$d((x',1),A) = \min\big(d(x',A^+),\, 1 + d(x',A^-)\big), \qquad d((x',-1),A) = \min\big(d(x',A^-),\, 1 + d(x',A^+)\big). \tag{3}$$

Since $A \neq \emptyset$, at least one section is nonempty. The map $(x', x_n) \mapsto (x', -x_n)$ preserves the uniform measure, distances and influences and swaps A^+ and A^-, so we may relabel freely.

Case 1: both sections nonempty. Put $\Phi_{\pm} = \mathbb{E}d(X', A^{\pm}) < \infty$ and, after relabeling, assume $\Phi_+ \ge \Phi_-$. Let $\delta = \min(\Phi_+ - \Phi_-, 1) \in [0,1]$. By (3) and $\mathbb{E}\min(U,V) \le \min(\mathbb{E}U, \mathbb{E}V)$,

$$\mathbb{E}d(X,A) = \tfrac{1}{2}\,\mathbb{E}\min\big(d(X',A^+), 1+d(X',A^-)\big) + \tfrac{1}{2}\,\mathbb{E}\min\big(d(X',A^-), 1+d(X',A^+)\big) \le \tfrac{1}{2}\big[\min(\Phi_+, 1+\Phi_-) + \Phi_-\big] = \Phi_- + \tfrac{\delta}{2}.$$

By the induction hypothesis, $I(A^+) \ge 2a\Phi_+$ and $I(A^-) \ge 2b\Phi_-$. With (2), $\Phi_+ \ge \Phi_- + \delta$ and $\mu(A^+ \triangle A^-) \ge |a-b|$,

$$I(A) \ge a\Phi_+ + b\Phi_- + |a-b| \ge (a+b)\Phi_- + a\delta + |a-b|.$$

Meanwhile $2\mathbb{P}(A)\,\mathbb{E}d(X,A) \le (a+b)\Phi_- + \frac{(a+b)\delta}{2}$. So (1) follows once

$$\frac{(a+b)\delta}{2} \le a\delta + |a-b|, \qquad \text{i.e.} \qquad \frac{(b-a)\delta}{2} \le |a-b|.$$

If $b \le a$ the left side is ≤ 0. If $b > a$, it is $\frac{(b-a)\delta}{2} \le \frac{b-a}{2} \le |a-b|$, because $\delta \le 1$.

Case 2: one section empty. After relabeling, $A^+ = \emptyset$ and $A^- \neq \emptyset$, so $a = 0$ and $b > 0$. By (3), $d((x', 1), A) = 1 + d(x', A^-)$ and $d((x', -1), A) = d(x', A^-)$. Hence, with $\Phi_- = \mathbb{E}d(X', A^-)$,

$$\mathbb{E}d(X, A) = \Phi_- + \tfrac{1}{2}, \qquad 2\mathbb{P}(A) = b, \qquad I(A) = \tfrac{1}{2}I(A^-) + \mu(A^-) = \tfrac{1}{2}I(A^-) + b$$

by (2). The induction hypothesis gives $b\Phi_- \leq \frac{1}{2}I(A^-)$, so

$$2\mathbb{P}(A)\,\mathbb{E}d(X, A) = b\Phi_- + \tfrac{b}{2} \leq \tfrac{1}{2}I(A^-) + b = I(A).$$

This completes the induction, so for every nonempty $A \subseteq \{-1, 1\}^n$,

$$\mathbb{E}d(X, A) \leq \frac{I(A)}{2\mathbb{P}(A)}.$$

Checks. For a single point $A = \{a\}$: $\mathbb{E}d = \frac{n}{2}$, $I_i(A) = 2 \cdot 2^{-n}$ and $\mathbb{P}(A) = 2^{-n}$, so the bound is n. For the half-cube $A = \{x_1 = 1\}$: $\mathbb{E}d = \frac{1}{2}$ and the bound is $\frac{1}{2 \cdot \frac{1}{2}} = 1$. An exhaustive computer check of (1) over all nonempty $A \subseteq \{-1, 1\}^n$ for $n \leq 4$ gives a largest ratio $\mathbb{E}d(X, A) \big/ \frac{I(A)}{2\mathbb{P}(A)}$ of exactly $\frac{1}{2}$ for each $n \leq 4$; on 3,000 uniformly random sets for $n = 6$ the ratio stays below $\frac{1}{2}$ (largest ≈ 0.23). ■

Chapter 10 — Isoperimetry on the Hypercube and Gaussian Spaces

Exercise 10.1

Central idea. Write $f(p) = P_p(A)$. By Russo's lemma $f'(p) = I^p(A)$ (total influence), and the mean value theorem gives $p^* \in (p_{1/2}, p_{1-\varepsilon})$ with $f'(p^*) = (1/2 - \varepsilon)/(p_{1-\varepsilon} - p_{1/2})$. Corollary 10.10 (isoperimetric inequality for monotone sets) lower-bounds the total influence by $\gamma(f(p^*))^2/(2p^* P_{p^*}(\partial_V A)\log(1/p^*))$. Solving for the threshold width and using $\gamma(f(p^*)) \geq \gamma(1-\varepsilon) \gtrsim \varepsilon\sqrt{\log(1/\varepsilon)}$ (Lemma below, with the sharp constant $c_0 = 2/(\pi \log 2)$) plus the global bound $p\log(1/p) \leq 1/e$ yields the width bound. The lower tail $p_{1/2} - p_\varepsilon$ follows by applying the same argument to $p \mapsto 1 - P_{1-p}(A)$.

Erratum. The statement is false as printed: the bound uses $\inf_{q \in (1/2, 1-\varepsilon)} P_q(\partial_V(A))$, but the critical window $(p_{1/2}, p_{1-\varepsilon})$ need not lie in $(1/2, 1-\varepsilon)$, and an infimum goes the wrong way in any case. Counterexample: the OR set $A = \{x : \text{some } x_i = 1\}$ with $n = 20$, $\varepsilon = 1/4$. Here $P_p(A) = 1 - (1-p)^{20}$, so $p_a = 1 - (1-a)^{1/20}$, $p_{1/2} \approx 0.03406$, $p_{3/4} \approx 0.06697$, and the width is $p_{3/4} - p_{1/2} \approx 0.0329$. The prefactor is $\frac{(\log 2)(1/4)}{(1/16)\log 4} = 2$, and $P_q(\partial_V(A))$ is decreasing on $(1/2, 3/4)$, so the printed right side is $2 \cdot 20 \cdot \frac{3}{4} \cdot (\frac{1}{4})^{19} \approx 1.1 \times 10^{-10}$ for the inner vertex boundary ($P_q = nq(1-q)^{n-1}$) and $2 \cdot (\frac{1}{4})^{20} \approx 1.8 \times 10^{-12}$ for the outer one ($P_q = (1-q)^n$). (Even a sup over the printed interval $(1/2, 3/4)$ gives only $2 \cdot 20 \cdot 2^{-20} \approx 3.8 \times 10^{-5}$.) Corrected statement (proved below, via Corollary 10.10):

$$p_{1-\varepsilon} - p_{1/2} \leq \frac{\pi \log 2}{e} \cdot \frac{1/2 - \varepsilon}{\varepsilon^2 \log(1/\varepsilon)} \sup_{q \in (p_{1/2},\, p_{1-\varepsilon})} P_q(\partial_V(A)),$$

and analogously $p_{1/2} - p_\varepsilon$ with the sup over $(p_\varepsilon, p_{1/2})$. In the example the corrected right side is ≈ 0.87 (inner) or ≈ 1.16 (outer), consistent with the width 0.0329. Whether the printed constant $\log 2$ suffices together with the sup is open: it is neither proved nor refuted here (the Corollary 10.10 route gives $\pi \log 2/e$; see the final remark).

Solution. Let $f(p) = P_p(A)$ for $p \in [0, 1]$, where P_p is the product of Bernoulli(p) measures on $\{-1, 1\}^n$ (with $P\{X_i = 1\} = p$). Since A is nonempty and monotone, $(1, \ldots, 1) \in A$; if also $(-1, \ldots, -1) \in A$ then $A = \{-1, 1\}^n$ and $P_p(A) \equiv 1$, contradicting the uniqueness of p_a. Hence $(-1, \ldots, -1) \notin A$, so f is continuous and strictly increasing with $f(0) = 0$, $f(1) = 1$. For each $a \in [0, 1]$ there is therefore a unique p_a with $f(p_a) = a$.

By Russo's lemma (Section 9.6), f is differentiable on $(0,1)$ with

$$f'(p) = I^p(A) := \sum_{i=1}^{n} I_i^p(A),$$

the total influence of A under P_p.

Fix $\varepsilon \in (0, 1/2)$. Applying the mean value theorem to f on $[p_{1/2}, p_{1-\varepsilon}]$, there exists $p^* \in (p_{1/2}, p_{1-\varepsilon})$ such that

$$f'(p^*) = \frac{f(p_{1-\varepsilon}) - f(p_{1/2})}{p_{1-\varepsilon} - p_{1/2}} = \frac{1/2 - \varepsilon}{p_{1-\varepsilon} - p_{1/2}}. \tag{1}$$

Corollary 10.10, in its "in particular" form, asserts that for every $p \in (0,1)$ and monotone A,

$$2\, P_p(\partial_V(A)) \cdot I^p(A) \cdot p \log \frac{1}{p} \;\geq\; \gamma(P_p(A))^2, \tag{2}$$

where $\gamma(s) = \phi(\Phi^{-1}(s))$ is the Gaussian isoperimetric function and $\partial_V(A)$ is the vertex boundary. Since $I^{p^*}(A) = f'(p^*)$, (2) at $p = p^*$ gives

$$f'(p^*) \;\geq\; \frac{\gamma(f(p^*))^2}{2\, p^*\, P_{p^*}(\partial_V(A))\, \log(1/p^*)}. \tag{3}$$

Combining (1) and (3) and solving for $p_{1-\varepsilon} - p_{1/2}$:

$$p_{1-\varepsilon} - p_{1/2} \;\leq\; \frac{2(1/2 - \varepsilon)\, p^* \log(1/p^*)}{\gamma(f(p^*))^2}\; P_{p^*}(\partial_V(A)). \tag{4}$$

We now bound the three factors. First, $f(p^*) \in (1/2, 1-\varepsilon)$ because $p^* \in (p_{1/2}, p_{1-\varepsilon})$ and f is strictly increasing. The function γ is symmetric about $1/2$ and strictly decreasing on $[1/2, 1]$, hence $\gamma(f(p^*)) \geq \gamma(1-\varepsilon)$.

Second, the function $p \mapsto p \log(1/p)$ attains its global maximum $1/e$ at $p = 1/e$ on $(0,1)$, so for every $p^* \in (0,1)$,

$$p^* \log(1/p^*) \leq \frac{1}{e}. \tag{5}$$

(Note that p^* need not lie in $[1/2, 1)$: e.g. for $A = \{x : \text{some } x_i = 1\} \subset \{-1,1\}^2$, $f(p) = 1 - (1-p)^2$ and $p_{1/2} = 1 - 1/\sqrt{2} \approx 0.293 < 1/2$, so the bound $\frac{1}{2}\log 2$ valid only on $[1/2, 1)$ does not apply; $1/e$ is the sharp uniform bound.)

Third, we use the corrected Gaussian tail estimate

$$\gamma(1-\varepsilon)^2 \;\geq\; c_0\, \varepsilon^2 \log \frac{1}{\varepsilon} \qquad (0 < \varepsilon < 1/2), \tag{6}$$

with the sharp constant $c_0 = \frac{2}{\pi \log 2} \approx 0.9184$. (The estimate with $c_0 = 1$ is false: $\gamma(1-\varepsilon)^2/(\varepsilon^2 \log(1/\varepsilon)) \to 2/(\pi \log 2)$ as $\varepsilon \to \frac{1}{2}^-$, and e.g. at $\varepsilon = 0.45$ the left side is $0.1567 < 0.1617$. Nor does it follow from Mill's ratio $\phi(t) \geq t(1 - \Phi(t))$, which would require $t^2 \geq -\log(1 - \Phi(t))$, false as $t \to 0$.)

Lemma. For $0 < \varepsilon < 1/2$, $\gamma(1-\varepsilon)^2 \geq c_0 \varepsilon^2 \log(1/\varepsilon)$ with $c_0 = 2/(\pi \log 2)$, best possible. *Proof.* Write $t = \Phi^{-1}(1-\varepsilon) > 0$, so $\gamma(1-\varepsilon) = \phi(t)$, $\varepsilon = 1 - \Phi(t)$. Put $L(t) = -\log(1 - \Phi(t))$, $\lambda(t) = \phi(t)/(1 - \Phi(t))$, and

$$F(t) = \phi(t)^2 - c_0 (1 - \Phi(t))^2 L(t), \qquad t \geq 0.$$

Then $F(0) = \frac{1}{2\pi} - c_0 \frac{\log 2}{4} = 0$. Differentiating, with $\lambda'(t) = \lambda(t)(\lambda(t) - t)$,

$$F'(t) = \phi(t)(1 - \Phi(t))\, G(t), \qquad G(t) = 2c_0 L(t) - 2t\lambda(t) - c_0.$$

Now $G(0) = c_0(2\log 2 - 1) > 0$; $G'(t) = 2(c_0 - 1)\lambda(t) - 2t\lambda(t)(\lambda(t) - t) < 0$ for $t > 0$ since $c_0 < 1$ and $\lambda(t) > t > 0$ (Mill's inequality $1 - \Phi(t) < \phi(t)/t$); and $G(t) \to -\infty$ as $t \to \infty$ (from $1 - \Phi(t) \geq t\phi(t)/(1 + t^2)$ we get $L(t) \leq \frac{1}{2}\log(2\pi) + \log(1 + t^2) - \log t + t^2/2$, while $-2t\lambda(t) \leq -2t^2$, so $G(t) \leq (c_0 - 2)t^2 + O(\log t)$ with $c_0 < 2$). Hence G has exactly one zero $t^* > 0$: F strictly increases on $[0, t^*]$ and strictly decreases on $[t^*, \infty)$. Since $F(0) = 0$ and $F(t) \to 0$ (as $\phi(t)^2 \to 0$ and $(1 - \Phi(t))^2 L(t) \leq (\phi(t)/t)^2 \cdot O(t^2) \to 0$), we obtain $F(t) \geq 0$ for all $t \geq 0$, i.e. (6). Sharpness follows from $\phi(t)^2/((1 - \Phi(t))^2 L(t)) \to \frac{1/(2\pi)}{(\log 2)/4} = c_0$ as $t \to 0^+$. ■

Inserting $\gamma(f(p^*)) \geq \gamma(1 - \varepsilon)$, (5), and (6) into (4), and using $\frac{2}{ec_0} = \frac{\pi \log 2}{e}$:

$$p_{1-\varepsilon} - p_{1/2} \;\leq\; \frac{\pi \log 2}{e} \cdot \frac{1/2 - \varepsilon}{\varepsilon^2 \log(1/\varepsilon)}\, P_{p^*}(\partial_V(A)). \tag{7}$$

Remark on inf *vs.* sup *and on the constant.* The estimate (7) bounds the width by the vertex boundary at the mean-value point $p^* \in (p_{1/2}, p_{1-\varepsilon})$, so

$$p_{1-\varepsilon} - p_{1/2} \;\leq\; \frac{\pi \log 2}{e} \cdot \frac{1/2 - \varepsilon}{\varepsilon^2 \log(1/\varepsilon)} \sup_{q \in (p_{1/2},\, p_{1-\varepsilon})} P_q(\partial_V(A)).$$

The problem as written has $\inf_{q \in (1/2, 1-\varepsilon)} P_q(\partial_V(A))$ with constant $\log 2$; neither follows from (7): inf $\leq$ $P_{p^*}(\partial_V(A))$ goes the wrong way, the interval $(p_{1/2}, p_{1-\varepsilon})$ need not lie in $(1/2, 1 - \varepsilon)$, and the constant $\log 2$ would require the false estimate (6) with $c_0 = 1$ together with $p^* \geq 1/2$. The corrected constant $\pi \log 2/e \approx 0.801$ exceeds the stated $\log 2 \approx 0.693$ by the factor $\pi/e \approx 1.156$ (from $1/c_0 = \pi \log 2/2$ and $(1/e)/(\frac{1}{2}\log 2)$). The inf/interval defect alone makes the printed statement false (see the Erratum); the constant shortfall is a limitation of this proof route, not a demonstrated falsity.

Lower tail. For $p_{1/2} - p_\varepsilon$, apply the same argument to $g(p) := 1 - f(1 - p)$. Then g is continuous strictly increasing with $g(0) = 0$, $g(1) = 1$; writing $g(p) = P_p(B)$ where $B = \{x : -x \notin A\}$ (nonempty and monotone, since A is, and $(-1, \ldots, -1) \notin B$ since $(1, \ldots, 1) \in A$), one checks $p_a^{(g)} = 1 - p_{1-a}$ (with $p_\cdot$ for f) and $P_q(\partial_V(B)) = P_{1-q}(\partial_V(A))$ (pivotality is preserved under $x \mapsto -x$, which maps P_q to P_{1-q}). Hence $p_{1-\varepsilon}^{(g)} - p_{1/2}^{(g)} = p_{1/2} - p_\varepsilon$, and (7) applied to B gives, with $r = 1 - q$,

$$p_{1/2} - p_\varepsilon \;\leq\; \frac{\pi \log 2}{e} \cdot \frac{1/2 - \varepsilon}{\varepsilon^2 \log(1/\varepsilon)} \sup_{r \in (p_\varepsilon,\, p_{1/2})} P_r(\partial_V(A)),$$

the analogous upper bound for the lower tail (with the same inf/sup and constant remarks as above). ■

Exercise 10.2

Central idea. In one dimension half-lines are isoperimetrically extremal. Testing the definition of α_P on $A = (-\infty, x]$ gives $\kappa(P) \leq \operatorname{ess\,inf} f / \min(F, 1 - F)$. For the reverse, the pointwise bound $f \geq c \min(F, 1 - F)$ (with c the essinf) implies the L^1 Poincaré inequality $\mathbb{E}_P|u'| \geq c\, \mathbb{E}_P|u - m|$ for Lipschitz u with median m: by the coarea formula this reduces to a level-set estimate, and since m is a median each component interval of $\{u \geq s\}$ carries P-mass at most $1/2$, where the elementary bound $\min(\alpha, 1 - \alpha) + \min(\beta, 1 - \beta) \geq \beta - \alpha$ applies. Transferring to sets through the Lipschitz approximants $u_t(x) = \min(1, (t - d(x, A))_+/t)$ of $\mathbf{1}_{\bar{A}}$ (with $\mathbb{E}|u_t'| \leq P(A_t \setminus A)/t$) yields $\alpha_P(p) \geq c \min(p, 1 - p)$.

Solution. Let P be absolutely continuous on $\mathbb{R}$ with density f and distribution function F, and put

$$c = \operatorname*{ess\,inf}_{a < x < b} \frac{f(x)}{\min(F(x), 1 - F(x))}, \qquad \psi(x) = \min(F(x), 1 - F(x)),$$

with $a = \inf\{x : F(x) > 0\}$, $b = \sup\{x : F(x) < 1\}$. For $x \in (a, b)$ one has $0 < F(x) < 1$. Since $f \geq c\psi$ a.e. on (a, b) and $f = \psi = 0$ a.e. outside $[a, b]$, we have $f \geq c\psi$ a.e. on $\mathbb{R}$. Below $\liminf_{t \to 0}$ means $t \downarrow 0$ (the t-enlargement).

Upper bound. Let $x \in (a,b)$ be a point of differentiability of F (a.e. x). For $A = (-\infty, x]$, $P(A) = F(x) \in (0,1)$ and $A_t = (-\infty, x+t)$, so

$$\liminf_{t \downarrow 0} \frac{P(A_t \setminus A)}{t} \le \lim_{t \downarrow 0} \frac{F(x+t) - F(x)}{t} = f(x).$$

Hence $\alpha_P(F(x)) \le f(x)$ for a.e. $x \in (a,b)$, and therefore

$$\kappa(P) = \inf_{p \in (0,1)} \frac{\alpha_P(p)}{\min(p, 1-p)} \le \frac{f(x)}{\min(F(x), 1-F(x))} \quad \text{for a.e. } x,$$

so $\kappa(P) \le c$.

Functional inequality. We claim that for every Lipschitz $u : \mathbb{R} \to \mathbb{R}$ with median m,

$$\int |u'| \, dP \ge c \int |u - m| \, dP. \tag{$*$}$$

Translating, assume $m = 0$. By the coarea formula,

$$\int_{\mathbb{R}} |u'(x)| \psi(x) \, dx = \int_{-\infty}^{\infty} \Big(\sum_{u(x)=t} \psi(x) \Big) \, dt,$$

and since $f \ge c\psi$ a.e., it suffices to show $\int |u'| \psi \, dx \ge \int |u| \, dP = \int_0^\infty P(|u| > s) \, ds$.

Fix $s > 0$. The closed set $\{u \ge s\}$ is a countable disjoint union of maximal closed intervals $[a_j, b_j]$ (points and rays allowed), with distinct finite endpoints: if e.g. $b_j = a_k$ for $j \ne k$, the union would be a strictly larger interval in $\{u \ge s\}$. By continuity and maximality, $u = s$ at each finite endpoint. Since 0 is a median, $P(u \ge s) \le P(u > 0) \le 1/2$, so each component has P-mass at most $1/2$. We claim

$$\psi(a_j) \mathbf{1}_{\{a_j > -\infty\}} + \psi(b_j) \mathbf{1}_{\{b_j < \infty\}} \ge P([a_j, b_j]). \tag{1}$$

If $[a_j, b_j]$ is bounded, $P([a_j, b_j]) = F(b_j) - F(a_j) \le 1/2$ by continuity of F; writing $\alpha = F(a_j)$, $\beta = F(b_j)$, the estimate $\min(\alpha, 1-\alpha) + \min(\beta, 1-\beta) \ge \beta - \alpha$ holds in all three cases: $\alpha + \beta \ge \beta - \alpha$ if $\beta \le 1/2$; $(1-\alpha) + (1-\beta) \ge \beta - \alpha$ if $\alpha \ge 1/2$; and $\alpha + (1-\beta) \ge \beta - \alpha \iff \beta - \alpha \le 1/2$ if $\alpha < 1/2 < \beta$. If $[a_j, \infty)$, then $P = 1 - F(a_j) \le 1/2$ forces $F(a_j) \ge 1/2$, so $\psi(a_j) = 1 - F(a_j) = P$; likewise for $(-\infty, b_j]$. The case $(-\infty, \infty)$ is impossible since $P(u \ge s) \le 1/2$.

Summing (1) over j gives $\sum_{u(x)=s} \psi(x) \ge P(u \ge s) \ge P(u > s)$. Applying the same argument to $\{u \le -s\}$ (using $P(u \le -s) \le P(u < 0) \le 1/2$) gives $\sum_{u(x)=-s} \psi(x) \ge P(u \le -s) \ge P(u < -s)$. Hence

$$\int |u'| \psi \, dx = \int_0^\infty \Big[\sum_{u=s} \psi + \sum_{u=-s} \psi \Big] \, ds \ge \int_0^\infty \left[P(u > s) + P(u < -s) \right] ds = \mathbb{E}|u|,$$

proving $(*)$.

From functions to sets. Let A be measurable with $P(A) = p \in (0,1)$ and put $\bar{p} = P(\bar{A}) \ge p$. If $\bar{p} > p$, then $\bar{A} \subseteq A_t$ for all $t > 0$ gives $P(A_t \setminus A) \ge \bar{p} - p > 0$, so $\liminf_{t \downarrow 0} P(A_t \setminus A)/t = +\infty$. Assume $\bar{p} = p$ and set $u_t(x) = \min\{1, (t - d(x, A))_+ / t\}$ with $d(x, A)$ the distance to A. Then u_t is $1/t$-Lipschitz, $u_t \to \mathbf{1}_{\bar{A}}$ pointwise, and $u_t' = 0$ a.e. outside $\{0 < d(\cdot, A) < t\} \subseteq A_t \setminus A$, so $\mathbb{E}|u_t'| \le P(A_t \setminus A)/t$. Applying $(*)$ with medians m_t,

$$\frac{P(A_t \setminus A)}{t} \ge c \, \mathbb{E}|u_t - m_t| = c \min_m \mathbb{E}|u_t - m| \xrightarrow[t \downarrow 0]{} c \min_m \mathbb{E}|\mathbf{1}_{\bar{A}} - m| = c \min(p, 1-p),$$

since $|\min_m \mathbb{E}|u_t - m| - \min_m \mathbb{E}|\mathbf{1}_{\bar{A}} - m|| \le \mathbb{E}|u_t - \mathbf{1}_{\bar{A}}| \to 0$ by dominated convergence. Thus $\alpha_P(p) \ge c \min(p, 1-p)$ for every p, i.e. $\kappa(P) \ge c$. With the upper bound, $\kappa(P) = c$.

Standard Gaussian. Here $f = \phi$, $F = \Phi$, and by symmetry $f(x)/\min(\Phi(x), 1 - \Phi(x))$ takes exactly the values $g(x) = \phi(x)/\Phi(-x)$ for $x \ge 0$. Since $(\log g)'(x) = -x + g(x) > 0$ for $x > 0$ by Mill's inequality $\Phi(-x) = \int_x^\infty \phi(t) \, dt \le \int_x^\infty \frac{t}{x} \phi(t) \, dt = \phi(x)/x$, g is strictly increasing, so $\kappa(P) = g(0) = \phi(0)/\Phi(0) = \sqrt{2/\pi}$.

Laplace. With $f(x) = \frac{1}{2} e^{-|x|}$ we have $F(x) = \frac{1}{2} e^x$ for $x \le 0$ and $F(x) = 1 - \frac{1}{2} e^{-x}$ for $x \ge 0$, whence $\min(F(x), 1 - F(x)) = f(x)$ for all x. The ratio is identically 1, so $\kappa(P) = 1$. ∎

Exercise 10.3

Central idea. Monotone transport from the Laplace law ν (density $\frac{1}{2}e^{-|y|}$) to P is $(1/\kappa(P))$-Lipschitz, because by Exercise 10-2 the density of P is at least $\kappa(P)\min(F, 1-F)$ while ν's density equals $\min(F_\nu, 1-F_\nu)$. The coordinatewise product map is then $(1/\kappa(P))$-Lipschitz on $\mathbb{R}^n$, and Lipschitz images lose at most the Lipschitz constant in the Cheeger constant, so $\kappa(P^n) \geq \kappa(P)\,\kappa(\nu^n)$. For $\kappa(\nu^n)$ we use Ledoux's semigroup (Buser-type) argument, which avoids any one-dimensional functional inequality. Smooth the convex potential $\sum_i |x_i|$ to $V_\varepsilon = \sum_i \sqrt{x_i^2+\varepsilon^2}$, which costs factors $e^{O(n\varepsilon)}$. For a diffusion with convex potential: - reverse Poincaré gives $|\nabla P_s g| \leq \|g\|_\infty/\sqrt{2s}$, hence $\|\mathbb{1}_A - P_t\mathbb{1}_A\|_1 \leq \sqrt{2t}\,\mu^+(A)$; - the spectral gap $\lambda \geq \frac{1}{4}$ (Laplace Poincaré constant 4, tensorized) gives $\|\mathbb{1}_A - P_t\mathbb{1}_A\|_1 \geq 2p(1-p)(1-e^{-\lambda t})$.

Optimizing in t yields $\kappa(\nu^n) \geq 0.2256 > \frac{1}{2\sqrt{6}} = 0.2041$.

Solution. We use the notation of Exercise 10-2: P is an absolutely continuous law on $\mathbb{R}$ with density f, distribution function F and support endpoints $a < b$. For a law μ on $\mathbb{R}^n$ and measurable A, put $\mu^+(A) = \liminf_{t\downarrow 0} \mu(A_t \setminus A)/t$, where $A_t = \{x : d(x,A) < t\}$. Then $\kappa(\mu) = \inf\{\mu^+(A)/\min(\mu(A), 1-\mu(A)) : 0 < \mu(A) < 1\}$. Let $k = \kappa(P)$; if $k = 0$ there is nothing to prove, so assume $k > 0$. Let ν be the Laplace law with distribution function F_ν, and write $\varphi(u) = \min(u, 1-u)$. As computed in Exercise 10-2, $f_\nu = \varphi \circ F_\nu$ everywhere, and $f \geq k\,\varphi \circ F$ a.e. on (a,b). In particular F is continuous and strictly increasing on (a,b).

Step 1: $\kappa(P^n) \geq k\,\kappa(\nu^n)$. Let $G = (F|_{(a,b)})^{-1} : (0,1) \to (a,b)$ and $T = G \circ F_\nu : \mathbb{R} \to (a,b)$. Since $F_\nu(Y)$ is uniform on $(0,1)$ for $Y \sim \nu$, we have $T_\# \nu = P$.

T is $(1/k)$-Lipschitz. Let $y_1 < y_2$, $x_j = T(y_j)$ and $u_j = F(x_j) = F_\nu(y_j)$. For an absolutely continuous nondecreasing H and Borel $h \geq 0$, $\int_{s_1}^{s_2} h(H(s))H'(s)\,ds = \int_{H(s_1)}^{H(s_2)} h(u)\,du$. This holds for indicators of intervals by the fundamental theorem of calculus and extends by monotone class. Applying it to $H = F_\nu$ and to $H = F$ with $h = 1/\varphi$ gives

$$y_2 - y_1 = \int_{y_1}^{y_2} \frac{f_\nu(y)}{\varphi(F_\nu(y))}\,dy = \int_{u_1}^{u_2} \frac{du}{\varphi(u)} = \int_{x_1}^{x_2} \frac{f(x)}{\varphi(F(x))}\,dx \geq k\,(x_2 - x_1).$$

Hence $T^{\otimes n}(y) = (T(y_1), \dots, T(y_n))$ satisfies $|T^{\otimes n}y - T^{\otimes n}y'|^2 = \sum_i |T(y_i) - T(y_i')|^2 \leq k^{-2}|y - y'|^2$, and $T^{\otimes n}_\# \nu^n = P^n$.

Lipschitz images. Let S be L-Lipschitz with $S_\#\mu = \mu'$, and let A satisfy $\mu'(A) = p \in (0,1)$. Put $B = S^{-1}A$, so $\mu(B) = p$. If $d(y,B) < t/L$, pick $y' \in B$ with $|y - y'| < t/L$; then $|Sy - Sy'| < t$, so $Sy \in A_t$. Thus $B_{t/L} \subseteq S^{-1}(A_t)$ and $\mu'(A_t \setminus A) \geq \mu(B_{t/L} \setminus B)$, giving $\mu'^+(A) \geq \mu^+(B)/L \geq \kappa(\mu)\min(p, 1-p)/L$. Hence $\kappa(\mu') \geq \kappa(\mu)/L$. With $S = T^{\otimes n}$ and $L = 1/k$,

$$\kappa(P^n) \geq k\,\kappa(\nu^n).$$

Step 2: reduction to closed sets and to a smooth potential. Let $A \subseteq \mathbb{R}^n$ with $\nu^n(A) = p \in (0,1)$. - If $\nu^n(\bar{A}) > p$, then $A_t \supseteq \bar{A}$ gives $\nu^n(A_t \setminus A) \geq \nu^n(\bar{A}) - p > 0$ for all t, so $(\nu^n)^+(A) = \infty$. - Otherwise $\nu^n(\bar{A}) = p$ and $A_t = \bar{A}_t$, so $(\nu^n)^+(A) = (\nu^n)^+(\bar{A})$.

So it suffices to treat closed A.

Fix $\varepsilon \in (0,1)$, let $V(x) = \sum_i |x_i|$ and $V_\varepsilon(x) = \sum_i \sqrt{x_i^2+\varepsilon^2}$, and let $\mu = \mu_\varepsilon = Z_\varepsilon^{-1} e^{-V_\varepsilon} dx$. Since $V \leq V_\varepsilon \leq V + n\varepsilon$, we have $Z_\varepsilon \in [2^n e^{-n\varepsilon}, 2^n]$, and the density $d\mu/d\nu^n = 2^n e^{-(V_\varepsilon - V)}/Z_\varepsilon$ lies in $[e^{-n\varepsilon}, e^{n\varepsilon}]$. Hence for closed A with $\nu^n(A) = p$ and $q = \mu(A)$:

$$(\nu^n)^+(A) \geq e^{-n\varepsilon}\mu^+(A), \qquad \min(q, 1-q) \geq e^{-n\varepsilon}\min(p, 1-p) > 0. \tag{1}$$

Poincaré inequality. First take $n = 1$, g Lipschitz on $[0,\infty)$ with $g(0) = 0$. Then $g^2 e^{-x} \to 0$ at ∞, and integrating by parts and applying Cauchy–Schwarz,

$$\int_0^\infty g^2 e^{-x} dx = 2\int_0^\infty g g' e^{-x} dx \leq 2\Big(\int_0^\infty g^2 e^{-x}\Big)^{1/2}\Big(\int_0^\infty g'^2 e^{-x}\Big)^{1/2},$$

so $\int_0^\infty g^2 e^{-x} \le 4\int_0^\infty g'^2 e^{-x}$. The same holds on $(-\infty,0]$. With $g = h - h(0)$ this gives $\mathrm{Var}_\nu(h) \le \int (h - h(0))^2 d\nu \le 4\int h'^2 d\nu$ for Lipschitz h.

By the Efron–Stein inequality (Theorem 3.1) applied coordinatewise, $\mathrm{Var}_{\nu^n}(h) \le 4\int |\nabla h|^2 d\nu^n$ for Lipschitz h on $\mathbb{R}^n$. Transferring to μ with the density bounds,

$$\mathrm{Var}_\mu(h) \le \int \Big(h - \int h\, d\nu^n\Big)^2 d\mu \le e^{n\varepsilon}\,\mathrm{Var}_{\nu^n}(h) \le 4e^{n\varepsilon}\int |\nabla h|^2 d\nu^n \le 4e^{2n\varepsilon}\int |\nabla h|^2 d\mu.$$

By density of Lipschitz functions in $H^1(\mu)$, the spectral gap of μ satisfies

$$\lambda \ge \tfrac{1}{4} e^{-2n\varepsilon}. \tag{2}$$

Step 3: the semigroup bound for μ. The potential V_ε is C^∞ and convex. Every derivative of $s \mapsto \sqrt{s^2+\varepsilon^2}$ of order ≥ 1 is bounded; in particular $|\nabla V_\varepsilon| \le \sqrt{n}$ and $0 \le \mathrm{Hess}\, V_\varepsilon \le \varepsilon^{-1} I$. We use the following standard facts (Bakry, Gentil and Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, 2014, Chapters 1–4). The SDE $dX_t = \sqrt{2}\, dB_t - \nabla V_\varepsilon(X_t)\, dt$ has unique non-exploding strong solutions X_t^x. The semigroup $P_t g(x) = \mathbb{E} g(X_t^x)$ is Markov and symmetric in $L^2(\mu)$. Its L^2 generator L (equal to $\Delta - \nabla V_\varepsilon \cdot \nabla$ on C_c^∞) is the self-adjoint operator of the Dirichlet form $\mathcal{E}(h,h) = \int |\nabla h|^2 d\mu$ on $H^1(\mu)$, which contains all bounded Lipschitz functions. Finally, $P_t g$ is C^1 for $g \in C_c^\infty$.

(a) Gradient bound. For $g \in C_c^\infty$ with $|g| \le 1$ and $s > 0$,

$$|\nabla P_s g| \le \frac{1}{\sqrt{2s}}. \tag{3}$$

Proof. Drive X^x and X^y by the same B. The noise cancels, so $\frac{d}{dt}|X_t^x - X_t^y|^2 = -2\langle \nabla V_\varepsilon(X_t^x) - \nabla V_\varepsilon(X_t^y), X_t^x - X_t^y\rangle \le 0$ by convexity, and $|X_t^x - X_t^y| \le |x-y|$. For $h \in C^1$ with bounded gradient, the quotient $|h(X_s^x) - h(X_s^y)|/|x-y|$ is at most the supremum of $|\nabla h|$ on the segment $[X_s^y, X_s^x]$, which shrinks to X_s^x as $y \to x$. By dominated convergence (the quotients are bounded by $\|\nabla h\|_\infty$), $|\nabla P_s h| \le P_s|\nabla h|$, hence $|\nabla P_s h|^2 \le P_s(|\nabla h|^2)$ by Jensen. Now fix $t > 0$ and let $\Psi(r) = P_r((P_{t-r} g)^2)$ for $r \in [0,t]$. The standard computation (BGL §4.7, the local Poincaré inequalities under $\mathrm{CD}(0,\infty)$) gives

$$\Psi'(r) = P_r(L(h^2) - 2hLh) = 2P_r(|\nabla h|^2) \ge 2|\nabla P_r h|^2 = 2|\nabla P_t g|^2, \qquad h = P_{t-r} g.$$

Integrating over $[0,t]$: $2t|\nabla P_t g|^2 \le P_t(g^2) - (P_t g)^2 \le 1$. □

(b) Upper bound. For bounded Lipschitz f and $t > 0$,

$$\|f - P_t f\|_{L^1(\mu)} \le \sqrt{2t}\, \|\nabla f\|_{L^1(\mu)}. \tag{4}$$

Proof. Let $g \in C_c^\infty$ with $|g| \le 1$. For $0 < \delta < t$, the map $s \mapsto P_s g$ is C^1 into $L^2(\mu)$ on $(0,\infty)$ with derivative $LP_s g$ (spectral theorem). Using symmetry, then $\int f(-Lw)\, d\mu = \mathcal{E}(f,w)$, then (3):

$$\int f\,(P_\delta g - P_t g)\, d\mu = \int_\delta^t \mathcal{E}(f, P_s g)\, ds = \int_\delta^t \int \langle \nabla f, \nabla P_s g\rangle\, d\mu\, ds \le \|\nabla f\|_1 \int_0^t \frac{ds}{\sqrt{2s}} = \sqrt{2t}\, \|\nabla f\|_1.$$

Let $\delta \downarrow 0$, using $P_\delta g \to g$ in L^2, and then use symmetry: $\int (f - P_t f) g\, d\mu = \int f(g - P_t g)\, d\mu \le \sqrt{2t}\|\nabla f\|_1$. Taking the supremum over such g (which computes the $L^1(\mu)$ norm) gives (4). □

Now let A be closed with $\mu(A) = q \in (0,1)$, and let $f_r = (1 - d(\cdot, A)/r)_+$. This function is $(1/r)$-Lipschitz, equals 1 on A and 0 off A_r. Since a Lipschitz function has zero gradient a.e. on any set where it is constant, $|\nabla f_r| \le r^{-1}\mathbb{1}_{A_r \setminus A}$ a.e., and so $\|\nabla f_r\|_1 \le \mu(A_r \setminus A)/r$. As $r \downarrow 0$, $f_r \to \mathbb{1}_A$ pointwise (as A is closed) and in $L^1(\mu)$, and P_t is an L^1 contraction. So $\|f_r - P_t f_r\|_1 \to \|\mathbb{1}_A - P_t \mathbb{1}_A\|_1$. Taking r along a sequence realizing the $\liminf$ in $\mu^+(A)$, (4) gives

$$\|\mathbb{1}_A - P_t \mathbb{1}_A\|_1 \le \sqrt{2t}\, \mu^+(A). \tag{5}$$

(c) Lower bound. Since $0 \le P_t\mathbb{1}_A \le 1$ and $\int P_t\mathbb{1}_A\,d\mu = q$ (invariance),

$$\|\mathbb{1}_A - P_t\mathbb{1}_A\|_1 = \int_A (1 - P_t\mathbb{1}_A)\,d\mu + \int_{A^c} P_t\mathbb{1}_A\,d\mu = 2\Big(q - \int \mathbb{1}_A P_t\mathbb{1}_A\,d\mu\Big) = 2(q - \|P_{t/2}\mathbb{1}_A\|_2^2),$$

using symmetry and the semigroup property. By the spectral gap (spectral theorem), $\|P_{t/2}(\mathbb{1}_A - q)\|_2^2 \le e^{-\lambda t}\|\mathbb{1}_A - q\|_2^2 = e^{-\lambda t}q(1-q)$. Since $P_{t/2}$ fixes constants and preserves means, $\|P_{t/2}\mathbb{1}_A\|_2^2 = q^2 + \|P_{t/2}(\mathbb{1}_A - q)\|_2^2$. Hence

$$\|\mathbb{1}_A - P_t\mathbb{1}_A\|_1 \ge 2q(1-q)(1 - e^{-\lambda t}). \tag{6}$$

(d) Optimization. Combining (5) and (6) with $t = s/\lambda$:

$$\mu^+(A) \ge 2q(1-q)\sqrt{\lambda}\,\frac{1 - e^{-s}}{\sqrt{2s}} \qquad (s > 0).$$

Take $s = \frac{5}{4}$; numerically $c_0 := \frac{1-e^{-5/4}}{\sqrt{5/2}} = 0.451254$. (The maximum over s is 0.451256, at $s \approx 1.2564$.) By (2), $\sqrt{\lambda} \ge \frac{1}{2}e^{-n\varepsilon}$, and $q(1-q) \ge \frac{1}{2}\min(q, 1-q)$. So

$$\mu^+(A) \ge \tfrac{c_0}{2}\,e^{-n\varepsilon}\min(q, 1-q).$$

Step 4: conclusion. By (1), for every closed A with $\nu^n(A) = p \in (0,1)$,

$$(\nu^n)^+(A) \ge e^{-n\varepsilon}\mu^+(A) \ge \tfrac{c_0}{2}e^{-2n\varepsilon}\min(q, 1-q) \ge \tfrac{c_0}{2}e^{-3n\varepsilon}\min(p, 1-p).$$

Let $\varepsilon \downarrow 0$ (n fixed); with Step 2 this gives $\kappa(\nu^n) \ge c_0/2 = 0.225627$. By Step 1,

$$\kappa(P^n) \ge 0.2256\,\kappa(P) \ge \frac{\kappa(P)}{2\sqrt{6}} \qquad \Big(\frac{1}{2\sqrt{6}} = 0.204124\Big).$$

Remarks. The constants $c_0 = 0.451254$, $\max_s(1-e^{-s})/\sqrt{2s} = 0.451256$ and $1/(2\sqrt{6}) = 0.204124$ were checked numerically. Bobkov and Houdré obtain $2\sqrt{6}$ from a one-dimensional square-root functional inequality that they tensorize. An earlier version of this file followed that route but left its one-dimensional base case unproved. The argument above does not use that inequality, nor the false variant with $\Phi(u) = \min(u, 1-u)^2$. ■

Exercise 10.4

Central idea. The upper bound is Lemma 10.9 (the $n = 1$ Tillich–Zémor inequality) applied to the indicator $f = \mathbf{1}_{\{X=1\}}$, which collapses the right-hand side to $p\sqrt{2\log(1/p)}$. For the lower bound, write $x = 1 - \Phi(t)$, $\gamma(x) = \varphi(t)$ with $t = -\Phi^{-1}(x) \ge 0$; Gordon's inequality gives $\gamma(x) \ge tx$, and the $\sqrt{\frac{1}{2}\log(1/x)}$ factor is obtained by a case analysis on t combining Gordon's two-sided Mills-ratio bounds with elementary estimates (the product $(1-\Phi(t))\sqrt{\frac{1}{2}\log(1/(1-\Phi(t)))}$ is decreasing, which localizes the hard regime to compact intervals). The limit follows from Gordon's inequality by squeeze, giving sharp Mills-ratio asymptotics.

Solution. Recall $\gamma(x) = \varphi(\Phi^{-1}(x))$ for $x \in (0,1)$, $\gamma(0) = \gamma(1) = 0$, where φ is the standard Gaussian density and Φ its cdf. Gordon's inequality (Exercise 7.8) states that for $t > 0$,

$$\frac{t}{t^2+1} \le \frac{1 - \Phi(t)}{\varphi(t)} \le \frac{1}{t}. \tag{G}$$

We use the numerical bounds $\pi \le 3.15$, $\pi \ge 3.125$, $e \ge 2.7083$ (from $e \ge \sum_{k=0}^{4} 1/k!$), $e \le 2.7183$, $e^{-u} \ge 1-u$ and $e^{-u} \le 1 - u + u^2/2$ for $u \ge 0$, whence $1/\sqrt{2\pi} \le 0.4$ and $1/\sqrt{2\pi} \ge 1/2.51$.

Upper bound. Apply Lemma 10.9 to the monotone $f : \{-1,1\} \to [0,1]$, $f(1) = 1$, $f(-1) = 0$, with $\mathbb{P}\{X = 1\} = p$. Since $\mathbb{E}f(X) = p$,

$$\gamma(p) \le \mathbb{E}\sqrt{\gamma(f(X))^2 + 2\log(1/p)\,(f(X) - f(-X))_+^2}.$$

At $X = 1$ the radicand is $0 + 2\log(1/p)$ (using $\gamma(1) = 0$); at $X = -1$ it is 0 (using $\gamma(0) = 0$ and $(-1)_+ = 0$). Hence $\gamma(p) \le p\sqrt{2\log(1/p)}$ for all $p \in (0,1)$, in particular $\gamma(x) \le x\sqrt{2\log(1/x)}$ for $x \in (0,1/2]$.

Lower bound. Let $x \in (0,1/2]$, $t = -\Phi^{-1}(x) \ge 0$, so $x = 1-\Phi(t)$ and $\gamma(x) = \varphi(t)$. Put $S(t) = \sqrt{\frac{1}{2}\log\frac{1}{1-\Phi(t)}}$; we prove $\varphi(t) \ge (1-\Phi(t))S(t)$. Note $p(t) := (1-\Phi(t))S(t)$ is decreasing: $p'(t) = \varphi(t)\frac{1-4S(t)^2}{4S(t)} \le 0$ since $1-\Phi(t) \le 1/2 < e^{-1/2}$ gives $S(t)^2 \ge 1/4$. Also φ is decreasing on $[0,\infty)$ and concave on $[0,1]$ (as $\varphi''(t) = (t^2-1)\varphi(t)$), so Hermite–Hadamard gives midpoint upper and trapezoidal lower bounds for its integrals.

First, $1-\Phi(t) \le \frac{1}{2}e^{-t^2/2}$ for $t \ge 0$: indeed $M(t) := \frac{1-\Phi(t)}{e^{-t^2/2}}$ satisfies $M(0) = 1/2$ and $M'(t) = \frac{-\varphi(t)+t(1-\Phi(t))}{e^{-t^2/2}} \le 0$ by (G).

Case $t \in [0,1/2]$: S is increasing, and $S(1/2) \le \sqrt{2/\pi}$ because $1-\Phi(1/2) \ge \frac{1}{2}(1-\frac{1}{\sqrt{2\pi}}) \ge 0.3 \ge e^{-4/\pi}$ (here $e^{-4/\pi} \le e^{-1.2698} \le 1/3.417 \le 0.3$ since $e^{1.2698} \ge 3.417$). Hence $(1-\Phi(t))S(t) \le \frac{1}{2}e^{-t^2/2}\sqrt{2/\pi} = \varphi(t)$.

Case $t \in [1/2,3/4]$: $\varphi(t) \ge \varphi(3/4) \ge \frac{1-0.28125}{2.51} \ge 0.286$, while $p(t) \le p(1/2) \le 0.3133 \cdot 0.782 \le 0.246$, using $1-\Phi(1/2) \le \frac{1}{2} - \frac{1}{4}(\varphi(0)+\varphi(1/2)) \le 0.3133$ (trapezoidal lower bound for the concave φ, with $\varphi(0) \ge 1/2.51$, $\varphi(1/2) \ge 0.875/2.51$) and $S(1/2) \le \sqrt{\frac{1}{2}\log\frac{1}{0.3}} \le \sqrt{0.61} \le 0.782$ (using $1-\Phi(1/2) \ge 0.3$ and $\log 3.34 \le 1.22$ as $e^{1.22} \ge 3.359$).

Case $t \in [3/4,1]$: $\varphi(t) \ge \varphi(1) \ge \frac{0.606}{2.51} \ge 0.241$ (using $e^{1/2} \le 1.65$, so $e^{-1/2} \ge 0.606$), while $p(t) \le p(3/4) \le 0.2433 \cdot 0.9 \le 0.219$, using $1-\Phi(3/4) \le \frac{1}{2} - \frac{3}{8}(\varphi(0)+\varphi(3/4)) \le 0.2433$ and $S(3/4) \le \sqrt{\frac{1}{2}\log 5} \le 0.9$ (using $1-\Phi(3/4) \ge \frac{1}{2} - \frac{3}{4}\varphi(0) \ge 0.2$ and $\log 5 \le 1.62$ as $e^{1.62} \ge 5.01$).

Case $t \ge 1$: by (G), $\varphi(t) \ge t(1-\Phi(t))$, so it suffices that $K(t) := t^2 - \frac{1}{2}\log\frac{1}{1-\Phi(t)} \ge 0$. Now $K'(t) = 2t - \frac{\varphi(t)}{2(1-\Phi(t))} \ge 2t - \frac{t^2+1}{2t} = \frac{3t^2-1}{2t} > 0$ for $t \ge 1$, using $\frac{1-\Phi(t)}{\varphi(t)} \ge \frac{t}{t^2+1}$ from (G). Moreover $K(1) \ge 0$ since $1-\Phi(1) = \frac{1}{2} - \int_0^1 \varphi \ge \frac{1}{2} - \frac{1}{2}(\varphi(1/4)+\varphi(3/4)) \ge 0.148 \ge e^{-2}$ (midpoint upper bound for concave φ, with $\varphi(1/4) \le 0.4$, $\varphi(3/4) \le 0.7584/2.5$, and $e^{-2} \le 1/7.33$). Hence $K(t) \ge 0$ for $t \ge 1$, giving $\varphi(t) \ge t(1-\Phi(t)) \ge S(t)(1-\Phi(t))$.

Thus $\gamma(x) \ge x\sqrt{\frac{1}{2}\log(1/x)}$ for all $x \in (0,1/2]$, and by continuity at $x = 0$.

Limit. From (G), $\frac{t^2}{t^2+1} \le \frac{t(1-\Phi(t))}{\varphi(t)} \le 1$, so with $t = -\Phi^{-1}(x) \to \infty$ as $x \to 0$, $\frac{\gamma(x)}{x} = \frac{\varphi(t)}{1-\Phi(t)} = t(1+o(1))$. Taking logs gives $-\log(1-\Phi(t)) = \frac{t^2}{2} + \log t + \frac{1}{2}\log(2\pi) + o(1)$, so $\frac{\log(1/x)}{t^2/2} \to 1$. Hence

$$\frac{\gamma(x)}{x\sqrt{2\log(1/x)}} = \frac{\varphi(t)/(1-\Phi(t))}{\sqrt{2\log(1/(1-\Phi(t)))}} = \frac{1+o(1)}{\sqrt{1+o(1)}} \to 1,$$

using also the upper bound for the ≤ 1 direction. This completes the proof. ■

Exercise 10.5

Central idea. By rotation invariance of the Gaussian law it suffices to treat the one-dimensional half-space $\{x_1 < \mu\}$; the distance from x to this half-space is $(\langle x, v\rangle - \mu)_+$, so enlarging by t just shifts the threshold to $\mu + t$, and Φ^{-1} converts the Gaussian tail back to the threshold.

Solution. Let $A_t = \{x : d(x,A) \le t\}$. If $u = 0$, then $A = \emptyset$ for $\lambda \le 0$ (so $A_t = \emptyset$, $P(A_t) = 0$) and $A = \mathbb{R}^n$ for $\lambda > 0$ (so $A_t = \mathbb{R}^n$, $P(A_t) = 1$); with $\Phi^{-1}(0) = -\infty$, $\Phi^{-1}(1) = +\infty$, the identity holds trivially in both cases. Assume $u \ne 0$ and set $v = u/|u|$, $\mu = \lambda/|u|$, so $A = \{x : \langle x, v\rangle < \mu\}$.

Since P is rotation invariant and v is a unit vector, $\langle X, v\rangle \sim N(0,1)$ for $X \sim P$, hence $P(A) = \Phi(\mu)$ and $\Phi^{-1}(P(A)) = \mu$. Moreover $d(x, A) = d(x, \bar{A})$ where $\bar{A} = \{y : \langle y, v\rangle \le \mu\}$; for $\langle x, v\rangle \ge \mu$ the nearest point is the orthogonal projection onto the boundary hyperplane, giving $d(x, A) = \langle x, v\rangle - \mu$, and $d(x, A) = 0$ for $x \in A$. Thus, up to a P-null boundary,

$$A_t = \{x : \langle x, v\rangle \le \mu + t\},$$

so $P(A_t) = \Phi(\mu + t)$ and $\Phi^{-1}(P(A_t)) = \mu + t = \Phi^{-1}(P(A)) + t$. ■

Exercise 10.6

Central idea. The statement is false as written, in two ways: absolute continuity alone does not suffice (a 1D kink when A_t exits the support), and even a smooth Gaussian density fails in dimension one (boundary points annihilate pairwise when balls merge, jumping the surface integral). The sharp picture: for $n \ge 2$ with a continuous density the claim is true — coarea writes $P(A_t)$ as an integral of surface integrals over ∂A_u, and each sphere's contribution is continuous in u by spherical parametrization plus dominated convergence (the "active" patch is cut out by inequalities with continuous coefficients, so in $n \ge 2$ its indicator converges a.e.). The book applies it to Gaussian P, where the $n = 1$ gap is a harmless erratum (1D Gaussian isoperimetry is elementary).

Solution. The claim is false for general absolutely continuous P. Counterexample 1: on $\mathbb{R}$, let P be uniform on $[0, 1]$ and $A = (-1/2, 1/2)$. Then $A_t = (-1/2 - t, 1/2 + t)$ and $P(A_t) = \min(1/2 + t, 1)$, not differentiable at $t = 1/2$ (left derivative 1, right derivative 0).

Even a smooth density does not save it in one dimension. Counterexample 2: on $\mathbb{R}$ with P standard Gaussian, $A = (-1, 1) \cup (3, 5)$. For $t < 1$, $A_t = (-1 - t, 1 + t) \cup (3 - t, 5 + t)$ (disjoint); for $t > 1$, $A_t = (-1 - t, 5 + t)$ (merged). With $F(t) = P(A_t)$,

$$F'(1^-) = 2\varphi(2) + \varphi(-2) + \varphi(6) \ne \varphi(-2) + \varphi(6) = F'(1^+),$$

so F is not differentiable at $t = 1$: the two boundary points $1 + t$, $3 - t$ collide and annihilate, jumping the surface sum.

What is true. Let $n \ge 2$, let P have a continuous density p (in particular Gaussian), and let $A = \bigcup_{i=1}^{m} B(c_i, r_i)$ be a finite union of distinct open balls ($m \ge 1$; $m = 0$ is trivial). Put $d(x) = d(x, A)$ and $F(t) = P(A_t)$. Since P charges no finite union of spheres, $F(t) = \int_{\{d<t\}} p$. By the coarea formula applied to the 1-Lipschitz d (with $|\nabla d| = 1$ a.e. on $\{d > 0\}$), for $0 \le s < t$,

$$F(t) - F(s) = \int_{\{s<d<t\}} p\, dx = \int_s^t G(u)\, du, \qquad G(u) := \int_{\{d=u\}} p\, d\mathcal{H}^{n-1},$$

and $\{d = u\} = \partial A_u = \bigsqcup_i S_i(u)$ up to $\mathcal{H}^{n-1}$-null sets, where $S_i(u) = \partial B(c_i, r_i + u) \setminus \bigcup_{j \ne i} B(c_j, r_j + u)$. Parametrize $\partial B(c_i, r_i + u)$ by $x = c_i + (r_i + u)\theta$, $\theta \in S^{n-1}$; then with $\alpha_{ij} = c_i - c_j$,

$$G_i(u) := \int_{S_i(u)} p\, d\mathcal{H}^{n-1} = (r_i + u)^{n-1} \int_{E_i(u)} p(c_i + (r_i + u)\theta)\, d\sigma(\theta),$$

where $E_i(u) = \{\theta : \langle \alpha_{ij}, \theta\rangle \ge \beta_{ij}(u)\ \forall j \ne i\}$ and $\beta_{ij}(u) = \frac{(r_j+u)^2 - (r_i+u)^2 - |\alpha_{ij}|^2}{2(r_i+u)}$ is continuous in $u > 0$ (for $\alpha_{ij} = 0$ the constraint is u-independent: it is vacuous or forces $G_i \equiv 0$). Fix $u_0 > 0$. For σ-a.e. θ (all but a finite union of great subspheres $\{\langle \alpha_{ij}, \theta\rangle = \beta_{ij}(u_0)\}$, null since $n \ge 2$), $\mathbf{1}_{E_i(u)}(\theta) \to \mathbf{1}_{E_i(u_0)}(\theta)$ as $u \to u_0$; with p uniformly continuous on compacta, dominated convergence gives continuity of $u \mapsto \int_{E_i(u)} p(c_i + (r_i + u)\theta) d\sigma$, hence of G_i. Thus $G = \sum_i G_i$ is continuous on $(0, \infty)$, and $F(t) = F(0) + \int_0^t G$ is C^1 there with $F'(t) = \int_{\partial A_t} p\, d\mathcal{H}^{n-1}$. ■

Exercise 10.7

Central idea. Apply the infinitesimal hypothesis to the enlarged sets A_t themselves (using $(A_t)_s = A_{t+s}$) to get a differential inequality for $p(t) = P(A_t)$; a first-crossing argument with $\psi(t) = F^{-1}(p(t))$ then forces $\psi(t) \geq \psi(0) + ct$, for if ψ dipped below the line $\alpha + (c-\delta)t$ at a first time T, the difference quotient along a sequence $t_n \searrow T$ of "bad" points would contradict the differential inequality at T.

Solution. Take closed enlargements $A_t = \{x : d(x, A) \leq t\}$, so $A_0 = \bar{A} \supseteq A$ and $p(t) := P(A_t)$ is nondecreasing and right-continuous. Since $d(x, A_t) = \max(0, d(x, A) - t)$ (nearest-point projection), $(A_t)_s = A_{t+s}$ for $s, t \geq 0$. Assume $F : \mathbb{R} \to (0, 1)$ is a C^1 diffeomorphism with $f = F' > 0$ (as for Φ), extended by $F^{-1}(0) = -\infty$, $F^{-1}(1) = +\infty$.

If $P(A) \in \{0, 1\}$ the claim is trivial ($p(t) \geq P(A)$ and $F(F^{-1}(P(A)) + ct) = P(A)$). Assume $0 < P(A) \leq p(0) \leq 1$; if $p(0) = 1$ then $p \equiv 1$ and both sides equal 1. Let $p(0) \in (0, 1)$, $\alpha = F^{-1}(p(0)) \geq F^{-1}(P(A))$; it suffices to show $p(t) \geq F(\alpha + ct)$.

Applying the hypothesis to the Borel set A_t and using $(A_t)_s = A_{t+s}$,

$$\liminf_{s \searrow 0} \frac{p(t+s) - p(t)}{s} \geq c f(F^{-1}(p(t))) \tag{$*$}$$

whenever $p(t) \in (0, 1)$. Put $\psi(t) = F^{-1}(p(t))$ (nondecreasing, right-continuous where finite). Fix $\delta > 0$ and let $S = \{t \geq 0 : \psi(t) < \alpha + (c-\delta)t\}$, $T = \inf S$ (with $\inf \emptyset = +\infty$). Suppose $T < \infty$. Choose $t_n \in S$ with $t_n \searrow T$. Right-continuity gives $\psi(T) = \lim_n \psi(t_n) \leq \alpha + (c-\delta)T$ (the limit is finite, so $p(T) \in (0, 1)$); monotonicity gives $\psi(T) \geq \psi(t) \geq \alpha + (c-\delta)t$ for $t < T$, whence $\psi(T) = \alpha + (c-\delta)T$.

Now $(*)$ at $t = T$ and the sequence $r_n = t_n - T \searrow 0$ give, since $\psi(t_n) < \alpha + (c-\delta)t_n$ and F is increasing,

$$c f(\psi(T)) \leq \liminf_{s \searrow 0} \frac{p(T+s) - p(T)}{s} \leq \liminf_n \frac{F(\alpha + (c-\delta)t_n) - F(\alpha + (c-\delta)T)}{t_n - T} = (c-\delta) f(\psi(T)).$$

As $f(\psi(T)) > 0$, this yields $c \leq c - \delta$, a contradiction. Hence $S = \emptyset$, i.e. $\psi(t) \geq \alpha + (c-\delta)t$ for all $t \geq 0$; letting $\delta \to 0$ gives $p(t) \geq F(\alpha + ct) \geq F(F^{-1}(P(A)) + ct)$. ■

Exercise 10.8

Central idea. The n-dimensional inequality tensorizes from a two-point ($n = 1$) inequality: condition on all but one coordinate, apply the 1D bound, and iterate, using Jensen twice (convexity of $(\cdot)_+$ for the gradient, convexity of the Euclidean norm to recombine). The 1D inequality for $a \geq b$ with $K = \kappa(p) = (1-p)/(1+p)$ is proved by squaring and factoring the difference as $\frac{p^2(p-1)(a-b)^2}{1+p} Q$ with an explicitly nonnegative Q. Monotonicity lets one use $\kappa(p)$ instead of $\max(\kappa(p), \kappa(1-p))$, and for $f = \mathbf{1}_A$ the $\alpha(f)^2$ term vanishes.

Solution. Write $\alpha(x) = x(1-x)$, $\kappa(p) = \frac{1-p}{1+p}$.

1D lemma. For $0 \leq b \leq a \leq 1$ and $K = \kappa(p)$,

$$p\sqrt{\alpha(a)^2 + K(a-b)^2} + (1-p)\alpha(b) \geq \alpha(pa + (1-p)b). \tag{1}$$

Let $m = pa + (1-p)b$. By concavity, $\alpha(m) - (1-p)\alpha(b) \geq p\alpha(a) \geq 0$, so with both sides nonnegative it suffices to square:

$$[\alpha(m) - (1-p)\alpha(b)]^2 - p^2\alpha(a)^2 - p^2 K(a-b)^2 \leq 0.$$

A direct expansion with $K = \frac{1-p}{1+p}$ gives the factorization

$$[\alpha(m) - (1-p)\alpha(b)]^2 - p^2\alpha(a)^2 - p^2 K(a-b)^2 = \frac{p^2(p-1)(a-b)^2}{1+p} Q,$$

where $Q = (1-p^2)b(2a-b) + [(1+p)a-1]^2$. Since $\frac{p^2(p-1)(a-b)^2}{1+p} \le 0$, it suffices that $Q \ge 0$. If $b(2a-b) \ge 0$ this is clear. Else $b > 2a \ge 0$ (hence $a < 1/2$), and

$$(1-(1+p)a)^2 - (1-p^2)(1-2a) = [p-(1+p)a]^2 \ge 0,$$

so $Q \ge (1-p^2)[b(2a-b)+(1-2a)]$. Since $0 < b-2a$ and $b \le 1$, $b(b-2a) \le b-2a$, giving $b(2a-b)+(1-2a) \ge 1-b \ge 0$. Hence $Q \ge 0$, proving (1).

Tensorization. We prove the first claim by induction on n with $K = \max(\kappa(p), \kappa(1-p))$. For $n = 1$, write $a = f(1)$, $b = f(-1)$; if $a \ge b$ the claim follows from (1) since $K \ge \kappa(p)$, and if $b > a$ it follows by swapping the roles of ± 1 (which swaps $p \leftrightarrow 1-p$ and leaves K invariant). Let $n \ge 2$, $x = (x', x_n)$, and $g(x') = \mathbb{E}_{X_n}[f(x', X_n)]$. By the induction hypothesis,

$$\alpha(\mathbb{E}f) = \alpha(\mathbb{E}_{X'} g) \le \mathbb{E}_{X'}\left[\sqrt{\alpha(g)^2 + K\|\nabla^+ g\|^2}\right].$$

For fixed x', (1) applied to $a = f(x',1)$, $b = f(x',-1)$ gives $\alpha(g(x')) \le \mathbb{E}_{X_n}[h]$ with $h = \sqrt{\alpha(f)^2 + K(\nabla_n^+ f)^2}$. For $i < n$, Jensen gives $\nabla_i^+ g(x') \le \mathbb{E}_{X_n}[\nabla_i^+ f(x', X_n)]$. Hence, with $w_i = \nabla_i^+ f$,

$$\alpha(g)^2 + K\|\nabla^+ g\|^2 \le (\mathbb{E}_{X_n} h)^2 + K\sum_{i<n} (\mathbb{E}_{X_n} w_i)^2,$$

and by convexity of the Euclidean norm,

$$\sqrt{\alpha(g)^2 + K\|\nabla^+ g\|^2} \le \mathbb{E}_{X_n}\left[\sqrt{h^2 + K\sum_{i<n} w_i^2}\right] = \mathbb{E}_{X_n}\left[\sqrt{\alpha(f)^2 + K\|\nabla^+ f\|^2}\right].$$

Taking $\mathbb{E}_{X'}$ and using Fubini completes the induction.

If f is increasing, so is g (expectation preserves order), and each slice satisfies $f(x',1) \ge f(x',-1)$; running the same induction with $K = \kappa(p)$ throughout gives the second claim.

For a monotone set A, apply the second claim to $f = \mathbf{1}_A$. Since $\alpha(\mathbf{1}_A) \equiv 0$,

$$P(A)(1-P(A)) = \alpha(\mathbb{E}f) \le \sqrt{\kappa(p)}\, \mathbb{E}\|\nabla^+ \mathbf{1}_A\|,$$

i.e. $\mathbb{E}\|\nabla^+ \mathbf{1}_A\| \ge \sqrt{\frac{1+p}{1-p}}\, P(A)(1-P(A))$.

Tightness. The inequality is not tight for any of the three. For a dictatorship $A = \{x_1 = 1\}$: $\mathbb{E}\|\nabla^+ \mathbf{1}_A\| = p$ while the right side is $p\sqrt{1-p^2}$, ratio $1/\sqrt{1-p^2} > 1$. For a singleton $A = \{(1,\dots,1)\}$: $\mathbb{E}\|\nabla^+ \mathbf{1}_A\| = p^n\sqrt{n}$, ratio $\sqrt{\frac{n(1-p)}{1+p}}/(1-p^n) \to \infty$ as $n \to \infty$. For majority $A = \{\sum x_i > 0\}$ ($n = 2m+1$, $p = 1/2$): $\mathbb{E}\|\nabla^+ \mathbf{1}_A\| = \binom{2m+1}{m+1} 2^{-(2m+1)}\sqrt{m+1} \to 1/\sqrt{\pi}$ while the right side is $\sqrt{3}/4$, ratio $\to 4/\sqrt{3\pi} \approx 1.30 > 1$. ■

Exercise 10.9

Central idea. The error set E is increasing, so Theorem 10.11 applies once every point of a vertex boundary has $\ge \Delta$ edges leaving the set. For codes this count is provable on the *outer* side (correctly decoded words adjacent to an error word), not on the inner side that Theorem 10.11 uses for E. Complementing coordinates turns the outer count into an inner count for an increasing set, but at parameter $1-p$; this proves the threshold bound with $\log(1-p)$ in place of $\log p$. The printed $\log p$ form survives every brute-force test but is not proved here.

Solution. Status: incomplete. Fully proved below: the outer-boundary lemma and the variant bound $(*)$ with $-\log(1-p)$. The printed bound with $-\log p$ is **not** proved; the exact missing step is inequality (D) in part 3. Part 4 documents why the natural routes fail (counterexamples re-verified numerically).

0. Setup. Write $|y|$ for Hamming weight, $y \cap c$ for the common support, and $\delta_c(y) = d(y,c) - d(y,0) = |c| - 2|y \cap c|$. ML decoding leaves ties open; take either translation-invariant convention: ties count as

errors, $A_0 = \{y : \delta_c(y) > 0 \ \forall c \neq 0\}$, or ties count as correct, $A_0 = \{y : \delta_c(y) \geq 0 \ \forall c \neq 0\}$. Since $d(v \oplus x, c) = d(x, c \oplus v)$ and $c \mapsto c \oplus v$ is a bijection of $\mathcal{C}$ sending v to 0, $A_v = A_0$, so $\mathrm{err}_v = \mathrm{err}_0 = \mathbb{P}_p(E)$ with $E = A_0^c$. Flipping a 0 to 1 at j changes δ_c by $-2 \cdot \mathbf{1}\{j \in c\}$, so every δ_c is nonincreasing upward: E is increasing and A_0 decreasing. Also $0 \notin E$ and $1^n \in E$ ($\delta_c(1^n) = -|c| < 0$). By Russo's lemma $\frac{d}{dp}\mathbb{P}_p(E) = I_p(E) > 0$ (some edge of the cube crosses ∂E and μ_p has full support), so err_0 increases continuously and strictly from 0 to 1 and p_c is unique. Assume $\mathcal{C} \neq \{0\}$ (otherwise $\mathrm{err}_v \equiv 0$ and p_c is undefined) and $\Delta \geq 1$ (if $\Delta = 0$ both claims read $\Phi(0) = 1/2$ and follow from monotonicity).

1. Lemma (outer count). If $y \in A_0$, $y_i = 0$ and $y \oplus e_i \in E$, then $g(y) := |\{j : y_j = 0,\ y \oplus e_j \in E\}| \geq \Delta$.

Proof. Ties as errors: some $c \neq 0$ has $\delta_c(y) - 2 \cdot \mathbf{1}\{i \in c\} \leq 0 < \delta_c(y)$, forcing $i \in c$ and $\delta_c(y) \in \{1, 2\}$. For every $j \in c \setminus y$, $\delta_c(y \oplus e_j) = \delta_c(y) - 2 \leq 0$, so $y \oplus e_j \in E$. Hence $g(y) \geq |c \setminus y| = |c| - |y \cap c| = \frac{1}{2}(|c| + \delta_c(y)) \geq \frac{1}{2}(2\Delta + 1)$, i.e. $g(y) \geq \Delta + 1$. Ties as correct: likewise $\delta_c(y) \in \{0, 1\}$, every $j \in c \setminus y$ gives $\delta_c(y \oplus e_j) < 0$, and $g(y) \geq \frac{1}{2}|c| \geq \Delta$. ■

2. Proved variant. For $0 < p < p_c$,

$$\mathrm{err}_v(p) \leq \Phi\Big(\sqrt{2\Delta}\big(\sqrt{-\log(1-p)} - \sqrt{-\log(1-p_c)}\big)\Big), \qquad (*)$$

and the reverse inequality holds for $p_c < p < 1$.

Proof. Let $\tau(y) = 1^n \oplus y$ and $B = \tau(A_0)$; B is increasing because A_0 is decreasing. If X has i.i.d. Bernoulli(p) coordinates, $\tau(X)$ has i.i.d. Bernoulli(q) coordinates, $q = 1 - p$, so $\mathbb{P}_q(B) = 1 - \mathrm{err}_0(p)$ and $q_{1/2} := 1 - p_c$ satisfies $\mathbb{P}_{q_{1/2}}(B) = 1/2$. For $z = \tau(y) \in B$ with a neighbour outside B, the edges leaving B from z (flips $z_j = 1 \to 0$) are exactly the flips $y_j = 0 \to 1$ entering E, so $h_B(z) = g(y) \geq \Delta$ by the Lemma: $h_B \geq \Delta$ on $\partial_V(B)$. Theorem 10.11 with $k = \Delta$ at parameter q gives, for $q > q_{1/2}$ (i.e. $p < p_c$),

$$1 - \mathrm{err}_0(p) \geq \Phi\big(\sqrt{2\Delta}\big(\sqrt{-\log(1-p_c)} - \sqrt{-\log(1-p)}\big)\big),$$

and $1 - \Phi(t) = \Phi(-t)$ yields $(*)$; the case $p > p_c$ is identical with inequalities reversed. (Under ties-as-errors, Δ may be replaced by $\Delta + 1$.) ■

Hassani–Kudekar–Ordentlich–Polyanskiy–Urbanke (arXiv:1801.09481, Thm. 1) restate Tillich–Zémor's BSC result with $\sqrt{d_{\min}} = \sqrt{2\Delta}$ and $\log(1 - \varepsilon)$, not $\log \varepsilon$, consistent with $(*)$ (original paper not consulted). The printed bound is not a consequence of $(*)$: for $\Delta = 2$, $p_c = 0.1$, $p = 0.01$ the printed right-hand side is 0.1044, that of $(*)$ is 0.3268; for $p, p_c < 1/2$ the printed form is the stronger one.

3. What would finish the printed bound. Let $P(p) = \mathbb{P}_p(E) \in (0, 1)$ and $\gamma = \varphi \circ \Phi^{-1}$. It suffices to prove

$$p\, I_p(E) = \mathbb{E}_p h_E(X) \ \geq \ \frac{\sqrt{\Delta}\,\gamma(P(p))}{\sqrt{2\log(1/p)}} \qquad \text{for all } p \in (0, 1). \qquad \text{(D)}$$

Indeed, by Russo's lemma (D) says $\frac{d}{dp}\Phi^{-1}(P(p)) = P'(p)/\gamma(P(p)) \geq \sqrt{2\Delta}\,\frac{d}{dp}(-\sqrt{-\log p})$; integrating from p to p_c (where $\Phi^{-1}(1/2) = 0$) gives $\Phi^{-1}(P(p)) \leq \sqrt{2\Delta}(\sqrt{-\log p_c} - \sqrt{-\log p})$ for $p < p_c$, and integrating from p_c to p gives the reverse for $p > p_c$ — exactly the printed statement. (D) is precisely what the proof of Theorem 10.11 extracts from "$h \geq k$ on ∂_V". **Missing step: a proof of (D) for the error set of a linear code.** (D) held in all 216 test cases below.

4. Why the natural routes fail (all re-checked numerically; scripts in `external_audit_2026-10-30/compare/repair_che`

(a) *Inner count $h_E \geq \Delta$ on $\partial_V(E)$ is false.* $n = 7$, $c = \mathbf{1}_{\{0,1,2,3\}}$, $c' = \mathbf{1}_{\{3,4,5,6\}}$; the code $\{0, c, c', c \oplus c'\}$ has $d_{\min} = 4$, $\Delta = 2$. Ties as errors, $x = \mathbf{1}_{\{0,3,4\}}$: $d(x, 0) = d(x, c) = d(x, c') = 3$, so $x \in E$; removing coordinate 0 leaves a tie with c', removing 4 a tie with c, and only removing 3 leaves E: $h_E(x) = 1 < 2$. Ties as correct also fails (random $n = 8$ code, generators $209, 92, 107$ with bit $i =$ coordinate i: minimum boundary count $1 < \Delta = 2$). The step "every $j \in S_1$ is pivotal" in the previous version of this solution is therefore false.

(b) *Complementing does not preserve* $\log p$. Part 2 is the correct use of the complement and gives $\log(1-p)$. Applying Theorem 10.11 verbatim to a decreasing set at parameter p is invalid: $A = \{x \in \{0,1\}^{40} : |x| \le 1\}$ has 39 outward edges at each boundary point, $p_{1/2} = 0.04160$, and at $p = 0.001$, $\Phi^{-1}(\mathbb{P}_p(A)) = 3.171 < 7.464 = \sqrt{78}(\sqrt{-\log p} - \sqrt{-\log p_{1/2}})$.

(c) *Averaged count* $\mathbb{E}_p h_E \ge \sqrt{\Delta}\,\mathbb{E}_p\sqrt{h_E}$ *is false.* Combined with Corollary 10.10 ($\mathbb{E}_p\sqrt{h_E} \ge \gamma(P)/\sqrt{2\log(1/p)}$) it would imply (D), but: $n = 9$, generators $326, 115, 314$ (bit i = coordinate i), $d_{\min} = 4$, $\Delta = 2$, ties as correct, $p = 0.695$: $\mathbb{E}h_E = 0.2232 < 0.2376 = \sqrt{2}\,\mathbb{E}\sqrt{h_E}$. (It failed in 7 of 216 cases.) So any proof of (D) must go beyond Corollary 10.10 combined with this averaged count.

5. Numerical support for the printed statement. For 216 (code, tie-convention) cases — random linear codes with $5 \le n \le 11$, dimension ≤ 4, $\Delta \ge 1$, exact probabilities on a 1999-point grid of p — the printed bound, $(*)$ and (D) all held (minimum margins ≥ 0); the outer-count Lemma held in every case, the inner count failed in 11. The independent audit found the same on $\{0, c, c', c \oplus c'\}$ codes and 60 further random codes. The printed statement is thus plausibly true but unproved here.

Exercise 10.10

Central idea. Apply Bobkov's Gaussian inequality (Theorem 10.13) to the rescaled function $g = \varepsilon f^2 \in [0,1]$ and let $\varepsilon \to 0$. Since $\gamma(u) = u\sqrt{2\log(1/u)}(1 + o(1))$ as $u \to 0^+$, both sides of Bobkov expand as $\varepsilon\sqrt{2\log(1/\varepsilon)}\,\mathbb{E}f^2$ plus a correction of order $\varepsilon/\sqrt{\log(1/\varepsilon)}$; the leading terms cancel, and the correction equals $\frac{1}{\sqrt{2}}(-\mathrm{Ent}(f^2) + 2\mathbb{E}\|\nabla f\|^2) \ge 0$, using the chain rule $\nabla f^2 = 2f\nabla f$. The same ε-scaling on the hypercube does not give Theorem 5.1, because the discrete gradient has no chain rule.

Solution. (a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth with bounded derivatives and $0 < c \le f \le C$; the general case follows by approximation (see below). For $0 < \varepsilon < 1/C^2$, $g = \varepsilon f^2$ maps $\mathbb{R}^n$ into $[0,1]$, so Theorem 10.13 gives

$$\gamma(\varepsilon\mathbb{E}f^2) \le \mathbb{E}\sqrt{\gamma(\varepsilon f^2)^2 + \varepsilon^2\|\nabla f^2\|^2}, \tag{$*$}$$

where $\gamma = \varphi \circ \Phi^{-1}$. By Lemma 10.3, $\gamma(u) = u\sqrt{2\log(1/u)}(1 + \eta(u))$ with $\eta(u) \to 0$ as $u \to 0^+$.

Write $L = \log(1/\varepsilon)$. For fixed x and $u = \varepsilon f^2(x)$,

$$\gamma(\varepsilon f^2) = \varepsilon f^2\sqrt{2L + 2\log(1/f^2)}\,(1 + \eta(\varepsilon f^2)) = \varepsilon f^2\sqrt{2L} + \varepsilon\frac{f^2\log(1/f^2)}{\sqrt{2L}} + o(\varepsilon/\sqrt{L}),$$

using $\sqrt{1+y} = 1 + y/2 + o(y)$. Since f and ∇f are bounded and $\eta(\varepsilon f^2) \to 0$ uniformly in x, every $o(\cdot)$ below is uniform in x, justifying passage of the limit $\varepsilon \to 0$ through $\mathbb{E}$.

Squaring and adding the gradient term,

$$\gamma(\varepsilon f^2)^2 + \varepsilon^2\|\nabla f^2\|^2 = 2L\varepsilon^2 f^4 + 2\varepsilon^2 f^4\log(1/f^2) + \varepsilon^2\|\nabla f^2\|^2 + o(\varepsilon^2),$$

and, since $f^2 \ge c^2 > 0$,

$$\sqrt{\gamma(\varepsilon f^2)^2 + \varepsilon^2\|\nabla f^2\|^2} = \varepsilon f^2\sqrt{2L} + \varepsilon\frac{f^2\log(1/f^2)}{\sqrt{2L}} + \varepsilon\frac{\|\nabla f^2\|^2}{2f^2\sqrt{2L}} + o(\varepsilon/\sqrt{L}).$$

The identical expansion with the constant $\mathbb{E}f^2$ in place of f^2 gives

$$\gamma(\varepsilon\mathbb{E}f^2) = \varepsilon\mathbb{E}f^2\sqrt{2L} + \varepsilon\frac{\mathbb{E}f^2\log(1/\mathbb{E}f^2)}{\sqrt{2L}} + o(\varepsilon/\sqrt{L}).$$

Inserting these into $(*)$, the leading terms $\varepsilon\sqrt{2L}\,\mathbb{E}f^2$ cancel. Multiplying by $\sqrt{L}/\varepsilon$ and letting $\varepsilon \to 0$ yields

$$0 \le \frac{1}{\sqrt{2}}\left(\mathbb{E}[f^2\log(1/f^2)] - \mathbb{E}f^2\log(1/\mathbb{E}f^2) + \mathbb{E}\frac{\|\nabla f^2\|^2}{2f^2}\right).$$

Now $\mathbb{E}[f^2\log(1/f^2)] - \mathbb{E}f^2\log(1/\mathbb{E}f^2) = -\mathrm{Ent}(f^2)$, and by the chain rule $\nabla f^2 = 2f\nabla f$, so $\|\nabla f^2\|^2/(2f^2) = 2\|\nabla f\|^2$. Hence $\mathrm{Ent}(f^2) \le 2\mathbb{E}\|\nabla f\|^2$, Theorem 5.4. For general C^1 f, apply this to $f_\delta = \sqrt{f^2+\delta}$ (bounded case first, then remove boundedness by truncation) and let $\delta \to 0$: $\mathrm{Ent}(f^2+\delta) \to \mathrm{Ent}(f^2)$ while $\|\nabla f_\delta\|^2 = \frac{f^2}{f^2+\delta}\|\nabla f\|^2 \uparrow \|\nabla f\|^2$.

(b) No. On $\{-1,1\}^n$ the discrete gradient $\nabla_i f(x) = (f(x) - f(x^{(i)}))/2$ is linear, so $\nabla_i(\varepsilon f^2) = \varepsilon\nabla_i(f^2)$, but it satisfies no chain rule:

$$\nabla_i(f^2) = \frac{f(x)^2 - f(x^{(i)})^2}{2} = 2\nabla_i f \cdot \frac{f(x)+f(x^{(i)})}{2}.$$

The identical ε-expansion therefore produces, in place of $2\|\nabla f\|^2$,

$$\frac{\|\nabla f^2\|^2}{2f^2} = \sum_{i=1}^n (\nabla_i f)^2 \, \frac{(f(x)+f(x^{(i)}))^2}{2f(x)^2},$$

and the limit $\varepsilon \to 0$ yields only $\mathrm{Ent}(f^2) \le \mathbb{E}\sum_i (\nabla_i f)^2 \frac{(f(x)+f(x^{(i)}))^2}{2f(x)^2}$, which is not Theorem 5.1. It is genuinely weaker: for $n=1$, $f(1)=2$, $f(-1)=1$, one has $\mathrm{Ent}(f^2) \approx 0.48$, Theorem 5.1 gives the bound $2\mathbb{E}\|\nabla f\|^2 = 0.50$, but the scaled Bobkov inequality gives only $\mathrm{Ent}(f^2) \le 0.70$. The chain rule $\nabla(f^2) = 2f\nabla f$ is essential to the computation in (a), and it has no discrete analogue.

Exercise 10.11

Central idea. $f_{t,\sigma}$ is the Gaussian mollification $f_t * p_\sigma$ of the $1/t$-Lipschitz ramp $f_t = (1 - d(\cdot, A)/t)_+$. Convolution with a smooth probability kernel preserves the Lipschitz constant and yields a C^∞ function; moreover $\nabla f_{t,\sigma} = (\nabla f_t) * p_\sigma \to \nabla f_t$ at every Lebesgue point, and the eikonal equation $\|\nabla d\| = 1$ a.e. on $\{d > 0\}$ turns this into the $1/t$ limit. The book's pointwise claim inside the strip is too strong as stated — an explicit two-ball counterexample gives limit 0 — but the a.e. version holds, which is all the dominated-convergence step needs.

Solution. Let $A \subset \mathbb{R}^n$ be Borel, $t > 0$, $d(x) = d(x,A) = \inf_{y\in A}\|x-y\|$. Put $\phi_t(s) = (1 - s/t)_+ = \max(0, 1 - s/t)$ and $f_t = \phi_t \circ d$. Then $f_t : \mathbb{R}^n \to [0,1]$, $f_t = 1$ on A, $f_t = 0$ outside $A_t = \{d \le t\}$, and f_t is $1/t$-Lipschitz, as the composition of the 1-Lipschitz d with the $1/t$-Lipschitz ϕ_t. For $\sigma > 0$ let p_σ be the $N(0, \sigma^2 I_n)$ density and

$$f_{t,\sigma}(x) = \int_{\mathbb{R}^n} f_t(y)\, p_\sigma(x-y)\, dy = (f_t * p_\sigma)(x) = \mathbb{E}[f_t(x + \sigma Z)], \qquad Z \sim N(0, I_n).$$

Since p_σ is a probability density and $0 \le f_t \le 1$, $f_{t,\sigma}$ maps $\mathbb{R}^n$ to $[0,1]$.

Smoothness. $p_\sigma \in C^\infty$ and $\partial^\alpha p_\sigma \in L^1$ for every multi-index α (each derivative is a polynomial times a Gaussian). For $f_t \in L^\infty$, differentiation under the integral sign is justified by dominated convergence, giving $\partial^\alpha f_{t,\sigma} = f_t * (\partial^\alpha p_\sigma)$, bounded and continuous. Hence $f_{t,\sigma}$ is infinitely many times differentiable.

$1/t$-Lipschitz. From the expectation form,

$$|f_{t,\sigma}(x) - f_{t,\sigma}(x')| \le \mathbb{E}|f_t(x+\sigma Z) - f_t(x'+\sigma Z)| \le \frac{1}{t}|x - x'|,$$

so $f_{t,\sigma}$ is $1/t$-Lipschitz for every $\sigma > 0$ (in particular $\|\nabla f_{t,\sigma}\| \le 1/t$ everywhere).

Gradient representation. Integrating by parts in y (boundary terms vanish since p_σ is Schwartz and f_t is bounded and Lipschitz),

$$\nabla f_{t,\sigma}(x) = \mathbb{E}[(\nabla f_t)(x+\sigma Z)] = ((\nabla f_t) * p_\sigma)(x),$$

with ∇f_t the a.e.-defined Rademacher gradient, $\|\nabla f_t\| \le 1/t$ a.e.

Limits outside the strip. If $x \in \mathrm{int}(A)$ then $\nabla f_t = 0$ a.e. on $\mathrm{int}(A)$, so with $\delta = d(x, A^c) > 0$,

$$\|\nabla f_{t,\sigma}(x)\| \le \frac{1}{t}\mathbb{P}(x + \sigma Z \notin A) \le \frac{1}{t}\mathbb{P}(\|Z\| \ge \delta/\sigma) \longrightarrow 0.$$

If $d(x) > t$ (i.e. x outside $\overline{A_t}$) then $\nabla f_t = 0$ a.e. on $\{d > t\}$, so

$$\|\nabla f_{t,\sigma}(x)\| \le \frac{1}{t}\mathbb{P}(d(x+\sigma Z) \le t) \longrightarrow 0,$$

since $d(x+\sigma Z) \to d(x) > t$ a.s.

Limit in the strip. Since $\nabla f_t \in L^\infty$, the approximate identity $(\nabla f_t) * p_\sigma(x) \to \nabla f_t(x)$ at every Lebesgue point of ∇f_t (split $\int |h(x-\sigma z) - h(x)| p_1(z)\,dz$ at $|z| \le M$ and let $\sigma \to 0$, then $M \to \infty$). Now $f_t = \phi_t \circ d$, so at every differentiability point y of d with $0 < d(y) < t$, $\nabla f_t(y) = -\frac{1}{t}\nabla d(y)$. The eikonal equation holds: if d is differentiable at y with $d(y) > 0$, then $\|\nabla d(y)\| = 1$. Indeed, take $a_k \in A$ with $|y - a_k| \to d(y)$ and $v_k = (y-a_k)/|y-a_k| \to v$, $\|v\| = 1$; for $0 < s < |y-a_k|$, $d(y - sv_k) \le |y-a_k| - s$, so $\frac{d(y)-d(y-sv_k)}{s} \ge 1 - \frac{|y-a_k|-d(y)}{s}$; letting $k \to \infty$ then $s \to 0$ gives $\langle \nabla d(y), v\rangle \ge 1$, whence $\|\nabla d(y)\| = 1$ since d is 1-Lipschitz. By Rademacher's theorem, $\|\nabla d\| = 1$ a.e. on $\{d > 0\}$, so $\|\nabla f_t\| = 1/t$ a.e. on $\{0 < d < t\}$. Consequently $\|\nabla f_{t,\sigma}(x)\| \to 1/t$ for a.e. $x \in A_t \setminus A$ (and $\to 0$ for a.e. x outside), and dominated convergence gives $\mathbb{E}_\gamma\|\nabla f_{t,\sigma}\| \to \frac{1}{t}\,\gamma_n(\{0 < d < t\})$.

Remark. The book's claim that $\|\nabla f_{t,\sigma}(x)\| \to 1/t$ at every interior point x of $A_t \setminus A$ is false as stated. Counterexample: $A = B(0,1) \cup B(10e_1, 1)$, $x = 5e_1$, $t = 6$. Then x is interior to $A_t \setminus A$, but reflection symmetry across $\{y_1 = 5\}$ forces the e_1-component of $\mathbb{E}[(\nabla f_t)(x + \sigma Z)]$ to vanish, and rotation symmetry about the e_1-axis forces it parallel to e_1; hence $\nabla f_{t,\sigma}(x) = 0$ for every $\sigma > 0$, not tending to $1/t$. The a.e. statement proved above is the correct version and suffices for the proof.

Exercise 10.12

Central idea. For closed convex A, enlargements are Minkowski sums, $A_s = A + sB_2^n$, and $\lambda A + (1-\lambda)A_s = A_{(1-\lambda)s}$; feeding this into Ehrhard's Gaussian Brunn–Minkowski inequality (10.4) makes $\varphi(s) = \Phi^{-1}(\gamma_n(A_s))$ concave. Since $\psi(t) := \varphi(t) - t$ is concave and bounded below on $[0,\infty)$ — boundedness below for large t follows from Gaussian concentration via a fixed ball $B(x_0, t) \subset A_t$ (using $\Phi^{-1}(1 - e^{-u^2/2}) - u$ bounded below on $[1,\infty)$, not the reversed estimate $\Phi^{-1}(1-\varepsilon) \ge \sqrt{2\log(1/\varepsilon)}$), and on compacta by continuity — concavity forces ψ to be nondecreasing, which is exactly $\Phi^{-1}(\gamma_n(A_t)) \ge \Phi^{-1}(\gamma_n(A)) + t$.

Solution. Let $P = \gamma_n$ be the canonical Gaussian measure on $\mathbb{R}^n$. The Gaussian isoperimetric theorem states that for Borel A with $P(A) = \Phi(a)$,

$$P(A_t) \ge \Phi(a + t), \qquad \text{i.e.} \quad \Phi^{-1}(P(A_t)) \ge \Phi^{-1}(P(A)) + t,$$

with the conventions $\Phi^{-1}(0) = -\infty$, $\Phi^{-1}(1) = +\infty$. Let A be convex. If $P(A) \in \{0, 1\}$ or $A = \emptyset$ the claim is trivial, so assume $A \neq \emptyset$ and $0 < P(A) < 1$. Replacing A by its closure does not change $P(A)$ (the boundary of a convex set is Lebesgue-null, hence P-null) nor the enlargements, since $d(\cdot, A) = d(\cdot, \bar{A})$; assume henceforth that A is closed and convex.

For $s \ge 0$, $A_s = \{x : d(x, A) \le s\} = A + sB_2^n$, where B_2^n is the closed Euclidean unit ball (the infimum defining d is attained since A is closed). Put $\varphi(s) = \Phi^{-1}(P(A_s))$, well-defined and finite: $0 < P(A) \le P(A_s)$, and $P(A_s) < 1$ because $\{d(\cdot, A) > s\} = d^{-1}((s,\infty))$ is a nonempty open set (d is continuous; e.g. for $y \notin A$ with projection a and $v = (y-a)/\|y-a\|$, convexity gives $d(y + rv, A) \ge \|y - a\| + r \to \infty$), and P charges every nonempty open set.

For $\lambda \in [0,1]$ and $s \ge 0$, convexity gives $\lambda A + (1-\lambda)A = A$, so

$$\lambda A + (1-\lambda)A_s = \lambda A + (1-\lambda)(A + sB_2^n) = A + (1-\lambda)sB_2^n = A_{(1-\lambda)s}.$$

Apply the Gaussian Brunn–Minkowski inequality (10.4) to the convex sets A and A_s:

$$\varphi((1-\lambda)s) = \Phi^{-1}(P(A_{(1-\lambda)s})) \ge \lambda\Phi^{-1}(P(A)) + (1-\lambda)\Phi^{-1}(P(A_s)) = \lambda\varphi(0) + (1-\lambda)\varphi(s).$$

Writing $u = (1-\lambda)s \in [0, s]$, i.e. $\lambda = 1 - u/s$, this is $\varphi(u) \ge (1-u/s)\varphi(0) + (u/s)\varphi(s)$: φ is concave on $[0,\infty)$.

Fix $x_0 \in A$; then $B(x_0, t) \subset A_t$. With $Z \sim P$,

$$P(A_t) \geq \mathbb{P}(\|Z + x_0\| \leq t) \geq \mathbb{P}(\|Z\| \leq t - \|x_0\|).$$

Gaussian concentration for the 1-Lipschitz norm gives $\mathbb{P}(\|Z\| \geq \mathbb{E}\|Z\| + u) \leq e^{-u^2/2}$, and $\mathbb{E}\|Z\| \leq (\mathbb{E}\|Z\|^2)^{1/2} = \sqrt{n}$, so for $u := t - \|x_0\| - \sqrt{n} \geq 0$,

$$P(A_t) \geq 1 - e^{-u^2/2}, \qquad \text{hence} \qquad \psi(t) := \varphi(t) - t \geq \left[\Phi^{-1}(1 - e^{-u^2/2}) - u\right] - \|x_0\| - \sqrt{n}.$$

We show $u \mapsto \Phi^{-1}(1 - e^{-u^2/2}) - u$ is bounded below on $[1, \infty)$. It is continuous on $(0, \infty)$. For $u \geq 3$, Gordon's lower bound $1 - \Phi(v) \geq v\phi(v)/(1 + v^2)$ at $v = u - 1 \geq 2$ gives $1 - \Phi(u - 1) \geq e^{-u^2/2}$: indeed this is $\frac{u-1}{1+(u-1)^2} e^{\,u-1/2} \geq \sqrt{2\pi}$, and $w \mapsto \frac{w}{1+w^2} e^{w+1/2}$ is increasing for $w \geq 2$ with value $\frac{2}{5} e^{2.5} \approx 4.87 \geq \sqrt{2\pi}$ at $w = 2$. Hence $\Phi^{-1}(1 - e^{-u^2/2}) \geq u - 1$ for $u \geq 3$. On the compact $[1, 3]$ continuity gives a finite minimum. Thus $\psi(t) \geq m_0 - \|x_0\| - \sqrt{n}$ for $t \geq \|x_0\| + \sqrt{n} + 1$ (so $u \geq 1$), with $m_0 > -\infty$. On $[0, \|x_0\| + \sqrt{n} + 1]$, ψ is continuous — $t \mapsto P(A_t)$ is: right-continuous by continuity of measure from above ($\{d \leq t_n\} \downarrow \{d \leq t\}$), left-continuous since $A_t \setminus \bigcup_{s<t} A_s = \{d(\cdot, A) = t\} \subset \partial A_t$ is P-null (the boundary of the convex set A_t is Lebesgue-null) — and Φ^{-1} is continuous on $(0, 1)$, so ψ attains a finite minimum there. Hence ψ is bounded below on $[0, \infty)$. (The estimate $\Phi^{-1}(1 - \varepsilon) \geq \sqrt{2\log(1/\varepsilon)}$ is false — the correct implication of $1 - \Phi(u) \leq e^{-u^2/2}$ is $\Phi^{-1}(1 - \varepsilon) \leq \sqrt{2\log(1/\varepsilon)}$ — and is not needed.)

But ψ is concave. If $\psi(s_1) > \psi(s_2)$ for some $s_1 < s_2$, concavity would give $\psi(t) \leq \psi(s_2) + m(t - s_2)$ for $t > s_2$ with $m = (\psi(s_2) - \psi(s_1))/(s_2 - s_1) < 0$, forcing $\psi(t) \to -\infty$, a contradiction. Hence ψ is nondecreasing, so $\psi(t) \geq \psi(0)$ for all $t \geq 0$, i.e.

$$\Phi^{-1}(P(A_t)) \geq \Phi^{-1}(P(A)) + t, \qquad t \geq 0,$$

which is the Gaussian isoperimetric inequality for the convex set A. ■

Chapter 11 — The Variance of Suprema of Empirical Processes

Exercise 11.1

Central idea. The wimpy variance sees only marginal variances, while $\mathrm{Var}(\sup_s \sum_i X_{i,s})$ also depends on the dependence across s. With $n = 1$, $\mathcal{T} = \{1, 2\}$, take Y_1, Y_2 Bernoulli(1/4) with disjoint supports: each has variance 3/16 (so $\sigma^2 = 3/16$), but $\max(Y_1, Y_2) \sim$ Bernoulli(1/2) has variance $1/4 > 3/16$.

Solution. Let $n = 1$, $\mathcal{T} = \{1, 2\}$, and write $Y_s = X_{1,s}$. The wimpy variance is $\sigma^2 = \max_{s=1,2} \mathrm{Var}(Y_s)$, and $Z = \max(Y_1, Y_2)$.

Define the joint law of (Y_1, Y_2) on $\{0, 1\}^2$ by

$$\mathbb{P}(Y_1 = 1, Y_2 = 0) = \mathbb{P}(Y_1 = 0, Y_2 = 1) = \tfrac{1}{4}, \quad \mathbb{P}(Y_1 = 0, Y_2 = 0) = \tfrac{1}{2}, \quad \mathbb{P}(Y_1 = 1, Y_2 = 1) = 0.$$

Each marginal is Bernoulli(1/4), so $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$, hence $\sigma^2 = 3/16$.

But $Z = \max(Y_1, Y_2)$ equals 1 iff at least one of Y_1, Y_2 is 1, so $\mathbb{P}(Z = 1) = 1/4 + 1/4 = 1/2$; i.e., $Z \sim$ Bernoulli(1/2) and

$$\mathrm{Var}(Z) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4} > \tfrac{3}{16} = \sigma^2.$$

Thus $\mathrm{Var}(Z) \leq \sigma^2$ fails in general. ■

Exercise 11.2

Central idea. Read the process diagonally, $X_{i,s} = Y_i \mathbb{1}\{i = s\}$ with Y_i i.i.d. Exp(1), so $Z = \max_s Y_s$ (the case the hint describes). The Rényi representation gives $\mathbb{E}Z = H_n$ and $\mathrm{Var}(Z) = H_n^{(2)} \leq \pi^2/6$. Theorem 11.1 bounds the variance by the strong variance V ($= 2n$, or n after centering) or, after centering, by $\Sigma^2 + \sigma^2$. Both are much too large: V grows linearly in n, and $\Sigma^2 + \sigma^2 \geq 1 + H_n^{(2)} + (H_n - 1)^2 \sim \log^2 n$, because the

weak variance is driven by the typical size of the maximum rather than its fluctuations. The leave-one-out Efron–Stein bound $\sum_i \mathbb{E}(Z - Z_i)^2$ equals exactly 2, within a bounded factor of the truth.

Solution. *Interpretation.* If all $X_{i,s}$ were literally i.i.d. Exp(1), each $\sum_{i=1}^n X_{i,s}$ would be Gamma$(n, 1)$, and Z would not be "the supremum of n i.i.d. exponentials" with mean H_n. We therefore take the diagonal process $X_{i,s} = Y_i \mathbb{1}\{i = s\}$, $1 \le i, s \le n$, with $Y_1, \dots, Y_n$ i.i.d. Exp(1). The vectors $X_i = (X_{i,s})_s$ are independent, $\sum_i X_{i,s} = Y_s$, and $Z = \max_{1 \le s \le n} Y_s$.

Mean and variance. $\mathbb{P}(Z > t) = 1 - (1 - e^{-t})^n$, so with $u = 1 - e^{-t}$,

$$\mathbb{E}Z = \int_0^\infty (1 - (1 - e^{-t})^n) dt = \int_0^1 \frac{1 - u^n}{1 - u}\, du = \sum_{k=1}^n \frac{1}{k} = H_n.$$

By the Rényi representation, the spacings of the order statistics are independent with $Y_{(n-k+1)} - Y_{(n-k)} \overset{d}{=} E_k/k$ ($Y_{(0)} = 0$, E_k i.i.d. Exp(1)). So $Z \overset{d}{=} \sum_{k=1}^n E_k/k$ and

$$\mathrm{Var}(Z) = \sum_{k=1}^n \frac{1}{k^2} =: H_n^{(2)} \le \sum_{k \ge 1} \frac{1}{k^2} = \frac{\pi^2}{6} \approx 1.645.$$

The bounds of Theorem 11.1. Recall $V = \sum_i \mathbb{E} \sup_s X_{i,s}^2$, $\Sigma^2 = \mathbb{E} \sup_s \sum_i X_{i,s}^2$ and $\sigma^2 = \sup_s \sum_i \mathbb{E} X_{i,s}^2$. Theorem 11.1 gives $\mathrm{Var}(Z) \le V$ always, and $\mathrm{Var}(Z) \le \Sigma^2 + \sigma^2$ when $\mathbb{E}X_{i,s} = 0$ for all i, s.

(a) For the process as given (not centered), $\sup_s X_{i,s}^2 = Y_i^2$, so

$$V = \sum_{i=1}^n \mathbb{E} Y_i^2 = 2n.$$

The second inequality does not apply, because $\mathbb{E}X_{i,i} = 1 \neq 0$.

(b) Centering. Put $\tilde{X}_{i,s} = (Y_i - 1)\mathbb{1}\{i = s\}$. These vectors are independent, $\mathbb{E}\tilde{X}_{i,s} = 0$ for all i, s, and $\tilde{Z} = \sup_s \sum_i \tilde{X}_{i,s} = \max_s (Y_s - 1) = Z - 1$, so $\mathrm{Var}(\tilde{Z}) = \mathrm{Var}(Z)$. For this process

$$\tilde{V} = \sum_i \mathbb{E}(Y_i - 1)^2 = n, \qquad \sigma^2 = \max_s \mathrm{Var}(Y_s) = 1, \qquad \Sigma^2 = \mathbb{E} \max_s (Y_s - 1)^2.$$

Taking s to be an index achieving $\max_s Y_s$ gives $\max_s (Y_s - 1)^2 \ge (Z - 1)^2$, so

$$\Sigma^2 \ge \mathbb{E}(Z - 1)^2 = \mathrm{Var}(Z) + (\mathbb{E}Z - 1)^2 = H_n^{(2)} + (H_n - 1)^2.$$

Conversely, fix s. If $Y_s \ge 1$, then $0 \le Y_s - 1 \le Z - 1$. If $0 \le Y_s < 1$, then $(Y_s - 1)^2 \le 1$. Hence $\max_s (Y_s - 1)^2 \le (Z - 1)^2 + 1$, and $\Sigma^2 \le 1 + H_n^{(2)} + (H_n - 1)^2$. Therefore

$$1 + H_n^{(2)} + (H_n - 1)^2 \;\le\; \Sigma^2 + \sigma^2 \;\le\; 2 + H_n^{(2)} + (H_n - 1)^2.$$

Comparison. Since $H_n = \log n + O(1)$, the bound $\Sigma^2 + \sigma^2$ is of exact order $\log^2 n$. The bound V is of order n ($2n$ uncentered, n centered). Meanwhile $\mathrm{Var}(Z) \le \pi^2/6$ stays bounded. Both bounds of Theorem 11.1 are off by an unbounded factor: Σ^2 is dominated by $(\mathbb{E}Z - 1)^2$, the squared typical size of the maximum, not by its fluctuations.

The leave-one-out bound. Here $Z_i = \sup_s \sum_{j \neq i} X_{j,s}$. For $s \neq i$ the inner sum is Y_s; for $s = i$ it is 0.

- If $n = 1$, then $Z_1 = 0$ and $\mathbb{E}(Z - Z_1)^2 = \mathbb{E}Y_1^2 = 2$.

- If $n \geq 2$, then $Z_i = \max(M_i, 0) = M_i$ with $M_i = \max_{j \neq i} Y_j \geq 0$. So $Z - Z_i = (Y_i - M_i)_+$, where M_i is independent of Y_i. For $m \geq 0$,

$$\mathbb{E}[(Y_i - m)_+^2] = \int_m^\infty (y-m)^2 e^{-y}\, dy = 2e^{-m},$$

so $\mathbb{E}[(Z - Z_i)^2] = 2\,\mathbb{E}[e^{-M_i}]$. Now $e^{-M_i} = \min_{j \neq i} e^{-Y_j}$, and the e^{-Y_j} are i.i.d. Uniform$[0,1]$, because $\mathbb{P}(e^{-Y_j} \leq t) = \mathbb{P}(Y_j \geq -\log t) = t$. Hence $\mathbb{E}[e^{-M_i}] = \int_0^1 (1-t)^{n-1} dt = 1/n$.

In both cases,

$$\sum_{i=1}^n \mathbb{E}\big[(Z - Z_i)^2\big] = n \cdot \frac{2}{n} = 2.$$

Equivalently, only the index of the maximum contributes, with $(Y_{(n)} - Y_{(n-1)})^2$, and the top spacing is Exp(1), whose second moment is 2. Since Z_i is a function of $X^{(i)}$, the Efron–Stein inequality gives $\mathrm{Var}(Z) \leq 2$.

Comparison with the truth. $\mathrm{Var}(Z) = H_n^{(2)} \in [1, \pi^2/6)$, while the leave-one-out bound is 2. The ratio $2/H_n^{(2)}$ decreases to $12/\pi^2 \approx 1.216$, so this bound is sharp up to a constant factor. The bounds of Theorem 11.1 ($2n$ or n, and $\gtrsim \log^2 n$) are not. The loss in the bound V of Theorem 11.1 comes from the step $(Z - Z_i)^2 \leq \sup_s X_{i,s}^2$ in its proof: it charges every coordinate its full size Y_i^2, instead of only the gap above the runner-up. ■

Exercise 11.3

Central idea. Separability makes every relevant supremum over $\mathcal{T}$ equal almost surely to the same supremum over a countable separant $\mathcal{S} = \{s_1, s_2, \dots\}$. Theorem 11.1 applies to the finite index sets $\{s_1, \dots, s_m\}$, and the finite-set quantities $V_m, \Sigma_m^2, \sigma_m^2$ are dominated by V, Σ^2, σ^2. The finite maxima $Z_m = \max_{k \leq m} \sum_i X_{i,s_k}$ increase to Z. Whenever the relevant bound (V, or $\Sigma^2 + \sigma^2$) is finite, $|Z_m|$ is dominated by a single square-integrable variable, so $\mathrm{Var}(Z_m) \to \mathrm{Var}(Z)$ and both inequalities pass to the limit.

Solution. *Setting.* $X_1, \dots, X_n$ are independent, $X_i = (X_{i,t})_{t \in \mathcal{T}}$, and $Z = \sup_{t \in \mathcal{T}} \sum_{i=1}^n X_{i,t}$. As in the text,

$$V = \sum_{i=1}^n \mathbb{E} \sup_{t \in \mathcal{T}} X_{i,t}^2, \qquad \Sigma^2 = \mathbb{E} \sup_{t \in \mathcal{T}} \sum_{i=1}^n X_{i,t}^2, \qquad \sigma^2 = \sup_{t \in \mathcal{T}} \sum_{i=1}^n \mathbb{E} X_{i,t}^2.$$

Separability means there is a countable $\mathcal{S} = \{s_1, s_2, \dots\} \subseteq \mathcal{T}$ and an event Ω_0 with $\mathbb{P}(\Omega_0) = 1$ such that on Ω_0, for every $t \in \mathcal{T}$, some sequence $(t_k) \subseteq \mathcal{S}$ satisfies $X_{i,t_k} \to X_{i,t}$ for all $i = 1, \dots, n$ (density of $\mathcal{S}$ for the topology of pointwise convergence). We must show: $\mathrm{Var}(Z) \leq V$, and $\mathrm{Var}(Z) \leq \Sigma^2 + \sigma^2$ if $\mathbb{E} X_{i,t} = 0$ for all i, t. We use Theorem 11.1 only for finite index sets.

Step 1: reduction to $\mathcal{S}$. Fix $\omega \in \Omega_0$ and $t \in \mathcal{T}$, and take $t_k \in \mathcal{S}$ as above. Then $\sum_i X_{i,t_k} \to \sum_i X_{i,t}$, $\sum_i X_{i,t_k}^2 \to \sum_i X_{i,t}^2$, and $X_{i,t_k}^2 \to X_{i,t}^2$ for each i. So each value at t is a limit of values at points of $\mathcal{S}$, hence at most the supremum over $\mathcal{S}$. The reverse inequalities are trivial because $\mathcal{S} \subseteq \mathcal{T}$. Hence, on Ω_0,

$$Z = Y := \sup_{s \in \mathcal{S}} \sum_i X_{i,s}, \qquad \sup_{t \in \mathcal{T}} \sum_i X_{i,t}^2 = \sup_{s \in \mathcal{S}} \sum_i X_{i,s}^2, \qquad \sup_{t \in \mathcal{T}} X_{i,t}^2 = \sup_{s \in \mathcal{S}} X_{i,s}^2 \quad (i \leq n).$$

The right-hand sides are countable suprema, hence measurable. So Z coincides a.s. with the random variable Y; $\mathrm{Var}(Z)$ means $\mathrm{Var}(Y)$; and V and Σ^2 are unchanged when $\mathcal{T}$ is replaced by $\mathcal{S}$. Also $\sigma_{\mathcal{S}}^2 := \sup_{s \in \mathcal{S}} \sum_i \mathbb{E} X_{i,s}^2 \leq \sigma^2$, and the centering hypothesis is inherited by $\mathcal{S}$.

Step 2: finite approximants. For $m \geq 1$ let $Z_m = \max_{k \leq m} \sum_i X_{i,s_k}$, and let $V_m, \Sigma_m^2, \sigma_m^2$ be the three quantities for the index set $\{s_1, \dots, s_m\}$. The vectors $(X_{i,s_k})_{k \leq m}$, $i = 1, \dots, n$, are independent, so Theorem 11.1 for a finite index set gives

$$\mathrm{Var}(Z_m) \leq V_m, \qquad \text{and, if centered,} \qquad \mathrm{Var}(Z_m) \leq \Sigma_m^2 + \sigma_m^2.$$

Maxima over subsets of $\mathcal{S}$ are smaller, so by Step 1 $V_m \le V$, $\Sigma_m^2 \le \Sigma^2$, and $\sigma_m^2 \le \sigma_{\mathcal{S}}^2 \le \sigma^2$. Let B denote the relevant bound ($B = V$, or $B = \Sigma^2 + \sigma^2$ in the centered case). Then

$$\mathrm{Var}(Z_m) \le B \qquad \text{for all } m. \tag{1}$$

If $B = \infty$ there is nothing to prove, so assume $B < \infty$. Moreover $Z_m \uparrow Y$ pointwise, since Y is the supremum over $\mathcal{S} = \bigcup_m \{s_1, \dots, s_m\}$.

Step 3: a square-integrable envelope. We find $U \ge 0$ with $\mathbb{E}U^2 < \infty$ and $|Z_m| \le U$ for all m; then also $|Y| \le U$.

- Case $B = V < \infty$. Put $U = \sum_{i=1}^n \sup_{s\in\mathcal{S}} |X_{i,s}|$. For every $s \in \mathcal{S}$, $|\sum_i X_{i,s}| \le U$, so $|Z_m| \le U$. By Cauchy–Schwarz, $U^2 \le n \sum_i \sup_s X_{i,s}^2$, so $\mathbb{E}U^2 \le nV < \infty$.
- Case $B = \Sigma^2 + \sigma^2 < \infty$. Put $U = \sqrt{n}\,\big(\sup_{s\in\mathcal{S}} \sum_i X_{i,s}^2\big)^{1/2}$. By Cauchy–Schwarz, $|\sum_i X_{i,s}| \le \sqrt{n}(\sum_i X_{i,s}^2)^{1/2} \le U$ for every $s \in \mathcal{S}$, so $|Z_m| \le U$, and $\mathbb{E}U^2 = n\Sigma^2 < \infty$.

Step 4: passage to the limit. $Z_m \to Y$ and $Z_m^2 \to Y^2$ pointwise, with $|Z_m| \le U$ and $Z_m^2 \le U^2$, where U and U^2 are integrable. Dominated convergence gives $\mathbb{E}Z_m \to \mathbb{E}Y$ and $\mathbb{E}Z_m^2 \to \mathbb{E}Y^2$, both finite. Hence $\mathrm{Var}(Z_m) \to \mathrm{Var}(Y)$. (Equivalently, as in the hint, monotone convergence applies to $0 \le Z_m - Z_1 \uparrow Y - Z_1$ and to its square; the envelope only guarantees finite limits.) Letting $m \to \infty$ in (1),

$$\mathrm{Var}(Z) = \mathrm{Var}(Y) \le V, \qquad \text{and, if } \mathbb{E}X_{i,t} = 0 \text{ for all } i,t, \qquad \mathrm{Var}(Z) \le \Sigma^2 + \sigma^2.$$

So both conclusions of Theorem 11.1 hold for separable processes indexed by an arbitrary set $\mathcal{T}$. As a by-product, $Z \in L^2$ whenever the relevant bound is finite; no integrability of Z had to be assumed. ■

Exercise 11.4

Central idea. The $-n$ spikes of $X_{i,s}$ flip to $+n$ under negative Rademachers. On $\{m \ge n/4\}$ (high probability, $m = \#\{i : \varepsilon_i = -1\}$), each coordinate's flipped-spike count $B_s \sim \mathrm{Bin}(m, 1/n)$ is independent across s, and $\mathbb{E}\max_s B_s \gtrsim \log(n/4)/\log\log n$ (binomial-max lemma); each spike contributes n. Maximizing over s via $s^* = \arg\max B_s$ keeps the competing $-n$ spikes C_{s^*} an independent $\mathrm{Bin}(n-m, 1/n)$ of mean < 1. The non-symmetrized process is $\le n^2/(n-1) = O(n)$ pointwise, so for large n half the symmetrized expectation exceeds it: the absolute values in Lemma 11.4's left symmetrization inequality cannot be dropped.

Solution. $\mathbb{E}X_{i,s} = \frac{n}{n-1} \cdot \frac{n-1}{n} - n \cdot \frac{1}{n} = 0$. Let $m = \sum_{i=1}^n \mathbf{1}_{\{\varepsilon_i = -1\}} \sim \mathrm{Bin}(n, 1/2)$; by Hoeffding, $\mathbb{P}\{m < n/4\} \le e^{-n/8}$. Fix ε with $m \ge n/4$ and write $I_- = \{i : \varepsilon_i = -1\}$, $I_+ = \{i : \varepsilon_i = +1\}$. For $s \in \mathcal{T}$ let $B_s = \#\{i \in I_- : X_{i,s} = -n\}$ and $C_s = \#\{i \in I_+ : X_{i,s} = -n\}$; conditional on ε, the B_s are iid $\mathrm{Bin}(m, 1/n)$, the C_s are iid $\mathrm{Bin}(n-m, 1/n)$, and the two families are independent.

Pointwise, $\varepsilon_i X_{i,s}$ equals n on the B_s good spikes, $-n$ on the C_s bad spikes, and is $\ge -n/(n-1)$ otherwise, so

$$\sum_{i=1}^n \varepsilon_i X_{i,s} \;\ge\; nB_s - nC_s - \frac{n^2}{n-1} \;\ge\; nB_s - nC_s - 2n.$$

Let $s^* \in \arg\max_s B_s$, measurable w.r.t. the I_--variables, hence independent of the I_+-variables; thus $C_{s^*} \sim \mathrm{Bin}(n-m, 1/n)$ still, with $\mathbb{E}[C_{s^*}] = (n-m)/n < 1$. Therefore

$$\mathbb{E}\Big[\sup_s \sum_i \varepsilon_i X_{i,s} \,\Big|\, \varepsilon\Big] \;\ge\; n\,\mathbb{E}[\max_s B_s \mid \varepsilon] - 3n.$$

Binomial-max lemma. If $Z_1, \dots, Z_n$ are iid $\mathrm{Bin}(N, 1/n)$ with $N = \lfloor n/4 \rfloor$, then $\mathbb{E}\max_s Z_s \ge (1 - o(1))\frac{\log(n/4)}{\log\log n}$. Indeed, for integer $k \ge 1$, $\mathbb{P}\{Z_1 \ge k\} \ge \binom{N}{k} n^{-k}(1-1/n)^N \ge \frac{1}{2}(5k)^{-k}$ (for large n), and with $k_n = \lfloor \log(n/4)/\log\log n \rfloor$ one checks $n\mathbb{P}\{Z_1 \ge k_n\} \to \infty$, so $\mathbb{P}\{\max_s Z_s \ge k_n\} \to 1$. Since $\mathrm{Bin}(m, 1/n)$ dominates $\mathrm{Bin}(N, 1/n)$ for $m \ge n/4$, $\mathbb{E}[\max_s B_s \mid \varepsilon] \ge (1 - o(1))\frac{\log(n/4)}{\log\log n}$.

Assembling, with $L_n = \frac{\log(n/4)}{\log\log n} \to \infty$,

$$\mathbb{E}\sup_s \sum_i \varepsilon_i X_{i,s} \;\geq\; (1-e^{-n/8})\, n((1-o(1))L_n - 3) - n^2 e^{-n/8} \;\geq\; \frac{n}{2} L_n$$

for n large.

For the non-symmetrized process, $\sum_i X_{i,s} = \frac{n}{n-1}(n - C'_s) - nC'_s = \frac{n^2}{n-1}(1 - C'_s)$ with $C'_s \sim \mathrm{Bin}(n, 1/n)$, so $\sup_s \sum_i X_{i,s} \leq \frac{n^2}{n-1}$ pointwise and $\mathbb{E}\sup_s \sum_i X_{i,s} \leq \frac{n^2}{n-1}$. Since $L_n \to \infty$, for large n,

$$\frac{1}{2}\,\mathbb{E}\sup_s \sum_{i=1}^n \varepsilon_i X_{i,s} \;\geq\; \frac{n}{4} L_n \;>\; \frac{n^2}{n-1} \;\geq\; \mathbb{E}\sup_s \sum_{i=1}^n X_{i,s}.$$

Comparison with Lemma 11.4. Lemma 11.4's left symmetrization inequality, $\frac{1}{2}\,\mathbb{E}\sup_s |\sum_i \varepsilon_i X_{i,s}| \leq \mathbb{E}\sup_s |\sum_i X_{i,s}|$, genuinely needs the absolute values: this example shows the one-sided analogue $\frac{1}{2}\,\mathbb{E}\sup_s \sum_i \varepsilon_i X_{i,s} \leq \mathbb{E}\sup_s \sum_i X_{i,s}$ is false — here the left side exceeds the right by an unbounded factor (at least $\frac{n-1}{4n} L_n \to \infty$). The heavy left tail ($-n$ spikes) keeps $\sup_s \sum_i X_{i,s} \leq n^2/(n-1)$ pointwise, but the Rademachers flip those spikes to $+n$, so the symmetrized process sees spikes the original process never sees. ■

Exercise 11.5

Central idea. Standard symmetrization with independent copies X'_i of X_i: centering lets Jensen move $\mathbb{E}_{X'}$ inside $\Psi(\sup_s |\cdot|)$; the identity $\sum_i \varepsilon_i(X_{i,s} - X'_{i,s}) \overset{d}{=} \sum_i (X_{i,s} - X'_{i,s})$ swaps Rademachers in and out; convexity of Ψ then splits $\Psi(\frac{A+B}{2}) \leq \frac{\Psi(A)+\Psi(B)}{2}$ (left inequality) and $\Psi(A+B) \leq \frac{\Psi(2A)+\Psi(2B)}{2}$ (right inequality), with Ψ increasing to pass through the triangle inequality.

Solution. Let $X'_1, \ldots, X'_n$ be independent copies of $X_1, \ldots, X_n$, independent of everything else. Each $X_{i,s} - X'_{i,s}$ is symmetric, so with ε_i independent Rademachers,

$$\sum_{i=1}^n \varepsilon_i (X_{i,s} - X'_{i,s}) \;\overset{d}{=}\; \sum_{i=1}^n (X_{i,s} - X'_{i,s}) \qquad \text{as processes in } s.$$

(The Ψ-expectations below are $+\infty$-valued where not integrable; each claimed inequality is trivial when its right-hand side is $+\infty$.)

Right inequality. Since $\mathbb{E}X'_{i,s} = 0$, $\sum_i X_{i,s} = \mathbb{E}_{X'}[\sum_i (X_{i,s} - X'_{i,s})]$. The map $(z_s) \mapsto \sup_s |z_s|$ is convex, so by Jensen (twice, using that Ψ is convex and increasing),

$$\Psi\left(\sup_s \Big|\sum_i X_{i,s}\Big|\right) \leq \mathbb{E}_{X'}\Psi\left(\sup_s \Big|\sum_i (X_{i,s} - X'_{i,s})\Big|\right).$$

Taking $\mathbb{E}_X$ and applying the symmetrization identity,

$$\mathbb{E}\Psi\left(\sup_s \Big|\sum_i X_{i,s}\Big|\right) \leq \mathbb{E}\Psi\left(\sup_s \Big|\sum_i \varepsilon_i (X_{i,s} - X'_{i,s})\Big|\right).$$

With $A = \sup_s |\sum_i \varepsilon_i X_{i,s}|$, $B = \sup_s |\sum_i \varepsilon_i X'_{i,s}|$, the triangle inequality and monotonicity give $\Psi(\sup_s |\sum_i \varepsilon_i (X_{i,s} - X'_{i,s})|) \leq \Psi(A+B)$, and convexity gives $\Psi(A+B) = \Psi(2 \cdot \frac{A+B}{2}) \leq \frac{\Psi(2A)+\Psi(2B)}{2}$. Since (X_i) and (X'_i) have the same law, $\mathbb{E}\Psi(2B) = \mathbb{E}\Psi(2A)$, whence

$$\mathbb{E}\Psi\left(\sup_s \Big|\sum_i X_{i,s}\Big|\right) \leq \mathbb{E}\Psi\left(2\sup_s \Big|\sum_i \varepsilon_i X_{i,s}\Big|\right).$$

Left inequality. Write $\sum_i \varepsilon_i X_{i,s} = \mathbb{E}_{X'}[\sum_i \varepsilon_i (X_{i,s} - X'_{i,s})]$. By Jensen as above,

$$\Psi\left(\tfrac{1}{2}\sup_s \Big|\sum_i \varepsilon_i X_{i,s}\Big|\right) \le \mathbb{E}_{X'}\Psi\left(\tfrac{1}{2}\sup_s \Big|\sum_i \varepsilon_i (X_{i,s} - X'_{i,s})\Big|\right).$$

Taking $\mathbb{E}_{X,\varepsilon}$ and using the symmetrization identity in reverse,

$$\mathbb{E}\Psi\left(\tfrac{1}{2}\sup_s \Big|\sum_i \varepsilon_i X_{i,s}\Big|\right) \le \mathbb{E}_{X,X'}\Psi\left(\tfrac{1}{2}\sup_s \Big|\sum_i (X_{i,s} - X'_{i,s})\Big|\right).$$

Now $\frac{1}{2}\sup_s|\sum_i(X_{i,s}-X'_{i,s})| \le \frac{A'+B'}{2}$ with $A' = \sup_s|\sum_i X_{i,s}|$, $B' = \sup_s|\sum_i X'_{i,s}|$; by monotonicity and convexity, $\Psi(\frac{A'+B'}{2}) \le \frac{\Psi(A')+\Psi(B')}{2}$, and $\mathbb{E}\Psi(B') = \mathbb{E}\Psi(A')$. Hence

$$\mathbb{E}\Psi\left(\tfrac{1}{2}\sup_s \Big|\sum_i \varepsilon_i X_{i,s}\Big|\right) \le \mathbb{E}\Psi\left(\sup_s \Big|\sum_i X_{i,s}\Big|\right). \qquad \blacksquare$$

Exercise 11.6

Central idea. The book's "Theorem 11.4" is Lemma 11.4 (the symmetrization inequalities) — no Theorem 11.4 exists in the numbering. If $S \subset T$ is a countable separant, the suprema over T agree a.s. with those over S (the same separant works for the Rademacher-symmetrized process, since multiplication by fixed signs is continuous), so Lemma 11.4 applied to the finite sets $S_m = \{s_1, \dots, s_m\} \uparrow S$ passes to the limit by monotone convergence, exactly as in Exercise 11.3.

Solution. (Remark on numbering: the chapter contains no Theorem 11.4; the symmetrization inequalities are Lemma 11.4, and Exercises 11.4 and 11.6 refer to it.) Let $X_1, \dots, X_n$ be independent centered random vectors $X_i = (X_{i,s})_{s\in T}$ with each $X_{i,s}$ integrable, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables independent of the X_i. Assume T is separable with countable separant S: a.s., for every $t \in T$ there is a sequence in S along which $(X_{1,t_n}, \dots, X_{n,t_n}) \to (X_{1,t}, \dots, X_{n,t})$ (the topology of pointwise convergence on $\mathbb{R}^n$). For fixed signs $\eta_i \in \{-1, 1\}$ the same sequence gives $\eta_i X_{i,t_n} \to \eta_i X_{i,t}$, so S is also a separant for every Rademacher-symmetrized process $(\varepsilon_i X_{i,s})_{s\in T}$. By continuity of the sum and of $|\cdot|$, a.s.

$$\sup_{s\in T}\sum_{i=1}^n X_{i,s} = \sup_{s\in S}\sum_{i=1}^n X_{i,s}, \qquad \sup_{s\in T}\Big|\sum_{i=1}^n X_{i,s}\Big| = \sup_{s\in S}\Big|\sum_{i=1}^n X_{i,s}\Big|,$$

and likewise with $\varepsilon_i X_{i,s}$ in place of $X_{i,s}$; this is the consequence of separability displayed in Section 11.1. In particular all suprema are a.s. equal to measurable (countable) suprema.

Enumerate $S = \{s_1, s_2, \dots\}$ and set $S_m = \{s_1, \dots, s_m\}$. Lemma 11.4 applied to the finite (hence countable) index set S_m gives

$$\tfrac{1}{2}\,\mathbb{E}\sup_{s\in S_m}\Big|\sum_i \varepsilon_i X_{i,s}\Big| \le \mathbb{E}\sup_{s\in S_m}\Big|\sum_i X_{i,s}\Big| \le 2\,\mathbb{E}\sup_{s\in S_m}\Big|\sum_i \varepsilon_i X_{i,s}\Big|$$

and

$$\mathbb{E}\sup_{s\in S_m}\sum_i X_{i,s} \le 2\,\mathbb{E}\sup_{s\in S_m}\sum_i \varepsilon_i X_{i,s}.$$

As $m \to \infty$, $\sup_{S_m} \uparrow \sup_S$ pointwise in each case. For the absolute-value suprema the terms are nonnegative, so monotone convergence gives convergence of the expectations to those over S, hence over T by the a.s. equalities above. For the signed supremum, $\sup_{S_m}\sum_i X_{i,s} \ge -\sum_i |X_{i,s_1}| \in L^1$, so monotone convergence applied to $\sup_{S_m}\sum_i X_{i,s} + \sum_i |X_{i,s_1}| \ge 0$ yields $\mathbb{E}\sup_{S_m}\sum_i X_{i,s} \to \mathbb{E}\sup_S \sum_i X_{i,s} = \mathbb{E}\sup_T \sum_i X_{i,s}$ (and similarly for the symmetrized version). Passing to the limit in the finite-index inequalities gives, for separable T,

$$\tfrac{1}{2}\,\mathbb{E}\sup_{s\in T}\Big|\sum_{i=1}^n \varepsilon_i X_{i,s}\Big| \le \mathbb{E}\sup_{s\in T}\Big|\sum_{i=1}^n X_{i,s}\Big| \le 2\,\mathbb{E}\sup_{s\in T}\Big|\sum_{i=1}^n \varepsilon_i X_{i,s}\Big|$$

and

$$\mathbb{E}\sup_{s\in T}\sum_{i=1}^{n} X_{i,s} \le 2\,\mathbb{E}\sup_{s\in T}\sum_{i=1}^{n}\varepsilon_i X_{i,s},$$

i.e. Lemma 11.4 holds for separable index sets. ■

Exercise 11.7

Central idea. Write $u=\Phi^{-1}(1-1/t)$, so that $\phi(u)=\gamma(1/t)$ where $\gamma(x)=\phi(\Phi^{-1}(x))$ is the Gaussian isoperimetric function (using $\Phi^{-1}(1/t)=-\Phi^{-1}(1-1/t)$ and evenness of ϕ). The two-sided bound of Lemma 10.3 on γ at $x=1/t$ translates directly into two-sided bounds on $u^2=2\log t\pm\log\log t\pm\log(\text{const})$. (The hint's "Lemma 10.1" refers to the isoperimetric-function material; Lemma 10.3 is its directly usable form.)

Solution. Let $u=\Phi^{-1}(1-1/t)$, so $\Phi(u)=1-1/t$. By symmetry $\Phi^{-1}(1/t)=-\Phi^{-1}(1-1/t)=-u$, and since ϕ is even,

$$\phi(u)=\phi(\Phi^{-1}(1/t))=\gamma(1/t),$$

where γ is the Gaussian isoperimetric function. For $t\ge 2$ we have $1/t\in(0,1/2]$, so Lemma 10.3 applies:

$$\frac{1}{t}\sqrt{\tfrac{1}{2}\log t}\le\gamma(1/t)=\frac{1}{\sqrt{2\pi}}e^{-u^2/2}\le\frac{1}{t}\sqrt{2\log t}.$$

Also $u=\Phi^{-1}(1-1/t)\ge\Phi^{-1}(1/2)=0$ for $t\ge 2$.

Lower bound. From the right-hand inequality, $e^{u^2/2}\ge t/\sqrt{4\pi\log t}$, i.e.

$$u^2\ge 2\log t-\log(4\pi)-\log\log t.$$

At $t=5$, $2\log 5-\log\log 5-\log(4\pi)\approx 0.21>0$, and the right-hand side is increasing for $t\ge 5$ (derivative $2/t-1/(t\log t)>0$), so it is positive for all $t\ge 5$. Since $u\ge 0$,

$$\Phi^{-1}(1-1/t)=u\ge\sqrt{2\log t-\log\log t-\log(4\pi)},\qquad t\ge 5.$$

Upper bound. From the left-hand inequality, $e^{u^2/2}\le t/\sqrt{\pi\log t}$, i.e.

$$u^2\le 2\log t-\log\pi-\log\log t.$$

The right-hand side is positive for $t\ge 2$ (at $t=2$ it equals $2\log 2-\log\log 2-\log\pi\approx 0.61>0$ and it is increasing), so with $u\ge 0$,

$$\Phi^{-1}(1-1/t)=u\le\sqrt{2\log t-\log\log t-\log\pi},\qquad t\ge 2.$$

■

Exercise 11.8

Central idea. Split $\mathbb{E}\|S_n\|_\infty^2=\mathrm{Var}(\|S_n\|_\infty)+(\mathbb{E}\|S_n\|_\infty)^2$ as in Theorem 11.2. The variance term needs no symmetry: Theorem 11.1 (centered case) gives $\mathrm{Var}(\|S_n\|_\infty)\le\Sigma^2+\sigma^2\le 2\Sigma^2$. Only the mean term needs symmetrization, via Lemma 11.4 ($\mathbb{E}\sup\le 2\,\mathbb{E}\sup$ of the Rademacher version), which contributes the factor 4 in front of $\log(2d)$ after the conditional sub-Gaussian maximal inequality (Theorem 2.5).

Solution. If $\Sigma^2=\mathbb{E}\max_{j\le d}\sum_{i=1}^n X_{i,j}^2=\infty$ there is nothing to prove; assume $\Sigma^2<\infty$. Write $\|\cdot\|$ for $\|\cdot\|_\infty$ and

$$\mathbb{E}\|S_n\|^2=\mathrm{Var}(\|S_n\|)+(\mathbb{E}\|S_n\|)^2.$$

Since $\|S_n\|=\max_{j\le d,\,b\in\{-1,1\}}\sum_{i=1}^n bX_{i,j}$, it is the supremum of the empirical process $(bX_{i,j})$ over the finite index set $\{(j,b)\}$. The variables $bX_{i,j}$ are centered (the X_i are centered) and no symmetry is assumed.

Variance term. By Theorem 11.1 (centered case), $\operatorname{Var}(\|S_n\|) \le \Sigma^2 + \sigma^2$ with $\sigma^2 = \max_{j,b} \sum_i \mathbb{E}[(bX_{i,j})^2] = \max_j \sum_i \mathbb{E}[X_{i,j}^2] \le \mathbb{E}\max_j \sum_i X_{i,j}^2 = \Sigma^2$. Hence

$$\operatorname{Var}(\|S_n\|) \le 2\Sigma^2.$$

Mean term. By Lemma 11.4 (symmetrization, finite index set),

$$\mathbb{E}\|S_n\| = \mathbb{E}\max_{j,b} \sum_i bX_{i,j} \le 2\,\mathbb{E}\max_{j,b} \sum_i \varepsilon_i bX_{i,j} = 2\,\mathbb{E}\max_{j\le d} \Big|\sum_i \varepsilon_i X_{i,j}\Big|,$$

with ε_i independent Rademacher variables independent of the X_i. Conditionally on the X_i, each $\sum_i \varepsilon_i X_{i,j}$ is a centered Rademacher sum, hence sub-Gaussian with variance factor $\sum_i X_{i,j}^2$. Theorem 2.5 (maximal inequality) gives, conditionally,

$$\mathbb{E}_\varepsilon \max_{j\le d} \Big|\sum_i \varepsilon_i X_{i,j}\Big| \le \sqrt{2\log(2d)\, \max_{j\le d} \sum_i X_{i,j}^2}.$$

Taking expectation over the X_i and using Jensen to bring $\mathbb{E}$ under the square root,

$$\mathbb{E}\max_{j\le d} \Big|\sum_i \varepsilon_i X_{i,j}\Big| \le \sqrt{2\log(2d)\,\Sigma^2}.$$

Therefore $\mathbb{E}\|S_n\| \le 2\sqrt{2\log(2d)}\,\Sigma$ and

$$(\mathbb{E}\|S_n\|)^2 \le 8\log(2d)\,\Sigma^2.$$

Combining,

$$\mathbb{E}\|S_n\|_\infty^2 \le 2\Sigma^2 + 8\log(2d)\,\Sigma^2 = 2(1 + 4\log(2d))\Sigma^2.$$

■

Exercise 11.9

Central idea. Take $\mathcal{T} = \{1, 2\}$ with deterministic $x_1 = x_2 = (1, -1)$, $\varphi_1(x) = x$, $\varphi_2(x) = -|x|$ (both 1-Lipschitz with $\varphi_i(0) = 0$), and Ψ the identity. Then $\sup_s |\varepsilon_1\varphi_1(x_{1,s}) + \varepsilon_2\varphi_2(x_{2,s})| = 2$ a.s. while $\sup_s |\varepsilon_1 x_{1,s} + \varepsilon_2 x_{2,s}| = |\varepsilon_1 + \varepsilon_2|$ has mean 1: both sides of the contraction inequality equal 1, so no constant larger than $1/2$ can work.

Solution. The second inequality of Theorem 11.6 states, for 1-Lipschitz φ_i with $\varphi_i(0) = 0$ and convex nondecreasing Ψ,

$$\mathbb{E}\,\Psi\left(\tfrac{1}{2}\sup_{s\in\mathcal{T}} \Big|\sum_{i=1}^n \varepsilon_i \varphi_i(x_{i,s})\Big|\right) \le \mathbb{E}\,\Psi\left(\sup_{s\in\mathcal{T}} \Big|\sum_{i=1}^n \varepsilon_i x_{i,s}\Big|\right).$$

Let $n = 2$, $\mathcal{T} = \{1, 2\}$, $x_1 = x_2 = (1, -1)$ (deterministic, in the hint's notation $X_{1,1} = X_{2,1} = 1$, $X_{1,2} = X_{2,2} = -1$), $\varphi_1(x) = x$, $\varphi_2(x) = -|x|$, and $\Psi(x) = x$. Both φ_i are 1-Lipschitz with $\varphi_i(0) = 0$.

Left-hand side. For $s = 1$: $\varepsilon_1\varphi_1(1) + \varepsilon_2\varphi_2(1) = \varepsilon_1 - \varepsilon_2$. For $s = 2$: $\varepsilon_1\varphi_1(-1) + \varepsilon_2\varphi_2(-1) = -\varepsilon_1 - \varepsilon_2$. Hence

$$\sup_{s\in\{1,2\}} \big|\varepsilon_1\varphi_1(x_{1,s}) + \varepsilon_2\varphi_2(x_{2,s})\big| = \max\left(|\varepsilon_1 - \varepsilon_2|,\, |\varepsilon_1 + \varepsilon_2|\right) = 2 \quad \text{a.s.},$$

since for $\varepsilon_i \in \{-1, 1\}$ one of $|\varepsilon_1 - \varepsilon_2|$, $|\varepsilon_1 + \varepsilon_2|$ equals 2 and the other 0. Thus the left-hand side equals $\frac{1}{2} \cdot 2 = 1$.

Right-hand side. For $s = 1$: $\varepsilon_1 x_{1,1} + \varepsilon_2 x_{2,1} = \varepsilon_1 + \varepsilon_2$. For $s = 2$: $\varepsilon_1 x_{1,2} + \varepsilon_2 x_{2,2} = -(\varepsilon_1 + \varepsilon_2)$. Hence

$$\sup_{s\in\{1,2\}} \big|\varepsilon_1 x_{1,s} + \varepsilon_2 x_{2,s}\big| = |\varepsilon_1 + \varepsilon_2| = \begin{cases} 2 & \text{with probability } 1/2, \\ 0 & \text{with probability } 1/2, \end{cases}$$

so the right-hand side equals 1.

Both sides equal 1: the inequality reads $1 \le 1$, i.e. it is tight. Replacing $1/2$ by any $c > 1/2$ would give $2c > 1$ on the left, violating the inequality. Hence the constant $1/2$ is optimal. ■

Exercise 11.10

Central idea. By homogeneity reduce to $\lambda_1 = 1$. Abel summation rewrites the weighted sum as a convex combination of the partial sums $S_j = \sum_{i\le j} \varepsilon_i v_i$ (with weights $b_j = \lambda_j - \lambda_{j+1} \ge 0$ summing to 1), so $\|\sum_i \lambda_i \varepsilon_i v_i\| \le \max_j \|S_j\|$. Lévy's maximal inequality — via first-passage sets and conditional symmetry of the increments — bounds $\mathbb{P}\{\max_j \|S_j\| > t\} \le 2\mathbb{P}\{\|S_n\| > t\}$.

Solution. If $\lambda_1 = 0$ both sides are 0. Otherwise divide by λ_1: with $\mu_i = \lambda_i/\lambda_1$ (still $1 = \mu_1 \ge \cdots \ge \mu_n \ge 0$),

$$\mathbb{P}\Big\{\Big\|\sum_i \lambda_i \varepsilon_i v_i\Big\|_{\mathbb{B}} > t\Big\} = \mathbb{P}\Big\{\Big\|\sum_i \mu_i \varepsilon_i v_i\Big\|_{\mathbb{B}} > t/\lambda_1\Big\},$$

so it suffices to prove the claim for $\lambda_1 = 1$ and then use $t/\lambda_1 \ge t$ (as $\lambda_1 \le 1$). Assume henceforth $\lambda_1 = 1$. Separability of $\mathbb{B}$ ensures all norms below are measurable.

Abel summation. Set $\lambda_{n+1} = 0$ and $b_j = \lambda_j - \lambda_{j+1} \ge 0$ for $j = 1, \dots, n$. With $S_j = \sum_{i=1}^{j} \varepsilon_i v_i$,

$$\sum_{j=1}^{n} b_j S_j = \sum_{j=1}^{n} (\lambda_j - \lambda_{j+1}) \sum_{i\le j} \varepsilon_i v_i = \sum_{i=1}^{n} \varepsilon_i v_i \sum_{j\ge i} (\lambda_j - \lambda_{j+1}) = \sum_{i=1}^{n} \lambda_i \varepsilon_i v_i.$$

Since $\sum_{j=1}^{n} b_j = \lambda_1 = 1$, convexity of the norm gives

$$\Big\|\sum_{i=1}^{n} \lambda_i \varepsilon_i v_i\Big\|_{\mathbb{B}} \le \sum_{j=1}^{n} b_j \|S_j\|_{\mathbb{B}} \le \max_{1\le j\le n} \|S_j\|_{\mathbb{B}},$$

hence $\{\|\sum_i \lambda_i \varepsilon_i v_i\|_{\mathbb{B}} > t\} \subset \{\max_{j\le n} \|S_j\|_{\mathbb{B}} > t\}$.

Lévy's maximal inequality. Let $\mathcal{F}_j = \sigma(\varepsilon_1, \dots, \varepsilon_j)$ and let $A_j = \{\|S_1\|_{\mathbb{B}} \le t, \dots, \|S_{j-1}\|_{\mathbb{B}} \le t, \|S_j\|_{\mathbb{B}} > t\}$ be the first-passage sets; they are disjoint, $\mathcal{F}_j$-measurable, and $\bigcup_j A_j = \{\max_{j\le n} \|S_j\|_{\mathbb{B}} > t\}$. Conditionally on $\mathcal{F}_j$, $Y := S_n - S_j$ is independent of $\mathcal{F}_j$ and symmetric. On A_j, $\|S_j\|_{\mathbb{B}} > t$, and since $\|S_j + Y\|_{\mathbb{B}} + \|S_j - Y\|_{\mathbb{B}} \ge 2\|S_j\|_{\mathbb{B}}$, at least one of $\|S_j + Y\|_{\mathbb{B}}$, $\|S_j - Y\|_{\mathbb{B}}$ exceeds t. By conditional symmetry these two have the same conditional law given $\mathcal{F}_j$, so on A_j,

$$\mathbb{P}\{\|S_n\|_{\mathbb{B}} > t \mid \mathcal{F}_j\} = \mathbb{P}\{\|S_j + Y\|_{\mathbb{B}} > t \mid \mathcal{F}_j\} \ge \tfrac{1}{2}.$$

Thus $\mathbb{P}(A_j \cap \{\|S_n\|_{\mathbb{B}} > t\}) = \mathbb{E}[1_{A_j} \mathbb{P}\{\|S_n\|_{\mathbb{B}} > t \mid \mathcal{F}_j\}] \ge \frac{1}{2}\mathbb{P}(A_j)$. Summing over j,

$$\mathbb{P}\Big\{\max_{1\le j\le n} \|S_j\|_{\mathbb{B}} > t\Big\} = \sum_{j=1}^{n} \mathbb{P}(A_j) \le 2 \sum_{j=1}^{n} \mathbb{P}(A_j \cap \{\|S_n\|_{\mathbb{B}} > t\}) \le 2\,\mathbb{P}\{\|S_n\|_{\mathbb{B}} > t\}.$$

Combining with the Abel step,

$$\mathbb{P}\Big\{\Big\|\sum_{i=1}^{n} \lambda_i \varepsilon_i v_i\Big\|_{\mathbb{B}} > t\Big\} \le \mathbb{P}\Big\{\max_{j\le n} \|S_j\|_{\mathbb{B}} > t\Big\} \le 2\,\mathbb{P}\Big\{\Big\|\sum_{i=1}^{n} \varepsilon_i v_i\Big\|_{\mathbb{B}} > t\Big\},$$

for all $t > 0$. ■

Exercise 11.11

Central idea. Condition on the X's to reduce to fixed vectors, then derive the Gaussian contraction from the Rademacher contraction: write $Y_i = \varepsilon_i |Y_i|$ with ε_i Rademacher independent of $|Y_i|$, and absorb the random amplitudes into new contractions $\psi_i(t) = |Y_i| \varphi_i(t/|Y_i|)$, so that $\sum_i Y_i \varphi_i(X_{i,s})$ becomes a Rademacher sum to which the contraction principle applies conditionally on $|Y|$.

Solution. *Convention.* "Lipschitz" means 1-Lipschitz here, by the book's own convention: p. 216 states "Recall that a function $f : \mathcal{X} \to \mathbb{R}$ is Lipschitz if for all $x, y \in \mathcal{X}$, $|f(x) - f(y)| \le d(x, y)$", and Theorem 11.6

(the contraction principle) uses the word in the same sense. The convention matters, since the dimension-free constants cannot hold for an arbitrary Lipschitz constant: with $n = 1$, $|\mathcal{T}| = 1$, $X_1 = \pm 1$ Rademacher, $\varphi_1(t) = 5t$ and $\Psi(x) = x$, the left side would be $\frac{5}{2}\,\mathbb{E}|Y_1| \approx 1.995$ and the right side $2\,\mathbb{E}|Y_1| \approx 1.596$. The Y_i are taken independent of the X_i, as is implicit in the statement. Condition on X; it suffices to prove the inequality for fixed vectors $x_i = (x_{i,s})_{s\in\mathcal{T}}$, then integrate over X.

Write $Y_i = \varepsilon_i A_i$ with $A_i = |Y_i| \geq 0$ and $\varepsilon_i = \operatorname{sign}(Y_i)$ Rademacher, ε_i independent of A_i, the pairs (ε_i, A_i) independent across i. For fixed $a_i \geq 0$ define $\psi_{i,a_i}(t) = a_i\varphi_i(t/a_i)$ (and $\psi_{i,0} \equiv 0$). Since φ_i is 1-Lipschitz, so is ψ_{i,a_i}: $|\psi_{i,a_i}(t) - \psi_{i,a_i}(s)| = a_i|\varphi_i(t/a_i) - \varphi_i(s/a_i)| \leq |t - s|$; moreover $\psi_{i,a_i}(0) = 0$ and $\psi_{i,A_i}(A_i x_{i,s}) = A_i\varphi_i(x_{i,s})$.

Hence, with $A = (A_i)$ fixed,

$$\sum_{i=1}^{n} Y_i\varphi_i(x_{i,s}) = \sum_{i=1}^{n} \varepsilon_i A_i\varphi_i(x_{i,s}) = \sum_{i=1}^{n} \varepsilon_i\, \psi_{i,A_i}(A_i x_{i,s}).$$

By the Rademacher contraction principle (Ledoux–Talagrand, Theorem 4.12) applied conditionally on A to the fixed vectors $(A_1 x_{1,s}, \dots, A_n x_{n,s})$ and the contractions ψ_{i,A_i} (with $\Psi - \Psi(0)$ in place of Ψ to ensure vanishing at 0, the constant cancelling),

$$\mathbb{E}_\varepsilon\Big[\Psi\Big(\sup_s\Big|\sum_i \varepsilon_i\psi_{i,A_i}(A_i x_{i,s})\Big|\Big)\Big] \leq \mathbb{E}_\varepsilon\Big[\Psi\Big(2\sup_s\Big|\sum_i \varepsilon_i A_i x_{i,s}\Big|\Big)\Big].$$

The left side equals $\mathbb{E}_\varepsilon[\Psi(\sup_s|\sum_i Y_i\varphi_i(x_{i,s})|)]$ given A, and the right side equals $\mathbb{E}_\varepsilon[\Psi(2\sup_s|\sum_i Y_i x_{i,s}|)]$ given A. Taking expectation over A,

$$\mathbb{E}_Y\Big[\Psi\Big(\sup_s\Big|\sum_i Y_i\varphi_i(x_{i,s})\Big|\Big)\Big] \leq \mathbb{E}_Y\Big[\Psi\Big(2\sup_s\Big|\sum_i Y_i x_{i,s}\Big|\Big)\Big].$$

Since Ψ is nondecreasing, $\Psi(u/2) \leq \Psi(u)$, so the left side dominates $\mathbb{E}_Y[\Psi(\frac{1}{2}\sup_s|\sum_i Y_i\varphi_i(x_{i,s})|)]$. Integrating over X gives the claim. ■

Exercise 11.12

Central idea. By homogeneity assume $|\lambda_i| \leq 1$. Symmetrize with independent Rademachers, then replace the coefficients λ_i by 1 one coordinate at a time: for fixed y, the map $a \mapsto \frac{1}{2}\|y + ax\|_{\mathbb{B}}^p + \frac{1}{2}\|y - ax\|_{\mathbb{B}}^p$ is convex and even in a, hence maximized on $[-1, 1]$ at $a = \pm 1$.

Solution. If $\|\lambda\|_\infty = 0$ there is nothing to prove; dividing by $\|\lambda\|_\infty$ we assume $|\lambda_i| \leq 1$ for all i. By symmetry, $(X_1, \dots, X_n) \overset{d}{=} (\varepsilon_1 X_1, \dots, \varepsilon_n X_n)$ for independent Rademachers ε_i independent of X, so

$$\mathbb{E}\Big\|\sum_{i=1}^{n}\lambda_i X_i\Big\|^p = \mathbb{E}_{X,\varepsilon}\Big\|\sum_{i=1}^{n}\varepsilon_i\lambda_i X_i\Big\|^p.$$

Fix deterministic $x_1, \dots, x_n \in \mathbb{B}$ and $|a_i| \leq 1$. We claim $\mathbb{E}_\varepsilon\|\sum_i \varepsilon_i a_i x_i\|^p \leq \mathbb{E}_\varepsilon\|\sum_i \varepsilon_i x_i\|^p$. Indeed, for fixed $y, x \in \mathbb{B}$ the function $f(a) = \frac{1}{2}\|y + ax\|^p + \frac{1}{2}\|y - ax\|^p$ is convex in a (as $\|\cdot\|^p \circ$ affine is convex for $p \geq 1$) and even, so $f(a) \leq f(1)$ for $|a| \leq 1$. Writing $S(a_1, \dots, a_n) = \mathbb{E}_\varepsilon\|\sum_i \varepsilon_i a_i x_i\|^p$ and conditioning on $\varepsilon_1, \dots, \varepsilon_{n-1}$,

$$S(a_1, \dots, a_n) = \mathbb{E}_{\varepsilon_{<n}}[f_n(a_n)] \leq \mathbb{E}_{\varepsilon_{<n}}[f_n(1)] = S(a_1, \dots, a_{n-1}, 1),$$

where $f_n(a) = \mathbb{E}_{\varepsilon_n}\|y + \varepsilon_n a x_n\|^p$ with $y = \sum_{i<n}\varepsilon_i a_i x_i$. Iterating over the coordinates gives $S(a_1, \dots, a_n) \leq S(1, \dots, 1)$.

Applying this with $a_i = \lambda_i$, $x_i = X_i$ conditionally on X,

$$\mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i\lambda_i X_i\Big\|^p \leq \mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i X_i\Big\|^p \quad \text{a.s.}$$

Taking $\mathbb{E}_X$ and using symmetry once more, $\mathbb{E}\|\sum_i \lambda_i X_i\|^p \leq \mathbb{E}\|\sum_i \varepsilon_i X_i\|^p = \mathbb{E}\|\sum_i X_i\|^p$. For general λ, apply to $\lambda_i/\|\lambda\|_\infty$ and multiply by $\|\lambda\|_\infty^p$. ■

Exercise 11.13

Central idea. Symmetrize: $X_i \overset{d}{=} \varepsilon_i|X_i|$, $Y_i \overset{d}{=} \varepsilon_i|Y_i|$ with ε_i Rademacher independent of the moduli. Condition on the moduli; with $0 \le |X_i| \le |Y_i|$ write $|X_i| = \lambda_i|Y_i|$ for $\lambda_i \in [0,1]$, and apply the deterministic contraction from Exercise 11-12 to the vectors $|Y_i|v_i$.

Solution. ("$|X_n| \le |Y_n|$ for all n" is read as $|X_i| \le |Y_i|$ for all i a.s.) By symmetry and independence, with $\varepsilon_1, \dots, \varepsilon_n$ iid Rademacher independent of all X_j, Y_j,

$$\sum_{i=1}^n X_i v_i \overset{d}{=} \sum_{i=1}^n \varepsilon_i |X_i| v_i, \qquad \sum_{i=1}^n Y_i v_i \overset{d}{=} \sum_{i=1}^n \varepsilon_i |Y_i| v_i,$$

so with $a_i = |X_i|$, $b_i = |Y_i|$ (hence $0 \le a_i \le b_i$ a.s.),

$$\mathbb{E}\Big\|\sum_i X_i v_i\Big\|^p = \mathbb{E}_{a,b}\,\mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i a_i v_i\Big\|^p,$$

and likewise for Y. Fix a_i, b_i with $0 \le a_i \le b_i$ and set $\lambda_i = a_i/b_i \in [0,1]$ ($\lambda_i = 0$ if $b_i = 0$). Then $\sum_i \varepsilon_i a_i v_i = \sum_i \varepsilon_i \lambda_i (b_i v_i)$, and by the deterministic contraction inequality proved in Exercise 11-12 applied to the fixed vectors $x_i = b_i v_i \in \mathbb{B}$,

$$\mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i \lambda_i (b_i v_i)\Big\|^p \le \mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i (b_i v_i)\Big\|^p = \mathbb{E}_\varepsilon\Big\|\sum_i \varepsilon_i b_i v_i\Big\|^p.$$

Integrating over $(a,b) = (|X|, |Y|)$ yields $\mathbb{E}\|\sum_i X_i v_i\|^p \le \mathbb{E}\|\sum_i Y_i v_i\|^p$. ■

Exercise 11.14

Central idea. Write $Y_i = |Y_i| X_i$ with $X_i = \operatorname{sign}(Y_i)$. Conditioning on the signs and applying Jensen's inequality to the convex map $z \mapsto \|z\|^p$ replaces each $|Y_i|$ by the deterministic $c_i = \mathbb{E}[|Y_i| \mid Y_i \ne 0] \ge \mathbb{E}|Y_i| \ge 1/C$ (the conditioning handles the possible atom at 0, where $|Y_i|$ and X_i are dependent); the contraction principle for the independent symmetric signs X_i then removes the multipliers c_i at the cost of C^p.

Solution. We read $X_i = \operatorname{sign}(Y_i)$ for every i (the 'X_n' is a typo; the Y_i are independent symmetric, hence so are the pairs $(|Y_i|, X_i)$). Then $Y_i = |Y_i| X_i$ a.s., each X_i is symmetric on $\{-1, 0, 1\}$, and $\mathbb{P}(Y_i \ne 0) > 0$ for all i (else $\mathbb{E}|Y_i| = 0 < 1/C$).

Put $c_i = \mathbb{E}[|Y_i| \mid Y_i \ne 0] = \mathbb{E}|Y_i|/\mathbb{P}(Y_i \ne 0) \ge \mathbb{E}|Y_i| \ge 1/C$. Since $|Y_i| \perp (X_j)_{j \ne i}$, $\mathbb{E}[|Y_i| \mid X] = \mathbb{E}[|Y_i| \mid X_i] = c_i \mathbf{1}_{\{X_i \ne 0\}}$, whence $\mathbb{E}[|Y_i| \mid X] X_i = c_i X_i$ a.s.

If $\mathbb{E}\|\sum_i Y_i v_i\|^p = \infty$ there is nothing to prove; assume it finite (then $\mathbb{E}\|\sum_i Y_i v_i\| < \infty$ as $p \ge 1$). By conditional Jensen,

$$\mathbb{E}\Big[\Big\|\sum_i Y_i v_i\Big\|^p\Big] = \mathbb{E}_X \mathbb{E}\Big[\Big\|\sum_i |Y_i| X_i v_i\Big\|^p \Big| X\Big] \ge \mathbb{E}\Big\|\sum_i c_i X_i v_i\Big\|^p.$$

It remains to show $\mathbb{E}\|\sum_i c_i X_i v_i\|^p \ge C^{-p}\mathbb{E}\|\sum_i X_i v_i\|^p$. Put $\alpha_i = 1/(C c_i) \in (0,1]$. The contraction principle for independent symmetric variables gives, with $w_i = c_i v_i$,

$$\mathbb{E}\Big\|\sum_i \alpha_i X_i w_i\Big\|^p \le \mathbb{E}\Big\|\sum_i X_i w_i\Big\|^p;$$

since $\alpha_i w_i = v_i/C$, this is $C^{-p}\mathbb{E}\|\sum_i X_i v_i\|^p \le \mathbb{E}\|\sum_i c_i X_i v_i\|^p$. For $X_i \in \{-1, 0, 1\}$ this follows from the one-step estimate: for $0 \le \alpha \le 1$,

$$\tfrac{1}{2}(\|v + \alpha x\|^p + \|v - \alpha x\|^p) \le (1-\alpha)\|v\|^p + \tfrac{\alpha}{2}(\|v + x\|^p + \|v - x\|^p) \le \tfrac{1}{2}(\|v + x\|^p + \|v - x\|^p),$$

using $\|v \pm \alpha x\| \le (1-\alpha)\|v\| + \alpha\|v \pm x\|$, convexity of $t \mapsto t^p$, and $\|v\|^p \le \frac{1}{2}(\|v + x\|^p + \|v - x\|^p)$; telescoping over i (the atom $\mathbb{P}(X_i = 0)$ contributes $\|v\|^p$ to both sides) yields the claim. ■

Exercise 11.15

Central idea. Write $Y_i = |Y_i|\varepsilon_i$ with $\varepsilon_i = \operatorname{sign}(Y_i)$ Rademacher independent of $|Y_i|$ (the Gaussian law is continuous symmetric). Conditioning on $|Y|$ and applying the contraction principle with $M = \max_i |Y_i|$ gives $\mathbb{E}[\|\sum_i Y_i v_i\| \mid |Y|] \le M\,\mathbb{E}\|\sum_i \varepsilon_i v_i\|$, so everything reduces to bounding $a_n := \mathbb{E}\max_{i\le n}|Y_i|$. The maximal inequality over the $2n$ variables $\pm Y_i$ gives $a_n \le \sqrt{2\log(2n)}$ at once; the printed constant $\sqrt{2\log n}$ needs the sharper fact $a_n < \sqrt{2\log n}$ for $n \ge 2$, proved analytically for large n (a tail split at $b = \sqrt{2\log n}$) and by a computer-assisted check for moderate n.

Erratum. As printed (statement, not hint), the inequality is false for $n = 1$: with $v_1 \ne 0$, the left side is $\mathbb{E}|Y_1|\,\|v_1\| = \sqrt{2/\pi}\,\|v_1\| \approx 0.798\,\|v_1\| > 0$, while $\sqrt{2\log 1} = 0$ makes the right side 0. This is the only failure: the printed bound holds for every $n \ge 2$, which is what we prove below. (For all $n \ge 1$ one may instead use the constant $\sqrt{2\log(2n)}$; see the Remark.)

Solution. *Step 1: reduction to a maximum of Gaussians.* The Y_i are continuous symmetric, so $\varepsilon_i := \operatorname{sign}(Y_i)$ are iid Rademacher, independent of the $|Y_i|$'s, and $Y_i = |Y_i|\varepsilon_i$ a.s.

Condition on $|Y| = (|Y_1|, \dots, |Y_n|)$ and put $M = \max_i |Y_i| \ge 0$, $\alpha_i = |Y_i|/M \in [0,1]$ (if $M = 0$ both sides below are 0). The ε_i remain iid Rademacher given $|Y|$, so by the contraction principle,

$$\mathbb{E}\Big[\Big\|\sum_i Y_i v_i\Big\| \Big| |Y|\Big] = \mathbb{E}\Big[\Big\|\sum_i \alpha_i\varepsilon_i (M v_i)\Big\| \Big| |Y|\Big] \le M\,\mathbb{E}\Big\|\sum_i \varepsilon_i v_i\Big\|,$$

the last expectation being deterministic since $\varepsilon \perp |Y|$. Taking expectations and using $\varepsilon_i \overset{d}{=} X_i$,

$$\mathbb{E}\Big\|\sum_i Y_i v_i\Big\| \le a_n\,\mathbb{E}\Big\|\sum_i X_i v_i\Big\|, \qquad a_n := \mathbb{E}\max_{i\le n}|Y_i|.$$

Step 2: $a_n < \sqrt{2\log n}$ for $n \ge 2$. Let Φ, φ be the standard normal cdf and density, $\bar\Phi = 1 - \Phi$, and $b = \sqrt{2\log n}$, so that $e^{-b^2/2} = 1/n$ and $2n\varphi(b) = \sqrt{2/\pi}$. Then

$$a_n = \int_0^\infty \mathbb{P}\{M > t\}\,dt, \qquad \mathbb{P}\{M > t\} = 1 - (2\Phi(t) - 1)^n \le \min\big(1,\, 2n\bar\Phi(t)\big),$$

and a_n is nondecreasing in n (the integrand is).

Large n: $b \ge 5$, i.e. $n \ge e^{12.5} \approx 2.68 \cdot 10^5$. Write $a_n = b - \int_0^b \mathbb{P}\{M \le t\}\,dt + \int_b^\infty \mathbb{P}\{M > t\}\,dt$.

Upper tail: using $\int_b^\infty \bar\Phi(t)\,dt = \varphi(b) - b\bar\Phi(b)$ and Gordon's lower bound $\bar\Phi(b) \ge \frac{b}{1+b^2}\varphi(b)$,

$$\int_b^\infty \mathbb{P}\{M > t\}\,dt \le 2n(\varphi(b) - b\bar\Phi(b)) \le \frac{2n\varphi(b)}{1+b^2} = \frac{\sqrt{2/\pi}}{1+b^2} < \frac{0.8}{b^2}.$$

Lower part: $t \mapsto \mathbb{P}\{M \le t\}$ is nondecreasing, so $\int_0^b \mathbb{P}\{M \le t\}\,dt \ge \frac{1}{b}\,\mathbb{P}\{M \le b - 1/b\} \ge \frac{1}{b}(1 - 2n\bar\Phi(b - 1/b))$. By the Mills bound $\bar\Phi(s) \le \varphi(s)/s$ and $\varphi(b - 1/b) = \varphi(b)\,e^{1-1/(2b^2)} \le e\,\varphi(b) = \frac{e}{n\sqrt{2\pi}}$,

$$2n\bar\Phi(b - 1/b) \le \frac{2e}{\sqrt{2\pi}\,(b - 1/b)} \le 0.452 \qquad (b \ge 5).$$

Hence

$$a_n \le b - \frac{0.548}{b} + \frac{0.8}{b^2} < b,$$

since $0.8/b \le 0.16 < 0.548$.

Moderate n: $2 \le n \le 323{,}714$ (computer-assisted). The integrand $t \mapsto 1 - (2\Phi(t) - 1)^n$ is decreasing, so the left-endpoint Riemann sum with step 10^{-4} on $[0, 12]$ is an upper bound for the integral over $[0, 12]$, and the rest is at most $\int_{12}^\infty 2n\bar\Phi = 2n(\varphi(12) - 12\bar\Phi(12))$, which is negligible. Call the resulting upper bound $\bar a_n \ge a_n$. We evaluated it in double precision (with `log_ndtr`, `log1p` and `expm1` for $(1 - 2\bar\Phi)^n$); this is floating point, not interval arithmetic, but every margin below exceeds rounding error by many orders of magnitude.

- For each $n = 2, \dots, 40$: $\sqrt{2\log n} - \bar{a}_n \geq 0.049$, the minimum being at $n = 2$ ($a_2 \approx 1.128 < \sqrt{2\log 2} \approx 1.177$).
- For $n \geq 40$: on the grid $N_0 = 40$, $N_{k+1} = \lfloor 1.15\, N_k \rfloor$ up to $N_{65} = 323{,}714$, we checked $\bar{a}_{N_{k+1}} < \sqrt{2\log N_k}$ for every k, with minimum margin 0.234. For $N_k \leq n \leq N_{k+1}$, monotonicity gives $a_n \leq a_{N_{k+1}} \leq \bar{a}_{N_{k+1}} < \sqrt{2\log N_k} \leq \sqrt{2\log n}$.

Sample values: $a_{10} \approx 1.881 < 2.146$, $a_{10^5} \approx 4.533 < 4.799$. The two ranges overlap ($323{,}714 > 2.68 \cdot 10^5$), so $a_n < \sqrt{2\log n}$ for all $n \geq 2$.

Conclusion. For $n \geq 2$,

$$\mathbb{E}\Big\|\sum_{i=1}^n Y_i v_i\Big\| \leq \sqrt{2\log n}\, \mathbb{E}\Big\|\sum_{i=1}^n X_i v_i\Big\|.$$

The $\sqrt{\log n}$ order is necessary: in ℓ_∞^n with $v_i = e_i$ the left side is $a_n \sim \sqrt{2\log n}$ while $\mathbb{E}\|\sum_i X_i e_i\|_\infty = 1$.

Remark (simple constant valid for all $n \geq 1$). $M = \max_{j \leq 2n} Z_j$ with Z_j the $2n$ standard Gaussians $\pm Y_i$. By the maximal inequality (Theorem 2.5) with $\psi(\lambda) = \lambda^2/2$,

$$a_n \leq \inf_{\lambda > 0} \frac{\log(2n) + \lambda^2/2}{\lambda} = \sqrt{2\log(2n)},$$

so $\mathbb{E}\|\sum_i Y_i v_i\| \leq \sqrt{2\log(2n)}\, \mathbb{E}\|\sum_i X_i v_i\|$ for every $n \geq 1$, including $n = 1$. ■

Exercise 11.16

Central idea. Apply the Efron–Stein inequality (Theorem 3.1) with the leave-one-out supremum $Z_i = \sup_s \sum_{j \neq i} \alpha_{j,s} Y_j$. From $\sup A - \sup B \leq \sup(A - B)$ we get $|Z - Z_i| \leq \sup_s |\alpha_{i,s} Y_i| \leq |Y_i|$ pointwise (no maximizers needed); since $|\alpha| \leq 1$, $(Z - Z_i)^2 \leq Y_i^2$, whence $\mathrm{Var}(Z) \leq n\sigma^2$. As $\mathbb{E}Z \geq 0$, the stated $n\sigma^2 + 2\mathbb{E}Z$ follows. This is the variance estimate underlying Bousquet's inequality, where $n\sigma^2 + 2\mathbb{E}Z$ is the natural variance proxy.

Solution. Assume $\mathcal{T}$ is countable (so suprema are measurable; the general separable case follows by approximation) and nonempty. Write $S_s = \sum_{i=1}^n \alpha_{i,s} Y_i$, so $Z = \sup_{s \in \mathcal{T}} S_s$. The Bernstein moment condition gives $\mathbb{E}Y_i^2 \leq \sigma^2$ (case $q = 2$) and exponential tails, so Z is integrable with $\mathbb{E}Z^2 \leq n \sum_i \mathbb{E}Y_i^2 < \infty$; moreover $\mathbb{E}Z \geq \sup_s \mathbb{E}S_s = 0$ since each Y_i is centered.

For $i = 1, \dots, n$ let $Z_i = \sup_{s \in \mathcal{T}} \sum_{j \neq i} \alpha_{j,s} Y_j$, which is $\sigma(Y^{(i)})$-measurable for $Y^{(i)} = (Y_j)_{j \neq i}$. By the Efron–Stein inequality (Theorem 3.1), in the form $\mathrm{Var}(Z) \leq \inf_{Z_i} \sum_i \mathbb{E}(Z - Z_i)^2$,

$$\mathrm{Var}(Z) \leq \sum_{i=1}^n \mathbb{E}\big[(Z - Z_i)^2\big].$$

For each s, $Z_i \geq \sum_{j \neq i} \alpha_{j,s} Y_j$, so

$$Z - Z_i = \sup_s \sum_j \alpha_{j,s} Y_j - Z_i \leq \sup_s \Big(\sum_j \alpha_{j,s} Y_j - \sum_{j \neq i} \alpha_{j,s} Y_j \Big) = \sup_s \alpha_{i,s} Y_i \leq |Y_i|.$$

Applying the same with Z and Z_i exchanged, $Z_i - Z \leq \sup_s(-\alpha_{i,s} Y_i) \leq |Y_i|$. Hence $|Z - Z_i| \leq |Y_i|$ pointwise (no maximizers needed), and

$$\mathbb{E}\big[(Z - Z_i)^2\big] \leq \mathbb{E}[Y_i^2] \leq \sigma^2.$$

Summing over i gives $\mathrm{Var}(Z) \leq n\sigma^2$. Since $\mathbb{E}Z \geq 0$, we obtain $\mathrm{Var}(Z) \leq n\sigma^2 + 2\mathbb{E}Z$. ■

Exercise 11.17

Central idea. Follow the Efron–Stein + self-bounding pattern of Theorem 11.10 (via Lemma 11.11), but center the leave-one-out approximators: use $\tilde{Z} = Z - nE$ and $\tilde{Z}_i = Z_i - (n-1)E$ with $E = \mathbb{E}Z/n$, so that $\tilde{Z} - \tilde{Z}_i = (Z - Z_i) - E$. The ϕ-argument of Lemma 11.11 then runs with linear coefficient $2(1-E)$ instead of 2, and the E^2 terms cancel, saving $(\mathbb{E}Z)^2/n$.

Solution. Use the notation of Theorem 11.10: $Z = \sup_{s\in T}\sum_{i=1}^n X_{i,s}$ with X_i iid, $|X_{i,s}| \le 1$, $\mathbb{E}X_{i,s} = 0$; $Z_i = \sup_{s\in T}\sum_{j\neq i} X_{j,s}$; $W_i = Z - Z_i$. Take T countable (the separable case follows by approximation). Fix $\varepsilon > 0$ and, enumerating $T = \{s_1, s_2, \dots\}$, choose $\hat{s}_i \in T$ $X^{(i)}$-measurably with $\sum_{j\neq i} X_{j,\hat{s}_i} \ge Z_i - \varepsilon$ (least such index); put $Y_i = X_{i,\hat{s}_i}$. Then $W_i \le 1$, $Y_i \le 1$, $\mathbb{E}^{(i)}Y_i = 0$ (with $\mathbb{E}^{(i)}$ conditional on $X^{(i)}$, as $X_i \perp X^{(i)}$), $Y_i \le W_i + \varepsilon$ (since $W_i = Z - Z_i \ge \sum_j X_{j,\hat{s}_i} - Z_i \ge Y_i - \varepsilon$), and $\sum_{i=1}^n W_i \le Z$. Let $\sigma^2 = \sup_{s\in T}\mathbb{E}X_{1,s}^2$ and $E = \mathbb{E}Z/n \ge 0$.

Put $\tilde{Z} = Z - nE$ and $\tilde{Z}_i = Z_i - (n-1)E$. Then $\mathrm{Var}(Z) = \mathrm{Var}(\tilde{Z})$, $\tilde{Z}_i$ is $X^{(i)}$-measurable, and

$$\tilde{Z} - \tilde{Z}_i = (Z - Z_i) - E = W_i - E.$$

By the Efron–Stein inequality (Theorem 3.1) and the L^2-projection property,

$$\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}\big[(\tilde{Z} - \mathbb{E}^{(i)}\tilde{Z})^2\big] \le \sum_{i=1}^n \mathbb{E}^{(i)}\big[(\tilde{Z} - \tilde{Z}_i)^2\big] = \sum_{i=1}^n \mathbb{E}\big[(W_i - E)^2\big].$$

Write $\tilde{W}_i = W_i - E$ and $\tilde{Y}_i = Y_i - E - \varepsilon$, so $\tilde{Y}_i \le \tilde{W}_i$. With $\phi(x) = x^2 - 2(1-E)x$,

$$\phi(\tilde{W}_i) - \phi(\tilde{Y}_i) = (\tilde{W}_i - \tilde{Y}_i)(\tilde{W}_i + \tilde{Y}_i - 2(1-E)) \le 0,$$

since $\tilde{W}_i - \tilde{Y}_i = W_i - Y_i + \varepsilon \ge 0$ and $\tilde{W}_i + \tilde{Y}_i - 2(1-E) = W_i + Y_i - 2 - \varepsilon \le 0$ (as $W_i \le 1$, $Y_i \le 1$). Hence, taking $\mathbb{E}^{(i)}$,

$$\mathbb{E}^{(i)}[\tilde{W}_i^2] \le 2(1-E)\,\mathbb{E}^{(i)}[\tilde{W}_i] + \mathbb{E}^{(i)}[\tilde{Y}_i^2] - 2(1-E)\,\mathbb{E}^{(i)}[\tilde{Y}_i].$$

Now $\mathbb{E}^{(i)}[\tilde{W}_i] = \mathbb{E}^{(i)}[W_i] - E$, $\mathbb{E}^{(i)}[\tilde{Y}_i] = -E - \varepsilon$ (since $\mathbb{E}^{(i)}Y_i = 0$), and $\mathbb{E}^{(i)}[\tilde{Y}_i^2] = \mathbb{E}^{(i)}[Y_i^2] + (E+\varepsilon)^2$. Thus

$$\begin{aligned}\mathbb{E}^{(i)}[(W_i - E)^2] &\le 2(1-E)(\mathbb{E}^{(i)}[W_i] - E) + \mathbb{E}^{(i)}[Y_i^2] + (E+\varepsilon)^2 + 2(1-E)(E+\varepsilon)\\ &= 2(1-E)\,\mathbb{E}^{(i)}[W_i] + \mathbb{E}^{(i)}[Y_i^2] + (E+\varepsilon)^2 + 2(1-E)\varepsilon,\end{aligned}$$

the $-2(1-E)E$ and $+2(1-E)E$ cancelling. Summing over i and taking expectations, with $\sum_i W_i \le Z$ and $\mathbb{E}[Y_i^2] = \mathbb{E}\big[\mathbb{E}[X_{i,s}^2]_{s=\hat{s}_i}\big] \le \sigma^2$,

$$\mathrm{Var}(Z) \le 2(1-E)\,\mathbb{E}Z + n\sigma^2 + n(E+\varepsilon)^2 + 2n(1-E)\varepsilon.$$

Letting $\varepsilon \to 0$ gives $\mathrm{Var}(Z) \le 2(1-E)\mathbb{E}Z + n\sigma^2 + nE^2$. Since $nE^2 = (\mathbb{E}Z)^2/n = E\,\mathbb{E}Z$, this is $n\sigma^2 + (2-E)\mathbb{E}Z$. ∎

Exercise 11.18

Central idea. Follow the pattern of Corollary 11.15: integrate the refined Hoffmann–Jørgensen tail bound against $px^{p-1}dx$, split the quadratic term $\mathbb{P}\{Z_n > \beta x\}\mathbb{P}\{Z_n > \gamma x\}$ at a quantile t (so the tail piece is absorbed), and bound t by Markov. The $O(p)$ constant comes from the parameter choice $\alpha = \beta \sim 1/p$, $\gamma \approx 1$: then $\gamma^p \to 1/e$ stays bounded away from zero, while $\alpha^{-1}, \beta^{-1} = O(p)$.

Solution. Let $Z_k = \sup_{s\in\mathcal{T}} |\sum_{i=1}^k X_{i,s}|$, $Z_n^* = \max_{k\le n} Z_k$, and $M = \max_{i\le n, s\in\mathcal{T}} |X_{i,s}|$. Then $Z = Z_n \le Z_n^*$, so it suffices to bound $\|Z_n^*\|_p$.

By the refined Hoffmann–Jørgensen inequality used in the proof of Corollary 11.15 (which follows from Lemma 11.13 and the symmetry of the $X_{i,s}$), for $\alpha, \beta, \gamma > 0$ with $\alpha + \beta + \gamma = 1$ and all $x > 0$,

$$\mathbb{P}\{Z_n^* > x\} \le \mathbb{P}\{M > \alpha x\} + 4\mathbb{P}\{Z_n > \beta x\}\mathbb{P}\{Z_n > \gamma x\}.$$

To justify the absorption below, truncate: $Z^*_{n,m} = \min(Z^*_n, m)$, $U_m = \mathbb{E}[(Z^*_{n,m})^p] < \infty$. Since $\mathbb{P}\{Z^*_{n,m} > x\} = \mathbb{P}\{Z^*_n > x\}$ for $x < m$,

$$\begin{aligned} U_m &= p\int_0^m x^{p-1}\mathbb{P}\{Z^*_n > x\}\,dx \\ &\le p\int_0^m x^{p-1}\mathbb{P}\{M > \alpha x\}\,dx + 4p\int_0^m x^{p-1}\mathbb{P}\{Z_n > \beta x\}\mathbb{P}\{Z_n > \gamma x\}\,dx \\ &\le \frac{\mathbb{E}[M^p]}{\alpha^p} + \frac{4p}{\beta^p}\int_0^{m\beta} y^{p-1}\mathbb{P}\{Z_n > y\}\mathbb{P}\{Z_n > \tfrac{\gamma}{\beta}y\}\,dy, \end{aligned}$$

via $y = \beta x$. Split the last integral at $0 < t < m\beta$ (if $t \ge m\beta$ the tail piece below is empty). On $[0, t]$, $\mathbb{P}\{\cdot\} \le 1$ gives $\frac{4p}{\beta^p} \cdot \frac{t^p}{p} = \frac{4t^p}{\beta^p}$. On $[t, m\beta]$,

$$\frac{4p}{\beta^p}\mathbb{P}\{Z_n > t\}\int_0^{m\beta} y^{p-1}\mathbb{P}\{Z_n > \tfrac{\gamma}{\beta}y\}\,dy = \frac{4\mathbb{P}\{Z_n > t\}}{\gamma^p}\,\mathbb{E}[\min(Z_n, m\gamma)^p] \le \frac{4\mathbb{P}\{Z_n > t\}}{\gamma^p}U_m,$$

using $z = \frac{\gamma}{\beta}y$ and $\min(Z_n, m\gamma) \le \min(Z^*_n, m)$ (since $Z_n \le Z^*_n$ and $\gamma \le 1$). Hence, with $V = \mathbb{E}[M^p]$,

$$U_m \le \frac{V}{\alpha^p} + \frac{4t^p}{\beta^p} + \frac{4\mathbb{P}\{Z_n > t\}}{\gamma^p}U_m.$$

Choose $t \ge 0$ with $\mathbb{P}\{Z_n > t\} \le \gamma^p/8$ (possible since $\mathbb{P}\{Z_n > t\} \to 0$); then the last coefficient is $\le \frac{1}{2}$ and

$$U_m \le \frac{2V}{\alpha^p} + \frac{8t^p}{\beta^p}.$$

By Markov, the quantile t satisfies $t \le \frac{8\mathbb{E}Z_n}{\gamma^p}$. Thus

$$U_m \le \frac{2V}{\alpha^p} + \frac{8^{p+1}(\mathbb{E}Z_n)^p}{\beta^p\gamma^{p^2}}, \qquad U_m^{1/p} \le \frac{2^{1/p}}{\alpha}V^{1/p} + \frac{8^{(p+1)/p}}{\beta\gamma^p}\mathbb{E}Z_n.$$

Take $\alpha = \beta = \frac{1}{2(p+1)}$, $\gamma = \frac{p}{p+1}$ (so $\alpha + \beta + \gamma = 1$). Then $\frac{2^{1/p}}{\alpha} \le 4(p+1)$, and since $(\frac{p}{p+1})^p = \frac{1}{(1+1/p)^p} > \frac{1}{e}$,

$$\frac{8^{(p+1)/p}}{\beta\gamma^p} \le \frac{24}{\frac{1}{2(p+1)} \cdot \frac{1}{e}} = 48e(p+1),$$

using $8^{(p+1)/p} = 8 \cdot 8^{1/p} \le 24$ for $p \ge 2$. Hence

$$\|Z^*_{n,m}\|_p \le 48e(p+1)(\|M\|_p + \mathbb{E}Z_n).$$

Letting $m \to \infty$, monotone convergence gives $\|Z^*_n\|_p \le \kappa_p(\|M\|_p + \mathbb{E}Z_n)$ with $\kappa_p = 48e(p+1)$. Since $Z = Z_n \le Z^*_n$,

$$\mathbb{E}[Z^p]^{1/p} \le \kappa_p(\mathbb{E}Z + \mathbb{E}[M^p]^{1/p}),$$

and $\kappa_p/p = 48e(1 + 1/p) \le 96e$ remains bounded as $p \to \infty$. ■

Exercise 11.19

Central idea. Apply Theorem 11.1 (second part) to the empirical process $X_{i,s} = \alpha_{i,s}X_i$: since the X_i are centered, $\mathrm{Var}(Z) \le \Sigma^2 + \sigma^2$ where the weak variance is exactly $\mathbb{E}\sup_s \sum_i \alpha^2_{i,s}X_i^2$ and the wimpy variance is at most $\sigma^2 \sup_s \sum_i \alpha^2_{i,s}/n$ (using $\mathbb{E}X_i^2 \le \sigma^2/n$ from the moment condition at $q = 2$). The crude bound $\mathrm{Var}(Z) \le 2\sigma^2$ follows by dropping the $\alpha^2_{i,s} \le 1$ weights. For the third inequality, symmetrize the weak-variance term (Lemma 11.9, the opening step of Theorem 11.17's proof): $\mathbb{E}\sup_s \sum_i \alpha^2_{i,s}X_i^2 \le \sigma^2 \sup_s \sum_i \alpha^2_{i,s}/n + 2\mathbb{E}\sup_s \sum_i \varepsilon_i\alpha^2_{i,s}X_i^2$.

Solution. Put $X_{i,s} = \alpha_{i,s} X_i$ for $i = 1, \dots, n$, $s \in \mathcal{T}$, so $Z = \sup_{s\in\mathcal{T}} \sum_{i=1}^n X_{i,s}$. Since the X_i are independent and centered, $\mathbb{E}X_{i,s} = \alpha_{i,s}\mathbb{E}X_i = 0$ for all i, s. (As in the book, we take $\mathcal{T}$ countable, or the process separable, so all suprema are measurable.) The moment hypothesis at $q = 2$ gives

$$\mathbb{E}X_i^2 = \mathrm{Var}(X_i) \le 2! \, \frac{c^0 \sigma^2/n}{2} = \frac{\sigma^2}{n}, \qquad i = 1, \dots, n.$$

By the second part of Theorem 11.1,

$$\mathrm{Var}(Z) \le \Sigma^2 + \sigma^2,$$

with weak variance $\Sigma^2 = \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n X_{i,s}^2$ and wimpy variance $\sigma^2 = \sup_{s\in\mathcal{T}} \sum_{i=1}^n \mathbb{E}X_{i,s}^2$. Here

$$\Sigma^2 = \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 X_i^2, \qquad \sigma^2 = \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 \, \mathbb{E}X_i^2 \le \sigma^2 \sup_{s\in\mathcal{T}} \sum_{i=1}^n \frac{\alpha_{i,s}^2}{n}.$$

This is the second claimed inequality:

$$\mathrm{Var}(Z) \le \sigma^2 \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2/n + \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 X_i^2.$$

For the first inequality, use $|\alpha_{i,s}| \le 1$: $\sup_s \sum_i \alpha_{i,s}^2/n \le 1$, and

$$\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 X_i^2 \le \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n X_i^2 = \mathbb{E}\sum_{i=1}^n X_i^2 = \sum_{i=1}^n \mathbb{E}X_i^2 \le \sigma^2,$$

since $\sum_i X_i^2$ does not depend on s. Hence $\mathrm{Var}(Z) \le \sigma^2 + \sigma^2 = 2\sigma^2$.

For the third inequality, bound the weak-variance term by symmetrization. Write $A = \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2/n$. Then

$$\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 X_i^2 \le \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 (X_i^2 - \mathbb{E}X_i^2) + \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 \, \mathbb{E}X_i^2 \le \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 (X_i^2 - \mathbb{E}X_i^2) + \sigma^2 A.$$

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables, independent of the X_i. By the symmetrization inequality (Lemma 11.4), applied to the independent summands $Y_{i,s} = \alpha_{i,s}^2 X_i^2$,

$$\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 (X_i^2 - \mathbb{E}X_i^2) \le 2\,\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \varepsilon_i \, \alpha_{i,s}^2 X_i^2.$$

(This is Lemma 11.9, the symmetrization step with which the proof of Theorem 11.17 opens; the full strength of Theorem 11.17, which further bounds the Rademacher term via contraction under a symmetry hypothesis, is not needed here.) Thus

$$\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2 X_i^2 \le \sigma^2 A + 2\,\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \varepsilon_i \, \alpha_{i,s}^2 X_i^2.$$

Substituting into the second inequality gives

$$\mathrm{Var}(Z) \le \sigma^2 A + \sigma^2 A + 2\,\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \varepsilon_i \, \alpha_{i,s}^2 X_i^2 = 2\sigma^2 \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha_{i,s}^2/n + 2\,\mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n \varepsilon_i \, \alpha_{i,s}^2 X_i^2,$$

as claimed. ■

Exercise 11.20

Central idea. Write $Z = \|X\|_2 = \sup_{\theta\in S^{n-1}} \sum_{i=1}^n \theta_i X_i$ as the supremum of a separable empirical process with centered summands $X_{i,\theta} = \theta_i X_i$. Jensen gives $\mathbb{E}Z \le \sqrt{n/3}$; the lower bound follows from $\mathrm{Var}(Z) = n/3 - (\mathbb{E}Z)^2$ together with Theorem 11.1, which yields $\mathrm{Var}(Z) \le \Sigma^2 + \sigma^2 \le 1 + 1/3 = 4/3$ (weak variance ≤ 1 since $X_i^2 \le 1$; wimpy variance $= 1/3$ since $\mathbb{E}[X_i^2] = 1/3$). The tail bound is Theorem 6.10 applied to $Z/2$, viewed as a separately convex 1-Lipschitz function of $Y_i = (X_i + 1)/2 \in [0, 1]$.

Solution. Since $\|X\|_2 = \sup_{\theta\in S^{n-1}} \langle\theta, X\rangle$ (attained at $\theta = X/\|X\|$ a.s.), and $\theta \mapsto \langle\theta, X\rangle$ is continuous, with $T \subset S^{n-1}$ countable dense,

$$Z = \sup_{\theta\in T} \sum_{i=1}^n \theta_i X_i \qquad \text{a.s.}$$

Put $X_{i,\theta} = \theta_i X_i$. For fixed θ the $X_{i,\theta}$ are independent and centered ($\mathbb{E}X_i = 0$), so Z is the supremum of a separable centered empirical process, and the book's results for countable T apply.

Since $\mathbb{E}[X_i^2] = \int_{-1}^1 \frac{x^2}{2}\,dx = 1/3$, Jensen's inequality gives

$$\mathbb{E}Z = \mathbb{E}\sqrt{\sum_{i=1}^n X_i^2} \le \sqrt{\sum_{i=1}^n \mathbb{E}[X_i^2]} = \sqrt{n/3}.$$

The wimpy variance is $\sigma^2 = \sup_{\theta\in S^{n-1}} \sum_{i=1}^n \mathbb{E}[(\theta_i X_i)^2] = \frac{1}{3}\sup_\theta \|\theta\|_2^2 = 1/3$, and the weak variance is $\Sigma^2 = \mathbb{E}[\sup_\theta \sum_i \theta_i^2 X_i^2] \le 1$, since $X_i^2 \le 1$ a.s. and $\sum_i \theta_i^2 = 1$. Theorem 11.1 (centered case) gives

$$\mathrm{Var}(Z) \le \Sigma^2 + \sigma^2 \le 1 + \frac{1}{3} = \frac{4}{3}.$$

As $\mathbb{E}[Z^2] = \sum_i \mathbb{E}[X_i^2] = n/3$,

$$(\mathbb{E}Z)^2 = \mathbb{E}[Z^2] - \mathrm{Var}(Z) \ge \frac{n}{3} - \frac{4}{3} = \frac{n-4}{3}.$$

For $n \ge 5$, $\sqrt{(n-4)/3} \ge \sqrt{n/3} - 1$: both sides are nonnegative and squaring gives the equivalent $2\sqrt{n/3} \ge 7/3$, i.e. $n \ge 49/12$. Hence $\mathbb{E}Z \ge \sqrt{n/3} - 1$. For $n \le 4$, $Z \ge |X_1|$ gives $\mathbb{E}Z \ge \mathbb{E}|X_1| = 1/2 > \sqrt{4/3} - 1 \ge \sqrt{n/3} - 1$.

For the tail bound, let $Y_i = (X_i + 1)/2$, iid uniform on $[0, 1]$, and $g(y) = \|2y - \mathbf{1}\|_2$ on $[0, 1]^n$, so $Z = g(Y)$. For fixed y_{-i}, $y_i \mapsto g(y) = \sqrt{(2y_i - 1)^2 + C}$ with $C = \sum_{j\ne i}(2y_j - 1)^2 \ge 0$ has second derivative $4C/((2y_i - 1)^2 + C)^{3/2} \ge 0$, so g is separately convex; by the reverse triangle inequality $|g(y) - g(y')| \le 2\|y - y'\|_2$. Hence $f = g/2$ is separately convex and 1-Lipschitz on $[0, 1]^n$. Theorem 6.10 applied to $\tilde{Z} = f(Y) = Z/2$ gives $\mathbb{P}\{\tilde{Z} > \mathbb{E}\tilde{Z} + t\} \le e^{-t^2/2}$ for $t > 0$, i.e. $\mathbb{P}\{Z > \mathbb{E}Z + s\} \le e^{-s^2/8}$ for $s = 2t > 0$. With $s = t\sqrt{n}$,

$$\mathbb{P}\{Z \ge \mathbb{E}Z + t\sqrt{n}\} \le e^{-nt^2/8}, \qquad t > 0.$$

Remark. Here $\Sigma^2 \le 1$ and $\sigma^2 = 1/3$ are both $O(1)$ while $\mathbb{E}Z \approx \sqrt{n/3}$; Theorem 11.8's estimate $\Sigma^2 \le \sigma^2 + 8\mathbb{E}Z$ is far from sharp in this case.

Chapter 12 — Suprema of Empirical Processes: Exponential Inequalities

Exercise 12.1

Central idea. Symmetrization replaces the unknown P by an independent ghost sample P_n' (this needs $n\varepsilon^2 \ge 1/2$, otherwise the right-hand side exceeds 1 and there is nothing to prove). Conditioning on the $2n$-sample, independently swapping each pair (X_i, X_{n+i}) randomizes the signs in $P_n(A) - P_n'(A)$; Hoeffding's inequality applied to each of the $2^{h(X_1,\dots,X_{2n})}$ distinct labelings, followed by a union bound, gives the result.

Erratum. The hint's second step (the hint, not the statement) is false as printed: it claims $\mathbb{P}\{\sup_{A\in\mathcal{C}}|P_n(A)-P'_n(A)|\ge\varepsilon\}\le 2\,\mathbb{E}[2^{h(X_1,\dots,X_{2n})}]e^{-2n\varepsilon^2}$. Counterexample: the single-set class $\mathcal{C}=\{A\}$ with $P(A)=1/2$, so every sample has one trace and $2^h\equiv 1$; take $n=450$, $\varepsilon=0.1$. Then $nP_n(A)$, $nP'_n(A)$ are independent $\mathrm{Bin}(450,1/2)$ and the exact probability is $\mathbb{P}\{|nP_n(A)-nP'_n(A)|\ge 45\}=0.00299$, which exceeds $2e^{-2n\varepsilon^2}=2e^{-9}\approx 2.47\times 10^{-4}$. The correct exponent is $n\varepsilon^2/2$: the permutation and Hoeffding argument below proves $\mathbb{P}\{\sup_A|P_n(A)-P'_n(A)|\ge\varepsilon\}\le 2\,\mathbb{E}[2^{h(X_1,\dots,X_{2n})}]e^{-n\varepsilon^2/2}$ (here $2e^{-2.25}\approx 0.211\ge 0.00299$), which is exactly what the exercise's main bound needs. The main statement itself is correct.

Solution. Write $\nu_n=P_n-P$. If $n\varepsilon^2<1/2$, then since $h\ge 0$ (so $2^h\ge 1$ and $\mathbb{E}[2^h]\ge 1$), the right-hand side is at least $4e^{-1/4}>1$ and the claim is trivial. Assume henceforth $n\varepsilon^2\ge 1/2$.

Symmetrization. For fixed A, Chebyshev's inequality gives $\mathbb{P}\{|P'_n(A)-P(A)|>\varepsilon\}\le P(A)(1-P(A))/(n\varepsilon^2)\le 1/(4n\varepsilon^2)\le 1/2$. On $\{\sup_A|\nu_n(A)|\ge 2\varepsilon\}$, pick A (depending only on the first sample) with $|\nu_n(A)|\ge 2\varepsilon$. Then $|P_n(A)-P'_n(A)|\ge|\nu_n(A)|-|P'_n(A)-P(A)|$, so

$$\mathbb{P}_{\text{ghost}}\{|P_n(A)-P'_n(A)|\ge\varepsilon\}\ge\mathbb{P}_{\text{ghost}}\{|P'_n(A)-P(A)|\le\varepsilon\}\ge 1/2.$$

Since $\sup_B|P_n(B)-P'_n(B)|\ge|P_n(A)-P'_n(A)|$, Fubini gives

$$\mathbb{P}\Big\{\sup_{A\in\mathcal{C}}|\nu_n(A)|\ge 2\varepsilon\Big\}\le 2\,\mathbb{P}\Big\{\sup_{A\in\mathcal{C}}|P_n(A)-P'_n(A)|\ge\varepsilon\Big\}.$$

Permutation bound. The $2n$ variables are i.i.d., so independently swapping each pair (X_i,X_{n+i}) $(i=1,\dots,n)$ preserves the joint law. Conditional on $X_1,\dots,X_{2n}$, for fixed A,

$$P_n(A)-P'_n(A)\ \overset{d}{=}\ \frac{1}{n}\sum_{i=1}^{n}\sigma_i(\mathbf{1}_{\{X_i\in A\}}-\mathbf{1}_{\{X_{n+i}\in A\}})$$

with σ_i i.i.d. Rademacher. Only $m\le n$ pairs contribute (the rest have $\mathbf{1}_{\{X_i\in A\}}=\mathbf{1}_{\{X_{n+i}\in A\}}$), so by Hoeffding's inequality,

$$\mathbb{P}_\sigma\Big\{\Big|\frac{1}{n}\sum_{i=1}^{m}\sigma_i d_i\Big|\ge\varepsilon\Big\}\le 2\exp\Big(-\frac{n^2\varepsilon^2}{2m}\Big)\le 2e^{-n\varepsilon^2/2},\qquad d_i\in\{-1,1\}.$$

Conditional on the $2n$-sample there are at most $2^{h(X_1,\dots,X_{2n})}$ distinct values of the signed sum as A varies, so a union bound gives

$$\mathbb{P}_\sigma\Big\{\sup_{A\in\mathcal{C}}|P_n(A)-P'_n(A)|\ge\varepsilon\,\Big|\,X_1,\dots,X_{2n}\Big\}\le 2^{h(X_1,\dots,X_{2n})}\cdot 2e^{-n\varepsilon^2/2}.$$

Taking expectations and combining with symmetrization,

$$\mathbb{P}\Big\{\sup_{A\in\mathcal{C}}|\nu_n(A)|\ge 2\varepsilon\Big\}\le 4\,\mathbb{E}\big[2^{h(X_1,\dots,X_{2n})}\big]\,e^{-n\varepsilon^2/2}.$$

Remark (improved bound). The improved symmetrization of Exercise 11.5 replaces the ghost sample by Rademacher variables on the original n-sample, giving $\mathbb{P}\{\sup_A|\nu_n(A)|\ge 2\varepsilon\}\le 2\,\mathbb{E}[2^{h(X_1,\dots,X_n)}]e^{-n\varepsilon^2/2}$: one conditions on $X_1,\dots,X_n$ and union-bounds over the $2^{h(X_1,\dots,X_n)}$ labelings of the n-sample, with Hoeffding's inequality supplying $e^{-n\varepsilon^2/2}$ per labeling.

Exercise 12.2

Central idea. By symmetry, $(X_1,\dots,X_n)\overset{d}{=}(\varepsilon_1|X_1|,\dots,\varepsilon_n|X_n|)$ with ε_i i.i.d. Rademacher independent of the $|X_i|$'s. Conditioning on the $|X_i|$'s turns the ratio into a Rademacher tail; the Chernoff bound with $\mathbb{E}_\varepsilon e^{\lambda\sum_i\varepsilon_i a_i}=\prod_i\cosh(\lambda a_i)\le e^{\lambda^2\sum_i a_i^2/2}$ (from $\cosh x\le e^{x^2/2}$) is optimized at $\lambda=t/\sqrt{\sum_i a_i^2}$, giving $e^{-t^2/2}$ uniformly.

Solution. Let $S = \sum_{i=1}^n X_i$ and $V = \sum_{i=1}^n X_i^2$. Since each X_i is symmetric, $(X_1, \dots, X_n)$ has the same law as $(\varepsilon_1|X_1|, \dots, \varepsilon_n|X_n|)$ where $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. Rademacher, independent of $(|X_1|, \dots, |X_n|)$. Define $S/\sqrt{V} = 0$ on $\{V = 0\}$ (then $S = 0$ too, so $\{S/\sqrt{V} \ge t\} = \emptyset$ for $t > 0$).

Condition on $\mathcal{F} = \sigma(|X_1|, \dots, |X_n|)$ and work on $\{V > 0\}$, writing $a_i = |X_i|$. For $\lambda > 0$, by Markov's inequality,

$$\mathbb{P}_\varepsilon\Big\{ \sum_{i=1}^n \varepsilon_i a_i \ge t\sqrt{\textstyle\sum_i a_i^2} \,\Big|\, \mathcal{F} \Big\} \le \frac{\mathbb{E}_\varepsilon \exp(\lambda \sum_i \varepsilon_i a_i)}{\exp(\lambda t \sqrt{\sum_i a_i^2})} = \frac{\prod_{i=1}^n \cosh(\lambda a_i)}{\exp(\lambda t \sqrt{\sum_i a_i^2})}.$$

Since $\cosh x = \sum_{k\ge 0} x^{2k}/(2k)! \le \sum_{k\ge 0} x^{2k}/(2^k k!) = e^{x^2/2}$ (as $(2k)! \ge 2^k k!$),

$$\mathbb{P}_\varepsilon\Big\{ \sum_{i=1}^n \varepsilon_i a_i \ge t\sqrt{\textstyle\sum_i a_i^2} \,\Big|\, \mathcal{F} \Big\} \le \exp\Big(\tfrac{\lambda^2}{2} \sum_i a_i^2 - \lambda t \sqrt{\textstyle\sum_i a_i^2} \Big).$$

With $s = \sqrt{\sum_i a_i^2} > 0$, the exponent $\lambda^2 s^2/2 - \lambda t s$ is minimized at $\lambda = t/s$, giving $-t^2/2$. Hence, on $\{V > 0\}$,

$$\mathbb{P}\Big\{ \frac{S}{\sqrt{V}} \ge t \,\Big|\, \mathcal{F} \Big\} = \mathbb{P}_\varepsilon\Big\{ \frac{\sum_i \varepsilon_i a_i}{\sqrt{\sum_i a_i^2}} \ge t \,\Big|\, \mathcal{F} \Big\} \le e^{-t^2/2}.$$

The same bound holds trivially on $\{V = 0\}$. Taking expectations over $\mathcal{F}$,

$$\mathbb{P}\Big\{ \frac{\sum_{i=1}^n X_i}{\sqrt{\sum_{i=1}^n X_i^2}} \ge t \Big\} \le e^{-t^2/2}, \qquad t > 0.$$

■

Exercise 12.3

Central idea. Symmetrize with a ghost sample, using the binomial median bound $\mathbb{P}\{nP_n'(A) \ge \lfloor nP(A) \rfloor\} \ge 1/2$ and the monotonicity of $f(x,q) = \frac{x-q}{\sqrt{(x+q)/2}}$. Done carefully, this reaches threshold $\sqrt{2}\,\varepsilon$ in the symmetrized event, not just the threshold ε of the hint. Conditionally on the $2n$-sample, random swaps of the pairs (X_i, X_{n+i}) plus Hoeffding's inequality, using $\sum_i a_i^2 \le 2nD(A)$, give $e^{-n\eta^2/4}$ per trace at threshold η. With $\eta = \sqrt{2}\varepsilon$ this is exactly the printed $e^{-n\varepsilon^2/2}$. **The printed bound is true.** Only the hint is too weak: its threshold ε must be $\sqrt{2}\,\varepsilon$.

Remark (hint erratum). The hinted inequality, with threshold ε on the right, is true whenever $n\varepsilon^2 > 1$ (the only nontrivial range): it follows from Step 1 below, since $\varepsilon < \sqrt{2}\varepsilon$. But it cannot lead to the printed exponent. Combined with the per-trace swap bound, threshold ε only yields $e^{-n\varepsilon^2/4}$, and the per-trace estimate at threshold ε really can exceed $e^{-n\varepsilon^2/2}$. Example: $n = 10$, $\varepsilon = 1$, a $2n$-sample in which A contains exactly one point of each of the 10 pairs. Then $D(A) = 1/2$, and after swaps $\frac{P_n'(A) - P_n(A)}{\sqrt{D(A)}} = \frac{\sum_{i=1}^{10} \sigma_i}{\sqrt{50}}$, so the conditional probability of exceeding 1 is $\mathbb{P}\{\sum_{i\le 10} \sigma_i \ge 8\} = 11/1024 \approx 0.0107 > e^{-5} \approx 0.0067$. This refutes only the step "threshold ε per trace $\le e^{-n\varepsilon^2/2}$", not the printed inequality. (An earlier version of this solution wrongly concluded from this example that the printed bound is false and "corrected" it to $e^{-n\varepsilon^2/4}$; that is retracted.) The fix is to symmetrize to threshold $\sqrt{2}\varepsilon$.

Solution. *Conventions.* As in Exercise 12.1, $X_1, \dots, X_n$ define P_n, the independent ghost sample $X_{n+1}, \dots, X_{2n}$ defines P_n', and $\mathcal{C}$ is countable (or regular enough that all suprema below are measurable). $h(X_1, \dots, X_{2n})$ is the VC-entropy, so $\{A \cap \{X_1, \dots, X_{2n}\} : A \in \mathcal{C}\}$ has at most 2^h elements ("traces"). Write $D(A) = (P_n(A) + P_n'(A))/2$. A ratio whose denominator vanishes is set to 0: if $P(A) = 0$ then $P_n(A) = 0$ a.s., and if $D(A) = 0$ then $P_n(A) = P_n'(A) = 0$. For $x, q \ge 0$ with $x + q > 0$ let

$$f(x,q) = \frac{x-q}{\sqrt{(x+q)/2}}, \qquad \partial_x f = \frac{x+3q}{\sqrt{2}\,(x+q)^{3/2}} \ge 0, \qquad \partial_q f = -\frac{3x+q}{\sqrt{2}\,(x+q)^{3/2}} \le 0,$$

so f is nondecreasing in x and nonincreasing in q.

Trivial range. Since $h \geq 0$, the right-hand side is at least $2e^{-n\varepsilon^2/2}$, which is > 1 when $n\varepsilon^2 \leq 1$ (as $2e^{-1/2} \approx 1.213$). Assume $n\varepsilon^2 > 1$.

Binomial median fact. If $S \sim \mathrm{Bin}(n,p)$, every median of S lies in $[\lfloor np \rfloor, \lceil np \rceil]$ (Kaas and Buhrman, 1980). Hence $\mathbb{P}\{S \geq \lfloor np \rfloor\} \geq 1/2$ and $\mathbb{P}\{S \leq \lceil np \rceil\} \geq 1/2$. (Both were checked numerically for all $n \leq 200$ on a grid of 400 values of p.)

Step 1 (symmetrization to threshold $\sqrt{2}\varepsilon$). We claim

$$\mathbb{P}\Big\{ \sup_{A \in \mathcal{C}} \frac{P(A) - P_n(A)}{\sqrt{P(A)}} \geq 2\varepsilon \Big\} \leq 2\,\mathbb{P}\Big\{ \sup_{A \in \mathcal{C}} \frac{P_n'(A) - P_n(A)}{\sqrt{D(A)}} \geq \sqrt{2}\,\varepsilon \Big\}. \tag{1}$$

Fix $\varepsilon' < \varepsilon$ with $n\varepsilon'^2 \geq 0.86$; this is possible since $n\varepsilon^2 > 1$. Let E' be the event that $\sup_A \frac{P(A)-P_n(A)}{\sqrt{P(A)}} > 2\varepsilon'$. It contains the event on the left of (1). On E', let A^* be the first set, in a fixed enumeration of $\mathcal{C}$, with $P(A^*) - P_n(A^*) > 2\varepsilon'\sqrt{P(A^*)}$. Then A^* is a measurable function of $X_1, \dots, X_n$ alone. Write $p = P(A^*)$, $q = P_n(A^*)$ and $x = P_n'(A^*)$. Then:

- $p > 0$, and since $q \geq 0$, $\sqrt{p} > 2\varepsilon'$, i.e. $p > 4\varepsilon'^2 \geq 3.44/n$;
- $q < q_0 := p - 2\varepsilon'\sqrt{p} = \sqrt{p}(\sqrt{p} - 2\varepsilon')$, where $q_0 > 0$;
- conditionally on $X_1, \dots, X_n$, $nx \sim \mathrm{Bin}(n,p)$. By the median fact and $\lfloor np \rfloor > np - 1$, the event $x > x_0 := p - 1/n$ has conditional probability at least $1/2$. Also $x_0 > 0$.

On that event, monotonicity of f gives $f(x,q) \geq f(x_0,q_0)$. Moreover

$$f(x_0,q_0) = \frac{2\varepsilon'\sqrt{p} - 1/n}{\sqrt{p - \varepsilon'\sqrt{p} - 1/(2n)}}.$$

Here the numerator is positive, because $2\varepsilon'\sqrt{p} > 4\varepsilon'^2 \geq 1/n$. The radicand lies in $(0,p]$, because $p - \varepsilon'\sqrt{p} = \sqrt{p}(\sqrt{p} - \varepsilon') > 2\varepsilon'^2 > 1/(2n)$. Hence

$$f(x,q) \;\geq\; \frac{2\varepsilon'\sqrt{p} - 1/n}{\sqrt{p}} = 2\varepsilon' - \frac{1}{n\sqrt{p}} \;\geq\; 2\varepsilon' - \frac{1}{2n\varepsilon'} \;\geq\; \sqrt{2}\,\varepsilon'.$$

The last inequality is equivalent to $n\varepsilon'^2 \geq \frac{1}{2(2-\sqrt{2})} \approx 0.8536$, which holds. Since $D(A^*) > 0$ on this event, $\sup_A \frac{P_n'(A)-P_n(A)}{\sqrt{D(A)}} \geq f(x,q) \geq \sqrt{2}\varepsilon'$. By Fubini (integrate first over the ghost sample),

$$\mathbb{P}\Big\{ \sup_A \frac{P_n'(A) - P_n(A)}{\sqrt{D(A)}} \geq \sqrt{2}\,\varepsilon' \Big\} \geq \mathbb{E}\Big[\mathbb{1}_{E'}\, \mathbb{P}\{x > x_0 \mid X_1, \dots, X_n\} \Big] \geq \frac{1}{2}\,\mathbb{P}(E') \geq \frac{1}{2}\,\mathbb{P}\Big\{ \sup_A \frac{P(A) - P_n(A)}{\sqrt{P(A)}} \geq 2\varepsilon \Big\}.$$

As $\varepsilon' \uparrow \varepsilon$, the events $\{\sup_A \cdots \geq \sqrt{2}\varepsilon'\}$ decrease to $\{\sup_A \cdots \geq \sqrt{2}\varepsilon\}$. Continuity from above gives (1). (A random check of the chain $x > p - 1/n,\ q < p - 2\varepsilon'\sqrt{p} \Rightarrow f(x,q) \geq \sqrt{2}\varepsilon'$ over $2 \cdot 10^5$ draws with $n\varepsilon'^2 \geq 0.86$ found no violation.)

Step 2 (random swaps). Let $\sigma_1, \dots, \sigma_n$ be i.i.d. Rademacher signs, independent of the data. Swapping $X_i \leftrightarrow X_{n+i}$ whenever $\sigma_i = -1$ preserves the joint law of $(X_1, \dots, X_{2n})$. Put $a_i(A) = \mathbb{1}_A(X_{n+i}) - \mathbb{1}_A(X_i)$. The swap turns $n(P_n'(A) - P_n(A)) = \sum_i a_i(A)$ into $\sum_i \sigma_i a_i(A)$ and leaves $D(A)$ unchanged. So, for $\eta > 0$,

$$\mathbb{P}\Big\{ \sup_A \frac{P_n'(A) - P_n(A)}{\sqrt{D(A)}} \geq \eta \Big\} = \mathbb{E}\,\mathbb{P}_\sigma\Big\{ \sup_{A:\, D(A) > 0} \frac{\sum_i \sigma_i a_i(A)}{n\sqrt{D(A)}} \geq \eta \Big\}.$$

Given $X_1, \dots, X_{2n}$, the quantities $a_i(A)$ and $D(A)$ depend on A only through its trace, so the inner supremum is a maximum over at most 2^h traces. Fix a trace with $D > 0$. Since $|a_i| \leq 1$ and $|a_i| \leq \mathbb{1}_A(X_i) + \mathbb{1}_A(X_{n+i})$,

$$\sum_i a_i^2 = \sum_i |a_i| \leq \sum_{j=1}^{2n} \mathbb{1}_A(X_j) = 2nD.$$

If $\sum_i a_i^2 = 0$, the event is empty. Otherwise Hoeffding's inequality for Rademacher sums gives

$$\mathbb{P}_\sigma\Big\{\sum_i \sigma_i a_i \ge n\eta\sqrt{D}\Big\} \le \exp\Big(-\frac{n^2\eta^2 D}{2\sum_i a_i^2}\Big) \le \exp\Big(-\frac{n^2\eta^2 D}{4nD}\Big) = e^{-n\eta^2/4}.$$

A union bound over traces, followed by expectation, gives

$$\mathbb{P}\Big\{\sup_A \frac{P_n'(A) - P_n(A)}{\sqrt{D(A)}} \ge \eta\Big\} \le \mathbb{E}\big[2^{h(X_1,\dots,X_{2n})}\big] e^{-n\eta^2/4}. \tag{2}$$

Conclusion. Taking $\eta = \sqrt{2}\varepsilon$ in (2) and combining with (1),

$$\mathbb{P}\Big\{\sup_{A\in\mathcal{C}} \frac{P(A) - P_n(A)}{\sqrt{P(A)}} \ge 2\varepsilon\Big\} \le 2\,\mathbb{E}\big[2^{h(X_1,\dots,X_{2n})}\big] e^{-n\varepsilon^2/2}.$$

This is the printed bound. (With the hint's threshold $\eta = \varepsilon$, (2) gives only $e^{-n\varepsilon^2/4}$. The example in the Remark shows that the per-trace step cannot do better at that threshold, so the gain has to come from Step 1.) ■

Exercise 12.4

Central idea. This mirrors Exercise 12.3 with the roles of P and P_n exchanged. Symmetrize with a ghost sample, using the upper binomial median bound $\mathbb{P}\{nP_n'(A) \le \lceil nP(A)\rceil\} \ge 1/2$ and the monotonicity of $g(x,q) = \frac{q-x}{\sqrt{(q+x)/2}}$, to reach threshold $\sqrt{2}\,\varepsilon$ in the symmetrized event. Random pair swaps plus Hoeffding's inequality, using $\sum_i a_i^2 \le 2nD(A)$, then give $e^{-n(\sqrt{2}\varepsilon)^2/4} = e^{-n\varepsilon^2/2}$ per trace. **The printed bound is true as stated.** (An earlier version of this solution claimed that $e^{-n\varepsilon^2/2}$ is "unattainable" and weakened it to $e^{-n\varepsilon^2/4}$. That claim is retracted. The $n = 10$ example of Exercise 12.3 only shows that symmetrizing to threshold ε is too weak; it does not refute the printed inequality.)

Solution. *Conventions.* Use the notation of Exercises 12.1 and 12.3: the ghost sample $X_{n+1},\dots,X_{2n}$ defines P_n', $\mathcal{C}$ is countable (or regular enough that all suprema are measurable), at most $2^{h(X_1,\dots,X_{2n})}$ traces $A\cap\{X_1,\dots,X_{2n}\}$ occur, and $D(A) = (P_n(A) + P_n'(A))/2$. Sets with $P_n(A) = 0$ contribute 0 to the left-hand supremum; a ratio with vanishing denominator is 0. For $q, x \ge 0$ with $q + x > 0$ let

$$g(x,q) = \frac{q-x}{\sqrt{(q+x)/2}}.$$

By the derivative computation of Exercise 12.3 (with the variables exchanged), g is nonincreasing in x and nondecreasing in q.

Trivial range. Since $h \ge 0$, the right-hand side is at least $2e^{-n\varepsilon^2/2} \ge 2e^{-1/2} > 1$ when $n\varepsilon^2 \le 1$. Assume $n\varepsilon^2 > 1$.

Step 1 (symmetrization to threshold $\sqrt{2}\varepsilon$). We claim

$$\mathbb{P}\Big\{\sup_{A\in\mathcal{C}} \frac{P_n(A) - P(A)}{\sqrt{P_n(A)}} \ge 2\varepsilon\Big\} \le 2\,\mathbb{P}\Big\{\sup_{A\in\mathcal{C}} \frac{P_n(A) - P_n'(A)}{\sqrt{D(A)}} \ge \sqrt{2}\,\varepsilon\Big\}. \tag{1}$$

Fix $\varepsilon' < \varepsilon$ with $r := n\varepsilon'^2 \ge 1$. Let E' be the event that some A with $P_n(A) > 0$ has $P_n(A) - P(A) > 2\varepsilon'\sqrt{P_n(A)}$. It contains the event on the left of (1). On E', let A^* be the first such set in a fixed enumeration of $\mathcal{C}$; it is a function of $X_1,\dots,X_n$ only. Write $q = P_n(A^*) > 0$, $s = \sqrt{q}$, $p = P(A^*)$ and $x = P_n'(A^*)$. Since $p \ge 0$, $q > 2\varepsilon' s$, i.e. $s > 2\varepsilon'$, and $p < q - 2\varepsilon' s$.

Conditionally on $X_1,\dots,X_n$, $nx \sim \mathrm{Bin}(n,p)$. Every median of $\mathrm{Bin}(n,p)$ is at most $\lceil np\rceil < np + 1$ (Kaas and Buhrman, 1980; checked numerically for all $n \le 200$ on a 400-point grid of p). So with conditional probability at least $1/2$,

$$x < p + \tfrac{1}{n} < x_1 := q - 2\varepsilon' s + \tfrac{1}{n}, \qquad x_1 = s(s - 2\varepsilon') + \tfrac{1}{n} > 0.$$

On that event, since g is nonincreasing in x,

$$g(x,q) \ge g(x_1,q) = \frac{2\varepsilon' s - 1/n}{\sqrt{s^2 - \varepsilon' s + 1/(2n)}} \;\ge\; F(s) := \frac{2\varepsilon' s - 1/n}{\sqrt{s^2 + 1/(2n)}}.$$

For the second inequality: the numerator is positive, because $2\varepsilon' s > 4\varepsilon'^2 \ge 4/n$, and $0 < s^2 - \varepsilon' s + \frac{1}{2n} \le s^2 + \frac{1}{2n}$, because $s > \varepsilon'$.

F is increasing on $(0,\infty)$: a direct computation gives $F'(s)\,(s^2 + \frac{1}{2n})^{3/2} = \frac{\varepsilon' + s}{n} > 0$. Since $s > 2\varepsilon'$, $F(s) \ge F(2\varepsilon')$. Now

$$F(2\varepsilon') = \frac{(4r-1)/n}{\sqrt{(8r+1)/(2n)}} = \sqrt{\frac{2}{n}}\,\frac{4r-1}{\sqrt{8r+1}}, \qquad \sqrt{2}\,\varepsilon' = \sqrt{\frac{2}{n}}\,\sqrt{r}.$$

Both sides are positive because $r \ge 1$, so

$$F(2\varepsilon') \ge \sqrt{2}\,\varepsilon' \iff (4r-1)^2 \ge r(8r+1) \iff 8r^2 - 9r + 1 \ge 0 \iff (8r-1)(r-1) \ge 0,$$

which holds for $r \ge 1$. Hence $g(x,q) \ge \sqrt{2}\,\varepsilon'$, and since $D(A^*) \ge q/2 > 0$,

$$\sup_A \frac{P_n(A) - P_n'(A)}{\sqrt{D(A)}} \ge g(x,q) \ge \sqrt{2}\,\varepsilon'$$

with conditional probability at least $1/2$ on E'. (A random check of this chain over $2 \cdot 10^5$ draws with $r \ge 1$ found no violation; the polynomial identity $(4r-1)^2 - r(8r+1) = (8r-1)(r-1)$ was checked symbolically.) Integrating over the first sample (Fubini),

$$\mathbb{P}\Big\{ \sup_A \frac{P_n(A) - P_n'(A)}{\sqrt{D(A)}} \ge \sqrt{2}\,\varepsilon' \Big\} \ge \tfrac{1}{2}\,\mathbb{P}(E') \ge \tfrac{1}{2}\,\mathbb{P}\Big\{ \sup_A \frac{P_n(A) - P(A)}{\sqrt{P_n(A)}} \ge 2\varepsilon \Big\}.$$

As $\varepsilon' \uparrow \varepsilon$, the left events decrease to $\{\sup_A \cdots \ge \sqrt{2}\varepsilon\}$, and continuity from above gives (1).

Step 2 (random swaps). As in Exercise 12.3, let $\sigma_1, \dots, \sigma_n$ be i.i.d. Rademacher signs independent of the data. Swapping $X_i \leftrightarrow X_{n+i}$ when $\sigma_i = -1$ preserves the joint law. With $a_i(A) = \mathbb{1}_A(X_i) - \mathbb{1}_A(X_{n+i})$, the swap turns $n(P_n(A) - P_n'(A)) = \sum_i a_i(A)$ into $\sum_i \sigma_i a_i(A)$ and leaves $D(A)$ unchanged. Conditionally on $X_1, \dots, X_{2n}$, only the at most 2^h traces matter. For a trace with $D > 0$, $\sum_i a_i^2 = \sum_i |a_i| \le \sum_{j \le 2n} \mathbb{1}_A(X_j) = 2nD$. If $\sum_i a_i^2 = 0$ the event below is empty. Otherwise Hoeffding's inequality gives

$$\mathbb{P}_\sigma\Big\{ \sum_i \sigma_i a_i \ge n\eta\sqrt{D} \Big\} \le \exp\Big(- \frac{n^2\eta^2 D}{2\sum_i a_i^2} \Big) \le e^{-n\eta^2/4}.$$

A union bound over traces and expectation over the $2n$-sample give, for $\eta > 0$,

$$\mathbb{P}\Big\{ \sup_A \frac{P_n(A) - P_n'(A)}{\sqrt{D(A)}} \ge \eta \Big\} \le \mathbb{E}\big[2^{h(X_1,\dots,X_{2n})}\big] e^{-n\eta^2/4}. \tag{2}$$

Conclusion. Taking $\eta = \sqrt{2}\varepsilon$ in (2) and combining with (1),

$$\mathbb{P}\Big\{ \sup_{A \in \mathcal{C}} \frac{P_n(A) - P(A)}{\sqrt{P_n(A)}} \ge 2\varepsilon \Big\} \le 2\,\mathbb{E}\big[2^{h(X_1,\dots,X_{2n})}\big] e^{-n\varepsilon^2/2}.$$

This is exactly the printed bound. (Symmetrizing only to threshold ε, as a literal transcription of the hint of Exercise 12.3 suggests, would give $e^{-n\varepsilon^2/4}$ after (2). The improvement comes entirely from Step 1.) ■

Exercise 12.5

Central idea. Apply the entropy method via the modified logarithmic Sobolev inequality (Theorem 6.6), as in the proof of Theorem 12.5, but replace the boundedness-dependent Lemma 12.6 by a direct estimate using the sub-gamma moment hypothesis. Writing $e^{\lambda Z}\phi(-\lambda(Z-Z_i)) = e^{\lambda Z_i}\psi(\lambda(Z-Z_i))$ with $\psi(u) = e^u\phi(-u)$, the bound $|Z-Z_i| \le |X_i|$ and monotonicity of ψ give $\psi(\lambda(Z-Z_i)) \le \psi(\lambda|X_i|)$; the power series $\psi(u) = \sum_{q\ge 2}\frac{q-1}{q!}u^q$ together with $\mathbb{E}|X_i|^q \le q!c^{q-2}\sigma^2/(2n)$ yields $\mathbb{E}\psi(\lambda|X_i|) \le \sigma^2\lambda^2/(2n(1-c\lambda)^2)$. This gives $\mathrm{Ent}(e^{\lambda Z}) \le \frac{\sigma^2\lambda^2}{2(1-c\lambda)^2}\mathbb{E}e^{\lambda Z}$, and the Herbst argument produces $\log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \sigma^2\lambda^2/(2(1-c\lambda)) \le v\lambda^2/(2(1-c\lambda))$ since $v = \sigma^2 + 2\mathbb{E}Z \ge \sigma^2$.

Solution. Let $\tilde{X}_{i,s} = \alpha_{i,s}X_i$, so $Z = \sup_{s\in\mathcal{T}}\sum_{i=1}^n \tilde{X}_{i,s}$ with $|\alpha_{i,s}| \le 1$, and let $Z_i = \sup_{s\in\mathcal{T}}\sum_{j\ne i}\tilde{X}_{j,s}$. Put $W_i = Z - Z_i$ and $\phi(u) = e^u - u - 1$. The modified logarithmic Sobolev inequality (Theorem 6.6) gives, for $\lambda \ge 0$,

$$\mathrm{Ent}(e^{\lambda Z}) \le \sum_{i=1}^n \mathbb{E}\left[e^{\lambda Z}\phi(-\lambda W_i)\right],$$

where $\mathrm{Ent}(Y) = \mathbb{E}[Y\log Y] - \mathbb{E}Y\log\mathbb{E}Y$ (all expectations finite for $0 \le \lambda < 1/c$ since $Z \le \sum_i |X_i|$ and each $|X_i|$ has an exponential moment).

Write $e^{\lambda Z}\phi(-\lambda W_i) = e^{\lambda Z_i}\psi(\lambda W_i)$ with $\psi(u) = e^u\phi(-u) = 1 - e^u + ue^u$. If $\hat{s}$ attains the supremum defining Z then $Z_i \ge \sum_{j\ne i}\tilde{X}_{j,\hat{s}}$, so $W_i \le \tilde{X}_{i,\hat{s}} \le |X_i|$; if $\hat{s}_i$ attains Z_i then $Z \ge \tilde{X}_{i,\hat{s}_i} + Z_i$, so $W_i \ge \tilde{X}_{i,\hat{s}_i} \ge -|X_i|$. Hence $|W_i| \le |X_i|$. Since $\psi'(u) = ue^u$, ψ is increasing on $[0,\infty)$, decreasing on $(-\infty,0]$, and $\psi(-u) \le \psi(u)$ for $u \ge 0$ (as $\psi(u) - \psi(-u)$ has nonnegative derivative $u(e^u - e^{-u})$ and vanishes at 0). Thus $\psi(\lambda W_i) \le \psi(\lambda|X_i|)$ pointwise, and, X_i being independent of $X^{(i)} = (X_j)_{j\ne i}$,

$$\mathbb{E}^{(i)}\left[e^{\lambda Z}\phi(-\lambda W_i)\right] = e^{\lambda Z_i}\mathbb{E}^{(i)}[\psi(\lambda W_i)] \le e^{\lambda Z_i}\mathbb{E}[\psi(\lambda|X_i|)],$$

where $\mathbb{E}^{(i)}$ conditions on $X^{(i)}$.

For $u \ge 0$, $\psi(u) = \sum_{q\ge 2}\frac{q-1}{q!}u^q$ (from $1 - e^u + ue^u$). By monotone convergence and the moment hypothesis,

$$\mathbb{E}[\psi(\lambda|X_i|)] = \sum_{q\ge 2}\frac{q-1}{q!}\lambda^q\mathbb{E}|X_i|^q \le \frac{\sigma^2}{2n}\sum_{q\ge 2}(q-1)(c\lambda)^{q-2}\lambda^2 = \frac{\sigma^2\lambda^2}{2n(1-c\lambda)^2}, \qquad 0 \le \lambda < 1/c,$$

using $\sum_{k\ge 0}(k+1)r^k = (1-r)^{-2}$. Since $\mathbb{E}^{(i)}[W_i] \ge \mathbb{E}^{(i)}[\tilde{X}_{i,\hat{s}_i}] = \alpha_{i,\hat{s}_i}\mathbb{E}X_i = 0$, Jensen gives $e^{\lambda Z_i} \le e^{\lambda\mathbb{E}^{(i)}Z} \le \mathbb{E}^{(i)}e^{\lambda Z}$, whence $\mathbb{E}e^{\lambda Z_i} \le \mathbb{E}e^{\lambda Z}$. Therefore

$$\mathrm{Ent}(e^{\lambda Z}) \le \frac{\sigma^2\lambda^2}{2(1-c\lambda)^2}\mathbb{E}e^{\lambda Z}, \qquad 0 \le \lambda < 1/c.$$

Put $G(\lambda) = \log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$. Then $\frac{\mathrm{Ent}(e^{\lambda Z})}{\mathbb{E}e^{\lambda Z}} = \lambda G'(\lambda) - G(\lambda)$, so

$$\lambda G'(\lambda) - G(\lambda) \le \frac{\sigma^2\lambda^2}{2(1-c\lambda)^2}, \qquad G(0) = G'(0) = 0.$$

Hence $\frac{d}{d\lambda}\frac{G(\lambda)}{\lambda} = \frac{\lambda G' - G}{\lambda^2} \le \frac{\sigma^2}{2(1-c\lambda)^2}$, and integrating,

$$\frac{G(\lambda)}{\lambda} \le \int_0^\lambda \frac{\sigma^2}{2(1-cs)^2}\,ds = \frac{\sigma^2\lambda}{2(1-c\lambda)},$$

i.e. $G(\lambda) \le \frac{\sigma^2\lambda^2}{2(1-c\lambda)}$. Since $\mathbb{E}Z \ge \sup_s\sum_i \alpha_{i,s}\mathbb{E}X_i = 0$, $v = \sigma^2 + 2\mathbb{E}Z \ge \sigma^2$, and

$$\log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \frac{v\lambda^2}{2(1-c\lambda)}, \qquad 0 \le \lambda < 1/c,$$

which is the sub-gamma bound with variance factor v and scale c (Bousquet 2002b, Theorem 2.12). The estimate is vacuous for $\lambda \ge 1/c$, where the right-hand side is nonpositive while the left-hand side is nonnegative.

Exercise 12.6

Central idea. Factor $e^{\lambda Z} - e^{\lambda Z_i} = e^{\lambda Z}(1 - e^{-\lambda(Z-Z_i)})$ and use the identity $1 - e^{-s} = s - \phi(-s)$ with $\phi(t) = e^t - t - 1$ to split the sum into a self-bounding term, controlled by $\lambda\mathbb{E}[Ze^{\lambda Z}]$, and a modified-log-Sobolev term, bounded below by the entropy $\mathrm{Ent}(e^{\lambda Z})$; the two $\lambda\mathbb{E}[Ze^{\lambda Z}]$ terms cancel, leaving $\mathbb{E}e^{\lambda Z}\log\mathbb{E}e^{\lambda Z}$.

Solution. Let $\lambda \geq 0$ and assume $\mathbb{E}e^{\lambda Z} < \infty$, the inequality being trivial otherwise. In Theorem 12.5, $Z = \sup_{t\in\mathcal{T}} \sum_{i=1}^n X_{i,t}$ with $X_{i,t} \geq 0$ and $Z_i = \sup_{t\in\mathcal{T}} \sum_{j\neq i} X_{j,t}$, so $Z \geq Z_i \geq 0$ and the self-bounding property $\sum_{i=1}^n (Z - Z_i) \leq Z$ holds.

Since $e^{\lambda Z_i} = e^{\lambda Z} e^{-\lambda(Z-Z_i)}$,

$$e^{\lambda Z} - e^{\lambda Z_i} = e^{\lambda Z}\left(1 - e^{-\lambda(Z-Z_i)}\right).$$

With $\phi(t) = e^t - t - 1$, setting $s = \lambda(Z - Z_i)$ gives

$$s - \phi(-s) = s - (e^{-s} + s - 1) = 1 - e^{-s},$$

hence $1 - e^{-\lambda(Z-Z_i)} = \lambda(Z - Z_i) - \phi(-\lambda(Z - Z_i))$. Summing over i and taking expectations,

$$\sum_{i=1}^n \mathbb{E}\left[e^{\lambda Z} - e^{\lambda Z_i}\right] = \lambda\mathbb{E}\left[e^{\lambda Z}\sum_{i=1}^n (Z - Z_i)\right] - \sum_{i=1}^n \mathbb{E}\left[e^{\lambda Z}\phi(-\lambda(Z - Z_i))\right].$$

By the self-bounding property and $\lambda \geq 0$,

$$\lambda\mathbb{E}\left[e^{\lambda Z}\sum_{i=1}^n (Z - Z_i)\right] \leq \lambda\mathbb{E}\left[Ze^{\lambda Z}\right].$$

Theorem 6.6 (the modified logarithmic Sobolev inequality) gives

$$\sum_{i=1}^n \mathbb{E}\left[e^{\lambda Z}\phi(-\lambda(Z - Z_i))\right] \geq \mathrm{Ent}(e^{\lambda Z}) = \lambda\mathbb{E}\left[Ze^{\lambda Z}\right] - \mathbb{E}e^{\lambda Z}\log\mathbb{E}e^{\lambda Z}.$$

Combining the two bounds,

$$\sum_{i=1}^n \mathbb{E}\left[e^{\lambda Z} - e^{\lambda Z_i}\right] \leq \lambda\mathbb{E}\left[Ze^{\lambda Z}\right] - \lambda\mathbb{E}\left[Ze^{\lambda Z}\right] + \mathbb{E}e^{\lambda Z}\log\mathbb{E}e^{\lambda Z} = \mathbb{E}e^{\lambda Z}\log\mathbb{E}e^{\lambda Z},$$

which is the claimed inequality. ■

Exercise 12.7

Central idea. Rio's trick is to apply the modified logarithmic Sobolev inequality (Theorem 6.6) not to the leave-one-out suprema Z_i but to constant shifts $\tilde{Z}_i = Z_i + c_i$, with c_i chosen so that $\mathbb{E}e^{\lambda\tilde{Z}_i} = \mathbb{E}e^{\lambda Z}$. Then the terms $\mathbb{E}[e^{\lambda\tilde{Z}_i} - e^{\lambda Z}]$ in Theorem 6.6 cancel, and the self-bounding property $\sum_i (Z - Z_i) \leq Z$ cancels $\lambda\mathbb{E}[Ze^{\lambda Z}]$ against the entropy, leaving $\sum_i \log(\mathbb{E}e^{\lambda Z}/\mathbb{E}e^{\lambda Z_i}) \leq \log\mathbb{E}e^{\lambda Z}$. Since the X_i are i.i.d., every Z_i has the law of Z_n, which gives the claim for every $\lambda \geq 0$. For the cancellation to work the shift must be $c_i = \lambda^{-1}\log(\mathbb{E}e^{\lambda Z}/\mathbb{E}e^{\lambda Z_i})$: the hint as printed is missing the factor $1/\lambda$.

Solution. *Erratum in the hint.* The hint's shift $\tilde{Z}_i = Z_i + \log\mathbb{E}e^{\lambda Z} - \log\mathbb{E}e^{\lambda Z_i}$ does not normalize the exponential moment unless $\lambda = 1$. Write $M = \mathbb{E}e^{\lambda Z}$ and $M_i = \mathbb{E}e^{\lambda Z_i}$. With the printed shift,

$$\mathbb{E}e^{\lambda\tilde{Z}_i} = \left(\frac{M}{M_i}\right)^{\lambda} M_i = M^{\lambda} M_i^{1-\lambda},$$

which differs from M whenever $\lambda \neq 1$ and $M_i \neq M$. For example, $M = 2$, $M_i = 1.5$, $\lambda = 0.5$ gives $\sqrt{3} \approx 1.732 \neq 2$. The correct shift is

$$\tilde{Z}_i = Z_i + c_i, \qquad c_i = \frac{1}{\lambda}\log\frac{M}{M_i},$$

for which $\mathbb{E}e^{\lambda\tilde{Z}_i} = e^{\lambda c_i}M_i = M$. (An earlier version of this solution used the printed shift, wrote $e^{\lambda c_i}M_i = M$ for all λ, and then could only conclude for $\lambda \geq 1$. Both the identity and that restriction were artifacts of the missing $1/\lambda$. The proof below holds for every $\lambda \geq 0$.)

Setting. As in Theorem 12.5, $X_1, \dots, X_n$ are i.i.d., $\mathbb{E}X_{i,s} = 0$ and $X_{i,s} \leq 1$ for all $s \in \mathcal{T}$. Let $\hat{s}$ attain Z and $\hat{s}_i$ attain Z_i (take $\mathcal{T}$ finite; the countable case follows by monotone convergence along finite $\mathcal{T}_k \uparrow \mathcal{T}$). As shown before Theorem 12.5, $Z \geq \sum_j X_{j,\hat{s}_i}$, and $Z_i \geq \sum_{j\neq i} X_{j,\hat{s}}$ gives $Z - Z_i \leq X_{i,\hat{s}}$. Summing,

$$\sum_{i=1}^{n}(Z - Z_i) \leq \sum_{i=1}^{n} X_{i,\hat{s}} = Z. \tag{1}$$

Since $Z \leq n$ and $Z \geq \sum_j X_{j,s_0}$ for any fixed s_0, all quantities below are finite for $\lambda \geq 0$. Because the X_i are i.i.d., each Z_i has the same distribution as Z_n, so $M_i = M_n := \mathbb{E}e^{\lambda Z_n}$ for all i.

Proof for $\lambda > 0$. (For $\lambda = 0$ both sides are 0.) Each $\tilde{Z}_i = Z_i + c_i$ is a function of $X^{(i)} = (X_j)_{j\neq i}$, so Theorem 6.6 applies with Z_i replaced by $\tilde{Z}_i$. With $\phi(u) = e^u - u - 1$,

$$\mathrm{Ent}(e^{\lambda Z}) = \lambda\mathbb{E}[Ze^{\lambda Z}] - M\log M \leq \sum_{i=1}^{n}\mathbb{E}[e^{\lambda Z}\phi(-\lambda(Z - \tilde{Z}_i))].$$

Since $e^{\lambda Z}\phi(-\lambda(Z - \tilde{Z}_i)) = e^{\lambda\tilde{Z}_i} - e^{\lambda Z} + \lambda(Z - \tilde{Z}_i)e^{\lambda Z}$, the right-hand side equals

$$\sum_{i=1}^{n}(\mathbb{E}e^{\lambda\tilde{Z}_i} - M) + \lambda\,\mathbb{E}\Big[e^{\lambda Z}\sum_{i=1}^{n}(Z - Z_i)\Big] - \lambda M\sum_{i=1}^{n} c_i.$$

The first sum is 0 by the choice of c_i. By (1) and $\lambda e^{\lambda Z} \geq 0$, the middle term is at most $\lambda\mathbb{E}[Ze^{\lambda Z}]$. Therefore

$$\lambda\mathbb{E}[Ze^{\lambda Z}] - M\log M \leq \lambda\mathbb{E}[Ze^{\lambda Z}] - \lambda M\sum_{i=1}^{n} c_i.$$

Cancel $\lambda\mathbb{E}[Ze^{\lambda Z}]$, divide by $M > 0$, and use $\lambda c_i = \log M - \log M_i$:

$$\sum_{i=1}^{n}(\log M - \log M_i) \leq \log M, \qquad \text{i.e.} \qquad (n-1)\log\mathbb{E}e^{\lambda Z} \leq \sum_{i=1}^{n}\log\mathbb{E}e^{\lambda Z_i}. \tag{2}$$

(Inequality (2) does not use identical distribution.) In the i.i.d. case the right-hand side of (2) is $n\log M_n$, so

$$\frac{\log\mathbb{E}e^{\lambda Z}}{n} \leq \frac{\log\mathbb{E}e^{\lambda Z_n}}{n-1}, \qquad \lambda \geq 0.$$

Pollard's inequality. Taking expectations in (1) gives $n\mathbb{E}Z - \sum_i \mathbb{E}Z_i \leq \mathbb{E}Z$, i.e. $(n-1)\mathbb{E}Z \leq n\mathbb{E}Z_n$, which is $\mathbb{E}Z/n \leq \mathbb{E}Z_n/(n-1)$. It also follows from the result: divide by λ and let $\lambda \downarrow 0$, using $\log\mathbb{E}e^{\lambda Z} = \lambda\mathbb{E}Z + o(\lambda)$.

Remark on $\lambda < 0$. The restriction $\lambda \geq 0$ (the range of Theorem 12.5) is needed: the step using (1) multiplies by $\lambda e^{\lambda Z}$, and the inequality itself can fail for $\lambda < 0$. Take $\mathcal{T} = \{1, 2\}$ and X_i uniform on $\{(1, -\frac{1}{2}), (-\frac{1}{2}, 1), (-\frac{1}{2}, -\frac{1}{2})\}$, which is centered with coordinates ≤ 1. For $n = 2$ and $\lambda = -0.1$, exact computation gives $\frac{1}{2}\log\mathbb{E}e^{\lambda Z} \approx -0.03153$, which is larger than $\log\mathbb{E}e^{\lambda Z_2} \approx -0.04746$. For the same distribution, exact enumeration confirms the inequality for $n \in \{3, 4, 5\}$ and $\lambda \in \{0.1, 0.5, 1, 2, 5\}$. ■

Exercise 12.8

Central idea. Since $\theta(y)$ is chosen so that equality holds at $x = y$, the inequality is equivalent to the ratio $\psi(t) = e^t\phi(-t)/(y\phi(t) + t^2/2)$ being nondecreasing in t. Differentiating, the numerator of ψ' splits as $yA(t) + B(t)$ with $A(t) = e^{2t} - (t^2+2)e^t + 1$ and $B(t) = t^3e^t/2 - th(t)$ (where $h(t) = e^t\phi(-t)$); both A and B are nonnegative for all $t \in \mathbb{R}$ (each is monotone on either side of a global minimum of 0 at $t = 0$), so $\psi' \geq 0$ for every $y \geq 0$.

Solution. Let $\phi(u) = e^u - u - 1 \geq 0$, $h(x) = e^x\phi(-x) = 1 - (1-x)e^x$, and $k_y(x) = y\phi(x) + x^2/2$. Then $\theta(y) = h(y)/k_y(y)$ for $y > 0$, and we show $h(x) \leq \theta(y)k_y(x)$ for $x \leq y$.

First, $k_y(x) \geq 0$ for all x (as $\phi \geq 0$, $y \geq 0$), and for $y > 0$, $k_y(x) > 0$ when $x \neq 0$ since $\phi(x) > 0$ for $x \neq 0$. Also $h(0) = 0$, $h'(x) = xe^x$ (from $h(x) = 1 - (1-x)e^x$), so h is strictly increasing on $[0, \infty)$ with $h(0) = 0$, hence $h(y) > 0$ and $\theta(y) > 0$ for $y > 0$.

Case $y = 0$. Here θ extends continuously with $\theta(0) = 1$ (since $h(y) = y^2/2 + O(y^3)$ and $k_y(y) = y^2/2 + O(y^3)$), and for $x \leq 0$ we need $h(x) \leq x^2/2$. With $d(x) = x^2/2 - h(x)$, $d(0) = 0$ and $d'(x) = x - xe^x = x(1-e^x) \leq 0$ for $x \leq 0$ (as $e^x \leq 1$), so d is decreasing on $(-\infty, 0]$ and $d(x) \geq 0$ for $x \leq 0$.

Case $y > 0$. For $x = 0$ both sides equal 0. For $x \neq 0$ put $\psi(t) = h(t)/k_y(t)$, continuously extended at $t = 0$ by $\psi(0) = \lim_{t\to 0} h(t)/k_y(t) = 1/(y+1)$ (using $h(t) = t^2/2 + O(t^3)$, $k_y(t) = (y+1)t^2/2 + O(t^3)$). Then

$$\psi'(t) = \frac{h'(t)k_y(t) - h(t)k_y'(t)}{k_y(t)^2}, \qquad k_y'(t) = y(e^t - 1) + t.$$

The numerator is $N(t) = te^t(y\phi(t) + t^2/2) - h(t)(y(e^t - 1) + t) = yA(t) + B(t)$ where

$$A(t) = te^t\phi(t) - h(t)(e^t - 1), \qquad B(t) = \frac{t^3e^t}{2} - th(t).$$

Expanding A: since $h(t) = 1 - (1-t)e^t$ and $\phi(t) = e^t - t - 1$,

$$\begin{aligned} A(t) &= te^t(e^t - t - 1) - (1 - (1-t)e^t)(e^t - 1) \\ &= e^{2t} - (t^2 + 2)e^t + 1. \end{aligned}$$

Now $A(0) = 0$ and $A'(t) = 2e^{2t} - (t^2+2t+2)e^t = e^t(2e^t - (t^2+2t+2))$. The bracket $n(t) = 2e^t - (t^2+2t+2)$ satisfies $n(0) = 0$ and $n'(t) = 2(e^t - t - 1) = 2\phi(t) \geq 0$, so $n(t) \leq 0$ for $t \leq 0$ and $n(t) \geq 0$ for $t \geq 0$. Hence A decreases on $(-\infty, 0]$, increases on $[0, \infty)$, with minimum $A(0) = 0$; thus $A(t) \geq 0$ for all t.

For B: $B(t) = t(\frac{t^2e^t}{2} - h(t)) =: t\,m(t)$ with $m(t) = \frac{t^2e^t}{2} - 1 + (1-t)e^t$. Then $m(0) = 0$ and $m'(t) = te^t + \frac{t^2e^t}{2} - te^t = \frac{t^2e^t}{2} \geq 0$, so m is nondecreasing with $m(0) = 0$, i.e. $m(t)$ has the sign of t. Consequently $B(t) = t\,m(t) \geq 0$ for all t.

Therefore $N(t) = yA(t) + B(t) \geq 0$ for every $y \geq 0$ and all t, so $\psi'(t) \geq 0$ on $(-\infty, 0)$ and $(0, \infty)$. With continuity at 0, ψ is nondecreasing on all of $\mathbb{R}$. For $x \leq y$ (and $y > 0$) we get $\psi(x) \leq \psi(y)$, i.e. $\frac{h(x)}{k_y(x)} \leq \frac{h(y)}{k_y(y)} = \theta(y)$ for $x \neq 0$ (where $k_y(x), k_y(y) > 0$), and hence $h(x) \leq \theta(y)k_y(x)$. For $x = 0$ this is $0 = 0$. In original notation,

$$e^x\phi(-x) \leq \theta(y)\left(y\phi(x) + \frac{x^2}{2}\right) \qquad \text{for } y \geq 0,\ x \leq y,$$

as claimed. ■

Exercise 12.9

Central idea. Refine Bousquet's inequality (Theorem 12.5) by applying the modified log-Sobolev inequality (Theorem 6.6) to the shifted leave-one-out variables $\widetilde{Z}_i = Z_i + E$ ($E = \mathbb{E}Z/n$) rather than Z_i. This yields an entropy bound governed by the improved variance proxy $\tilde{v} = n\sigma^2 + (2-E)\mathbb{E}Z$. The entropy bound becomes a first-order differential inequality for $G(\lambda) = \log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)}$; since θ is the log-derivative of $(\phi(x) + x/2)/x$, it integrates to $G(\lambda) \leq \frac{\tilde{v}}{(1-E)^2}\phi((1-E)\lambda)$. Large λ follows from the trivial $Z \leq n$.

Solution. Let $Z = \sup_s \sum_{i=1}^n X_{i,s}$ as in Theorem 12.5 ($X_{i,s} \leq 1$, $\mathbb{E}X_{i,s} = 0$), $E = \mathbb{E}Z/n < 1$, $\tilde{v} = n\sigma^2 + (2-E)\mathbb{E}Z$, $\phi(x) = e^x - x - 1$, and $G(\lambda) = \log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)}$.

Large λ. Since $Z \leq n$, $G(\lambda) \leq \lambda(n - \mathbb{E}Z) = \lambda n(1-E)$ for $\lambda \geq 0$. For $\lambda \geq 2n(1-E)/\tilde{v}$, we claim $\lambda n(1-E) \leq \frac{\tilde{v}}{(1-E)^2}\phi((1-E)\lambda)$. With $x = (1-E)\lambda$, this is $\phi(x)/x \geq n(1-E)^2/\tilde{v}$. Since $\phi(x) \geq x^2/2$, $\phi(x)/x \geq x/2$, and $x \geq 2n(1-E)^2/\tilde{v}$ gives the result. Henceforth $0 \leq \lambda \leq 2n(1-E)/\tilde{v}$, i.e., $0 \leq \lambda\tilde{v}/(2n) \leq (1-E)$.

Entropy bound. Let $Z_i = \sup_s \sum_{j\neq i} X_{j,s}$ and $\tilde{Z}_i = Z_i + E$, still independent of X_i. Theorem 6.6 gives $\mathrm{Ent}(e^{\lambda Z}) \le \sum_i \mathbb{E}[e^{\lambda Z}\phi(-\lambda(Z-\tilde{Z}_i))]$. Since $Z - Z_i \le \sup_s X_{i,s} \le 1$, we have $Z - \tilde{Z}_i \le 1 - E$. By Exercise 12.8, with $\theta(y) = e^y\phi(-y)/(y\phi(y) + y^2/2)$,

$$\phi(-\lambda(Z-\tilde{Z}_i))e^{\lambda Z} \le \lambda\theta(\lambda(1-E))\left((1-E)e^{\lambda Z} + e^{\lambda\tilde{Z}_i}\left[\lambda\left(\frac{(Z-\tilde{Z}_i)^2}{2} - (1-E)(Z-\tilde{Z}_i)\right) - (1-E)\right]\right).$$

Taking expectations and treating the quadratic term by the Theorem 11.10/Exercise 11.17 pattern (producing the variance proxy $\tilde{v}$),

$$\mathbb{E}[\phi(-\lambda(Z-\tilde{Z}_i))e^{\lambda Z}] \le \lambda\theta(\lambda(1-E))\,\mathbb{E}\left[(1-E)e^{\lambda Z} + e^{\lambda\tilde{Z}_i}\left(\frac{\lambda\tilde{v}}{2n} - (1-E)\right)\right].$$

Decoupling. Exercise 12.7 gives $\mathbb{E}e^{\lambda\tilde{Z}_i} \ge \mathbb{E}e^{\lambda Z}(1 - G(\lambda)/n)$. Since $\frac{\lambda\tilde{v}}{2n} - (1-E) \le 0$,

$$\mathbb{E}e^{\lambda\tilde{Z}_i}\left(\frac{\lambda\tilde{v}}{2n} - (1-E)\right) \le \left(\frac{\lambda\tilde{v}}{2n} - (1-E)\right)\mathbb{E}e^{\lambda Z}\left(1 - \frac{G(\lambda)}{n}\right).$$

Summing over i and dividing by $\mathbb{E}e^{\lambda Z}$,

$$\frac{\mathrm{Ent}(e^{\lambda Z})}{\mathbb{E}e^{\lambda Z}} \le n\lambda\theta\left((1-E) + \left(\frac{\lambda\tilde{v}}{2n} - (1-E)\right)\left(1 - \frac{G}{n}\right)\right),$$

where $\theta = \theta(\lambda(1-E))$.

Differential inequality. Since $\mathrm{Ent}(e^{\lambda Z})/\mathbb{E}e^{\lambda Z} = \lambda G' - G$, expanding the right side gives

$$\lambda G' - G \le \theta\left[\frac{\lambda^2\tilde{v}}{2} - \lambda\left(\frac{\lambda\tilde{v}}{2n} - (1-E)\right)G\right] = \lambda\theta\left[\frac{\lambda\tilde{v}}{2} + \left((1-E) - \frac{\lambda\tilde{v}}{2n}\right)G\right].$$

As G is convex with $G(0) = G'(0) = 0$, $G \ge 0$; replacing $(1-E) - \lambda\tilde{v}/(2n)$ by $(1-E)$,

$$\lambda G' - G \le \lambda\theta(\lambda(1-E))\left((1-E)G + \frac{\lambda\tilde{v}}{2}\right), \qquad 0 \le \lambda \le \frac{2n(1-E)}{\tilde{v}}.$$

Integration. One verifies $\theta(x) = \frac{d}{dx}\log\frac{\phi(x)+x/2}{x}$. Moreover $G_0(\lambda) := \frac{\tilde{v}}{(1-E)^2}\phi((1-E)\lambda)$ satisfies the ODE with equality: with $u = (1-E)\lambda$,

$$\lambda G_0' - G_0 = \frac{\tilde{v}}{(1-E)^2}[u\phi'(u) - \phi(u)], \quad \lambda\theta(u)\left((1-E)G_0 + \frac{\lambda\tilde{v}}{2}\right) = \frac{\tilde{v}}{(1-E)^2}u\theta(u)\left(\phi(u) + \frac{u}{2}\right),$$

and $u\phi'(u) - \phi(u) = ue^u - e^u + 1 = u\theta(u)(\phi(u) + u/2)$ from the definition of θ. For $D = G_0 - G$,

$$\lambda D' - D \ge \lambda(1-E)\theta((1-E)\lambda)D,$$

i.e., $\frac{d}{d\lambda}(D/\lambda) \ge (1-E)\theta((1-E)\lambda)(D/\lambda)$. Multiplying by the integrating factor $\frac{(1-E)\lambda}{\phi((1-E)\lambda)+(1-E)\lambda/2}$ (whose log-derivative is $-(1-E)\theta((1-E)\lambda)$), the product is nondecreasing, vanishes at $\lambda = 0$ (since $D(0) = D'(0) = 0$), hence is ≥ 0; thus $D \ge 0$. So $G \le G_0$ on $[0, 2n(1-E)/\tilde{v}]$, and with the large-λ case, for all $\lambda \ge 0$:

$$\log\mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \frac{\tilde{v}}{(1-E)^2}\phi((1-E)\lambda). \qquad \blacksquare$$

Exercise 12.10

Central idea. Apply Theorem 6.6 with the leave-one-out suprema Z_i and write $e^{\lambda Z}\phi(-\lambda(Z-Z_i)) = e^{\lambda Z_i}\psi(\lambda w_i)$ with $\psi(u) = e^u\phi(-u)$ and $w_i = Z - Z_i$. Since $\psi'(r) = re^r \le \lambda e^r$ for $r \le \lambda$, and $X_{i,\hat{s}_i} \le w_i \le 1$, we can trade $\psi(\lambda w_i)$ for $\lambda(e^{\lambda w_i}-1)$ plus a term in $x_i = X_{i,\hat{s}_i}$ alone. The $\lambda(e^{\lambda w_i}-1)$ part is controlled by Exercise 12.6. The x_i part is controlled, after conditioning on $X^{(i)}$ and using centering, by the pointwise bound $\psi(u) - u_+\phi(u) \le u^2/2$. This gives $\lambda G' - (1+\lambda)G \le v\lambda^2/2$ for $G(\lambda) = \log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)}$, which integrates to $G(\lambda) \le v\lambda(e^\lambda - 1)/2$.

Solution. (This replaces an earlier version that asserted the differential inequality without derivation and used the false claim $Z_i \le Z$. Here the $X_{i,s}$ are centered, so $Z - Z_i$ can be negative: with $|\mathcal{T}| = 1$, $Z - Z_i = X_i$.)

Notation. As in Theorem 12.5, $X_1, \dots, X_n$ are i.i.d., $\mathbb{E}X_{i,s} = 0$, $X_{i,s} \le 1$, $\sigma^2 = \sup_s \sum_{i=1}^n \mathbb{E}X_{i,s}^2$ and $v = 2\mathbb{E}Z + \sigma^2$. Take $\mathcal{T}$ finite. (For countable $\mathcal{T}$, apply the finite case to $\mathcal{T}_k \uparrow \mathcal{T}$. Then $Z^{(k)} \uparrow Z$, $\mathbb{E}Z^{(k)} \to \mathbb{E}Z$ and $\mathbb{E}e^{\lambda Z^{(k)}} \to \mathbb{E}e^{\lambda Z}$ by monotone convergence, while $v^{(k)} \le v$ since $\mathbb{E}Z^{(k)} \le \mathbb{E}Z$ and the wimpy variance can only grow with $\mathcal{T}$.) Let $\hat{s}_i$ attain $Z_i = \sup_s \sum_{j\neq i} X_{j,s}$, chosen as a function of $X^{(i)} = (X_j)_{j\neq i}$, and let $\hat{s}$ attain Z. Put

$$x_i = X_{i,\hat{s}_i}, \qquad w_i = Z - Z_i.$$

From $Z \ge \sum_j X_{j,\hat{s}_i} = Z_i + x_i$ and $Z_i \ge \sum_{j\neq i} X_{j,\hat{s}} = Z - X_{i,\hat{s}}$,

$$x_i \le w_i \le X_{i,\hat{s}} \le 1.$$

Write $\mathbb{E}^{(i)}$ for integration over X_i with $X^{(i)}$ fixed. Since $\hat{s}_i$ depends only on $X^{(i)}$ and X_i is independent of $X^{(i)}$,

$$\mathbb{E}^{(i)}x_i = 0, \qquad \mathbb{E}^{(i)}x_i^2 \le \sigma_1^2 := \sup_s \mathbb{E}X_{1,s}^2, \qquad n\sigma_1^2 = \sigma^2,$$

where the last identity uses that the X_i are identically distributed. Since $Z \le n$ and $Z \ge \sum_j X_{j,s_0}$, every exponential moment for $\lambda \ge 0$ is finite. Let $\phi(u) = e^u - u - 1$ and $\psi(u) = e^u\phi(-u) = 1 + (u-1)e^u$, so $\psi'(u) = ue^u$. Fix $\lambda > 0$.

Step 1: entropy bound (the hint). Theorem 6.6 gives

$$\mathrm{Ent}(e^{\lambda Z}) \le \sum_{i=1}^n \mathbb{E}[e^{\lambda Z}\phi(-\lambda w_i)] = \sum_{i=1}^n \mathbb{E}[e^{\lambda Z_i}\psi(\lambda w_i)].$$

Since $\lambda x_i \le \lambda w_i \le \lambda$ and $\psi'(r) = re^r \le \lambda e^r$ for $r \le \lambda$,

$$\psi(\lambda w_i) - \psi(\lambda x_i) = \int_{\lambda x_i}^{\lambda w_i} re^r\,dr \le \lambda(e^{\lambda w_i} - e^{\lambda x_i}),$$

that is,

$$\psi(\lambda w_i) \le \lambda(e^{\lambda w_i} - 1) + \psi(\lambda x_i) - \lambda(e^{\lambda x_i} - 1).$$

Multiply by $e^{\lambda Z_i} \ge 0$ and note that $e^{\lambda Z_i}(e^{\lambda w_i} - 1) = e^{\lambda Z} - e^{\lambda Z_i}$. By Exercise 12.6, $\lambda\sum_i \mathbb{E}[e^{\lambda Z} - e^{\lambda Z_i}] \le \lambda\,\mathbb{E}e^{\lambda Z}\log \mathbb{E}e^{\lambda Z}$. Since Z_i is $X^{(i)}$-measurable,

$$\mathrm{Ent}(e^{\lambda Z}) \le \lambda\,\mathbb{E}e^{\lambda Z}\log \mathbb{E}e^{\lambda Z} + \sum_{i=1}^n \mathbb{E}\Big[e^{\lambda Z_i}\,\mathbb{E}^{(i)}[\psi(\lambda x_i) - \lambda(e^{\lambda x_i} - 1)]\Big].$$

By centering, $\mathbb{E}^{(i)}[\lambda(e^{\lambda x_i} - 1)] = \lambda\,\mathbb{E}^{(i)}[\phi(\lambda x_i) + \lambda x_i] = \lambda\,\mathbb{E}^{(i)}\phi(\lambda x_i)$. Since $\phi \ge 0$ and $(\lambda x_i)_+ \le \lambda$, this is at least $\mathbb{E}^{(i)}[\phi(\lambda x_i)(\lambda x_i)_+]$. This gives the inequality displayed in the hint:

$$\mathrm{Ent}(e^{\lambda Z}) \le \lambda\,\mathbb{E}e^{\lambda Z}\log \mathbb{E}e^{\lambda Z} + \sum_{i=1}^n \mathbb{E}\Big[e^{\lambda Z_i}\,\mathbb{E}^{(i)}[\psi(\lambda x_i) - \phi(\lambda x_i)(\lambda x_i)_+]\Big]. \tag{1}$$

Step 2: the elementary inequality $\psi(u) - u_+\phi(u) \le u^2/2$ for all $u \in \mathbb{R}$. For $u \le 0$, set $g(u) = u^2/2 - \psi(u)$. Then $g(0) = 0 - \psi(0) = 0$ and $g'(u) = u - ue^u = u(1 - e^u) \le 0$ on $(-\infty, 0]$, so g is nonincreasing there and $g(u) \ge g(0) = 0$. For $u \ge 0$,

$$\psi(u) - u\phi(u) = 1 + (u-1)e^u - ue^u + u^2 + u = u^2 - \phi(u) \le \frac{u^2}{2},$$

because $\phi(u) = \sum_{k\ge 2} u^k/k! \ge u^2/2$. Applying this with $u = \lambda x_i$,

$$\mathbb{E}^{(i)}\big[\psi(\lambda x_i) - \phi(\lambda x_i)(\lambda x_i)_+\big] \le \frac{\lambda^2}{2}\,\mathbb{E}^{(i)} x_i^2 \le \frac{\lambda^2 \sigma_1^2}{2}.$$

Moreover $\mathbb{E}e^{\lambda Z_i} \le \mathbb{E}e^{\lambda Z}$. Indeed, $Z \ge Z_i + x_i$ and Jensen's inequality give $\mathbb{E}^{(i)} e^{\lambda Z} \ge e^{\lambda Z_i}\,\mathbb{E}^{(i)} e^{\lambda x_i} \ge e^{\lambda Z_i} e^{\lambda \mathbb{E}^{(i)} x_i} = e^{\lambda Z_i}$. Plugging into (1) and using $n\sigma_1^2 = \sigma^2$,

$$\operatorname{Ent}(e^{\lambda Z}) \le \lambda\,\mathbb{E}e^{\lambda Z} \log \mathbb{E}e^{\lambda Z} + \frac{\lambda^2 \sigma^2}{2}\,\mathbb{E}e^{\lambda Z}. \tag{2}$$

Step 3: the differential inequality. Let $L(\lambda) = \log \mathbb{E}e^{\lambda Z}$ and $G(\lambda) = L(\lambda) - \lambda \mathbb{E}Z = \log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)}$. These are differentiable on $(0, \infty)$ because Z is bounded above and integrable. Then $\operatorname{Ent}(e^{\lambda Z})/\mathbb{E}e^{\lambda Z} = \lambda L' - L = \lambda G' - G$. Dividing (2) by $\mathbb{E}e^{\lambda Z}$ and substituting $L = G + \lambda \mathbb{E}Z$,

$$\lambda G' - G \le \lambda G + \lambda^2 \mathbb{E}Z + \frac{\lambda^2 \sigma^2}{2}, \qquad \text{i.e.} \qquad \lambda G' - (1+\lambda) G \le \frac{\lambda^2(2\mathbb{E}Z + \sigma^2)}{2} = \frac{v\lambda^2}{2}.$$

Let $H(\lambda) = G(\lambda)/\lambda$. Dividing by λ^2 gives $H' - H \le v/2$, i.e.

$$\big(e^{-\lambda} H(\lambda)\big)' \le \frac{v}{2} e^{-\lambda}.$$

Also $H(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$. To see this, let $Y = Z - \mathbb{E}Z$, so $Y \le n - \mathbb{E}Z =: b$ and $\mathbb{E}|Y| < \infty$. For $0 < \varepsilon \le 1$, $|e^{\varepsilon Y} - 1|/\varepsilon \le |Y| e^{\varepsilon Y_+} \le |Y| e^{|b|}$. By dominated convergence, $(\mathbb{E}e^{\varepsilon Y} - 1)/\varepsilon \to \mathbb{E}Y = 0$, hence $G(\varepsilon)/\varepsilon \to 0$. Integrating over $[\varepsilon, \lambda]$ and letting $\varepsilon \downarrow 0$,

$$e^{-\lambda} H(\lambda) \le \frac{v}{2}(1 - e^{-\lambda}), \qquad \text{so} \qquad G(\lambda) = \lambda H(\lambda) \le \frac{v}{2}\lambda(e^\lambda - 1).$$

Hence, for all $\lambda > 0$,

$$\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \le \frac{v\lambda(e^\lambda - 1)}{2}.$$

Why it is sub-optimal. As $\lambda \to 0$ the bound is $v\lambda^2/2 + O(\lambda^3)$, matching Bousquet's $v\phi(\lambda)$ (Theorem 12.5) to leading order. For large λ it is of order λe^λ rather than e^λ. (Numerical sanity checks: $\psi(u) - u_+\phi(u) - u^2/2 \le 0$ on a grid over $[-30, 30]$ with maximum 0 at $u = 0$; on an exactly enumerated example with $|\mathcal{T}| = 2$ and $n = 4$, the final bound holds for $\lambda \in \{0.1, 0.5, 1, 2, 4\}$.) ■

Exercise 12.11

Central idea. The operator norm of a sum of PSD matrices is monotone in each summand, so the leave-one-out differences satisfy $0 \le Z - Z^{(i)} \le \|X_i\| \le a$ and, via the maximizing unit vector, $\sum_i (Z - Z^{(i)}) \le Z$; thus Z/a is self-bounding and Theorem 6.12 gives both claims after rescaling.

Solution. Let $S = \sum_{i=1}^n X_i$ and $Z = \|S\|$. For each i let $S^{(i)} = \sum_{j \ne i} X_j$ and $Z^{(i)} = \|S^{(i)}\|$. Since $X_i \succeq 0$, $S^{(i)} \preceq S$, and the operator norm is monotone on the PSD order, $Z^{(i)} \le Z$. By the triangle inequality,

$$0 \le Z - Z^{(i)} = \|S\| - \|S^{(i)}\| \le \|S - S^{(i)}\| = \|X_i\| \le a \quad \text{a.s.}$$

Let v be a unit vector with $Z = v^T S v$. Then $Z^{(i)} \ge v^T S^{(i)} v$, so

$$Z - Z^{(i)} \le v^T S v - v^T S^{(i)} v = v^T X_i v,$$

and summing over i,

$$\sum_{i=1}^{n}(Z - Z^{(i)}) \le v^T\Big(\sum_{i=1}^{n} X_i\Big)v = v^T S v = Z.$$

Hence $W := Z/a$ satisfies $0 \le W - W^{(i)} \le 1$ and $\sum_i (W - W^{(i)}) \le W$, i.e. W is self-bounding. By Theorem 6.12,

$$\mathrm{Var}(W) \le \mathbb{E}W, \qquad \log \mathbb{E}e^{\lambda(W-\mathbb{E}W)} \le (\mathbb{E}W)\,\phi(\lambda) \quad \forall\, \lambda \in \mathbb{R},$$

where $\phi(\lambda) = e^\lambda - \lambda - 1$. Since $\mathrm{Var}(W) = a^{-2}\mathrm{Var}(Z)$ and $\mathbb{E}W = a^{-1}\mathbb{E}Z$, the first inequality gives $\mathrm{Var}(Z) \le a\,\mathbb{E}Z$. For the second, $\lambda(W - \mathbb{E}W) = (\lambda/a)(Z - \mathbb{E}Z)$; replacing λ by $a\lambda$ yields

$$\log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le (\mathbb{E}Z/a)\,\phi(a\lambda) \qquad \forall\, \lambda \in \mathbb{R},$$

as claimed.

Exercise 12.12

Central idea. The nonzero eigenvalues of the Gram matrix G coincide with those of $S = \sum_i X_i X_i^T$. By the Ky Fan variational principle, $Z = \sum_{j=1}^{k} \mu_j(S) = \max_P \mathrm{tr}(PS)$ over rank-k orthogonal projectors. With P^* the maximizer, $0 \le Z - Z^{(i)} \le \|P^* X_i\|^2 \le M^2$ and $\sum_i (Z - Z^{(i)}) \le \mathrm{tr}(P^* S) = Z$. So Z/M^2 is self-bounding, and Exercise 12.11 (Theorem 6.12) gives the mgf bound $(\mathbb{E}Z/M^2)\phi(M^2\lambda)$. Its Chernoff transform is the Bennett tail $\exp(-\frac{\mathbb{E}Z}{M^2} h(\frac{t}{\mathbb{E}Z}))$.

Solution. Write $G = XX^T$ where X is the $n \times d$ matrix with rows X_i^T. Then G is PSD with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$. Its nonzero eigenvalues coincide with the nonzero eigenvalues of the $d \times d$ matrix $S := X^T X = \sum_{i=1}^{n} X_i X_i^T$ (a standard fact: XX^T and $X^T X$ have the same nonzero eigenvalues). Let $\mu_1 \ge \cdots \ge \mu_d \ge 0$ be the eigenvalues of S. At most d eigenvalues of G are nonzero and zeros sort last, so $\lambda_j = \mu_j$ for $j \le d$. As $k \le d$,

$$Z = \sum_{j=1}^{k} \lambda_j = \sum_{j=1}^{k} \mu_j(S).$$

By the Courant–Fisher (Ky Fan) variational characterization,

$$Z = \max\{\mathrm{tr}(PS) : P \text{ an orthogonal projector of rank } k\} = \max_P \sum_{i=1}^{n} \mathrm{tr}(PX_iX_i^T).$$

The maximum is attained because the set of rank-k projectors is compact.

Let $M := \operatorname{ess\,sup}\|X_1\| < \infty$ (the X_i are i.i.d. and a.s. bounded). If $M = 0$ then $Z \equiv 0$ and there is nothing to prove, so assume $M > 0$. For each i, let $S^{(i)} = S - X_i X_i^T$ and $Z^{(i)} = \sum_{j=1}^{k} \mu_j(S^{(i)}) = \max_P \mathrm{tr}(PS^{(i)})$, a function of $(X_j)_{j \ne i}$. Since $\mathrm{tr}(PX_iX_i^T) = \|PX_i\|^2 \ge 0$ for every P, we have $\mathrm{tr}(PS) \ge \mathrm{tr}(PS^{(i)})$ for each P, hence $Z \ge Z^{(i)}$. Let P^* attain the maximum for S. Then

$$Z - Z^{(i)} = \mathrm{tr}(P^* S) - \max_P \mathrm{tr}(PS^{(i)}) \le \mathrm{tr}(P^* S) - \mathrm{tr}(P^* S^{(i)}) = \|P^* X_i\|^2 \le \|X_i\|^2 \le M^2 \quad \text{a.s.}$$

Summing over i with the same P^*,

$$\sum_{i=1}^{n}(Z - Z^{(i)}) \le \sum_{i=1}^{n} \mathrm{tr}(P^* X_i X_i^T) = \mathrm{tr}(P^* S) = Z.$$

Thus $W := Z/M^2$ satisfies $0 \le W - W^{(i)} \le 1$ and $\sum_i (W - W^{(i)}) \le W$, so W is self-bounding. Applying Exercise 12.11 (i.e. Theorem 6.12) to W and rescaling as there gives

$$\mathrm{Var}(Z) \le M^2\,\mathbb{E}Z, \qquad \log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le \frac{\mathbb{E}Z}{M^2}\,\phi(M^2\lambda) \quad \forall\, \lambda \in \mathbb{R},$$

with $\phi(u) = e^u - u - 1$.

Bennett-type tails. Write $a = M^2$, $m = \mathbb{E}Z$ (assume $m > 0$; if $m = 0$ then $Z = 0$ a.s. since $Z \geq 0$), and $h(x) = (1+x)\log(1+x) - x$ for $x \geq -1$. For $t > 0$ and $\lambda > 0$, Markov's inequality gives $\mathbb{P}\{Z \geq m + t\} \leq \exp(-\lambda t + (m/a)\phi(a\lambda))$. Substituting $u = a\lambda$, so that $\lambda t = (m/a)\, u\, (t/m)$,

$$\sup_{\lambda > 0} \Big[\lambda t - \frac{m}{a}\phi(a\lambda)\Big] = \frac{m}{a} \sup_{u > 0} \Big[u\frac{t}{m} - \phi(u)\Big] = \frac{m}{a}\, h\Big(\frac{t}{m}\Big),$$

where the last step is the standard identity $\sup_{u>0}[ux - \phi(u)] = h(x)$ for $x > 0$, attained at $u = \log(1+x)$. Hence

$$\mathbb{P}\{Z \geq \mathbb{E}Z + t\} \leq \exp\Big(-\frac{\mathbb{E}Z}{M^2}\, h\Big(\frac{t}{\mathbb{E}Z}\Big)\Big), \qquad t > 0.$$

Using $h(x) \geq \frac{x^2}{2(1+x/3)}$ for $x \geq 0$ gives the Bernstein form

$$\mathbb{P}\{Z \geq \mathbb{E}Z + t\} \leq \exp\left(-\frac{t^2}{2M^2(\mathbb{E}Z + t/3)}\right).$$

For the lower tail, apply the mgf bound with $\lambda < 0$. Since $\phi(u) \leq u^2/2$ for $u \leq 0$, for $0 < t \leq \mathbb{E}Z$ we get

$$\mathbb{P}\{Z \leq \mathbb{E}Z - t\} \leq \exp\Big(-\frac{\mathbb{E}Z}{M^2} h\Big(-\frac{t}{\mathbb{E}Z}\Big)\Big) \leq \exp\left(-\frac{t^2}{2M^2\,\mathbb{E}Z}\right).$$

(Here $h(-x) \geq x^2/2$ for $x \in [0,1]$, and the lower tail is trivially 0 for $t > \mathbb{E}Z$ since $Z \geq 0$.) These are Bennett-like concentration inequalities for the sum of the top k eigenvalues.

Correction note. An earlier version of this solution displayed the upper tail as $\exp(-\frac{\mathbb{E}Z}{M^2} h(\frac{M^2 t}{\mathbb{E}Z}))$, with an extra factor M^2 inside h. That does not follow from the mgf bound and is false. Take $d = k = 1$ and $X_i = 2\,\mathrm{Bern}(0.1)$ with $n = 100$, so $Z = \sum_i X_i^2 = 4\,\mathrm{Bin}(100, 0.1)$, $M^2 = 4$ and $\mathbb{E}Z = 40$. At $t = 20$ the true probability is $\mathbb{P}\{Z \geq 60\} = \mathbb{P}\{\mathrm{Bin}(100, 0.1) \geq 15\} \approx 0.0726$. The wrong display gives $e^{-10h(2)} \approx 2.36 \cdot 10^{-6}$, which is below the true probability, so it is not a valid upper bound. The correct bound gives $e^{-10h(1/2)} \approx 0.339$, and the Bernstein form gives ≈ 0.343. (Numbers checked in Python.)

Exercise 12.13

Central idea. A quantile coupling lets us replace the X_i by independent majorants Y_i with exactly the sub-gamma tail, so $Y_i \overset{d}{=} g(T_i)$ with $g(s) = \mu + \sqrt{2vs} + cs$ and T_i i.i.d. Exp(1). Then $\max_i Y_i = g(T_{(n)})$, and Rényi's representation $T_{(n)} \overset{d}{=} \sum_{j=1}^n E_j/j$ gives exact moments: the mean bound follows by Jensen. The printed second-moment bound is false because it involves M^2, which sees the lower tail that upper-tail hypotheses do not control. It holds for M_+ (with $\mu \geq 0$), or for $(M-\mu)_+$. For the mgf, linearizing the concave g at H_n and using Rényi gives the printed shape with an extra factor $\sum_j 1/j^2 \leq \pi^2/6$ in the quadratic term. Without that factor, the printed mgf bound fails under the tail-form reading of the hypothesis (which is what the exercise's parenthetical and hint use). Under the literal mgf sub-gamma hypothesis we leave it unresolved.

Solution. *Reduction.* As the hint allows (it also follows from a quantile coupling, since $\mathbb{P}\{X_i \geq x\} \leq \mathbb{P}\{Y_i \geq x\}$ for every x), take independent $Y_1, \ldots, Y_n$ with $X_i \leq Y_i$ a.s. and $\mathbb{P}\{Y_i \geq g(t)\} = e^{-t}$ for $t \geq 0$, where $g(s) = \mu + \sqrt{2vs} + cs$. Since g is continuous and strictly increasing on $[0, \infty)$ (if $v = c = 0$ then $Y_i \equiv \mu$ and everything below is trivial), $Y_i \overset{d}{=} g(T_i)$ with T_i i.i.d. Exp(1). Also, g is concave. Put $\tilde{M} = \max_i Y_i$. Then $M \leq \tilde{M}$ a.s. and, g being increasing,

$$\tilde{M} = g(T_{(n)}) = \mu + \sqrt{2vT_{(n)}} + cT_{(n)}, \qquad T_{(n)} = \max_{1 \leq i \leq n} T_i.$$

By Rényi's representation of exponential order statistics (de Haan–Ferreira, Ch. 2), $T_{(n)} \overset{d}{=} \sum_{j=1}^n E_j/j$ with E_j i.i.d. Exp(1). Hence $\mathbb{E}T_{(n)} = H_n := \sum_{j=1}^n 1/j$ and $\mathrm{Var}(T_{(n)}) = \zeta_n(2) := \sum_{j=1}^n 1/j^2 < \pi^2/6$.

Mean (true as stated). By Jensen's inequality and the concavity of g, $\mathbb{E}M \le \mathbb{E}g(T_{(n)}) \le g(H_n) = \mu + \sqrt{2vH_n} + cH_n$. This is the Theorem 2.6 bound.

Second moment: false as stated. Take $n = 2$, $X_1 = X_2 \equiv -100$, $\mu = 0$, $v = c = 1$. Then $\mathbb{E}X_i = -100 \le \mu$ and $\mathbb{P}\{X_i \ge \sqrt{2t} + t\} = 0 \le e^{-t}$ for all $t \ge 0$. Constants are also sub-gamma in the mgf sense, so the hypotheses hold under every reading. But $M \equiv -100$ gives $\mathbb{E}M^2 = 10000$, whereas the claimed bound is $(\sqrt{3} + 3/2)^2 + 10/\log 2 + \pi^2/3 \approx 28.16$. The reason: $\mathbb{E}M^2$ depends on the lower tail of M, which upper-tail hypotheses do not control.

Corrected second-moment bounds ($n \ge 2$). Let $W := \sqrt{2vT_{(n)}} + cT_{(n)} \ge 0$, so $\tilde{M} = \mu + W$. First, $\mathbb{E}W \le \sqrt{2vH_n} + cH_n$ by Jensen. Second, $\mathrm{Var}(W) \le 2\mathrm{Var}(\sqrt{2vT_{(n)}}) + 2c^2\mathrm{Var}(T_{(n)})$, where

$$\mathrm{Var}(\sqrt{T_{(n)}}) \le \mathbb{E}[(\sqrt{T_{(n)}} - \sqrt{H_n})^2] = \mathbb{E}\frac{(T_{(n)} - H_n)^2}{(\sqrt{T_{(n)}} + \sqrt{H_n})^2} \le \frac{\mathrm{Var}(T_{(n)})}{H_n} \le \frac{\pi^2}{6H_n}.$$

So $2\mathrm{Var}(\sqrt{2vT_{(n)}}) = 4v\,\mathrm{Var}(\sqrt{T_{(n)}}) \le \frac{2\pi^2 v}{3H_n} \le \frac{10v}{\log n}$, using $H_n \ge \log n$ and $2\pi^2/3 < 10$. Also $2c^2\mathrm{Var}(T_{(n)}) \le c^2\pi^2/3$. Hence

$$\mathrm{Var}(W) \le \frac{10v}{\log n} + \frac{c^2\pi^2}{3}. \tag{V}$$

Two corrected statements follow.

(a) Since $(M - \mu)_+ \le (\tilde{M} - \mu)_+ = W$,

$$\mathbb{E}[(M - \mu)_+^2] \le \mathbb{E}W^2 = (\mathbb{E}W)^2 + \mathrm{Var}(W) \le (\sqrt{2vH_n} + cH_n)^2 + \frac{10v}{\log n} + \frac{c^2\pi^2}{3}.$$

(b) (The form used by the independent audit.) If $\mu \ge 0$, then $\tilde{M} = \mu + W \ge 0$ and $M_+ := \max(M, 0) \le \tilde{M}$, so

$$\mathbb{E}M_+^2 \le \mathbb{E}\tilde{M}^2 = (\mu + \mathbb{E}W)^2 + \mathrm{Var}(W) \le \left(\mu + \sqrt{2vH_n} + cH_n\right)^2 + \frac{10v}{\log n} + \frac{c^2\pi^2}{3}.$$

This is the printed bound with M^2 replaced by M_+^2. If $\mu < 0$, the hypotheses still hold with μ replaced by 0, so (b) holds with μ_+ in place of μ. The two forms are consistent: (b) is (a) shifted by the constant $\mu \ge 0$, because $\tilde{M} - \mu = W$ in both. In particular, if $X_i \ge 0$ a.s. and $\mu \ge 0$ (e.g. gamma X_i), then $M = M_+$ and the printed $\mathbb{E}M^2$ bound holds.

Corrected mgf bound. Let $B_1 := c + \sqrt{v/(2H_n)}$ and $a := g(H_n) = \mu + \sqrt{2vH_n} + cH_n$. By concavity of $\sqrt{\cdot}$ at H_n, $\sqrt{2vT_{(n)}} \le \sqrt{2vH_n} + \sqrt{v/(2H_n)}\,(T_{(n)} - H_n)$. Hence $\tilde{M} \le a + B_1(T_{(n)} - H_n)$. For $\lambda > 0$, $\mathbb{E}e^{\lambda M} \le \mathbb{E}e^{\lambda\tilde{M}}$. By Rényi, for $0 < \lambda < 1/B_1$,

$$\mathbb{E}e^{\lambda B_1(T_{(n)} - H_n)} = \prod_{j=1}^{n} \frac{e^{-\lambda B_1/j}}{1 - \lambda B_1/j}.$$

For $x \in [0, 1)$, $-x - \log(1 - x) = \sum_{k\ge2} x^k/k \le \frac{x^2}{2}\sum_{k\ge0} x^k = \frac{x^2}{2(1-x)}$. Applying this with $x = \lambda B_1/j \le \lambda B_1$,

$$\log \mathbb{E}e^{\lambda B_1(T_{(n)} - H_n)} = \sum_{j=1}^{n}\Big[-\tfrac{\lambda B_1}{j} - \log\Big(1 - \tfrac{\lambda B_1}{j}\Big)\Big] \le \frac{\lambda^2 B_1^2}{2(1 - \lambda B_1)}\,\zeta_n(2).$$

Therefore, for $0 < \lambda < 1/B_1$,

$$\log \mathbb{E}e^{\lambda M} \le \lambda(\mu + \sqrt{2vH_n} + cH_n) + \zeta_n(2)\,\frac{\lambda^2 B_1^2}{2(1 - \lambda B_1)} \le \lambda(\mu + \sqrt{2vH_n} + cH_n) + \frac{\pi^2}{6} \cdot \frac{\lambda^2 B_1^2}{2(1 - \lambda B_1)}.$$

Comparison with the printed shape. Let $B := c + \sqrt{v/H_n} \ge B_1$. On $\lambda B < 1$, the map $b \mapsto \frac{\lambda^2 b^2}{2(1-\lambda b)}$ is increasing in $b \in [0, B]$. Also $\sqrt{2vH_n} \le 2\sqrt{vH_n}$ and $\lambda > 0$. So, for $0 < \lambda < 1/B$,

$$\log \mathbb{E}e^{\lambda M} \le \lambda(\mu + 2\sqrt{vH_n} + cH_n) + \frac{\pi^2}{6} \cdot \frac{\lambda^2 (c + \sqrt{v/H_n})^2}{2(1 - \lambda(c + \sqrt{v/H_n}))}.$$

This is the printed bound with the extra factor $\pi^2/6$ (more precisely $\zeta_n(2)$) in the quadratic term.

Is the factor $\pi^2/6$ needed? Under the tail-form hypothesis, yes. The parenthetical "thus, for $t \geq 0$, $\mathbb{P}\{X_i \geq \mu+\sqrt{2vt}+ct\} \leq e^{-t}$" and the hint use only the tail form. Under that reading, the printed mgf bound (without the factor) is false, i.e. it cannot be derived from the tail assumptions. Take $v = 0$, $c = 1$, $\mu = 0$, $n = 2$, so the claim reads $\log \mathbb{E}e^{\lambda M} \leq \frac{3}{2}\lambda + \frac{\lambda^2}{2(1-\lambda)}$. Let $Y_i \sim \mathrm{Exp}(1)$ be independent and $X_i^{(K)} = Y_i \mathbf{1}_{\{Y_i \leq K\}} - C_K \mathbf{1}_{\{Y_i > K\}}$, where $C_K = e^K - K - 1 > 0$ makes $\mathbb{E}X_i^{(K)} = 0$. For $t \geq 0$, $\mathbb{P}\{X_i^{(K)} \geq t\} \leq \mathbb{P}\{Y_i \geq t\} = e^{-t}$, so the tail-form hypotheses hold. Also $M^{(K)} = \max_i X_i^{(K)} \to T_{(2)} := \max(Y_1, Y_2)$ a.s., and $e^{\lambda M^{(K)}} \leq e^{\lambda T_{(2)}} \in L^1$ for $0 < \lambda < 1$. By dominated convergence, $\log \mathbb{E}e^{\lambda M^{(K)}} \to \log \mathbb{E}e^{\lambda T_{(2)}} = -\log(1-\lambda) - \log(1-\lambda/2)$. At $\lambda = 0.01$ this limit is $0.0150629 > 0.0150505 = \frac{3}{2}(0.01) + \frac{0.01^2}{2(0.99)}$, so the printed bound is violated for large K. By continuity it is also violated for small $v > 0$: with $v = 10^{-8}$ the printed right-hand side is 0.0150530. The limit does satisfy the corrected bound, $0.0150629 \leq 0.0150831$. (All numbers checked in Python.)

Scope (unresolved case). The $X_i^{(K)}$ have variance of order e^K, so they are not sub-gamma in the mgf sense, $\log \mathbb{E}e^{\lambda(X_i - \mathbb{E}X_i)} \leq \frac{\lambda^2 v}{2(1-c\lambda)}$, with $v = 0$ or small v. Under that literal mgf hypothesis, the counterexample is not admissible. Whether the printed bound without the factor $\pi^2/6$ then holds is **not resolved here**. Our corrected bound holds under both readings, because mgf sub-gamma with $\mathbb{E}X_i \leq \mu$ implies the tail form (Section 2.4). (An earlier version of this solution called the printed mgf bound "false as stated" and argued that $B^2 \to c^2$ contradicts the Gumbel variance $\pi^2 c^2/6$. That argument is invalid. The printed linear term exceeds $\mathbb{E}M$ by a margin of order $\lambda(\mu + \sqrt{vH_n})$, so for small λ no comparison of λ^2-coefficients follows. For instance, for i.i.d. Exp(1) variables, where $\mu = v = c = 1$, the printed bound holds numerically for $n \in \{2, 3, 5, 10, 10^2, 10^3, 10^5\}$ on a grid of $\lambda \in (0, 1/B)$.)

Tightness (gamma case). If $X_i \sim \Gamma(\alpha, \beta)$ i.i.d. (shape α, rate β), then $-\alpha \log(1-\lambda/\beta) - \alpha\lambda/\beta \leq \frac{(\alpha/\beta^2)\lambda^2}{2(1-\lambda/\beta)}$. So the X_i are sub-gamma in the mgf sense, and hence in the tail sense, with $v = \alpha/\beta^2$, $c = 1/\beta$, $\mu = \alpha/\beta \geq 0$, and $X_i \geq 0$. Classical extreme-value theory gives $\beta \max_i X_i - \log n - (\alpha - 1)\log\log n + \log\Gamma(\alpha) \Rightarrow$ Gumbel, with convergence of the variance, so $\mathrm{Var}(M) \to \pi^2/(6\beta^2) = \pi^2 c^2/6$. The excess $\frac{10v}{\log n} + \frac{c^2\pi^2}{3}$ in the corrected second-moment bound tends to $\frac{\pi^2}{3}c^2$, within a factor 2 of this. For the mgf, consider the exponential case $\alpha = 1$ and $\beta = 1/c$. Then $M \overset{d}{=} cT_{(n)}$ and $\log \mathbb{E}e^{\lambda(M - \mathbb{E}M)} = \sum_{j=1}^n [-\lambda c/j - \log(1 - \lambda c/j)] \geq \frac{\lambda^2 c^2}{2}\zeta_n(2)$. So the factor $\zeta_n(2)$ in the quadratic term cannot be improved for the centered maximum. Also $B_1 = c + c/\sqrt{2H_n} \to c$ as $n \to \infty$. Thus the right tail of the maximum is no heavier than a Gumbel tail with scale c, up to the constants above.

Exercise 12.14

Central idea. The quantile coupling $|X_i| \leq \sqrt{2vY_i} + cY_i$ with Y_i i.i.d. Exp(1) turns the truncation $\{|X_i| \geq \tau_n\}$ into $\{Y_i \geq \log n\}$ (by monotonicity of $s \mapsto \sqrt{2vs} + cs$); by memorylessness, $Y_i = \log n + E_i$ on this event with $E_i \sim \mathrm{Exp}(1)$, reducing every computation to a single exponential variable. A convexity estimate $\sqrt{2vY} + cY \leq \tau_n(1 + (Y - \log n)/\log n)$ for $Y \geq \log n$ gives the mgf in closed form, and Chernoff with $\lambda = 1/(4\tau_n)$ yields the tail.

Erratum. Two defects, one in the statement and one in the hint. (i) *Statement:* the range "$0 \leq \lambda \leq \tau_n/2$" should read $0 \leq \lambda \leq 1/(2\tau_n)$. As printed, whenever $\tau_n > \sqrt{2}$ we have $\tau_n/2 > 1/\tau_n$, so the range contains values $\lambda \geq 1/\tau_n$ at which the denominator $1 - \lambda\tau_n$ of the bound vanishes or is negative, so the printed inequality is undefined or false there. For instance, with standard Gaussian X_i ($v = 1$, $c = 0$) and $n \geq 3$, $\tau_n = \sqrt{2\log n} > \sqrt{2}$ and Z is symmetric, so $\log \mathbb{E}e^{\lambda Z} \geq \lambda \mathbb{E}Z = 0$ by Jensen, whereas for $1/\tau_n < \lambda \leq \tau_n/2$ the printed right side $\frac{e^{\lambda\tau_n} + \lambda\tau_n - 1}{1 - \lambda\tau_n}$ is negative. (ii) *Hint:* the indicator must sit inside the exponential, $n \log \mathbb{E} \exp\left(\lambda(\sqrt{2vY_1} + cY_1)\mathbb{1}_{\{Y_1 \geq \log n\}}\right)$. As written, $n \log \mathbb{E}\left[e^{\lambda(\sqrt{2vY_1} + cY_1)}\mathbb{1}_{\{Y_1 \geq \log n\}}\right] \to n \log \mathbb{P}\{Y_1 \geq \log n\} = n \log(1/n) < 0$ as $\lambda \to 0$, so it cannot bound $\log \mathbb{E}e^{\lambda Z} \to 0$. We prove the mgf bound for $0 \leq \lambda < 1/\tau_n$ (indeed for $0 \leq \lambda < \log n/\tau_n$ in the sharper form), which contains the corrected range, and the mean and tail bounds as printed.

Solution. Put $L = \log n$ (assume $n \geq 3$ so $L > 1$) and $\tau_n = \sqrt{2vL} + cL$. By the hint's quantile coupling there are independent $Y_i \sim \mathrm{Exp}(1)$ with $|X_i| \leq \sqrt{2vY_i} + cY_i$ a.s. (this uses the two-sided sub-gamma tail for $|X_i|$). If $|X_i| \geq \tau_n$ then $\sqrt{2vY_i} + cY_i \geq \sqrt{2vL} + cL$, whence $Y_i \geq L$ by strict monotonicity. Thus with $V_i := X_i \mathbf{1}_{\{|X_i| \geq \tau_n\}}$ and $W_i := (\sqrt{2vY_i} + cY_i)\mathbf{1}_{\{Y_i \geq L\}} \geq 0$,

$$V_i \leq W_i \quad \text{a.s.}, \qquad Z = \sum_{i=1}^{n} V_i.$$

Mean. $\mathbb{E}Z \leq n\mathbb{E}W_1 = n\mathbb{E}[(\sqrt{2vY} + cY)\mathbf{1}_{\{Y \geq L\}}]$. By memorylessness, on $\{Y \geq L\}$, $Y \overset{d}{=} L + E$ with $E \sim \mathrm{Exp}(1)$ independent, and $\mathbb{P}\{Y \geq L\} = 1/n$, so

$$\mathbb{E}Z \leq \mathbb{E}[\sqrt{2vL + 2vE}] + cL + c.$$

Since $\sqrt{2vL + 2vE} - \sqrt{2vL} = \frac{2vE}{\sqrt{2vL+2vE}+\sqrt{2vL}} \leq \frac{2vE}{2\sqrt{2vL}}$, taking expectations gives $\mathbb{E}\sqrt{2vL + 2vE} \leq \sqrt{2vL} + \sqrt{v/(2L)}$. Hence $\mathbb{E}Z \leq \tau_n + c + \sqrt{v/(2\log n)}$.

Moment-generating function. For $\lambda \geq 0$, $e^{\lambda V_i} \leq e^{\lambda W_i}$ pointwise, so by independence $\mathbb{E}e^{\lambda Z} = \prod_i \mathbb{E}e^{\lambda V_i} \leq (\mathbb{E}e^{\lambda W_1})^n$, i.e.

$$\log \mathbb{E}e^{\lambda Z} \leq n \log \mathbb{E} \exp(\lambda(\sqrt{2vY} + cY)\mathbf{1}_{\{Y \geq L\}})$$

(the indicator inside the exponential, as corrected in the Erratum). For $Y \geq L$,

$$\sqrt{2vY} + cY \leq \tau_n\Big(1 + \frac{Y - L}{L}\Big) :$$

indeed $h(Y) := \tau_n(1 + (Y - L)/L) - (\sqrt{2vY} + cY)$ satisfies $h(L) = 0$, $h'(L) = \sqrt{v/(2L)} \geq 0$, $h''(Y) = \sqrt{2v}/(4Y^{3/2}) > 0$, so h is convex and nondecreasing on $[L, \infty)$. Hence, with $x := \lambda\tau_n$ and memorylessness,

$$\mathbb{E}[e^{\lambda(\sqrt{2vY}+cY)}\mathbf{1}_{\{Y \geq L\}}] \leq e^x\, \mathbb{E}[e^{x(Y-L)/L}\mathbf{1}_{\{Y \geq L\}}] = \frac{e^x}{n} \cdot \frac{1}{1 - x/L}, \qquad x < L.$$

Thus $\mathbb{E}e^{\lambda W_1} \leq 1 + \frac{1}{n}(\frac{e^x}{1-x/L} - 1)$ and, with $\log(1 + u) \leq u$,

$$\log \mathbb{E}e^{\lambda Z} \leq \frac{e^{\lambda\tau_n}}{1 - \lambda\tau_n/\log n} - 1, \qquad 0 \leq \lambda < \frac{\log n}{\tau_n}.$$

For $L \geq 1$ this implies the stated bound $\frac{e^{\lambda\tau_n} + \lambda\tau_n - 1}{1 - \lambda\tau_n}$, since $\frac{e^x}{1-x/L} - 1 \leq \frac{e^x}{1-x} - 1$. This covers the corrected range $0 \leq \lambda \leq 1/(2\tau_n)$ of the Erratum.

Tail. By Chernoff, for $0 \leq \lambda < (\log n)/\tau_n$,

$$\mathbb{P}\{Z \geq t\} \leq \exp\Big(-\lambda t + \frac{e^{\lambda\tau_n}}{1 - \lambda\tau_n/L} - 1\Big).$$

Take $\lambda = 1/(4\tau_n)$ (so $x = 1/4$): $\mathbb{P}\{Z \geq t\} \leq e^{-t/(4\tau_n)} \exp(\frac{e^{1/4}}{1-1/(4L)} - 1)$. Since $\frac{e^{1/4}}{1-1/(4L)} - 1 \leq \log 2$ for $L \geq 1.0347$ (in particular for $n \geq 3$),

$$\mathbb{P}\{Z \geq t\} \leq 2\exp(-t/(4\tau_n)), \qquad t \geq 0.$$

Note the 2 genuinely needs the L-dependent mgf above: the loosened stated mgf $\frac{e^x + x - 1}{1-x}$ only gives $e^{0.712}e^{-t/(4\tau_n)} \approx 2.04\, e^{-t/(4\tau_n)}$.

Remark. For small λ, $\frac{e^{\lambda\tau_n} + \lambda\tau_n - 1}{1 - \lambda\tau_n} = 2\lambda\tau_n + O(\lambda^2)$, while Exercise 12.13's (corrected) mgf is $\lambda(\mu + \sqrt{2vH_n} + cH_n) + O(\lambda^2)$ with $H_n \sim \log n$ — both of order $\lambda\tau_n$, as expected since, with overwhelming probability, at most one $|X_i|$ exceeds τ_n and Z is then essentially $\max_i X_i$ (the setting of Lemma 11.16).

Exercise 12.15

Central idea. Write $\{Z \ge (1+\eta)\mathbb{E}Z + t\} = \{Z \ge \mathbb{E}Z + s\}$ with $s = \eta\mathbb{E}Z + t$ and apply Bousquet's inequality (Theorem 12.5), then convert the Bennett function to Bernstein form via $h(u) \ge u^2/(2(1+u/3))$. Since $s \ge \eta\mathbb{E}Z$, the mean is absorbed into the variance proxy as $2\mathbb{E}Z + s/3 \le (2/\eta + 1/3)s$, leaving an exponent of at least $s^2/(2(\sigma^2 + cs))$ with $c = 2/\eta + 1/3$. A two-case analysis on whether $cs \le \delta\sigma^2$ (Gaussian regime) or not (exponential regime) produces the two terms; the hint justifies passing from a maximum of exponentials to their sum.

Solution. Let $m = \mathbb{E}Z \ge 0$ and $s = \eta m + t \ge 0$. Then $\{Z \ge (1+\eta)m + t\} = \{Z \ge m + s\}$, so by Theorem 12.5 (Bousquet's inequality),

$$\mathbb{P}\{Z \ge (1+\eta)m + t\} \le \exp(-vh(s/v)),$$

where $v = \sigma^2 + 2m$ with $\sigma^2 = \sup_{f\in\mathcal{F}} \sum_{i=1}^n \mathrm{Var}(f(X_i))$, and $h(u) = (1+u)\log(1+u) - u$.

Lemma (hint). For $u, v > 0$ and $\lambda \in [0,1]$, $\exp(-(1/u + v)) \le \max(e^{-\lambda/u},\, e^{-(1-\lambda)/v})$. *Proof.* Since $1/u + v \ge 1/u \ge \lambda/u \ge \min(\lambda/u, (1-\lambda)/v)$, negating gives $-(1/u+v) \le \max(-\lambda/u, -(1-\lambda)/v)$, and exponentiating yields the claim. ■

For $u \ge 0$, $h(u) \ge u^2/(2(1+u/3))$ (the standard Bennett-to-Bernstein bound, cf. Exercise 2.8), so

$$vh(s/v) \ge \frac{s^2}{2(v + s/3)}.$$

Since $s = \eta m + t \ge \eta m$, we have $m \le s/\eta$, whence

$$v + s/3 = \sigma^2 + 2m + s/3 \le \sigma^2 + (2/\eta + 1/3)s =: \sigma^2 + cs,$$

with $c = 2/\eta + 1/3 > 0$. (Assume $\sigma^2 > 0$; if $\sigma^2 = 0$ the first term below is 0 for $t > 0$.) Put $E = vh(s/v) \ge s^2/(2(\sigma^2 + cs))$.

If $cs \le \delta\sigma^2$, then $\sigma^2 + cs \le (1+\delta)\sigma^2$, so using $s \ge t$,

$$E \ge \frac{s^2}{2(1+\delta)\sigma^2} \ge \frac{t^2}{2(1+\delta)\sigma^2} =: A.$$

If $cs > \delta\sigma^2$, then $\sigma^2 + cs < (1 + 1/\delta)cs$, so

$$E \ge \frac{s^2}{2(1+1/\delta)cs} = \frac{\delta s}{2(1+\delta)c} \ge \frac{\delta t}{2(1+\delta)c} =: B.$$

In either case $\exp(-E) \le \max(e^{-A}, e^{-B}) \le e^{-A} + e^{-B}$, the last step being the content of the Lemma. That is,

$$\mathbb{P}\{Z \ge (1+\eta)\mathbb{E}Z + t\} \le \exp\!\left(-\frac{t^2}{2(1+\delta)\sigma^2}\right) + \exp\!\left(-\frac{\delta t}{2(1+\delta)(2/\eta + 1/3)}\right),$$

as claimed. ■

Exercise 12.16

Central idea. Truncate each coordinate at τ_n: $X_{i,s} = V_{i,s} + W_{i,s}$ with $|V_{i,s}| \le \tau_n$ bounded and $W_{i,s}$ the heavy tail. Then $Z \le Z_1 + Z_2$, and symmetry gives $\mathbb{E}Z \ge \mathbb{E}Z_1 - \mathbb{E}Z_2$. The bounded part Z_1 is controlled by Exercise 12.15 (Bousquet's inequality, Theorem 12.5) after scaling by τ_n, using $\mathrm{Var}(V_{i,s}) \le 8\sigma^2 + 32c^2$; the tail part Z_2 is controlled by Exercise 12.14 via $Z_2 \le \sum_i Y_i \mathbf{1}_{\{Y_i \ge \tau_n\}}$. A union-bound split of $\{Z \ge (1+\eta)\mathbb{E}Z + t\}$ into $\{Z_1 \ge \cdots\}$ and $\{Z_2 \ge \varepsilon t\}$ yields the three-term bound.

Solution. Write $X_{i,s} = V_{i,s} + W_{i,s}$ with $V_{i,s} = X_{i,s}\mathbf{1}_{\{|X_{i,s}| < \tau_n\}}$ and $W_{i,s} = X_{i,s}\mathbf{1}_{\{|X_{i,s}| \ge \tau_n\}}$. Since $|X_{i,s}|$ is invariant under the sign flip $X_{i,s} \mapsto -X_{i,s}$, both $V_{i,s}$ and $W_{i,s}$ are symmetric, hence centered. Also $Z = \sup_s \sum_i X_{i,s} \le \sup_s \sum_i V_{i,s} + \sup_s \sum_i W_{i,s} = Z_1 + Z_2$.

Check 1: $\mathbb{E}Z \geq \mathbb{E}Z_1 - \mathbb{E}Z_2$. Since $V_{i,s} = X_{i,s} - W_{i,s}$,

$$\sup_s \sum_i V_{i,s} \leq \sup_s \sum_i X_{i,s} + \sup_s \sum_i (-W_{i,s}) = Z + \sup_s \sum_i (-W_{i,s}).$$

By symmetry $-W_{i,s} \overset{d}{=} W_{i,s}$, so $\sup_s \sum_i (-W_{i,s}) \overset{d}{=} Z_2$. Taking expectations, $\mathbb{E}Z_1 \leq \mathbb{E}Z + \mathbb{E}Z_2$.

Check 2: $\mathbb{E}Z_2 \leq \tau_n(1 + 1/\log n)$. Since $|W_{i,s}| \leq Y_i \mathbf{1}_{\{Y_i \geq \tau_n\}}$,

$$Z_2 \leq \sum_{i=1}^n \sup_s |W_{i,s}| \leq \sum_{i=1}^n Y_i \mathbf{1}_{\{Y_i \geq \tau_n\}},$$

so $\mathbb{E}Z_2 \leq n\,\mathbb{E}[Y_1 \mathbf{1}_{\{Y_1 \geq \tau_n\}}]$. By the layer-cake identity and the sub-gamma tail bound $\mathbb{P}\{Y_1 \geq \sqrt{2\sigma^2 u} + cu\} \leq e^{-u}$,

$$\begin{aligned}
\mathbb{E}[Y_1 \mathbf{1}_{\{Y_1 \geq \tau_n\}}] &= \tau_n \mathbb{P}\{Y_1 \geq \tau_n\} + \int_{\tau_n}^\infty \mathbb{P}\{Y_1 \geq t\}\, dt \\
&\leq \frac{\tau_n}{n} + \int_{\log n}^\infty e^{-u} \Big(\sqrt{\tfrac{\sigma^2}{2u}} + c \Big)\, du \qquad (t = \sqrt{2\sigma^2 u} + cu) \\
&\leq \frac{1}{n} \Big(\tau_n + \sqrt{\tfrac{\sigma^2}{2\log n}} + c \Big),
\end{aligned}$$

using $\tau_n = \sqrt{2\sigma^2 \log n} + c \log n$ (i.e. $u = \log n$) and $\int_{\log n}^\infty u^{-1/2} e^{-u} du \leq (\log n)^{-1/2}/n$. Hence

$$\mathbb{E}Z_2 \leq \tau_n + c + \sqrt{\tfrac{\sigma^2}{2\log n}} \leq \tau_n + c + \sqrt{\tfrac{2\sigma^2}{\log n}} = \tau_n \Big(1 + \tfrac{1}{\log n} \Big).$$

Check 3: $\mathbb{P}\{Z_2 \geq t\} \leq 2\exp(-t/(8\tau_n))$. From $Z_2 \leq \sum_i Y_i \mathbf{1}_{\{Y_i \geq \tau_n\}}$, Chernoff's bound applied to this nonnegative sum — exactly the proof of 12.14's tail, which uses only the envelope mgf — gives $\mathbb{P}\{Z_2 \geq t\} \leq \mathbb{P}\{\sum_i Y_i \mathbf{1}_{\{Y_i \geq \tau_n\}} \geq t\} \leq 2\exp(-t/(4\tau_n)) \leq 2\exp(-t/(8\tau_n))$.

Check 4: the decomposition. From $Z \leq Z_1 + Z_2$ and $\mathbb{E}Z \geq \mathbb{E}Z_1 - \mathbb{E}Z_2$,

$$\{Z \geq (1+\eta)\mathbb{E}Z + t\} \subseteq \{Z_1 + Z_2 \geq (1+\eta)\mathbb{E}Z_1 - (1+\eta)\mathbb{E}Z_2 + t\}.$$

For any $\varepsilon > 0$, $\{Z_1 + Z_2 \geq A + t\} \subseteq \{Z_1 \geq A + (1-\varepsilon)t\} \cup \{Z_2 \geq \varepsilon t\}$ (contrapositive: $Z_1 < A + (1-\varepsilon)t$ and $Z_2 < \varepsilon t$ imply $Z_1 + Z_2 < A + t$). With $A = (1+\eta)\mathbb{E}Z_1 - (1+\eta)\mathbb{E}Z_2$,

$$\mathbb{P}\{Z \geq (1+\eta)\mathbb{E}Z + t\} \leq \mathbb{P}\{Z_1 \geq (1+\eta)\mathbb{E}Z_1 - (1+\eta)\mathbb{E}Z_2 + (1-\varepsilon)t\} + \mathbb{P}\{Z_2 \geq \varepsilon t\}.$$

Main bound. Apply Exercise 12.15 to Z_1. The summands $V_{i,s}$ satisfy $|V_{i,s}| \leq \tau_n$ and, by Theorem 2.3, $\mathrm{Var}(V_{i,s}) \leq \mathbb{E}[V_{i,s}^2] \leq \mathbb{E}[X_{i,s}^2] = \mathrm{Var}(X_{i,s}) \leq 8\sigma^2 + 32c^2$. Scaling by τ_n, $V_{i,s}/\tau_n$ are bounded by 1 with variance factor $8n(\sigma^2 + 4c^2)/\tau_n^2$. Exercise 12.15 (with $\kappa = 2/\eta + 1/3$) gives, for deviation $(1-2\varepsilon)t$ (the $-(1+\eta)\mathbb{E}Z_2$ term being absorbed into the passage from $(1-\varepsilon)$ to $(1-2\varepsilon)$),

$$\mathbb{P}\{Z_1 \geq (1+\eta)\mathbb{E}Z_1 + (1-2\varepsilon)t\} \leq \exp\Big(-\frac{(1-2\varepsilon)^2 t^2}{16(1+\delta)n(\sigma^2+4c^2)} \Big) + \exp\Big(-\frac{\delta(1-2\varepsilon)t}{2(1+\delta)\kappa\tau_n} \Big).$$

Indeed, with $u = (1-2\varepsilon)t/\tau_n$ in 12.15: the Gaussian exponent is $\frac{u^2}{2(1+\delta)\cdot 8n(\sigma^2+4c^2)/\tau_n^2} = \frac{(1-2\varepsilon)^2 t^2}{16(1+\delta)n(\sigma^2+4c^2)}$, and the exponential exponent is $\frac{\delta u}{2(1+\delta)\kappa} = \frac{\delta(1-2\varepsilon)t}{2(1+\delta)\kappa\tau_n}$. Combining with Check 3 for $\mathbb{P}\{Z_2 \geq \varepsilon t\} \leq 2\exp(-\varepsilon t/(8\tau_n))$ yields

$$\mathbb{P}\{Z \geq (1+\eta)\mathbb{E}Z + t\} \leq \exp\Big(-\frac{(1-2\varepsilon)^2 t^2}{16(1+\delta)n(\sigma^2+4c^2)} \Big) + \exp\Big(-\frac{\delta(1-2\varepsilon)t}{2(1+\delta)\kappa\tau_n} \Big) + 2\exp\Big(-\frac{\varepsilon t}{8\tau_n} \Big).$$

The last term is trivial (exceeds 1) when $\varepsilon t \leq 4\tau_n$, since then $\varepsilon t/(8\tau_n) \leq 1/2$ and $2e^{-1/2} > 1$. (The passage from $(1-\varepsilon)t - (1+\eta)\mathbb{E}Z_2$ to $(1-2\varepsilon)t$ above needs $\varepsilon t \geq (1+\eta)\mathbb{E}Z_2$; otherwise $\varepsilon t < (1+\eta)\mathbb{E}Z_2 \leq (1+\eta)(1+1/\log n)\tau_n \leq 4\tau_n$ for $\eta \leq 1$, $n \geq 3$, so the third term already exceeds 1 and the bound holds trivially.) ■

Chapter 13 — The Expected Value of Suprema of Empirical Processes

Exercise 13.1

Central idea. Mimic Lemma 13.1's chaining with nets at the entropy-number scales $e_n(S)$, but extract a subsequence n_k with $e_{n_{k+1}} \le e_{n_k}/2$: then the drift ($a$-) terms sum geometrically to $O(a\delta)$, while each subsequence block's $\sqrt{v}/c$ contribution is dominated by the corresponding block of the full sums $\sum_n e_n 2^{n/2}$, $\sum_n e_n 2^n$ (geometric-series comparison). Each link is controlled by the Bernstein maximal inequality (Corollary 2.6).

Solution. We use the notation of Lemma 13.1: S finite, $\log \mathbb{E} e^{\lambda(X_s - X_{s'})} \le a\lambda d(s,s') + \frac{v\lambda^2 d(s,s')^2}{2(1-c\lambda d(s,s'))}$ for $0 < \lambda < (cd(s,s'))^{-1}$, $\delta = \sup_{s\in S} d(s,s_0)$, $e_n = e_n(S)$. (The sums in the statement mix indices j and n; read $e_j(S)$, $2^{j/2}$, 2^j. Universal constants are implied though not written.)

For $n \ge 1$ with $e_n > 0$, let $S_n \subset S$ be a minimal $(e_n + \eta_n)$-net, so $|S_n| = N(e_n + \eta_n, S) \le 2^{2^n}$ by definition of e_n as an infimum, and choose $\Pi_n(s) \in S_n$ with $d(s, \Pi_n(s)) \le e_n + \eta_n$. (This is the rigorous form of the hint's maximal e_n-packing: maximality gives the net property, but the cardinality $|S_n| \le 2^{2^{n+1}}$ as stated needs this η-regularization, since $N(e_n, S) \le 2^{2^n}$ may fail at the infimum itself.) If $e_n = 0$, take $S_n = S$, $\Pi_n = \mathrm{id}$ (valid as S is finite). Put $S_0 = \{s_0\}$, $\Pi_0 \equiv s_0$.

Since $e_n \to 0$, choose $1 = n_1 < n_2 < \cdots$ with $e_{n_{k+1}} \le e_{n_k}/2$ (stop if some $e_{n_k} = 0$). Telescoping,

$$X_s - X_{s_0} = \sum_{k\ge 1} \big(X_{\Pi_{n_k}(s)} - X_{\Pi_{n_{k-1}}(s)}\big), \qquad \Pi_{n_0} := \Pi_0,$$

a finite sum since $\Pi_{n_k} = \mathrm{id}$ eventually. Hence

$$\mathbb{E}\sup_{s\in S}(X_s - X_{s_0}) \le \sum_{k\ge 1} \mathbb{E}\max_{s\in S}\big(X_{\Pi_{n_k}(s)} - X_{\Pi_{n_{k-1}}(s)}\big).$$

At level k there are at most $M_k := |S_{n_{k-1}}| \cdot |S_{n_k}| \le 2^{2^{n_{k-1}} + 2^{n_k}}$ distinct pairs, each at distance at most $\bar d_k := e_{n_{k-1}} + e_{n_k} + 2\eta \le 3e_{n_{k-1}}$ for $k \ge 2$ (with $e_{n_0} := \delta$), and $\bar d_1 \le \delta + e_{n_1} + \eta \le 2\delta$. Moreover $\log M_k \le 2(\log 2) 2^{n_k}$.

For fixed pairs with $\max d \le \bar d$, $\mathbb{E}\max_{\text{pairs}}(X_t - X_{t'}) \le \max \mathbb{E}[X_t - X_{t'}] + \mathbb{E}\max(\text{centered})$. From the mgf bound, $\mathbb{E}[X_t - X_{t'}] \le a\bar d$ (let $\lambda \downarrow 0$), and centered increments satisfy $\log \mathbb{E} e^{\lambda W} \le 2a\lambda\bar d + \frac{v\lambda^2 \bar d^2}{2(1-c\lambda\bar d)}$, whence by the Chernoff maximal bound $\mathbb{E}\max_i W_i \le \inf_{0<\lambda<1/(c\bar d)}[\frac{\log M_k}{\lambda} + 2a\bar d + \frac{v\lambda\bar d^2}{2(1-c\lambda\bar d)}] \le 2a\bar d + \sqrt{2v\bar d^2 \log M_k} + c\bar d \log M_k$ (Bernstein optimization, as in Lemma 2.4). Altogether, level k contributes at most $3a\bar d_k + C_1\sqrt{v}\,\bar d_k 2^{n_k/2} + C_2 c\,\bar d_k 2^{n_k}$.

Summing: the a-terms give $3a\sum_k \bar d_k \lesssim a\delta$, since $\bar d_1 \lesssim \delta$ and $\bar d_k \lesssim e_{n_{k-1}} \le 2^{-(k-2)}\delta$ for $k \ge 2$ sum geometrically. For the $\sqrt{v}$-terms ($k \ge 2$), $e_{n_{k-1}} 2^{n_k/2} \le 2\sqrt{2}\sum_{n=n_{k-1}}^{n_k-1} e_n 2^{n/2}$: indeed, minimality of n_k gives $e_n > e_{n_{k-1}}/2$ for $n_{k-1} \le n < n_k$, and $\sum_{n=n_{k-1}}^{n_k-1} 2^{n/2} \ge 2^{n_k/2}/\sqrt{2}$. Summing over the disjoint blocks, $\sum_{k\ge 2} e_{n_{k-1}} 2^{n_k/2} \lesssim \sum_{n\ge 1} e_n 2^{n/2}$; similarly $\sum_{k\ge 2} e_{n_{k-1}} 2^{n_k} \lesssim \sum_{n\ge 1} e_n 2^n$. Letting $\eta \downarrow 0$,

$$\mathbb{E}\left[\sup_{s\in S} X_s - X_{s_0}\right] \lesssim a\delta + \sqrt{v}\,\delta + c\,\delta + \sqrt{v}\sum_{n\ge 1} e_n 2^{n/2} + c\sum_{n\ge 1} e_n 2^n.$$

Remark (inaccuracy in the statement). As written—sums from $j = 1$ with no base-scale term—the inequality is false. Counterexample: $S = \{s_0, s_1\}$, $X_{s_1} - X_{s_0} \sim N(0, v\delta^2)$ (so $a = c = 0$; the hypothesis holds as $\log \mathbb{E} e^{\lambda(X_{s_1} - X_{s_0})} = \frac{v\lambda^2\delta^2}{2}$). Here $e_n(S) = 0$ for all $n \ge 1$ (since $N(\varepsilon, S) \le 2 \le 4 = 2^{2^1}$ for every $\varepsilon > 0$), so the right-hand side is 0, but $\mathbb{E}[\sup_s X_s - X_{s_0}] = \mathbb{E}[(X_{s_1} - X_{s_0})^+] = \sqrt{v}\,\delta/\sqrt{2\pi} > 0$. The corrected form above (equivalently, sums from $n = 0$ with $e_0(S) := \delta$) holds. ■

Exercise 13.2

Central idea. Condition on the $X_{i,s}$: the $2|\mathcal{T}|$ variables $\pm\sum_i \varepsilon_i X_{i,s}$ are sub-Gaussian, so Theorem 2.5 bounds the Rademacher supremum by $\sqrt{2\max_s \sum_i X_{i,s}^2 \cdot \log(2|\mathcal{T}|)}$. The random variance proxy $W = \max_s \sum_i X_{i,s}^2$ is controlled via symmetrization and the absolute-value form of the contraction principle (Theorem 11.6 with the 1-Lipschitz map $\varphi(x) = x^2/2$ on $[-1,1]$), which feeds W back into the original quantity and yields a quadratic inequality for $R = \mathbb{E}\sup_s |\sum_i \varepsilon_i X_{i,s}|$ solving to the claimed bound with $\kappa = 34$. (See Koltchinskii, 2008.)

Remark (repair). An earlier version bounded $\mathbb{E}\max_s |\sum_i \varepsilon_i X_{i,s}^2|$ by $2R$ by applying the one-sided contraction principle on $\mathcal{T} \times \{-1,1\}$ to the vectors $\tau\varphi(X_{i,s})$. That step is unjustified: $\tau\varphi(X_{i,s})$ is not φ applied to the coordinates $\tau X_{i,s}$ (here φ is even), so Theorem 11.6 does not apply in that form. The proof below uses the absolute-value statement of Theorem 11.6 instead; this costs a factor 2 and changes only the value of κ.

Solution. Assume $|\mathcal{T}| \geq 2$: for $|\mathcal{T}| = 1$ the right-hand side is 0, so the statement implicitly requires $|\mathcal{T}| \geq 2$. We take $\mathcal{T}$ finite, as $\log|\mathcal{T}|$ in the statement presupposes. Let

$$R = \mathbb{E}\sup_{s\in\mathcal{T}} \Big|\sum_{i=1}^n \varepsilon_i X_{i,s}\Big|, \qquad W = \max_{s\in\mathcal{T}} \sum_{i=1}^n X_{i,s}^2, \qquad L = \log(2|\mathcal{T}|),$$

expectations being over both ε and X (which are independent).

Step 1 (conditional sub-Gaussian bound). Fix the $X_{i,s}$. For $\tau \in \{-1,1\}$, $Z_{s,\tau} = \tau\sum_i \varepsilon_i X_{i,s}$ satisfies $\mathbb{E}_\varepsilon e^{\lambda Z_{s,\tau}} = \prod_i \cosh(\lambda X_{i,s}) \leq \exp(\frac{\lambda^2}{2}\sum_i X_{i,s}^2) \leq \exp(\frac{\lambda^2}{2}W)$, using $\cosh y \leq e^{y^2/2}$. So the $2|\mathcal{T}|$ variables $Z_{s,\tau}$ are sub-Gaussian with variance factor W, and Theorem 2.5 gives

$$\mathbb{E}_\varepsilon \sup_s \Big|\sum_i \varepsilon_i X_{i,s}\Big| = \mathbb{E}_\varepsilon \max_{s,\tau} Z_{s,\tau} \leq \sqrt{2WL}.$$

Taking $\mathbb{E}_X$ and using Jensen's inequality ($\sqrt{\cdot}$ is concave), $R \leq \sqrt{2L\,\mathbb{E}W}$.

Step 2 (controlling $\mathbb{E}W$). Since $\sup_s \sum_i \mathbb{E}X_{i,s}^2 \leq n\sigma^2$,

$$\mathbb{E}W \leq n\sigma^2 + \mathbb{E}\max_s \sum_i (X_{i,s}^2 - \mathbb{E}X_{i,s}^2).$$

Symmetrize as in the proof of Lemma 11.4: let X' be an independent copy of X and $\mathbb{E}'$ the expectation in X'. By Jensen's inequality (a maximum of linear functions is convex),

$$\mathbb{E}\max_s \sum_i (X_{i,s}^2 - \mathbb{E}'X_{i,s}'^2) \leq \mathbb{E}\max_s \sum_i (X_{i,s}^2 - X_{i,s}'^2) = \mathbb{E}\max_s \sum_i \varepsilon_i (X_{i,s}^2 - X_{i,s}'^2),$$

the equality because the vectors $(X_{i,s}^2 - X_{i,s}'^2)_s$, $i \leq n$, are independent and symmetric. Splitting the maximum and using that $-\varepsilon$ has the law of ε,

$$\mathbb{E}\max_s \sum_i (X_{i,s}^2 - \mathbb{E}X_{i,s}^2) \leq 2\,\mathbb{E}\max_s \sum_i \varepsilon_i X_{i,s}^2 \leq 2\,\mathbb{E}\max_s \Big|\sum_i \varepsilon_i X_{i,s}^2\Big|.$$

Let $\varphi(y) = y^2/2$ for $|y| \leq 1$ and $\varphi(y) = |y| - \frac{1}{2}$ for $|y| > 1$. Then $\varphi : \mathbb{R} \to \mathbb{R}$ is 1-Lipschitz ($|\varphi'| \leq 1$ everywhere and φ is C^1), $\varphi(0) = 0$, and $\varphi(X_{i,s}) = X_{i,s}^2/2$ because $|X_{i,s}| \leq 1$. Condition on X and apply the second (absolute-value) inequality of Theorem 11.6 to the vectors $x_i = (X_{i,s})_{s\in\mathcal{T}}$ with $\varphi_i = \varphi$ and $\Psi(t) = t$ (nondecreasing and convex on $[0,\infty)$):

$$\frac{1}{2}\,\mathbb{E}_\varepsilon \max_s \Big|\sum_i \varepsilon_i \frac{X_{i,s}^2}{2}\Big| \leq \mathbb{E}_\varepsilon \max_s \Big|\sum_i \varepsilon_i X_{i,s}\Big|.$$

Multiplying by 4 and taking $\mathbb{E}_X$ gives $\mathbb{E}\max_s |\sum_i \varepsilon_i X_{i,s}^2| \leq 4R$. Hence

$$\mathbb{E}W \leq n\sigma^2 + 8R.$$

Step 3 (quadratic inequality). Combining Steps 1 and 2, $R^2 \le 2L(n\sigma^2 + 8R)$, i.e. $R^2 - 16LR - 2Ln\sigma^2 \le 0$, so

$$R \le 8L + \sqrt{64L^2 + 2Ln\sigma^2} \le 16L + \sigma\sqrt{2nL},$$

using $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. For $|\mathcal{T}| \ge 2$ we have $\log 2 \le \log|\mathcal{T}|$, so $L = \log 2 + \log|\mathcal{T}| \le 2\log|\mathcal{T}|$. Therefore

$$\mathbb{E}\sup_{s\in\mathcal{T}}\Big|\sum_{i=1}^{n}\varepsilon_i X_{i,s}\Big| \le 32\log|\mathcal{T}| + 2\sigma\sqrt{n\log|\mathcal{T}|} \le 34\max\Big(\sigma\sqrt{n\log|\mathcal{T}|},\ \log|\mathcal{T}|\Big),$$

which is the claim with the universal constant $\kappa = 34$. (The argument never uses $\mathbb{E}X_{i,s} = 0$: the symmetrization in Step 2 is applied to the uncentered vectors $(X_{i,s}^2)_s$ via an independent copy.) ■

Exercise 13.3

Central idea. In one dimension, $\phi'(x) = -x\phi(x)$ turns $\int xF(x)\phi(x)dx$ into a boundary term (killed by the growth hypothesis) plus $\int F'(x)\phi(x)dx$ via integration by parts. For a general centered Gaussian vector write $X = AZ$ with Z standard Gaussian in $\mathbb{R}^r$ ($r = \operatorname{rank}\Sigma$); conditioning on Z_{-k} reduces $\mathbb{E}[Z_k F(AZ)]$ to the one-dimensional identity, and $\partial_k[F(A\,\cdot)] = \sum_j A_{jk}(\partial_j F)(A\,\cdot)$ produces the covariance weights $\Sigma_{ij} = \sum_k A_{ik}A_{jk}$.

Solution. (We write d for the dimension; the "$\mathbb{R}^n$" in the statement is a typo.)

Dimension one. Let $Z \sim N(0,1)$ with density ϕ. The growth hypothesis gives $\mathbb{E}|ZF(Z)| < \infty$: indeed $|F(x)| \le e^{x^2/8}$ for large $|x|$ (take $a = 1/8$), so $|xF(x)|\phi(x) \le |x|e^{-3x^2/8}/\sqrt{2\pi}$, integrable. For $M > 0$,

$$\int_{-M}^{M} xF(x)\phi(x)\,dx = \big[-F(x)\phi(x)\big]_{-M}^{M} + \int_{-M}^{M} F'(x)\phi(x)\,dx,$$

since $(F\phi)' = F'\phi - xF\phi$. The boundary term vanishes as $M \to \infty$: $|F(\pm M)|\phi(M) \le e^{-3M^2/8}/\sqrt{2\pi} \to 0$. Hence, with the symmetric improper integral,

$$\mathbb{E}[ZF(Z)] = \lim_{M\to\infty}\int_{-M}^{M} F'(x)\phi(x)\,dx,$$

and if $\mathbb{E}|F'(Z)| < \infty$ this equals $\mathbb{E}[F'(Z)]$. For $X \sim N(\mu,\sigma^2)$, $\sigma > 0$, write $X = \mu + \sigma Z$ and apply this to $\tilde{F}(z) = F(\mu + \sigma z)$ (the growth hypothesis transfers: $|\tilde{F}(z)|e^{-az^2} \le e^{2a'\mu^2}|F(\mu+\sigma z)|e^{-a'|\mu+\sigma z|^2} \to 0$ with $a' = a/(2\sigma^2)$); if $\sigma = 0$ both sides below are 0. This gives $\mathbb{E}[(X-\mu)F(X)] = \sigma^2\mathbb{E}[F'(X)]$.

General d. Let $\Sigma = \mathbb{E}[XX^T]$. If $\Sigma = 0$ both sides are 0. Otherwise write $X = AZ$ with $Z \sim N(0, I_r)$, $r = \operatorname{rank}(\Sigma) \ge 1$, A $d \times r$ of full column rank, $\Sigma = AA^T$. Then

$$\mathbb{E}[X_i F(X)] = \sum_{k=1}^{r} A_{ik}\,\mathbb{E}[Z_k F(AZ)].$$

Fix k and put $G(z) = F(Az)$. Conditioning on Z_{-k}: given Z_{-k}, $Z_k \sim N(0,1)$ is independent of Z_{-k}, and $t \mapsto G(Z_{-k}, t)$ is C^1 with $|G(z)|e^{-a\|z\|^2} \to 0$ (since $\|Az\| \ge \sigma_{\min}(A)\|z\| \to \infty$ and $|G(z)|e^{-a\|z\|^2} \le |F(Az)|e^{-a'\|Az\|^2}$ with $a' = a/\|A\|_{op}^2$). The one-dimensional identity gives $\mathbb{E}[Z_k G(Z) \mid Z_{-k}] = \mathbb{E}[\partial_k G(Z) \mid Z_{-k}]$ a.s. By the chain rule $\partial_k G(z) = \sum_{j=1}^{d} A_{jk}(\partial_j F)(Az)$, so

$$\mathbb{E}[Z_k F(AZ)] = \sum_{j=1}^{d} A_{jk}\,\mathbb{E}[(\partial_j F)(X)].$$

Therefore $\mathbb{E}[X_i F(X)] = \sum_{k,j} A_{ik}A_{jk}\mathbb{E}[(\partial_j F)(X)] = \sum_{j=1}^{d}\Sigma_{ij}\,\mathbb{E}[(\partial_j F)(X)]$, as claimed. ■

Remark (integrability subtlety). The growth hypothesis implies $\mathbb{E}|X_i F(X)| < \infty$ and kills the boundary terms, but it does not imply $\mathbb{E}|\partial_j F(X)| < \infty$, so the right-hand side need not exist as a Lebesgue integral. E.g., in one dimension $F(x) = \sin(e^{x^2})$ satisfies the growth hypothesis ($|F| \le 1$) and $\mathbb{E}[ZF(Z)] = 0$ by symmetry, yet $\mathbb{E}|F'(Z)| = \infty$ since $|F'(x)| \approx 2xe^{x^2}$ and $e^{x^2}\phi(x) = e^{x^2/2}/\sqrt{2\pi}$ is not integrable at infinity. The identity holds as stated whenever $\mathbb{E}|\nabla F(X)| < \infty$ (true in all applications, e.g. Slepian's lemma, Theorem 13.3, where F has bounded derivatives); otherwise it holds with $\mathbb{E}[\partial_j F(X)]$ read as the corresponding improper integral.

Exercise 13.4

Central idea. The metric $d(s,s') = \sqrt{1/\log s + 1/\log s'}$ makes $\{s : s \geq e^{2/\delta^2}\}$ a single δ-ball while $\{2, \dots, e^{2/\delta^2}\}$ needs $\asymp e^{2/\delta^2}$ balls, so $H(\delta, \mathbb{N}) \asymp 1/\delta^2$. Then Dudley's integral $\int_0 \sqrt{H(u)}du \asymp \int_0 du/u$ diverges (chaining gives nothing), Sudakov gives only $\mathbb{E}Z \gtrsim \delta\sqrt{H(\delta)} \asymp 1$ (a constant), while slicing into dyadic blocks $\{2^m \leq i < 2^{m+1}\}$—inside which the variances $1/\log i$ are comparable—and a union bound per block yields $\mathbb{P}\{Z > t\} \lesssim 2^{-t^2/2}$ for $t \geq 2$, hence $\mathbb{E}Z \leq 10$.

Solution. Write $X_i = Y_i/\sqrt{\log\max(i,2)}$, so $Z = \sup_{i\geq 1} X_i$ is a centered Gaussian process with natural distance $d(i,j)^2 = \mathbb{E}[(X_i - X_j)^2] = 1/\log\max(i,2) + 1/\log\max(j,2)$ for $i \neq j$ (independence), i.e. $d(s,s') = \sqrt{1/\log s + 1/\log s'}$ for $s \neq s'$, $s, s' \geq 2$.

Entropy. Let $\delta < 1/2$. If $s, s' \geq e^{2/\delta^2}$ then $1/\log s + 1/\log s' \leq \delta^2$, so all such s lie in one δ-ball; covering $\{1, \dots, \lceil e^{2/\delta^2}\rceil\}$ needs at most $e^{2/\delta^2} + 1$ further balls. Hence $N(\delta, \mathbb{N}) \leq e^{2/\delta^2} + 2$ and $H(\delta, \mathbb{N}) \leq 2/\delta^2 + \log 2 \leq 3/\delta^2 =: \kappa'/\delta^2$ (using $\log 2 \leq 1/\delta^2$ for $\delta < 1/2$). Conversely, $S_\delta := \{s : 2 \leq s \leq e^{2/\delta^2}\}$ is $(\delta/\sqrt{2})$-separated since $1/\log s \geq \delta^2/2$ there gives $d(s,s') \geq \delta/\sqrt{2}$; thus $N(\delta/\sqrt{2}, \mathbb{N}) \geq |S_\delta| \geq e^{2/\delta^2} - 1$ and $H(\delta/\sqrt{2}, \mathbb{N}) \geq 2/\delta^2 - \log 2 \geq 1/\delta^2$, i.e. $H(u, \mathbb{N}) \geq \frac{1}{2u^2} =: \kappa/u^2$.

Dudley (Corollary 13.2). $\mathbb{E}Z \lesssim \int_0^D \sqrt{H(u,\mathbb{N})}du$ with $D = \operatorname{diam}(\mathbb{N}) \leq \sqrt{2/\log 2}$. Since $\sqrt{H(u)} \gtrsim 1/u$ near 0, the integral diverges: Dudley's bound is $+\infty$, i.e. vacuous. Classical chaining cannot even show $\mathbb{E}Z < \infty$ here.

Sudakov (Theorem 13.4). For $\varepsilon < D$, $\mathbb{E}Z \geq \frac{1}{2}\varepsilon\sqrt{H(\varepsilon, \mathbb{N})} \geq \frac{\sqrt{\kappa}}{2}$, a positive constant independent of everything—true but uninformative (no rate; consistent with $\mathbb{E}Z$ being bounded, as shown next).

Finiteness via slicing. $Z \leq \max_{m\geq 0} W_m$ where $W_0 = \max_{i\in\{1,2\}} |Y_i|/\sqrt{\log 2}$ and $W_m = \max_{2^m \leq i < 2^{m+1}} |Y_i|/\sqrt{m\log 2}$ for $m \geq 1$ (using $\log i \geq m \log 2$ on the block, as hinted). By the union bound, for $m \geq 1$,

$$\mathbb{P}\{W_m > t\} \leq 2^m \cdot 2e^{-t^2 m\log 2/2} = 2 \cdot 2^{m(1-t^2/2)},$$

and $\mathbb{P}\{W_0 > t\} \leq 4 \cdot 2^{-t^2/2}$. For $t \geq 2$, $1 - t^2/2 \leq -1$, so $\sum_{m\geq 1} \mathbb{P}\{W_m > t\} \leq 2\sum_{m\geq 1} 2^{-m} \cdot 2^{1-t^2/2}/(1 - 2^{1-t^2/2}) \leq 8 \cdot 2^{-t^2/2}$ (using $2^{1-t^2/2} \leq 1/2$). Hence $\mathbb{P}\{Z > t\} \leq 12 \cdot 2^{-t^2/2}$ for $t \geq 2$. Therefore

$$\mathbb{E}Z = \int_0^\infty \mathbb{P}\{Z > t\}dt \leq 2 + 12\int_2^\infty e^{-(\log 2)t^2/2}dt \leq 2 + 12 \cdot \frac{e^{-2\log 2}}{2\log 2} = 2 + \frac{3}{2\log 2} \leq 5 < 10,$$

using $\int_x^\infty e^{-at^2/2}dt \leq e^{-ax^2/2}/(ax)$. The point of the hint: the X_i have wildly different variances ($1/\log 2$ down to 0), so a single-scale chaining argument drowns; slicing into blocks of comparable variance and bounding each block sharply recovers finiteness. ■

Exercise 13.5

Central idea. By Cauchy–Schwarz, $\sup_{s\in\mathcal{T}} \sum_i s_i X_i = (\sum_i a_i^2 X_i^2)^{1/2}$ exactly, and Jensen gives $\mathbb{E}Z \leq (\sum_i a_i^2)^{1/2}$. For the modulus, differences $s - s'$ lie in $2\mathcal{T}$ ($\mathcal{T}$ convex symmetric) intersected with the Euclidean c-ball; since $\sum_i t_i^2/\min(a_i^2, c^2) \leq \sum_i t_i^2/a_i^2 + \sum_i t_i^2/c^2 \leq 5$, this set sits inside the ellipsoid $\{\sum_i t_i^2/\min(a_i^2, c^2) \leq 8\}$, whose Gaussian supremum is $\sqrt{8}(\sum_i \min(a_i^2, c^2)X_i^2)^{1/2}$. Both bounds are sharp up to universal constants, whereas Dudley's entropy bound (Corollary 13.2) is not sharp for ellipsoids in general.

Solution. Let $X = (X_1, \dots, X_n)$ be standard Gaussian and $\mathcal{T} = \{s : \sum_i s_i^2/a_i^2 \leq 1\}$, $a_i > 0$.

Global bound. For fixed X, $\sum_i s_i X_i = \sum_i (s_i/a_i)(a_i X_i) \leq (\sum_i s_i^2/a_i^2)^{1/2}(\sum_i a_i^2 X_i^2)^{1/2} \leq (\sum_i a_i^2 X_i^2)^{1/2}$ by Cauchy–Schwarz (equality for $s_i \propto a_i^2 X_i$). Hence $Z = (\sum_i a_i^2 X_i^2)^{1/2}$ a.s., and by Jensen ($\sqrt{\cdot}$ concave),

$$\mathbb{E}Z = \mathbb{E}\Big[\Big(\sum_i a_i^2 X_i^2\Big)^{1/2}\Big] \leq \Big(\sum_i a_i^2 \mathbb{E}X_i^2\Big)^{1/2} = \Big(\sum_i a_i^2\Big)^{1/2}.$$

Modulus of continuity. Put $t = s - s'$. Since $\mathcal{T}$ is convex and symmetric, $\{s - s' : s, s' \in \mathcal{T}\} = \mathcal{T} - \mathcal{T} = 2\mathcal{T} = \{t : \sum_i t_i^2/a_i^2 \le 4\}$. With $\|t\| \le c$,

$$\sup_{\substack{s,s'\in\mathcal{T}\\ \|s-s'\|\le c}} \sum_i (s_i - s_i')X_i \le \sup_{\substack{t\in 2\mathcal{T}\\ \|t\|\le c}} \langle t, X\rangle.$$

Let $m_i = \min(a_i^2, c^2)$. For $t \in 2\mathcal{T}$, $\|t\| \le c$:

$$\sum_i \frac{t_i^2}{m_i} = \sum_i t_i^2 \max\Big(\frac{1}{a_i^2}, \frac{1}{c^2}\Big) \le \sum_i \frac{t_i^2}{a_i^2} + \frac{1}{c^2}\sum_i t_i^2 \le 4 + 1 = 5 \le 8.$$

Thus $\{t \in 2\mathcal{T} : \|t\| \le c\} \subseteq \mathcal{E} := \{t : \sum_i t_i^2/m_i \le 8\}$, and $\sup_{t\in\mathcal{E}}\langle t, X\rangle = \sqrt{8}(\sum_i m_i X_i^2)^{1/2}$ (maximizer $t_i \propto m_i X_i$). By Jensen,

$$\mathbb{E}\sup_{\substack{s,s'\in\mathcal{T}\\ \|s-s'\|\le c}} \sum_i (s_i - s_i')X_i \le \sqrt{8}\,\mathbb{E}\Big[\Big(\sum_i m_i X_i^2\Big)^{1/2}\Big] \le \sqrt{8\sum_{i=1}^n \min(a_i^2, c^2)}.$$

On "Theorem 13.2". There is no Theorem 13.2 in the book (the numbering runs Lemma 13.1, Corollary 13.2, Theorem 13.3, Theorem 13.4); the intended reference is the Gaussian chaining bound, Corollary 13.2 (Dudley). With the natural metric $d(s,s') = \|s - s'\|_2$ (since $\mathbb{E}[(\sum_i (s_i - s_i')X_i)^2] = \|s - s'\|^2$), it gives $\mathbb{E}Z \le 12\int_0^{\max_i a_i} \sqrt{H(u,\mathcal{T})}du$. The volumetric estimate $N(\mathcal{T}, \|\cdot\|_2, u) \le \prod_{i:a_i>u}(Ca_i/u)$ yields $H(u,\mathcal{T}) \le \sum_{i:a_i>u}\log_+(Ca_i/u)$, i.e.

$$\mathbb{E}Z \lesssim \int_0^{\max_i a_i} \sqrt{\sum_{i:a_i>u} \log_+ \frac{a_i}{u}}\,du.$$

As the book notes (p. 377), chaining is not sharp for ellipsoids in general; the sharp bound requires generic chaining (Talagrand, 1996a). By contrast, the two direct bounds above are sharp up to universal constants: e.g. $\mathbb{E}Z = \mathbb{E}[(\sum_i a_i^2 X_i^2)^{1/2}] \ge \sum_i a_i^2 \mathbb{E}|X_i|/(\sum_j a_j^2)^{1/2} = \sqrt{2/\pi}\,(\sum_i a_i^2)^{1/2}$, using $\|v\|_2 \ge \langle v, a\rangle/\|a\|_2$ with $v_i = a_i|X_i|$. ■

Exercise 13.6

Central idea. The stated constant $1/\sqrt{2}$ is false (see the remark below), but the $\sqrt{\log N}$ rate is correct. For $N = 2$ the maximum has the exact value $1/\sqrt{\pi}$; for large N, $t_N = \bar{\Phi}^{-1}(1/N)$ satisfies $t_N \ge \sqrt{\log N}$ (via a lower Mill's-ratio bound), and $\mathbb{E}M \ge t_N(1 - e^{-1}) - \mathbb{E}[M_-]$ with $\mathbb{E}[M_-] \le 2^{-(N+1)/2}$ negligible. Interpolating gives $\mathbb{E}\max \ge \frac{1}{4}\sqrt{\log N}$ for all $N \ge 2$.

Remark (erratum). As stated the inequality is false: for $N = 2$, $\mathbb{E}\max(X_1, X_2) = 1/\sqrt{\pi} \approx 0.564 < \frac{1}{\sqrt{2}}\sqrt{\log 2} \approx 0.589$. The attempt in the problem's hint has two gaps: (i) $\mathbb{E}M \ge t\,\mathbb{P}\{M \ge t\}$ is not valid for the (possibly negative) maximum M — one must pass through $\mathbb{E}[M_+]$ and control $\mathbb{E}[M_-]$; (ii) Mill's ratio $\bar{\Phi}(t) \le \phi(t)/t$ is an upper bound on the tail, hence gives an upper bound on the quantile t_N, not the lower bound $t_N \ge \sqrt{2\log N}$ that the hint claims. We prove the corrected bound $\mathbb{E}\max_{s\in\mathcal{T}} X_s \ge \frac{1}{4}\sqrt{\log|\mathcal{T}|}$.

Solution. Let $N = |\mathcal{T}| \ge 2$, $M_N = \max_{1\le i\le N} X_i$ with X_i i.i.d. $N(0,1)$.

For $N = 2$, $\max(X_1, X_2) = \frac{1}{2}(X_1 + X_2 + |X_1 - X_2|)$, and $X_1 - X_2 \sim N(0,2)$, so

$$\mathbb{E}M_2 = \tfrac{1}{2}\mathbb{E}|X_1 - X_2| = \tfrac{1}{2}\cdot\sqrt{2}\sqrt{\tfrac{2}{\pi}} = \frac{1}{\sqrt{\pi}} \approx 0.5642.$$

Since $M_N \ge \max(X_1, X_2)$ pointwise, $\mathbb{E}M_N \ge 1/\sqrt{\pi}$ for all $N \ge 2$. As $\frac{1}{4}\sqrt{\log 100} \approx 0.5365 < 1/\sqrt{\pi}$, this gives

$$\mathbb{E}M_N \ge \frac{1}{4}\sqrt{\log N} \qquad \text{for } 2 \le N \le 100. \tag{1}$$

Let now $N \ge 101$ and put $t_N = \bar\Phi^{-1}(1/N)$, so $\bar\Phi(t_N) = 1/N$. Write $M = M_N$. Since $\mathbb{E}M = \mathbb{E}[M_+] - \mathbb{E}[M_-]$ and, by Cauchy–Schwarz, $\mathbb{E}[M_-] = \mathbb{E}[(-M)_+] \le (\mathbb{E}[(-M)_+^2])^{1/2}\mathbb{P}\{M<0\}^{1/2} \le (1/2)^{1/2}(2^{-N})^{1/2} = 2^{-(N+1)/2}$ (using $(-M)_+ \le (-X_1)_+$ and $\mathbb{E}[(-X_1)_+^2] = 1/2$),

$$\mathbb{E}M \ge \mathbb{E}[M_+] - 2^{-(N+1)/2}.$$

Moreover, with $(1-x)^N \le e^{-Nx}$,

$$\mathbb{E}[M_+] = \int_0^\infty \mathbb{P}\{M > t\}\,dt \ge \int_0^{t_N} (1 - e^{-N\bar\Phi(t)})dt \ge t_N(1-e^{-1}),$$

since $\bar\Phi(t) \ge \bar\Phi(t_N) = 1/N$ for $t \le t_N$. Hence

$$\mathbb{E}M_N \ge (1-e^{-1})t_N - 2^{-(N+1)/2}. \tag{2}$$

We show $t_N \ge \sqrt{\log N}$ for $N \ge 101$. Since $\bar\Phi$ is decreasing, it suffices that $\bar\Phi(\sqrt{\log N}) \ge 1/N$. By the lower Mill's-ratio bound $\bar\Phi(t) \ge \phi(t)\,\frac{t}{t^2+1}$ (Birnbaum),

$$\bar\Phi(\sqrt{\log N}) \ge \frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{N}}\frac{\sqrt{\log N}}{\log N + 1},$$

so it suffices that $\frac{\sqrt{N}}{\sqrt{2\pi}} \ge \frac{\log N + 1}{\sqrt{\log N}} = \sqrt{\log N} + \frac{1}{\sqrt{\log N}}$. Since $\frac{1}{\sqrt{\log N}} \le 1$ for $N \ge 3$, it suffices that $\sqrt{N} \ge \sqrt{2\pi}\,(\sqrt{\log N} + 1)$, i.e. $p(N) := \sqrt{N} - \sqrt{2\pi\log N} \ge \sqrt{2\pi}$. Now $p'(N) = \frac{1}{2\sqrt{N}}(1 - \frac{\sqrt{2\pi}}{\sqrt{N\log N}}) \ge 0$ for $N \ge 101$ (as $\sqrt{N\log N} \ge \sqrt{101\log 101} > 21 > \sqrt{2\pi}$), and $p(101) = \sqrt{101} - \sqrt{2\pi\log 101} \approx 4.66 \ge \sqrt{2\pi} \approx 2.51$. Thus $p(N) \ge \sqrt{2\pi}$ for all $N \ge 101$, and $t_N \ge \sqrt{\log N}$.

From (2), for $N \ge 101$,

$$\mathbb{E}M_N \ge (1-e^{-1})\sqrt{\log N} - 2^{-51} \ge 0.63\sqrt{\log N} \ge \frac{1}{4}\sqrt{\log N},$$

since $1 - e^{-1} \approx 0.6321$ and $2^{-51} < 10^{-15}$. Together with (1) this proves

$$\mathbb{E}\max_{s\in\mathcal{T}} X_s \ge \frac{1}{4}\sqrt{\log|\mathcal{T}|} \qquad \text{for } |\mathcal{T}| \ge 2,$$

the $\sqrt{\log N}$ rate with a valid constant. ■

Exercise 13.7

Central idea. Both suprema are over Gaussian processes on the same index set K; the Sudakov–Fernique inequality (Theorem 13.3) bounds $\mathbb{E}Z$ by $\mathbb{E}U$ provided the increments of the Z-process are dominated by those of the U-process, which follows from the identity $u^TXv = \langle X, vu^T\rangle_{HS}$ and the elementary estimate $\|vu^T - ts^T\|_{HS}^2 \le \|u-s\|^2 + \|v-t\|^2$ (Davidson–Szarek (2001)); the second inequality is Exercise 13.5.

Solution. Write $u^TXv = \sum_{i,j} u_iX_{ij}v_j = \langle X, vu^T\rangle_{HS}$, the Frobenius inner product, where vu^T is the rank-one matrix with entries v_iu_j. Since the entries X_{ij} are i.i.d. $N(0,1)$, the process $Z_{(u,v)} = u^TXv$ is centered Gaussian, and for $(u,v),(s,t) \in K$,

$$\mathbb{E}[(Z_{(u,v)} - Z_{(s,t)})^2] = \big\|\langle X, vu^T - ts^T\rangle_{HS} \text{ as a Gaussian form}\big\|^2 = \|vu^T - ts^T\|_{HS}^2,$$

because $\langle X, M\rangle_{HS} \sim N(0, \|M\|_{HS}^2)$ for any fixed matrix M.

Now $\|vu^T\|_{HS}^2 = \|u\|^2\|v\|^2 = 1$ (and likewise for ts^T), while $\langle vu^T, ts^T\rangle_{HS} = \operatorname{trace}(uv^Tts^T) = \langle u,s\rangle\langle v,t\rangle$. Hence

$$\|vu^T - ts^T\|_{HS}^2 = 2 - 2\langle u,s\rangle\langle v,t\rangle.$$

We claim this is at most $\|u-s\|^2+\|v-t\|^2$. Indeed, with $a=\langle u,s\rangle$, $b=\langle v,t\rangle$ (so $|a|,|b|\le 1$),

$$\|u-s\|^2+\|v-t\|^2-\|vu^T-ts^T\|_{HS}^2=(2-2a)+(2-2b)-(2-2ab)=2(1-a)(1-b)\ge 0.$$

The process $U_{(u,v)}=\langle u,Y\rangle+\langle v,Y'\rangle$ is centered Gaussian with, by independence of Y,Y',

$$\mathbb{E}\big[(U_{(u,v)}-U_{(s,t)})^2\big]=\mathbb{E}\big[\langle u-s,Y\rangle^2\big]+\mathbb{E}\big[\langle v-t,Y'\rangle^2\big]=\|u-s\|^2+\|v-t\|^2.$$

Thus $\mathbb{E}[(Z_{(u,v)}-Z_{(s,t)})^2]\le\mathbb{E}[(U_{(u,v)}-U_{(s,t)})^2]$ for all $(u,v),(s,t)\in K$, and the Sudakov–Fernique inequality (Theorem 13.3) gives $\mathbb{E}\sup_K Z\le\mathbb{E}\sup_K U$.

Finally, $\sup_{(u,v)\in K}(\langle u,Y\rangle+\langle v,Y'\rangle)=\sup_u\langle u,Y\rangle+\sup_v\langle v,Y'\rangle=\|Y\|+\|Y'\|$, since $\sup_{u\in S^{n-1}}\langle u,Y\rangle=\|Y\|$. Hence $\mathbb{E}U=2\mathbb{E}\|Y\|\le 2\sqrt{\mathbb{E}\|Y\|^2}=2\sqrt{n}$ by Jensen's inequality (Exercise 13.5), completing the proof. ■

Exercise 13.8

Central idea. The largest GUE eigenvalue is $\sup_{u\in S^{n-1}}u^*Xu$, the supremum of a centered real Gaussian process. By the hint its canonical metric satisfies $\mathbb{E}[(u^*Xu-v^*Xv)^2]=\frac{1}{n}\|uu^*-vv^*\|_{\mathrm{HS}}^2\le\frac{4}{n}\|u-v\|^2$, so Slepian's lemma (Theorem 13.3) compares it with the isonormal process $H_u=\frac{2}{\sqrt{n}}\langle u,Y\rangle$ on $\mathbb{R}^{2n}$, whose supremum equals $\frac{2}{\sqrt{n}}\|Y\|$.

Solution. Let X be an $n\times n$ GUE matrix: Hermitian with centered Gaussian entries. For $u\in S^{n-1}=\{u\in\mathbb{C}^n:\|u\|=1\}$, $G_u:=u^*Xu$ is real-valued (as X is Hermitian) and, being a linear functional of the Gaussian entries of X, $(G_u)_{u\in S^{n-1}}$ is a centered Gaussian process. Its supremum $Z=\sup_{u\in S^{n-1}}G_u$ is the largest eigenvalue of X.

By the hint, $\mathbb{E}[(G_u-G_v)^2]=\frac{1}{n}\|uu^*-vv^*\|_{\mathrm{HS}}^2$. Since $\|uu^*\|_{\mathrm{HS}}^2=\mathrm{tr}(uu^*uu^*)=(u^*u)^2=1$ and $\langle uu^*,vv^*\rangle_{\mathrm{HS}}=\mathrm{tr}(uu^*vv^*)=|u^*v|^2$,

$$\|uu^*-vv^*\|_{\mathrm{HS}}^2=2-2|u^*v|^2.$$

Writing $u^*v=re^{i\theta}$ with $0\le r\le 1$, $1-r^2=(1-r)(1+r)\le 2(1-r)\le 2(1-r\cos\theta)$, whence $\|uu^*-vv^*\|_{\mathrm{HS}}^2\le 8(1-\mathrm{Re}(u^*v))=4\|u-v\|^2$. Therefore

$$\mathbb{E}[(G_u-G_v)^2]\le\frac{4}{n}\|u-v\|^2\qquad u,v\in S^{n-1}.$$

Identify $\mathbb{C}^n\cong\mathbb{R}^{2n}$ and let Y be a standard Gaussian vector in $\mathbb{R}^{2n}$. The process $H_u:=\frac{2}{\sqrt{n}}\langle u,Y\rangle_{\mathbb{R}^{2n}}$ is centered Gaussian with $\mathbb{E}[(H_u-H_v)^2]=\frac{4}{n}\|u-v\|^2\ge\mathbb{E}[(G_u-G_v)^2]$ for all u,v. By Theorem 13.3 (Slepian's lemma) applied to finite subsets, and separability of S^{n-1},

$$\mathbb{E}\sup_{u\in S^{n-1}}G_u\le\mathbb{E}\sup_{u\in S^{n-1}}H_u.$$

But $\sup_{\|u\|=1}\langle u,Y\rangle=\|Y\|$, so $\mathbb{E}\sup_u H_u=\frac{2}{\sqrt{n}}\mathbb{E}\|Y\|\le\frac{2}{\sqrt{n}}\sqrt{\mathbb{E}\|Y\|^2}=\frac{2}{\sqrt{n}}\sqrt{2n}=2\sqrt{2}$.

Hence $\mathbb{E}Z\le 2\sqrt{2}$, a universal constant $C=2\sqrt{2}\ge 1$, independent of n.

Exercise 13.9

Central idea. Chaining telescopes $\sup_s(X_s-X_{s_0})$ into increments $X_{\Pi_{j+1}s}-X_{\Pi_j s}$ over refining nets. If the supremum exceeds uE, some increment must exceed its level threshold, and the thresholds are chosen to telescope to at most uE. Bernstein tails bound each increment; a union bound over at most $e^{2H(\delta_{j+1})}$ net pairs (as in Lemma 13.1) and summation over j gives κe^{-u}.

Solution. Work in the setting of Lemma 13.1 (the exercise's "Theorem 13.1"): T finite, $\log\mathbb{E}e^{\lambda(X_t-X_{t'})}\le a\lambda d(t,t')+\frac{v\lambda^2d^2(t,t')}{2(1-c\lambda d(t,t'))}$. Assume $\delta:=\max_t d(t,s_0)>0$ (if $\delta=0$ then $E=0$ and $X\equiv X_{s_0}$ a.s., a degenerate

case). Let $\delta_j = 2^{-j}\delta$, let T_j be δ_j-nets with $T_0 = \{s_0\}$, $\Pi_j : T \to T_j$ with $d(t, \Pi_j t) \le \delta_j$, and $H(\delta) = \log N(\delta, T)$ nonincreasing. Since T is finite, for J large $\Pi_{J+1}(s) = s$ and $X_s - X_{s_0} = \sum_{j=0}^{J}(X_{\Pi_{j+1}s} - X_{\Pi_j s})$ exactly, with $d(\Pi_{j+1}s, \Pi_j s) \le 3\delta_{j+1}$.

Remark on the hint. The threshold $\delta_{j+1}(3ua + 2\sqrt{uvL} + cuL)$ in the hint has slightly optimistic constants in the v, c terms to follow directly from Bernstein's inequality with $d \le 3\delta_{j+1}$. We use the corrected threshold $\mathrm{thr}_j' := 3\delta_{j+1}(a + \sqrt{2uvL_j} + cuL_j)$ with $L_j := j + 2H(\delta_{j+1})$, which rigorously yields the per-pair bound below and still telescopes to at most uE.

Per-pair tail. From the mgf hypothesis, $Z := X_s - X_t - ad(s,t)$ satisfies $\log \mathbb{E}e^{\lambda Z} \le \frac{(vd^2)\lambda^2}{2(1-(cd)\lambda)}$, so Bernstein's inequality (Chernoff bound) gives $\mathbb{P}\{X_s - X_t \ge ad(s,t) + d\sqrt{2v\ell} + cd\ell\} \le e^{-\ell}$. With $d \le 3\delta_{j+1}$ and $\ell = uL_j$,

$$\mathbb{P}\{X_{\Pi_{j+1}s} - X_{\Pi_j s} \ge \mathrm{thr}_j'\} \le e^{-uL_j},$$

since $\mathrm{thr}_j' \ge 3a\delta_{j+1} + 3\delta_{j+1}\sqrt{2uvL_j} + 3c\delta_{j+1}uL_j$.

Inclusion. If $X_s - X_{s_0} \ge uE$ then some increment exceeds its threshold, because $\sum_{j=0}^{J}\mathrm{thr}_j' \le uE$: indeed, with $H_j := H(\delta_j)$,

$$\begin{aligned}\sum_{j=0}^{J}\mathrm{thr}_j' &= 3a\sum_{j=0}^{J}\delta_{j+1} + 3\sqrt{2}\sum_{j=0}^{J}\delta_{j+1}\sqrt{uv(j+2H_{j+1})} + 3cu\sum_{j=0}^{J}\delta_{j+1}(j+2H_{j+1}) \\ &\le 3a\delta + 3\sqrt{2}\,\delta\sum_{k\ge 1}2^{-k}\sqrt{v(k+2H_k)} + 3cu\,\delta\sum_{k\ge 1}2^{-k}(k+2H_k) \\ &\le uE,\end{aligned}$$

reindexing $k = j+1$ (using $\sqrt{(k-1)+2H_k} \le \sqrt{k+2H_k}$) and, for $u \ge 1$, $3a \le 3ua$, $3\sqrt{2} < 6 \le 6u$, $3c \le 6cu$. Thus

$$\{\exists s : X_s - X_{s_0} \ge uE\} \subseteq \bigcup_{j=0}^{J}\bigcup_{s\in T}\{X_{\Pi_{j+1}s} - X_{\Pi_j s} \ge \mathrm{thr}_j'\}.$$

Union bound and summation. For fixed j there are at most $e^{H(\delta_j)+H(\delta_{j+1})} \le e^{2H_{j+1}}$ distinct pairs $(\Pi_{j+1}s, \Pi_j s)$ (as in Lemma 13.1's proof), so with the per-pair bound,

$$\mathbb{P}\{\sup_s X_s - X_{s_0} \ge uE\} \le \sum_{j=0}^{J} e^{2H_{j+1}}e^{-uL_j} = \sum_{j=0}^{J} e^{-uj}e^{-2(u-1)H_{j+1}}.$$

For $u > 1$: $= e^{-u}\big[e^{\,u-2(u-1)H_1} + \sum_{j\ge 1} e^{-u(j-1)}e^{-2(u-1)H_{j+1}}\big]$. Since $H_{j+1} \ge 0$, the sum over $j \ge 1$ is at most $\sum_{j\ge 1} e^{-u(j-1)} = \frac{1}{1-e^{-u}} \le \frac{1}{1-e^{-1}}$. Moreover $H_1 = H(\delta/2) \ge \log 2$: as $\delta > 0$, two balls of radius $\delta/4$ (around s_0 and a point at distance δ) are disjoint, so $N(\delta/2, T) \ge 2$. Then $u \mapsto u - 2(u-1)H_1$ is decreasing (as $H_1 > \frac{1}{2}$), hence $e^{\,u-2(u-1)H_1} \le e$ for $u \ge 1$.

Therefore $\mathbb{P}\{\sup_s X_s - X_{s_0} \ge uE\} \le \kappa e^{-u}$ with the universal constant $\kappa = e + \frac{1}{1-e^{-1}}$. Up to constants, E matches Lemma 13.1's expectation bound (the dyadic sums approximate the entropy integrals).

Exercise 13.10

Central idea. Each increment $X^T(A-B)X$ of the chaos, with $A-B$ symmetric of zero trace, is sub-gamma with variance factor $2\|A-B\|_{\mathrm{HS}}^2$ and scale $2\|A-B\|_{\mathrm{op}}$, so it has a mixed Gaussian/exponential tail in the two norms. Chaining along partitions generated by packing nets for both norms (as in the hint), with level-m thresholds of order $\varepsilon_{\mathrm{HS},m}\sqrt{u2^m} + u\,\varepsilon_{\mathrm{op},m}2^m$ and a union bound over the $e^{O(2^m)}$ links at level m, gives $Z \le KuE$ outside an event of probability at most κe^{-u}, after comparing the dyadic sums with the entropy integrals. However, the bound **as printed (with $K = 1$) is false**: a two-point index set has a tail decaying

like $e^{-0.763u}$ only. A universal factor K in front of E is necessary; we prove the statement with $K = 137$, $\kappa = 1.42$.

Erratum. The printed claim $\mathbb{P}\{Z \ge uE\} \le \kappa e^{-u}$ for all $u > 1$, with κ universal, is false. Take $n = 2$, $M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\mathcal{T} = \{0, M\}$, $A_0 = 0$. Then $\delta_{\mathrm{HS}} = \|M\|_{\mathrm{HS}} = \sqrt{2}$, $\delta_{\mathrm{op}} = \|M\|_{\mathrm{op}} = 1$, and (with the book's packing entropy, $N(t)$ the maximal number of points with pairwise distances $> t$) $H_\nu(t) = \log 2$ for $t < \delta_\nu$. Hence

$$E = \frac{\sqrt{2}}{2}\sqrt{2\log 2} + \frac{1}{2}\cdot 2\log 2 = \sqrt{\log 2} + \log 2 = 1.52570\ldots$$

Now $Z = \max(0, 2X_1X_2)$, and with $W_1 = (X_1 + X_2)/\sqrt{2}$, $W_2 = (X_1 - X_2)/\sqrt{2}$ (independent standard Gaussians) $2X_1X_2 = W_1^2 - W_2^2$. Therefore, with $t = \sqrt{uE+1}$ and the Mills-ratio lower bound $\mathbb{P}\{|g| \ge t\} \ge \sqrt{2/\pi}\,\frac{t}{1+t^2}e^{-t^2/2}$,

$$\mathbb{P}\{Z \ge uE\} \ge \mathbb{P}\{W_2^2 \le 1\}\,\mathbb{P}\{W_1^2 \ge uE + 1\} \ge 0.68 \cdot \sqrt{2/\pi}\,\frac{t}{1+t^2}\,e^{-1/2}e^{-uE/2}.$$

Since $E/2 = 0.7629 < 1$, $e^u\,\mathbb{P}\{Z \ge uE\} \to \infty$ as $u \to \infty$, so no universal (indeed no finite) κ exists. Numerically (exact law $\mathbb{P}\{X_1X_2 \ge a\} = \pi^{-1}\int_a^\infty K_0$), $e^u\,\mathbb{P}\{Z \ge uE\} = 0.588,\ 1.44,\ 11.3,\ 933,\ 8.8\times 10^6$ at $u = 5, 10, 20, 40, 80$. The same example shows that in a corrected statement $\mathbb{P}\{Z \ge KuE\} \le \kappa e^{-u}$ one needs $KE/2 \ge 1$, i.e. $K \ge 2/E = 1.31$.

Corrected statement. There are universal constants K, κ (one may take $K = 137$, $\kappa = 1.42$) such that, whenever $\mathcal{T}$ has at least two elements, for all $u > 1$,

$$\mathbb{P}\{Z \ge KuE\} \le \kappa e^{-u}.$$

Equivalently $\mathbb{P}\{Z \ge uE\} \le 1.42\,e^{-u/137}$ for $u > 137$. (If $|\mathcal{T}| = 1$ then $Z = 0 = E$ and $\mathbb{P}\{Z \ge uE\} = 1$, so $|\mathcal{T}| \ge 2$ is needed in any version.)

Solution. Throughout, $\nu \in \{\mathrm{HS}, \mathrm{op}\}$, $N_\nu(t)$ is the maximal cardinality of a subset of $\mathcal{T}$ with pairwise ν-distances $> t$, $H_\nu(t) = \log N_\nu(t)$ (nonincreasing in t), δ_ν is the ν-diameter, and

$$I_{\mathrm{HS}} = \int_0^{\delta_{\mathrm{HS}}/2} \sqrt{2H_{\mathrm{HS}}(t)}\,dt, \qquad I_{\mathrm{op}} = \int_0^{\delta_{\mathrm{op}}/2} 2H_{\mathrm{op}}(t)\,dt, \qquad E = I_{\mathrm{HS}} + I_{\mathrm{op}}.$$

We may assume $\mathcal{T}$ bounded and $E < \infty$ (otherwise there is nothing to prove, Z being finite when $\mathcal{T}$ is bounded since $|X^TAX| \le \|A\|_{\mathrm{op}}|X|^2$).

Step 1 (increments). Let M be symmetric with zero diagonal, eigenvalues λ_j, so $\sum_j \lambda_j = \mathrm{tr}M = 0$ and, by rotation invariance, $X^TMX \overset{d}{=} \sum_j \lambda_j(g_j^2 - 1)$ with g_j i.i.d. $N(0,1)$. For $y < \frac{1}{2}$, $\log \mathbb{E}e^{y(g^2-1)} = \phi(y) := -\frac{1}{2}\log(1-2y) - y$. For $0 \le y < \frac{1}{2}$, $\phi(y) = \sum_{k\ge 2}\frac{(2y)^k}{2k} \le \sum_{k\ge 2}\frac{(2y)^k}{4} = \frac{y^2}{1-2y}$; for $y < 0$, $\phi(0) = \phi'(0) = 0$ and $\phi''(y) = 2/(1-2y)^2 \le 2$, so $\phi(y) \le y^2$. Hence for $0 \le \theta < 1/(2\|M\|_{\mathrm{op}})$,

$$\log \mathbb{E}e^{\theta X^TMX} = \sum_j \phi(\theta\lambda_j) \le \frac{\theta^2\sum_j \lambda_j^2}{1 - 2\theta\|M\|_{\mathrm{op}}} = \frac{\theta^2 \cdot 2\|M\|_{\mathrm{HS}}^2}{2(1 - 2\|M\|_{\mathrm{op}}\theta)},$$

i.e. X^TMX is sub-gamma on the right tail with variance factor $v = 2\|M\|_{\mathrm{HS}}^2$ and scale $c = 2\|M\|_{\mathrm{op}}$ (Section 2.4, as in Example 2.12). The sub-gamma tail bound $\mathbb{P}\{Y > \sqrt{2vx} + cx\} \le e^{-x}$ gives, whenever $\|M\|_{\mathrm{HS}} \le a$ and $\|M\|_{\mathrm{op}} \le b$,

$$\mathbb{P}\big\{X^TMX > 2a\sqrt{x} + 2bx\big\} \le e^{-x} \qquad (x > 0). \tag{1}$$

For $A, B \in \mathcal{T}$, $A - B$ is symmetric with zero diagonal, so (1) applies to $Y_A - Y_B$ where $Y_A = X^TAX$.

Step 2 (reduction to finite $\mathcal{T}$). Suppose the corrected statement holds for every finite $\mathcal{T}'$ with $|\mathcal{T}'| \ge 2$. The space of symmetric $n \times n$ matrices is finite-dimensional, so $\mathcal{T}$ has a countable dense subset $\mathcal{T}_\infty \ni A_0$, and

$A \mapsto X^T(A-A_0)X$ is continuous, so $Z = \sup_{\mathcal{T}_\infty}$. Let $\mathcal{T}_k \uparrow \mathcal{T}_\infty$ be finite with $A_0 \in \mathcal{T}_k$, $|\mathcal{T}_k| \ge 2$, and Z_k the corresponding suprema, so $Z_k \uparrow Z$. Packing numbers and diameters are monotone under inclusion, hence $E(\mathcal{T}_k) \le E$. Thus $\mathbb{P}\{Z > KuE\} = \lim_k \mathbb{P}\{Z_k > KuE\} \le \limsup_k \mathbb{P}\{Z_k \ge KuE(\mathcal{T}_k)\} \le \kappa e^{-u}$, and $\mathbb{P}\{Z \ge KuE\} \le \mathbb{P}\{Z > Ku'E\} \le \kappa e^{-u'}$ for every $u' \in (1,u)$; letting $u' \uparrow u$ gives the claim. So assume from now on that $\mathcal{T}$ is finite with $|\mathcal{T}| \ge 2$; then $\delta_\nu > 0$ and $E > 0$.

Step 3 (nets and partitions). For $m \ge 0$ let $\varepsilon_{\nu,m} = \inf\{\varepsilon > 0 : H_\nu(\varepsilon) \le 2^m\}$; this is nonincreasing in m, and by monotonicity of H_ν: $H_\nu(t) > 2^m$ for $0 < t < \varepsilon_{\nu,m}$, while $H_\nu(\eta) \le 2^m$ for every $\eta > \varepsilon_{\nu,m}$. Also $\varepsilon_{\nu,m} = 0$ as soon as $2^m \ge \log|\mathcal{T}|$; let m^* be the least such m. Fix $\gamma > 0$ smaller than every nonzero ν-distance between points of $\mathcal{T}$ (both ν), and put $\eta_{\nu,m} = \varepsilon_{\nu,m} + \gamma$. Let $S_{\nu,m} \subset \mathcal{T}$ be a maximal $\eta_{\nu,m}$-separated set: $|S_{\nu,m}| \le N_\nu(\eta_{\nu,m}) \le e^{2^m}$, and by maximality every $A \in \mathcal{T}$ is within ν-distance $\eta_{\nu,m}$ of $S_{\nu,m}$. For $m \ge m^*$, $\eta_{\nu,m} = \gamma$ and $S_{\nu,m} = \mathcal{T}$. Let $\pi_{\nu,m} : \mathcal{T} \to S_{\nu,m}$ be a fixed nearest-point map, $\mathcal{B}_m, \mathcal{C}_m$ the partitions of $\mathcal{T}$ into the fibres of $\pi_{\mathrm{op},m}$, $\pi_{\mathrm{HS},m}$, $\mathcal{A}_0 = \{\mathcal{T}\}$ and $\mathcal{A}_m = \mathcal{B}_{m-1} \wedge \mathcal{C}_{m-1}$ for $m \ge 1$. Then $|\mathcal{A}_m| \le e^{2^{m-1}} e^{2^{m-1}} = e^{2^m}$. Let $\Pi_m A \in \mathcal{T}$ be a distinguished point of the cell of $\mathcal{A}_m$ containing A, with $\Pi_0 A = A_0$. For $m \ge 1$, A and $\Pi_m A$ have the same image under $\pi_{\nu,m-1}$, so $\|\Pi_m A - A\|_\nu \le 2\eta_{\nu,m-1}$. With $J = m^* + 1$, the cells of $\mathcal{A}_J$ are singletons, so $\Pi_J A = A$ and

$$Y_A - Y_{A_0} = \sum_{m=0}^{J-1} (Y_{\Pi_{m+1}A} - Y_{\Pi_m A}).$$

Link sizes: $\|\Pi_1 A - A_0\|_\nu \le \delta_\nu$; for $m \ge 1$, $\|\Pi_{m+1}A - \Pi_m A\|_\nu \le 2\eta_{\nu,m} + 2\eta_{\nu,m-1} \le 4\eta_{\nu,m-1}$. Link counts: at level 0 the pair $(\Pi_0 A, \Pi_1 A)$ takes at most $|\mathcal{A}_1| \le e^2$ values; at level $m \ge 1$, $(\Pi_m A, \Pi_{m+1} A)$ takes at most $|\mathcal{A}_m||\mathcal{A}_{m+1}| \le e^{3\cdot 2^m}$ values.

Step 4 (union bound). Fix $u \ge 1$. Put $x_0 = u + 2$ and $x_m = (u+3)2^m$ for $m \ge 1$, and thresholds

$$t_0 = 2\delta_{\mathrm{HS}}\sqrt{x_0} + 2\delta_{\mathrm{op}} x_0, \qquad t_m = 8\eta_{\mathrm{HS},m-1}\sqrt{x_m} + 8\eta_{\mathrm{op},m-1} x_m \ (m \ge 1).$$

Let Ω_{bad} be the event that some link $Y_{\Pi_{m+1}A} - Y_{\Pi_m A}$ ($A \in \mathcal{T}$, $0 \le m < J$) exceeds t_m. By (1) and the link counts,

$$\mathbb{P}(\Omega_{\mathrm{bad}}) \le e^2 e^{-(u+2)} + \sum_{m\ge1} e^{3\cdot 2^m} e^{-(u+3)2^m} = e^{-u} + \sum_{m\ge1} e^{-u2^m} \le e^{-u}\Big(1 + \sum_{m\ge1} e^{-(2^m-1)}\Big) \le 1.42\, e^{-u},$$

using $u(2^m - 1) \ge 2^m - 1$ for $u \ge 1$ and $\sum_{m\ge1} e^{-(2^m-1)} = 0.41858\ldots$

Step 5 (the good event). Outside Ω_{bad}, $Z \le \sum_{m=0}^{J-1} t_m$. For $u \ge 1$: $x_0 \le 3u$ and $x_m \le 4u2^m$, and $\sqrt{u} \le u$, so $t_0 \le u(2\sqrt{3}\,\delta_{\mathrm{HS}} + 6\delta_{\mathrm{op}})$ and, for $m \ge 1$, with $k = m - 1$,

$$t_m \le 8\eta_{\mathrm{HS},k} \cdot 2\sqrt{u}\, 2^{(k+1)/2} + 8\eta_{\mathrm{op},k} \cdot 4u\, 2^{k+1} \le u(16\sqrt{2}\,\eta_{\mathrm{HS},k} 2^{k/2} + 64\,\eta_{\mathrm{op},k} 2^k).$$

Hence, outside Ω_{bad}, $Z \le uB_\gamma$, where

$$B_\gamma = 2\sqrt{3}\,\delta_{\mathrm{HS}} + 6\delta_{\mathrm{op}} + \sum_{k=0}^{m^*} \Big(16\sqrt{2}\,(\varepsilon_{\mathrm{HS},k} + \gamma)2^{k/2} + 64(\varepsilon_{\mathrm{op},k} + \gamma)2^k\Big).$$

So $\mathbb{P}\{Z > uB_\gamma\} \le 1.42e^{-u}$ for every admissible γ. As $\gamma \downarrow 0$, $B_\gamma \downarrow B_0$ (the same expression with $\gamma = 0$; m^* does not depend on γ), and $\{Z > uB_0\} = \bigcup_\gamma \{Z > uB_\gamma\}$ is an increasing union, so $\mathbb{P}\{Z > uB_0\} \le 1.42e^{-u}$ for all $u \ge 1$.

Step 6 (comparison with the integrals). Diameters: two points at distance δ_ν are t-separated for $t < \delta_\nu$, so $H_\nu(t) \ge \log 2$ there. Hence $I_{\mathrm{HS}} \ge \frac{\delta_{\mathrm{HS}}}{2}\sqrt{2\log 2}$ and $I_{\mathrm{op}} \ge \delta_{\mathrm{op}} \log 2$, i.e.

$$\delta_{\mathrm{HS}} \le \sqrt{2/\log 2}\; I_{\mathrm{HS}} \le 1.699\, I_{\mathrm{HS}}, \qquad \delta_{\mathrm{op}} \le I_{\mathrm{op}}/\log 2 \le 1.443\, I_{\mathrm{op}}.$$

Sums: write $\varepsilon_k = \varepsilon_{\nu,k}$, which vanishes for $k \ge m^*$. Since $H_\nu(t) > 2^j$ on $(0, \varepsilon_j)$, $\int_{\varepsilon_{j+1}}^{\varepsilon_j} \sqrt{H_\nu} \ge (\varepsilon_j - \varepsilon_{j+1})2^{j/2}$ and $\int_{\varepsilon_{j+1}}^{\varepsilon_j} H_\nu \ge (\varepsilon_j - \varepsilon_{j+1})2^j$. Also $\varepsilon_0 \le \delta_\nu$ (as $H_\nu(t) = 0$ for $t \ge \delta_\nu$), and since H_ν is nonincreasing,

$\int_0^{\delta_\nu} f(H_\nu) \le 2\int_0^{\delta_\nu/2} f(H_\nu)$ for f nondecreasing. Summation by parts ($\varepsilon_k = \sum_{j\ge k}(\varepsilon_j - \varepsilon_{j+1})$) gives

$$\sum_{k\ge 0}\varepsilon_k 2^{k/2} = \sum_{j\ge 0}(\varepsilon_j - \varepsilon_{j+1})\sum_{k\le j} 2^{k/2} \le \frac{1}{1-2^{-1/2}}\sum_j (\varepsilon_j-\varepsilon_{j+1})2^{j/2} \le \frac{1}{1-2^{-1/2}}\int_0^{\delta_{\mathrm{HS}}}\sqrt{H_{\mathrm{HS}}} \le \frac{\sqrt{2}}{1-2^{-1/2}} I_{\mathrm{HS}} \le 4.829\, I_{\mathrm{HS}}$$

for $\nu = \mathrm{HS}$, and for $\nu = \mathrm{op}$, using $\sum_{k\le j} 2^k \le 2^{j+1}$,

$$\sum_{k\ge 0}\varepsilon_k 2^k \le 2\int_0^{\delta_{\mathrm{op}}} H_{\mathrm{op}} \le 4\int_0^{\delta_{\mathrm{op}}/2} H_{\mathrm{op}} = 2I_{\mathrm{op}}.$$

Therefore

$$B_0 \le (2\sqrt{3}\cdot 1.699 + 16\sqrt{2}\cdot 4.829) I_{\mathrm{HS}} + (6\cdot 1.443 + 64\cdot 2) I_{\mathrm{op}} \le 115.2\, I_{\mathrm{HS}} + 136.7\, I_{\mathrm{op}} \le 137\, E.$$

Conclusion. For $u > 1$ and any $u' \in (1, u)$, by Steps 5–6, $\mathbb{P}\{Z \ge 137uE\} \le \mathbb{P}\{Z \ge uB_0\} \le \mathbb{P}\{Z > u'B_0\} \le 1.42e^{-u'}$ (using $B_0 > 0$); letting $u' \uparrow u$,

$$\mathbb{P}\{Z \ge 137\, uE\} \le 1.42\, e^{-u} \qquad (u > 1),$$

which is the corrected statement with $K = 137$, $\kappa = 1.42$. By the erratum, some factor $K \ge 1.31$ is unavoidable. ■

Exercise 13.11

Central idea. Haussler's probabilistic method: if $A_1, \dots, A_N$ are δ-separated in $L_2(Q)$, then $n \asymp (\log N)/\delta^2$ random samples from Q distinguish all N traces with positive probability, so $\log N$ is bounded by the VC-entropy $V\log(en/V)$ (Sauer). Unwinding the implicit inequality $L \le V\log(4eL/(V\delta^2))$ for $L = \log N$ gives an explicit bound — but the V in the denominator inside the logarithm as stated makes the claimed inequality false.

Status: repaired. The bound as originally stated is false — counterexample: $\mathcal{X} = [10]$, $\mathcal{A} = 2^{[10]}$ (VC-dimension $V = 10$), Q uniform, $\delta = 1$, $\kappa = 2$: then $\emptyset$ and $[10]$ satisfy $\|1_\emptyset - 1_{[10]}\|^2_{L_2(Q)} = 1$, so $H(1,\mathcal{A}) \ge \log 2 > 0$, while the stated bound gives $20\log(0.4) < 0$. The V in the denominator of $\frac{2\kappa}{V\delta^2}$ is a typo; the argument below proves the corrected bound $H(\delta,\mathcal{A}) \le \frac{\kappa}{\kappa-1} V\log\left(\frac{4e\kappa}{\delta^2}\right)$ for all $\kappa > 1$, which is sound.

Solution. (of the corrected bound) Let $H(\delta,\mathcal{A}) = \sup_Q \log N(\delta,\mathcal{A},L_2(Q))$ be the universal entropy, $N(\delta,\cdot)$ the covering number. If $\delta > 1$ then no two distinct sets are δ-separated (as $\|1_{A_i} - 1_{A_j}\|^2_{L_2(Q)} = Q(A_i \triangle A_j) \le 1 < \delta^2$), so $H(\delta,\mathcal{A}) = 0$ and there is nothing to prove; assume $\delta \le 1$. A maximal δ-separated set is a δ-cover, so it suffices to bound $\log N$ for N δ-separated sets $A_1, \dots, A_N$. If $N = 1$ there is nothing to prove; assume $N \ge 2$ (hence $V \ge 1$, as two distinct sets shatter the singleton $\{x\}$ for any $x \in A_1 \triangle A_j$).

Fix Q and let $A_1, \dots, A_N \in \mathcal{A}$ be δ-separated: $Q(A_i \triangle A_j) = \|1_{A_i} - 1_{A_j}\|^2_{L_2(Q)} \ge \delta^2$ for $i \ne j$. Put $n = \lfloor 4\log N/\delta^2 \rfloor \ge 2$ and draw $X_1, \dots, X_n$ i.i.d. $\sim Q$, $S = \{X_1, \dots, X_n\}$. For fixed $i \ne j$,

$$\mathbb{P}\{A_i \cap S = A_j \cap S\} = (1 - Q(A_i\triangle A_j))^n \le (1-\delta^2)^n \le e^{-n\delta^2} \le \frac{e^{\delta^2}}{N^4} \le \frac{e}{N^4},$$

since $n \ge 4\log N/\delta^2 - 1$. By the union bound over $\binom{N}{2}$ pairs,

$$\mathbb{P}\{\text{all } A_i \cap S \text{ distinct}\} \ge 1 - \binom{N}{2}\frac{e}{N^4} \ge 1 - \frac{e}{2N^2} > 0 \qquad (N \ge 2).$$

Hence some realization has $|\{A \cap S : A \in \mathcal{A}\}| \ge N$. If $n \ge V$, Sauer's lemma gives $|\{A \cap S\}| \le (en/V)^V$, so with $n \le 4\log N/\delta^2$,

$$\log N \le V\log(en/V) \le V\log\left(\frac{4e\log N}{V\delta^2}\right). \qquad (*)$$

Put $L = \log N$ and $t = L/V$, so $t \le \log(4et/\delta^2) = \log(4e/\delta^2) + \log t$. Fix $\kappa > 1$. If $t \le 1$ then $L \le V \le \frac{\kappa}{\kappa-1} V \log(4e\kappa/\delta^2)$ since $\frac{\kappa}{\kappa-1} > 1$ and $\log(4e\kappa/\delta^2) > 1$ (as $\delta \le 1$, $\kappa > 1$). If $t > 1$, use $\log t \le t/\kappa + \log\kappa - 1$:

$$t \le \log(4e/\delta^2) + t/\kappa + \log\kappa - 1, \quad \text{so} \quad t\left(1 - \frac{1}{\kappa}\right) \le \log(4e\kappa/\delta^2),$$

giving $t \le \frac{\kappa}{\kappa-1}\log(4e\kappa/\delta^2)$, i.e. $L \le \frac{\kappa}{\kappa-1} V \log(4e\kappa/\delta^2)$. If $n < V$ then $4L/\delta^2 < V + 1$ (as $n = \lfloor 4L/\delta^2 \rfloor$), so $L < (V+1)\delta^2/4 \le (V+1)/4 \le V \le \frac{\kappa}{\kappa-1} V \log(4e\kappa/\delta^2)$ as above (using $V \ge 1$). Thus in all cases $\log N \le \frac{\kappa}{\kappa-1} V \log(4e\kappa/\delta^2)$, and taking $\sup_Q$ gives $H(\delta, \mathcal{A}) \le \frac{\kappa}{\kappa-1} V \log(4e\kappa/\delta^2)$ for all $\kappa > 1$. ■

Exercise 13.12

Central idea. Identify $\mathrm{tr}(x)$ with a set $S \subseteq \{0,1\}^n$; the 1-inclusion graph is the subgraph of the cube induced by S. Every bound comes from splitting S along one coordinate: $S_0, S_1 \subseteq \{0,1\}^{n-1}$ (the two fibres), the cross edges are in bijection with $S_0 \cap S_1$. For (a) this gives a recursion closed by an elementary calculus inequality. For (b) the overlap $S_0 \cap S_1$ has VC-dimension at most $V-1$, which gives the Haussler–Littlestone–Warmuth bound $|E| \le V(|S| - 1)$, i.e. dens $< V$. **The printed claim** dens $\le V/2$ **is false** (thresholds give a path with density $n/(n+1)$, VC-dimension 1); the correct bound V is asymptotically tight but never attained. **The printed claim that Z is self-bounding is also false**: the leave-one-out density can exceed Z, and a single change of one sample point can lower dens by $1.5 > 1$, which no self-bounding function allows. What is true is that the VC-entropy $\log_2 |\mathrm{tr}(x)|$ is self-bounding, and it dominates $2\,\mathrm{dens}$ by (a).

Errata.

1. dens $\le V/2$ *is false.* Let $\mathcal{X} = \mathbb{R}$, $\mathcal{C} = \{(-\infty, a] : a \in \mathbb{R}\}$ (thresholds; VC-dimension 1: one point is shattered, and for $y < z$ no set contains z but not y). For $x_1 < \cdots < x_n$ the trace consists of the $n+1$ prefixes $\{1, \ldots, k\}$, $0 \le k \le n$, and consecutive prefixes differ in one coordinate, so the 1-inclusion graph is a path: dens $= n/(n+1) > 1/2$ for $n \ge 2$ (e.g. 2/3, 3/4, 10/11 for $n = 2, 3, 10$; checked by enumeration). Similarly $\mathcal{C} = \{\emptyset\} \cup \{\{y\} : y \in \mathcal{X}\}$ (VC-dimension 1) gives a star with density $n/(n+1)$. The correct bound is dens $\le V$ (proved below, in the sharper form dens $\le V(1 - 1/|\mathrm{tr}(x)|)$). The value $V/2$ is the density of the full V-cube, which is not extremal.

2. *Z is not self-bounding.* Recall (Section 3.3) that $f : \mathcal{X}^n \to [0, \infty)$ is self-bounding if there are $f_i : \mathcal{X}^{n-1} \to \mathbb{R}$ with $0 \le f(x) - f_i(x^{(i)}) \le 1$ and $\sum_i (f(x) - f_i(x^{(i)})) \le f(x)$ for all x.

   - *The natural leave-one-out choice fails.* Take $f_i =$ dens on the sample with x_i deleted. Let $\mathcal{X} = \{1,2,3\}$, $x = (1,2,3)$, and let $\mathcal{C}$ consist of the five sets with indicator patterns $S = \{000, 001, 011, 110, 111\}$. Its edges are $000{-}001$, $001{-}011$, $011{-}111$, $110{-}111$, so $Z = 4/5 = 0.8$. Deleting coordinate 1 or 2 leaves the full square $\{00, 01, 10, 11\}$ (density 1), deleting coordinate 3 leaves the path $\{00, 01, 11\}$ (density 2/3). So $Z_i = (1, 1, 0.667)$ and $Z - Z_1 = -0.2 < 0$, violating $f - f_i \ge 0$.
   - *No choice of f_i works.* If f is self-bounding and x, x' differ only in coordinate i, then $x^{(i)} = x'^{(i)}$ and $f(x) - f(x') = (f(x) - f_i(x^{(i)})) - (f(x') - f_i(x'^{(i)})) \le 1 - 0 = 1$. Let $\mathcal{X} = \{1, 2, a, b\}$ and $\mathcal{C} = \{B \cup T \cup P(B) : B \subseteq \{1,2\}, T \subseteq \{a\}\}$ with $P(B) = \{b\}$ if $|B|$ is odd and $P(B) = \emptyset$ otherwise. On $x = (1, 2, a)$ the trace is all of $\{0,1\}^3$: dens $= 12/8 = 1.5$. On $x' = (1, 2, b)$ the trace is $\{000, 011, 101, 110\}$ (third bit = parity of the first two), which has no edges: dens $= 0$. So $f(x) - f(x') = 1.5 > 1$, and dens is not self-bounding for any choice of the f_i. If X_1, X_2, X_3 are i.i.d. uniform on $\mathcal{X}$, both samples have positive probability, so the failure cannot be avoided by ignoring null sets. (All numbers above were checked by direct enumeration.)

Solution. Throughout, $S \subseteq \{0,1\}^n$ is nonempty, $E(S)$ is the set of edges of the subgraph of the cube induced by S (pairs of elements of S at Hamming distance 1), and dens $= |E(S)|/|S|$. For $S = \mathrm{tr}(x)$ this is $\mathrm{dens}(x)$ (we assume $\mathcal{C} \ne \emptyset$ so that $\mathrm{tr}(x) \ne \emptyset$). For $n \ge 1$ put

$$S_b = \{y \in \{0,1\}^{n-1} : (y, b) \in S\} \ (b = 0, 1), \qquad U = S_0 \cup S_1, \qquad C = S_0 \cap S_1.$$

An edge of S either changes the last coordinate — it is then $\{(y,0), (y,1)\}$ with $y \in C$ — or it joins $(y, b), (y', b)$ with $\{y, y'\} \in E(S_b)$. Hence

$$|E(S)| = |E(S_0)| + |E(S_1)| + |C|, \qquad |S| = |S_0| + |S_1| = |U| + |C|. \tag{1}$$

(a) $\mathrm{dens}(x) \le \frac{1}{2}\log_2|\mathrm{tr}(x)|$**.** We prove $|E(S)| \le \frac{1}{2}|S|\log_2|S|$ for every $S \subseteq \{0,1\}^n$ (with $0\log 0 = 0$) by induction on n. For $n=0$, $|S| \le 1$ and $|E(S)| = 0$. Let $n \ge 1$, $m_b = |S_b|$, $m = m_0 + m_1$. If $m_0 = 0$ or $m_1 = 0$, $|E(S)| = |E(S_{1-b})|$ and the claim is the induction hypothesis. Otherwise, by symmetry assume $1 \le m_0 \le m_1$; then $|C| \le m_0$ and by (1) and induction

$$|E(S)| \le \tfrac{m_0}{2}\log_2 m_0 + \tfrac{m_1}{2}\log_2 m_1 + m_0.$$

With $r = m_1/m_0 \ge 1$,

$$\tfrac{m}{2}\log_2 m - \tfrac{m_0}{2}\log_2 m_0 - \tfrac{m_1}{2}\log_2 m_1 = \tfrac{m_0}{2}\log_2(1+r) + \tfrac{m_1}{2}\log_2(1+1/r) = \tfrac{m_0}{2}f(r),$$

where $f(r) = \log_2(1+r) + r\log_2(1+1/r)$. Now $f(1) = 2$ and

$$f'(r) = \frac{1}{(1+r)\ln 2} + \log_2(1+1/r) - \frac{1}{(1+r)\ln 2} = \log_2(1+1/r) > 0,$$

so $f(r) \ge 2$ for $r \ge 1$, i.e. $\frac{m_0}{2}f(r) \ge m_0$. Hence $|E(S)| \le \frac{m}{2}\log_2 m$. Applied to $S = \mathrm{tr}(x)$ this is $\mathrm{dens}(x) \le \frac{1}{2}\log_2|\mathrm{tr}(x)|$.

(b) VC classes: $\mathrm{dens}(x) \le V(1 - 1/|\mathrm{tr}(x)|) < V$**.** Say S *shatters* $Y \subseteq \{1,\dots,n\}$ if $\{(y_i)_{i\in Y} : y \in S\} = \{0,1\}^Y$, and let $\mathrm{vc}(S)$ be the largest size of a shattered set.

Trace. If $\mathcal{C}$ has VC-dimension V, then $\mathrm{vc}(\mathrm{tr}(x)) \le V$: if $\mathrm{tr}(x)$ shatters Y, the points x_i, $i \in Y$, are distinct (if $x_i = x_j$ with $i \ne j$ in Y, no pattern has i in and j out), and $\{x_i : i \in Y\}$ is shattered by $\mathcal{C}$, so $|Y| \le V$.

Claim. For every nonempty $S \subseteq \{0,1\}^n$ with $\mathrm{vc}(S) \le V$, $|E(S)| \le V(|S|-1)$.

Proof by induction on n, for all $V \ge 0$ simultaneously. If $V = 0$, $|S| = 1$ (two distinct patterns differ in some coordinate i, and then $\{i\}$ is shattered), so $|E(S)| = 0$. If $n = 0$, $|S| = 1$ and $|E(S)| = 0$. Let $n \ge 1$, $V \ge 1$. Three facts:

- $|E(S_0)| + |E(S_1)| \le |E(U)| + |E(C)|$. Indeed, for an edge e of the $(n-1)$-cube let $a = \mathbf{1}[e \subseteq S_0]$, $b = \mathbf{1}[e \subseteq S_1]$; then $\mathbf{1}[e \subseteq U] \ge \max(a,b)$ and $\mathbf{1}[e \subseteq C] = ab = \min(a,b)$, so $a+b \le \mathbf{1}[e \subseteq U] + \mathbf{1}[e \subseteq C]$. Sum over e.
- $\mathrm{vc}(U) \le V$: a set $Y \subseteq \{1,\dots,n-1\}$ shattered by U is shattered by S.
- $\mathrm{vc}(C) \le V - 1$ when $C \ne \emptyset$: if C shatters Y, then for every pattern on Y there is $y \in C$ realizing it, and $(y,0),(y,1) \in S$, so S shatters $Y \cup \{n\}$; thus $|Y| + 1 \le V$.

U is nonempty. If $C = \emptyset$, then by (1) and the first fact $|E(S)| \le |E(U)| \le V(|U|-1) = V(|S|-1)$. If $C \ne \emptyset$, the induction hypothesis (dimension $n-1$; parameters V and $V-1 \ge 0$) gives

$$|E(S)| \le |E(U)| + |E(C)| + |C| \le V(|U|-1) + (V-1)(|C|-1) + |C| = V|S| - 2V + 1 \le V(|S|-1),$$

using $|S| = |U| + |C|$ and $V \ge 1$. $\square$

Applying the claim to $S = \mathrm{tr}(x)$: $\mathrm{dens}(x) \le V(1 - 1/|\mathrm{tr}(x)|) < V$. In particular $\mathrm{dens}(x) \le V$ (Haussler, Littlestone and Warmuth, 1994). The weaker bound $|E(S)| \le V|S|$ can also be obtained by down-shifting: shifts do not decrease the edge count or increase vc, and for a down-closed S, $|E(S)| = \sum_{y\in S}|y|$ with all weights $|y| \le V$.

Is the bound tight? The printed bound $V/2$ is not even valid (Erratum 1). For the correct bound:

- *V is asymptotically tight.* Let $\mathcal{X}$ be infinite and $\mathcal{C} = \{A \subseteq \mathcal{X} : |A| \le V\}$, which has VC-dimension V (every V-set is shattered; a $(V+1)$-set is not, since it is not itself traced). On $n \ge V$ distinct points the trace is $\{J : |J| \le V\}$; every J with $|J| = k \ge 1$ has its k down-neighbours in the trace, so counting edges by their upper endpoint,

$$\mathrm{dens} = \frac{\sum_{k=0}^{V} k\binom{n}{k}}{\sum_{k=0}^{V}\binom{n}{k}} \xrightarrow[n\to\infty]{} V,$$

because $\binom{n}{k}/\binom{n}{V} \to 0$ for $k < V$. (Numerically: 0.9995, 1.9990, 2.9985, 4.9975 for $V = 1,2,3,5$ at $n = 2000$; for $V = 1$ this is the star, $n/(n+1)$.) So no bound cV with $c < 1$ holds, and $\sup_x \mathrm{dens}(x) = V$ for this class.

- *V is never attained*: by the claim, dens $\le V - V/|\mathrm{tr}(x)| < V$ for every finite sample. For $V = 1$ the claim says the 1-inclusion graph is a forest (a cycle on k vertices of S would give a subset with vc ≤ 1 and at least $k > k-1$ induced edges), and the path/star examples show $1 - 1/|\mathrm{tr}(x)|$ is attained.
- *The full V-cube* ($\mathcal{C} = 2^{\{1,\dots,V\}}$, $x = (1, \dots, V)$) has density exactly $V/2$; this is the source of the printed value, but it is not the extremal configuration.

(Exhaustive enumeration of all nonempty $S \subseteq \{0,1\}^n$ for $n \le 4$, plus 15,000 random subsets for $n = 5, 6$: no violation of $|E(S)| \le \mathrm{vc}(S)(|S| - 1)$ or of (a).)

(c) Self-bounding. As shown in Erratum 2, $Z = \mathrm{dens}(X_1, \dots, X_n)$ is in general *not* self-bounding: a self-bounding function changes by at most 1 when one coordinate changes, while dens can drop from 1.5 to 0. What does hold is the following, which controls Z from above.

The VC-entropy $h(x) = \log_2 |\mathrm{tr}(x)|$ is self-bounding, with $h_i(x^{(i)}) = \log_2 |\mathrm{tr}(x^{(i)})|$, where $\mathrm{tr}(x^{(i)})$ is the trace on the sample with x_i deleted, i.e. the projection of $\mathrm{tr}(x)$ deleting coordinate i.

- $0 \le h - h_i \le 1$: the projection $\mathrm{tr}(x) \to \mathrm{tr}(x^{(i)})$ is onto and each fibre has one or two elements, so $|\mathrm{tr}(x^{(i)})| \le |\mathrm{tr}(x)| \le 2|\mathrm{tr}(x^{(i)})|$.
- $\sum_i (h - h_i) \le h$: let $Y = (Y_1, \dots, Y_n)$ be uniform on $\mathrm{tr}(x)$, so its Shannon entropy (base 2) is $H(Y) = h(x)$, and $Y^{(i)}$ takes values in $\mathrm{tr}(x^{(i)})$, so $H(Y^{(i)}) \le h_i$. Han's inequality (Theorem 4.1), $\sum_{i=1}^n (H(Y) - H(Y^{(i)})) \le H(Y)$, gives $\sum_i (h - h_i) \le \sum_i (H(Y) - H(Y^{(i)})) \le h$.

So $h(X_1, \dots, X_n)$ is a self-bounding random variable, and by (a) $Z \le h/2$ pointwise. Consequently the upper-tail inequalities for self-bounding functions apply to h and transfer to Z through $\{Z \ge t\} \subseteq \{h \ge 2t\}$; for a VC class additionally $Z < V$ by (b). Nothing in this solution establishes a lower-tail or variance bound for Z itself. ■

Exercise 13.13

Central idea. Turn $L_2(Q)$-packing of $\mathcal{F}$ into L_1-packing of the subgraphs under one explicit probability measure on $\mathcal{X} \times \mathbb{R}$, then apply Haussler's bound (Lemma 13.6) to the subgraph class, which is a VC class. The measure is $R(dx, dt) = \frac{F(x)\mathbb{1}\{|t| \le F(x)\}}{2\|F\|_Q^2}\, Q(dx)\, dt$. It turns $R(C_f \triangle C_{f'})$ into $Q[F|f - f'|]/(2\|F\|_Q^2)$, and the envelope inequality $|f - f'|^2 \le 2F|f - f'|$ then gives $\|f - f'\|_Q^2 \le 4\|F\|_Q^2\, R(C_f \triangle C_{f'})$. So $L_2(Q)$-separation $\delta\|F\|_Q$ forces $L_2(R)$-separation $\delta/2$ of the subgraph indicators. Haussler's bound at scale $\delta/2$ gives $e(V+1)(2e/\delta)^{2V} \le e(V+1)(16e)^{V+1}\delta^{-2V}$.

Solution. *Conventions and remarks on the statement.* As in Section 13.3, $N(\varepsilon, \mathcal{F}, Q)$ is the packing number: the largest N for which there are $f_1, \dots, f_N \in \mathcal{F}$ with $\|f_i - f_j\|_Q > \varepsilon$ for all $i \ne j$. The members of $\mathcal{F}$ and F are measurable, which makes every subgraph $C_f = \{(x,t) : t < f(x)\}$ a measurable subset of $\mathcal{X} \times \mathbb{R}$. The exercise has three printing slips. "Theorem 13.6" is the book's Lemma 13.6 (Haussler's VC bound for universal entropy). The "$\wedge$" in the hint should be "$\vee$", because the set in the hint is $C_f \triangle C_{f'}$. And κ is a universal constant: we show that $\kappa = e$ works for $0 < \delta \le 2$. For $\delta > 2$, $N = 1$ trivially, since $\|f - f'\|_Q \le 2\|F\|_Q$. (With $V \ge 1$ the printed right side drops below 1 once δ is very large, for example $\delta > 101.5$ when $V = 1$ and $\kappa = e$. So the bound only makes sense for bounded δ, which is where it matters.)

If $\|F\|_Q = 0$, then every $f \in \mathcal{F}$ vanishes Q-a.e., all $L_2(Q)$ distances are 0, and $N = 1$. If $\|F\|_Q = \infty$, then no pair is ∞-separated and again $N = 1$. Both cases satisfy the bound, so assume $0 < \|F\|_Q < \infty$ and $0 < \delta \le 2$.

Step 1: an auxiliary probability measure. On $\mathcal{X} \times \mathbb{R}$ define

$$R(dx, dt) = \frac{F(x)\,\mathbb{1}\{|t| \le F(x)\}}{2\|F\|_Q^2}\, Q(dx)\, dt.$$

By Fubini its total mass is $\int F(x) \cdot 2F(x)\, Q(dx)/(2\|F\|_Q^2) = 1$, so R is a probability measure. Take $f, f' \in \mathcal{F}$. The x-section of $C_f \triangle C_{f'}$ is the interval between $f(x)$ and $f'(x)$ (closed at the lower end, open at the upper end). It has Lebesgue length $|f(x) - f'(x)|$ and lies in $[-F(x), F(x)]$ because $|f|, |f'| \le F$. On that interval

the density of R in t is the constant $F(x)/(2\|F\|_Q^2)$. Fubini therefore gives

$$R(C_f \triangle C_{f'}) = \frac{Q[F\,|f-f'|]}{2\|F\|_Q^2}.$$

This is the hint's layer-cake identity, weighted by F.

Step 2: comparing $L_2(Q)$ with $L_1(R)$. Since $|f-f'| \le 2F$ pointwise,

$$\|f-f'\|_Q^2 = Q|f-f'|^2 \le 2\,Q[F|f-f'|] = 4\|F\|_Q^2\, R(C_f \triangle C_{f'}).$$

So $\|f-f'\|_Q > \delta\|F\|_Q$ implies $R(C_f \triangle C_{f'}) > \delta^2/4 = (\delta/2)^2$. Now let $f_1, \dots, f_N$ be $\delta\|F\|_Q$-separated in $L_2(Q)$. Their subgraphs $C_i = C_{f_i}$ satisfy $R(C_i \triangle C_j) > (\delta/2)^2$ for $i \ne j$, and in particular they are distinct.

Step 3: passing to a finitely supported measure. Lemma 13.6 bounds $H(\cdot, \mathcal{A}) = \sup_{Q'} \log N(\cdot, \mathcal{A}, Q')$, where the supremum runs over probability measures concentrated on finite sets, and R is not one of these. Let $Y_1, Y_2, \dots$ be i.i.d. with law R, and let $R_m = \frac{1}{m}\sum_{k \le m} \delta_{Y_k}$. By the strong law of large numbers, $R_m(C_i \triangle C_j) \to R(C_i \triangle C_j)$ almost surely for each of the finitely many pairs $i < j$. So for some realization and some m, $R_m(C_i \triangle C_j) > (\delta/2)^2$ holds for all $i \ne j$ at once. Here R_m is concentrated on a finite subset of $\mathcal{X} \times \mathbb{R}$. Write $\mathcal{S} = \{C_f : f \in \mathcal{F}\}$. Then

$$N \le N(\delta/2, \mathcal{S}, R_m) \le \exp H(\delta/2, \mathcal{S}).$$

Step 4: Haussler's bound. By hypothesis $\mathcal{S}$ is a VC class of subsets of $\mathcal{X} \times \mathbb{R}$ with VC dimension V. Lemma 13.6 at scale $\delta/2 \in (0,1]$ gives

$$H(\delta/2, \mathcal{S}) \le 2V \log \frac{2e}{\delta} + \log\big(e(V+1)\big).$$

This holds for every finite family of $\delta\|F\|_Q$-separated functions, so

$$N(\delta\|F\|_Q, \mathcal{F}, Q) \le e(V+1)\Big(\frac{2e}{\delta}\Big)^{2V} = e(V+1)\,(4e^2)^V\,\delta^{-2V} \le e(V+1)(16e)^{V+1}\Big(\frac{1}{\delta}\Big)^{2V},$$

using $4e^2 \approx 29.56 \le 16e \approx 43.49$ and $16e \ge 1$. This is the claim with $\kappa = e$, uniformly in Q. ■

Note (repair, 2026-10-30). An earlier version had a gap at Step 3. It claimed $N \le D(c\delta^2, \mathcal{S}, L_1(P))$ "for a suitable P" by citing van der Vaart–Wellner (Thm. 2.6.7), without constructing P, although the measure it used, $\tilde{Q} \otimes \lambda$, is not a probability measure. It also misquoted Haussler's bound. Steps 1–3 above give the missing construction ($c = 1/4$).

Exercise 13.14

Central idea. The layer-cake identity writes a $[0,1]$-valued ϕ as a pointwise limit of uniform averages of the super-level indicators $\mathbf{1}_{\{\phi > t\}}$, which lie in the VC-class $\mathcal{I} = \{\mathbf{1}_{\{g>t\}} : g \in \mathcal{F},\, t \in \mathbb{R}\}$. Decomposing $|f| \le 1$ as a difference $g - h$ of two $[0,1]$-valued functions, the sub-level indicators needed for h are handled by complements, $\mathbf{1}_{\{f<c\}} = \mathbf{1}_{\mathcal{X}} - \lim_k \mathbf{1}_{\{f > c-1/k\}}$, at the cost of total absolute weight 2; balancing the weights exhibits $f/3$ as a pointwise limit of signed combinations from $\mathcal{I}$ of total absolute weight at most 1.

Solution. Let $M = \sup_{f \in \mathcal{F},\, x \in \mathcal{X}} |f(x)| < \infty$; dividing by M we assume $M = 1$. Put $\mathcal{C} = \{\{g > t\} : g \in \mathcal{F},\, t \in \mathbb{R}\}$, a VC-class of sets of dimension at most V by hypothesis, and $\mathcal{I} = \{\mathbf{1}_C : C \in \mathcal{C}\}$, a VC-class of indicator functions. Let $\operatorname{sconv}(\mathcal{I}) = \{\sum_k \nu_k \mathbf{1}_{E_k} : \mathbf{1}_{E_k} \in \mathcal{I},\ \sum_k |\nu_k| \le 1\}$ and denote by $\overline{\operatorname{sconv}}(\mathcal{I})$ its pointwise closure.

For $0 \le \phi \le 1$, the hint gives $\phi(x) = \lim_{m\to\infty} \frac{1}{m}\sum_{i=1}^m \mathbf{1}_{\{\phi(x) > i/m\}}$ for every x: the sum counts $|\{1 \le i \le m : i < m\phi(x)\}| = \max(0, \lceil m\phi(x)\rceil - 1)$, and $(\lceil m\phi(x)\rceil - 1)/m \to \phi(x)$.

Let $f \in \mathcal{F}$, $|f| \le 1$, and write $f = g - h$ with $g = \frac{1+f}{2}$, $h = \frac{1-f}{2} \in [0,1]$. Since $\mathbf{1}_{\{g > i/m\}} = \mathbf{1}_{\{f > 2i/m - 1\}} \in \mathcal{I}$, the averages $g_m = \frac{1}{m}\sum_{i=1}^m \mathbf{1}_{\{f > 2i/m-1\}}$ lie in $\operatorname{conv}(\mathcal{I}) \subseteq \operatorname{sconv}(\mathcal{I})$, and $g = \lim_m g_m$ pointwise, so $g \in \overline{\operatorname{sconv}}(\mathcal{I})$.

For h, $\mathbf{1}_{\{h>i/m\}} = \mathbf{1}_{\{f<1-2i/m\}}$. For any c, $\mathbf{1}_{\{f<c\}} = \mathbf{1}_{\mathcal{X}} - \mathbf{1}_{\{f\geq c\}}$ with $\mathbf{1}_{\mathcal{X}} = \mathbf{1}_{\{f>-2\}} \in \mathcal{J}$ and $\mathbf{1}_{\{f\geq c\}} = \lim_{k\to\infty} \mathbf{1}_{\{f>c-1/k\}}$ pointwise, each $\mathbf{1}_{\{f>c-1/k\}} \in \mathcal{J}$. Hence $\mathbf{1}_{\{f<c\}}$ is a pointwise limit of the signed combinations $\mathbf{1}_{\{f>-2\}} - \mathbf{1}_{\{f>c-1/k\}}$, of total absolute weight 2; i.e. $\mathbf{1}_{\{f<c\}} \in \overline{2\,\mathrm{sconv}(\mathcal{J})}$. Thus $h_m = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{\{f<1-2i/m\}} \in \overline{2\,\mathrm{sconv}(\mathcal{J})}$ (a convex combination inside the convex set $\overline{2\,\mathrm{sconv}(\mathcal{J})}$) and $h = \lim_m h_m$ pointwise.

Finally $f/3 = g/3 - h/3$, and $\frac{1}{3}\mathrm{sconv}(\mathcal{J}) - \frac{2}{3}\mathrm{sconv}(\mathcal{J}) \subseteq \mathrm{sconv}(\mathcal{J})$, since for $a = \frac{1}{3}\sum_i \nu_i \mathbf{1}_{E_i}$, $b = \frac{2}{3}\sum_j \mu_j \mathbf{1}_{F_j}$ the difference has total absolute weight $\frac{1}{3}\sum_i |\nu_i| + \frac{2}{3}\sum_j |\mu_j| \leq 1$. Hence $f/3 \in \overline{\mathrm{sconv}}(\mathcal{J})$. For general M, $\mathcal{F} \subseteq 3M \cdot \overline{\mathrm{sconv}}(\mathcal{J})$. The closure is inherent in the hint's limit representation and is the form used in applications. (If $\mathcal{F}$ is symmetric, $\mathbf{1}_{\{f<c\}} = \mathbf{1}_{\{-f>-c\}} \in \mathcal{J}$ directly and the constant improves to $2M$; if $\mathcal{F} \geq 0$, to M.)

Exercise 13.15

Central idea. Symmetrize, then chain conditionally on the sample (Dudley, Corollary 13.2), measuring distances in $L_2(P_n)$. The chain runs from the empty set, so the bound is $\mathbb{E}Z^{\pm} \leq C\,\mathbb{E}F(\delta_n)$, where $F(r) = \int_0^r \sqrt{\psi}$ and $\delta_n^2 = \sup_s P_n(A_s)$. The map $t \mapsto F(\sqrt{t})$ is concave, so Jensen moves the expectation inside, and $\mathbb{E}\delta_n^2 \leq \sigma^2 + \mathbb{E}Z^+/\sqrt{n}$. Karamata's theorem gives $F(\rho) \asymp \rho\sqrt{\psi(\rho)}$. Concavity of F gives $F(r) \leq (r/\sigma)F(\sigma)$ for $r \geq \sigma$. The fixed-point inequality that results has solution $\mathbb{E}Z^{\pm} \lesssim \sigma\sqrt{\psi(\sigma/2)} + \psi(\sigma/2)/\sqrt{n}$. **The claim as printed is false:** a bound uniform in n is impossible. It holds under a condition $n\sigma^2 \geq \psi(\sigma/2)$, the analogue of the proviso in the book's Lemma 13.5 and Theorem 13.7. The hint's first display is also false as printed: $\sqrt{\sigma^2 + \mathbb{E}Z^+/\sqrt{n}}$ belongs in the upper limit of the integral, not in front of it.

Solution.

Conventions. As in Section 13.3, $N(\delta, \mathcal{T}, Q)$ is the largest number of indices that are pairwise δ-separated in $d_Q(s,t) = Q(A_s \triangle A_t)^{1/2}$. The universal entropy $H(\delta, \mathcal{T})$ is the supremum of $\log N(\delta, \mathcal{T}, Q)$ over probability measures Q concentrated on finite sets, such as $P_n = \frac{1}{n}\sum_i \delta_{X_i}$. Since ψ is nonincreasing, $\psi(1/x)$ is nondecreasing in x, so its index satisfies $\alpha \in [0, 2)$. Write $Z^{\pm} = Z^{\pm}_{\mathcal{T}}$ and $F(r) = \int_0^r \sqrt{\psi(u)}\,du$ for $r \in [0, 1]$.

Erratum 1: the bound cannot be uniform in n. Let P be uniform on $[0,1]$, and let $\mathcal{T}$ index the intervals $[a, a+\sigma^2)$ with $a \in \mathbb{Q}$. This class is countable, $P(A_s) \leq \sigma^2$, and it is contained in the class of intervals, which has VC dimension at most 2. The right side of Lemma 13.6 increases in V for $\delta \leq e$, so the lemma gives $H(\delta, \mathcal{T}) \leq \psi(\delta) := 4\log(e/\delta) + \log(3e)$. This ψ is nonincreasing, $\psi \geq \psi(1) \approx 6.10 \geq 1$, and $\psi(1/x) = 4\log(ex) + \log(3e)$ is regularly varying of index $0 < 2$. Every sample has some rational $a \in (X_1 - \sigma^2, X_1]$, and for it $\sum_i \mathbb{1}\{X_i \in [a, a+\sigma^2)\} \geq 1$ while $nP(A_s) \leq n\sigma^2$. Hence

$$Z^+ \geq \frac{1 - n\sigma^2}{\sqrt{n}} \quad \text{always.}$$

Take $n = 1$. Then $\mathbb{E}Z^+ \geq 1 - \sigma^2$, whereas $\sigma\sqrt{\psi(\sigma/2)}$ equals 0.0522 at $\sigma = 10^{-2}$, 6.76×10^{-4} at $\sigma = 10^{-4}$, and tends to 0 as $\sigma \to 0$. So no constant K works. (Here $n\sigma^2 \geq \psi(\sigma/2)$ fails, which is consistent with the corrected statement below.)

Erratum 2: the hint's first display is false as printed. Take a single set A with $P(A) = \sigma^2$, and take $\psi \equiv 1$. Then $H \equiv 0 \leq \psi$, and $\psi(1/x)$ is regularly varying of index 0. The display reads $\mathbb{E}Z^+ \leq K\sqrt{\sigma^2 + \mathbb{E}Z^+/\sqrt{n}} \cdot \sigma/2$, where $\mathbb{E}Z^+ = \mathbb{E}(B - n\sigma^2)_+/\sqrt{n}$ and $B \sim \mathrm{Bin}(n, \sigma^2)$. For $n = 10^6$ (exact binomial computation): - at $\sigma = 0.1$, $\mathbb{E}Z^+ = 0.0397$ while the right side is $0.00501\,K$, a ratio of about 7.9 when $K = 1$; - at $\sigma = 0.03$, the values are 0.0120 against $0.000453\,K$; - at $\sigma = 0.01$, they are 0.00399 against $0.000051\,K$.

By the CLT (with uniform integrability) the ratio tends to $2\sqrt{1-\sigma^2}/(\sqrt{2\pi}K\sigma)$ as $n \to \infty$, and this is unbounded as $\sigma \to 0$. So the display fails for every fixed K. The correct form, proved in Step 1, is $\mathbb{E}Z^{\pm} \leq C\,F(\sqrt{\sigma^2 + \mathbb{E}Z^+/\sqrt{n}})$. In addition, the hint's "$\lim_{\sigma\to 0}\int_0^{\sigma/2}\sqrt{\psi} = \ldots$" should be a limit of the ratio of the two sides (Step 2).

Corrected statement. Assume in addition $\psi \geq 1$ on $(0,1]$. Then there is K depending only on ψ such that for all $n \geq 1$ and $\sigma \in (0,1)$,

$$\max(\mathbb{E}Z^-, \mathbb{E}Z^+) \leq K\Big(\sigma\sqrt{\psi(\sigma/2)} + \frac{\psi(\sigma/2)}{\sqrt{n}}\Big),$$

and therefore

$$\max(\mathbb{E}Z^-, \mathbb{E}Z^+) \leq 2K\sigma\sqrt{\psi(\sigma/2)} \qquad \text{whenever } n\sigma^2 \geq \psi(\sigma/2).$$

On the assumption $\psi \geq 1$. Replacing ψ by $\psi_1 = \psi \vee 1$ keeps every hypothesis. It is still nonincreasing and still dominates H. If $\psi(1/x) \to \infty$, then $\psi_1(1/x) = \psi(1/x)$ for large x. Otherwise $\psi(1/x)$ increases to a finite limit $L > 0$, so $\alpha = 0$ and $\psi_1(1/x) \to L \vee 1$, which also has index 0. If moreover $\psi(1/2) > 0$, then $\psi_1(\sigma/2) \leq \max(1, 1/\psi(1/2))\,\psi(\sigma/2)$ for $\sigma < 1$, so the bound transfers to ψ with a larger K. When $\mathcal{T}$ contains two distinct sets, $\psi(\sigma/2) \geq H(\sigma/2, \mathcal{T}) \geq \log 2$, because the two sets are at d_{δ_x}-distance 1 for a point x of their symmetric difference. Some positivity is genuinely needed. Take a single set with $P(A) = 0.64$, $\psi(\delta) = (\log \frac{1}{4\delta})_+$, $\sigma = 0.8$ and $n = 1$. Then $\psi(\sigma/2) = 0$, yet $\mathbb{E}Z^+ = 0.64 \cdot 0.36 = 0.2304 > 0$.

Proof of the corrected statement. Assume $\psi \geq 1$.

Step 0: finiteness of F and the Karamata estimate. Substituting $u = 1/x$ gives $F(\rho) = \int_{1/\rho}^{\infty} g(x)\,dx$ with $g(x) = \sqrt{\psi(1/x)}\,x^{-2}$. For $t > 0$,

$$\frac{g(tx)}{g(x)} = \sqrt{\frac{\psi(1/(tx))}{\psi(1/x)}}\,t^{-2} \to t^{\alpha/2-2} \quad (x \to \infty),$$

so g is regularly varying of index $\beta = \alpha/2 - 2 < -1$. It is also locally bounded on $[1,\infty)$, because $\psi(1/x)$ is monotone and finite there. Karamata's theorem (Bingham–Goldie–Teugels, Prop. 1.5.10) says that $\int_y^{\infty} g$ converges and $\int_y^{\infty} g \sim y\,g(y)/(-\beta-1)$ as $y \to \infty$. With $y = 1/\rho$ this gives $F(1) < \infty$ and

$$\lim_{\rho \to 0^+} \frac{F(\rho)}{\rho\sqrt{\psi(\rho)}} = \frac{1}{1-\alpha/2},$$

which is the hint's second display, read as a ratio. So $F(\rho) \leq K_1\rho\sqrt{\psi(\rho)}$ on $(0,\rho_0]$ for some $\rho_0 > 0$. On $[\rho_0, 1]$ we have $F(\rho) \leq F(1)$ and $\rho\sqrt{\psi(\rho)} \geq \rho_0$. Hence, enlarging K_1 (and taking $K_1 \geq 1$),

$$F(\rho) \leq K_1\,\rho\sqrt{\psi(\rho)} \qquad (0 < \rho \leq 1). \tag{0}$$

Since $F' = \sqrt{\psi}$ is nonincreasing, F is concave with $F(0) = 0$. So $F(\rho)/\rho$ is nonincreasing, and F is nondecreasing.

Step 1: symmetrization and conditional chaining. First, one-sided symmetrization. Let (X_i') be an independent copy of the sample and (ε_i) independent Rademacher signs. For bounded g_s indexed by the countable set $\mathcal{T}$, Jensen's inequality (conditionally on X) gives $\mathbb{E}\sup_s \sum_i (g_s(X_i) - Pg_s) \leq \mathbb{E}\sup_s \sum_i (g_s(X_i) - g_s(X_i'))$. The summands are symmetric, so the right side equals $\mathbb{E}\sup_s \sum_i \varepsilon_i(g_s(X_i) - g_s(X_i'))$. Splitting the supremum and using that $-\varepsilon$ has the law of ε, this is at most $2\,\mathbb{E}\sup_s \sum_i \varepsilon_i g_s(X_i)$. (This is the last inequality of Lemma 11.4, used in the same form in the proof of Lemma 13.5.) Applied to $g_s = \pm\mathbb{1}_{A_s}$, and again using that $-\varepsilon$ has the law of ε,

$$\mathbb{E}Z^{\pm} \leq 2\,\mathbb{E}\sup_s Y_s, \qquad Y_s = \frac{1}{\sqrt{n}}\sum_i \varepsilon_i \mathbb{1}_{A_s}(X_i).$$

Next, chaining given X. Adjoin an index s_0 with $A_{s_0} = \emptyset$, so that $Y_{s_0} = 0$. For any indices s, t, Hoeffding's lemma gives $\log \mathbb{E}_\varepsilon e^{\lambda(Y_s - Y_t)} \leq \lambda^2 d_n(s,t)^2/2$ with $d_n = d_{P_n}$. Also $d_n(s, s_0) = P_n(A_s)^{1/2} \leq \delta_n$, where $\delta_n^2 := \sup_s P_n(A_s) \leq 1$. For a finite $T' \subset \mathcal{T}$, the packing numbers of $T' \cup \{s_0\}$ in d_n are at most $N(u, \mathcal{T}, P_n) + 1 \leq e^{\psi(u)} + 1 \leq e^{2\psi(u)}$ (since $\psi \geq 1$). Corollary 13.2 therefore gives

$$\mathbb{E}_\varepsilon \max_{s \in T'} Y_s \leq \mathbb{E}_\varepsilon \max_{s \in T' \cup \{s_0\}} (Y_s - Y_{s_0}) \leq 12 \int_0^{\delta_n/2} \sqrt{2\psi(u)}\,du \leq 12\sqrt{2}\,F(\delta_n).$$

Letting $T' \uparrow \mathcal{T}$ (monotone convergence; $|Y_s| \le \sqrt{n}$) yields $\mathbb{E}Z^{\pm} \le 24\sqrt{2}\,\mathbb{E}F(\delta_n)$.

Then Jensen. The function $G(t) = F(\sqrt{t}) = \int_0^t \frac{\sqrt{\psi(\sqrt{v})}}{2\sqrt{v}}\,dv$ is the integral of a nonincreasing function, so it is concave on $[0,1]$. Hence $\mathbb{E}F(\delta_n) = \mathbb{E}G(\delta_n^2) \le F(\rho)$ with $\rho^2 := \mathbb{E}\delta_n^2 \le 1$. Since $P(A_s) \le \sigma^2$, we have $P_n(A_s) \le \sigma^2 + n^{-1/2} \cdot n^{-1/2} \sum_i (\mathbb{1}_{A_s}(X_i) - P(A_s))$, so

$$\rho^2 \le r^2 := \sigma^2 + \frac{\mathbb{E}Z^+}{\sqrt{n}}.$$

Let $\bar{r} = \max(\sigma, \rho) \in [\sigma, 1]$. By monotonicity of F and of $F(\rho)/\rho$, followed by (0) and the monotonicity of ψ,

$$\mathbb{E}Z^{\pm} \le 24\sqrt{2}\,F(\bar{r}) \le 24\sqrt{2}\,\frac{\bar{r}}{\sigma}F(\sigma) \le A\,r, \qquad A := 24\sqrt{2}\,K_1\sqrt{\psi(\sigma/2)}. \tag{1}$$

(This is the corrected first display of the hint, $\mathbb{E}Z^{\pm} \le 24\sqrt{2}\,F(r)$, combined with $F(r) \le (r/\sigma)F(\sigma)$.)

Step 2: the fixed point. Let $x = \mathbb{E}Z^+$, which is finite because $Z^+ \le \sqrt{n}$. Since $r \le \sigma + (x/\sqrt{n})^{1/2}$, (1) gives $y^2 \le by + c$ with $y = \sqrt{x}$, $b = An^{-1/4}$ and $c = A\sigma$. Hence $y \le \frac{b+\sqrt{b^2+4c}}{2}$ and

$$x \le \frac{b^2 + (b^2 + 4c)}{2} = 2A\sigma + \frac{A^2}{\sqrt{n}}.$$

Next, $\mathbb{E}Z^- \le A\sigma + An^{-1/4}\sqrt{x}$. Here $\sqrt{x} \le \sqrt{2A\sigma} + An^{-1/4}$, and $An^{-1/4}\sqrt{2A\sigma} = \sqrt{2 \cdot A\sigma \cdot A^2/\sqrt{n}} \le A\sigma + A^2/\sqrt{n}$. So

$$\max(\mathbb{E}Z^+, \mathbb{E}Z^-) \le 2A\sigma + \frac{2A^2}{\sqrt{n}} = 48\sqrt{2}K_1\,\sigma\sqrt{\psi(\sigma/2)} + 2304K_1^2\,\frac{\psi(\sigma/2)}{\sqrt{n}}.$$

This is the general bound with $K = 2304K_1^2$. If $n\sigma^2 \ge \psi(\sigma/2)$, then $\psi(\sigma/2)/\sqrt{n} = \sqrt{\psi(\sigma/2)} \cdot \sqrt{\psi(\sigma/2)/n} \le \sigma\sqrt{\psi(\sigma/2)}$, and the bound becomes $\max(\mathbb{E}Z^-, \mathbb{E}Z^+) \le 2K\sigma\sqrt{\psi(\sigma/2)}$. For VC classes, $\psi(\delta) = 2V\log(e/\delta) + \log(e(V+1))$ by Lemma 13.6, which recovers the form of Theorem 13.7. ■

Note (repair, 2026-10-30). An earlier version used the hint's first display literally and concluded a bound uniform in n. Erratum 2 shows that display is false, and Erratum 1 shows that conclusion is false. The numbers above were re-checked in Python (exact binomial sums). The proof follows the independent audit's argument, with the symmetrization written out and the fixed-point constant recomputed.

Exercise 13.16

Central idea. The proof of Exercise 13-15 carries over to $[0,1]$-valued functions once the chain starts at the zero function. Its Step 1 gives $\mathbb{E}Z^{\pm} \le A\max(\sigma, \sqrt{\mathbb{E}\delta_n^2})$, where $\delta_n^2 = \sup_s P_n f_s^2$. The only new ingredient is a bound on $\mathbb{E}\delta_n^2$ through σ^2 and $\mathbb{E}Z^{\pm}$. Symmetrization and the contraction principle (Theorem 11.6, applied to $x \mapsto x^2/2$, which is 1-Lipschitz on $[0,1]$) reduce $\mathbb{E}\sup_s(P_n - P)f_s^2$ to the uncentered Rademacher average $\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i)$. The functions are not centered, so this average has to be recentered before desymmetrizing. That gives $\mathbb{E}\delta_n^2 \le \sigma^2 + 2\sigma/\sqrt{n} + 4(\mathbb{E}Z^+ + \mathbb{E}Z^-)/\sqrt{n}$, and the fixed-point argument of 13-15 then closes. As in 13-15, **the comparable bound $K\sigma\sqrt{\psi(\sigma/2)}$ cannot be uniform in n.** It holds under $n\sigma^2 \ge \psi(\sigma/2)$, and in general one has $K(\sigma\sqrt{\psi(\sigma/2)} + \psi(\sigma/2)/\sqrt{n})$.

Solution.

Setting. $(f_s)_{s\in\mathcal{T}}$ is a countable family of measurable functions $\mathcal{X} \to [0,1]$ with $Pf_s^2 \le \sigma^2$ for all s, where $\sigma \in (0,1)$. This is the analogue of $P(A_s) \le \sigma^2$. It is also implied by $Pf_s \le \sigma^2$, because $f_s^2 \le f_s$, so the results below cover that reading too. $N(\delta, \mathcal{T}, Q)$ is the largest number of indices that are pairwise δ-separated in $L_2(Q)$, and the universal entropy $H(\delta, \mathcal{T}) = \sup_Q \log N(\delta, \mathcal{T}, Q)$ (over finitely supported Q) satisfies $H \le \psi$. Here ψ is as in 13-15, and we assume $\psi \ge 1$ (see 13-15 for the reduction $\psi \to \psi \vee 1$). Let

$$Z^{\pm} = \sup_s \pm\frac{1}{\sqrt{n}}\sum_{i=1}^n (f_s(X_i) - Pf_s), \qquad \delta_n^2 = \sup_s P_n f_s^2, \qquad S = \mathbb{E}Z^+ + \mathbb{E}Z^-.$$

Erratum (the statement needs a condition linking n and σ). Indicator functions are $[0,1]$-valued, with $P\mathbb{1}_A^2 = P(A)$ and $\|\mathbb{1}_A - \mathbb{1}_B\|_{L_2(Q)} = Q(A\triangle B)^{1/2}$. So the counterexample of 13-15 (Erratum 1) applies verbatim. There, P is uniform on $[0,1]$, the f_s are the indicators of $[a, a+\sigma^2)$ with $a \in \mathbb{Q}$, and $\psi(\delta) = 4\log(e/\delta) + \log(3e)$. With $n = 1$, $\mathbb{E}Z^+ \geq 1 - \sigma^2$, while $\sigma\sqrt{\psi(\sigma/2)} = 0.0522$ at $\sigma = 10^{-2}$ and tends to 0. So no bound $K\sigma\sqrt{\psi(\sigma/2)}$ uniform in n can hold. The hint's display (1) of 13-15 is false as printed (13-15, Erratum 2), and we do not use it.

Corrected statement (the "comparable bound"). There is K depending only on ψ such that for all $n \geq 1$ and $\sigma \in (0,1)$,

$$\max(\mathbb{E}Z^+, \mathbb{E}Z^-) \leq K\Big(\sigma\sqrt{\psi(\sigma/2)} + \frac{\psi(\sigma/2)}{\sqrt{n}}\Big),$$

and $\max(\mathbb{E}Z^+, \mathbb{E}Z^-) \leq K\sigma\sqrt{\psi(\sigma/2)}$ whenever $n\sigma^2 \geq \psi(\sigma/2)$. These are the same bounds as for classes of sets.

Step 1: chaining (as in 13-15, Steps 0–1). Put $F(r) = \int_0^r \sqrt{\psi}$. By 13-15 Step 0, F is finite, concave and nondecreasing on $[0,1]$, $F(\rho)/\rho$ is nonincreasing, and $F(\rho) \leq K_1\rho\sqrt{\psi(\rho)}$ for some $K_1 \geq 1$. One-sided symmetrization (13-15 Step 1) gives $\mathbb{E}Z^\pm \leq 2\,\mathbb{E}\sup_s Y_s$ with $Y_s = n^{-1/2}\sum_i \varepsilon_i f_s(X_i)$. Given X, adjoin the zero function ($Y_0 = 0$). Hoeffding's lemma gives $\log \mathbb{E}_\varepsilon e^{\lambda(Y_s - Y_t)} \leq \lambda^2\|f_s - f_t\|^2_{L_2(P_n)}/2$. Also $\|f_s - 0\|_{L_2(P_n)} = (P_n f_s^2)^{1/2} \leq \delta_n \leq 1$, and the packing numbers of a finite subfamily together with 0 are at most $e^{\psi(u)}+1 \leq e^{2\psi(u)}$. Corollary 13.2, followed by monotone convergence over finite subfamilies, gives $\mathbb{E}_\varepsilon \sup_s Y_s \leq 12\sqrt{2}\,F(\delta_n)$. Since $t \mapsto F(\sqrt{t})$ is concave, Jensen gives $\mathbb{E}Z^\pm \leq 24\sqrt{2}\,F(\rho)$ with $\rho^2 = \mathbb{E}\delta_n^2 \leq 1$. Now let R be any number with $R \geq \max(\sigma^2, \rho^2)$. Then $F(\rho) \leq F(\max(\sigma,\rho)) \leq \frac{\max(\sigma,\rho)}{\sigma}F(\sigma) \leq \frac{\sqrt{R}}{\sigma}K_1\sigma\sqrt{\psi(\sigma/2)}$, so

$$\mathbb{E}Z^\pm \leq A\sqrt{R}, \qquad A := 24\sqrt{2}\,K_1\sqrt{\psi(\sigma/2)} \ \ (\geq 1). \tag{1}$$

Step 2: bounding $\mathbb{E}\delta_n^2$. Since $Pf_s^2 \leq \sigma^2$,

$$\mathbb{E}\delta_n^2 \leq \sigma^2 + \frac{1}{n}\mathbb{E}\sup_s \sum_i \big(f_s(X_i)^2 - Pf_s^2\big) \leq \sigma^2 + \frac{2}{n}\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i)^2.$$

The second inequality is one-sided symmetrization (13-15 Step 1, with $g_s = f_s^2$). Let $\phi(x) = x^2/2$ for $|x| \leq 1$ and $\phi(x) = |x| - 1/2$ for $|x| > 1$. This ϕ is 1-Lipschitz with $\phi(0) = 0$, and $\phi(f_s(X_i)) = f_s(X_i)^2/2$. Theorem 11.6, with the identity as the convex nondecreasing function (as in the proof of Theorem 11.8; the theorem is stated for Ψ on $[0,\infty)$, but its proof via Lemma 11.7 works for any convex nondecreasing $\Psi : \mathbb{R} \to \mathbb{R}$, so the sign of the supremum does not matter), applied conditionally on X to finite subfamilies and extended by monotone convergence, gives

$$\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i)^2 = 2\,\mathbb{E}\sup_s \sum_i \varepsilon_i \phi(f_s(X_i)) \leq 2\,\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i).$$

Recentering. Write $f_s(X_i) = h_s(X_i) + Pf_s$ with $h_s = f_s - Pf_s$. Since $0 \leq Pf_s \leq (Pf_s^2)^{1/2} \leq \sigma$,

$$\sup_s \sum_i \varepsilon_i f_s(X_i) \leq \sup_s \sum_i \varepsilon_i h_s(X_i) + \sigma\Big(\sum_i \varepsilon_i\Big)_+,$$

and $\mathbb{E}(\sum_i \varepsilon_i)_+ = \frac{1}{2}\mathbb{E}|\sum_i \varepsilon_i| \leq \frac{\sqrt{n}}{2}$. For the centered part, let X' be an independent copy of the sample. Since $\mathbb{E}h_s(X_i') = 0$, we can write $\sum_i \varepsilon_i h_s(X_i) = \mathbb{E}'\big[\sum_i \varepsilon_i(h_s(X_i) - h_s(X_i'))\big]$, and $\sup_s \mathbb{E}' \leq \mathbb{E}'\sup_s$. Conditionally on ε, swapping $X_i \leftrightarrow X_i'$ for every i with $\varepsilon_i = -1$ preserves the joint law. Hence

$$\mathbb{E}\sup_s \sum_i \varepsilon_i h_s(X_i) \leq \mathbb{E}\sup_s \sum_i \big(h_s(X_i) - h_s(X_i')\big) \leq \mathbb{E}\sup_s \sum_i h_s(X_i) + \mathbb{E}\sup_s \sum_i \big(-h_s(X_i')\big) = \sqrt{n}\,S.$$

Combining these,

$$\mathbb{E}\delta_n^2 \leq \sigma^2 + \frac{2}{n}\cdot 2\Big(\frac{\sigma\sqrt{n}}{2} + \sqrt{n}\,S\Big) = \sigma^2 + \frac{2\sigma}{\sqrt{n}} + \frac{4S}{\sqrt{n}}. \tag{2}$$

(For indicators one has directly $\delta_n^2 \le \sigma^2 + Z^+/\sqrt{n}$. The extra terms appear because Pf_s need not equal Pf_s^2. The hint's sharper $\mathbb{E}\delta_n^2 \le \sigma^2 + 8\mathbb{E}Z^+/\sqrt{n}$ is neither proved nor refuted here, and it is not needed.)

Step 3: the fixed point. Take $R = \sigma^2 + 2\sigma/\sqrt{n} + 4S/\sqrt{n}$, which is at least $\max(\sigma^2, \rho^2)$ by (2). Summing (1) over $\pm$ gives $S \le 2A\sqrt{R}$, where $S \le 2\sqrt{n} < \infty$.

Under $n\sigma^2 \ge \psi(\sigma/2)$. Since $\psi \ge 1$, $1/\sqrt{n} \le \sigma$, so $2\sigma/\sqrt{n} \le 2\sigma^2$ and $\sqrt{R} \le \sqrt{3}\,\sigma + 2n^{-1/4}\sqrt{S}$. With $y = \sqrt{S}$ this gives $y^2 \le by + c$, where $b = 4An^{-1/4}$ and $c = 2\sqrt{3}A\sigma$. Hence $S \le b^2 + 2c = 4\sqrt{3}\,A\sigma + 16A^2/\sqrt{n}$. Also $A^2/\sqrt{n} = (24\sqrt{2}K_1)^2\psi(\sigma/2)/\sqrt{n} \le (24\sqrt{2}K_1)^2\sigma\sqrt{\psi(\sigma/2)}$. So

$$\max(\mathbb{E}Z^+, \mathbb{E}Z^-) \le S \le (96\sqrt{6}\,K_1 + 18432\,K_1^2)\,\sigma\sqrt{\psi(\sigma/2)}.$$

In general. $2\sigma/\sqrt{n} \le \sigma^2 + 1/n$, so $\sqrt{R} \le \sqrt{2}\,\sigma + n^{-1/2} + 2n^{-1/4}\sqrt{S}$. The same quadratic argument gives $S \le 4\sqrt{2}A\sigma + 4A/\sqrt{n} + 16A^2/\sqrt{n} \le 4\sqrt{2}A\sigma + 20A^2/\sqrt{n}$ (using $A \ge 1$). This is at most $K(\sigma\sqrt{\psi(\sigma/2)} + \psi(\sigma/2)/\sqrt{n})$ with $K = 23040K_1^2$. ■

Note (repair, 2026-10-30). An earlier version claimed a bound uniform in n (false, by the Erratum). It also contained a false step: "desymmetrization gives $\mathbb{E}\sup_s \frac{1}{n}\sum_i \varepsilon_i f_s(X_i) \le \frac{2}{\sqrt{n}}\mathbb{E}Z^+$", which fails for uncentered f_s. For the class $\{f \equiv \sigma,\ f \equiv 0\}$, $Z^+ \equiv 0$, yet $\mathbb{E}\sup_s \frac{1}{n}\sum_i \varepsilon_i f_s(X_i) = \sigma\,\mathbb{E}(\sum_i \varepsilon_i)_+/n$, which is about $0.398\,\sigma/\sqrt{n} > 0$ (for example 0.0398σ at $n = 100$, exact binomial computation). The recentering in Step 2 repairs this, following the independent audit.

Exercise 13.17

Central idea. Symmetrize and chain conditionally on the sample, adjoining the zero function. The Dudley integral is then the perspective $\Phi(a, c) = aH(c/a)$ of a concave entropy integral H, evaluated at $a = \|g\|_{L_2(P_n)}$ and $c = \delta_n = \sup_s \|f_s\|_{L_2(P_n)}$. $\Phi(\sqrt{s}, \sqrt{t})$ is jointly concave and monotone, so Jensen's inequality replaces both random quantities by $\|g\|_{L_2(P)}$ and $\sqrt{\mathbb{E}\delta_n^2}$ at no cost. Contraction bounds $\mathbb{E}\delta_n^2$ by $\sigma^2 + O(b(\sigma + \mathbb{E}Z + \mathbb{E}Z')/\sqrt{n})$. Karamata's theorem gives $H(y) \lesssim y\sqrt{\psi(\kappa/y)}$. Solving the resulting quadratic fixed-point inequality yields $\sigma\sqrt{\psi} + b\psi/\sqrt{n}$. As printed, the exercise has a garbled entropy hypothesis and a proviso without n; both are corrected below.

Solution.

Erratum (statement as printed).

(i) The entropy hypothesis is garbled. As printed, ψ is nonincreasing, so $\psi(\kappa\|g\|_{L_2(Q)}/\delta) \le \psi(1)$ whenever $\delta \le \kappa\|g\|_{L_2(Q)}$. The hypothesis would then force $\log N(\delta, \mathcal{T}, L_2(Q)) \le \psi(1)$ for all small δ, i.e. every class would be finite in each $L_2(Q)$. The regular-variation assumption on $\psi(1/x)$ would only constrain ψ near 0, where it plays no role. VC classes, with $\log N \asymp V\log(1/\delta)$, would be excluded, although the exercise is meant to generalize Theorem 13.7. The consistent reading is the one of Exercise 13.15 with the envelope scaling: $\psi : (0, \infty) \to [1, \infty)$ **is nondecreasing and regularly varying at ∞ with index $\alpha < 2$.** Equivalently, $u \mapsto \psi(1/u)$ is nonincreasing and satisfies the hypothesis of Exercise 13.15. With $\psi \ge 1$, this is a harmless normalization (replace ψ by $\psi \vee 1$). The entropy hypothesis is then $\log N(\delta, \mathcal{T}, L_2(Q)) \le \psi(\kappa\|g\|_{L_2(Q)}/\delta)$, as printed.

(ii) The proviso must involve n. Even under the corrected reading, "$\sigma \ge \sqrt{\psi(\kappa\|g\|_{L_2(P)}/\sigma)}$" does not suffice; compare Theorem 13.7, whose proviso $\sigma \ge 24\sqrt{V\log(4e^2/\sigma)/(5n)}$ contains n. Counterexample: take P uniform on $[0, 1]$, $b \ge 1$, $p \in (0, 1)$, and the countable class $f_a = b\,\mathbb{1}_{[a, a+p]}$, $a \in \mathbb{Q} \cap [0, 1-p]$, with envelope $g \equiv b$.

- *Entropy.* For any probability Q and $\eta \in (0, 1]$, let $\nu(B) = Q(B) + Q(B + p)$, a measure of total mass 2. Choose cut points $0 = a_0 < a_1 < \cdots < a_m = 1 - p$ greedily, with $a_{j+1} = \min\{1 - p, \sup\{t : \nu((a_j, t)) \le \eta\}\}$. Every open cell then has $\nu((a_j, a_{j+1})) \le \eta$, and every cell but the last has $\nu((a_j, a_{j+1}]) \ge \eta$, so $m \le 2/\eta + 1$. If $a < a'$ lie in the same open cell, then $[a, a+p] \triangle [a', a'+p] \subseteq [a, a') \cup (a+p, a'+p]$, which has Q-mass at most $\nu((a_j, a_{j+1})) \le \eta$. One rational center per open cell plus the cut points therefore gives $N(\sqrt{\eta}, \{\mathbb{1}_{[a, a+p]}\}, L_2(Q)) \le 2m + 1 \le 4/\eta + 3$. Since $\|b\mathbb{1}_I - b\mathbb{1}_J\|_{L_2(Q)} = b\sqrt{Q(I \triangle J)}$

and $\|g\|_{L_2(Q)} = b$, we get $N(\delta) \le 4b^2/\delta^2 + 3 \le 7b^2/\delta^2$ for $\delta \le b$, and $N(\delta) = 1$ for $\delta \ge b$. So the hypothesis holds with $\kappa = 1$ and $\psi(x) = 2 + 2\log(1+x)$, since $\log 7 < 2$. This ψ is nondecreasing, ≥ 1, and regularly varying with index 0.

- *Choice of parameters.* Let $\sigma = b^{1/10}$ and $p = \sigma^2/b^2 = b^{-9/5}$, so that $Pf_a^2 = b^2 p = \sigma^2$. The printed proviso reads $b^{1/5} \ge 2 + 2\log(1 + b^{9/10})$. It holds for all $b \ge 4.31 \times 10^7$ (checked numerically up to $b = 10^{20}$; for larger b it follows because $b^{1/5}$ grows faster than $\log b$).
- *Failure of the bound.* Almost surely, some $a \in \mathbb{Q} \cap [0, 1-p]$ has $X_1 \in [a, a+p]$. Hence $Z \ge (b - nbp)/\sqrt{n} = b(1-np)/\sqrt{n}$ a.s. The claimed bound is $\kappa' \sigma \sqrt{\psi(b/\sigma)} = \kappa' b^{1/10} \sqrt{2 + 2\log(1 + b^{9/10})}$. For fixed n, the ratio of the two tends to ∞ as $b \to \infty$. For example, with $n = 1$ and $b = 10^{12}$: $\mathbb{E}Z \ge 10^{12}(1 - 10^{-21.6})$, while $\sigma\sqrt{\psi(b/\sigma)} \approx 114.0$, a ratio of about 8.8×10^9. So no constant κ' works. (The corrected proviso below fails here, as it must.)

Corrected statement. Let ψ be as in (i), and suppose $\log N(\delta, \mathcal{T}, L_2(Q)) \le \psi(\kappa \|g\|_{L_2(Q)}/\delta)$ for every probability Q with $\|g\|_{L_2(Q)} > 0$ and every $\delta > 0$. Let $Z = \sup_s n^{-1/2} \sum_i (f_s(X_i) - Pf_s)$ and $Z' = \sup_s n^{-1/2} \sum_i (Pf_s - f_s(X_i))$, as in Theorem 13.7. Write $G = \|g\|_{L_2(P)}$ and $L = \sqrt{\psi(\kappa G/\sigma)}$. There are constants κ', κ'' depending only on ψ such that

$$\max(\mathbb{E}Z, \mathbb{E}Z') \le \kappa'' \Big(\sigma L + \frac{bL^2}{\sqrt{n}} \Big) \quad \text{always, and} \quad \max(\mathbb{E}Z, \mathbb{E}Z') \le \kappa' \sigma \sqrt{\psi\Big(\frac{\kappa \|g\|_{L_2(P)}}{\sigma}\Big)} \quad \text{if} \quad \sqrt{n}\,\sigma \ge b \sqrt{\psi\Big(\frac{\kappa \|g\|_{L_2(P)}}{\sigma}\Big)}.$$

With $b = 1$ and $g \equiv 1$, this is the form of Exercises 13.15–13.16: the bound $\sigma\sqrt{\psi} + \psi/\sqrt{n}$, and the condition $n\sigma^2 \ge \psi$.

Proof.

Step 0: reductions and two constants. By pointwise separability, there is a countable $\mathcal{T}_0 \subseteq \mathcal{T}$ such that every f_s is a pointwise limit of f_{s_k} with $s_k \in \mathcal{T}_0$. Since $|f_s| \le b$, dominated convergence gives $Pf_{s_k} \to Pf_s$. So Z, Z' and δ_n (below) are unchanged when $\mathcal{T}$ is replaced by $\mathcal{T}_0$; assume $\mathcal{T}$ countable. If $\sigma = 0$ or $G = 0$, then $f_s(X_i) = 0$ a.s. for all s, i, and everything vanishes. So assume $\sigma > 0$ and $G > 0$. Note $0 \le \mathbb{E}Z, \mathbb{E}Z' \le 2b\sqrt{n} < \infty$.

Let $F(z) = \int_0^z \sqrt{2\psi(1/v)}\, dv = \int_{1/z}^\infty \ell(x)\, dx$, where $\ell(x) = \sqrt{2\psi(x)}\, x^{-2}$. The function ℓ is regularly varying at ∞ with index $\alpha/2 - 2 < -1$. By Karamata's theorem (Bingham–Goldie–Teugels, Prop. 1.5.10), $F(z) < \infty$ and $F(z) \sim z\sqrt{2\psi(1/z)}/(1 - \alpha/2)$ as $z \to 0^+$. We claim that

$$K := \sup_{z > 0} \frac{F(z)}{z\sqrt{\psi(1/z)}} < \infty.$$

Choose $z_0 \in (0, 1]$ such that the ratio is at most $2\sqrt{2}/(1 - \alpha/2)$ on $(0, z_0]$. On $[z_0, 1]$, the ratio is at most $F(1)/z_0$, because $\psi \ge 1$. For $z \ge 1$, $F(z) \le F(1) + z\sqrt{2\psi(1)}$ and $z\sqrt{\psi(1/z)} \ge z \ge 1$, so the ratio is at most $F(1) + \sqrt{2\psi(1)}$. We also need

$$c_\psi := \sup_{x > 0} \psi(2x)/\psi(x) < \infty.$$

Since $\psi(2x)/\psi(x) \to 2^\alpha$, the ratio is at most $2^\alpha + 1$ for $x \ge x_0$. For $x < x_0$, it is at most $\psi(2x_0)$, because $\psi \ge 1$ and ψ is nondecreasing. Both K and c_ψ depend only on ψ.

Step 1: symmetrization. Let X' be an independent copy of the sample, and let ε_i be independent Rademacher variables independent of everything else. By Jensen's inequality with respect to X', and since swapping $X_i \leftrightarrow X_i'$ preserves the joint law,

$$\mathbb{E}Z \le \mathbb{E} \sup_s \frac{1}{\sqrt{n}} \sum_i (f_s(X_i) - f_s(X_i')) = \mathbb{E} \sup_s \frac{1}{\sqrt{n}} \sum_i \varepsilon_i (f_s(X_i) - f_s(X_i')) \le 2R,$$

where $R = \mathbb{E} \sup_s n^{-1/2} \sum_i \varepsilon_i f_s(X_i)$. By the symmetry of ε, also $\mathbb{E}Z' \le 2R$.

Step 2: chaining given the sample. Fix X. Let $\|g\|_n = \|g\|_{L_2(P_n)}$ and $\delta_n = \sup_s \|f_s\|_{L_2(P_n)} \le \|g\|_n$. If $\|g\|_n = 0$, the conditional Rademacher average is 0; assume $\|g\|_n > 0$. Let $\mathcal{T}_m$ be the first m indices, and adjoin an

index 0 with $f_0 \equiv 0$. The Rademacher process $s \mapsto n^{-1/2}\sum_i \varepsilon_i f_s(X_i)$ satisfies the condition of Corollary 13.2 with $d(s,t) = \|f_s - f_t\|_{L_2(P_n)}$, and $\sup_t d(t,0) \le \delta_n$. A $u/2$-cover of $\mathcal{T}$ yields a u-cover of $\mathcal{T}_m$ by points of $\mathcal{T}_m$. Adding 0 and using $e^{\psi} + 1 \le e^{2\psi}$ for $\psi \ge 1$,

$$\log N(u, \mathcal{T}_m \cup \{0\}) \le \log\left(N(u/2, \mathcal{T}, L_2(P_n)) + 1\right) \le 2\psi(2\kappa\|g\|_n/u).$$

Corollary 13.2 with $t_0 = 0$, followed by monotone convergence as $m \to \infty$, gives

$$\mathbb{E}_\varepsilon \sup_s \frac{1}{\sqrt{n}}\sum_i \varepsilon_i f_s(X_i) \le 12\int_0^{\delta_n} \sqrt{2\psi\Big(\frac{2\kappa\|g\|_n}{u}\Big)}\,du = 12\,\|g\|_n\, H\Big(\frac{\delta_n}{\|g\|_n}\Big) =: 12\,\Phi(\|g\|_n, \delta_n),$$

where $H(y) = \int_0^y \sqrt{2\psi(2\kappa/w)}\,dw = 2\kappa F(y/(2\kappa))$, after substituting $u = \|g\|_n w$ and then $w = 2\kappa v$. The integrand of H is nonincreasing, because ψ is nondecreasing. So H is finite, nondecreasing and concave with $H(0) = 0$, and $H(y)/y$ is nonincreasing. By the definition of K, with $z = y/(2\kappa)$,

$$\frac{H(y)}{y} = \frac{F(z)}{z} \le K\sqrt{\psi(2\kappa/y)} \qquad (y > 0). \tag{1}$$

Step 3: Jensen with a random envelope norm. Put $\Phi(a,c) = aH(c/a)$ for $a > 0$, $c \ge 0$, and $\Phi(0,0) = 0$, on the convex set $D = \{a > 0, c \ge 0\} \cup \{(0,0)\}$.

- Φ *is concave on* D. For $a = \lambda a_1 + (1-\lambda)a_2 > 0$, concavity of H gives

$$\Phi(a, \lambda c_1 + (1-\lambda)c_2) = aH\Big(\tfrac{\lambda a_1}{a}\tfrac{c_1}{a_1} + \tfrac{(1-\lambda)a_2}{a}\tfrac{c_2}{a_2}\Big) \ge \lambda\Phi(a_1,c_1) + (1-\lambda)\Phi(a_2,c_2).$$

  Mixtures with the origin are covered by $\Phi(\lambda a, \lambda c) = \lambda\Phi(a,c)$.
- Φ *is nondecreasing in each coordinate.* It is nondecreasing in c because H is. It is nondecreasing in a because $\Phi = c \cdot H(y)/y$ with $y = c/a$ decreasing in a.
- *Hence* $\Psi(s,t) = \Phi(\sqrt{s}, \sqrt{t})$ *is concave* on the convex set $\{s > 0, t \ge 0\} \cup \{(0,0)\}$, since $\sqrt{\cdot}$ is concave and Φ is monotone and concave.

The random point $W = (\|g\|_n^2, \delta_n^2)$ lies in this set, and $\mathbb{E}W = (G^2, \mathbb{E}\delta_n^2)$. If $\mathbb{E}\delta_n^2 = 0$, then $\delta_n = 0$ a.s. and $\Phi(\|g\|_n, \delta_n) = 0$. Otherwise $\mathbb{E}W$ is an interior point, where Ψ has a supergradient, so Jensen's inequality applies:

$$\mathbb{E}\,\Phi(\|g\|_n, \delta_n) \le \Phi\Big(G, \sqrt{\mathbb{E}\delta_n^2}\Big).$$

Step 4: bounding $\mathbb{E}\delta_n^2$. Throughout Steps 4–6, adjoin the index 0 with $f_0 \equiv 0$ to $\mathcal{T}$. This changes neither σ nor δ_n. It makes every supremum below nonnegative, as Theorem 11.6 requires (its convex function is defined on $[0,\infty)$; we use it with the identity). It replaces Z, Z' by $Z_+ = \max(Z,0)$ and $Z'_+ = \max(Z',0)$. Steps 1–3 hold verbatim for the enlarged class, since Step 2 already chains over $\mathcal{T}_m \cup \{0\}$; hence $\max(\mathbb{E}Z_+, \mathbb{E}Z'_+) \le 24\,\mathbb{E}\Phi(\|g\|_n, \delta_n)$. We have $\delta_n^2 \le \sup_s Pf_s^2 + \sup_s(P_n - P)f_s^2 \le \sigma^2 + \sup_s(P_n - P)f_s^2$. The argument of Step 1, applied to f_s^2, gives $\mathbb{E}\sup_s(P_n - P)f_s^2 \le \frac{2}{n}\mathbb{E}\sup_s\sum_i \varepsilon_i f_s(X_i)^2$. The function $\varphi(x) = \min(|x|, b)^2/(2b)$ is 1-Lipschitz with $\varphi(0) = 0$ and equals $x^2/(2b)$ on $[-b, b]$. Theorem 11.6, applied conditionally on X to finite subclasses and then in the limit, gives

$$\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i)^2 \le 2b\,\mathbb{E}\sup_s \sum_i \varepsilon_i f_s(X_i).$$

Next, $\sum_i \varepsilon_i f_s(X_i) = \sum_i \varepsilon_i(f_s(X_i) - Pf_s) + Pf_s\sum_i \varepsilon_i$, and $|Pf_s| \le \sigma$. Also $\mathbb{E}|\sum_i \varepsilon_i| \le \sqrt{n}$, and, as in Step 1 (Jensen in X', then removing ε by the swap),

$$\mathbb{E}\sup_s \sum_i \varepsilon_i(f_s(X_i) - Pf_s) \le \mathbb{E}\sup_s \sum_i (f_s(X_i) - f_s(X'_i)) \le \sqrt{n}\,(\mathbb{E}Z_+ + \mathbb{E}Z'_+).$$

With $S = \mathbb{E}Z_+ + \mathbb{E}Z'_+$, this gives

$$\mathbb{E}\delta_n^2 \le r^2 := \sigma^2 + \frac{4b\sigma}{\sqrt{n}} + \frac{4bS}{\sqrt{n}}.$$

Step 5: regular variation. Since $r \ge \sigma$, $H(y)/y$ is nonincreasing, and by (1),

$$\Phi(G, r) = r\,\frac{H(r/G)}{r/G} \le r\,\frac{H(\sigma/G)}{\sigma/G} \le rK\sqrt{\psi(2\kappa G/\sigma)} \le rK\sqrt{c_\psi}\,L.$$

Combining Steps 1–3 with the monotonicity of Φ in c, and taking expectations over X in Step 2,

$$\max(\mathbb{E}Z, \mathbb{E}Z') \le \max(\mathbb{E}Z_+, \mathbb{E}Z'_+) \le 24\,\mathbb{E}\Phi(\|g\|_n, \delta_n) \le 24\,\Phi(G, r) \le Ar, \qquad A := c_0 L, \quad c_0 := 24K\sqrt{c_\psi}.$$

Step 6: fixed point. Let $\beta = b/\sqrt{n}$. Then $r \le \sigma + 2\sqrt{\beta\sigma} + 2\sqrt{\beta S} \le 2\sigma + \beta + 2\sqrt{\beta S}$. From $S \le 2Ar$,

$$S \le 2A(2\sigma + \beta) + 4A\sqrt{\beta}\,\sqrt{S}.$$

Since $x^2 \le p + qx$ implies $x^2 \le 2p + q^2$, and $L \ge 1$,

$$S \le 8A\sigma + 4A\beta + 16A^2\beta \le 8c_0\,\sigma L + (4c_0 + 16c_0^2)\frac{bL^2}{\sqrt{n}}.$$

This is the general bound, with $\kappa'' = 4c_0 + 16c_0^2$. Under the corrected proviso, $bL/\sqrt{n} \le \sigma$, so $bL^2/\sqrt{n} \le \sigma L$, and

$$\max(\mathbb{E}Z, \mathbb{E}Z') \le S \le (12c_0 + 16c_0^2)\,\sigma\sqrt{\psi\Big(\frac{\kappa\|g\|_{L_2(P)}}{\sigma}\Big)}.$$

So $\kappa' = 12c_0 + 16c_0^2$, which depends only on ψ through K and c_ψ, and not on σ, P, $\mathcal{T}$, n, b or κ.

Remark. With $g \equiv b$, this recovers Theorem 13.7 for VC classes, up to constants. When $\|g\|_{L_2(P)} \ll b$, the logarithmic factor improves to $\psi(\kappa\|g\|_{L_2(P)}/\sigma)$ (Giné–Koltchinskii 2006, Theorem 3.1). ■

Exercise 13.18

Central idea. Symmetrize to a Rademacher process, then condition on the U_i's: only the $N \sim \mathrm{Bin}(n, \sigma^2)$ points falling in $[0, \sigma^2]$ matter, and the supremum becomes $\max_{k \le N} S_k$ for a simple random walk. Lévy's maximal inequality (Lemma 11.12) plus Hoeffding bounds the tail of this maximum by $2e^{-t^2/(2N)}$ with no union bound over the n order statistics — that is where the $\sqrt{\log n}$ factor is avoided — and integrating the tail gives $2\sqrt{N}$, whence Jensen over N yields 4σ.

Solution. The class $\{A_s = [0, s] : s \in [0, \sigma^2]\}$ has VC-dimension 1: for $x_1 < x_2$ the traces are $\emptyset$, $\{x_1\}$, $\{x_1, x_2\}$ (the set $\{x_2\}$ never occurs), so no two-point set is shattered, while one point is. Put $f_s(u) = \mathbb{1}_{\{u \le s\}}$ so that $X_{i,s} = f_s(U_i) - \mathbb{E}f_s(U_i)$ with the U_i i.i.d. uniform on $[0, 1]$; the f_s are centered in the sense of Theorem 11.4. By the symmetrization inequality (Theorem 11.4), with i.i.d. Rademacher ε_i independent of the U_i,

$$\mathbb{E}\sup_{s \in [0,\sigma^2] \cap \mathbb{Q}} \sum_{i=1}^n X_{i,s} \le 2\,\mathbb{E}_{U,\varepsilon} \sup_{s \in [0,\sigma^2] \cap \mathbb{Q}} \sum_{i=1}^n \varepsilon_i f_s(U_i),$$

the restriction to rational s being immaterial by right-continuity of $s \mapsto \sum_i \varepsilon_i f_s(U_i)$.

Condition on $U_1, \dots, U_n$ and let $N = \sum_{i=1}^n \mathbb{1}_{\{U_i \le \sigma^2\}}$, so $N \sim \mathrm{Bin}(n, \sigma^2)$. Order the $U_i \le \sigma^2$ as $V_1 \le \cdots \le V_N$ with paired Rademachers $\eta_1, \dots, \eta_N$ (still i.i.d. Rademacher given U). For $s \in [0, \sigma^2]$,

$$\sum_{i=1}^n \varepsilon_i f_s(U_i) = \sum_{j : V_j \le s} \eta_j,$$

a step function whose supremum over s equals $\max_{0 \le k \le N} S_k$ with $S_k = \sum_{j=1}^k \eta_j$, $S_0 = 0$.

Fix U (hence N). Lévy's maximal inequality (Lemma 11.12) applied to the symmetric variables η_j gives, for $t \ge 0$,

$$\mathbb{P}_\varepsilon\Big\{\max_{k \le N} S_k \ge t\Big\} \le 2\,\mathbb{P}_\varepsilon\{S_N \ge t\} \le 2e^{-t^2/(2N)},$$

the last step by Hoeffding's inequality (for $N = 0$ both sides are 0). Hence, with $M_N = \max_{0\le k\le N} S_k \ge 0$,

$$\mathbb{E}_\varepsilon[M_N^2] = \int_0^\infty 2t\,\mathbb{P}_\varepsilon\{M_N \ge t\}\,dt \le \int_0^\infty 4te^{-t^2/(2N)}\,dt = 4N,$$

so $\mathbb{E}_\varepsilon M_N \le \sqrt{\mathbb{E}_\varepsilon M_N^2} \le 2\sqrt{N}$. Taking expectation over U and using Jensen,

$$\mathbb{E}_U[2\sqrt{N}] \le 2\sqrt{\mathbb{E}N} = 2\sqrt{n\sigma^2} = 2\sigma\sqrt{n}.$$

Therefore

$$\mathbb{E}\left[\frac{1}{\sqrt{n}}\sup_{s\in[0,\sigma^2]}\sum_{i=1}^n X_{i,s}\right] \le \frac{2}{\sqrt{n}}\cdot 2\sigma\sqrt{n} = 4\sigma,$$

as claimed. The logarithmic factor of Theorem 13.7 never appears because Lévy's inequality controls the maximum of the partial sums directly, without a union bound over the n jump points.

Exercise 13.19

Central idea. Symmetrize, then fix the sample and expand in an $L_2(P)$-orthonormal basis of $\mathcal{T}$: the supremum over the L_2-ball of radius R becomes R times the Euclidean norm of the d-vector $(\sum_i \varepsilon_i e_j(X_i))_j$ by Cauchy–Schwarz. Jensen (twice) and orthonormality then give $R\sqrt{nd}$.

Solution. Since $\mathcal{T}$ consists of centered functions, $\mathbb{E}f(X_i) = 0$ for $f \in \mathcal{T}$, and the symmetrization inequality (Theorem 11.4) gives

$$\mathbb{E}\sup_{\substack{f\in\mathcal{T}\\ \mathbb{E}f(X)^2\le R^2}}\sum_{i=1}^n f(X_i) = \mathbb{E}\sup_{\substack{f\in\mathcal{T}\\ \|f\|_{L_2(P)}\le R}}\sum_{i=1}^n \big(f(X_i)-\mathbb{E}f\big) \le 2\,\mathbb{E}_{X,\varepsilon}\sup_{\substack{f\in\mathcal{T}\\ \|f\|_{L_2(P)}\le R}}\sum_{i=1}^n \varepsilon_i f(X_i),$$

with i.i.d. Rademacher ε_i independent of the X_i.

Fix $X_1,\dots,X_n$ and let $e_1,\dots,e_d$ be an orthonormal basis of $\mathcal{T}$ in $L_2(P)$. Every $f \in \mathcal{T}$ with $\|f\|_{L_2(P)} \le R$ writes $f = \sum_{j=1}^d \theta_j e_j$ with $\|\theta\|_2 \le R$, and

$$\sum_{i=1}^n \varepsilon_i f(X_i) = \sum_{j=1}^d \theta_j \sum_{i=1}^n \varepsilon_i e_j(X_i) = \langle \theta, v\rangle, \qquad v_j := \sum_{i=1}^n \varepsilon_i e_j(X_i).$$

Maximizing over $\|\theta\|_2 \le R$ (Cauchy–Schwarz) yields

$$\sup_{\substack{f\in\mathcal{T}\\ \|f\|_{L_2(P)}\le R}}\sum_{i=1}^n \varepsilon_i f(X_i) = R\|v\|_2 = R\left(\sum_{j=1}^d\Big(\sum_{i=1}^n \varepsilon_i e_j(X_i)\Big)^2\right)^{1/2}.$$

Taking $\mathbb{E}_\varepsilon$ and applying Jensen,

$$\mathbb{E}_\varepsilon \sup_f \sum_{i=1}^n \varepsilon_i f(X_i) \le R\left(\sum_{j=1}^d \mathbb{E}_\varepsilon\Big(\sum_{i=1}^n \varepsilon_i e_j(X_i)\Big)^2\right)^{1/2} = R\left(\sum_{i=1}^n\sum_{j=1}^d e_j(X_i)^2\right)^{1/2},$$

since $\mathbb{E}_\varepsilon(\sum_i \varepsilon_i a_i)^2 = \sum_i a_i^2$. Finally, Jensen over X and orthonormality $\mathbb{E}e_j(X)^2 = 1$ give

$$\mathbb{E}_X\left(\sum_{i=1}^n\sum_{j=1}^d e_j(X_i)^2\right)^{1/2} \le \left(\sum_{i=1}^n\sum_{j=1}^d \mathbb{E}e_j(X_i)^2\right)^{1/2} = \sqrt{nd}.$$

Combining the steps,

$$\mathbb{E}\sup_{\substack{f\in\mathcal{T}\\ \mathbb{E}[f(X)^2]\le R^2}}\sum_{i=1}^n f(X_i) \le 2R\sqrt{nd},$$

as required. The same symmetrization-plus-basis argument is the content of Exercise 13.5.

Exercise 13.20

Central idea. Since the g_j are centered, symmetrization (Lemma 11.4) reduces the problem to a Rademacher process $\sup_f \sum_i \varepsilon_i f(X_i)$; writing $f = \sum_j \alpha_j g_j$ turns this into a linear program over the coefficient set $A = \{\sum_j \alpha_j^2/\lambda_j \le 1,\ \sum_j \alpha_j^2 \le R^2\}$. The key observation is that the intersection of the two ellipsoids sits inside a single ellipsoid with effective axes $\mu_j = \lambda_j \wedge R^2$ (up to a factor 2), since $\alpha_j^2/\mu_j \le \alpha_j^2/\lambda_j + \alpha_j^2/R^2$. Cauchy–Schwarz then bounds the supremum by $\sqrt{2\sum_j \mu_j S_j^2}$, and Jensen in ε and in X (using orthonormality $\mathbb{E} g_j^2 = 1$) closes the estimate — exactly the mechanism of Exercise 13.5.

Solution. Let $\mathcal{F} = \mathcal{E}_1(1) \cap \mathcal{E}_2(R)$. Each $f \in \mathcal{F}$ is a linear combination of the centered functions g_j, hence centered: $\mathbb{E} f(X) = 0$. The class $\mathcal{F}$ is symmetric ($f \in \mathcal{F}$ implies $-f \in \mathcal{F}$), so $\sup_{f\in\mathcal{F}} \sum_i f(X_i) = \sup_{f\in\mathcal{F}} |\sum_i f(X_i)|$. By the symmetrization lemma (Lemma 11.4), with i.i.d. Rademachers $\varepsilon_1, \dots, \varepsilon_n$ independent of the X_i,

$$\mathbb{E} \sup_{f\in\mathcal{F}} \sum_{i=1}^n f(X_i) \le 2\, \mathbb{E}_{X,\varepsilon} \sup_{f\in\mathcal{F}} \sum_{i=1}^n \varepsilon_i f(X_i).$$

Write $f = \sum_{j=1}^d \alpha_j g_j$ and set $S_j = \sum_{i=1}^n \varepsilon_i g_j(X_i)$. Then

$$\sum_{i=1}^n \varepsilon_i f(X_i) = \sum_{j=1}^d \alpha_j S_j,$$

and we must bound $\sup_{\alpha\in A} \sum_j \alpha_j S_j$ where

$$A = \left\{ \alpha : \sum_{j=1}^d \frac{\alpha_j^2}{\lambda_j} \le 1,\ \sum_{j=1}^d \alpha_j^2 \le R^2 \right\}.$$

Put $\mu_j = \lambda_j \wedge R^2$. Since $\frac{1}{\mu_j} = \max\left(\frac{1}{\lambda_j}, \frac{1}{R^2}\right) \le \frac{1}{\lambda_j} + \frac{1}{R^2}$, for $\alpha \in A$,

$$\sum_{j=1}^d \frac{\alpha_j^2}{\mu_j} \le \sum_{j=1}^d \frac{\alpha_j^2}{\lambda_j} + \frac{1}{R^2} \sum_{j=1}^d \alpha_j^2 \le 1 + 1 = 2.$$

Hence $A \subseteq \{\alpha : \sum_j \alpha_j^2/\mu_j \le 2\}$, and by Cauchy–Schwarz, with $\beta_j = \alpha_j/\sqrt{\mu_j}$,

$$\sup_{\alpha\in A} \sum_{j=1}^d \alpha_j S_j \le \sup_{\|\beta\|_2 \le \sqrt{2}} \sum_{j=1}^d \beta_j \sqrt{\mu_j}\, S_j \le \sqrt{2} \left(\sum_{j=1}^d \mu_j S_j^2 \right)^{1/2}.$$

Fix the X_i and take expectation over ε. By Jensen's inequality,

$$\mathbb{E}_\varepsilon \sup_{\alpha\in A} \sum_j \alpha_j S_j \le \sqrt{2} \left(\sum_{j=1}^d \mu_j\, \mathbb{E}_\varepsilon S_j^2 \right)^{1/2} = \sqrt{2} \left(\sum_{j=1}^d \mu_j \sum_{i=1}^n g_j(X_i)^2 \right)^{1/2},$$

using $\mathbb{E}_\varepsilon S_j^2 = \sum_{i=1}^n g_j(X_i)^2$. Taking expectation over X and applying Jensen again,

$$\mathbb{E}_X \left(\sum_{j=1}^d \mu_j \sum_{i=1}^n g_j(X_i)^2 \right)^{1/2} \le \left(\sum_{j=1}^d \mu_j \sum_{i=1}^n \mathbb{E} g_j(X_i)^2 \right)^{1/2} = \sqrt{n \sum_{j=1}^d \mu_j},$$

since $\mathbb{E} g_j(X)^2 = 1$ by orthonormality. Combining with the symmetrization factor 2,

$$\mathbb{E} \sup_{f\in\mathcal{E}_1(1)\cap\mathcal{E}_2(R)} \sum_{i=1}^n f(X_i) \le 2\sqrt{2} \sqrt{n \sum_{j=1}^d (\lambda_j \wedge R^2)} \le 4 \sqrt{n \sum_{j=1}^d \lambda_j \wedge R^2},$$

as claimed.

Exercise 13.21

Central idea. The domination is pointwise, not probabilistic: for every realization of the X_i and ε_i, the triangle inequality gives $|\sum_i \varepsilon_i g(X_i)| \le \sup_{f\in\mathcal{F}} |\sum_i \varepsilon_i f(X_i)|$ for each $g \in \overline{\mathrm{sconv}}(\mathcal{F})$, because the weights satisfy $\sum_j |\lambda_j| \le 1$. Hence the inequality holds with $\kappa = 1$ (no chaining or VC theory needed); we assume the suprema are measurable, e.g. $\mathcal{F}$ countable, or interpret $\mathbb{E}$ as outer expectation.

Solution. Fix a realization of $X_1, \dots, X_n$ and $\varepsilon_1, \dots, \varepsilon_n$. Let $g \in \overline{\mathrm{sconv}}(\mathcal{F})$, so $g = \sum_{j=1}^k \lambda_j f_j$ for some $k \in \mathbb{N}$, $f_j \in \mathcal{F}$, and scalars with $\sum_{j=1}^k |\lambda_j| \le 1$. Then

$$\begin{aligned}\left|\sum_{i=1}^n \varepsilon_i g(X_i)\right| &= \left|\sum_{j=1}^k \lambda_j \sum_{i=1}^n \varepsilon_i f_j(X_i)\right| \\ &\le \sum_{j=1}^k |\lambda_j| \left|\sum_{i=1}^n \varepsilon_i f_j(X_i)\right| \\ &\le \left(\sum_{j=1}^k |\lambda_j|\right) \sup_{f\in\mathcal{F}} \left|\sum_{i=1}^n \varepsilon_i f(X_i)\right| \\ &\le \sup_{f\in\mathcal{F}} \left|\sum_{i=1}^n \varepsilon_i f(X_i)\right|.\end{aligned}$$

Since $g \in \overline{\mathrm{sconv}}(\mathcal{F})$ was arbitrary, taking the supremum over g on the left yields the pointwise inequality

$$\sup_{g\in\overline{\mathrm{sconv}}(\mathcal{F})} \left|\sum_{i=1}^n \varepsilon_i g(X_i)\right| \le \sup_{f\in\mathcal{F}} \left|\sum_{i=1}^n \varepsilon_i f(X_i)\right|,$$

valid for every realization. Taking expectations (assuming the suprema are measurable, or using outer expectations, under which the pointwise domination is preserved) gives

$$\mathbb{E} \sup_{g\in\overline{\mathrm{sconv}}(\mathcal{F})} \left|\sum_{i=1}^n \varepsilon_i g(X_i)\right| \le \mathbb{E} \sup_{f\in\mathcal{F}} \left|\sum_{i=1}^n \varepsilon_i f(X_i)\right|,$$

i.e. the claimed bound with the universal constant $\kappa = 1$. ■

Exercise 13.22

Central idea. A volume comparison in both directions: a maximal ε-separated set in B_p^d yields disjoint $\varepsilon/2$-balls packed inside $(1+\varepsilon/2)B_p^d$ (upper bound on the covering number), while any ε-cover of B_p^d must have total volume at least that of B_p^d (lower bound). Since $\mathrm{vol}(rB_p^d) = r^d \mathrm{vol}(B_p^d)$ and volume is translation invariant, both bounds follow from counting.

Solution. Let $N(\varepsilon) = N(\varepsilon, B_p^d, \|\cdot\|_p)$ be the minimal number of ℓ_p^d-balls of radius ε covering B_p^d, so $H(\varepsilon, B_p^d) = \log N(\varepsilon)$. Fix $\varepsilon > 0$.

Upper bound. Let $\{x_1, \dots, x_M\} \subset B_p^d$ be a maximal ε-separated set, i.e. $\|x_i - x_j\|_p > \varepsilon$ for $i \ne j$; it is finite since B_p^d is compact. The balls $B(x_i, \varepsilon/2)$ are pairwise disjoint by the triangle inequality. Moreover each is contained in $(1+\varepsilon/2)B_p^d$: if $\|y - x_i\|_p \le \varepsilon/2$ and $\|x_i\|_p \le 1$, then $\|y\|_p \le 1 + \varepsilon/2$. Comparing volumes (using translation invariance and $\mathrm{vol}(rB_p^d) = r^d \mathrm{vol}(B_p^d)$),

$$M\left(\tfrac{\varepsilon}{2}\right)^d \mathrm{vol}(B_p^d) = \sum_{i=1}^M \mathrm{vol}\big(B(x_i, \tfrac{\varepsilon}{2})\big) \le \mathrm{vol}\big((1+\tfrac{\varepsilon}{2})\, B_p^d\big) = \left(1+\tfrac{\varepsilon}{2}\right)^d \mathrm{vol}(B_p^d),$$

so $M \le \left(\frac{1+\varepsilon/2}{\varepsilon/2}\right)^d = \left(1+\frac{2}{\varepsilon}\right)^d$. A maximal ε-separated set is an ε-net (any $x \in B_p^d$ at distance $> \varepsilon$ from all x_i could be adjoined, contradicting maximality), hence $N(\varepsilon) \le M$, and

$$H(\varepsilon, B_p^d) = \log N(\varepsilon) \le d \log\left(1+\frac{2}{\varepsilon}\right).$$

Lower bound. Let $\{y_1, \dots, y_K\} \subset \mathbb{R}^d$ with $K = N(\varepsilon)$ be centers of an ε-cover of B_p^d, so $B_p^d \subseteq \bigcup_{j=1}^K B(y_j, \varepsilon)$. By subadditivity and translation invariance of volume,

$$\operatorname{vol}(B_p^d) \le \sum_{j=1}^K \operatorname{vol}(B(y_j, \varepsilon)) = K\, \varepsilon^d \operatorname{vol}(B_p^d),$$

whence $K \ge \varepsilon^{-d}$ and

$$H(\varepsilon, B_p^d) = \log K \ge d \log \frac{1}{\varepsilon}.$$

Both bounds hold for every $\varepsilon > 0$ (and hence in particular for $0 < \varepsilon < 1$). ■

Exercise 13.23

Central idea. The case $p = 2$ is an identity: independence and $\mathbb{E}X_i = 0$ give $\mathbb{E}\|S_n\|_2^2 = \sum_i \mathbb{E}\|X_i\|_2^2$. The ℓ_p-norm comparison inequalities $\|x\|_q \le \|x\|_p \le d^{1/p-1/q}\|x\|_q$ for $1 \le p \le q \le \infty$ then transfer this to general p, producing $K(p,d) = d^{2/p-1}$ for $p \le 2$ and $d^{1-2/p}$ for $p \ge 2$. (This crude dimension-dependent bound is what the sharper Nemirovski inequality, Theorem 13.10, improves upon.)

Solution. Let $X_1, \dots, X_n$ be independent mean-zero random vectors in $\mathbb{R}^d$ and $S_n = \sum_{i=1}^n X_i$. For $p = 2$, expanding coordinatewise and using independence with $\mathbb{E}X_i = 0$,

$$\mathbb{E}\|S_n\|_2^2 = \sum_{j=1}^d \mathbb{E}\left(\sum_{i=1}^n X_{i,j}\right)^2 = \sum_{j=1}^d \sum_{i=1}^n \mathbb{E}X_{i,j}^2 = \sum_{i=1}^n \mathbb{E}\|X_i\|_2^2,$$

since $\mathbb{E}[X_{i,j}X_{k,j}] = \mathbb{E}X_{i,j}\,\mathbb{E}X_{k,j} = 0$ for $i \ne k$.

Recall the norm comparisons on $\mathbb{R}^d$: for $1 \le p \le q \le \infty$,

$$\|x\|_q \le \|x\|_p \le d^{1/p-1/q}\,\|x\|_q,$$

by Hölder's inequality (power-mean monotonicity).

Case $1 \le p \le 2$. From $\|S_n\|_p \le d^{1/p-1/2}\|S_n\|_2$,

$$\mathbb{E}\|S_n\|_p^2 \le d^{2/p-1}\,\mathbb{E}\|S_n\|_2^2 = d^{2/p-1}\sum_{i=1}^n \mathbb{E}\|X_i\|_2^2 \le d^{2/p-1}\sum_{i=1}^n \mathbb{E}\|X_i\|_p^2 = d^{2/p-1}V,$$

using $\|X_i\|_2 \le \|X_i\|_p$ for $p \le 2$.

Case $2 \le p \le \infty$. From $\|S_n\|_p \le \|S_n\|_2$,

$$\mathbb{E}\|S_n\|_p^2 \le \mathbb{E}\|S_n\|_2^2 = \sum_{i=1}^n \mathbb{E}\|X_i\|_2^2 \le d^{1-2/p}\sum_{i=1}^n \mathbb{E}\|X_i\|_p^2 = d^{1-2/p}V,$$

using $\|X_i\|_2 \le d^{1/2-1/p}\|X_i\|_p$ for $p \ge 2$ (with the convention $1/\infty = 0$).

Thus $\mathbb{E}\|S_n\|_p^2 \le K(p,d)V$ with $K(p,d) = d^{2/p-1}$ for $1 \le p \le 2$ and $K(p,d) = d^{1-2/p}$ for $2 \le p \le \infty$, noting both formulas give $K(2,d) = 1$. ■

Exercise 13.24

Central idea. For $p \geq 2$ the crude bound $K(p,d) = d^{1-2/p}$ of Exercise 13.23 comes from comparing ℓ_p directly to ℓ_2; interpolating through an intermediate exponent $q \in [2,p]$ and using the sharp type-2 inequality $\mathbb{E}\|S_n\|_q^2 \leq (q-1)\sum_{i=1}^n \mathbb{E}\|X_i\|_q^2$ for ℓ_q ($q \geq 2$) — proved by induction from a Hessian bound on $z \mapsto \|z\|_q^2$ — gives $\mathbb{E}\|S_n\|_p^2 \leq (q-1)d^{2/q-2/p}V$ for every $q \in [2,p]$, and optimizing over q yields the infimum.

Solution. Let $p \geq 2$, $\mathbb{E}X_i = 0$, and $V = \sum_{i=1}^n \mathbb{E}\|X_i\|_p^2 < \infty$. We first prove a type-2 inequality for ℓ_q, $q \geq 2$.

Lemma. For $q \geq 2$, $\mathbb{E}\|S_n\|_q^2 \leq (q-1)\sum_{i=1}^n \mathbb{E}\|X_i\|_q^2$.

Proof. It suffices that for deterministic x and mean-zero Y with $\mathbb{E}\|Y\|_q^2 < \infty$,

$$\mathbb{E}\|x+Y\|_q^2 \leq \|x\|_q^2 + (q-1)\mathbb{E}\|Y\|_q^2, \tag{$*$}$$

for then conditioning on $X_1, \dots, X_{n-1}$ and inducting gives the claim. Let $f(z) = \|z\|_q^2$. Where f is C^2,

$$\nabla^2 f(z) = 2(2-q)\|z\|_q^{2-2q} vv^T + 2(q-1)\|z\|_q^{2-q}\mathrm{diag}(|z_j|^{q-2}), \qquad v_j = |z_j|^{q-1}\mathrm{sgn}(z_j).$$

For $q \geq 2$ the first term is negative semidefinite, and by Hölder,

$$h^T\nabla^2 f(z)h \leq 2(q-1)\|z\|_q^{2-q}\sum_{j=1}^d |z_j|^{q-2}h_j^2 \leq 2(q-1)\|h\|_q^2 \qquad \forall\, h \in \mathbb{R}^d,$$

since $\sum_j |z_j|^{q-2}h_j^2 \leq \|z\|_q^{q-2}\|h\|_q^2$. Taylor's theorem with integral remainder gives, where defined,

$$f(x+y) \leq f(x) + \langle \nabla f(x), y\rangle + (q-1)\|y\|_q^2,$$

because $\int_0^1 (1-t)\,dt = 1/2$. Taking expectations and using $\mathbb{E}Y = 0$ yields $(*)$; the points where f fails to be C^2 form a null set, and $(*)$ extends past them by continuity and approximation. ■

Now fix $q \in [2,p]$. Since $q \leq p$, $\|z\|_p \leq \|z\|_q$ for all z (as $\sum_j |z_j|^p \leq (\sum_j |z_j|^q)^{p/q}$), so $\mathbb{E}\|S_n\|_p^2 \leq \mathbb{E}\|S_n\|_q^2$. By the Lemma,

$$\mathbb{E}\|S_n\|_q^2 \leq (q-1)\sum_{i=1}^n \mathbb{E}\|X_i\|_q^2.$$

Since $q \leq p$, $\|z\|_q \leq d^{1/q-1/p}\|z\|_p$ (Hölder), so $\sum_{i=1}^n \mathbb{E}\|X_i\|_q^2 \leq d^{2/q-2/p}V$. Combining,

$$\mathbb{E}\|S_n\|_p^2 \leq (q-1)d^{2/q-2/p}V \qquad \forall\, q \in [2,p],$$

and taking the infimum over $q \in [2,p]$ gives $K(p,d) \leq \inf_{q\in[2,p]}(q-1)d^{2/q-2/p}$. At $q = 2$ this recovers the bound $d^{1-2/p}$ of Exercise 13.23, so it is a genuine improvement.

Remark. The book writes the infimum over "$q \in [2,p] \cup \mathbb{R}$", which equals $\mathbb{R}$ and is a typo; the argument gives the infimum over $[2,p]$.

Exercise 13.25

Central idea. The Banach–Mazur distance measures how far $\mathbb{B}$ is from a Hilbert space $\mathbb{H}$; pushing the sum through a near-optimal isomorphism $T : \mathbb{B} \to \mathbb{H}$ reduces the problem to the Hilbert case, where independence and centering give the exact identity $\mathbb{E}\|TS_n\|_{\mathbb{H}}^2 = \sum_{i=1}^n \mathbb{E}\|TX_i\|_{\mathbb{H}}^2$ (all cross terms vanish), and the operator norms $\|T\|$, $\|T^{-1}\|$ convert back to $\mathbb{B}$ at the cost of $(\|T\|\|T^{-1}\|)^2$; optimizing over T yields $d_{\mathrm{bm}}(\mathbb{B},\mathbb{H})^2$.

Solution. Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$ and put $V = \sum_{i=1}^n \mathbb{E}\|X_i\|_{\mathbb{B}}^2 < \infty$ (the "strong variance", as in Exercise 13.23; centering is needed for the conclusion). Let $T : \mathbb{B} \to \mathbb{H}$ be a linear isomorphism. Then

$$\|S_n\|_{\mathbb{B}} = \|T^{-1}(TS_n)\|_{\mathbb{B}} \leq \|T^{-1}\| \cdot \|TS_n\|_{\mathbb{H}},$$

so $\mathbb{E}\|S_n\|_{\mathbb{B}}^2 \le \|T^{-1}\|^2\,\mathbb{E}\|TS_n\|_{\mathbb{H}}^2$. Now $TS_n = \sum_{i=1}^n TX_i$ with the TX_i independent and $\mathbb{E}[TX_i] = T(\mathbb{E}X_i) = 0$. In a Hilbert space,

$$\mathbb{E}\Big\|\sum_{i=1}^n Y_i\Big\|_{\mathbb{H}}^2 = \sum_{i=1}^n \mathbb{E}\|Y_i\|_{\mathbb{H}}^2 + 2\sum_{i<j}\mathbb{E}\langle Y_i, Y_j\rangle = \sum_{i=1}^n \mathbb{E}\|Y_i\|_{\mathbb{H}}^2$$

for independent mean-zero Y_i, since $\mathbb{E}\langle Y_i, Y_j\rangle = \langle \mathbb{E}Y_i, \mathbb{E}Y_j\rangle = 0$. Hence

$$\mathbb{E}\|TS_n\|_{\mathbb{H}}^2 = \sum_{i=1}^n \mathbb{E}\|TX_i\|_{\mathbb{H}}^2 \le \|T\|^2 \sum_{i=1}^n \mathbb{E}\|X_i\|_{\mathbb{B}}^2 = \|T\|^2 V,$$

using $\|Tx\|_{\mathbb{H}} \le \|T\|\,\|x\|_{\mathbb{B}}$. Combining,

$$\mathbb{E}\|S_n\|_{\mathbb{B}}^2 \le \|T\|^2\|T^{-1}\|^2 V.$$

Taking the infimum over all isomorphisms $T : \mathbb{B} \to \mathbb{H}$ gives $\mathbb{E}\|S_n\|_{\mathbb{B}}^2 \le d_{\mathrm{bm}}(\mathbb{B},\mathbb{H})^2 V$.

Exercise 13.26

Central idea. For unit vectors the sum and difference $W_i(\alpha) \pm W_i(\alpha')$ are independent centered Gaussians (the cross-covariance $\|\alpha\|^2 - \|\alpha'\|^2$ vanishes), so $W_i(\alpha)^2 - W_i(\alpha')^2$ factors as a product of two independent Gaussians with variances $2(1 \mp \cos\theta)$; the polarization identity $Z_1 Z_2 = \frac{1}{2}(Y_1 - Y_2)$ for the rotated independent χ_1^2 variables Y_1, Y_2 turns this product into $\sin\theta\,(Y_1 - Y_2)$, whose mgf is explicit; the Bernstein-type bound then follows from $-\log(1-x) \le x/(1-x)$ and $\sin^2\theta \le \|\alpha - \alpha'\|^2$.

Solution. In Section 13.6, T consists of unit vectors, so let $\|\alpha\| = \|\alpha'\| = 1$ and let θ be the angle between them, i.e. $\langle \alpha, \alpha'\rangle = \cos\theta$. Write $W_i(\alpha) = \sum_{j=1}^D \alpha_j X_{i,j}$ with $X_{i,j}$ iid $N(0,1)$.

Fix i. Then $(W_i(\alpha), W_i(\alpha'))$ is bivariate Gaussian with $\mathrm{Var}(W_i(\alpha)) = \mathrm{Var}(W_i(\alpha')) = 1$ and $\mathrm{Cov}(W_i(\alpha), W_i(\alpha')) = \cos\theta$. Put $A_i = W_i(\alpha) - W_i(\alpha')$ and $B_i = W_i(\alpha) + W_i(\alpha')$. Then $A_i \sim N(0, 2-2\cos\theta)$, $B_i \sim N(0, 2+2\cos\theta)$, and $\mathrm{Cov}(A_i, B_i) = \|\alpha\|^2 - \|\alpha'\|^2 = 0$, so A_i and B_i are independent. Since $W_i(\alpha)^2 - W_i(\alpha')^2 = A_i B_i$ and $2 \mp 2\cos\theta = 4\sin^2(\theta/2),\ 4\cos^2(\theta/2)$ respectively, writing $A_i = 2\sin(\theta/2) Z_1$, $B_i = 2\cos(\theta/2) Z_2$ with Z_1, Z_2 iid $N(0,1)$ gives

$$W_i(\alpha)^2 - W_i(\alpha')^2 = 2\sin\theta \cdot Z_1 Z_2.$$

With $U = (Z_1 + Z_2)/\sqrt{2}$, $V = (Z_1 - Z_2)/\sqrt{2}$ iid $N(0,1)$, we have $U^2 - V^2 = 2Z_1 Z_2$; setting $Y_1 = U^2$, $Y_2 = V^2$, which are independent χ_1^2, yields $Z_1 Z_2 = (Y_1 - Y_2)/2$. Hence

$$W_i(\alpha)^2 - W_i(\alpha')^2 = \sin\theta\, Y_1 - \sin\theta\, Y_2,$$

as claimed.

Now $\|W(\alpha)\|^2 - \|W(\alpha')\|^2 = \frac{1}{d}\sum_{i=1}^d \sin\theta\,(Y_{1,i} - Y_{2,i})$, a sum of d iid copies. For $|t| < 1/2$ the χ_1^2 mgf is $(1-2t)^{-1/2}$, so for $2|\lambda|\sin\theta/d < 1$,

$$\mathbb{E}e^{\lambda \sin\theta (Y_1 - Y_2)/d} = (1 - 2\lambda\sin\theta/d)^{-1/2}(1 + 2\lambda\sin\theta/d)^{-1/2} = (1 - 4\lambda^2\sin^2\theta/d^2)^{-1/2}.$$

Thus, with $\sin^2\theta = 1 - \langle\alpha,\alpha'\rangle^2$,

$$\log \mathbb{E}e^{\lambda(\|W(\alpha)\|^2 - \|W(\alpha')\|^2)} = -\frac{d}{2}\log\left(1 - \frac{4\lambda^2(1 - \langle\alpha,\alpha'\rangle^2)}{d^2}\right).$$

For the bound, put $x = 4\lambda^2\sin^2\theta/d^2$ and use $-\log(1-x) \le x/(1-x)$:

$$-\frac{d}{2}\log(1-x) \le \frac{d}{2}\cdot\frac{x}{1-x} = \frac{2\lambda^2\sin^2\theta/d}{1-x}.$$

Since $\sin^2\theta = (1-\cos\theta)(1+\cos\theta) = \|\alpha-\alpha'\|^2\frac{1+\cos\theta}{2} \le \|\alpha-\alpha'\|^2$, the numerator is at most $2\lambda^2\|\alpha-\alpha'\|^2/d$. Moreover $1-x \ge 1-8\lambda\|\alpha-\alpha'\|/d$: indeed $x \le 4\lambda^2\|\alpha-\alpha'\|^2/d^2$, so it suffices that $\lambda\|\alpha-\alpha'\|/d \le 2$, which follows from $\lambda < d/(8\|\alpha-\alpha'\|)$ (this also gives $x < 1/16$, so the mgf is valid). Hence

$$-\frac{d}{2}\log(1-x) \le \frac{2\lambda^2\|\alpha-\alpha'\|^2/d}{1-8\lambda\|\alpha-\alpha'\|/d} \le \frac{\lambda^2 8\|\alpha-\alpha'\|^2/d}{2(1-8\lambda\|\alpha-\alpha'\|/d)},$$

the last step using $2 \le 4$. This is the claimed inequality (in fact with a slightly sharper constant).

Exercise 13.27

Central idea. The constraint set $\mathcal{F}_r = \{(s,\alpha) : \alpha \in [0,1],\ \alpha L(s) \le r\}$ is star-shaped in α. Shrinking α by $c = r/r'$ maps $\mathcal{F}_{r'}$ into $\mathcal{F}_r$, so $\psi^*(r)/r$ is nonincreasing. Splitting α proportionally gives subadditivity. Monotonicity is immediate. These facts force continuity on $(0,\infty)$. But the book's definition of sub-linear (p. 388) also asks for continuity on $[0,\infty)$ and $\psi(1) \ge 1$, and both can fail for ψ^*. So the first claim holds only after a harmless modification. The weighted-supremum bound holds exactly as printed: $\alpha(s) = r/\max(r, L(s))$ is a feasible choice.

Solution.

Erratum (statement as printed). The book (p. 388; Exercise 13.41) calls $\psi : [0,\infty) \to [0,\infty)$ *sub-linear* if it is nondecreasing, continuous, $\psi(x)/x$ is nonincreasing, and $\psi(1) \ge 1$. In this sense, ψ^* need **not** be sub-linear:

- *Normalization fails.* If $X_{i,s} \equiv 0$ (with any admissible L), then $\psi^* \equiv 0$, so $\psi^*(1) = 0 < 1$. A non-degenerate example: $|\mathcal{T}| = 1$, $n = 1$, $X_{1,s} = \pm\varepsilon$ with probability $1/2$ each, $L(s) = \varepsilon < 1$. Then $\psi^*(1) = \mathbb{E}(X_{1,s})_+ = \varepsilon/2 < 1$.
- *Continuity at 0 fails.* Take $n \ge 1$ and $\mathcal{T} = \{1, 2, \dots\}$. Let all $X_{i,s}$ be independent (over i and s), with $X_{i,s} = +1$ and $X_{i,s} = -1$ each with probability $p_s/2$, and $X_{i,s} = 0$ otherwise, where $p_s = 1/(s+1)$. These are centered and $|X_{i,s}| \le 1$. Put $L(s) = \sqrt{p_s} = (\mathbb{E}X_{i,s}^2)^{1/2}$. Since every $L(s) > 0$, $r = 0$ forces $\alpha = 0$, so $\psi^*(0) = 0$. Now fix $r > 0$ and let $S_r = \{s : p_s \le \min(r^2, 1/2)\}$, an infinite tail of $\mathcal{T}$. For $s \in S_r$, $\alpha = 1$ is feasible, so $Y_r \ge \sup_{s\in S_r}(\sum_i X_{i,s})_+$. The events $E_s = \{\sum_i X_{i,s} \ge 1\}$ are independent across s, and
$$\mathbb{P}(E_s) \ge \mathbb{P}\{X_{1,s} = 1,\ X_{i,s} = 0\ (i \ge 2)\} = \tfrac{p_s}{2}(1-p_s)^{n-1} \ge 2^{-n}p_s,$$
so $\sum_{s\in S_r}\mathbb{P}(E_s) = \infty$. By the second Borel–Cantelli lemma, some E_s with $s \in S_r$ occurs almost surely. The sums are integers, so $Y_r \ge 1$ a.s. Hence $\psi^*(r) \ge 1$ for every $r > 0$, while $\psi^*(0) = 0$: ψ^* jumps at 0. (Numerically, for $n = 1$, $r = 0.1$, the probability bound $\prod_s(1-\mathbb{P}(E_s))$ that no E_s occurs with $99 \le s \le S$ decreases toward 0 as S grows. It equals $\exp(\sum\log(1-p_s/2))$, which behaves like $(100/S)^{1/2}$: about 0.0996 at $S = 10^4$, 0.00996 at $S = 10^6$ and 0.000996 at $S = 10^8$.)

Corrected statement. Assume, as in Lemma 13.18 and Theorem 13.19, that $\mathcal{T}$ is countable, so that all suprema below are measurable. Then:

(a) $0 \le \psi^* \le n$, $\psi^*(0) = 0$, and ψ^* is nondecreasing and subadditive on $[0,\infty)$. On $(0,\infty)$, $\psi^*(r)/r$ is nonincreasing and ψ^* is continuous.

(b) Let $\tilde\psi(0) = \psi^*(0+) = \lim_{r\downarrow 0}\psi^*(r)$ and $\tilde\psi(r) = \psi^*(r)$ for $r > 0$. Then $\bar\psi(r) = \tilde\psi(r) + r$ is sub-linear in the book's sense and $\bar\psi \ge \psi^*$. (The property actually used in slicing arguments, $\psi(2^k r) \le 2^k\psi(r)$ for $r > 0$ as in the proof of Lemma 13.18, already holds for ψ^* itself by (a).)

(c) For every $r > 0$, $\mathbb{E}\sup_{s\in\mathcal{T}}\frac{r}{\max(r,L(s))}\sum_{i=1}^n X_{i,s} \le \psi^*(r)$, as printed.

The counterexamples above show that (a)–(b) cannot be upgraded to "ψ^* is sub-linear".

Proof. Note that $L(s) \ge (\mathbb{E}X_{i,s}^2)^{1/2} \ge 0$. Let

$$Y_r = \sup_{(s,\alpha)\in\mathcal{F}_r}\sum_{i=1}^n \alpha X_{i,s}, \qquad \psi^*(r) = \mathbb{E}Y_r.$$

For fixed s, the feasible α form an interval $[0, a_r(s)]$ with $a_r(s) = \min(1, r/L(s))$ (and $a_r(s) = 1$ if $L(s) = 0$). Hence $Y_r = \sup_s a_r(s)(\sum_i X_{i,s})_+$, a countable supremum, so Y_r is measurable.

Bounds and $\psi^*(0) = 0$. Since $\alpha = 0$ is always feasible and $|X_{i,s}| \le 1$, we have $0 \le Y_r \le n$. At $r = 0$: if $L(s) > 0$, then $a_0(s) = 0$; if $L(s) = 0$, then $\mathbb{E}X_{i,s}^2 = 0$, so $X_{i,s} = 0$ a.s. As $\mathcal{T}$ is countable, $Y_0 = 0$ a.s., and $\psi^*(0) = 0$.

Monotonicity. For $r \le r'$, $\mathcal{F}_r \subseteq \mathcal{F}_{r'}$, so $Y_r \le Y_{r'}$ and $\psi^*(r) \le \psi^*(r')$.

$\psi^*(r)/r$ nonincreasing on $(0, \infty)$. Let $0 < r < r'$ and $c = r/r' \in (0, 1)$. If $(s, \alpha) \in \mathcal{F}_{r'}$, then $(s, c\alpha) \in \mathcal{F}_r$, because $c\alpha \in [0, 1]$ and $c\alpha L(s) \le cr' = r$. Hence $Y_r \ge cY_{r'}$ pointwise. Taking expectations, $\psi^*(r)/r \ge \psi^*(r')/r'$.

Continuity on $(0, \infty)$. For $0 < r < r'$, the last two properties give $\psi^*(r) \le \psi^*(r') \le \frac{r'}{r}\psi^*(r)$. Letting $r' \downarrow r$ or $r \uparrow r'$, and using that ψ^* is finite, gives continuity at every point of $(0, \infty)$.

Subadditivity. Let $r_1, r_2 \ge 0$ with $r_1 + r_2 > 0$. For $(s, \alpha) \in \mathcal{F}_{r_1+r_2}$, put $\alpha_j = \alpha r_j/(r_1 + r_2)$. Then $\alpha_j \in [0, 1]$, $\alpha_j L(s) \le r_j$, and $\alpha = \alpha_1 + \alpha_2$. So $\sum_i \alpha X_{i,s} = \sum_i \alpha_1 X_{i,s} + \sum_i \alpha_2 X_{i,s} \le Y_{r_1} + Y_{r_2}$. Taking the supremum over $\mathcal{F}_{r_1+r_2}$ and then expectations, $\psi^*(r_1 + r_2) \le \psi^*(r_1) + \psi^*(r_2)$. The case $r_1 = r_2 = 0$ is trivial. This proves (a).

(b). $\tilde{\psi}(0) = \inf_{r>0} \psi^*(r)$ exists by monotonicity, and $0 \le \tilde{\psi}(0) \le \tilde{\psi}(r)$. So $\tilde{\psi}$ is nondecreasing, nonnegative, and continuous on $[0, \infty)$: at 0 by construction, elsewhere by (a). Also $\tilde{\psi} \ge \psi^*$, since they agree on $(0, \infty)$ and $\tilde{\psi}(0) \ge 0 = \psi^*(0)$. Therefore $\bar{\psi} = \tilde{\psi} + r$ is nondecreasing and continuous. On $(0, \infty)$, $\bar{\psi}(x)/x = \psi^*(x)/x + 1$ is nonincreasing, and $\bar{\psi}(1) = \psi^*(1) + 1 \ge 1$. So $\bar{\psi}$ is sub-linear and $\bar{\psi} \ge \psi^*$.

(c) The star-shaping inequality. Fix $r > 0$ and, for each s, put $\alpha(s) = r/\max(r, L(s))$. This is well defined, and $\alpha(s) \in (0, 1]$ because $r \le \max(r, L(s))$. Also

$$\alpha(s)L(s) = \frac{rL(s)}{\max(r, L(s))} \le r,$$

so $(s, \alpha(s)) \in \mathcal{F}_r$ for every s. Hence, pointwise,

$$\sup_{s\in\mathcal{T}} \frac{r}{\max(r, L(s))} \sum_{i=1}^n X_{i,s} = \sup_{s\in\mathcal{T}} \sum_{i=1}^n \alpha(s) X_{i,s} \le Y_r.$$

Taking expectations,

$$\mathbb{E} \sup_{s\in\mathcal{T}} \frac{r}{\max(r, L(s))} \sum_{i=1}^n X_{i,s} \le \psi^*(r) \ (\le \bar{\psi}(r)). \qquad \blacksquare$$

Exercise 13.28

Central idea. The m-th power map is locally Lipschitz with constant $m \max(\|X\|, \|Y\|)^{m-1}$ (telescoping sum), while $\exp(A/m)\exp(B/m)$ and $\exp((A+B)/m)$ agree to first order in $1/m$, differing only by $O(1/m^2)$. The $O(1/m^2)$ discrepancy, amplified by the Lipschitz constant m, vanishes as $m \to \infty$.

Solution. Let $\|\cdot\|$ be a submultiplicative matrix norm. For any $n \times n$ matrices X, Y,

$$X^m - Y^m = \sum_{j=0}^{m-1} X^{m-1-j}(X - Y)Y^j,$$

whence, with $M = \max(\|X\|, \|Y\|)$,

$$\|X^m - Y^m\| \le \sum_{j=0}^{m-1} \|X\|^{m-1-j}\|X - Y\|\|Y\|^j \le mM^{m-1}\|X - Y\|. \qquad (*)$$

Also $\|\exp(Z)\| \le e^{\|Z\|}$ for any Z.

Set $X_m = \exp((A+B)/m)$ and $Y_m = \exp(A/m)\exp(B/m)$. Then $\|X_m\| \le e^{\|A+B\|/m}$ and $\|Y_m\| \le e^{(\|A\|+\|B\|)/m}$, so with $M_m = \max(\|X_m\|, \|Y_m\|)$,

$$M_m^{m-1} \le e^{(\|A\|+\|B\|)(m-1)/m} \le e^{\|A\|+\|B\|}.$$

For any Z, writing $\exp(Z) = I + Z + R(Z)$,

$$\|R(Z)\| \le \sum_{k=2}^{\infty} \frac{\|Z\|^k}{k!} = \|Z\|^2 \sum_{k=0}^{\infty} \frac{\|Z\|^k}{(k+2)!} \le \|Z\|^2 e^{\|Z\|}.$$

Hence, for $m \ge 1$,

$$\Big\| \exp\frac{A}{m} - \Big(I + \frac{A}{m}\Big) \Big\| \le \frac{\|A\|^2 e^{\|A\|}}{m^2},$$

and similarly for B and $A+B$. Expanding the product,

$$Y_m - \Big(I + \frac{A+B}{m}\Big) = R_A + R_B + R_A R_B + \frac{A}{m} R_B + R_A \frac{B}{m} + \frac{AB}{m^2},$$

each term of which is $O(1/m^2)$ with constant depending only on A, B. Since $X_m - (I + (A+B)/m) = R_{A+B} = O(1/m^2)$ as well,

$$\|X_m - Y_m\| = O(1/m^2).$$

Applying $(*)$ to X_m, Y_m gives

$$\|X_m^m - Y_m^m\| \le m\, e^{\|A\|+\|B\|} \cdot O(1/m^2) = O(1/m) \xrightarrow[m\to\infty]{} 0.$$

Finally $X_m^m = (\exp((A+B)/m))^m = \exp(A+B)$, because $(A+B)/m$ commutes with itself. Therefore

$$\Big(\exp\frac{A}{m}\exp\frac{B}{m}\Big)^m = Y_m^m \longrightarrow \exp(A+B) \quad \text{as } m \to \infty,$$

which is the Lie product formula. ∎

Exercise 13.29

Central idea. Golden–Thompson follows from the Lie product formula once we can "unsplit" $\operatorname{trace}((XY)^{2^{k+1}})$ into $\operatorname{trace}(X^{2^{k+1}} Y^{2^{k+1}})$ by iterating the hint. The hint itself is Weyl's eigenvalue–singular-value majorization for XY combined with convexity of $t \mapsto t^{2m}$: $\sum \lambda_i^{2m} \le \sum \sigma_i^{2m} = \operatorname{trace}((X^2Y^2)^m)$.

Solution. We first prove the hint. Let X, Y be Hermitian positive semi-definite and $m \ge 1$ an integer.

Lemma. $\operatorname{trace}((XY)^{2m}) \le \operatorname{trace}((X^2Y^2)^m)$.

Proof of Lemma. The eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$ of XY are real and nonnegative (since XY has the same nonzero eigenvalues as $X^{1/2}YX^{1/2} \ge 0$), so $\operatorname{trace}((XY)^{2m}) = \sum_{i=1}^n \lambda_i^{2m}$. Let $\sigma_1 \ge \cdots \ge \sigma_n$ be the singular values of XY. By Weyl's majorization inequalities, $\sum_{i=1}^k \lambda_i \le \sum_{i=1}^k \sigma_i$ for all k. Since $\phi(t) = t^{2m}$ is convex, increasing on $[0,\infty)$ with $\phi(0) = \phi'(0) = 0$, we have $\phi(t) = \int_0^\infty (t-s)_+ \phi''(s)\, ds$, and for $c \ge 0$,

$$\sum_{i=1}^k (\lambda_i - c)_+ = \max_{0\le j\le k} \Big(\sum_{i=1}^j \lambda_i - jc \Big) \le \max_{0\le j\le k} \Big(\sum_{i=1}^j \sigma_i - jc \Big) = \sum_{i=1}^k (\sigma_i - c)_+,$$

whence integrating against $\phi''(s)\, ds \ge 0$ gives $\sum_{i=1}^k \lambda_i^{2m} \le \sum_{i=1}^k \sigma_i^{2m}$ for all k. For $k = n$,

$$\operatorname{trace}((XY)^{2m}) = \sum_{i=1}^n \lambda_i^{2m} \le \sum_{i=1}^n \sigma_i^{2m} = \operatorname{trace}(((XY)^*(XY))^m) = \operatorname{trace}((YX^2Y)^m).$$

Cyclically permuting the leading Y to the end gives $\operatorname{trace}((YX^2Y)^m) = \operatorname{trace}((X^2Y^2)^m)$. ∎

Now let A, B be Hermitian and put $m = 2^{k+1}$, $X = \exp(A/m)$, $Y = \exp(B/m)$, which are Hermitian positive definite. For $j = 0, \dots, k$, apply the Lemma to the positive semi-definite pair (X^{2^j}, Y^{2^j}) with exponent 2^{k-j}:

$$\operatorname{trace}((X^{2^j}Y^{2^j})^{2\cdot 2^{k-j}}) \le \operatorname{trace}(((X^{2^j})^2(Y^{2^j})^2)^{2^{k-j}}).$$

With $S_j = \operatorname{trace}((X^{2^j}Y^{2^j})^{2^{k+1-j}})$ this reads $S_j \le S_{j+1}$, so telescoping,

$$\operatorname{trace}((XY)^{2^{k+1}}) = S_0 \le S_{k+1} = \operatorname{trace}(X^{2^{k+1}}Y^{2^{k+1}}).$$

Since $X^m = \exp(A)$ and $Y^m = \exp(B)$ (as A/m commutes with itself),

$$\operatorname{trace}\Big(\big(\exp(A/m)\exp(B/m)\big)^m\Big) \le \operatorname{trace}(\exp(A)\exp(B)) \qquad (m = 2^{k+1}).$$

By the Lie product formula (Exercise 13.28), $(\exp(A/m)\exp(B/m))^m \to \exp(A+B)$ as $m \to \infty$, hence along $m = 2^{k+1}$ as $k \to \infty$. The trace is continuous, so passing to the limit yields

$$\operatorname{trace}(\exp(A+B)) \le \operatorname{trace}(\exp(A)\exp(B)),$$

the Golden–Thompson inequality. ■

Exercise 13.30

Central idea. For the variance, use the Efron–Stein inequality (Theorem 3.1) with the i-th sign resampled, as the hint suggests (Example 3.6 replacing the Gaussian Poincaré inequality). Resampling a Rademacher variable either does nothing or flips it. A flip can lower the norm by at most $2|\langle u, A_i u\rangle|$, where u is the top eigenvector of the *unflipped* sum, and $\sum_i \langle u, A_i u\rangle^2 \le \sigma^2$. Only the positive part of $Z - Z_i'$ enters, so this one-sided control is enough. The result is $\operatorname{Var}(Z) \le 2\sigma^2$; this method does not reach the printed σ^2 (which we neither prove nor refute). For the expectation, compare the Rademacher sum with a Gaussian sum (Exercise 11.14, constant $\sqrt{\pi/2}$) and apply the Gaussian bound $\sqrt{2\log(2d)}\,\sigma$ of Theorem 13.8.

Solution.

Status: incomplete — the printed variance bound $\operatorname{Var}(Z) \le \sigma^2$ is **not proved** here; only $\operatorname{Var}(Z) \le 2\sigma^2$ is. The expectation bound $\mathbb{E}Z \le \sqrt{\pi\log(2d)}\,\sigma$ is proved in full.

Erratum (to an earlier version of this solution). An earlier version claimed $\operatorname{Var}(Z) \le \sigma^2$ through the pointwise bound $\sum_i (Z - Z_i)^2 \le \sum_i \langle u, A_i u\rangle^2 \le \sigma^2$, with $Z_i = \|\sum_{j\ne i} X_j A_j\|$. That step is invalid. The argument only controls $Z - Z_i$ from above, by $sX_i\langle u, A_i u\rangle$. But Z_i can exceed Z, and the size of $Z_i - Z$ is governed by the top eigenvector of $\sum_{j\ne i} X_j A_j$, not by u. Counterexample to the pointwise bound: $d = 2$, $n = 3$, with

$$A_1 = \begin{pmatrix} -0.9 & -0.2 \\ -0.2 & 0.4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0.4 & -0.1 \\ -0.1 & 0.4 \end{pmatrix}, \quad A_3 = \begin{pmatrix} -0.5 & -0.3 \\ -0.3 & 0.7 \end{pmatrix}, \quad x = (-1, -1, 1).$$

Here $Z = 0.1$, $\sum_i \langle u, A_i u\rangle^2 = 0.81$, and $\sigma^2 \approx 1.3639$, but $\sum_i (Z - Z_i)^2 \approx 1.4105 > \sigma^2$. These values were checked numerically.

Notation. Write $\|\cdot\|$ for the operator norm. For symmetric M, $\|M\| = \max_{\|v\|=1} |\langle v, Mv\rangle|$, so $\|M\| \ge \pm\langle v, Mv\rangle$ for every unit vector v. For $x \in \{-1, 1\}^n$, put $M(x) = \sum_i x_i A_i$ and $Z = \|M(X)\|$. Let $\bar{x}^{(i)}$ denote x with its i-th coordinate replaced by $-x_i$.

Key estimate. For each x, choose a unit vector $u = u(x)$ and a sign $s = s(x) \in \{-1, 1\}$ with $\|M(x)\| = s\langle u, M(x)u\rangle$: take an eigenvector for an eigenvalue of largest absolute value. Then

$$\sum_i \langle u, A_i u\rangle^2 \le \sum_i \|A_i u\|^2 = \Big\langle u, \sum_i A_i^2\, u\Big\rangle \le \sigma^2, \tag{1}$$

by Cauchy–Schwarz ($\langle u, A_i u\rangle^2 \le \|u\|^2\|A_i u\|^2$) and the symmetry of A_i. Moreover, for each i,

$$\|M(\bar{x}^{(i)})\| \ge s\langle u, M(\bar{x}^{(i)})u\rangle = s\langle u, M(x)u\rangle - 2sx_i\langle u, A_i u\rangle = \|M(x)\| - 2sx_i\langle u, A_i u\rangle.$$

Hence

$$\big(\|M(x)\| - \|M(\bar{x}^{(i)})\|\big)_+ \le 2|\langle u(x), A_i u(x)\rangle|. \tag{2}$$

Variance: $\mathrm{Var}(Z) \le 2\sigma^2$. Let X_i' be an independent copy of X_i, and let Z_i' be Z computed with X_i' in place of X_i. The Efron–Stein inequality (Theorem 3.1) gives $\mathrm{Var}(Z) \le \sum_i \mathbb{E}(Z - E^{(i)}Z)^2 = \frac{1}{2}\sum_i \mathbb{E}(Z - Z_i')^2$. Since (Z, Z_i') is exchangeable, $\mathbb{E}(Z - Z_i')^2 = 2\,\mathbb{E}(Z - Z_i')_+^2$, so

$$\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}(Z - Z_i')_+^2.$$

With probability $1/2$, independently of X, $X_i' = X_i$ and $Z_i' = Z$; otherwise $Z_i' = \|M(\bar{X}^{(i)})\|$. Therefore, by (2) and then (1),

$$\mathrm{Var}(Z) \le \frac{1}{2}\sum_{i=1}^n \mathbb{E}\big(\|M(X)\| - \|M(\bar{X}^{(i)})\|\big)_+^2 \le \frac{1}{2}\,\mathbb{E}\sum_{i=1}^n 4\langle u(X), A_i u(X)\rangle^2 \le 2\sigma^2.$$

Why this stops at $2\sigma^2$. In the Gaussian case (Theorem 13.8), the Poincaré inequality uses $\|\nabla Z\|^2 = \sum_i \langle u, A_i u\rangle^2 \le \sigma^2$ with constant 1. With signs, a resampled coordinate moves by 2, not infinitesimally. Efron–Stein then costs exactly the factor $\frac{1}{2}\cdot 2^2 = 2$. We could not find a valid argument giving constant 1.

Numerical evidence on the constant 1 *(not a proof).* We computed $\mathrm{Var}(Z)/\sigma^2$ exactly by enumerating all 2^n sign patterns. This covered 4,000 random instances with $n \le 6$, $d \le 3$ and random symmetric Gaussian matrices with random scales, plus 60 hill-climbing searches that maximized the ratio. The largest ratio found was 0.4996. The value $1/2$ is attained at $d = 1$, $n = 2$, $A_1 = A_2 = 1$: there $Z \in \{0, 2\}$ with equal probability, so $\mathrm{Var}(Z) = 1$, while $\sigma^2 = 2$. So the evidence suggests $\mathrm{Var}(Z) \le \sigma^2/2$, which would imply the printed bound. Whether $\mathrm{Var}(Z) \le \sigma^2$ holds in general remains **open** here.

Expectation: $\mathbb{E}Z \le \sqrt{\pi \log(2d)}\,\sigma$. Let $Y_1, \dots, Y_n$ be independent standard Gaussians. Then $(\operatorname{sign} Y_i)_i$ are independent Rademacher variables, independent of $(|Y_i|)_i$, and $\mathbb{E}|Y_i| = \sqrt{2/\pi}$. Symmetric $d \times d$ matrices with the operator norm form a separable Banach space. Exercise 11.14 with $p = 1$, $v_i = A_i$ and $C = \sqrt{\pi/2}$ gives

$$\mathbb{E}\Big\|\sum_{i=1}^n X_i A_i\Big\| \le \sqrt{\frac{\pi}{2}}\;\mathbb{E}\Big\|\sum_{i=1}^n Y_i A_i\Big\|.$$

Theorem 13.8 gives $\mathbb{E}\|\sum_i Y_i A_i\| \le \sqrt{2\log(2d)}\,\sigma$. Combining,

$$\mathbb{E}Z \le \sqrt{\frac{\pi}{2}}\cdot\sqrt{2\log(2d)}\,\sigma = \sqrt{\pi\log(2d)}\,\sigma.$$

Summary. Proved: $\mathbb{E}Z \le \sqrt{\pi\log(2d)}\,\sigma$ and $\mathrm{Var}(Z) \le 2\sigma^2$. Not proved: the printed $\mathrm{Var}(Z) \le \sigma^2$. Exact enumeration on small cases never exceeded $\sigma^2/2$.

Exercise 13.31

Central idea. Symmetrization (Lemma 11.4) reduces the expectation to the Rademacher sum $\sum_i \varepsilon_i X_i$. Two facts give the matrix mgf bound $\mathbb{E}_\varepsilon \operatorname{tr}\exp(\theta\sum_i \varepsilon_i X_i) \le \operatorname{tr}\exp(\frac{\theta^2}{2}\sum_i X_i^2)$: the scalar bound $\cosh t \le e^{t^2/2}$, applied through $\mathbb{E}_{\varepsilon_i} e^{\theta\varepsilon_i X_i} = \cosh(\theta X_i)$, and the Golden–Thompson telescoping argument of Section 13.4 (proof of Theorem 13.8). Combine this with $e^{\theta\|Y\|} \le \operatorname{tr} e^{\theta Y} + \operatorname{tr} e^{-\theta Y}$ and with $X_i^2 \preceq A_i^2$, which gives $\|\sum_i X_i^2\| \le \sigma^2$. Chernoff optimization then yields $\mathbb{E}Z \le 2\sqrt{2\log(2d)}\,\sigma$. For the variance and the tail,

write $Z = |u^T(\sum_j X_j)u|$ for an X-measurable unit vector u. Replacing one X_i by an independent copy changes Z by at most $|u^T(X_i - X_i')u|$ from above. Since X_i' is centered, this gives $V_+ \le 2\sigma^2$ almost surely. Efron–Stein then gives $\mathrm{Var}(Z) \le 2\sigma^2$. The exponential Efron–Stein inequality (Theorem 6.16) gives $\mathbb{P}\{Z \ge \mathbb{E}Z + t\} \le e^{-t^2/(8\sigma^2)}$, with no dimensional prefactor.

Solution. Corrections to the statement. - $\log d$ must be $\log(2d)$: the printed expectation bound is false for $d = 1$. - The tail must read $e^{-t^2/(8\sigma^2)}$. The printed $e^{-t^2/8}$ is the case $\sigma = 1$, and a σ^2 is forced by scaling.

A pre-repair version of this solution also claimed that the tail needs a $2d$ prefactor. That claim is false and is withdrawn: the bound below has none.

Status: incomplete (variance constant only). We prove $\mathrm{Var}(Z) \le 2\sigma^2$, not the printed $\mathrm{Var}(Z) \le \sigma^2$. The independent audit also proves only $2\sigma^2$. - In the special case $X_i = \varepsilon_i A_i$ with Rademacher ε_i, the book asserts after Theorem 13.8 that Rudelson's bound $\mathrm{Var}(Z) \le \sigma^2$ remains valid. It leaves the details to the reader, and we do not prove it here. - For general centered X_i with $X_i^2 \preceq A_i^2$, we have neither a proof of $\mathrm{Var}(Z) \le \sigma^2$ nor a counterexample. - A numerical search over Rademacher sums $\sum_i \varepsilon_i B_i$ with $d \in \{2, 3\}$ and $n \le 6$ found $\mathrm{Var}(Z)/\|\sum_i B_i^2\| \le 0.5$. This family is the Rademacher special case just mentioned, so the search says nothing about general X_i.

The expectation and tail bounds below are complete.

Preliminaries. "Symmetric" means self-adjoint, $X_i^T = X_i$. The X_i are centered but need not be symmetrically distributed, hence the need for symmetrization. $A_i^2 - X_i^2 \succeq 0$ a.s. means $X_i^2 \preceq A_i^2$ in the PSD order. Summing, $\sum_i X_i^2 \preceq \sum_i A_i^2$ a.s. Both sides are PSD, so $\|\sum_i X_i^2\| \le \|\sum_i A_i^2\| = \sigma^2$ a.s. Also $\mathbb{E}[X_i^2] \preceq A_i^2$. For a unit vector u and symmetric B, Cauchy–Schwarz gives $(u^TBu)^2 \le \|Bu\|^2 = u^TB^2u$.

Matrix mgf lemma. Let $X_1, \dots, X_n$ be fixed symmetric $d \times d$ matrices and ε_i i.i.d. Rademacher variables. For all $\theta \in \mathbb{R}$,

$$\mathbb{E}_\varepsilon \mathrm{tr} \exp\Big(\theta \sum_{i=1}^n \varepsilon_i X_i\Big) \le \mathrm{tr} \exp\Big(\frac{\theta^2}{2} \sum_{i=1}^n X_i^2\Big).$$

To see this, follow the proof of Theorem 13.8 in Section 13.4, with Rademacher signs in place of Gaussians. Put $D_0 = \frac{\theta^2}{2}\sum_i X_i^2$ and $D_j = D_{j-1} + Y_j$ with $Y_j = \theta\varepsilon_j X_j - \frac{\theta^2}{2}X_j^2$, so $D_n = \theta\sum_i \varepsilon_i X_i$. The two terms of Y_j commute, so $e^{Y_j} = e^{\theta\varepsilon_j X_j} e^{-\theta^2 X_j^2/2}$ and

$$C_j := \mathbb{E}_{\varepsilon_j} e^{Y_j} = \cosh(\theta X_j)\, e^{-\theta^2 X_j^2/2}.$$

By the scalar inequality $\cosh t \le e^{t^2/2}$ and the spectral mapping theorem, $0 \preceq C_j \preceq I$. By the Golden–Thompson inequality (Exercise 13.29), $\mathrm{tr}\, e^{D_{j-1}+Y_j} \le \mathrm{tr}(e^{D_{j-1}} e^{Y_j})$. Since ε_j is independent of D_{j-1}, linearity of the trace gives

$$\mathbb{E}_\varepsilon \mathrm{tr}\, e^{D_j} \le \mathrm{tr}(\mathbb{E}_\varepsilon[e^{D_{j-1}}]\, C_j) \le \mathbb{E}_\varepsilon \mathrm{tr}\, e^{D_{j-1}}.$$

The last step uses $\mathrm{tr}(PC) = \mathrm{tr}(P^{1/2}CP^{1/2}) \le \mathrm{tr}P$ for PSD P and $C \preceq I$. Iterating from $j = n$ down to $j = 1$ gives $\mathbb{E}_\varepsilon \mathrm{tr}\, e^{D_n} \le \mathrm{tr}\, e^{D_0}$, which is the lemma. Moreover, $\mathrm{tr}\exp(\frac{\theta^2}{2}\sum_i X_i^2) \le d\exp(\frac{\theta^2}{2}\|\sum_i X_i^2\|)$.

Expectation. By symmetrization (Lemma 11.4), $\mathbb{E}Z \le 2\,\mathbb{E}_{X,\varepsilon}\|\sum_i \varepsilon_i X_i\|$. Fix X and put $Y = \sum_i \varepsilon_i X_i$. Since

$$e^{\theta\|Y\|} = \max(e^{\theta\lambda_{\max}(Y)}, e^{\theta\lambda_{\max}(-Y)}) \le \mathrm{tr}\, e^{\theta Y} + \mathrm{tr}\, e^{-\theta Y},$$

the lemma gives, for $\theta > 0$,

$$\mathbb{E}_\varepsilon e^{\theta\|Y\|} \le 2d \exp\Big(\frac{\theta^2\sigma^2}{2}\Big) \quad \text{a.s.}$$

By Jensen, $\mathbb{E}_\varepsilon\|Y\| \le \frac{1}{\theta}\log \mathbb{E}_\varepsilon e^{\theta\|Y\|} \le \frac{\log(2d)}{\theta} + \frac{\theta\sigma^2}{2}$. Take $\mathbb{E}_X$ and choose $\theta = \sqrt{2\log(2d)}/\sigma$. This gives $\mathbb{E}_{X,\varepsilon}\|\sum_i \varepsilon_i X_i\| \le \sqrt{2\log(2d)}\,\sigma$, and hence

$$\mathbb{E}Z \le 2\sqrt{2\log(2d)}\,\sigma.$$

Remark. The printed $2\sqrt{2\log d}\,\sigma$ is false for $d = 1$. Take $d = 1$, $n = 1$, $X_1 = \pm 1$ with probability $1/2$ each, and $A_1 = 1$. Then $\mathbb{E}Z = 1$ and $\sigma = 1$, but the printed right-hand side is 0.

A one-coordinate perturbation bound. Let $S = \sum_j X_j$. Choose an X-measurable unit vector u and sign $\eta \in \{\pm 1\}$ with $Z = \|S\| = \eta\, u^T S u$. For instance, take a unit eigenvector for the eigenvalue of largest modulus, chosen by a fixed measurable rule. Let X_i' be an independent copy of X_i, independent of X. Let $Z_i' = \|S - X_i + X_i'\|$. Then

$$Z_i' \geq \eta\, u^T (S - X_i + X_i') u = Z - \eta\, u^T (X_i - X_i') u,$$

so $(Z - Z_i')_+ \leq |u^T X_i u - u^T X_i' u|$. Let $\mathbb{E}'$ denote expectation over X' only. Since u is X-measurable and $\mathbb{E}X_i' = 0$, the cross term vanishes:

$$\mathbb{E}'\big[(u^T X_i u - u^T X_i' u)^2\big] = (u^T X_i u)^2 + \mathbb{E}'(u^T X_i' u)^2 \leq u^T X_i^2 u + u^T \mathbb{E}[X_i^2] u \leq 2\, u^T A_i^2 u.$$

Summing over i,

$$V_+ := \sum_{i=1}^n \mathbb{E}'\big[(Z - Z_i')_+^2\big] \leq 2\, u^T \Big(\sum_i A_i^2 \Big) u \leq 2\sigma^2 \quad \text{a.s.}$$

Variance. The Efron–Stein inequality in the form $\mathrm{Var}(Z) \leq \mathbb{E}V_+$ (Theorem 3.1, in its form with independent copies X_i') gives

$$\mathrm{Var}(Z) \leq 2\sigma^2.$$

(See the status note above for the printed σ^2.)

Tail. Theorem 6.16 states: for $\theta, \lambda > 0$ with $\theta\lambda < 1$ and $\mathbb{E}e^{\lambda V_+/\theta} < \infty$,

$$\log \mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \frac{\lambda\theta}{1 - \lambda\theta} \log \mathbb{E}e^{\lambda V_+/\theta}.$$

Since $V_+ \leq 2\sigma^2$ a.s., the right-hand side is at most $\frac{\lambda\theta}{1-\lambda\theta} \cdot \frac{2\lambda\sigma^2}{\theta} = \frac{2\lambda^2\sigma^2}{1-\lambda\theta}$. Letting $\theta \downarrow 0$ gives $\log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \leq 2\lambda^2\sigma^2$ for all $\lambda > 0$. By the Chernoff bound with $\lambda = t/(4\sigma^2)$,

$$\mathbb{P}\{Z \geq \mathbb{E}Z + t\} \leq \exp\Big(-\lambda t + 2\lambda^2\sigma^2 \Big) = \exp\Big(-\frac{t^2}{8\sigma^2} \Big), \qquad t > 0.$$

The same bound also follows from Theorem 6.7, the bounded-differences-type inequality of the hint. Let $Z_i = \inf_{x_i'} f(X_1, \ldots, x_i', \ldots, X_n)$, the infimum taken over values x_i' with $x_i'^2 \preceq A_i^2$. The argument above gives $Z - Z_i \leq \sup_{x_i'} |u^T X_i u - u^T x_i' u| \leq 2\sqrt{u^T A_i^2 u}$. Hence $\sum_i (Z - Z_i)^2 \leq 4\sigma^2$, and Theorem 6.7 gives $e^{-t^2/(2\cdot 4\sigma^2)}$.

Remark. The printed $e^{-t^2/8}$ cannot hold as written. Scaling $(X_i, A_i, t) \mapsto (cX_i, cA_i, ct)$ leaves the left-hand side unchanged but sends $e^{-t^2/8}$ to $e^{-c^2t^2/8}$. So σ^2 must appear in the exponent. No dimensional prefactor is needed. Plain bounded differences would give $\sum_i \|A_i\|^2$ in place of $\sigma^2 = \|\sum_i A_i^2\|$, which can be much larger. ■

Exercise 13.32

Central idea. The map $Z = \|\sum_{i=1}^n X_i\|$ is self-bounding: removing one PSD summand can only decrease the operator norm, by at most $\|X_i\| \leq 1$, and the leave-one-out deficits sum to at most Z via the variational representation of the norm. Theorem 6.12 then yields $\mathrm{Var}(Z) \leq \mathbb{E}Z$ and the Bennett inequalities. For the expectation bound, split off the mean and symmetrize; conditional on the X_i, Rudelson's inequality (Theorem 13.8) controls the Rademacher sum by $\|\sum_i X_i^2\|^{1/2}$, and $X_i^2 \preceq X_i$ (since $0 \preceq X_i \preceq I$) turns this into $\sqrt{\mathbb{E}Z}$.

Solution. Write $S = \sum_{i=1}^n X_i$ and $Z = \|S\|$, and put $Z_i = \|\sum_{j \neq i} X_j\|$.

Self-bounding. Since each X_i is positive semidefinite, $\sum_{j\neq i} X_j \preceq \sum_{j=1}^n X_j$, and both sums are PSD. The operator norm is monotone for the PSD order ($\lambda_{\max}(A) \leq \lambda_{\max}(B)$ when $0 \preceq A \preceq B$), so $Z_i \leq Z$, i.e. $Z - Z_i \geq$

0. By the triangle inequality, $Z \le Z_i + \|X_i\| \le Z_i + 1$ almost surely, so $0 \le Z - Z_i \le 1$ for all i. Let v be a unit vector with $Z = v^T S v$ (possible since S is symmetric). Then $Z_i = \|\sum_{j\neq i} X_j\| \ge v^T(\sum_{j\neq i} X_j)v$, whence $Z - Z_i \le v^T X_i v$. Summing over i,

$$\sum_{i=1}^n (Z - Z_i) \le \sum_{i=1}^n v^T X_i v = v^T S v = Z.$$

Thus Z is self-bounding. By the Efron–Stein inequality, $\mathrm{Var}(Z) \le \sum_{i=1}^n \mathbb{E}[(Z - Z_i)^2]$; since $0 \le Z - Z_i \le 1$, $(Z - Z_i)^2 \le Z - Z_i$, so $\mathrm{Var}(Z) \le \mathbb{E}[\sum_i (Z - Z_i)] \le \mathbb{E}Z$. Theorem 6.12 applied to Z gives, with $h(u) = (1+u)\log(1+u) - u$,

$$\mathbb{P}\{Z \ge \mathbb{E}Z + t\} \le \exp\!\left(-\mathbb{E}Z\, h\!\left(\frac{t}{\mathbb{E}Z}\right)\right), \qquad t \ge 0,$$

$$\mathbb{P}\{Z \le \mathbb{E}Z - t\} \le \exp\!\left(-\mathbb{E}Z\, h\!\left(\frac{-t}{\mathbb{E}Z}\right)\right), \qquad 0 \le t \le \mathbb{E}Z,$$

as claimed.

Expectation bound. By the triangle inequality,

$$\mathbb{E}\|S\| \le \|\mathbb{E}S\| + \mathbb{E}\left\|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right\|.$$

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables, independent of the X_i. Symmetrization (Lemma 11.4) gives

$$\mathbb{E}\left\|\sum_{i=1}^n (X_i - \mathbb{E}X_i)\right\| \le 2\,\mathbb{E}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\|.$$

Condition on $X_1, \dots, X_n$. By the Rademacher version of Rudelson's inequality (Exercise 13.30, with $C = \sqrt{\pi/2}$) applied to the fixed symmetric matrices X_i,

$$\mathbb{E}_\varepsilon\left[\left\|\sum_{i=1}^n \varepsilon_i X_i\right\| \,\middle|\, X\right] \le \sqrt{\frac{\pi}{2}}\,\sqrt{2\log(2d)}\,\left\|\sum_{i=1}^n X_i^2\right\|^{1/2}.$$

Taking expectations and using Jensen's inequality (concavity of $\sqrt{\cdot}$),

$$\mathbb{E}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\| \le \sqrt{\frac{\pi}{2}}\,\sqrt{2\log(2d)}\,\sqrt{\mathbb{E}\left\|\sum_{i=1}^n X_i^2\right\|}.$$

Since $0 \preceq X_i \preceq I$ (as $\|X_i\| \le 1$ and $X_i \succeq 0$), the eigenvalues of X_i lie in $[0,1]$, so $X_i^2 \preceq X_i$. Hence $\sum_i X_i^2 \preceq \sum_i X_i = S$ and, both being PSD, $\|\sum_i X_i^2\| \le \|S\| = Z$. Therefore $\mathbb{E}\|\sum_i X_i^2\| \le \mathbb{E}Z$, and with $m = \mathbb{E}Z$,

$$\mathbb{E}\left\|\sum_{i=1}^n \varepsilon_i X_i\right\| \le \sqrt{\pi\log(2d)}\sqrt{m}.$$

Combining,

$$\mathbb{E}Z \le \left\|\mathbb{E}\sum_{i=1}^n X_i\right\| + 2\sqrt{\pi\log(2d)}\sqrt{\mathbb{E}Z}.$$

Remark on the constant. The stated $2\sqrt{2\log d}$ follows from the same argument with a one-sided refinement of Rudelson's inequality (bounding $\mathbb{E}[\lambda_{\max}(\cdot)_+]$ directly, which replaces $\log(2d)$ by $\log d$ and absorbs the Rademacher-to-Gaussian conversion factor $\sqrt{\pi/2}$ of Exercise 13.30); the $\log(2d)$ above is otherwise the direct consequence of Theorem 13.8 as stated in the book. Either way, with $x = \sqrt{\mathbb{E}Z}$ and $\mu = \|\mathbb{E}\sum_i X_i\|$, $x^2 \le Cx + \mu$ with the book's constant $C = 2\sqrt{2\log d}$ gives, by $Cx \le x^2/2 + C^2/2$,

$$\mathbb{E}Z \le 8\log d + 2\left\|\mathbb{E}\sum_{i=1}^n X_i\right\|,$$

as noted in the hint. Jensen's inequality gives the matching lower bound $\mathbb{E}Z \ge \|\mathbb{E}\sum_i X_i\|$.

Exercise 13.33

Central idea. Write the operator norm as the supremum of an empirical process. With $\mathcal{T}$ a countable dense subset of $\{(u,v) : \|u\| = \|v\| = 1\}$, we have $Z = \sup_{(u,v)\in\mathcal{T}} \sum_{i=1}^n u^T X_i v$. The summands $f_{u,v}(X_i) = u^T X_i v$ are centered and take values in $[-1,1]$. So when the X_i are i.i.d., Bousquet's inequality (Theorem 12.5) applies. When they are only independent, Klein and Rio's variance bound applies. Symmetry of the X_i gives the wimpy-variance bound $\sum_i \mathbb{E}[(u^T X_i v)^2] \le v^T(\sum_i \mathbb{E}X_i^2)v$, hence $\sigma^2 \le \|\mathbb{E}\sum_{i=1}^n X_i^2\|$.

Solution. *Scope.* We prove the variance bound $\mathrm{Var}(Z) \le v$ for independent X_i, exactly as printed. We prove the Bennett and Bernstein inequalities with variance factor v and scale factor 1 when the X_i are i.i.d. That is the setting of Bousquet's inequality (Theorem 12.5), which the hint points to. For independent but non-identically distributed X_i, we do **not** prove the scale-1 exponential bound. That case remains an open caveat of this solution. All we can prove there is the weaker Klein–Rio tail at the end.

Status: incomplete (non-identical summands, exponential part only). The printed exercise assumes only independent X_i. For non-identically distributed X_i we do not prove the Bennett or Bernstein inequality with scale factor 1; the book itself notes (Section 12.5) that Bousquet's inequality requires identically distributed summands. The variance bound and the i.i.d. exponential bounds are complete.

Representation as a supremum. For a $d \times d$ matrix A, $\|A\| = \sup_{\|u\|=\|v\|=1} u^T A v$. Let $\mathcal{T} \subset \{(u,v) \in \mathbb{R}^d \times \mathbb{R}^d : \|u\| = \|v\| = 1\}$ be countable and dense. The map $(u,v) \mapsto u^T S v$ is continuous and the product of spheres is compact, so

$$Z = \Big\|\sum_{i=1}^n X_i\Big\| = \sup_{(u,v)\in\mathcal{T}} \sum_{i=1}^n f_{u,v}(X_i), \qquad f_{u,v}(X) := u^T X v.$$

For each $(u,v) \in \mathcal{T}$ and each i,

$$\mathbb{E} f_{u,v}(X_i) = u^T(\mathbb{E}X_i)v = 0, \qquad |f_{u,v}(X_i)| \le \|X_i\|\,\|u\|\,\|v\| \le 1 \text{ a.s.}$$

Wimpy variance. Since X_i is symmetric, $(u^T X_i v)^2 \le \|X_i v\|^2 = v^T X_i^T X_i v = v^T X_i^2 v$. Hence, for every $(u,v) \in \mathcal{T}$,

$$\sum_{i=1}^n \mathbb{E}[(u^T X_i v)^2] \le v^T\Big(\sum_{i=1}^n \mathbb{E}X_i^2\Big)v \le \Big\|\mathbb{E}\sum_{i=1}^n X_i^2\Big\|,$$

using $\|v\| = 1$. Therefore

$$\sigma^2 := \sup_{(u,v)\in\mathcal{T}} \sum_{i=1}^n \mathbb{E}[f_{u,v}(X_i)^2] \le \Big\|\mathbb{E}\sum_{i=1}^n X_i^2\Big\|.$$

This holds whether or not the X_i are identically distributed. Because the $f_{u,v}(X_i)$ are centered, $\sigma^2 = \sup_{(u,v)} \sum_i \mathrm{Var}(f_{u,v}(X_i))$. This is the quantity V_n of Klein and Rio.

Variance, independent X_i (as printed). Klein and Rio (2005), Corollary 1.1, covers this case. (The paper is *Concentration around the mean for maxima of empirical processes*, Ann. Probab. 33, 1060–1077, arXiv math/0506594.) Its hypotheses are: - the random variables are independent, and need not be identically distributed; - the class of functions is countable; - each coordinate function s_k takes values in $[-1,1]$, with $\mathbb{E}s_k(X_k) = 0$.

Under these hypotheses it states

$$\mathrm{Var}(Z) \le V_n + 2\mathbb{E}Z, \qquad V_n = \sup_s \sum_{k=1}^n \mathrm{Var}\, s_k(X_k).$$

The class $\{(f_{u,v}, \dots, f_{u,v}) : (u,v) \in \mathcal{T}\}$ satisfies these hypotheses, and for it $V_n = \sigma^2$. Therefore

$$\mathrm{Var}(Z) \le \sigma^2 + 2\mathbb{E}Z \le \Big\|\mathbb{E}\sum_{i=1}^n X_i^2\Big\| + 2\mathbb{E}Z = v.$$

In the i.i.d. case, the same bound also follows from Theorem 12.5 (or Theorem 11.10). The book's Theorem 12.9 (Section 12.5, stated there without proof and attributed to Klein and Rio) also gives $\mathrm{Var}(Z) \le 2\mathbb{E}Z + \sigma^2$ for independent, non-identically distributed summands with $|X_{i,s}| \le 1$. The pre-repair version of this solution restricted the variance bound to i.i.d. X_i. Corollary 1.1 makes that restriction unnecessary.

Exponential bounds, i.i.d. X_i. Let the X_i be i.i.d. random matrices. Then the vectors $(f_{u,v}(X_i))_{(u,v)\in\mathcal{T}}$, $i = 1, \dots, n$, are i.i.d., centered, and bounded above by 1. So Bousquet's inequality (Theorem 12.5) applies. With $v' := 2\mathbb{E}Z + \sigma^2 \le v$, it gives, for $\lambda \ge 0$,

$$\log \mathbb{E}e^{\lambda(Z-\mathbb{E}Z)} \le v'\phi(\lambda) \le v\,\phi(\lambda), \qquad \phi(\lambda) = e^\lambda - \lambda - 1 \ge 0.$$

The Chernoff argument, as in Theorem 12.5, yields the Bennett inequality

$$\mathbb{P}\{Z \ge \mathbb{E}Z + t\} \le e^{-vh(t/v)}, \qquad h(u) = (1+u)\log(1+u) - u, \quad t \ge 0.$$

Using $h(u) \ge u^2/(2(1+u/3))$, as in Section 2.7, gives the Bernstein inequality

$$\mathbb{P}\{Z \ge \mathbb{E}Z + t\} \le \exp\left(-\frac{t^2}{2(v + t/3)}\right), \quad t \ge 0.$$

Both inequalities have variance factor v and scale factor 1. The scale factor comes from $f_{u,v} \le 1$.

Open for non-identical X_i. Theorem 12.5 is stated and proved only for identically distributed summands. Klein–Rio, Theorem 1.1, handles independent non-identical summands under the hypotheses listed above, with $v = 2\mathbb{E}Z + V_n$. In our setting that v is at most our v. Their part (c) gives

$$\mathbb{P}\{Z \ge \mathbb{E}Z + x\} \le \exp\left(-\frac{x^2}{2v + 3x}\right), \qquad x \ge 0.$$

This is a Bernstein-type inequality with variance factor v. Its denominator $2v + 3x$ is larger than the scale-1 denominator $2v + 2x/3$. Their part (a) bound, $\log \mathbb{E}e^{t(Z-\mathbb{E}Z)} \le t\,v\,(\exp((e^{2t}-1)/2) - 1)/2$, is similarly weaker than $v\phi(t)$.

We therefore do not prove the Bennett bound $e^{-vh(t/v)}$, or the Bernstein bound with $t/3$, for non-identically distributed X_i. The independent audit claimed that Klein–Rio gives "the same bound with the same v". That claim is incorrect: as shown above, Klein–Rio's bound is weaker. ■

Exercise 13.34

Central idea. Poissonize with i.i.d. $Y_i \sim \mathrm{Poisson}(1)$ independent of the X_i: the constant $1 - 1/e$ is exactly $\mathbb{E}[Y_i \wedge 1] = \mathbb{P}\{Y_i \ge 1\}$. Move the Y-expectation outside the supremum (Jensen, since $\sup_s \mathbb{E}[\cdot] \le \mathbb{E}[\sup_s \cdot]$), then condition on the Y's: the active index set has size $K \le \sum_{i=1}^n Y_i \sim \mathrm{Poisson}(n)$, and $k \mapsto \mathbb{E}\sup_s \sum_{i=1}^k X_{i,s}$ is nondecreasing by centeredness, so the thinned sum is dominated by the Poissonized sum.

Solution. Let $Y_1, \dots, Y_n$ be i.i.d. Poisson(1), independent of the X_i. Since $Y_i \wedge 1 = \mathbf{1}_{\{Y_i \ge 1\}}$,

$$c := \mathbb{E}[Y_i \wedge 1] = \mathbb{P}\{Y_i \ge 1\} = 1 - e^{-1} = 1 - \frac{1}{e}.$$

Hence, with $c > 0$ pulled inside the supremum,

$$\left(1 - \frac{1}{e}\right) \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^n X_{i,s} = \mathbb{E}_X\left[\sup_{s\in\mathcal{T}} \sum_{i=1}^n \mathbb{E}_Y[Y_i \wedge 1]\, X_{i,s}\right].$$

For fixed values $x_{i,s}$ of the $X_{i,s}$, Jensen's inequality for the convex supremum gives

$$\sup_{s\in\mathcal{T}} \mathbb{E}_Y\left[\sum_{i=1}^n (Y_i \wedge 1) x_{i,s}\right] \le \mathbb{E}_Y\left[\sup_{s\in\mathcal{T}} \sum_{i=1}^n (Y_i \wedge 1) x_{i,s}\right].$$

Taking $\mathbb{E}_X$ yields

$$\left(1-\frac{1}{e}\right)\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{n}X_{i,s}\le\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{n}(Y_i\wedge 1)X_{i,s}. \tag{$*$}$$

Write $(Y_i\wedge 1)=\mathbf{1}_{\{Y_i\ge 1\}}$ and condition on $\mathbf{Y}=(Y_1,\dots,Y_n)$. Let $K=\sum_{i=1}^{n}\mathbf{1}_{\{Y_i\ge 1\}}$. Given $\mathbf{Y}$, the set $\{i: Y_i\ge 1\}$ is fixed with cardinality K, and since the X_i are i.i.d. and independent of $\mathbf{Y}$,

$$\mathbb{E}\left[\sup_{s\in\mathcal{T}}\sum_{i=1}^{n}\mathbf{1}_{\{Y_i\ge 1\}}X_{i,s}\,\middle|\,\mathbf{Y}\right]=\phi(K),\quad\text{where }\phi(k):=\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{k}X_{i,s}.$$

The map ϕ is nondecreasing: for $a_s=\sum_{i=1}^{k}X_{i,s}$ fixed,

$$\mathbb{E}_{X_{k+1}}\left[\sup_{s\in\mathcal{T}}(a_s+X_{k+1,s})\right]\ge\sup_{s\in\mathcal{T}}\left(a_s+\mathbb{E}X_{k+1,s}\right)=\sup_{s\in\mathcal{T}}a_s,$$

by Jensen and $\mathbb{E}X_{k+1,s}=0$ (centeredness). Since $\mathbf{1}_{\{Y_i\ge 1\}}\le Y_i$,

$$K\le M:=\sum_{i=1}^{n}Y_i,$$

and $M\sim\text{Poisson}(n)$ is independent of the X_i. Thus, from $(*)$,

$$\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{n}(Y_i\wedge 1)X_{i,s}=\mathbb{E}[\phi(K)]\le\mathbb{E}[\phi(M)]=\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{M}X_{i,s}=\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{N_n}X_{i,s},$$

identifying $M\overset{d}{=}N_n$. ($\mathcal{T}$ finite, so all suprema are maxima.) ■

Exercise 13.35

Central idea. Condition on N_n and split the variance: $\text{Var}(Z)=\mathbb{E}[\text{Var}(Z\mid N_n)]+\text{Var}(\mathbb{E}[Z\mid N_n])$. The first term is handled by Theorem 11.10 applied to Z_k (giving $k\sigma^2+2\mathbb{E}Z_k$); the second is $\text{Var}(h(N_n))$ for the deterministic $h(k)=\mathbb{E}Z_k$, bounded by the Poisson Poincaré inequality (Exercise 3.21). The increments $h(k+1)-h(k)$ are controlled by the leave-one-out symmetrization $(k+1)h(k)\ge k\,h(k+1)$.

Solution. Write $h(k)=\mathbb{E}[Z_k]=\mathbb{E}\sup_{s\in\mathcal{T}}\sum_{i=1}^{k}X_{i,s}$ (deterministic; $h(0)=0$). By independence of N_n and the X_i, $\mathbb{E}[Z\mid N_n]=h(N_n)$, and the variance decomposition gives

$$\text{Var}(Z)=\mathbb{E}[\text{Var}(Z\mid N_n)]+\text{Var}(h(N_n)).$$

Given $N_n=k$, $Z=Z_k$. Theorem 11.10 applies to Z_k (the X_i are i.i.d., centered, $|X_{i,s}|\le 1$): with the problem's $\sigma^2=\sup_s\mathbb{E}X_{1,s}^2$,

$$\text{Var}(Z_k)\le 2\mathbb{E}Z_k+k\sigma^2=2h(k)+k\sigma^2.$$

Hence $\mathbb{E}[\text{Var}(Z\mid N_n)]\le 2\mathbb{E}Z+\mathbb{E}N_n\,\sigma^2$.

For the second term, the Poisson Poincaré inequality (Exercise 3.21) with $Df(k)=f(k+1)-f(k)$ gives

$$\text{Var}(h(N_n))\le\mathbb{E}N_n\cdot\mathbb{E}\big[(h(N_n+1)-h(N_n))^2\big].$$

Given $N_n=k$, $\mathbb{E}[Z_{N_n+1}-Z_{N_n}\mid N_n=k]=\mathbb{E}[Z_{k+1}-Z_k]=h(k+1)-h(k)$ (here N_{n+1} denotes N_n+1), so

$$\text{Var}(Z)\le\mathbb{E}N_n\,\sigma^2+2\mathbb{E}Z+\mathbb{E}N_n\,\mathbb{E}\Big[\big(\mathbb{E}[Z_{N_n+1}-Z_{N_n}\mid N_n]\big)^2\Big],$$

the first claimed inequality.

It remains to bound the increments. Fix $k \geq 1$ and, for $j = 1, \dots, k+1$, let $Z_{k+1}^{(j)} = \max_{s \in \mathcal{T}} \sum_{i \neq j} X_{i,s}$ be the leave-one-out supremum; $\mathbb{E} Z_{k+1}^{(j)} = h(k)$ by identical distribution. Let s^* achieve $Z_{k+1} = \sum_{i=1}^{k+1} X_{i,s^*}$ ($\mathcal{T}$ finite). Then $Z_{k+1}^{(j)} \geq \sum_{i \neq j} X_{i,s^*} = Z_{k+1} - X_{j,s^*}$, and summing over j,

$$\sum_{j=1}^{k+1} Z_{k+1}^{(j)} \geq (k+1) Z_{k+1} - \sum_{j=1}^{k+1} X_{j,s^*} = k Z_{k+1}.$$

Taking expectations, $(k+1)h(k) \geq k\, h(k+1)$, i.e.

$$h(k+1) - h(k) \leq \frac{h(k)}{k}, \qquad k \geq 1,$$

and for $k = 0$, $h(1) - h(0) = h(1) = h(\max(0,1))/\max(0,1)$. Thus, with $\mathbb{E}[Z_{\max(N_n,1)} \mid N_n] = h(\max(N_n, 1))$,

$$\Big(\mathbb{E}[Z_{N_n+1} - Z_{N_n} \mid N_n]\Big)^2 = \big(h(N_n+1) - h(N_n)\big)^2 \leq \left(\frac{\mathbb{E}[Z_{\max(N_n,1)} \mid N_n]}{\max(N_n, 1)}\right)^2,$$

and taking expectations and multiplying by $\mathbb{E} N_n$ yields the second inequality. ■

Exercise 13.36

Central idea. Binomial approximation of the Poisson law: for $M > n$, replace N_n by $N^{(M)} \sim \text{Binomial}(M, n/M)$ and Z by $Z^{(M)} = \sup_s \sum_{k=1}^{M} B_k X_{k,s}$ with $B_k \sim \text{Bernoulli}(n/M)$ independent. For fixed M the pairs (B_k, X_k) are independent across k, so Efron–Stein applies directly: the resampling cost splits into $B_k X_{k,\hat{s}}^2$ (telescoping to the weak variance) plus $(n/M)\sigma^2$ per coordinate. Letting $M \to \infty$ (Skorokhod coupling + uniform integrability) recovers the Poissonized bound. This avoids Poisson thinning of a single draw $Y_i X_{i,s}$, which does not have the claimed law.

Remark (erratum fixed). The identity $(\sum_{i=1}^{m} Y_i X_{i,s})_s \overset{d}{=} (\sum_{k=1}^{N_n} X_{k,s})_s$ with $Y_i \sim \text{Poisson}(n/m)$ is false: e.g. with Rademacher V_i, $n = 6$, $m = 4$, $\mathbb{E}[(\sum_i Y_i V_i)^2] = n(1 + n/m) = 15 \neq 6 = \mathbb{E}[(\sum_{k \leq N_n} V_k)^2]$. The correct thinning is $\sum_{k \leq N_n} X_{k,s} \overset{d}{=} \sum_i \sum_{j \leq Y_i} X_{i,j,s}$ (independent copies per group). The characteristic-function computation in the earlier version also dropped the rate n/m and moved an expectation inside an exponential. The proof below establishes the claimed bound without any thinning identity.

Solution. Use the notation of Exercise 13-35: $\mathcal{T}$ is finite, the $X_i = (X_{i,s})_{s \in \mathcal{T}}$ are i.i.d. centered with $|X_{i,s}| \leq 1$, $N_n \sim \text{Poisson}(n)$ is independent of the X_i, $Z = \sup_{s \in \mathcal{T}} \sum_{i=1}^{N_n} X_{i,s}$, and $\sigma^2 = \sup_{s \in \mathcal{T}} \mathbb{E} X_{1,s}^2$.

Fix an integer $M > n$. Let $B_1, \dots, B_M$ be i.i.d. Bernoulli(n/M), independent of the X_i, and put

$$N^{(M)} = \sum_{k=1}^{M} B_k \sim \text{Binomial}(M, n/M), \qquad Z^{(M)} = \sup_{s \in \mathcal{T}} \sum_{k=1}^{M} B_k X_{k,s}.$$

The pairs (B_k, X_k) are independent across k. By the Efron–Stein inequality,

$$\text{Var}(Z^{(M)}) \leq \sum_{k=1}^{M} \mathbb{E}\big[(Z^{(M)} - Z_k^{(M)})_+^2\big],$$

where $Z_k^{(M)}$ replaces (B_k, X_k) by an independent copy (B_k', X_k'). Let $\hat{s} \in \mathcal{T}$ attain the supremum in $Z^{(M)}$ ($\mathcal{T}$ finite). Then

$$(Z^{(M)} - Z_k^{(M)})_+ \leq \big(B_k X_{k,\hat{s}} - B_k' X_{k,\hat{s}}'\big)_+,$$

and, with $\mathbb{E}'$ over the primed variables ($\hat{s}$ fixed),

$$\mathbb{E}'\big[(B_k X_{k,\hat{s}} - B_k' X_{k,\hat{s}}')_+^2\big] \leq \mathbb{E}'\big[(B_k X_{k,\hat{s}} - B_k' X_{k,\hat{s}}')^2\big] = B_k X_{k,\hat{s}}^2 + \tfrac{n}{M} \mathbb{E}[X_{1,\hat{s}}^2],$$

since $\mathbb{E}'[B'_k X'_{k,\hat{s}}] = \mathbb{E}[B'_k]\mathbb{E}[X_{1,\hat{s}}] = 0$ (centered) and $\mathbb{E}'[(B'_k X'_{k,\hat{s}})^2] = \mathbb{E}[(B'_k)^2]\mathbb{E}[X^2_{1,\hat{s}}] = \frac{n}{M}\mathbb{E}[X^2_{1,\hat{s}}]$. With $\mathbb{E}[X^2_{1,\hat{s}}] \le \sigma^2$ and taking $\mathbb{E}$ over the unprimed variables,

$$\mathrm{Var}(Z^{(M)}) \le \mathbb{E}\Big[\sum_{k=1}^{M} B_k X^2_{k,\hat{s}}\Big] + n\sigma^2 \le \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{k=1}^{M} B_k X^2_{k,s} + n\sigma^2, \tag{$*$}$$

using $\sum_k B_k X^2_{k,\hat{s}} \le \sup_s \sum_k B_k X^2_{k,s}$ pointwise.

Given $N^{(M)} = m$, the set $\{k : B_k = 1\}$ is a uniform m-subset independent of the X_k, so $(\sum_k B_k X_{k,s})_{s\in\mathcal{T}} \overset{d}{=} (\sum_{j=1}^{N^{(M)}} X_{j,s})_{s\in\mathcal{T}}$ with $N^{(M)}$ independent of the X_j on the right. Hence, $\mathcal{T}$ being finite,

$$Z^{(M)} \overset{d}{=} \sup_{s\in\mathcal{T}} \sum_{j=1}^{N^{(M)}} X_{j,s}, \qquad \mathbb{E}\sup_s \sum_k B_k X^2_{k,s} = \mathbb{E}\sup_s \sum_{j=1}^{N^{(M)}} X^2_{j,s}.$$

Let $M \to \infty$. By Skorokhod's representation, couple $\tilde{N}^{(M)} \overset{d}{=} N^{(M)}$ and $\tilde{N} \overset{d}{=} N_n$ on one space with $\tilde{N}^{(M)} \to \tilde{N}$ a.s.; fix i.i.d. (X_j) independent of these. Then $\tilde{Z}^{(M)} := \sup_s \sum_{j\le\tilde{N}^{(M)}} X_{j,s} \overset{d}{=} Z^{(M)}$ converges a.s. to $\tilde{Z} := \sup_s \sum_{j\le\tilde{N}} X_{j,s} \overset{d}{=} Z$ (finite $\mathcal{T}$; the integer upper limits are eventually equal). Since $|\tilde{Z}^{(M)}| \le \tilde{N}^{(M)}$, $|X_{j,s}| \le 1$, and $\mathbb{E}[(\tilde{N}^{(M)})^2] = n^2 + n(1 - n/M) \to n^2 + n = \mathbb{E}[\tilde{N}^2]$, uniform integrability gives $\mathrm{Var}(\tilde{Z}^{(M)}) \to \mathrm{Var}(\tilde{Z})$. Likewise $\tilde{W}^{(M)} := \sup_s \sum_{j\le\tilde{N}^{(M)}} X^2_{j,s} \to \sup_s \sum_{j\le\tilde{N}} X^2_{j,s}$ a.s. with $0 \le \tilde{W}^{(M)} \le \tilde{N}^{(M)}$, so its expectations converge. Passing to the limit in $(*)$,

$$\mathrm{Var}(Z) \le \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{j=1}^{N_n} X^2_{j,s} + n\sigma^2 = \mathbb{E}\sup_{s\in\mathcal{T}} \sum_{i=1}^{N_n} X^2_{i,s} + \mathbb{E}N_n \sup_{s\in\mathcal{T}} \mathbb{E}X^2_{1,s},$$

as claimed. ■

Exercise 13.37

Central idea. Apply Exercise 13.35 to $X_{i,t} = \mathbb{1}_{\{U_i\in A_t\}} - P(A_t)$, where the $U_i \sim P$ are i.i.d. This gives $\mathrm{Var}(Z) \le n\sigma^2 + 2\mathbb{E}Z + n\,\mathbb{E}[(g(N')/N')^2]$, with $g(k) = \mathbb{E}Z_k$ and $N' = \max(N_n, 1)$. The chaining bound behind Theorem 13.7 holds at every sample size k in the form $g(k) \le \kappa_0(\sigma\sqrt{kV\ell} + V\ell)$, $\ell = \log(e/\sigma)$; this is the quadratic inequality of Lemma 13.5 before its proviso is used. Feeding it through $\mathbb{E}\sqrt{N_n} \le \sqrt{n}$, $\mathbb{E}[1/N'] \le 2/n$ and $\mathbb{E}[1/N'^2] \le 6/n^2$ gives the stated bound plus $\kappa(V\ell + V^2\ell^2/n)$. These extra terms are absorbed exactly when $n\sigma^2 \ge V\ell$, which is the regime of Theorem 13.7. Without that regime the printed bound is false. A class of $1/\sigma^2$ disjoint sets with $n = 1$ shows the ratio of the variance to the bound is unbounded.

Solution. Erratum (statement needs Theorem 13.7's proviso). Theorem 13.7 is proved only under the proviso $\sigma \ge 24\sqrt{V\log(4e^2/\sigma)/(5n)}$, that is, $n \gtrsim V\log(e/\sigma)/\sigma^2$. If "the notation of Theorem 13.7" is read as importing that proviso, the printed statement is correct and is proved below; the weaker condition $n\sigma^2 \ge V\log(e/\sigma)$ suffices. Read literally, for all n, the statement is false. (The pre-repair version of this solution said that outside the regime "the trivial bound is absorbed into the universal constant". That is wrong, as the following counterexample shows.)

Counterexample. Fix $\sigma \in (0,1)$ and let $m = \lfloor 1/\sigma^2 \rfloor$. Let $\mathcal{A} = \{A_1, \dots, A_m\}$ be disjoint sets with $P(A_t) = \sigma^2$. The class has VC dimension $V = 1$: any point of A_1 is shattered, since A_1 contains it and A_2 does not ($m \ge 2$), but no two points are, since no set contains both. By Poisson thinning, the counts $N_t = \#\{i \le N_n : U_i \in A_t\}$ are i.i.d. Poisson($n\sigma^2$). The Kac supremum is

$$Z = \max_t N_t - \sigma^2 N_n.$$

Take $n = 1$. Then $\max_t N_t$ is the maximum of $m \approx \sigma^{-2}$ i.i.d. Poisson(σ^2) variables. It is nonzero with probability $\approx 1 - e^{-1}$, so $\mathrm{Var}(Z) \approx (1 - e^{-1})e^{-1} \approx 0.2325$ for all small σ. With $\ell = \log(e/\sigma)$ and $V = 1$, the bound without κ is

$$B = n\sigma^2 + \sigma\sqrt{Vn\ell} + V\sigma^2\ell = \sigma^2 + \sigma\sqrt{\ell} + \sigma^2\ell \to 0.$$

We computed $\mathrm{Var}(Z)$ exactly, with Poisson sums at 40-digit precision, keeping the $-\sigma^2 N_n$ term. The results:

σ	0.1	0.03	0.01	0.003	0.001
$\mathrm{Var}(Z)$	0.2337	0.2327	0.2326	0.2325	0.2325
$\mathrm{Var}(Z)/B$	1.04	3.39	9.56	29.4	82.4

A Monte Carlo check with 4×10^5 replications agrees: 0.2336 at $\sigma = 0.1$ and 0.2328 at $\sigma = 0.03$.

The ratio grows like $1/(\sigma\sqrt{\ell})$, so no universal κ makes the bound hold. The term $n\sigma^2$ cannot help here: it equals σ^2. The same failure occurs when the Poisson mean is not an integer. With mean $n = \sigma$, the ratio is $0.92, 1.89, 3.36, 6.09, 10.3$ at the same five values of σ (exact; growth $\asymp \sigma^{-1/2}\ell^{-1/2}$). Here $n\sigma^2 = \sigma^3 \ll V\ell$, far outside the regime.

Corrected statement. Let $\sigma \in (0, 1]$ and $\ell = \log(e/\sigma) \geq 1$. There is a universal constant κ such that, for every $n \geq 1$,

$$\mathrm{Var}(Z) \leq n\sigma^2 + \kappa\sigma\sqrt{Vn\ell} + \kappa V\sigma^2\ell + \kappa V\ell + \kappa\frac{V^2\ell^2}{n}.$$

In particular, if $n\sigma^2 \geq V\ell$, then

$$\mathrm{Var}(Z) \leq n\sigma^2 + \kappa\sigma\sqrt{Vn\log\frac{e}{\sigma}} + \kappa V\sigma^2\log\frac{e}{\sigma}.$$

The condition $n\sigma^2 \geq V\ell$ is implied by Theorem 13.7's proviso, since that proviso gives $n\sigma^2 \geq \frac{576}{5}V\log(4e^2/\sigma) \geq V\ell$.

Proof. **Setting.** Let $U_1, U_2, \ldots$ be i.i.d. with law P, and let $N_n \sim \mathrm{Poisson}(n)$ be independent of them. Let $\mathcal{A} = \{A_t : t \in \mathcal{T}\}$ be a VC class of dimension V with $P(A_t) \leq \sigma^2$. Exercise 13.35 is stated for finite $\mathcal{T}$. We prove the bound for every finite subclass, whose VC dimension is at most V. It then passes to countable $\mathcal{T}$ by monotone convergence of the suprema along an increasing sequence of finite subclasses, with $\mathbb{E}Z^2 < \infty$ since $|Z| \leq N_n + n$.

If $V = 0$, no point is shattered. Then all A_t have the same trace on every point, so the class is a single set A. In that case $Z = \sum_{i \leq N_n}(\mathbb{1}_A(U_i) - P(A))$ is compound Poisson with variance $n\,\mathbb{E}(\mathbb{1}_A - P(A))^2 = nP(A)(1 - P(A)) \leq n\sigma^2$. So assume $V \geq 1$.

Put $X_{i,t} = \mathbb{1}_{\{U_i \in A_t\}} - P(A_t)$. The vectors X_i are i.i.d., centered, and satisfy $|X_{i,t}| \leq 1$. Moreover $\sup_t \mathbb{E}X_{1,t}^2 = \sup_t P(A_t)(1 - P(A_t)) \leq \sigma^2$. Let $Z_k = \sup_t \sum_{i \leq k} X_{i,t}$, $g(k) = \mathbb{E}Z_k$ for $k \geq 1$, $g(0) = 0$, and $N' = \max(N_n, 1)$. Exercise 13.35 gives

$$\mathrm{Var}(Z) \leq n\sigma^2 + 2\mathbb{E}Z + n\,\mathbb{E}\Big[\Big(\frac{g(N')}{N'}\Big)^2\Big]. \tag{1}$$

Step 1: $g(k) \leq \kappa_0(\sigma\sqrt{kV\ell} + V\ell)$ **for every** $k \geq 1$**.** Let $\mathcal{A}' = \mathcal{A} \cup \{\emptyset\}$. Then $P(\emptyset) = 0 \leq \sigma^2$, and $\sup_{\mathcal{A}'} \sum_{i \leq k} X_{i,t} = (Z_k)_+ \geq Z_k$, so $g(k) \leq \mathbb{E}(Z_k)_+$.

Adding $\emptyset$ raises the VC dimension by at most one. Suppose $\mathcal{A}'$ shatters a set S with $|S| \geq 1$, and fix $x \in S$. Put $S' = S \setminus \{x\}$. For any $T \subseteq S'$, some $B \in \mathcal{A}'$ has $B \cap S = T \cup \{x\}$. Then $B \neq \emptyset$, so $B \in \mathcal{A}$ and $B \cap S' = T$. Thus $\mathcal{A}$ shatters S'. Hence $V' \leq V + 1 \leq 2V$.

Write $Y_k = k^{-1/2}(Z_k)_+$ for the normalized supremum over $\mathcal{A}'$, and $y = \mathbb{E}Y_k \geq 0$. The proof of Lemma 13.5 shows, without using its proviso, that

$$y \leq D\sqrt{\sigma^2 + y/\sqrt{k}}.$$

The ingredients are symmetrization, Lemma 13.1 applied conditionally, $\delta_k^2 \le \sigma^2 + Y_k/\sqrt{k}$, and Jensen. Here $D = D_\sigma$ is the entropy integral of Lemma 13.5 for $\mathcal{A}'$. As in the proof of Theorem 13.7, Haussler's bound (Lemma 13.6) gives $D \le 24\sqrt{V' \log(4e^2/\sigma)}$. Since $\log(4e^2/\sigma) = \log 4 + 1 + \ell \le (2 + \log 4)\ell$ for $\ell \ge 1$,

$$D^2 \le 576 \cdot 2(2 + \log 4)\, V\ell =: c_0 V\ell.$$

Squaring, $y^2 \le D^2\sigma^2 + D^2 y/\sqrt{k}$. Hence

$$y \le \frac{D^2}{2\sqrt{k}} + \sqrt{\frac{D^4}{4k} + D^2\sigma^2} \le \frac{D^2}{\sqrt{k}} + D\sigma.$$

Therefore $g(k) \le \sqrt{k}\, y \le D^2 + D\sigma\sqrt{k} \le \kappa_0(V\ell + \sigma\sqrt{kV\ell})$ with $\kappa_0 = c_0$, using $c_0 \ge \sqrt{c_0}$. The bound also holds trivially for $k = 0$.

Step 2: the mean. By Step 1 and Jensen ($\mathbb{E}\sqrt{N_n} \le \sqrt{n}$),

$$\mathbb{E}Z = \mathbb{E}\, g(N_n) \le \kappa_0(\sigma\sqrt{nV\ell} + V\ell).$$

Step 3: the Poisson term. By Step 1 and $(a+b)^2 \le 2a^2 + 2b^2$,

$$\Big(\frac{g(N')}{N'}\Big)^2 \le 2\kappa_0^2\Big(\frac{\sigma^2 V\ell}{N'} + \frac{V^2\ell^2}{N'^2}\Big).$$

Write $N = N_n$. Two elementary inequalities hold for all integers $N \ge 0$: - $1/N' \le 2/(N+1)$, since $1 \le 2$ for $N = 0$ and $N + 1 \le 2N$ for $N \ge 1$; - $1/N'^2 \le 6/((N+1)(N+2))$, since $1 \le 3$ for $N = 0$, and $(N+1)(N+2) \le 6N^2$ for $N \ge 1$, with equality at $N = 1$.

For $N \sim \mathrm{Poisson}(n)$,

$$\mathbb{E}\frac{1}{N+1} = \frac{1 - e^{-n}}{n} \le \frac{1}{n}, \qquad \mathbb{E}\frac{1}{(N+1)(N+2)} = \frac{1}{n^2}\sum_{j\ge 2} e^{-n}\frac{n^j}{j!} \le \frac{1}{n^2}.$$

Hence $\mathbb{E}[1/N'] \le 2/n$, $\mathbb{E}[1/N'^2] \le 6/n^2$, and

$$n\,\mathbb{E}\Big[\Big(\frac{g(N')}{N'}\Big)^2\Big] \le 4\kappa_0^2\sigma^2 V\ell + 12\kappa_0^2\frac{V^2\ell^2}{n}.$$

Step 4: conclusion. Insert Steps 2 and 3 into (1):

$$\mathrm{Var}(Z) \le n\sigma^2 + 2\kappa_0\sigma\sqrt{nV\ell} + 2\kappa_0 V\ell + 4\kappa_0^2\sigma^2 V\ell + 12\kappa_0^2\frac{V^2\ell^2}{n}.$$

This is the corrected general bound with $\kappa = 12\kappa_0^2$.

If $n\sigma^2 \ge V\ell$, then $V\ell = \sqrt{V\ell}\sqrt{V\ell} \le \sqrt{V\ell}\,\sigma\sqrt{n}$ and $V^2\ell^2/n \le \sigma^2 V\ell$. The last bound then becomes

$$\mathrm{Var}(Z) \le n\sigma^2 + \kappa\,\sigma\sqrt{Vn\log\frac{e}{\sigma}} + \kappa\, V\sigma^2\log\frac{e}{\sigma},$$

with κ a universal constant. ■

Remark. The leading term $n\sigma^2$ is $\mathbb{E}N_n \cdot \sigma^2$, the Poisson-count variance of a single set of mass σ^2. In the counterexample above, the variance is instead driven by the event that some set is hit at all, which has probability of order $1 - e^{-nm\sigma^2}$. This is exactly the small-n effect that the terms $V\ell + V^2\ell^2/n$ track.

Exercise 13.38

Central idea. Write $Z = \sqrt{n}\,\|F_n - F\|_{L^2[0,1]}$ with $F_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{X_i \le x\}}$, $F(x) = x$; by Riesz–Fischer $\|g\|_2 = \sup_{\|h\|_2 \le 1}\langle h, g\rangle$, so Z is a supremum of the centered empirical process $(1/\sqrt{n})\sum_i \int_0^1 h(x)(\mathbb{1}_{\{X_i \le x\}} - x)\,dx$ over a countable dense subset of the L^2 unit ball. Then $\mathbb{E}Z^2 = n\int_0^1 \mathrm{Var}(F_n(x))dx = 1/6$ gives $\mathbb{E}Z \le 1/\sqrt{6}$, and Efron–Stein with $|Z - Z^{(i)}| \le \sqrt{|X_i - X_i'|/n}$ gives $\mathrm{Var}(Z) \le \frac{1}{2}\mathbb{E}|X_1 - X_1'| = 1/6$.

Solution. The stated $P_n([0,x]) = \sum_{i=1}^n \mathbb{1}_{\{X_i \le x\}}$ is missing the factor $1/n$: the empirical measure is $P_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$, so $P_n([0,x]) = F_n(x) := \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{X_i \le x\}}$. With this correction,

$$Z^2 = n\int_0^1 (F_n(x) - x)^2 dx, \qquad Z = \sqrt{n}\,\|F_n - F\|_2, \quad F(x) = x.$$

By Riesz–Fischer, $\|g\|_2 = \sup_{\|h\|_2 \le 1}\langle h, g\rangle$ for $g \in L^2[0,1]$. Since $L^2[0,1]$ is separable, fix a countable dense $\mathcal{D}$ in the unit ball; by continuity of $h \mapsto \langle h, g\rangle$,

$$Z = \sqrt{n}\sup_{h\in\mathcal{D}} \int_0^1 h(x)(F_n(x) - x)dx = \sup_{h\in\mathcal{D}} \sum_{i=1}^n Y_{i,h},$$

where $Y_{i,h} = \frac{1}{\sqrt{n}}\int_0^1 h(x)(\mathbb{1}_{\{X_i \le x\}} - x)dx$. For fixed h the $Y_{i,h}$ are independent across i and centered, since $\mathbb{E}[\mathbb{1}_{\{X_i \le x\}}] = x$. Thus Z is the supremum of an empirical process.

Remark on the hint. With $e_i(x) = \sqrt{2}\sin(2\pi i x)$ and $\mathcal{T}$ the rational unit sequences, $\sup_{s\in\mathcal{T}} \sum_{i\ge 1} s_i\sqrt{n}\langle e_i, F_n - F\rangle = \sqrt{n}\,(\sum_{i\ge 1}\langle e_i, F_n - F\rangle^2)^{1/2} = \sqrt{n}\,\|P(F_n - F)\|_2$, where P projects onto $\overline{\mathrm{span}}\{e_i\}$. But $\{e_i\}$ is *not* complete in $L^2[0,1]$ (e.g. it is orthogonal to constants), and $F_n - F$ has nonzero mean a.s., so this supremum is *strictly smaller* than Z in general: the stated identity is inaccurate as written (with $\sin(\pi i x)$ in place of $\sin(2\pi i x)$ the system $\{\sqrt{2}\sin(\pi i x)\}$ is complete and the identity would hold by Parseval).

Mean. $\mathbb{E}[(F_n(x) - x)^2] = \mathrm{Var}(F_n(x)) = x(1-x)/n$, so

$$\mathbb{E}Z^2 = n\int_0^1 \frac{x(1-x)}{n}dx = \int_0^1 x(1-x)dx = \frac{1}{6},$$

hence $\mathbb{E}Z \le \sqrt{\mathbb{E}Z^2} = 1/\sqrt{6}$ by Jensen.

Variance (Efron–Stein). Let X_i' be an independent copy of X_i and $Z^{(i)}$ the statistic with X_i replaced by X_i'. By the reverse triangle inequality,

$$|Z - Z^{(i)}| = \sqrt{n}\,\big|\|F_n - F\|_2 - \|F_n^{(i)} - F\|_2\big| \le \sqrt{n}\,\|F_n - F_n^{(i)}\|_2.$$

Now $F_n - F_n^{(i)} = \frac{1}{n}(\mathbb{1}_{\{X_i' \le \cdot\}} - \mathbb{1}_{\{X_i \le \cdot\}})$, and since the difference of indicators takes values in $\{-1, 0, 1\}$,

$$\|F_n - F_n^{(i)}\|_2^2 = \frac{1}{n^2}\int_0^1 \big|\mathbb{1}_{\{X_i' \le x\}} - \mathbb{1}_{\{X_i \le x\}}\big|dx = \frac{|X_i - X_i'|}{n^2},$$

because $\int_0^1 |\mathbb{1}_{\{a \le x\}} - \mathbb{1}_{\{b \le x\}}|dx = |a - b|$ for $a, b \in [0,1]$. Hence $(Z - Z^{(i)})^2 \le |X_i - X_i'|/n$, and by Efron–Stein,

$$\mathrm{Var}(Z) \le \frac{1}{2}\sum_{i=1}^n \mathbb{E}\big[(Z - Z^{(i)})^2\big] \le \frac{1}{2}\sum_{i=1}^n \frac{\mathbb{E}|X_i - X_i'|}{n} = \frac{1}{2}\,\mathbb{E}|X_1 - X_1'|.$$

For $X_1, X_1' \overset{iid}{\sim} \mathrm{Uniform}[0,1]$, $\mathbb{E}|X_1 - X_1'| = \int_0^1\int_0^1 |x - y|dxdy = 1/3$. Therefore

$$\mathrm{Var}(Z) \le \frac{1}{6}.$$

In particular $\mathrm{Var}(Z) \le \mathbb{E}Z^2 = 1/6$, with equality in the Efron–Stein bound up to the $(EZ)^2$ gap. ■

Exercise 13.39

Central idea. Write $Z = \sqrt{n}\,\|g\|_2$ with $g(x) = (F_n(x) - x)/\sqrt{x(1-x)} \in L^2[0,1]$ (the empirical measure needs its $1/n$ normalization, as in Exercise 13-38); by Riesz–Fischer $\|g\|_2$ is the supremum of $\langle h, g\rangle$ over a countable dense subset of the L^2 unit ball, so Z is the supremum of the centered empirical process $\sum_i Y_{i,h}$. For the variance, Efron–Stein with $|Z - Z^{(i)}| \le \sqrt{n}\,\|g - g^{(i)}\|_2$ reduces to $\frac{1}{n^2}\int_{X_i \wedge X_i'}^{X_i \vee X_i'} \frac{dx}{x(1-x)} = \frac{1}{n^2}|\mathrm{logit}(X_i') - \mathrm{logit}(X_i)|$; since $\mathrm{logit}(X)$ is standard logistic and $\mathbb{E}|L - L'| = 2$ for iid logistics, Efron–Stein gives $\mathrm{Var}(Z) \le 1 = \mathbb{E}Z^2$ — no better than the trivial bound, the heavy-tailed logit influence being the price of the $1/(x(1-x))$ weight.

Solution. As in Exercise 13-38, $P_n([0,x])$ must be read as the empirical distribution function $F_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\{X_i \le x\}}$ with $X_1, \dots, X_n \overset{iid}{\sim} \mathrm{Uniform}[0,1]$; without the $1/n$ the stated $\mathbb{E}Z^2 = 1$ fails. Put

$$Z^2 = n\int_0^1 \frac{(F_n(x) - x)^2}{x(1-x)}dx = n\|g\|_2^2, \qquad g(x) = \frac{F_n(x) - x}{\sqrt{x(1-x)}}.$$

$Z < \infty$ a.s.: on $[0, X_{(1)})$ we have $F_n = 0$ and the integrand is $x/(1-x)$; on $[X_{(1)}, \frac{1}{2}]$ it is at most $1/(x(1-x)) \le 2/x$, integrable since $X_{(1)} > 0$ a.s.; the right half is symmetric, using $X_{(n)} < 1$ a.s. Hence $g \in L^2[0,1]$ a.s.

Supremum representation. By Riesz–Fischer, $\|g\|_2 = \sup_{\|h\|_2 \le 1}\langle h, g\rangle$. Fix a countable dense $\mathcal{D}$ in the L^2 unit ball ($L^2[0,1]$ is separable); by continuity of $h \mapsto \langle h, g\rangle$,

$$Z = \sqrt{n}\sup_{h \in \mathcal{D}} \int_0^1 h(x)g(x)dx = \sup_{h \in \mathcal{D}} \sum_{i=1}^n Y_{i,h},$$

where $Y_{i,h} = \frac{1}{\sqrt{n}}\int_0^1 h(x)\,\frac{\mathbb{1}_{\{X_i \le x\}} - x}{\sqrt{x(1-x)}}dx$. For fixed h the $Y_{i,h}$ are independent across i (each depends only on X_i), integrable by Cauchy–Schwarz since both factors are in L^2, and centered because $\mathbb{E}[\mathbb{1}_{\{X_i \le x\}}] = x$. Thus Z is the supremum of an empirical process.

Mean. By Tonelli, $\mathbb{E}Z^2 = n\int_0^1 \frac{\mathbb{E}[(F_n(x)-x)^2]}{x(1-x)}dx$. Since $\mathbb{E}[(F_n(x) - x)^2] = \mathrm{Var}(F_n(x)) = x(1-x)/n$,

$$\mathbb{E}Z^2 = n\int_0^1 \frac{x(1-x)/n}{x(1-x)}dx = 1,$$

the usual Anderson–Darling normalization.

Variance (Efron–Stein). Let X_i' be an independent copy of X_i and $Z^{(i)}$ the statistic with X_i replaced by X_i'. By the reverse triangle inequality,

$$|Z - Z^{(i)}| = \sqrt{n}\,\big|\|g\|_2 - \|g^{(i)}\|_2\big| \le \sqrt{n}\,\|g - g^{(i)}\|_2.$$

Since $F_n - F_n^{(i)} = \frac{1}{n}(\mathbb{1}_{\{X_i \le \cdot\}} - \mathbb{1}_{\{X_i' \le \cdot\}})$ takes values in $\{-\frac{1}{n}, 0, \frac{1}{n}\}$,

$$\|g - g^{(i)}\|_2^2 = \frac{1}{n^2}\int_0^1 \frac{|\mathbb{1}_{\{X_i \le x\}} - \mathbb{1}_{\{X_i' \le x\}}|}{x(1-x)}dx = \frac{1}{n^2}\int_{X_i \wedge X_i'}^{X_i \vee X_i'} \frac{dx}{x(1-x)},$$

the integrand being $\mathbb{1}_{[X_i \wedge X_i',\, X_i \vee X_i')}(x)$. With $\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$,

$$\int \frac{dx}{x(1-x)} = \log\frac{x}{1-x} =: \mathrm{logit}(x),$$

so $(Z - Z^{(i)})^2 \le \frac{1}{n}\,|\mathrm{logit}(X_i') - \mathrm{logit}(X_i)|$. Efron–Stein (Theorem 3.1) gives

$$\mathrm{Var}(Z) \le \frac{1}{2}\sum_{i=1}^n \mathbb{E}[(Z - Z^{(i)})^2] \le \frac{1}{2n}\sum_{i=1}^n \mathbb{E}\big|\mathrm{logit}(X_i') - \mathrm{logit}(X_i)\big|.$$

The logit computation. For $X \sim \text{Uniform}[0,1]$, $L := \text{logit}(X)$ satisfies $\mathbb{P}\{L \le l\} = \mathbb{P}\{X \le e^l/(1+e^l)\} = 1/(1+e^{-l})$: L is standard logistic. For iid L, L' with cdf F,

$$|l - l'| = \int_{-\infty}^{\infty} \big|\mathbb{1}_{\{l>t\}} - \mathbb{1}_{\{l'>t\}}\big| dt,$$

and $\mathbb{E}|\mathbb{1}_{\{L>t\}} - \mathbb{1}_{\{L'>t\}}| = 2F(t)(1-F(t))$ (difference of iid Bernoullis), so by Tonelli $\mathbb{E}|L-L'| = 2\int_{-\infty}^{\infty} F(t)(1-F(t))dt$. For the standard logistic, $F(t)(1-F(t)) = e^t/(1+e^t)^2$, and $\int_{-\infty}^{\infty} \frac{e^t}{(1+e^t)^2} dt = 1$ (substitute $u = e^t$). Hence

$$\mathbb{E}\big|\text{logit}(X_i') - \text{logit}(X_i)\big| = 2.$$

Therefore

$$\text{Var}(Z) \le \frac{1}{2} \cdot 2 = 1.$$

Comparison. $\text{Var}(Z) = \mathbb{E}Z^2 - (\mathbb{E}Z)^2 \le \mathbb{E}Z^2 = 1$, so the Efron–Stein bound $\text{Var}(Z) \le 1$ coincides with the trivial bound and yields no improvement — the $1/(x(1-x))$ weight makes the single-observation influence $|\text{logit}(X_i') - \text{logit}(X_i)|$ heavy-tailed (its mean is 2, versus $\mathbb{E}|X_i - X_i'| = 1/3$ for Cramér–von Mises), which is exactly what the Anderson–Darling weight pays for its $\mathbb{E}Z^2 = 1$ normalization. ■

Exercise 13.40

Central idea. On $[0,1)$, the process $M(s) = (F_n(s) - s)/(1-s)$ is a martingale in s. Its variance at time t is v_t/n with $v_t = t/(1-t)$. On $[0, 1/2]$ its summands are bounded by 1. Doob's inequality, applied to $e^{\lambda\sqrt{n}M}$, turns Bernstein's bound at a fixed time into a sub-Gaussian bound for the running supremum.

Peel $[n^{-1/2}, 1/2]$ into geometric shells $[t_{k+1}, t_k]$ with $t_k = \rho^{-k}/2$. On each shell, $|V(s)|$ is at most $\sqrt{\rho/(1-t_k)}$ times a normalized running supremum. There are $K \approx \log n/(2\log\rho)$ shells. A maximal inequality over $2K$ nearly-Gaussian variables then gives $\mathbb{E}Z \le \sqrt{2\log\log n}\,(1+o(1))$.

The exact inequality $\mathbb{E}Z \le \sqrt{2\log\log n}$ is false. An exact finite computation refutes it at $n = 16$, and the Gumbel limit says it fails for all large n. The Efron–Stein inequality gives $\text{Var}(Z) \le 3 + o(1)$. The $1/\log\log n$ variance rate is not proved here.

Solution. Notation: - $F_n(s) = P_n([0,s])$; - $V(s) = \sqrt{n}\,(F_n(s) - s)/\sqrt{s(1-s)}$; - $J = [n^{-1/2}, 1 - n^{-1/2}]$, so that $Z = \sup_{s\in J} |V(s)|$; - $\ell_n = \log\log n$.

All processes here are right-continuous with left limits, so the suprema are measurable.

Erratum (the exact inequality is false). As printed, $\mathbb{E}Z \le \sqrt{2\log\log n}$ fails. Here are three pieces of evidence.

Exact finite computation at $n = 16$. For any points $s_1 < \cdots < s_k$ in J, we have $Z \ge \max_j |V(s_j)|$. The vector $(nF_n(s_j))_j$ is a function of multinomial cell counts, so $\mathbb{E}\max_j |V(s_j)|$ is a finite sum. For $n = 16$ and the 7 equally spaced points $s_j = \frac{1}{4} + \frac{j}{12}$, $j = 0, \ldots, 6$, which fill $J = [\frac{1}{4}, \frac{3}{4}]$, enumerating all cell-count vectors gives

$$\mathbb{E}Z \ge \mathbb{E}\max_j |V(s_j)| = 1.44118 > 1.42813 = \sqrt{2\log\log 16}.$$

The probabilities sum to 1 to twelve digits.

Simulation. We computed the supremum exactly from the order statistics and checked both one-sided values at every jump inside J and at both endpoints:

n	16	10^2	10^3	10^4	10^5
replications	2×10^5	10^5	2×10^4	4000	1500
$\mathbb{E}Z$ (s.e.)	1.672 (.001)	1.980 (.002)	2.177 (.004)	2.320 (.009)	2.408 (.014)
$\sqrt{2\log\log n}$	1.428	1.748	1.966	2.107	2.211

Asymptotically. Assume uniform integrability, which we do not prove. The Gumbel limit in the hint then gives

$$\mathbb{E}Z - \sqrt{2\ell_n} \sim \frac{\log\sqrt{\ell_n/(2\pi)} + \gamma}{\sqrt{2\ell_n}}.$$

This is positive once $\ell_n > 2\pi e^{-2\gamma} \approx 1.98$. So the inequality is not merely off at small n; under that assumption it fails for every large n.

Corrected statement. We prove

$$\mathbb{E}Z \le \sqrt{2\log\log n}\,(1+o(1)), \qquad \text{i.e.} \quad \limsup_{n\to\infty} \frac{\mathbb{E}Z}{\sqrt{2\log\log n}} \le 1.$$

We also prove $\mathrm{Var}(Z) \le 3 + n^{-1/4}\mathbb{E}Z = 3 + o(1)$.

Step 1: a martingale and its maximal inequality. For a single uniform X and $s \in [0,1)$, put $m(s) = (\mathbb{1}\{X \le s\} - s)/(1-s)$. Let $\mathcal{G}_s = \sigma(\mathbb{1}\{X \le r\} : r \le s)$ and take $s < s' < 1$. - On $\{X \le s\}$: $m(s') = 1 = m(s)$. - On $\{X > s\}$: $\mathbb{P}(X \le s' \mid X > s) = (s'-s)/(1-s)$, so

$$\mathbb{E}[m(s') \mid \mathcal{G}_s] = \frac{\frac{s'-s}{1-s} - s'}{1-s'} = \frac{-s(1-s')}{(1-s)(1-s')} = -\frac{s}{1-s} = m(s).$$

So m is a martingale, with $\mathbb{E}m(t) = 0$ and $\mathrm{Var}\, m(t) = t(1-t)/(1-t)^2 = v_t$. For $t \le 1/2$, $m(t) \in [-t/(1-t), 1] \subset [-1,1]$.

Summing over the sample, $\sqrt{n}M(t) = n^{-1/2}\sum_{i=1}^n m_i(t)$ is a martingale in t for the joint filtration. At each $t \le \frac{1}{2}$ it is a sum of independent centered terms, each bounded by $n^{-1/2}$ in absolute value, with total variance v_t. Bernstein's moment-generating-function bound (Section 2.8) gives, for $0 \le \lambda < 3\sqrt{n}$,

$$\log \mathbb{E}e^{\pm\lambda\sqrt{n}M(t)} \le \frac{\lambda^2 v_t}{2(1-\lambda/(3\sqrt{n}))}.$$

Since $e^{\pm\lambda\sqrt{n}M(\cdot)}$ is a nonnegative right-continuous submartingale, Doob's maximal inequality gives

$$\mathbb{P}\Big\{\sup_{s\le t} \pm\sqrt{n}M(s) \ge x\Big\} \le e^{-\lambda x}\mathbb{E}e^{\pm\lambda\sqrt{n}M(t)}.$$

Optimizing over λ as in the proof of Bernstein's inequality and adding the two signs:

$$\mathbb{P}\Big\{\sup_{s\le t} \sqrt{n}|M(s)| \ge x\Big\} \le 2\exp\Big(-\frac{x^2}{2(v_t + x/(3\sqrt{n}))}\Big), \qquad t \le \tfrac{1}{2}.$$

Define $Y(t) = \sup_{s\le t}\sqrt{n}|M(s)|/\sqrt{v_t}$. If $t \ge n^{-1/2}$, then $nv_t \ge nt \ge \sqrt{n}$. Putting $x = y\sqrt{v_t}$ gives, for all $y \ge 0$,

$$\mathbb{P}\{Y(t) \ge y\} \le 2\exp\Big(-\frac{y^2}{2(1+y\delta)}\Big), \qquad \delta := \frac{n^{-1/4}}{3}. \tag{1}$$

Step 2: peeling the left half. Fix $\eta \in (0, \frac{1}{2})$ and put $\rho = 1+\eta$, $t_k = \rho^{-k}/2$. Let K be the least integer with $t_K \le n^{-1/2}$; then $K \le 1 + \log(\sqrt{n}/2)/\log\rho$. The shells $[t_{k+1}, t_k]$, $0 \le k < K$, cover $[n^{-1/2}, \frac{1}{2}]$, and every t_k with $k < K$ satisfies $t_k > n^{-1/2}$. So (1) applies to each $Y_k := Y(t_k)$.

For $s \in [t_{k+1}, t_k]$,

$$|V(s)| = \sqrt{n}|M(s)|\sqrt{\frac{1-s}{s}} \le \sqrt{n}|M(s)|\,\frac{1}{\sqrt{t_{k+1}}} \le Y_k\sqrt{\frac{\rho\, v_{t_k}}{t_k}} = c_k Y_k, \qquad c_k = \sqrt{\frac{\rho}{1-t_k}}.$$

There are two kinds of shells: - *Small shells*, $t_k \le \eta$: here $c_k \le c_\eta := \sqrt{(1+\eta)/(1-\eta)}$. - *Large shells*, $t_k > \eta$: here $c_k \le \sqrt{2\rho}$, since $t_k \le \frac{1}{2}$. Since $t_k > \eta$ forces $k < \log(1/(2\eta))/\log\rho$, there are at most $L_0 := 1 + \log(1/(2\eta))/\log\rho$ large shells, a number independent of n.

Step 3: the right half. Use the reflection $X_i \mapsto 1 - X_i$. Then $nF_n(s) = n - \#\{i : 1 - X_i < 1 - s\}$. The weight $\sqrt{s(1-s)}$ is symmetric. Almost surely no X_i equals $n^{-1/2}$, $\frac{1}{2}$, or $1 - n^{-1/2}$. Hence, almost surely, $\sup_{s\in[1/2,\,1-n^{-1/2}]} |V(s)|$ equals the left-half supremum computed from the reflected sample. That sample has the same law.

So $Z \le \max(\max_k c_k Y_k,\ \max_k c_k \tilde{Y}_k)$, where the $\tilde{Y}_k$ are the reflected copies and also satisfy (1). Since the maximum of nonnegative quantities is at most their sum,

$$Z \le c_\eta \max_{\text{small } k,\ Y \text{ or } \tilde{Y}} Y + \sqrt{2\rho} \max_{\text{large } k,\ Y \text{ or } \tilde{Y}} Y. \tag{2}$$

The first maximum is over at most $2K$ variables, the second over at most $2L_0$.

Step 4: a maximal inequality. Let $Y_1, \dots, Y_m \ge 0$ each satisfy (1), and put $Y^* = \delta^{-1/2}$ and $a = 1 + \delta^{1/2}$. - For $y \le Y^*$, the exponent in (1) is at least $y^2/(2a)$. - For $y > Y^*$, we have $1 + y\delta \le y(\delta^{1/2} + \delta) \le 2y\delta^{1/2}$, using $\delta \le 1$. So the exponent is at least $y/(4\delta^{1/2})$.

Hence, for any $y_0 > 0$,

$$\mathbb{E}\max_j Y_j \le y_0 + \sum_{j=1}^m \int_{y_0}^\infty \mathbb{P}\{Y_j \ge y\}\, dy \le y_0 + 2m \int_{y_0}^\infty e^{-y^2/(2a)} dy + 2m \int_{Y^*}^\infty e^{-y/(4\sqrt{\delta})} dy.$$

Use $\int_{y_0}^\infty e^{-y^2/(2a)} dy \le \frac{a}{y_0} e^{-y_0^2/(2a)}$ and choose $y_0 = \sqrt{2a\log(2m)}$. This gives

$$\mathbb{E}\max_{j\le m} Y_j \le \sqrt{2a\log(2m)} + \sqrt{\frac{a}{2\log(2m)}} + 8m\sqrt{\delta}\, e^{-1/(4\delta)}. \tag{3}$$

Step 5: conclusion for $\mathbb{E}Z$. Apply (3) to both maxima in (2), with $m = 2K$ and $m = 2L_0$, and take expectations:

$$\mathbb{E}Z \le c_\eta\Big(\sqrt{2a\log(4K)} + 1 + 16K\sqrt{\delta}e^{-1/(4\delta)}\Big) + \sqrt{2\rho}\Big(\sqrt{2a\log(4L_0)} + 1 + 16L_0\sqrt{\delta}e^{-1/(4\delta)}\Big).$$

Here we used $\sqrt{a/(2\log 2m)} \le 1$, since $a \le 2$ and $m \ge 2$.

Now fix η and let $n \to \infty$: - $a = 1 + \delta^{1/2} \to 1$; - $\log(4K) \le \log\log n + \log(2/\log\rho) + o(1)$, so $\sqrt{2a\log(4K)}/\sqrt{2\ell_n} \to 1$; - $K\sqrt{\delta}e^{-1/(4\delta)} = O(\log n)\, n^{-1/8} e^{-3n^{1/4}/4} \to 0$; - the second bracket is $O(1)$, because L_0 does not depend on n.

Therefore $\limsup_n \mathbb{E}Z/\sqrt{2\ell_n} \le c_\eta = \sqrt{(1+\eta)/(1-\eta)}$. Since $\eta \in (0, \frac{1}{2})$ is arbitrary,

$$\limsup_{n\to\infty} \frac{\mathbb{E}Z}{\sqrt{2\log\log n}} \le 1, \qquad \text{that is,} \qquad \mathbb{E}Z \le \sqrt{2\log\log n}\,(1 + o(1)).$$

The same display with η fixed, for example $\eta = \frac{1}{4}$, gives an explicit non-asymptotic bound $\mathbb{E}Z \le C_1\sqrt{\log\log n} + C_2$ for $n \ge 16$.

Step 6: an $O(1)$ variance bound. Write $f_s(x) = (\mathbb{1}\{x \le s\} - s)/\sqrt{s(1-s)}$. Then $\mathbb{E}f_s(X) = 0$, $\mathbb{E}f_s(X)^2 = 1$, and $V(s) = n^{-1/2}\sum_i f_s(X_i)$. Note that $|V(s)| \le \sqrt{2n}\, n^{1/4}$ on J, so Z is bounded.

Fix $\varepsilon > 0$ and choose $s^* \in J$ and a sign $\theta \in \{\pm 1\}$, depending on X, with $\theta V(s^*) \ge Z - \varepsilon$. Replace X_i by an independent copy X_i' and call the resulting statistic Z_i'. Then

$$Z_i' \ge \theta V(s^*) + \theta\, \frac{f_{s^*}(X_i') - f_{s^*}(X_i)}{\sqrt{n}},$$

so $(Z - Z_i')_+ \le \varepsilon + D_i/\sqrt{n}$ with $D_i = |f_{s^*}(X_i) - f_{s^*}(X_i')|$. Since s^* depends only on X,

$$\mathbb{E}' D_i^2 = f_{s^*}(X_i)^2 + 1.$$

For any $\varepsilon' > 0$, $(\varepsilon + D_i/\sqrt{n})^2 \le (1+\varepsilon')D_i^2/n + (1+1/\varepsilon')\varepsilon^2$. Hence

$$V_+ := \sum_i \mathbb{E}'(Z - Z_i')_+^2 \le (1+\varepsilon')\Big(1 + \frac{1}{n}\sum_i f_{s^*}(X_i)^2\Big) + (1+1/\varepsilon')n\varepsilon^2.$$

Next,

$$\frac{1}{n}\sum_i f_s(X_i)^2 = \frac{F_n(s)(1-s)}{s} + \frac{(1-F_n(s))s}{1-s}.$$

- If $s \le \frac{1}{2}$: the second term is at most 1. For the first,

$$\frac{F_n(s)}{s} = 1 + \frac{F_n(s)-s}{s} \le 1 + \frac{|V(s)|}{\sqrt{ns}} \le 1 + Zn^{-1/4},$$

because $s \ge n^{-1/2}$. - If $s > \frac{1}{2}$: symmetrically, the first term is at most 1 and the second is at most $1 + Zn^{-1/4}$.

Thus $\frac{1}{n}\sum_i f_{s^*}(X_i)^2 \le 2 + Zn^{-1/4}$. Let $\varepsilon \to 0$ and then $\varepsilon' \to 0$; the left side V_+ does not depend on them. This gives $V_+ \le 3 + Zn^{-1/4}$ a.s. The Efron–Stein inequality (Theorem 3.1) then gives

$$\mathrm{Var}(Z) \le \mathbb{E}V_+ \le 3 + n^{-1/4}\,\mathbb{E}Z = 3 + o(1).$$

Variance rate.

Status: incomplete. We do not prove that $\mathrm{Var}(Z)$ is bounded by a function of $1/(2\log\log n)$ that tends to 0. The Gumbel limit suggests $\mathrm{Var}(Z) \approx \frac{\pi^2}{6}\cdot\frac{1}{2\log\log n}$.

The gap. Steps 1–5 control $\mathbb{E}Z$ only up to a factor $1 + o(1)$, and the Efron–Stein bound of Step 6 is $O(1)$. Getting the rate requires a superconcentration argument, which we do not have. Possible routes are a Talagrand L^1–L^2 (hypercontractive) bound exploiting the small influence of each shell, or a quantitative extreme-value argument. The Gumbel convergence in law alone does not give convergence of variances: uniform integrability of $\sqrt{2\ell_n}(Z - b_n)$ would have to be proved.

The pre-repair version of this solution derived the rate from a peeled tail bound with constant $c_3 \le 1/27$. That derivation needed $c_3 \ge \frac{1}{2}$ to absorb the $\log n$ shell count, and it assumed this uniform integrability, so it is withdrawn.

For orientation only, the simulation above gives $\mathrm{Var}(Z) \approx 0.32, 0.32, 0.30, 0.30, 0.29$ at $n = 16, 10^2, 10^3, 10^4, 10^5$; an independent rerun gave 0.326, 0.323, 0.306, 0.276 at $n = 16, \dots, 10^4$, so the second digit is noisy for $n \ge 10^4$. This decreases slowly. It is below the heuristic $\frac{\pi^2}{6}/(2\log\log n)$, which takes the values 0.81, 0.54, 0.43, 0.37, 0.34 at the same n. ■

Exercise 13.41

Central idea. The engine behind all four parts is the single hypothesis that $\psi(x)/x$ is nonincreasing: it yields subadditivity via the weighted-average trick $\psi(u+v) = (u+v)\cdot\frac{\psi(u+v)}{u+v}$, the bound $\mathbb{E}\psi(X) \le 2\psi(\mathbb{E}X)$ by splitting at $\mathbb{E}X$ and dominating $\psi(X)/X$ by $\psi(\mathbb{E}X)/\mathbb{E}X$ on $\{X > \mathbb{E}X\}$, strict decrease of $\psi(r)/r^2$ for the fixed-point uniqueness, and closure under composition and sums via products and sums of nonincreasing ratios (using that a sub-linear function is strictly positive on $(0,\infty)$).

Solution. Let $\psi : [0,\infty) \to [0,\infty)$ be non-decreasing, continuous, with $\psi(x)/x$ nonincreasing on $(0,\infty)$ and $\psi(1) \ge 1$. First, $\psi(x) > 0$ for every $x > 0$: if $\psi(x_0) = 0$ for some $x_0 > 0$, then monotonicity gives $\psi \equiv 0$ on $[0, x_0]$, and $\psi(x)/x \le \psi(x_0)/x_0 = 0$ gives $\psi \equiv 0$ on $[x_0, \infty)$; either way $\psi(1) = 0 < 1$, a contradiction. Also, for $0 < x \le 1$, $\psi(x)/x \ge \psi(1) \ge 1$, i.e. $\psi(x) \ge x$.

(a) $\psi + \rho$ is non-decreasing and continuous, $(\psi+\rho)(x)/x = \psi(x)/x + \rho(x)/x$ is a sum of nonincreasing functions hence nonincreasing, and $(\psi+\rho)(1) = \psi(1) + \rho(1) \ge 2 \ge 1$. For $\psi \circ \rho$: it is non-decreasing and continuous as a composition of such. For $x > 0$, $\rho(x) > 0$ by the remark above, so

$$\frac{(\psi\circ\rho)(x)}{x} = \frac{\psi(\rho(x))}{\rho(x)}\cdot\frac{\rho(x)}{x},$$

a product of two nonnegative nonincreasing functions of x (the first because $x \mapsto \rho(x)$ is non-decreasing and $y \mapsto \psi(y)/y$ is nonincreasing; the second by hypothesis), hence nonincreasing. Finally $(\psi \circ \rho)(1) = \psi(\rho(1)) \geq \psi(1) \geq 1$ since $\rho(1) \geq 1$ and ψ is non-decreasing.

(b) For $r > 0$, $\alpha r^2 = \psi(r)$ is equivalent to $\alpha = h(r)$ where $h(r) = \psi(r)/r^2$. Write $h(r) = \frac{\psi(r)}{r} \cdot \frac{1}{r}$: for $0 < r_1 < r_2 \leq 1$, $\frac{\psi(r_1)}{r_1} \geq \frac{\psi(r_2)}{r_2} > 0$ and $\frac{1}{r_1} > \frac{1}{r_2} > 0$, so $h(r_1) > h(r_2)$; i.e. h is continuous and strictly decreasing on $(0, 1]$. Moreover $h(r) \geq \psi(1)/r \to \infty$ as $r \to 0^+$, while $h(1) = \psi(1) \leq \alpha$. If $\alpha = \psi(1)$ then $r = 1$ is the (unique) solution; otherwise $h(1) < \alpha < h(\delta)$ for small $\delta > 0$, and the intermediate value theorem on $[\delta, 1]$ gives a root in $(\delta, 1) \subset (0, 1]$. Strict monotonicity gives uniqueness.

(c) Let $u, v \geq 0$; if either is 0 the claim is trivial since $\psi \geq 0$. For $u, v > 0$, since $u \leq u + v$ and $\psi(x)/x$ is nonincreasing,

$$\frac{\psi(u+v)}{u+v} \leq \frac{\psi(u)}{u}, \qquad \frac{\psi(u+v)}{u+v} \leq \frac{\psi(v)}{v}.$$

Hence $\psi(u+v) = (u+v)\frac{\psi(u+v)}{u+v} = u\frac{\psi(u+v)}{u+v} + v\frac{\psi(u+v)}{u+v} \leq u\frac{\psi(u)}{u} + v\frac{\psi(v)}{v} = \psi(u) + \psi(v)$.

(d) Let $X \geq 0$ with $\mathbb{E}X < \infty$. If $\mathbb{E}X = 0$ then $X = 0$ a.s. and $\mathbb{E}\psi(X) = \psi(0) \leq 2\psi(0) = 2\psi(\mathbb{E}X)$. Assume $\mathbb{E}X > 0$ and split

$$\mathbb{E}\psi(X) = \mathbb{E}[\psi(X)\mathbf{1}_{\{X \leq \mathbb{E}X\}}] + \mathbb{E}[\psi(X)\mathbf{1}_{\{X > \mathbb{E}X\}}].$$

On $\{X \leq \mathbb{E}X\}$, $\psi(X) \leq \psi(\mathbb{E}X)$ by monotonicity, so the first term is at most $\psi(\mathbb{E}X)$. On $\{X > \mathbb{E}X\}$, write $\psi(X) = X \cdot \frac{\psi(X)}{X} \leq X \cdot \frac{\psi(\mathbb{E}X)}{\mathbb{E}X}$ since $\psi(x)/x$ is nonincreasing; thus the second term is at most $\frac{\psi(\mathbb{E}X)}{\mathbb{E}X}\mathbb{E}[X\mathbf{1}_{\{X > \mathbb{E}X\}}] \leq \psi(\mathbb{E}X)$. Adding gives $\mathbb{E}\psi(X) \leq 2\psi(\mathbb{E}X)$.

Exercise 13.42

Central idea. The stated inequality is false: the fixed point r_* controls only $\psi \circ \rho$, not ρ alone, so the term $\frac{b}{\sqrt{n}}\rho(r)$ can be arbitrarily large compared to r_*. An explicit sub-linear counterexample ($\psi(s) = \min(s, \sqrt{s})$ in effect, $\rho(r) = 100r$) gives $u^2 \approx 10200$ while $2(a^2 + b^2)r_*^2 + 2c \approx 86$. Under the additional hypothesis $\psi(t) \geq t$ for all $t \geq 0$ (so that $\rho \leq \psi \circ \rho$), both terms are controlled by the key sub-linearity bound $\psi(\rho(r)) \leq \sqrt{n}\, r_* r$, and AM-GM yields the claimed inequality.

Solution. We use the book's definition: $\psi : [0, \infty) \to [0, \infty)$ is sub-linear if it is nondecreasing, continuous, $\psi(x)/x$ is nonincreasing on $(0, \infty)$, and $\psi(1) \geq 1$ (p. 399). By Exercise 13.41(a), $\phi := \psi \circ \rho$ is sub-linear.

The statement is false as stated. Let $n = 1$, $a = b = 1$, $c = 0$,

$$\psi(s) = \begin{cases} s & 0 \leq s \leq 1 \\ \sqrt{s} & s \geq 1 \end{cases} \qquad \text{and} \qquad \rho(r) = 100r.$$

Then ψ is nondecreasing, continuous, $\psi(x)/x$ equals 1 on $(0, 1]$ and $x^{-1/2}$ on $[1, \infty)$ (hence nonincreasing), and $\psi(1) = 1$; ρ is nondecreasing, continuous, $\rho(x)/x \equiv 100$, and $\rho(1) = 100 \geq 1$. Both are non-trivial sub-linear functions. Now $\psi(100r) = 100r$ for $r \leq 0.01$ and $10\sqrt{r}$ for $r \geq 0.01$. The function $r \mapsto \psi(100r)/r^2$ is continuous and strictly decreasing on $(0, \infty)$ (it equals $100/r$ on $(0, 0.01]$ and $10/r^{3/2}$ on $[0.01, \infty)$, matching at $r = 0.01$), tends to $+\infty$ as $r \downarrow 0$ and to 0 as $r \to \infty$; hence $r^2 = \psi(100r)$ has a unique positive solution $r_* = 10^{2/3}$ (since $(10^{2/3})^2 = 10^{4/3} = 10\sqrt{10^{2/3}}$), with $r_*^2 = 10^{4/3} < 21.6$. Similarly $r \mapsto (\psi(100r) + 100r)/r^2$ is continuous and strictly decreasing with the same limits, so $r^2 = \psi(100r) + 100r$ has a unique positive solution u. Since $\psi \geq 0$, $u^2 \geq 100u$, so $u \geq 100$; then $u^2 = 10\sqrt{u} + 100u \geq 100u \geq 10000$. But $2(a^2 + b^2)r_*^2 + 2c = 4 \cdot 10^{4/3} < 4 \cdot 21.6 = 86.4 < 10000 \leq u^2$, contradicting the claimed bound. Thus the inequality does not follow from sub-linearity alone; the term $\frac{b}{\sqrt{n}}\rho(r)$ is invisible to the fixed point r_*.

Corrected version. Add the hypothesis $\psi(t) \geq t$ for all $t \geq 0$ (equivalently $\rho \leq \psi \circ \rho$). Then $u^2 \leq 2(a^2 + b^2)r_*^2 + 2c$.

Indeed, $\phi = \psi \circ \rho$ is sub-linear with $\phi(r_*) = \sqrt{n}\, r_*^2$. For every $r > 0$,

$$\phi(r) \leq \sqrt{n}\, r_* r :$$

indeed $\phi(r)/r$ is nonincreasing; if $0 < r \le r_*$, then $\phi(r) \le \phi(r_*) = \sqrt{n}\, r_*^2 \le \sqrt{n}\, r_* r$; if $r \ge r_*$, then $\phi(r)/r \le \phi(r_*)/r_* = \sqrt{n}\, r_*$. Since $\psi \ge \mathrm{id}$, $\rho(r) \le \psi(\rho(r)) = \phi(r) \le \sqrt{n}\, r_* r$ for all $r > 0$. Evaluating the equation for u at $r = u$,

$$u^2 = \frac{a}{\sqrt{n}}\phi(u) + \frac{b}{\sqrt{n}}\rho(u) + c \le a r_* u + b r_* u + c = (a+b) r_* u + c.$$

By AM-GM, $(a+b) r_* u \le \frac{(a+b)^2 r_*^2 + u^2}{2}$, and $(a+b)^2 \le 2(a^2+b^2)$, so $(a+b) r_* u \le (a^2+b^2) r_*^2 + u^2/2$. Hence $u^2 \le (a^2+b^2) r_*^2 + u^2/2 + c$, i.e.

$$u^2 \le 2(a^2+b^2) r_*^2 + 2c,$$

as claimed. (The hypothesis $a \ge 1$ is not needed for this argument.)

Exercise 13.43

Central idea. The geometric engine is the inclusion $A \subseteq 2\,\mathrm{conv}(A_\varepsilon)$, proved by duality: for every unit vector w, some net point p satisfies $\langle w, p\rangle \ge 1 - \varepsilon^2/2$, so by a separation argument $\mathrm{conv}(A_\varepsilon)$ contains the ball of radius $1-\varepsilon^2/2$. Writing $\alpha \in A$ as twice a convex combination of net points, linearity of W and the triangle inequality dominate $\|W(\alpha)\|$ by twice the maximum of $\|W\|$ over the net — the deterministic core of the contraction principle. For Lemma 13.13, expanding the centered quantity $\|W(\alpha)\|^2 - 1 = \langle \alpha, (W^*W - I)\alpha\rangle$ bilinearly in the net coefficients preserves centering and reduces the supremum over T to a finite maximum of averages of i.i.d. centered sub-exponential variables, controlled by Bernstein's inequality and a union bound. No chaining is needed.

Solution. Let $E = \mathrm{span}\{e_1, \dots, e_k\} \subset \mathbb{R}^D$, so that $A = \{x \in E : \|x\|_2 = 1\}$ is the unit sphere of E. Let A_ε be a finite ε-net of A with $0 < \varepsilon < 1$ (possible since A is compact).

The convex-hull inclusion. Let $w \in E$ with $\|w\|_2 = 1$ and choose $p \in A_\varepsilon$ with $\|w - p\|_2 \le \varepsilon$. Then

$$\langle w, p\rangle = \tfrac{1}{2}(\|w\|_2^2 + \|p\|_2^2 - \|w-p\|_2^2) = 1 - \tfrac{1}{2}\|w-p\|_2^2 \ge 1 - \varepsilon^2/2.$$

For $q = \sum_i \mu_i p_i \in \mathrm{conv}(A_\varepsilon)$ (finite convex combination), $\langle w, q\rangle = \sum_i \mu_i \langle w, p_i\rangle \le \max_{p \in A_\varepsilon} \langle w, p\rangle$, so

$$\max_{q \in \mathrm{conv}(A_\varepsilon)} \langle w, q\rangle = \max_{p \in A_\varepsilon} \langle w, p\rangle \ge 1 - \varepsilon^2/2 \qquad \text{for every unit } w \in E.$$

We claim $(1-\varepsilon^2/2) B_E \subseteq \mathrm{conv}(A_\varepsilon)$, where B_E is the unit ball of E. Otherwise, since $\mathrm{conv}(A_\varepsilon)$ is a compact convex polytope, strict separation in E yields a unit vector $w \in E$ and $z \notin \mathrm{conv}(A_\varepsilon)$ with $\|z\|_2 \le 1 - \varepsilon^2/2$ such that $\langle w, z\rangle > \max_{q \in \mathrm{conv}(A_\varepsilon)} \langle w, q\rangle \ge 1 - \varepsilon^2/2$, contradicting $\langle w, z\rangle \le \|z\|_2$. Since $A \subset B_E$,

$$A \subseteq \frac{1}{1 - \varepsilon^2/2}\,\mathrm{conv}(A_\varepsilon) = \frac{2}{2 - \varepsilon^2}\,\mathrm{conv}(A_\varepsilon) \subseteq 2\,\mathrm{conv}(A_\varepsilon),$$

as $\varepsilon^2 < 1$. The same argument applies in any k-dimensional coordinate subspace.

Domination by the net. Put $\mathcal{A} = A$ and $\mathcal{A}_\varepsilon = A_\varepsilon$. Let $\alpha \in \mathcal{A}$. By the inclusion, $\alpha = 2\sum_{r=1}^m \lambda_r \alpha_r$ for some $m < \infty$, $\alpha_r \in \mathcal{A}_\varepsilon$, $\lambda_r \ge 0$, $\sum_r \lambda_r = 1$. By linearity of W,

$$W(\alpha) = 2\sum_{r=1}^m \lambda_r W(\alpha_r), \qquad \text{hence} \qquad \|W(\alpha)\| \le 2\sum_{r=1}^m \lambda_r \|W(\alpha_r)\| \le 2 \max_{1 \le r \le m} \|W(\alpha_r)\| \le 2 \sup_{\beta \in \mathcal{A}_\varepsilon} \|W(\beta)\|.$$

The right-hand side does not depend on α; taking $\sup_{\alpha \in \mathcal{A}}$ and then expectations,

$$\mathbb{E} \sup_{\alpha \in \mathcal{A}} \|W(\alpha)\| \le 2\,\mathbb{E} \sup_{\beta \in \mathcal{A}_\varepsilon} \|W(\beta)\|.$$

This domination of a convex combination by the maximum over its extreme points is the deterministic mechanism behind the contraction principle (Theorem 11.5, cited as Lemma 11.5 in the exercise; the Gaussian form is Exercise 11.11).

Another proof of Lemma 13.13. Let T be the set of $2k$-sparse unit vectors of $\mathbb{R}^D$, $V = \sup_{\alpha\in T}(\|W(\alpha)\|^2 - 1)$ and $V' = \sup_{\alpha\in T}(1 - \|W(\alpha)\|^2)$. Partition $T = \bigcup_{|S|=2k} A^{(S)}$, where $A^{(S)}$ is the unit sphere of the coordinate subspace indexed by $S \subset [D]$. Fix $\varepsilon \in (0,1)$ and let $A^{(S)}_\varepsilon$ be an ε-net of $A^{(S)}$ with $|A^{(S)}_\varepsilon| \le (1 + 2/\varepsilon)^{2k}$ (Lemma 13.11); by the inclusion proved above, $A^{(S)} \subseteq 2\,\mathrm{conv}(A^{(S)}_\varepsilon)$.

For $\alpha \in A^{(S)}$, write $\alpha = 2\sum_r \lambda_r \alpha^{(r)}$ as above and put $M = W^*W - I$. Since $\|\alpha\|_2 = 1$,

$$\|W(\alpha)\|^2 - 1 = \langle \alpha, M\alpha\rangle = 4\sum_{r,s} \lambda_r\lambda_s \langle \alpha^{(r)}, M\alpha^{(s)}\rangle \le 4\max_{r,s} \langle \alpha^{(r)}, M\alpha^{(s)}\rangle,$$

because $\sum_{r,s}\lambda_r\lambda_s = 1$ with $\lambda_r\lambda_s \ge 0$. Moreover,

$$\langle \alpha^{(r)}, M\alpha^{(s)}\rangle = \frac{1}{d}\sum_{i=1}^{d} W_i(\alpha^{(r)})W_i(\alpha^{(s)}) - \langle \alpha^{(r)}, \alpha^{(s)}\rangle =: \frac{1}{d}\sum_{i=1}^{d} \zeta_i^{(rs)},$$

where, the X_{ij} being independent across i with $\mathbb{E}X_{ij} = 0$ and $\mathbb{E}X_{ij}^2 = 1$, the variables $\zeta_i^{(rs)}$ are i.i.d. across i and centered, since $\mathbb{E}[W_i(\alpha^{(r)})W_i(\alpha^{(s)})] = \langle \alpha^{(r)}, \alpha^{(s)}\rangle$. Each $W_i(\alpha^{(r)})$ is sub-gaussian with $O(1)$ ψ_2-norm (for X standard Gaussian or Rademacher), so $\|\zeta_i^{(rs)}\|_{\psi_1} \le K_0$ for an absolute constant K_0; i.e., the $\zeta_i^{(rs)}$ are sub-exponential. Hence, with $Z_{rs} = \frac{1}{d}\sum_i \zeta_i^{(rs)}$,

$$V \le 4\max_{(r,s)} Z_{rs}, \qquad V' \le 4\max_{(r,s)}(-Z_{rs}),$$

the maxima over the at most $M := \binom{D}{2k}(1 + 2/\varepsilon)^{4k}$ pairs (r,s) from $\bigcup_S A^{(S)}_\varepsilon$. Each Z_{rs} is an average of d i.i.d. centered sub-exponential variables, hence sub-gamma with variance factor $O(1/d)$ and scale $O(1/d)$; by Bernstein's inequality (Theorem 2.10) and the maximal inequality for sub-gamma variables (cf. Corollary 2.6),

$$\mathbb{E}\max_{(r,s)} Z_{rs} \le C_1\sqrt{\frac{\log M}{d}} + C_2\frac{\log M}{d},$$

and the same bound holds for $-Z_{rs}$. Take $\varepsilon = \varepsilon_*$ (admissible since $\varepsilon_* < 1/2$). Then

$$\log M \le 2k\log\frac{eD}{2k} + 4k\log\left(1 + \frac{2}{\varepsilon_*}\right) \le 2k\log\frac{2eD}{k\varepsilon_*} + 4k\log\frac{3}{\varepsilon_*} \le 3d\varepsilon_*^2,$$

using $d\varepsilon_*^2 = 2k\log\frac{2eD}{k\varepsilon_*}$ together with $\varepsilon_* \le 4$ and $3 \le 2eD/k$ (as $D \ge 2k$). Therefore

$$\max(\mathbb{E}V, \mathbb{E}V') \le 4C_1\sqrt{3}\,\varepsilon_* + 12C_2\varepsilon_*^2 \le C\varepsilon_*$$

for an absolute constant C, using $\varepsilon_* \le 1/2$. This is Lemma 13.13 up to the value of the absolute constant, proved without chaining and without Lemma 13.12.

Chapter 14 — Φ-Entropies

Exercise 14.1

Central idea. Mirror the proof of Theorem 14.6 with η nonincreasing: the Bregman divergence $D_\Phi(\eta(y), \eta(x))$ is bounded by $\frac{1}{2}(x-y)^2\varphi(\tilde z)$ where $\varphi = \eta'^2\Phi''(\eta)$ is nonincreasing (as $|\eta'|$, $|g'|$ are, for $g = \Phi' \circ \eta$ convex), evaluated at $\tilde z \le \min(x,y)$. Sub-additivity (Theorem 14.1) plus the variational formula (Lemma 14.4) give the V-bound; symmetrization gives the V_+/V_- bounds. The stated hypothesis is insufficient as written: $\tilde Z$ must also lie below Z (counterexample below).

Solution. Let $\Phi \in \mathcal{C}$ (so $\Phi'' > 0$, Φ' concave by Lemma 14.5) and let $\eta \ge 0$ be nonincreasing, differentiable and convex on I; hence $\eta' \le 0$ is nondecreasing and $|\eta'|$ is nonincreasing. Put $g = \Phi' \circ \eta$; since Φ' is nondecreasing and η nonincreasing, g is nonincreasing.

Remark on the hypothesis. The condition "$\tilde{Z} \le \min_i Z_i$" does not imply the conclusion. Counterexample: $n=1$, $\mathbb{P}\{X_1=0\}=\mathbb{P}\{X_1=10\}=1/2$, $Z=X_1$, $Z_1\equiv 5$, $\tilde{Z}\equiv 5$, $\eta(z)=e^{-z}$, $\Phi(x)=x\log x$. Then $\Phi\in\mathcal{C}$, η satisfies the hypotheses, $g(z)=1-z$ is convex, and $\tilde{Z}\le\min_i Z_i$ holds, but $H_\Phi(\eta(Z))\approx 0.346$ while $\frac{1}{2}\mathbb{E}[V\eta'(\tilde{Z})^2\Phi''(\eta(\tilde{Z}))]=\frac{1}{2}\cdot 25\cdot e^{-5}\approx 0.084$. We therefore prove the theorem under the corrected hypothesis that $\tilde{Z}$ is an a.s. lower bound for Z and all Z_i (resp. Z_i').

Pointwise lemma. Assume g convex (hence g' nondecreasing, $g'\le 0$, so $|g'|$ is nonincreasing). Then $\varphi(t):=\eta'(t)^2\Phi''(\eta(t))=|\eta'(t)|\cdot|g'(t)|$ is nonincreasing, and for $x,y\in I$, $\tilde{z}\le\min(x,y)$,

$$D_\Phi(\eta(y),\eta(x)):=\Phi(\eta(y))-\Phi(\eta(x))-(\eta(y)-\eta(x))\Phi'(\eta(x))\le\tfrac{1}{2}(x-y)^2\varphi(\tilde{z}).$$

Indeed, for $x<y$, with $h(t)=\Phi(\eta(y))-\Phi(\eta(t))-(\eta(y)-\eta(t))g(t)$, $h'(t)=-g'(t)(\eta(y)-\eta(t))$, so $D_\Phi=h(x)=\int_x^y|g'(t)|(\eta(t)-\eta(y))\,dt$; for $t\in[x,y]$, $|g'(t)|\le|g'(\tilde{z})|$ and $\eta(t)-\eta(y)=\int_t^y|\eta'(s)|ds\le(y-t)|\eta'(\tilde{z})|$, giving $\le\frac{1}{2}(y-x)^2\varphi(\tilde{z})$. For $y<x$, $D_\Phi=\int_y^x(\eta(y)-\eta(t))\Phi''(\eta(t))|\eta'(t)|dt$ (via $D_\Phi(a,b)=\int_b^a(a-s)\Phi''(s)ds$ and $s=\eta(t)$); for $t\in[y,x]$, $\eta(y)-\eta(t)\le(t-y)|\eta'(\tilde{z})|$, $|\eta'(t)|\le|\eta'(\tilde{z})|$, $\Phi''(\eta(t))\le\Phi''(\eta(\tilde{z}))$ (as Φ'' is nonincreasing), giving $\le\frac{1}{2}(x-y)^2\varphi(\tilde{z})$.

First inequality. By sub-additivity (Theorem 14.1) and Lemma 14.4, $H_\Phi(\eta(Z))\le\sum_i\mathbb{E}[H_\Phi^{(i)}(\eta(Z))]$ with $H_\Phi^{(i)}(\eta(Z))\le\mathbb{E}^{(i)}[D_\Phi(\eta(Z),\eta(Z_i))]$ (variational formula at $u=\eta(Z_i)$). The lemma with $(x,y)=(Z_i,Z)$, $\tilde{z}=\tilde{Z}$ gives $\le\frac{1}{2}(Z-Z_i)^2\varphi(\tilde{Z})$; summing yields $\frac{1}{2}\mathbb{E}[V\eta'(\tilde{Z})^2\Phi''(\eta(\tilde{Z}))]$.

For the ψ-bounds, $\psi(x)=(\Phi(x)-\Phi(0))/x$ satisfies $\psi'(x)=[x\Phi'(x)-\Phi(x)+\Phi(0)]/x^2\ge 0$ (by convexity of Φ), so $\psi\circ\eta$ is nonincreasing; if convex, $|(\psi\circ\eta)'|$ is nonincreasing and $\chi(t):=\eta'(t)^2\psi'(\eta(t))=|\eta'(t)|\cdot|(\psi\circ\eta)'(t)|$ is nonincreasing. For $x<y$, $\tilde{z}\le x$,

$$(\eta(x)-\eta(y))(\psi(\eta(x))-\psi(\eta(y)))\le(y-x)^2\eta'(\tilde{z})^2\psi'(\eta(\tilde{z})),$$

since $\eta(x)-\eta(y)=\int_x^y|\eta'|\le(y-x)|\eta'(\tilde{z})|$ and similarly for $\psi\circ\eta$.

Second inequality. By Lemma 14.4 and symmetry, $H_\Phi^{(i)}(\eta(Z))\le\mathbb{E}^{(i)}[(\eta(Z_i')-\eta(Z))_+(\psi(\eta(Z_i'))-\psi(\eta(Z)))]$. If $Z_i'<Z$ this is $(\eta(Z_i')-\eta(Z))(\psi(\eta(Z_i'))-\psi(\eta(Z)))\le(Z-Z_i')^2\eta'(\tilde{Z})^2\psi'(\eta(\tilde{Z}))$ (lemma with $x=Z_i'$); if $Z_i'\ge Z$ it is 0. Hence $\le(Z-Z_i')_+^2\eta'(\tilde{Z})^2\psi'(\eta(\tilde{Z}))$, and summing gives $\mathbb{E}[V_+\eta'(\tilde{Z})^2\psi'(\eta(\tilde{Z}))]$.

Third inequality. Unswapped, $H_\Phi^{(i)}(\eta(Z))\le\mathbb{E}^{(i)}[(\eta(Z)-\eta(Z_i'))_+(\psi(\eta(Z))-\psi(\eta(Z_i')))]$; for $Z<Z_i'$ this is $\le(Z_i'-Z)^2\eta'(Z)^2\psi'(\eta(Z))$ (lemma at the left endpoint $x=Z$), i.e. $(Z-Z_i')_-^2\eta'(Z)^2\psi'(\eta(Z))$, whence $\mathbb{E}[V_-\eta'(Z)^2\psi'(\eta(Z))]$. ■

Exercise 14.2

Central idea. All six are the convexity of $(u,v)\mapsto\Phi''(u)v^2$ in disguise: A^Φ and B^Φ are averages of C^Φ along lines $(u+tv,v)$ (giving i) $\iff$ iii) $\iff$ ii) by integration and a $h\to 0$ scaling limit), the Hessian of C^Φ gives v), $(-1/\Phi'')''$ computes to the same inequality (giving iv) $\iff$ v)), and the Jensen gap J_t is a convex combination of A^Φ-terms (giving i) $\iff$ vi)). Φ is assumed C^4 on an open interval.

Solution. Write $C(u,v)=\Phi''(u)v^2$. Its Hessian is

$$\nabla^2 C=\begin{pmatrix}\Phi''''(u)v^2 & 2\Phi'''(u)v\\ 2\Phi'''(u)v & 2\Phi''(u)\end{pmatrix}.$$

Hence (for C^2 C) iii) holds iff this is PSD everywhere, i.e. $\Phi''(u)\ge 0$, $\Phi''''(u)\ge 0$ and $\det=2v^2(\Phi''(u)\Phi''''(u)-2\Phi'''(u)^2)\ge 0$ for all u,v.

iii) $\iff$ v): If C is convex and $\Phi''\not\equiv 0$, Φ is affine. Otherwise $\Phi''(u_0)>0$ for some u_0; let J be the maximal open interval about u_0 with $\Phi''>0$. On J, $\psi:=-1/\Phi''$ satisfies $\psi''=(\Phi''\Phi''''-2\Phi'''^2)/(\Phi'')^3\ge 0$ (by $\det\ge 0$), so ψ is convex on J. If J were not the whole domain it would have a finite endpoint u_1 with $\Phi''(u_1)=0$, whence $\psi\to-\infty$ there — impossible for a convex function (tangent lower bound at any interior point). Thus $\Phi''>0$ everywhere and $\Phi''\Phi''''\ge 2\Phi'''^2$, i.e. v). Conversely, if Φ is affine, $C\equiv 0$ is convex; if

$\Phi'' > 0$ and $\Phi''\Phi'''' \geq 2\Phi'''^2$, then $\Phi'''' \geq 2\Phi'''^2/\Phi'' \geq 0$, so $\nabla^2 C$ has nonnegative diagonal and determinant, hence is PSD and C is convex.

iv) $\iff$ v): If Φ is affine both hold. Else $\Phi'' > 0$ and $(-1/\Phi'')'' = (\Phi''\Phi'''' - 2\Phi'''^2)/(\Phi'')^3$; since $(\Phi'')^3 > 0$, $-1/\Phi''$ is convex iff $\Phi''\Phi'''' \geq 2\Phi'''^2$.

i) $\iff$ iii): $A^\Phi(u,v) = \Phi(u+v) - \Phi(u) - \Phi'(u)v = \int_0^1 (1-t)\Phi''(u+tv)v^2 dt = \int_0^1 (1-t)C(u+tv,v)dt$ (Taylor with integral remainder). If C is convex, $(u,v) \mapsto C(u+tv,v)$ is convex for each t (linear precomposition), and A^Φ, an integral thereof against $(1-t) \geq 0$, is convex. Conversely, $(u,v) \mapsto A^\Phi(u,hv)/h^2$ is convex and tends to $\frac{1}{2}\Phi''(u)v^2 = \frac{1}{2}C(u,v)$ as $h \to 0$; a pointwise limit of convex functions is convex.

ii) $\iff$ iii): $B^\Phi(u,v) = (\Phi'(u+v) - \Phi'(u))v = \int_0^1 \Phi''(u+tv)v^2 dt = \int_0^1 C(u+tv,v)dt$, so iii)$\Rightarrow$ii) as above; and $B^\Phi(u,hv)/h^2 \to \Phi''(u)v^2 = C(u,v)$, giving ii)$\Rightarrow$iii).

i) $\iff$ vi): Put $J_t(a,b) = t\Phi(a) + (1-t)\Phi(b) - \Phi(ta+(1-t)b)$ and $u = ta+(1-t)b$. Since $t(a-u)+(1-t)(b-u) = 0$,

$$J_t(a,b) = t\, A^\Phi(u, (1-t)(a-b)) + (1-t)\, A^\Phi(u, -t(a-b)),$$

so i)$\Rightarrow$vi) (convexity preserved under linear precomposition and positive combinations; $J_0 = J_1 \equiv 0$). Conversely, $J_t(a,b)/t \to \Phi(a) - \Phi(b) - \Phi'(b)(a-b) = A^\Phi(b, a-b)$ as $t \to 0^+$; if each J_t is convex so is the limit, and $(a,b) \mapsto (b, a-b)$ is a linear isomorphism, giving i). ■

Exercise 14.3

Central idea. Differentiate θ: with $m(r) = \|f\|_r^2 = (\mathbb{E}[f^r])^{2/r}$, one gets $\theta'(r) \geq 0 \iff \mathbb{E}[f^2] - m(r) \geq \frac{2-r}{r} m(r) H_r$ where $H_r = D(\mu_r \| \mathbb{P}) \geq 0$ is the KL divergence for $d\mu_r = (f^r/\mathbb{E}[f^r])d\mathbb{P}$ (this is exactly $m'(r) = m(r) \cdot 2H_r/r^2$). The stronger exponential form follows from $1 + x \leq e^x$ and Jensen, collapsing to $\mathbb{E}[f^2] \geq \mathbb{E}[f^r] \exp((2-r)\mathbb{E}_{\mu_r} \log f)$.

Solution. Assume $0 < \mathbb{E}[f^2] < \infty$ (else $\theta \equiv \infty$); then $\mathbb{E}[f^r] < \infty$ for $r < 2$. Put $A = \mathbb{E}[f^2]$, $m(r) = (\mathbb{E}[f^r])^{2/r}$. Then

$$\theta'(r) = \frac{2\big[\, 2(A - m(r)) - r(2-r)m'(r) \,\big]}{(2-r)^2},$$

so $\theta'(r) \geq 0 \iff 2(A - m(r)) \geq r(2-r)m'(r)$.

Let $d\mu_r = (f^r/\mathbb{E}[f^r])\, d\mathbb{P}$ (if $\mathbb{E}[f^r] = 0$ there is nothing to prove) and $H_r = \mathbb{E}_{\mu_r}[\log(f^r/\mathbb{E}[f^r])] \geq 0$, the KL divergence. Since $\log m(r) = \frac{2}{r} \log \mathbb{E}[f^r]$,

$$\frac{d}{dr} \log m(r) = \frac{2}{r} \left(\frac{\mathbb{E}[f^r \log f]}{\mathbb{E}[f^r]} - \frac{1}{r} \log \mathbb{E}[f^r] \right) = \frac{2H_r}{r^2},$$

i.e. $m'(r) = m(r) \cdot 2H_r/r^2$. Thus we need $A - m(r) \geq \frac{2-r}{r} m(r) H_r$.

Since $1 + x \leq e^x$, it suffices that $A \geq m(r) \exp(\frac{2-r}{r} H_r)$. Now

$$\log\Big(m(r) e^{\frac{2-r}{r} H_r} \Big) = \frac{2}{r} \log \mathbb{E}[f^r] + \frac{2-r}{r} H_r = \log \mathbb{E}[f^r] + (2-r)\mathbb{E}_{\mu_r}[\log f],$$

using $H_r = r\, \mathbb{E}_{\mu_r}[\log f] - \log \mathbb{E}[f^r]$. Hence $m(r) \exp(\frac{2-r}{r} H_r) = \mathbb{E}[f^r] \exp((2-r)\mathbb{E}_{\mu_r}[\log f])$, and by Jensen,

$$\exp\left((2-r)\mathbb{E}_{\mu_r}[\log f]\right) = \exp\left(\mathbb{E}_{\mu_r}[\log f^{\,2-r}]\right) \leq \mathbb{E}_{\mu_r}[f^{2-r}] = \frac{\mathbb{E}[f^2]}{\mathbb{E}[f^r]},$$

so $m(r) \exp(\frac{2-r}{r} H_r) \leq A$, as required. Thus $\theta'(r) \geq 0$ on $[1,2)$. ■

Exercise 14.4

Central idea. For $r = 1$, $\rho \equiv 4$. For $r \in (1, 2]$, write $\rho(z) = g(\sqrt{z})$ with $g(t) = ((1+t)^r + (1-t)^r)^{2/r}$; then $\rho''(z) = [t\,g''(t) - g'(t)]/(4t^3)$, so convexity is $t\,g''(t) \geq g'(t)$. Expanding with $u = 1+t$, $v = 1-t$ collapses this to $(v^{2r-2} - u^{2r-2}) + 4(r-1)t\,u^{r-2}v^{r-2} \geq 0$, which after $a = u/v$, $p = r-1$ becomes $p(a^2-1) \geq a^{p+1} - a^{1-p}$, proved by two differentiations.

Solution. For $r = 1$, $\rho(z) = ((1+\sqrt{z}) + (1-\sqrt{z}))^2 \equiv 4$. Let $r \in (1, 2]$ and $g(t) = A(t)^{2/r}$, $A(t) = (1+t)^r + (1-t)^r$, $t \in [0, 1]$, so $\rho(z) = g(\sqrt{z})$. Put $B(t) = (1+t)^{r-1} - (1-t)^{r-1}$, $C(t) = (1+t)^{r-2} + (1-t)^{r-2}$. Then $g'(t) = 2A^{2/r-1}B$ and, with $A' = rB$,

$$g''(t) = 2A^{2/r-2}[(2-r)B^2 + (r-1)AC].$$

On $(0, 1)$, $\rho'(z) = g'(\sqrt{z})/(2\sqrt{z})$; at 0, L'Hôpital gives $\rho'_+(0) = \lim_{t\to 0^+} g'(t)/(2t) = 2^{2/r}(r-1)$ (since $B(t)/t \to 2(r-1)$); at 1, ρ is left-differentiable with derivative $g'(1)/2$ by the MVT. Hence ρ is differentiable on $[0, 1]$.

Moreover, with $t = \sqrt{z}$, $\rho''(z) = [t\,g''(t) - g'(t)]/(4t^3)$ on $(0, 1)$, so convexity follows from $t\,g''(t) - g'(t) = 2A^{2/r-2}E_0 \geq 0$ where $E_0 := t(2-r)B^2 + t(r-1)AC - AB$. With $u = 1+t$, $v = 1-t$:

$$(2-r)B^2 + (r-1)AC = (u^{2r-2} + v^{2r-2}) - 2(2-r)u^{r-1}v^{r-1} + (r-1)u^{r-2}v^{r-2}(u^2+v^2),$$
$$AB = u^{2r-1} - v^{2r-1} - 2t\,u^{r-1}v^{r-1},$$

whence, using $t - u = -1$, $t + v = 1$, $(u+v)^2 = 4$,

$$E_0 = (v^{2r-2} - u^{2r-2}) + 4(r-1)t\,u^{r-2}v^{r-2}.$$

Put $p = r - 1 \in (0, 1]$, $a = u/v \geq 1$ (so $v = 2/(a+1)$). Dividing by $u^{p-1}v^{p-1} > 0$, $E_0 \geq 0 \iff 2p(a-1) \geq v(a^{p+1} - a^{1-p}) \iff p(a^2-1) \geq a^{p+1} - a^{1-p}$. For $L(a) = p(a^2-1) - a^{p+1} + a^{1-p}$: $L(1) = L'(1) = 0$ and $L''(a) = p[2 - (p+1)a^{p-1} - (1-p)a^{-p-1}] \geq 0$ on $[1, \infty)$, because with $b = 1/a \in (0, 1]$, $M(b) := (p+1)b^{1-p} + (1-p)b^{p+1}$ satisfies $M(1) = 2$ and $M'(b) = (p+1)(1-p)(b^{-p} + b^p) \geq 0$. Thus $L' \geq 0$, $L \geq 0$, so $E_0 \geq 0$, $\rho'' \geq 0$ on $(0, 1)$, and ρ (continuous on $[0, 1]$) is convex. ■

Exercise 14.5

Central idea. Mimic the proof of the Gaussian log-Sobolev inequality (Theorem 5.4): approximate the standard Gaussian vector by normalized Rademacher sums, apply the Bernoulli Φ-Sobolev inequality (Theorem 14.8) to the approximating functions on the hypercube, and pass to the limit via the CLT. The Dirichlet form $\mathcal{E}$ converges to $\mathbb{E}\|\nabla f\|^2$ while the Φ-entropy converges to $H_\Phi(f^r)$.

Solution. Let $r \in [1, 2)$ and $\Phi(z) = z^{2/r}$. For $r = 1$, $H_\Phi(f) = \mathrm{Var}(f)$ and the claim is the Gaussian Poincaré inequality, so assume $1 < r < 2$. By a standard density argument it suffices to prove the inequality for $f : \mathbb{R}^n \to [0, \infty)$ twice continuously differentiable with bounded support (hence all derivatives bounded).

Let $(\varepsilon_{ij})_{i\leq n,\, j\geq 1}$ be independent Rademacher variables and put, for $m \geq 1$,

$$S_i^{(m)} = \frac{1}{\sqrt{m}} \sum_{j=1}^{m} \varepsilon_{ij}, \qquad S^{(m)} = (S_1^{(m)}, \dots, S_n^{(m)}) \in \mathbb{R}^n.$$

Define $g_m : \{-1, 1\}^{nm} \to [0, \infty)$ by $g_m(\varepsilon) = f(S^{(m)})$. By the multivariate CLT, $S^{(m)} \Rightarrow X$ where X is a standard Gaussian vector in $\mathbb{R}^n$, and since f is bounded with bounded support,

$$\mathbb{E}[g_m(\varepsilon)^2] \to \mathbb{E}[f(X)^2], \qquad \mathbb{E}[g_m(\varepsilon)^r] \to \mathbb{E}[f(X)^r],$$

hence with $H_\Phi(h^r) = \mathbb{E}[h^2] - (\mathbb{E}[h^r])^{2/r}$,

$$H_\Phi(g_m^r) \longrightarrow H_\Phi(f^r). \tag{1}$$

Apply Theorem 14.8 to g_m on $\{-1,1\}^{nm}$:

$$H_\Phi(g_m^r) \le (2-r)\,\mathcal{E}(g_m), \qquad \mathcal{E}(g_m) = \frac{1}{4}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbb{E}\big[(g_m(\varepsilon)-g_m(\varepsilon^{(ij)}))^2\big],$$

where $\varepsilon^{(ij)}$ flips ε_{ij}. Flipping ε_{ij} replaces $S_i^{(m)}$ by $S_i^{(m)} - 2\varepsilon_{ij}/\sqrt{m}$. By Taylor's formula with integral remainder, for $f \in C_b^2$,

$$f(S^{(m)}) - f\big(S^{(m)} - \tfrac{2\varepsilon_{ij}}{\sqrt{m}}e_i\big) = \tfrac{2\varepsilon_{ij}}{\sqrt{m}}\,\partial_i f(S^{(m)}) + R_{ij}, \qquad |R_{ij}| \le \tfrac{C}{m}$$

with C depending only on $\|f''\|_\infty$. Hence

$$(g_m(\varepsilon) - g_m(\varepsilon^{(ij)}))^2 = \tfrac{4}{m}(\partial_i f(S^{(m)}))^2 + O(m^{-3/2}),$$

and therefore

$$\begin{aligned}\mathcal{E}(g_m) &= \frac{1}{4}\sum_{i=1}^{n}\sum_{j=1}^{m}\big[\tfrac{4}{m}\mathbb{E}\big[(\partial_i f(S^{(m)}))^2\big] + O(m^{-3/2})\big] \\ &= \sum_{i=1}^{n}\mathbb{E}\big[(\partial_i f(S^{(m)}))^2\big] + O(m^{-1/2}).\end{aligned}$$

Since $\partial_i f$ is bounded with bounded support, the CLT gives $\mathbb{E}[(\partial_i f(S^{(m)}))^2] \to \mathbb{E}[(\partial_i f(X))^2]$, whence

$$\mathcal{E}(g_m) \longrightarrow \sum_{i=1}^{n}\mathbb{E}\big[(\partial_i f(X))^2\big] = \mathbb{E}\big[\|\nabla f(X)\|^2\big]. \tag{2}$$

Combining Theorem 14.8 with (1)–(2) and letting $m \to \infty$ yields

$$H_\Phi(f^r) \le (2-r)\,\mathbb{E}\big[\|\nabla f(X)\|^2\big],$$

as claimed. ■

Exercise 14.6

Central idea. The integrand on the right is the Bregman divergence $A^\Phi(u,v) = \Phi(u+v) - \Phi(u) - \Phi'(u)v$ evaluated at $u = f(x)$, $v = Df(x)$. For a single Bernoulli variable, convexity of Φ gives the exact two-point bound $H_\Phi(y_B) \le p\,A^\Phi(y_0, y_1 - y_0)$. Since $\Phi(x) = x^{2/r}$ belongs to the class $\mathcal{C}$, H_Φ tensorizes (Theorem 14.1); applying the two-point bound coordinatewise to $\mathrm{Bin}(m,\mu/m)$ *before* letting $m \to \infty$ keeps the sharp Bregman form, and the Poisson limit follows for bounded f. Unbounded f is handled by truncation $f \wedge M$, using that the Bregman integrand can only decrease under truncation.

Solution. Let $\mu = \mathbb{E}X$. If $\mu = 0$ then $X \equiv 0$ and both sides vanish, so assume $\mu > 0$. Throughout, $\Phi(x) = x^{2/r}$ on $[0,\infty)$ with $2/r \in (1,2)$, so Φ is C^1 on $[0,\infty)$ with $\Phi'(x) = \frac{2}{r}x^{2/r-1}$ (and $\Phi'(0) = 0$), convex, and C^2 on $(0,\infty)$. We read $H_\Phi(f(X)) = \mathbb{E}\Phi(f(X)) - \Phi(\mathbb{E}f(X))$ for $\mathbb{E}f(X) < \infty$ (with value $+\infty$ if $\mathbb{E}\Phi(f(X)) = \infty$); this is the setting in which the left side is defined.

Rewriting the right-hand side. With $A^\Phi(u,v) = \Phi(u+v) - \Phi(u) - \Phi'(u)v$,

$$D(\Phi\circ f)(x) - \tfrac{2}{r}f(x)^{2/r-1}Df(x) = A^\Phi(f(x), Df(x)),$$

so the claim is $H_\Phi(f(X)) \le \mu\,\mathbb{E}[A^\Phi(f(X), Df(X))]$. By convexity $A^\Phi \ge 0$, so the right side is a well-defined element of $[0,\infty]$.

$\Phi \in \mathcal{C}$. Φ is convex, and $1/\Phi''(x) = c\,x^{2-2/r}$ with $c > 0$ and $2 - 2/r \in (0,1)$ is concave on $(0,\infty)$, so $-1/\Phi''$ is convex. Hence the tensorization inequality of Theorem 14.1 applies to H_Φ.

Step 1 (two-point bound). Let $B \sim \text{Bernoulli}(p)$, $y_0, y_1 \geq 0$, $\Delta = y_1 - y_0$, and y_B the variable equal to y_1 if $B = 1$ and y_0 if $B = 0$. Then $\mathbb{E}y_B = y_0 + p\Delta$, and convexity gives $\Phi(y_0 + p\Delta) \geq \Phi(y_0) + p\Phi'(y_0)\Delta$. Therefore

$$H_\Phi(y_B) = p\Phi(y_1) + (1-p)\Phi(y_0) - \Phi(y_0 + p\Delta) \leq p\big[\Phi(y_1) - \Phi(y_0) - \Phi'(y_0)\Delta\big] = p\, A^\Phi(y_0, \Delta).$$

Step 2 (binomial approximation, bounded f). Assume $0 \leq f \leq K$. For $m > \mu$ let $B_1, \dots, B_m$ be i.i.d. Bernoulli(μ/m), $X_m = \sum_{j \leq m} B_j$ and $S_{-j} = X_m - B_j \sim \text{Bin}(m-1, \mu/m)$. Conditionally on $(B_k)_{k \neq j}$, $f(X_m) = f(S_{-j} + B_j)$ is a two-point variable with $y_0 = f(S_{-j})$, $y_1 = f(S_{-j} + 1)$, so by Step 1 the conditional Φ-entropy in coordinate j is at most $\frac{\mu}{m} A^\Phi(f(S_{-j}), Df(S_{-j}))$. Theorem 14.1 then gives

$$H_\Phi(f(X_m)) \leq \sum_{j=1}^{m} \frac{\mu}{m}\, \mathbb{E}\big[A^\Phi(f(S_{-j}), Df(S_{-j}))\big] = \mu\, \mathbb{E}\big[A^\Phi(f(X'_{m-1}), Df(X'_{m-1}))\big], \qquad X'_{m-1} \sim \text{Bin}(m-1, \mu/m).$$

Step 3 (the limit, bounded f). For each fixed k, $\mathbb{P}\{X_m = k\} \to e^{-\mu}\mu^k/k!$ and $\mathbb{P}\{X'_{m-1} = k\} \to e^{-\mu}\mu^k/k!$; since all are probability mass functions on $\mathbb{N}$, Scheffé's lemma gives convergence of both laws to Poisson(μ) in total variation. The functions f, $\Phi \circ f$ and $x \mapsto A^\Phi(f(x), Df(x))$ are bounded (by K, $\Phi(K)$ and $2\Phi(K) + \Phi'(K)K$), so their expectations converge; Φ is continuous, so $\Phi(\mathbb{E}f(X_m)) \to \Phi(\mathbb{E}f(X))$. Letting $m \to \infty$ in Step 2,

$$H_\Phi(f(X)) \leq \mu\, \mathbb{E}\big[A^\Phi(f(X), Df(X))\big] \qquad (f \text{ bounded}).$$

Step 4 (unbounded f). Let $f_M = f \wedge M$. We claim that pointwise, for $u, w \geq 0$,

$$A^\Phi(u \wedge M,\ w \wedge M - u \wedge M) \leq A^\Phi(u,\ w - u). \tag{$*$}$$

Write $D_\Phi(w, u) = A^\Phi(u, w-u) = \Phi(w) - \Phi(u) - \Phi'(u)(w-u)$. For fixed u, $\partial_w D_\Phi(w, u) = \Phi'(w) - \Phi'(u) \geq 0$ for $w \geq u$, so $D_\Phi(\cdot, u)$ is nondecreasing on $[u, \infty)$. For fixed w and $u > 0$, $\partial_u D_\Phi(w, u) = \Phi''(u)(u - w) \geq 0$ for $u \geq w$, so $D_\Phi(w, \cdot)$ is nondecreasing on $[w, \infty)$ (continuity at $u = 0$ covers $w = 0$). Now: if $u, w \leq M$ the two sides of $(*)$ are equal; if $u \leq M < w$, the left side is $D_\Phi(M, u) \leq D_\Phi(w, u)$ by the first monotonicity; if $w \leq M < u$, it is $D_\Phi(w, M) \leq D_\Phi(w, u)$ by the second; if $u, w > M$, it is $0 \leq D_\Phi(w, u)$. This proves $(*)$.

Since $Df_M(x) = f_M(x+1) - f_M(x)$, Step 3 applied to the bounded f_M and $(*)$ give

$$H_\Phi(f_M(X)) \leq \mu\, \mathbb{E}\big[A^\Phi(f_M(X), Df_M(X))\big] \leq \mu\, \mathbb{E}\big[A^\Phi(f(X), Df(X))\big].$$

As $M \uparrow \infty$, monotone convergence gives $\mathbb{E}f_M(X) \uparrow \mathbb{E}f(X) < \infty$ and $\mathbb{E}\Phi(f_M(X)) \uparrow \mathbb{E}\Phi(f(X)) \in [0, \infty]$; by continuity of Φ, $\Phi(\mathbb{E}f_M(X)) \to \Phi(\mathbb{E}f(X))$. Hence $H_\Phi(f_M(X)) \to H_\Phi(f(X))$ (including the case $H_\Phi(f(X)) = \infty$, in which the right side is then infinite as well), and

$$H_\Phi(f(X)) \leq (\mathbb{E}X)\, \mathbb{E}\Big[D(\Phi \circ f)(X) - \tfrac{2}{r} f(X)^{2/r-1} Df(X)\Big].$$

■

Remarks. (i) The argument uses only that $\Phi \in \mathcal{C}$ is C^1 and convex; with $\Phi(x) = x^2$ it gives the Poisson Poincaré inequality ($A^{x^2}(u, v) = v^2$, Exercise 3.21), and with $\Phi(x) = x \log x$ the Poisson modified log-Sobolev inequality (Exercise 6.12). (ii) The bound is at least as sharp as $H_\Phi(f(X)) \leq \mu\, \mathbb{E}[(f(X+1)^{1/r} - f(X)^{1/r})^2]$, which is what the binomial limit of Theorem 14.9 applied to $f^{1/r}$ gives: with $a = u^{1/r}$, $b = w^{1/r}$, the inequality $A^\Phi(u, w-u) \leq (b-a)^2$ is equivalent (for $a > 0$, $\tau = b/a$) to $\tau - 1 \leq (\tau^r - 1)/r$, i.e. convexity of τ^r; equality holds when $u = 0$ or $u = w$, so the comparison is "$\leq$", not strict in general. (iii) Numerical check (Python): the inequality holds in all 45 cases $r \in \{1.2, 1.5, 1.8\}$, $\mu \in \{0.5, 2, 5\}$, $f \in \{x+1, (x+1)^2, \mathbb{1}_{\{x \geq 1\}}, \sqrt{x+1}, e^{x/3}\}$ (largest ratio LHS/RHS ≈ 0.998), and $(*)$ and Remark (ii) had no violations in $2 \cdot 10^5$ random draws.

Note on the earlier version. A previous version of this file was marked incomplete: it applied Theorem 14.9 to $f^{1/r}$ in the binomial limit, which yields only the weaker bound of Remark (ii), and it described the Bregman form as "strictly smaller", which is false in general (equality for $f = \mathbb{1}_{\{x \geq 1\}}$). Applying the exact two-point bound per coordinate before the limit, as above, closes the gap.

Exercise 14.7

Central idea. Prove a hypercontractive estimate for the biased cube: the noise operator T_σ, which multiplies every centered coordinate function by σ, maps L^2 contractively into L^r for $\sigma = 1/(10C_{r,p})$ when $r \geq \log(1/p)$. Since $T_\sigma Z = \sigma Z$, this is exactly $\|Z\|_r \leq 10C_{r,p}\|Z\|_2$. By Minkowski's inequality the contraction tensorizes, so it reduces to a two-point inequality $\mathbb{E}|1+tX|^r \leq (1+Dpq\,t^2)^{r/2}$, proved by a second-order Taylor bound and a two-case analysis. The range $2 < r \leq \log(1/p)$ follows by Hölder interpolation between L^2 and $L^{\log(1/p)}$ at the cost of a factor e. (This hypercontractivity plays the role of the Φ-Sobolev input of Theorem 14.9; Theorem 14.9 itself is not invoked.)

Solution. The coefficients are $\alpha_i \in \mathbb{R}$; the "$\mathbb{R}^n$" in the statement is a typo. Write $q = 1-p \in (1/2, 1)$, $L = \log(1/p) > \log 2$, and let X denote a generic X_i. Then $\mathbb{E}X = 0$, $\mathbb{E}X^2 = pq$, $|X| \leq 1$, and $\mathbb{E}Z^2 = pq\sum_i \alpha_i^2$. We prove - $\|Z\|_r \leq 10\,C_{r,p}\|Z\|_2$ for $r > 2$, $r \geq L$ (Steps 1–2); - $\|Z\|_r \leq 10e\,C_{r,p}\|Z\|_2$ for $2 < r \leq L$ (Step 3).

So $\kappa = 10e$ works. (At $r = L$ the two formulas for $C_{r,p}$ differ by the factor e, namely $(1/p)^{1/2-1/L} = e^{-1}p^{-1/2}$ versus $p^{-1/2}$; Step 2 gives $10p^{-1/2}$, which equals $10e\cdot e^{-1}p^{-1/2}$ and is at most $10e\,p^{-1/2}$, so it is within $\kappa C_{L,p}$ under either formula.)

Step 1 (reduction to a two-point inequality). All functions below are functions of $X = (X_1, \dots, X_n)$ on a finite product space. For $\sigma \in (0,1]$ and a coordinate i, let $T_\sigma^{(i)} g = \mathbb{E}_i g + \sigma(g - \mathbb{E}_i g)$, where $\mathbb{E}_i$ integrates out X_i, and let $T_\sigma = T_\sigma^{(1)} \cdots T_\sigma^{(n)}$ (the factors act on different coordinates and commute). For $j \neq i$, $\mathbb{E}_i(\alpha_j X_j) = \alpha_j X_j$, so $T_\sigma^{(i)}(\alpha_j X_j) = \alpha_j X_j$; and $\mathbb{E}_i(\alpha_i X_i) = 0$, so $T_\sigma^{(i)}(\alpha_i X_i) = \sigma\alpha_i X_i$. By linearity, $T_\sigma Z = \sigma Z$.

Tensorization. Fix $r \geq 2$ and suppose that, on one coordinate, $\|T_\sigma^{(1)} g\|_r \leq \|g\|_2$ for every $g = g(X_1)$. We show $\|T_\sigma f\|_r \leq \|f\|_2$ for every f, by induction on n. Write $x = (x_1, x')$ and $T_\sigma = T_\sigma^{(1)} T'$ with T' acting on x', and put $h = T'f$, so $h(x_1, \cdot) = T'[f(x_1, \cdot)]$. Applying the one-coordinate bound for each fixed x', then Minkowski's inequality in $L^{r/2}(x')$ (valid as $r/2 \geq 1$), then the induction hypothesis:

$$\|T_\sigma f\|_r^r = \mathbb{E}_{x'}\mathbb{E}_{x_1}|T_\sigma^{(1)}h|^r \leq \mathbb{E}_{x'}\big(\mathbb{E}_{x_1}h^2\big)^{r/2} = \big\|\mathbb{E}_{x_1}h^2\big\|_{L^{r/2}(x')}^{r/2} \leq \Big(\mathbb{E}_{x_1}\|h(x_1,\cdot)\|_{L^r(x')}^2\Big)^{r/2} \leq \Big(\mathbb{E}_{x_1}\|f(x_1,\cdot)\|_{L^2(x')}^2\Big)^{r/2} = \|f\|_2^r.$$

Applied to $f = Z$: $\sigma\|Z\|_r = \|T_\sigma Z\|_r \leq \|Z\|_2$.

The one-coordinate statement. Every $g = g(X_1)$ is of the form $a + bX_1$ (two points), and $T_\sigma^{(1)} g = a + \sigma b X_1$. The required bound is $\mathbb{E}|a + \sigma bX|^r \leq (a^2 + b^2pq)^{r/2}$. For $a \neq 0$ divide by $|a|^r$ and set $t = \sigma b/a$; it becomes

$$F(t) := \mathbb{E}|1+tX|^r \leq (1 + Dpq\,t^2)^{r/2} \quad (t \in \mathbb{R}), \qquad D = \sigma^{-2}. \tag{$\star$}$$

For $a = 0$, divide $(\star)$ by $|t|^r$ and let $|t| \to \infty$: $\mathbb{E}|X|^r \leq (Dpq)^{r/2}$, which is the required bound $\sigma^r\mathbb{E}|bX|^r \leq (b^2pq)^{r/2}$. So $(\star)$ for all t implies the one-coordinate contraction.

Step 2 (proof of $(\star)$ for $r > 2$, $r \geq L$, with $D = 100r/(pL)$). This choice gives $\sigma = \sqrt{pL/(100r)} = 1/(10C_{r,p})$, with $C_{r,p} = \sqrt{r/(pL)}$ the second formula. Put $c := Dpq = 100qr/L \geq 50$ (as $q \geq 1/2$, $r \geq L$).

A bound on F. The map $x \mapsto |x|^r$ is C^2 with second derivative $r(r-1)|x|^{r-2}$. Taylor's formula with Lagrange remainder, and $|\xi| \leq \max(1, |1+u|)$ for ξ between 1 and $1+u$, give

$$|1+u|^r \leq 1 + ru + \tfrac{r(r-1)}{2}u^2\max(1, |1+u|)^{r-2}.$$

Take $u = tX$. Then $\mathbb{E}[ru] = 0$, $\mathbb{E}X^2 = pq$, and $\max(1, |1+tX|) \leq 1 + |t|$ since $|X| \leq 1$. With $r(r-1) \leq r^2$,

$$F(t) \leq 1 + \tfrac{r^2}{2}pq\,t^2(1+|t|)^{r-2},$$

so it suffices to show

$$\tfrac{r^2}{2}pq\,t^2(1+|t|)^{r-2} \leq (1+ct^2)^{r/2} - 1. \tag{1}$$

Case $ct^2 \leq 1$. Since $\log(1+y) \geq y - y^2/2 \geq y/2$ on $[0,1]$, we have $(1+ct^2)^{r/2} \geq e^{rct^2/4}$. Since $e^x - 1 = 2e^{x/2}\sinh(x/2) \geq xe^{x/2}$ for $x \geq 0$, the right side of (1) is at least $\frac{rct^2}{4}e^{rct^2/8} = \frac{25qr^2t^2}{L}e^{rct^2/8}$. On the left,

$(1+|t|)^{r-2} \le e^{|t|(r-2)} \le e^s$ with $s = r|t|$. Dividing by qr^2t^2 (the case $t=0$ is trivial), (1) follows from $\frac{p}{2}e^s \le \frac{25}{L}e^{rct^2/8}$. Since $rct^2/8 = \frac{100q\,s^2}{8L} \ge \frac{100s^2}{16L}$ and $p = e^{-L}$, it suffices that

$$s - L - \frac{100s^2}{16L} \le \log\frac{50}{L}.$$

The maximum over s of $s - \frac{100s^2}{16L}$ is $\frac{16L}{100} = 0.04L$, so the left side is at most $-0.96L$. Finally $0.96L + \log(50/L) = \log 50 + (0.96L - \log L) \ge \log 50 + 1 + \log 0.96 > 0$ for every $L > 0$, since $0.96L - \log L$ is minimized at $L = 1/0.96$.

Case $ct^2 > 1$. Then $(1+ct^2)^{r/2} \ge 2^{r/2} \ge 2$, so the right side of (1) is at least $\frac{1}{2}(1+ct^2)^{r/2}$. Write $|t| = aw$ with $a = c^{-1/2} \le 1/\sqrt{50}$ and $w = \sqrt{ct^2} > 1$. Then $g(w) = (1+aw)^2/(1+w^2)$ has $g'(w) = 2(1+aw)(a-w)/(1+w^2)^2 < 0$ for $w > a$, so on $w \ge 1$ it is decreasing, and

$$\frac{(1+|t|)^2}{1+ct^2} \le g(1) = \frac{(1+a)^2}{2} \le \frac{1 + 2/\sqrt{50} + 1/50}{2} = 0.6514\ldots < 0.66.$$

Hence $(1+|t|)^{r-2} \le 0.66^{(r-2)/2}(1+ct^2)^{(r-2)/2}$. Then (1) follows from $r^2pq\,t^2\,0.66^{(r-2)/2} \le 1 + ct^2$, which is implied by $r^2pq\,0.66^{(r-2)/2} \le c = 100qr/L$, i.e.

$$r\,pL\,0.66^{(r-2)/2} \le 100.$$

This holds because $pL = p\log(1/p) \le 1/e$ and $\max_{r\ge 2} r\,0.66^{(r-2)/2} = 2.683\ldots$ (attained at $r = 2/\log(1/0.66) \approx 4.81$), so the left side is below 1.

Thus $(\star)$ holds, and Step 1 gives $\|Z\|_r \le \sigma^{-1}\|Z\|_2 = 10\,C_{r,p}\|Z\|_2$ for $r > 2$, $r \ge L$.

Step 3 (the range $2 < r \le L$*; nonempty only if* $L > 2$*).* Define $\theta \in (0,1]$ by $\frac{1}{r} = \frac{1-\theta}{2} + \frac{\theta}{L}$, i.e. $\theta = \frac{1/2-1/r}{1/2-1/L}$. Hölder's (log-convexity) inequality and Step 2 at $r = L > 2$, where $C_{L,p} = \sqrt{L/(pL)} = p^{-1/2}$, give

$$\|Z\|_r \le \|Z\|_2^{1-\theta}\|Z\|_L^{\theta} \le 10^{\theta}p^{-\theta/2}\|Z\|_2 \le 10\,p^{-\theta/2}\|Z\|_2.$$

Now, with $C_{r,p} = (1/p)^{1/2-1/r} = e^{L(1/2-1/r)}$,

$$p^{-\theta/2} = \exp\Big(L(\tfrac12-\tfrac1r)\tfrac{L}{L-2}\Big) = C_{r,p}\exp\Big(L(\tfrac12-\tfrac1r)\tfrac{2}{L-2}\Big) = C_{r,p}\exp\Big(\frac{L(r-2)}{r(L-2)}\Big) \le e\,C_{r,p},$$

because $\frac{r-2}{r} \le \frac{L-2}{L}$ for $r \le L$. Hence $\|Z\|_r \le 10e\,C_{r,p}\|Z\|_2$.

Conclusion. For all $r > 2$ and all real $\alpha_1, \ldots, \alpha_n$,

$$\mathbb{E}\big[|Z|^r\big]^{1/r} \le \kappa\,C_{r,p}\,\mathbb{E}[Z^2]^{1/2}, \qquad \kappa = 10e.$$

■

Remarks. (i) For $r \le L$ the order of $C_{r,p}$ is attained by a single summand: $\|X\|_r^r = qp^r + pq^r \ge p\,2^{-r}$ and $\|X\|_2 \le \sqrt{p}$, so $\|X\|_r/\|X\|_2 \ge \frac12 p^{1/r-1/2} = \frac12 C_{r,p}$. (ii) Numerical checks (Python): $(\star)$ with $D = 100r/(pL)$ has no violation on a grid of 192,000 points (40 values $p \in [10^{-12}, 0.49]$, 40 values $r \in [\max(2,L), 50\max(2,L)]$, $|t| \in [10^{-6}, 10^6]$ of both signs); the constants 0.6514, 2.683 and $\min_L(0.96L + \log(50/L)) = 4.87 > 0$ were recomputed; exact enumeration for $n \le 10$, $p \in \{0.3, 0.1, 10^{-2}, 10^{-4}, 10^{-8}\}$, $r \in [2.5, 100]$ and random α gives $\max \|Z\|_r/(C_{r,p}\|Z\|_2) \approx 1.00$, well below $10e$.

Erratum (previous version of this file). The earlier version applied Theorem 14.9 to $|Z|^{r/2}$ and claimed the Dirichlet-form bound $\mathcal{E}(|Z|^{r/2}) \le r^2B^2A_r^{r-2}$ ($B = \|Z\|_2$, $A_r = \|Z\|_r$), justified "by symmetry and Jensen", and from it $A_r \lesssim rB$ uniformly in p. Both are false for biased bits: the flip of a rare coordinate is weighted by its probability, so symmetry does not hold. Counterexample: $n = 1$, $\alpha_1 = 1$, $r = 3$, $p = 10^{-6}$. Then $\mathcal{E}(|X|^{3/2}) = pq\,(q^{3/2} - p^{3/2})^2 = 1.0 \cdot 10^{-6}$, whereas $r^2B^2A_3 = 9pq\,(qp^3 + pq^3)^{1/3} = 9.0 \cdot 10^{-8}$. Moreover $A_3/B = 3.16,\ 10.0,\ 100.0$ at $p = 10^{-3}, 10^{-6}, 10^{-12}$ (it grows like $p^{-1/6}$), so no bound $A_r \lesssim rB$ uniform in p exists. The earlier proof of the regime $r \le \log(1/p)$ rested on this, and the regime $r \ge \log(1/p)$ was left unproved. The proof above replaces both.

Chapter 15 — Moment Inequalities

Exercise 15.1

Central idea. Extend x_q to a smooth function $X(t)$ for real $t \geq 4$ and show $X'(t) \geq 0$ by bounding the logarithmic derivative $X'/X = a(t) - \frac{Cw(t)}{1+Cw(t)} b(t)$ with explicit elementary Taylor bounds; since $\lim_{t\to\infty} X(t) = 1$, monotonicity forces $x_q \leq 1$ for all integers $q \geq 4$.

Solution. Recall $K = 1/(e - \sqrt{e})$ (p. 444), so with $C := 1/K = e - \sqrt{e}$,

$$x_q = \left(\frac{q-1}{q}\right)^{q/2} \left(1 + C\left(\frac{q-2}{q-1}\right)^{(q-2)/2}\right), \qquad q \geq 4.$$

For real $t \geq 4$ define

$$X(t) = \left(1 - \frac{1}{t}\right)^{t/2} \left[1 + C\left(1 - \frac{1}{t-1}\right)^{(t-2)/2}\right],$$

so $X(q) = x_q$ for integer $q \geq 4$. Write $X(t) = e^{u(t)}(1 + Ce^{v(t)})$ with

$$u(t) = \tfrac{t}{2} \log\left(1 - \tfrac{1}{t}\right), \qquad v(t) = \tfrac{t-2}{2} \log\left(1 - \tfrac{1}{t-1}\right).$$

Then, with $w(t) = e^{v(t)}$,

$$\frac{X'(t)}{X(t)} = u'(t) + \frac{Cw(t)}{1+Cw(t)} v'(t) = a(t) - \frac{Cw(t)}{1+Cw(t)} b(t),$$

where

$$a(t) = \tfrac{1}{2} \log\left(1 - \tfrac{1}{t}\right) + \tfrac{1}{2(t-1)}, \qquad b(t) = -\tfrac{1}{2} \log\left(1 - \tfrac{1}{t-1}\right) - \tfrac{1}{2(t-1)},$$

using $u'(t) = \frac{1}{2}\log(1 - \frac{1}{t}) + \frac{1}{2(t-1)}$ and $v'(t) = \frac{1}{2}\log(1 - \frac{1}{t-1}) + \frac{1}{2(t-1)} = -b(t)$. Since $\log(1-x) < -x$ for $x \in (0,1)$, $v'(t) < 0$, so w is decreasing and $w(t) \leq w(4) = 2/3$ for $t \geq 4$; hence, $s \mapsto Cs/(1+Cs)$ being increasing,

$$\frac{Cw(t)}{1+Cw(t)} \leq \frac{\frac{2}{3}C}{1+\frac{2}{3}C}.$$

Now $\frac{\frac{2}{3}C}{1+\frac{2}{3}C} \leq \frac{7}{12} \iff C \leq \frac{21}{10}$, which holds since $C = e - \sqrt{e} < 2.8 - 1.5 < 2.1$ (using $e < 2.8$ and $e > 2.25$).

We bound a from below and b from above. For $x \in (0, \frac{1}{3}]$, $\log(1+x) \leq x - \frac{x^2}{2} + \frac{x^3}{3}$ (alternating Taylor series with decreasing terms), so with $x = \frac{1}{t-1}$,

$$\log\left(1 - \tfrac{1}{t}\right) = -\log\left(1 + \tfrac{1}{t-1}\right) \geq -\tfrac{1}{t-1} + \tfrac{1}{2(t-1)^2} - \tfrac{1}{3(t-1)^3},$$

giving $a(t) \geq \frac{1}{4(t-1)^2} - \frac{1}{6(t-1)^3} = \frac{1}{(t-1)^2}\left(\frac{1}{4} - \frac{1}{6(t-1)}\right) \geq \frac{7}{36(t-1)^2}$ for $t \geq 4$. For b, write $y = \frac{1}{t-1} \in (0, \frac{1}{3}]$; then $-\log(1-y) - y = \sum_{k\geq 2} y^k/k \leq \frac{y^2}{2} + \frac{y^3}{2} \leq \frac{2y^2}{3}$ (since $\sum_{k\geq 3} y^k/k \leq \frac{y^3}{3(1-y)} \leq \frac{y^3}{2}$), so $b(t) \leq \frac{1}{3(t-1)^2}$.

Consequently $\frac{a(t)}{b(t)} \geq \frac{7/36}{1/3} = \frac{7}{12} \geq \frac{Cw(t)}{1+Cw(t)}$, whence $\frac{X'(t)}{X(t)} \geq 0$ for all $t \geq 4$; i.e., X is nondecreasing.

Finally,

$$\lim_{q\to\infty} \left(1 - \tfrac{1}{q}\right)^{q/2} = e^{-1/2}, \qquad \lim_{q\to\infty} \left(1 - \tfrac{1}{q-1}\right)^{(q-2)/2} = e^{-1/2},$$

since $\frac{q-2}{2(q-1)} \to \frac{1}{2}$. Hence

$$\lim_{q\to\infty} x_q = e^{-1/2}\left(1 + Ce^{-1/2}\right) = e^{-1/2} + (e - \sqrt{e})e^{-1} = 1.$$

As X is nondecreasing with limit 1, $X(t) \leq 1$ for all $t \geq 4$; in particular $x_q = X(q) \leq 1$ for every integer $q \geq 4$. ■

Exercise 15.2

Central idea. Theorem 15.6 is proved by the same induction on q as Theorem 15.5, but with V in place of V^+: under the hypothesis $Z_i \le Z$ we have $(Z - Z_i)_+ = Z - Z_i$, so $V = \sum_i (Z - Z_i)_+^2$ is a genuine upper-tail variance proxy. Lemma 15.1's V-inequality carries the smaller coefficient $q/2$ (rather than $q-1$), which is exactly what removes the factor 2 from the constant. The recursion $x_q \le (1-1/q)^{q/2} + \frac{1}{2\kappa_q} x_q^{1-2/q}$ is then closed by $x_q \le 1$ using strict concavity, just as in Theorem 15.5.

Solution. We prove Theorem 15.6: assume $Z_i \le Z$ for all $1 \le i \le n$. Then for any real $q \ge 2$,

$$\|(Z - \mathbb{E}Z)_+\|_q \le \sqrt{\kappa_q q \, \|V\|_{q/2}} \le \sqrt{\kappa q \, \|V\|_{q/2}},$$

where $V = \sum_{i=1}^n (Z - Z_i)^2$, $\kappa_1 = 1$, and for $q \ge 2$,

$$\kappa_q = \frac{1}{2(1 - (1-1/q)^{q/2})},$$

so that $\kappa_q \uparrow \kappa = \frac{\sqrt{e}}{2(\sqrt{e}-1)} < 1.271$ and in particular $\kappa_q \ge 1$.

Since $Z_i \le Z$, we have $(Z - Z_i)_+ = Z - Z_i \ge 0$, hence $V = \sum_i (Z - Z_i)_+^2$. We argue by induction on $k \ge 1$ that for all $q \in (k, k+1]$,

$$\|(Z - \mathbb{E}Z)_+\|_q^q \le \left(q\kappa_q \|V\|_{q/2}\right)^{q/2}, \tag{$*$}$$

i.e., with $x_q := \|(Z - \mathbb{E}Z)_+\|_q^q / (q\kappa_q \|V\|_{q/2})^{q/2}$, that $x_q \le 1$.

For $k = 1$ ($1 < q \le 2$), H"older's inequality and Efron–Stein give

$$\|(Z - \mathbb{E}Z)_+\|_q^2 \le \|(Z - \mathbb{E}Z)_+\|_2^2 = \mathbb{E}[(Z - \mathbb{E}Z)_+^2] \le \mathrm{Var}(Z) \le \mathbb{E}V = \|V\|_1 \le \|V\|_{q/2},$$

using Lyapunov's inequality ($1 \le q/2$). Since $q\kappa_q \ge 1$, this yields $(*)$.

Let $k > 1$ and $q \in (k, k+1]$. Apply Lemma 15.1 (first inequality) with $\alpha = q - 1$:

$$\|(Z - \mathbb{E}Z)_+\|_q^q \le \|(Z - \mathbb{E}Z)_+\|_{q-1}^q + \frac{q}{2} \, \mathbb{E}[V (Z - \mathbb{E}Z)_+^{q-2}].$$

By H"older's inequality with exponents $q/2$ and $q/(q-2)$,

$$\mathbb{E}[V (Z - \mathbb{E}Z)_+^{q-2}] \le \|V\|_{q/2} \, \|(Z - \mathbb{E}Z)_+\|_q^{q-2}.$$

Dividing by $(q\kappa_q \|V\|_{q/2})^{q/2}$ and writing $\|(Z-\mathbb{E}Z)_+\|_{q-1}^q = x_{q-1}^{q/(q-1)} \left((q-1)\kappa_{q-1} \|V\|_{(q-1)/2}\right)^{q/2}$, we obtain, using $\|V\|_{(q-1)/2} \le \|V\|_{q/2}$ and $\kappa_{q-1} \le \kappa_q$,

$$x_q \le x_{q-1}^{q/(q-1)} \Big(1 - \frac{1}{q}\Big)^{q/2} + \frac{1}{2\kappa_q} \, x_q^{1-2/q}.$$

Indeed, the second term equals $\frac{q}{2} \|V\|_{q/2} \, x_q^{1-2/q} (q\kappa_q \|V\|_{q/2})^{(q-2)/2} / (q\kappa_q \|V\|_{q/2})^{q/2} = \frac{1}{2\kappa_q} x_q^{1-2/q}$; the absence of the factor 2 (present in Theorem 15.5) is due to the coefficient $q/2$ in Lemma 15.1's V-inequality versus $q-1$ in the V^+-inequality.

By the induction hypothesis $x_{q-1} \le 1$ (as $q - 1 \in (k-1, k]$), so

$$x_q \le \Big(1 - \frac{1}{q}\Big)^{q/2} + \frac{1}{2\kappa_q} \, x_q^{1-2/q} = 1 - \frac{1}{2\kappa_q} + \frac{1}{2\kappa_q} \, x_q^{1-2/q},$$

where we used the defining identity $(1-1/q)^{q/2} = 1 - \frac{1}{2\kappa_q}$. The function $f_q(x) = 1 - \frac{1}{2\kappa_q} + \frac{1}{2\kappa_q} x^{1-2/q} - x$ is strictly concave on $[0, \infty)$ (since $0 < 1 - 2/q < 1$), satisfies $f_q(0) = 1 - \frac{1}{2\kappa_q} > 0$ and $f_q(1) = 0$; hence $f_q(x_q) \ge 0$ forces $x_q \le 1$. This proves $(*)$.

Taking q-th roots gives $\|(Z - \mathbb{E}Z)_+\|_q \le \sqrt{\kappa_q q \, \|V\|_{q/2}}$, and $\kappa_q \le \kappa$ yields the second inequality. ∎

Exercise 15.3

Central idea. Under $0 \le Z - Z_i \le M$, the V-version of Lemma 15.3 applies, giving the recurrence $m_q^q \le m_{q-1}^q + \frac{q}{2}\mathbb{E}[V((Z-\mathbb{E}Z)_- + M)^{q-2}]$ with coefficient $q/2$ instead of the q in the V^+-version used for Theorem 15.7. The base case $q \in [2,3)$ follows by subadditivity as in Theorem 15.7. For $q \ge 3$, the θ-splitting of $((Z-\mathbb{E}Z)_- + M)^{q-2}$ does NOT close the induction: its last-term factor $(1 + 1/\sqrt{C_2 q})^{q-3}$ is unbounded in q (it grows like $e^{\sqrt{q/C_2}}$), so no finite C_2 works that way. Instead, split on the size of m_q itself: if $m_q \le \sqrt{C_2}\, q\|M\|_q$ then $x_q \le 1$ directly; otherwise $(1+\|M\|_q/m_q)^{q-2} \le e^{1/\sqrt{C_2}}$, and the concavity argument closes with C_2 solving $C_2 = e^{1/\sqrt{C_2}}/(2(1-e^{-1/2})) < 2.42$.

Solution. Write $m_q = \|(Z-\mathbb{E}Z)_-\|_q$ and $c_q = \|V\|_{1\vee q/2} \vee q\|M\|_q^2$. Since $0 \le Z - Z_i \le M$, Lemma 15.3 (V-version) with $\alpha = q-1$ gives, for $q \ge 2$,

$$m_q^q \le m_{q-1}^q + \frac{q}{2}\mathbb{E}\big[V\big((Z-\mathbb{E}Z)_- + M\big)^{q-2}\big]. \tag{$*$}$$

For $q \in [1,2]$, H"older and Efron–Stein ($\mathrm{Var}(Z) \le \mathbb{E}V$) give $m_q \le \sqrt{\|V\|_1}$; hence $m_{q-1} \le \sqrt{c_{q-1}} \le \sqrt{c_q}$ for $q \in [2,3)$. If $c_q = 0$ then $(*)$ and the base case give $m_q = 0$; assume $c_q > 0$ and put $x_q = (m_q/\sqrt{C_2 q c_q})^q$. Note c_q is nondecreasing in q (Lyapunov's inequality for $\|V\|_{q/2}$, and $q\|M\|_q^2$ is a product of nondecreasing factors).

Base case $q \in [2,3)$. By subadditivity of $x \mapsto x^{q-2}$ and H"older applied to $(*)$,

$$m_q^q \le m_{q-1}^q + \frac{q}{2}\|V\|_{q/2}\|M\|_q^{q-2} + \frac{q}{2}\|V\|_{q/2} m_q^{q-2}.$$

Since $\|V\|_{q/2} \le c_q$ and $\|M\|_q^{q-2} \le (c_q/q)^{(q-2)/2}$, the middle term is at most $\frac{1}{2}q^{(4-q)/2}c_q^{q/2} \le \frac{3}{2}c_q^{q/2}$ for $q \in [2,3)$, because $\frac{4-q}{2}\ln q < \ln 3$ there, so $q^{(4-q)/2} < 3$. Dividing by $(C_2 q c_q)^{q/2}$ and using $(C_2 q)^{-q/2} \le 1/(2C_2)$ for $q \ge 2$,

$$x_q \le \frac{1}{2C_2} + \frac{3}{4C_2} + \frac{1}{2C_2}x_q^{1-2/q}.$$

The function $g_q(x) = \frac{5}{4C_2} + \frac{1}{2C_2}x^{1-2/q} - x$ is strictly concave on $[0,\infty)$, $g_q(0) > 0$, and $g_q(1) = \frac{7}{4C_2} - 1 < 0$ since $C_2 > 2.4 > 7/4$; since $g_q(x_q) \ge 0$, concavity forces $x_q \le 1$, i.e. $m_q \le \sqrt{C_2 q c_q}$ for $q \in [2,3)$.

Induction step $q \ge 3$. By H"older (exponents $q/2$, $q/(q-2)$) and Minkowski applied to $(*)$,

$$m_q^q \le m_{q-1}^q + \frac{q}{2}\|V\|_{q/2}\big\|(Z-\mathbb{E}Z)_- + M\big\|_q^{q-2} \le m_{q-1}^q + \frac{q}{2}c_q\big(m_q + \|M\|_q\big)^{q-2}. \tag{$**$}$$

We argue by induction on $k \ge 2$ for $q \in [k, k+1)$, assuming $m_{q-1} \le \sqrt{C_2(q-1)c_{q-1}}$. Then

$$\frac{m_{q-1}^q}{(C_2 q c_q)^{q/2}} \le \left(\frac{q-1}{q}\right)^{q/2} \le e^{-1/2}. \tag{$***$}$$

If $m_q \le \sqrt{C_2}\, q\|M\|_q$, then since $\|M\|_q \le \sqrt{c_q/q}$,

$$m_q \le \sqrt{C_2}\, q\sqrt{c_q/q} = \sqrt{C_2 q c_q}, \qquad \text{i.e. } x_q \le 1,$$

directly. Otherwise $\|M\|_q/m_q < 1/(\sqrt{C_2}q)$, and

$$\big(m_q + \|M\|_q\big)^{q-2} = m_q^{q-2}\left(1 + \frac{\|M\|_q}{m_q}\right)^{q-2} \le m_q^{q-2}\left(1 + \frac{1}{\sqrt{C_2}q}\right)^{q-2} \le e^{1/\sqrt{C_2}} m_q^{q-2},$$

using $1 + x \le e^x$. Inserting into $(**)$ and dividing by $(C_2 q c_q)^{q/2}$, with $(***)$,

$$x_q \le \left(1 - \frac{1}{q}\right)^{q/2} + \frac{e^{1/\sqrt{C_2}}}{2C_2}\, x_q^{1-2/q} =: f_q(x_q),$$

since $\frac{q}{2}c_q e^{1/\sqrt{C_2}} m_q^{q-2}/(C_2 q c_q)^{q/2} = \frac{e^{1/\sqrt{C_2}}}{2C_2} x_q^{(q-2)/q}$. The function f_q is strictly concave on $[0,\infty)$ (as $0 < (q-2)/q < 1$) with $f_q(0) = (1-1/q)^{q/2} > 0$. Defining C_2 by $e^{-1/2} + e^{1/\sqrt{C_2}}/(2C_2) = 1$, i.e.

$$C_2 = \frac{e^{1/\sqrt{C_2}}}{2(1-e^{-1/2})} \approx 2.4176 < 2.42,$$

gives $f_q(1) \le e^{-1/2} + e^{1/\sqrt{C_2}}/(2C_2) - 1 = 0$. Since $g_q := f_q - \mathrm{id}$ is strictly concave with $g_q(0) > 0 \ge g_q(1)$ and $g_q(x_q) \ge 0$, strict concavity forces $x_q \le 1$ (past the unique zero of g_q in $(0,1]$, g_q is strictly negative).

Thus $m_q \le \sqrt{C_2 q c_q}$ for all $q \ge 2$, i.e.

$$\|(Z-\mathbb{E}Z)_-\|_q \le \sqrt{C_2 q(\|V\|_{q/2} \vee q\|M\|_q^2)}, \qquad q \ge 2,$$

with $C_2 < 2.42$. ■

Remark (audit note). An earlier draft bounded the $q \ge 3$ step by the θ-split $\left((Z-\mathbb{E}Z)_- + M\right)^{q-2} \le \theta^{-q+3} M^{q-2} + (1-\theta)^{-q+3}(Z-\mathbb{E}Z)_-^{q-2}$ with $\theta = 1/(1+\sqrt{C_2 q})$, claiming $(1+1/\sqrt{C_2 q})^{q-2} \le e^{1/\sqrt{C_2}}$ and dropping the unbounded last-term factor $(1-\theta)^{-q+3} = (1+1/\sqrt{C_2 q})^{q-3}$ from the fixed-point equation. Both claims are false: $(1+1/\sqrt{C_2 q})^{q-3} \sim e^{\sqrt{q/C_2}} \to \infty$, so no finite C_2 closes the induction that way. The case split above repairs this. (The threshold $\sqrt{C_2}$ is optimal: with threshold $\beta q\|M\|_q$ the fixed point is $C = e^{1/\beta}/(2(1-e^{-1/2}))$ subject to $\beta \le \sqrt{C}$, minimized at $\beta = \sqrt{C}$.)

Exercise 15.4

Central idea. Exercise 15.3 bounds the lower-tail moment by $\|V\|_{q/2}$; from $V \le WZ$, Hölder gives $\|V\|_{q/2} \le \|W\|_q \|Z\|_q$, and mimicking the proof of Corollary 15.8—Theorem 15.6 for the upper tail, then closing the loop on $\|Z\|_q$ with $\sqrt{ab} \le (a+b)/2$—yields $\|Z\|_q \le 2\mathbb{E}Z + 2q\|W\|_q$, using $\kappa_q < 1.271$ from Theorem 15.5. Substituting back into Exercise 15.3 gives the estimate.

Solution. Let $q \ge 2$ and assume $Z \ge 0$, the setting of Corollary 15.8. Since $0 \le Z - Z_i \le M$ for all i, Exercise 15.3 applies and gives

$$\|(Z-\mathbb{E}Z)_-\|_q \le \sqrt{C_2 q\left(\|V\|_{q/2} \vee q\|M\|_q^2\right)}, \qquad C_2 < 2.42. \tag{1}$$

Because $0 \le V \le WZ$ with $W \ge 0$, Hölder's inequality (with $\frac{1}{q/2} = \frac{1}{q} + \frac{1}{q}$) yields

$$\|V\|_{q/2} \le \|WZ\|_{q/2} \le \|W\|_q \|Z\|_q. \tag{2}$$

Since $Z_i \le Z$ for every i, Theorem 15.6 applies to the upper tail. With $\kappa_q = \frac{1}{2(1-(1-1/q)^{q/2})} \uparrow \frac{\sqrt{e}}{2(\sqrt{e}-1)} < 1.271$,

$$\|(Z-\mathbb{E}Z)_+\|_q \le \sqrt{\kappa_q q\|V\|_{q/2}} \overset{(2)}{\le} \sqrt{\kappa_q q\|W\|_q\|Z\|_q}. \tag{3}$$

As $Z \ge 0$, $Z \le \mathbb{E}Z + (Z-\mathbb{E}Z)_+$ holds pointwise, so Minkowski's inequality gives

$$\|Z\|_q \le \mathbb{E}Z + \|(Z-\mathbb{E}Z)_+\|_q \le \mathbb{E}Z + \sqrt{\kappa_q q\|W\|_q\|Z\|_q}. \tag{4}$$

Using $\sqrt{ab} \le \frac{a+b}{2}$ with $a = \|Z\|_q$ and $b = \kappa_q q\|W\|_q$,

$$\|Z\|_q \le \mathbb{E}Z + \tfrac{1}{2}\|Z\|_q + \tfrac{1}{2}\kappa_q q\|W\|_q,$$

and since $\kappa_q < 1.271 < 2$,

$$\|Z\|_q \le 2\mathbb{E}Z + \kappa_q q\|W\|_q \le 2\mathbb{E}Z + 2q\|W\|_q. \tag{5}$$

This is the first assertion of Corollary 15.8. Combining (2) and (5),

$$\|V\|_{q/2} \le \|W\|_q \left(2\mathbb{E}Z + 2q\|W\|_q\right). \tag{6}$$

Inserting (6) into (1) yields

$$\|(Z-\mathbb{E}Z)_-\|_q \le \sqrt{C_2 q \left(\|W\|_q \left(2\mathbb{E}Z + 2q\|W\|_q\right) \vee q\|M\|_q^2\right)}. \tag{7}$$

Remark on the stated form. The inequality as printed has $\|M\|_q$ in place of the outer $\|W\|_q$ in (7); the two coincide when $\|W\|_q \le \|M\|_q$, e.g. when $W \le M$ a.s., as in Theorem 15.13 where the choice $W = M$ is made. Equation (7) is the precise form delivered by combining Exercise 15.3 with the proof of Corollary 15.8. ■

Exercise 15.5

Central idea. Symmetrization replaces the centered sum $\sum_i X_i$ by the Rademacher sum $\sum_i \varepsilon_i X_i$ at the cost of a factor 2; conditioning on the X_i, Theorem 15.9 (Khinchine's inequality) bounds the q-th moment of the Rademacher sum by $(2^{1/q}\sqrt{2Kq})^q (\sum_i X_i^2)^{q/2}$, and taking expectations over X turns $(\sum_i X_i^2)^{q/2}$ into $\|\sum_i X_i^2\|_{q/2}^{q/2}$. The constant $2^{1+1/q}$ is exactly the symmetrization factor 2 times Khinchine's $2^{1/q}$.

Solution. Let $X_1', \dots, X_n'$ be independent copies of $X_1, \dots, X_n$ and let $\varepsilon_1, \dots, \varepsilon_n$ be i.i.d. Rademacher variables independent of all the X_i, X_i'. If $\|\sum_i X_i^2\|_{q/2} = \infty$ there is nothing to prove, so assume it is finite; then $\mathbb{E}|X_i|^q \le \mathbb{E}[(\sum_j X_j^2)^{q/2}] < \infty$ for each i, so all moments below exist. Theorem 15.9 is stated for integer $q \ge 2$, but its proof (via Theorem 15.4 and Lemma 15.1, the latter stated for real $q > 2$) works verbatim for every real $q \ge 2$, which we use.

Since $\mathbb{E}X_i = 0$, writing $\mathbb{E}_{X'}$ for expectation over the copies,

$$\sum_{i=1}^n X_i = \mathbb{E}_{X'}\Big[\sum_{i=1}^n (X_i - X_i')\Big].$$

By Jensen's inequality for conditional expectations ($|\cdot|^q$ is convex for $q \ge 1$),

$$\Big|\sum_i X_i\Big|^q \le \mathbb{E}_{X'}\Big|\sum_i (X_i - X_i')\Big|^q,$$

and taking $\mathbb{E}_X$ gives $\|\sum_i X_i\|_q \le \|\sum_i (X_i - X_i')\|_q$. Each $X_i - X_i'$ is symmetric, so $\sum_i (X_i - X_i') \overset{d}{=} \sum_i \varepsilon_i (X_i - X_i')$, and by Minkowski's inequality,

$$\Big\|\sum_i X_i\Big\|_q \le \Big\|\sum_i \varepsilon_i (X_i - X_i')\Big\|_q \le \Big\|\sum_i \varepsilon_i X_i\Big\|_q + \Big\|\sum_i \varepsilon_i X_i'\Big\|_q = 2\Big\|\sum_i \varepsilon_i X_i\Big\|_q. \tag{1}$$

Condition on $X_1, \dots, X_n$. Theorem 15.9 applied to the Rademacher sum $\sum_i \varepsilon_i a_i$ with the (nonrandom, given the conditioning) coefficients $a_i = X_i$ gives, for $q \ge 2$,

$$\mathbb{E}_\varepsilon\Big|\sum_i \varepsilon_i X_i\Big|^q \le \Big(2^{1/q}\sqrt{2Kq}\Big)^q \Big(\sum_i X_i^2\Big)^{q/2} = 2(2Kq)^{q/2}\Big(\sum_i X_i^2\Big)^{q/2},$$

where $K = 1/(e - \sqrt{e}) < 0.935$ is the constant from Theorems 15.4 and 15.9. (The proof of Theorem 15.9 uses only a_i^2, via $V^+ \le 2\sum_i a_i^2$, so it applies to arbitrary real a_i, not just $a_i > 0$.) Taking expectations over X,

$$\mathbb{E}\Big|\sum_i \varepsilon_i X_i\Big|^q \le 2(2Kq)^{q/2}\,\mathbb{E}\Big[\Big(\sum_i X_i^2\Big)^{q/2}\Big] = 2(2Kq)^{q/2}\Big\|\sum_i X_i^2\Big\|_{q/2}^{q/2},$$

and taking q-th roots,

$$\Big\|\sum_i \varepsilon_i X_i\Big\|_q \le 2^{1/q}\sqrt{2Kq}\,\sqrt{\Big\|\sum_i X_i^2\Big\|_{q/2}}. \tag{2}$$

Combining (1) and (2) yields, for every $q \geq 2$,

$$\left\| \sum_{i=1}^{n} X_i \right\|_q \leq 2^{1+1/q} \sqrt{2Kq} \sqrt{\left\| \sum_{i=1}^{n} X_i^2 \right\|_{q/2}},$$

which is Marcinkiewicz' inequality. ■

Exercise 15.6

Central idea. Write $Z = \sum_{i=1}^{n} Y_i$ with $Y_i = a_i(X_i^4 - 3)$ independent and centered: the variance is then exact by independence, and for $\|Z_+\|_q$ the right tool is the Rosenthal-type inequality of Theorem 15.11, which bounds $\|Z_+\|_q$ by the variance term $\sigma\sqrt{q}$ plus a maximum term $q\|Y\|_q$ with $Y = \max_i |Y_i|$; the only remaining work is the Gaussian moment $\|X^4 - 3\|_q$, controlled via $(4q-1)!! \leq (4q)^{2q}$.

Solution. Put $Y_i = a_i(X_i^4 - 3)$, so the Y_i are independent, $\mathbb{E}Y_i = 0$, and $Z = \sum_{i=1}^{n} Y_i$ with $\mathbb{E}Z = 0$. For a standard Gaussian, $\mathbb{E}X^{2k} = (2k-1)!!$, hence $\mathbb{E}X^4 = 3!! = 3 = \mu$ and $\mathbb{E}X^8 = 7!! = 105$. Therefore

$$\operatorname{Var}(Y_i) = a_i^2(\mathbb{E}X_i^8 - (\mathbb{E}X_i^4)^2) = 96a_i^2,$$

and by independence

$$\operatorname{Var}(Z) = \sum_{i=1}^{n} \operatorname{Var}(Y_i) = 96 \sum_{i=1}^{n} a_i^2.$$

For the q-th moment of the positive part, apply Theorem 15.11 to the centered independent variables $Y_1, \dots, Y_n$. With $\sigma^2 = \sum_{i=1}^{n} \mathbb{E}Y_i^2 = 96 \sum_{i=1}^{n} a_i^2$ and $Y = \max_{1 \leq i \leq n} |Y_i|$, for every integer $q \geq 2$,

$$\|Z_+\|_q \leq \sigma\sqrt{6\kappa q} + q\kappa\sqrt{2}\,\|Y\|_q,$$

where κ is the universal constant from Theorem 15.5. Now

$$\|Y\|_q^q = \mathbb{E} \max_{1 \leq i \leq n} |a_i|^q |X_i^4 - 3|^q \leq \sum_{i=1}^{n} |a_i|^q \, \mathbb{E}|X^4 - 3|^q,$$

so $\|Y\|_q \leq \big(\sum_{i=1}^{n} |a_i|^q\big)^{1/q} \|X^4 - 3\|_q$. By Minkowski's inequality,

$$\|X^4 - 3\|_q \leq \|X^4\|_q + 3 = \big((4q-1)!!\big)^{1/q} + 3.$$

Since $(2m-1)!! = \prod_{j=1}^{m}(2j-1) \leq \prod_{j=1}^{m} 2j = 2^m m! \leq (2m)^m$, with $2m = 4q$,

$$\big((4q-1)!!\big)^{1/q} \leq \big((4q)^{2q}\big)^{1/q} = 16q^2,$$

hence $\|X^4 - 3\|_q \leq 16q^2 + 3 \leq 17q^2$ for $q \geq 2$. Consequently, for integer $q \geq 2$,

$$\|Z_+\|_q \leq 24\sqrt{\kappa q \sum_{i=1}^{n} a_i^2} + 17\sqrt{2}\,\kappa q^3 \Big(\sum_{i=1}^{n} |a_i|^q \Big)^{1/q}.$$

For real $q \geq 2$, monotonicity of L_q norms gives $\|Z_+\|_q \leq \|Z_+\|_{\lceil q \rceil}$, and the same bound holds with q replaced by $\lceil q \rceil \leq q + 1$.

Exercise 15.7

Central idea. This is the Gluskin–Kwapień splitting: a Bernstein-type tail for sums of symmetric exponentials gives the crude moment bound $\|Z\|_p \lesssim \sqrt{p}\,\sigma + pM$; splitting the sum at the p-th largest weight, the

first block is controlled by the exponential term ($ps_1 \le p\sum_{i\le p} s_i$) and the remaining block of small weights by the Gaussian term, with monotonicity of s absorbing the leftover ps_{p+1} into $\sum_{i\le p} s_i$.

Solution. The X_i are Laplace(1): density $\frac12 e^{-|x|}$, $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 2$, and $\mathbb{E}e^{\lambda X_i} = (1-\lambda^2)^{-1}$ for $|\lambda| < 1$. For real $p \ge 2$ interpret $\sum_{i=1}^{p} s_i = \sum_{1\le i\le p} s_i$ and $\sum_{i=p+1}^{n} s_i^2 = \sum_{p<i\le n} s_i^2$; put $m = \lfloor p\rfloor \wedge n \ge 1$.

Bernstein bound. Let $Z = \sum_{i=1}^n a_i X_i$, $\sigma^2 = 2\sum a_i^2 = \mathrm{Var}(Z)$, $M = \max_i |a_i|$ (if $M = 0$ there is nothing to prove). For $0 \le \lambda \le 1/(2M)$ we have $\lambda^2 a_i^2 \le 1/4$, and $-\log(1-x) \le 2x$ on $[0, 1/4]$ (since $x \mapsto \log(1-x)+2x$ has derivative $(1-2x)/(1-x) \ge 0$ there), so

$$\log \mathbb{E}e^{\lambda Z} = \sum_i -\log(1-\lambda^2 a_i^2) \le 2\lambda^2 \sum_i a_i^2 = \lambda^2\sigma^2.$$

Chernoff's bound $\mathbb{P}\{Z > t\} \le e^{-\lambda t + \lambda^2\sigma^2}$ with $\lambda = t/(2\sigma^2)$ for $t \le \sigma^2/M$ and $\lambda = 1/(2M)$ for $t > \sigma^2/M$ gives, using symmetry for the lower tail,

$$\mathbb{P}\{|Z| > t\} \le 2\exp\Big(-\tfrac14 \min\left(t^2/\sigma^2,\ t/M\right)\Big), \qquad t \ge 0. \tag{1}$$

Tail to moment. For $q \ge 1$, $\|Z\|_q^q = \int_0^\infty q t^{q-1}\mathbb{P}\{|Z| > t\}\,dt$. With $t_0 = \sigma^2/M$ and (1),

$$\begin{aligned}\|Z\|_q^q &\le 2q\int_0^{t_0} t^{q-1}e^{-t^2/(4\sigma^2)}dt + 2q\int_{t_0}^\infty t^{q-1}e^{-t/(4M)}dt \\ &\le 2q\int_0^\infty t^{q-1}e^{-t^2/(4\sigma^2)}dt + 2q\int_0^\infty t^{q-1}e^{-t/(4M)}dt.\end{aligned}$$

With $u = t^2/(4\sigma^2)$ the first integral is $q2^q\sigma^q\Gamma(q/2)$; with $u = t/(4M)$ the second is $2q(4M)^q\Gamma(q)$ (the prefactor is $2q$, not q: the integrand carries the full $2q$ from the tail bound). Using $\Gamma(x) \le x^x$ for $x \ge 1$ (standard, from log-convexity of Γ and the integer case),

$$q2^q\sigma^q\Gamma(q/2) \le q(\sigma\sqrt{2q})^q, \qquad 2q(4M)^q\Gamma(q) \le 2q(4qM)^q.$$

Since $q^{1/q} \le 2$ for $q \ge 1$, taking q-th roots and using $(a^q + b^q)^{1/q} \le a + b$: the first term contributes at most $q^{1/q}2\sigma\sqrt{q/2} \le 2\sqrt2\sqrt{q}\,\sigma$, the second at most $(2q)^{1/q}4Mq \le 8qM$, so

$$\|Z\|_q \le C_0(\sqrt{q}\,\sigma + qM) \tag{2}$$

for an absolute constant C_0 (e.g. $C_0 = 16$), valid for every real $q \ge 1$.

Splitting. Write $Z = \sum_{i=1}^n s_i X_i = Z_1 + Z_2$ with $Z_1 = \sum_{i=1}^m s_i X_i$, $Z_2 = \sum_{i=m+1}^n s_i X_i$ (empty sum $= 0$). Apply (2) with $q = p$ to Z_1: $\sigma_1^2 = 2\sum_{i\le m} s_i^2 \le 2ms_1^2$ and $M_1 = s_1$, so with $m \le p$,

$$\|Z_1\|_p \le C_0(\sqrt{p}\sqrt{2m}\,s_1 + ps_1) \le C_0(\sqrt2 + 1)\,p\sum_{i=1}^m s_i,$$

using $s_1 \le \sum_{i\le m} s_i$ (as $s_i \ge 0$, $m \ge 1$). Apply (2) to Z_2 (if $m < n$; otherwise $Z_2 = 0$): $\sigma_2^2 = 2\sum_{i>m} s_i^2$, $M_2 = s_{m+1}$, so

$$\|Z_2\|_p \le C_0\sqrt{2p\Big(\sum_{i=m+1}^n s_i^2\Big)} + C_0 ps_{m+1}.$$

Since s is nonincreasing, $s_{m+1} \le s_m \le \frac1m\sum_{i=1}^m s_i$, whence $ps_{m+1} \le \frac{p}{m}\sum_{i\le m} s_i \le 2\sum_{i\le m} s_i$, because $m = \lfloor p\rfloor \ge p - 1 \ge p/2$ for $p \ge 2$ (when $\lfloor p\rfloor \le n$). Thus

$$\|Z_2\|_p \le C_0\sqrt{2p\Big(\sum_{i=m+1}^n s_i^2\Big)} + 2C_0\sum_{i=1}^m s_i.$$

Finally, by the triangle inequality and $p \geq 1$,

$$\begin{aligned}\|Z\|_p &\leq \|Z_1\|_p + \|Z_2\|_p \\ &\leq \kappa \left(p \sum_{i=1}^{m} s_i + \sqrt{p \sum_{i=m+1}^{n} s_i^2} \right)\end{aligned}$$

with $\kappa = C_0(\sqrt{2}+3) + C_0\sqrt{2}$ absolute, i.e.

$$\mathbb{E}\Big[\Big|\sum_{i=1}^{n} s_i X_i\Big|^p\Big]^{1/p} \leq \kappa \left(p \sum_{i=1}^{p} s_i + \sqrt{p \sum_{i=p+1}^{n} s_i^2} \right),$$

with the stated convention for non-integer p. ■

Exercise 15.8

Central idea. Apply the Efron–Stein inequality and the generalized Efron–Stein moment inequality (Theorem 15.5) to the supremum Z. The key step is controlling the Efron–Stein proxy V^+: resampling X_i can raise the supremum by at most $A_i|X_i^4 - (X_i')^4|$ with $A_i = \sup_s |a_{i,s}|$. Gaussian moments ($\mathbb{E}X^8 = 105$, $\|X\|_p \leq \sqrt{p}$) then give explicit constants for $\mathrm{Var}(Z)$ and for the centered moments $\|(Z - \mathbb{E}Z)_+\|_q$. The uncentered moments follow from $Z_+ \leq (Z - \mathbb{E}Z)_+ + \mathbb{E}Z$, and the mean $\mathbb{E}Z$ must be kept as a separate term (it is bounded by $\mathbb{E}|X^4 - 3| \sum_i A_i$, and it cannot be absorbed into the centered moments).

Solution. Put $\mu = \mathbb{E}X_i^4 = 3$ and $Y_s = \sum_{i=1}^n a_{i,s}(X_i^4 - 3)$, so $Z = \sup_{s\in\mathcal{T}} Y_s$ with $\mathcal{T}$ countable. Let $A_i := \sup_{s\in\mathcal{T}} |a_{i,s}|$ and $S = (\sum_i A_i^2)^{1/2}$; we assume all $A_i < \infty$ (otherwise the bounds below are trivial). Then $Y_{s_0} \leq Z \leq \sum_i A_i|X_i^4 - 3|$ for any fixed s_0, so Z is integrable, and $\mathbb{E}Z \geq \mathbb{E}Y_{s_0} = 0$.

For each i, let X_i' be an independent copy of X_i, $X^{(i)}$ the vector X with X_i replaced by X_i', and $Z^{(i)} = \sup_s Y_s^{(i)}$ defined analogously. Since $\sup_s Y_s \leq \sup_s (Y_s - Y_s^{(i)}) + \sup_s Y_s^{(i)}$,

$$(Z - Z^{(i)})_+ \leq \Big(\sup_{s\in\mathcal{T}}(Y_s - Y_s^{(i)})\Big)_+ = \sup_{s\in\mathcal{T}}(Y_s - Y_s^{(i)})_+,$$

and $Y_s - Y_s^{(i)} = a_{i,s}(X_i^4 - (X_i')^4)$. Therefore

$$(Z - Z^{(i)})_+ \leq A_i\,|X_i^4 - (X_i')^4|, \qquad (Z - Z^{(i)})_+^2 \leq A_i^2\,(X_i^4 - (X_i')^4)^2.$$

Hence, with $V^+ = \sum_{i=1}^n \mathbb{E}[(Z - Z^{(i)})_+^2 \mid X]$ as in Theorem 15.5,

$$V^+ \leq \sum_{i=1}^{n} A_i^2\,\mathbb{E}[(X_i^4 - (X_i')^4)^2 \mid X_i] = \sum_{i=1}^{n} A_i^2\,(X_i^8 - 6X_i^4 + 105),$$

since $\mathbb{E}[(X_i')^4] = 3$ and $\mathbb{E}[(X_i')^8] = 105$.

Variance. By the Efron–Stein inequality (Theorem 3.1), $\mathrm{Var}(Z) \leq \mathbb{E}V^+$, and $\mathbb{E}[X_i^8 - 6X_i^4 + 105] = 105 - 18 + 105 = 192$, so

$$\mathrm{Var}(Z) \leq 192 \sum_{i=1}^{n} \sup_{s\in\mathcal{T}} a_{i,s}^2.$$

Centered moments. For $q \geq 2$, Theorem 15.5 gives $\|(Z - \mathbb{E}Z)_+\|_q \leq \sqrt{2\kappa_q q}\,\|\sqrt{V^+}\|_q$ with $\kappa_q = \frac{1}{2}(1 - (1 - 1/q)^{q/2})^{-1} \uparrow \kappa = \frac{\sqrt{e}}{2(\sqrt{e}-1)} < 1.271$ (the $\sqrt{q}$ is essential: e.g. for a Rademacher sum, $\|Z_+\|_q \asymp \sqrt{qn}$, so no q-independent multiple of $\|\sqrt{V^+}\|_q$ can work). By Minkowski in $L^{q/2}$ and the triangle inequality,

$$\|V^+\|_{q/2} \leq \sum_{i=1}^{n} A_i^2\,\|X_i^8 - 6X_i^4 + 105\|_{q/2} \leq \sum_{i=1}^{n} A_i^2(\|X_i^8\|_{q/2} + 6\|X_i^4\|_{q/2} + 105).$$

For $X \sim N(0,1)$, $\|X\|_p \le \sqrt{p}$ $(p \ge 1)$, so $\|X_i^8\|_{q/2} = \|X_i\|_{4q}^8 \le (4q)^4 = 256q^4$ and $\|X_i^4\|_{q/2} = \|X_i\|_{2q}^4 \le (2q)^2 = 4q^2$. For $q \ge 2$, $256q^4 + 24q^2 + 105 \le 269q^4$, whence $\|\sqrt{V^+}\|_q = \|V^+\|_{q/2}^{1/2} \le \sqrt{269}\, q^2 S$ and

$$\|(Z - \mathbb{E}Z)_+\|_q \le \sqrt{538\kappa_q q}\; q^2 \Big(\sum_{i=1}^n \sup_{s\in\mathcal{T}} a_{i,s}^2\Big)^{1/2} \le \sqrt{538\kappa}\; q^{5/2} S, \qquad q \ge 2.$$

Uncentered moments. Since $\mathbb{E}Z \ge 0$, $Z_+ \le (Z - \mathbb{E}Z)_+ + \mathbb{E}Z$ pointwise (trivial if $Z \le 0$; otherwise $Z = (Z - \mathbb{E}Z) + \mathbb{E}Z$). By Minkowski,

$$\|Z_+\|_q \le \mathbb{E}Z + \sqrt{538\kappa_q q}\; q^2 S, \qquad q \ge 2.$$

Moreover $\mathbb{E}Z \le \mathbb{E}\sum_i A_i |X_i^4 - 3| = c_4 \sum_{i=1}^n A_i$ with $c_4 := \mathbb{E}|X^4 - 3| = 4.1801\ldots$, so explicitly

$$\|Z_+\|_q \le 4.181 \sum_{i=1}^n \sup_{s\in\mathcal{T}} |a_{i,s}| + \sqrt{538\kappa_q q}\; q^2 \Big(\sum_{i=1}^n \sup_{s\in\mathcal{T}} a_{i,s}^2\Big)^{1/2}, \qquad q \ge 2.$$

■

Remark (the mean term is necessary). No bound of the form $\|Z_+\|_q \le C(q)\, S$ holds with $C(q)$ independent of n. Take $\mathcal{T} = \{\pm 1\}^n$ and $a_{i,s} = s_i/\sqrt{n}$. Then $S = 1$ and $Z = n^{-1/2}\sum_i |X_i^4 - 3|$, so $\mathbb{E}Z = c_4\sqrt{n}$ is unbounded in n. In this example the $\mathbb{E}Z$ bound $c_4 \sum_i A_i = c_4\sqrt{n}$ is attained with equality.

Erratum. A previous version of this file concluded

$$\|Z_+\|_q \le 2\sqrt{538\kappa_q q}\; q^2 S \le C q^{5/2} S, \qquad C = 2\sqrt{538\kappa} < 53,$$

via the step $(\mathbb{E}Z)_+ \le \mathbb{E}[(Z - \mathbb{E}Z)_+]$. That step is false: the right side depends only on fluctuations, the left side on the location. Counterexample (the example of the Remark, $n = 10^4$, $q = 2$): $c_4 = 4.180129$ and $\mathbb{E}(X^4 - 3)^2 = 96$ (numerical integration), so $\mathbb{E}Z = 418.013$, $\mathrm{Var}(Z) = 96 - c_4^2 = 78.53$, and $\|Z_+\|_2 = \sqrt{(\mathbb{E}Z)^2 + \mathrm{Var}(Z)} = 418.107$ (Monte Carlo with 2000 draws: 418.10). The old bound gives $2\sqrt{538 \cdot \kappa_2 \cdot 2} \cdot 2^2 = 262.42$ (with $\kappa_2 = 1$), and $53 \cdot 2^{5/2} = 299.8$; both are below 418.1. The false step itself fails by a factor of about 47: $(\mathbb{E}Z)_+ = 418.0$, whereas $\mathbb{E}[(Z - \mathbb{E}Z)_+] \le \sqrt{\mathrm{Var}(Z)} = 8.86$. The corrected bound above gives $418.01 + \sqrt{538 \cdot 2} \cdot 4 = 549.2 \ge 418.1$. The variance bound and the centered bound of the earlier version are unaffected and are retained above.

Exercise 15.9

Central idea. Apply Bernstein's inequality at the inflated deviation $s = \theta\mathbb{E}Z + t$; the $\mathbb{E}Z$-dependent part of the variance factor is absorbed into an effective scale $A = 1/3 + 2/\theta$ via $2\mathbb{E}Z \le 2s/\theta$, and the resulting exponent $s^2/(2(\sigma^2 + As)) = 1/(u+w)$ is split into a Gaussian term and an exponential term by Adamczak's elementary estimate $\exp(-1/(u+w)) \le \exp(-\lambda/u) \vee \exp(-(1-\lambda)/w)$. One caveat: as stated the exercise omits the hypothesis $\mathbb{E}Z \ge 0$, without which it is false — see the Solution.

Solution. The claimed inequality is false without the additional hypothesis $\mathbb{E}Z \ge 0$ (which holds in the intended application, e.g. Adamczak (2008), where $Z \ge 0$ a.s.). Counterexample: let $Z \equiv -1$, so $\mathbb{E}Z = -1$, and take $\sigma^2 = 2$, so the variance factor is $\sigma^2 + 2\mathbb{E}Z = 0$. Then Z satisfies Bernstein's inequality with variance factor 0 and scale factor $1/3$: for $u > 0$, $\mathbb{P}\{Z - \mathbb{E}Z \ge u\} = \mathbb{P}\{0 \ge u\} = 0 \le e^{-3u/2}$, and both sides equal 1 at $u = 0$. But with $\theta = 1$, $t = 1$, $\lambda = 1/2$, the deviation is $\theta\mathbb{E}Z + t = 0$, so the left-hand side equals $\mathbb{P}\{0 \ge 0\} = 1$, while the right-hand side is $e^{-1/8} \vee e^{-3/28} \approx 0.898 < 1$. We prove the corrected statement under $\mathbb{E}Z \ge 0$.

Hint. Let $u, v > 0$ and $0 \le \lambda \le 1$. If $\exp(-1/(u+v)) > \exp(-\lambda/u)$ and $\exp(-1/(u+v)) > \exp(-(1-\lambda)/v)$, then $1/(u+v) < \lambda/u$ and $1/(u+v) < (1-\lambda)/v$, i.e. $\lambda(u+v) > u$ and $(1-\lambda)(u+v) > v$; adding gives the contradiction $u + v > u + v$. Hence $\exp(-1/(u+v)) \le \exp(-\lambda/u) \vee \exp(-(1-\lambda)/v)$. The same

conclusion holds for $u = 0$ (with $\exp(-\lambda/0) := 0$ for $\lambda > 0$ and $:= 1$ for $\lambda = 0$): if $\lambda > 0$ the right side is $\exp(-(1-\lambda)/v) \geq \exp(-1/v)$, and if $\lambda = 0$ it equals 1.

Now fix $\theta > 0$ and $\lambda \in [0,1]$. If $t \leq 0$, then $-(1-\lambda)t \geq 0$, so $\exp(-(1-\lambda)t/(2(1/3+2/\theta))) \geq 1$ and the claimed bound is trivial. Assume $t > 0$ and put $s = \theta\mathbb{E}Z + t$. Since $\mathbb{E}Z \geq 0$ and $\theta > 0$, $s \geq t > 0$. Bernstein's inequality with variance factor $v := \sigma^2 + 2\mathbb{E}Z$ and scale factor $c := 1/3$ gives

$$\mathbb{P}\{Z - \mathbb{E}Z \geq s\} \leq \exp\left(-\frac{s^2}{2(v+cs)}\right).$$

Since $2s/\theta - 2\mathbb{E}Z = 2t/\theta \geq 0$,

$$v + cs = \sigma^2 + 2\mathbb{E}Z + cs \leq \sigma^2 + \frac{2s}{\theta} + cs = \sigma^2 + As, \qquad A := \frac{1}{3} + \frac{2}{\theta},$$

so that

$$\mathbb{P}\{Z - \mathbb{E}Z \geq \theta\mathbb{E}Z + t\} \leq \exp\left(-\frac{s^2}{2(\sigma^2 + As)}\right).$$

Writing $\frac{s^2}{2(\sigma^2+As)} = \frac{1}{u+w}$ with $u := \frac{2\sigma^2}{s^2} \geq 0$ and $w := \frac{2A}{s} > 0$, the hint yields

$$\exp\left(-\frac{s^2}{2(\sigma^2 + As)}\right) \leq \exp\left(-\frac{\lambda s^2}{2\sigma^2}\right) \vee \exp\left(-\frac{(1-\lambda)s}{2A}\right).$$

Finally $s \geq t$ gives $\exp(-\lambda s^2/(2\sigma^2)) \leq \exp(-\lambda t^2/(2\sigma^2))$ and $\exp(-(1-\lambda)s/(2A)) \leq \exp(-(1-\lambda)t/(2A))$, which is the claimed bound. ■

Exercise 15.10

Central idea. Truncate each summand at level $\rho = 8b\log n$. The truncated, recentered process has summands bounded by 2ρ, so Bousquet's inequality in the Bernstein form of Exercise 12.15 applies to it; this produces the Gaussian term $\exp(-t^2/(2(1+\delta)\sigma^2))$ and an exponential term at scale $\rho \asymp b\log n$. The remainder has an exponential moment at scale $2b$, which costs only $O(b)$ in the mean and a tail $2e^{-x/(2b)}$. Adding the pieces proves Adamczak's inequality in its original form, with "+" between the two terms. The printed "$\vee$" form follows from this only up to a factor 2 on the right-hand side; it is not proved here.

Solution. Status: incomplete. What is proved below is the "+" form (Adamczak 2008), for $n \geq 2$: for all $\varepsilon \in (0,1)$ and $\delta > 0$ there is $\kappa = \kappa(\varepsilon,\delta)$ such that for all $t > 0$,

$$\mathbb{P}\{Z \geq (1+\varepsilon)\mathbb{E}Z + t\} \leq \exp\Big(-\frac{t^2}{2(1+\delta)\sigma^2}\Big) + 3\exp\Big(-\frac{t}{\kappa\, b\log n}\Big). \tag{A}$$

Since $x + y \leq 2(x \vee y)$, (A) gives the printed inequality with its right-hand side multiplied by 2. The printed "$\vee$" form itself, with the factor 1 and a κ depending only on (ε,δ), is **not** proved; see the Gap section at the end. ($n \geq 2$ is needed for $\log n > 0$; for $n = 1$ the printed right-hand side reads $\exp(-t^2/(2(1+\delta)\sigma^2)) \vee 0$ and the claim is not meaningful in general.)

Reduction. Index over $\mathcal{T}' = \mathcal{T} \times \{\pm 1\}$ with $X_{i,(s,\epsilon)} = \epsilon X_{i,s}$. Then $\mathcal{T}'$ is countable, the $X_{i,(s,\epsilon)}$ are centered, Y_i and σ^2 are unchanged, and $Z = \sup_{\mathcal{T}'}\sum_i X_{i,(s,\epsilon)}$. Also $Z \leq \sum_i Y_i$ and $\mathbb{E}Y_i \leq b\log\mathbb{E}e^{Y_i/b} \leq b\log 2$ by Jensen, so $\mathbb{E}Z < \infty$.

Step 1 (truncation). Let $\rho = 8b\log n$ and decompose $X_{i,s} = U_{i,s} + R_{i,s}$ with

$$U_{i,s} = X_{i,s}\mathbb{1}\{Y_i \leq \rho\} - \mathbb{E}\big[X_{i,s}\mathbb{1}\{Y_i \leq \rho\}\big], \qquad R_{i,s} = X_{i,s}\mathbb{1}\{Y_i > \rho\} - \mathbb{E}\big[X_{i,s}\mathbb{1}\{Y_i > \rho\}\big].$$

The identity holds because the two subtracted means add up to $\mathbb{E}X_{i,s} = 0$. The vectors $U_i = (U_{i,s})_s$ are i.i.d. in i (each is the same fixed function of X_i), centered, satisfy $|U_{i,s}| \leq \rho + \rho = 2\rho$, and $\mathbb{E}U_{i,s}^2 = \mathrm{Var}(X_{i,s}\mathbb{1}\{Y_i \leq \rho\}) \leq \mathbb{E}X_{i,s}^2$. Put

$$Z_1 = \sup_s\Big|\sum_i U_{i,s}\Big|, \qquad Z_2 = \sup_s\Big|\sum_i R_{i,s}\Big|, \qquad R = \sum_i Y_i\mathbb{1}\{Y_i > \rho\}.$$

Then $|R_{i,s}| \le Y_i \mathbb{1}\{Y_i > \rho\} + \mathbb{E}[Y_i \mathbb{1}\{Y_i > \rho\}]$, so $Z_2 \le R + \mathbb{E}R$. Moreover $Z \le Z_1 + Z_2$ and $Z_1 \le Z + Z_2$, so $\mathbb{E}Z_1 \le \mathbb{E}Z + \mathbb{E}Z_2$.

Step 2 (the remainder). By Markov, $\mathbb{P}\{Y_1 > \rho\} \le e^{-\rho/b}\mathbb{E}e^{Y_1/b} \le 2n^{-8}$. By independence and Cauchy–Schwarz,

$$\mathbb{E}e^{R/(2b)} = \Big(\mathbb{P}\{Y_1 \le \rho\} + \mathbb{E}\big[e^{Y_1/(2b)}\mathbb{1}\{Y_1 > \rho\}\big]\Big)^n \le \Big(1 + \sqrt{\mathbb{E}e^{Y_1/b}}\sqrt{\mathbb{P}\{Y_1 > \rho\}}\Big)^n \le (1+2n^{-4})^n \le e^{2n^{-3}} \le e^{1/4}$$

for $n \ge 2$. Consequences: - By Jensen, $\mathbb{E}R \le 2b \log \mathbb{E}e^{R/(2b)} \le b/2$, hence $\mathbb{E}Z_2 \le 2\mathbb{E}R \le b$. - For $x \ge b/2$: $\mathbb{P}\{Z_2 \ge x\} \le \mathbb{P}\{R \ge x - b/2\} \le e^{-(x-b/2)/(2b)}\mathbb{E}e^{R/(2b)} \le e^{1/2}e^{-x/(2b)} \le 2e^{-x/(2b)}$. For $x < b/2$ the bound $2e^{-x/(2b)} > 1$ holds trivially.

Step 3 (splitting the event). Fix $\gamma \in (0, 1/2)$. Since $\varepsilon < 1$ and $\mathbb{E}Z_2 \le b$, we have $(1+\varepsilon)\mathbb{E}Z \ge (1+\varepsilon)(\mathbb{E}Z_1 - b) \ge (1+\varepsilon)\mathbb{E}Z_1 - 2b$. Using $Z \le Z_1 + Z_2$,

$$\{Z \ge (1+\varepsilon)\mathbb{E}Z + t\} \subseteq \{Z_1 \ge (1+\varepsilon)\mathbb{E}Z_1 + (1-\gamma)t - 2b\} \cup \{Z_2 \ge \gamma t\}.$$

Small t. If $t \le 2b/\gamma$, then $3\exp(-t/(\kappa b \log n)) \ge 3\exp(-2/(\gamma\kappa\log 2)) \ge 1$ as soon as $\kappa \ge 2/(\gamma \log 2 \log 3)$. In that case (A) is trivial. *Large t.* If $t > 2b/\gamma$, then $(1-\gamma)t - 2b \ge (1-2\gamma)t$, and

$$\mathbb{P}\{Z \ge (1+\varepsilon)\mathbb{E}Z + t\} \le \mathbb{P}\{Z_1 \ge (1+\varepsilon)\mathbb{E}Z_1 + (1-2\gamma)t\} + 2e^{-\gamma t/(2b)}.$$

Step 4 (the truncated process). Let $W = Z_1/(2\rho) = \sup_{(s,\epsilon)\in\mathcal{T}'} \sum_i \epsilon U_{i,s}/(2\rho)$. Its summands are i.i.d. in i and centered, and they are ≤ 1, so Bousquet's inequality (Theorem 12.5) and hence Exercise 12.15 apply. The wimpy variance of W is at most $\sigma^2/(4\rho^2)$, and the bound of Exercise 12.15 is increasing in the wimpy variance. Take $\eta = \varepsilon \in (0,1)$, any $\delta' > 0$, and $u = (1-2\gamma)t/(2\rho)$:

$$\mathbb{P}\{Z_1 \ge (1+\varepsilon)\mathbb{E}Z_1 + (1-2\gamma)t\} \le \exp\Big(-\frac{(1-2\gamma)^2 t^2}{2(1+\delta')\sigma^2}\Big) + \exp\Big(-\frac{\delta'(1-2\gamma)t}{32(1+\delta')(2/\varepsilon + 1/3)\, b\log n}\Big),$$

since $2(1+\delta')(2/\varepsilon + 1/3)\cdot 2\rho = 32(1+\delta')(2/\varepsilon+1/3) b\log n$.

Conclusion. Choose $\gamma = \min\big(\frac{1}{8}, \frac{1}{2}(1-(1+\delta)^{-1/4})\big)$ and $\delta' = (1+\delta)(1-2\gamma)^2 - 1$. Then $1 - 2\gamma \ge (1+\delta)^{-1/4}$ gives $\delta' \ge (1+\delta)^{1/2} - 1 > 0$, and $(1+\delta')/(1-2\gamma)^2 = 1+\delta$, so the first term equals $\exp(-t^2/(2(1+\delta)\sigma^2))$. Since $n \ge 2$, $e^{-\gamma t/(2b)} \le e^{-\gamma \log 2\, t/(2b\log n)}$. With

$$\kappa = \max\Big(\frac{32(1+\delta')(2/\varepsilon+1/3)}{\delta'(1-2\gamma)},\ \frac{2}{\gamma\log 2}\Big)$$

(note $2/(\gamma\log 2) \ge 2/(\gamma\log 2\log 3)$, so the small-$t$ condition holds as well), Steps 3–4 give, for all $t > 0$,

$$\mathbb{P}\{Z \ge (1+\varepsilon)\mathbb{E}Z + t\} \le \exp\Big(-\frac{t^2}{2(1+\delta)\sigma^2}\Big) + 3\exp\Big(-\frac{t}{\kappa\, b\log n}\Big).$$

Here κ depends only on (ε, δ). This proves (A), and hence the printed inequality with its right-hand side doubled. ■ (for (A))

Numerical checks (Python). $\max_{n\ge 2} n\log(1+2n^{-4}) = 0.236 \le 1/4$ and $e^{1/2} = 1.649 \le 2$.

Gap: the "$\vee$" form. Write $G_\delta = \exp(-t^2/(2(1+\delta)\sigma^2))$ and $E_\kappa = 3\exp(-t/(\kappa b\log n))$. The printed claim is $\mathbb{P}\{\cdot\} \le G_\delta \vee E_\kappa$. Proving (A) with any constants (δ', κ'), $\delta' \le \delta$, does not yield it by comparing right-hand sides, for any κ: - If $\delta' = \delta$, then $G_\delta + E_{\kappa'} > G_\delta \vee E_\kappa$ whenever $G_\delta \ge E_\kappa$. - If $\delta' < \delta$, fix $t > \kappa \log 3\, b\log n$, so that $E_\kappa < 1$, and let $\sigma \to \infty$. Then $G_{\delta'} + E_{\kappa'} \to 1 + E_{\kappa'} > 1$, while $G_\delta \vee E_\kappa \le 1$. Numerically, with $\delta = 1$, $\delta' = 0.5$, $\kappa' = 1$, $\kappa = 10$, $b\log n = 1$, $\sigma = 10^3$ and $t = 11.10$, the "$+$" side is 1.0000045 and the "$\vee$" side is 0.99997.

So the "$\vee$" form would need κ to depend on $\sigma/(b\log n)$, or a genuinely different argument; neither is given here. This does not show that the "$\vee$" form is false. Adamczak's theorem, which the problem cites, is

the "+" form (A). The earlier version of this file wrongly called the printed "$\vee$" statement "a true theorem (Adamczak 2008)".

Why the earlier moment-method route was abandoned. The earlier version tried Theorem 15.14, $\|(Z - \mathbb{E}Z)_+\|_q \lesssim \sqrt{q}\,\sigma + q\|\max_i Y_i\|_q$, with $\|\max_i Y_i\|_q \le 9b(q + \log n)$. That lemma is correct and follows from $\mathbb{P}\{\max_i Y_i > u\} \le \min(1, 2ne^{-u/b})$. But it gives a bq^2 term in the q-th moment bound. With Markov's inequality this yields at best $\exp(-O(\sqrt{t/b}))$ tails, never $\exp(-t/(\kappa b \log n))$. The truncation argument above avoids this, because it applies Bousquet's inequality to a bounded process.